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(54) **METHODS FOR ENHANCING MICROBIAL PRODUCTION OF SPECIFIC LENGTH FATTY ALCOHOLS IN THE PRESENCE OF METHANOL**

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(57) **ABSTRACT**

The invention provides non-naturally occurring microbial organisms having a formaldehyde fixation pathway, a formate assimilation pathway, and/or a methanol metabolic pathway in combination with a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway, wherein the microbial organisms selectively produce a fatty alcohol, fatty aldehyde or fatty acid of a specified length or isopropanol. The microbial organisms provided advantageously enhance the production of substrates and/or pathway intermediates for the production of chain length specific fatty alcohols, fatty aldehydes, fatty acids or isopropanol. In some aspects, the microbial organisms of the invention have select gene disruptions or enzyme attenuations that increase production of fatty alcohols, fatty aldehydes or fatty acids. The invention additionally provides methods of using the above microbial organisms to produce a fatty alcohol, a fatty aldehyde, a fatty acid or isopropanol.

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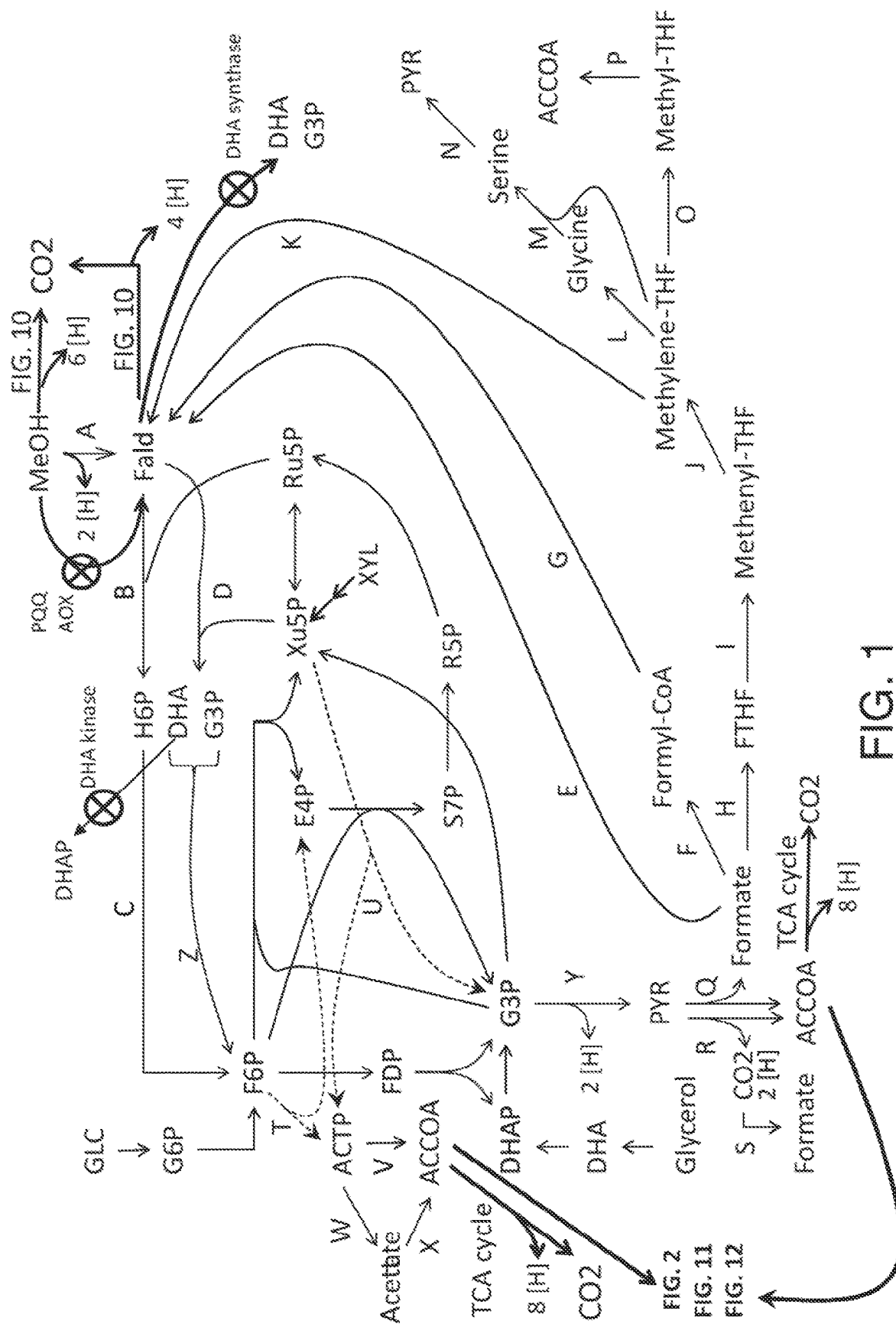
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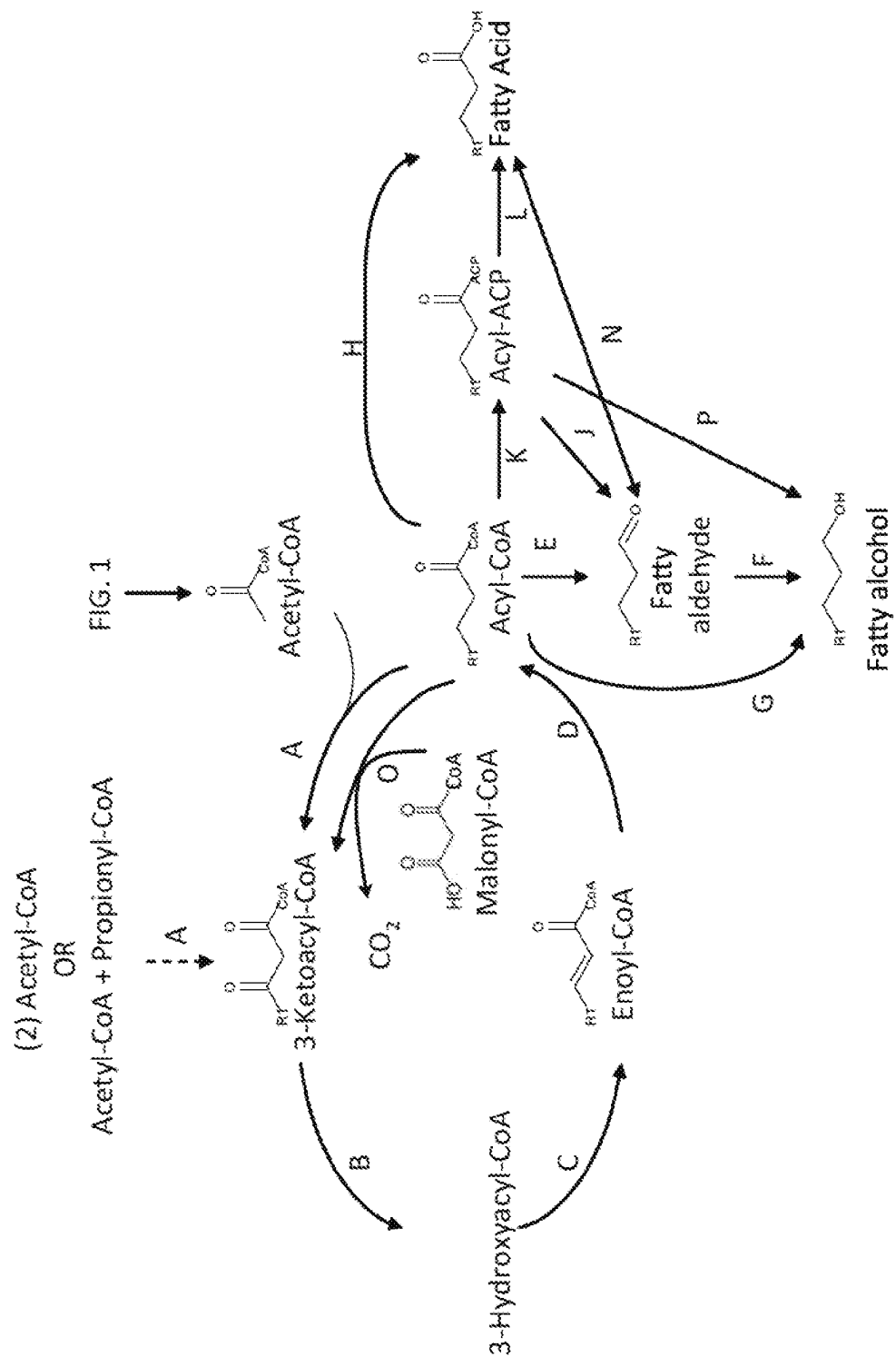
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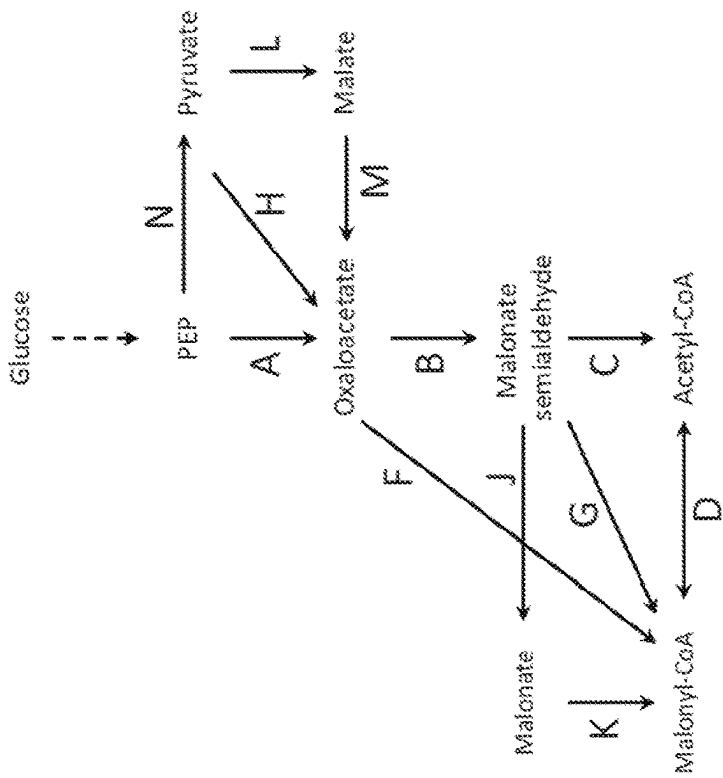


FIG. 4

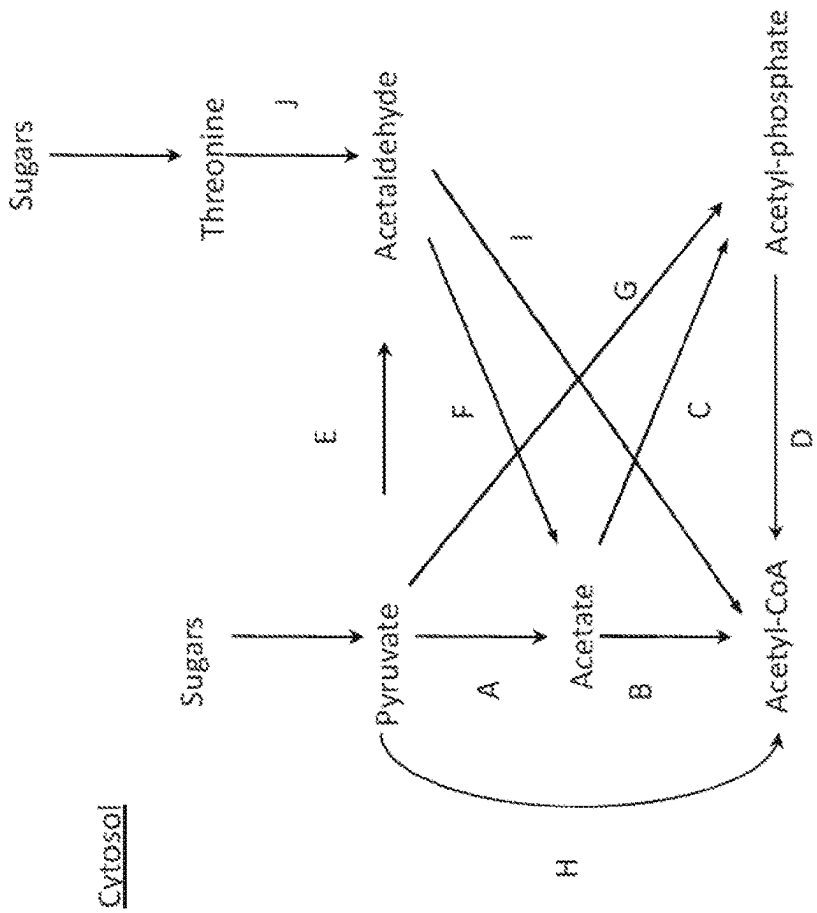


FIG. 3

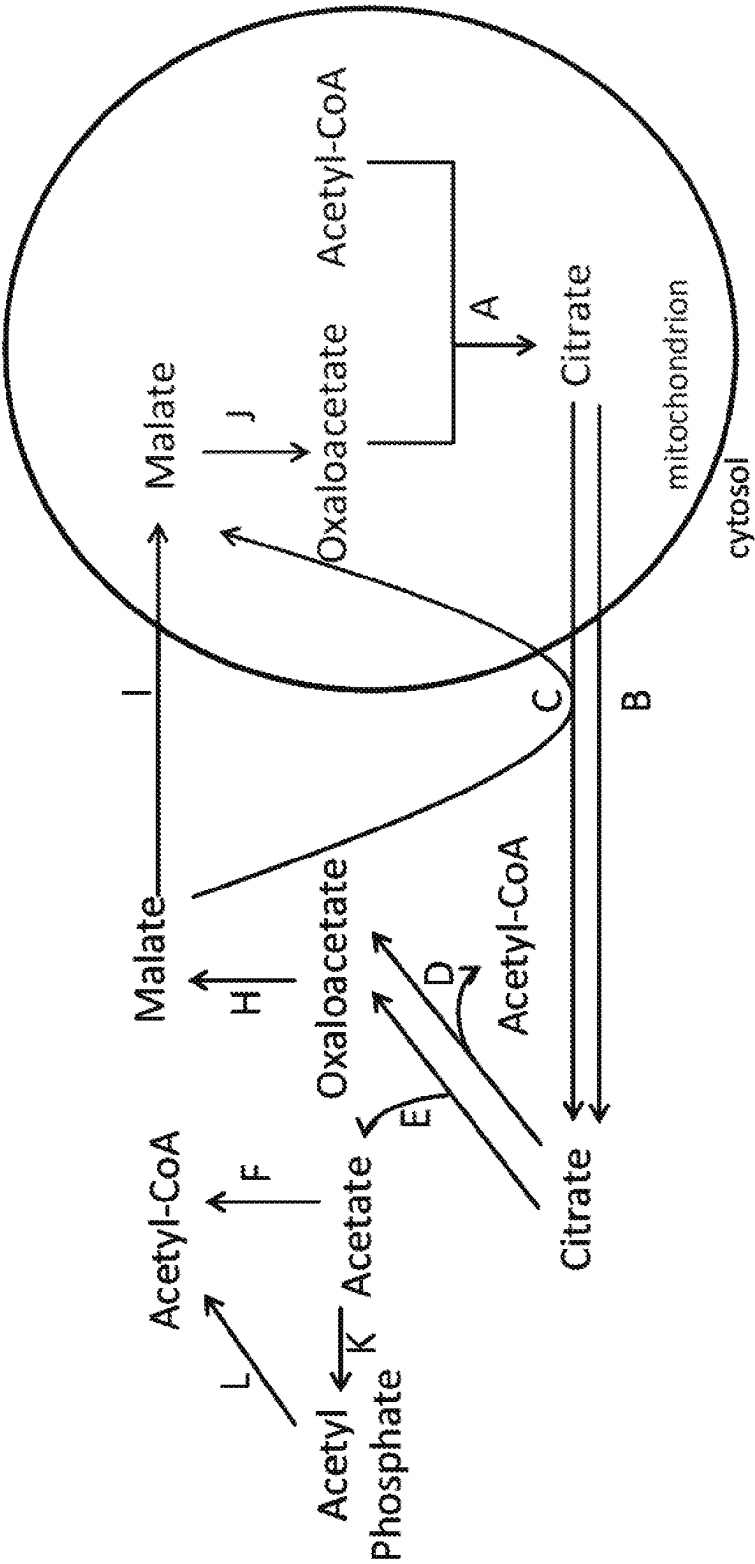


FIG. 5

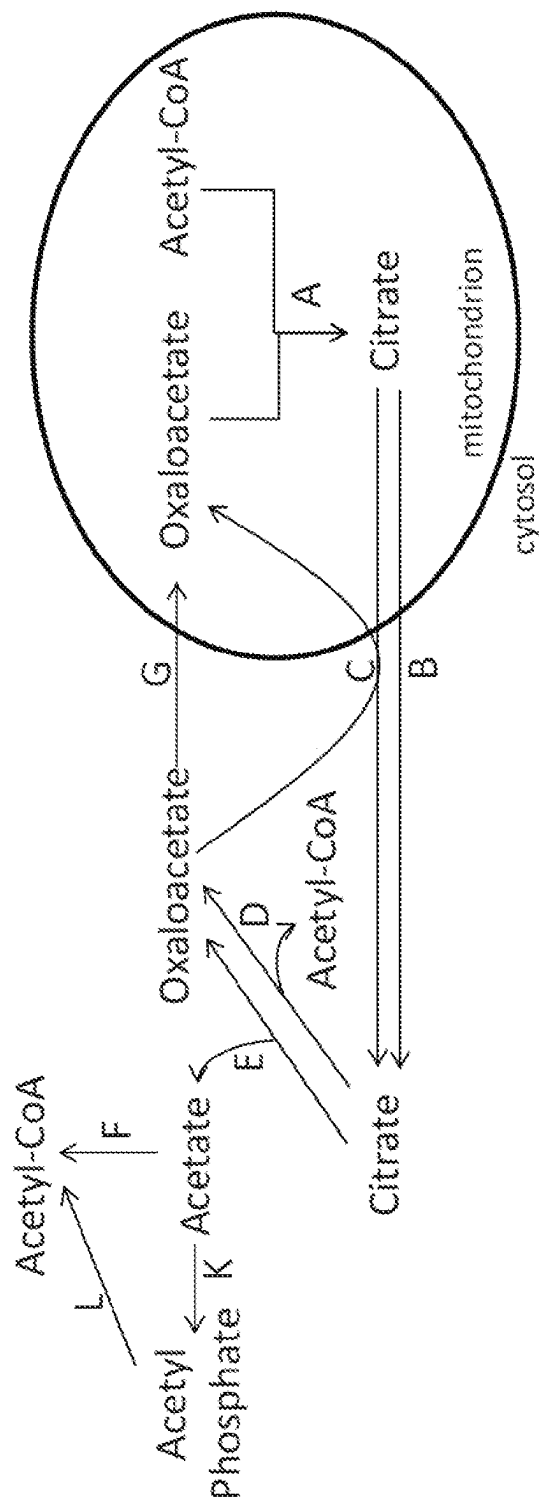


FIG. 6

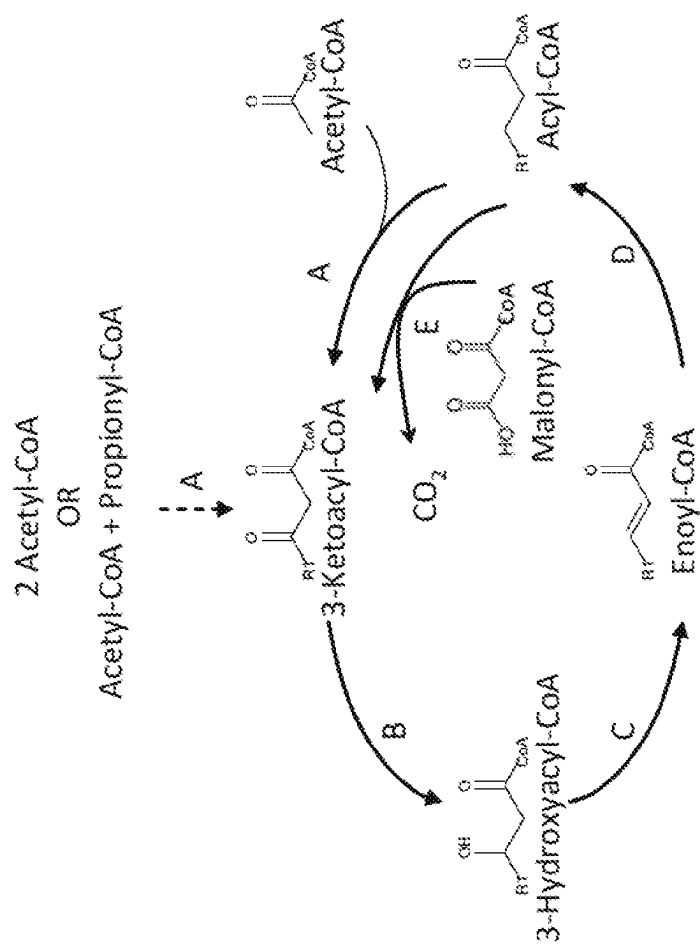


FIG. 7

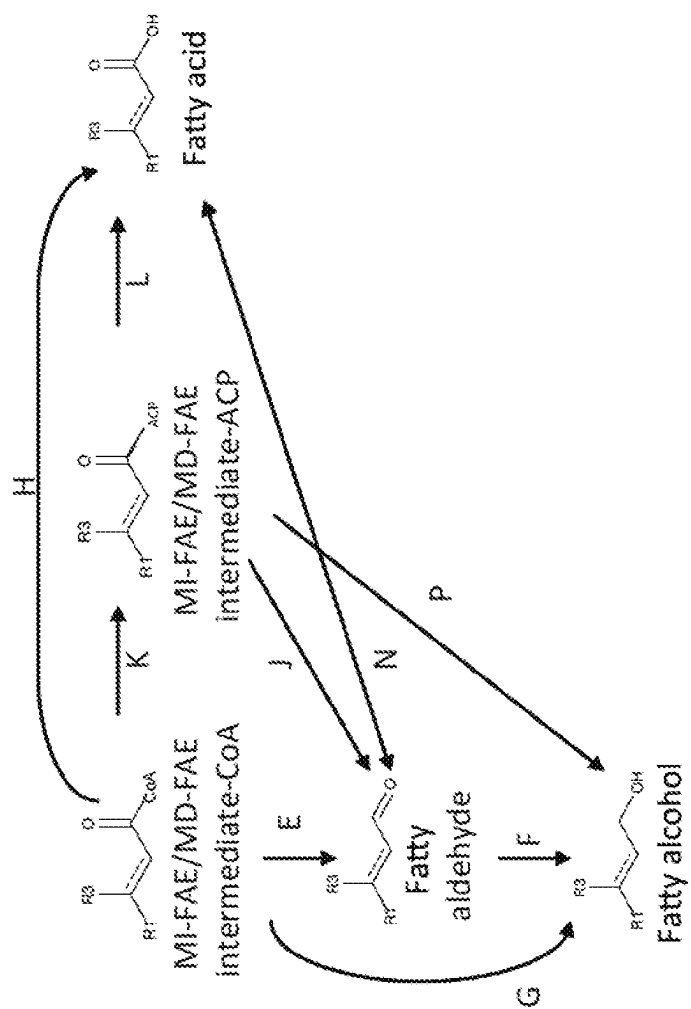


FIG. 8

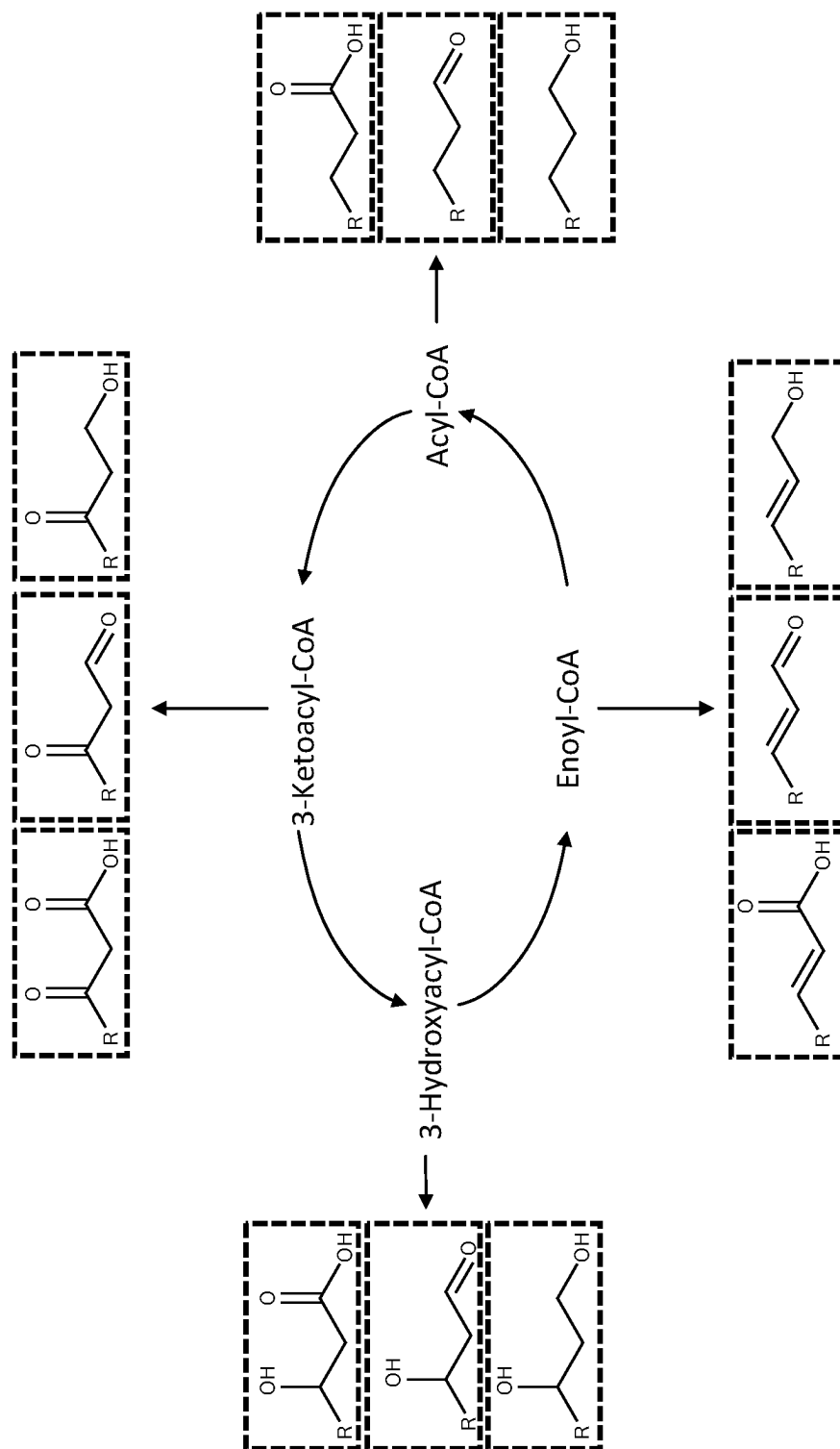


FIG. 9

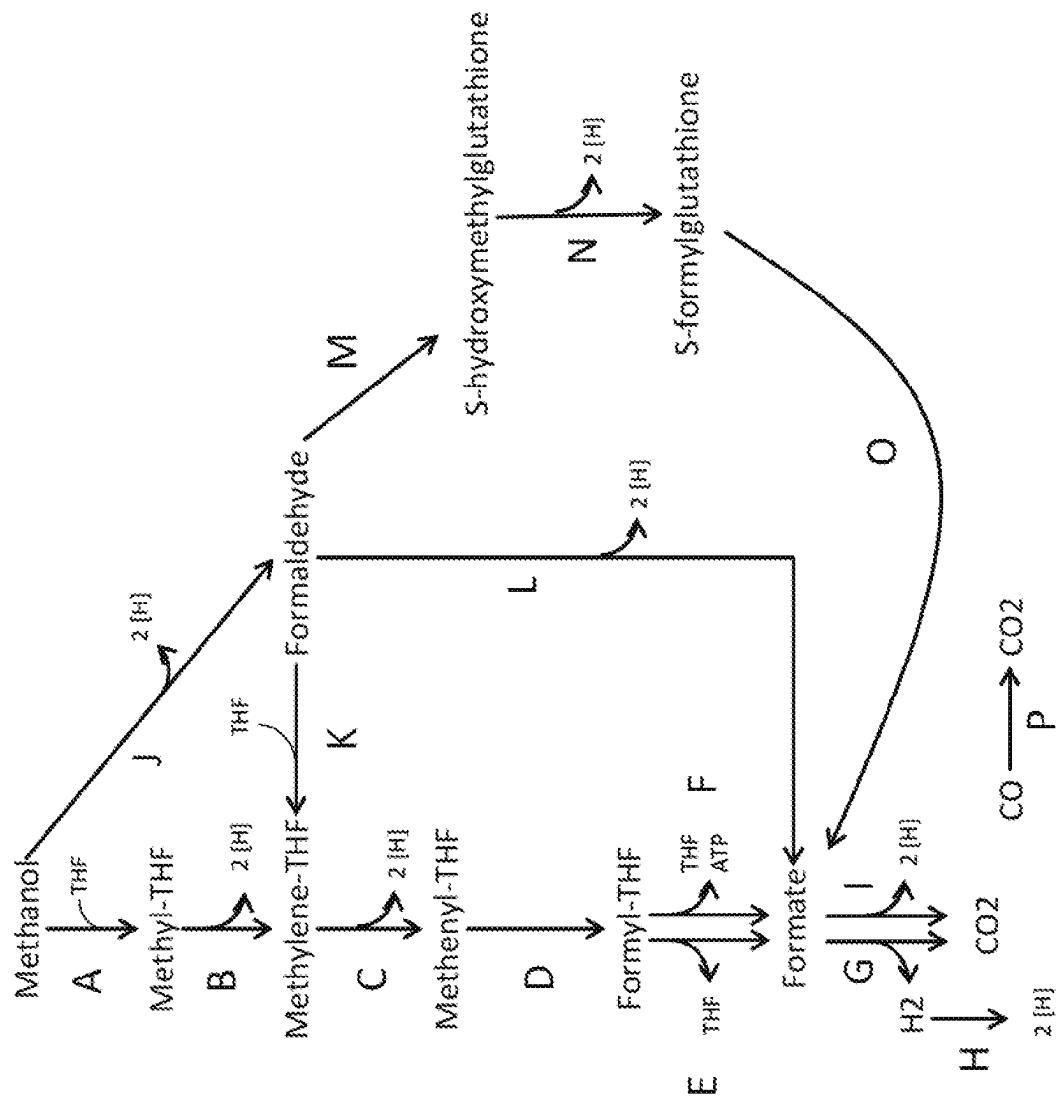


FIG. 10

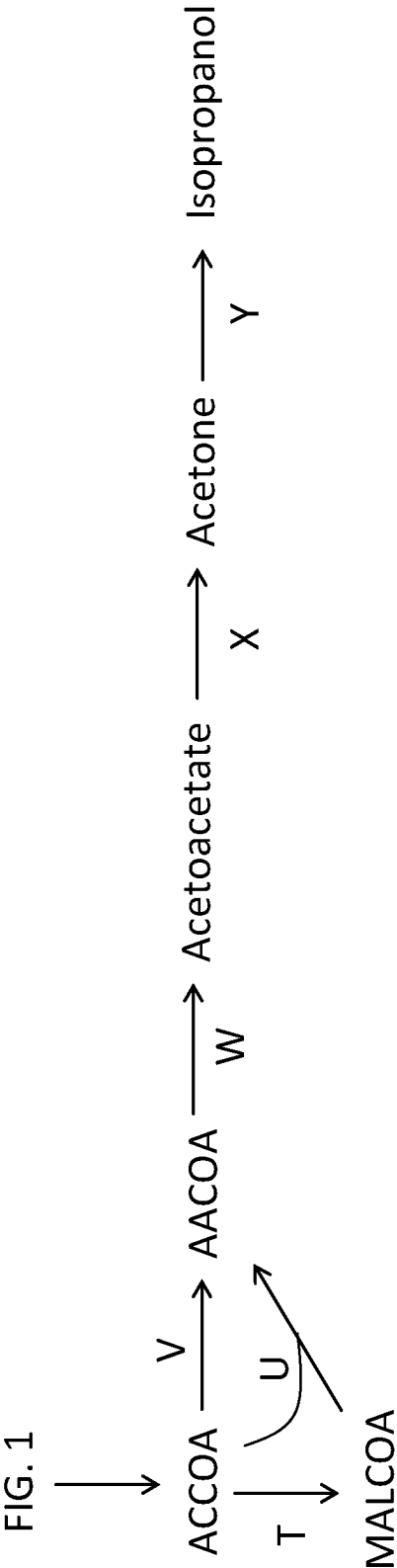
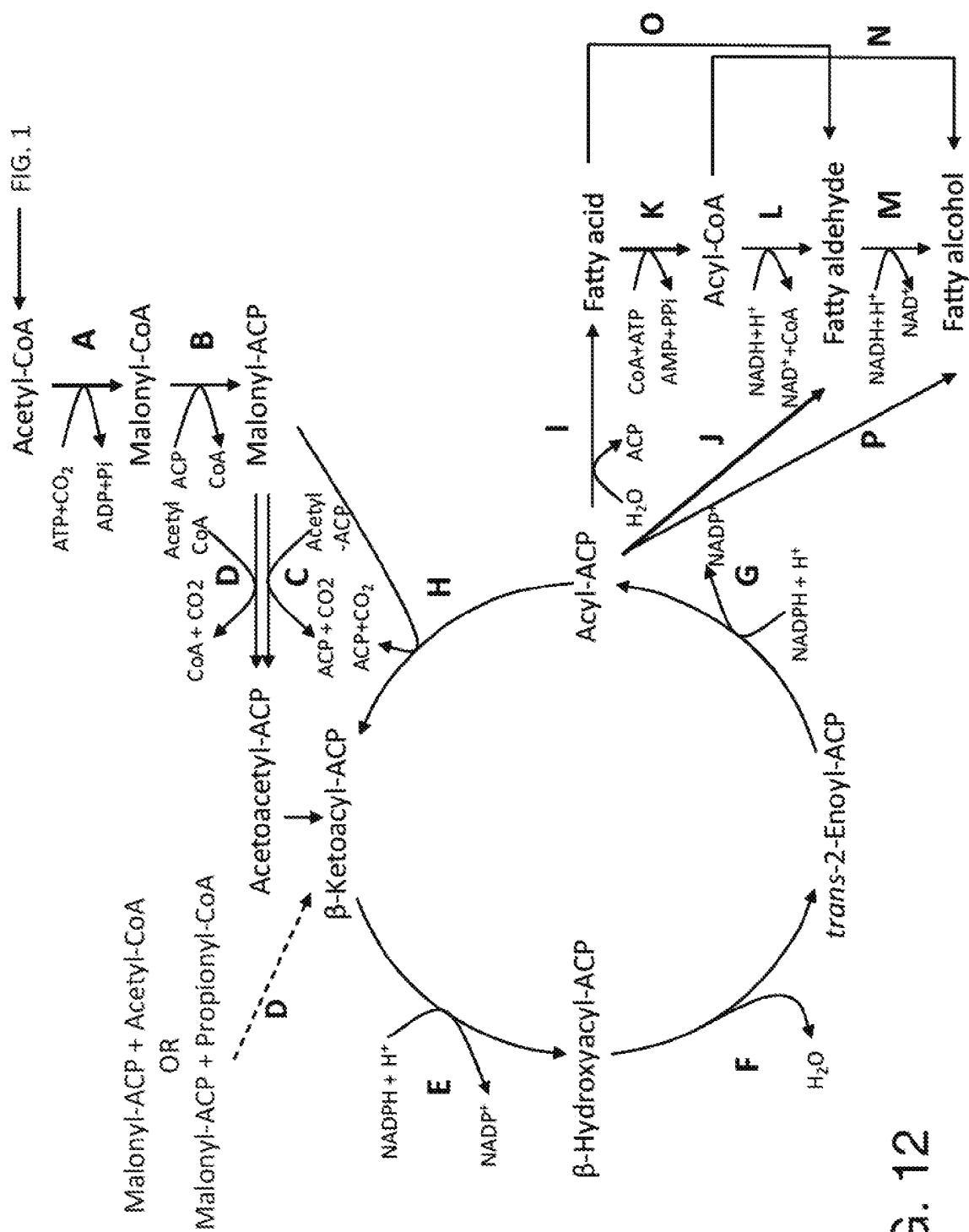


FIG. 11



EV2: two empty plasmid control
EV3: three empty plasmid control
Thl: 1491/560
Hbd: 1495
Ald: 707
Adh: 28
PDH: ALD6+Acs_m

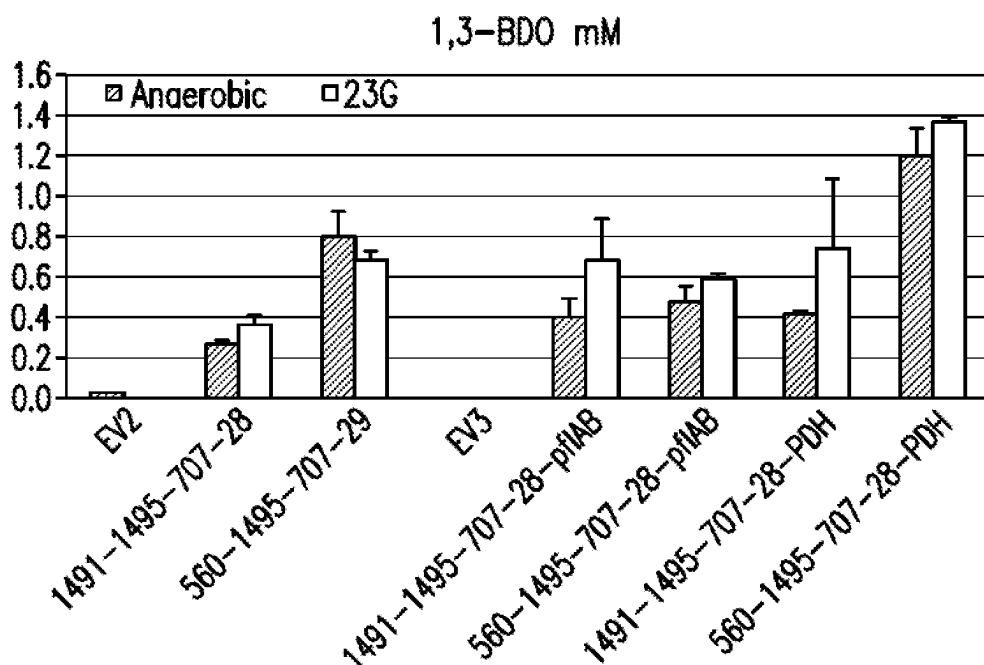


FIG. 13A

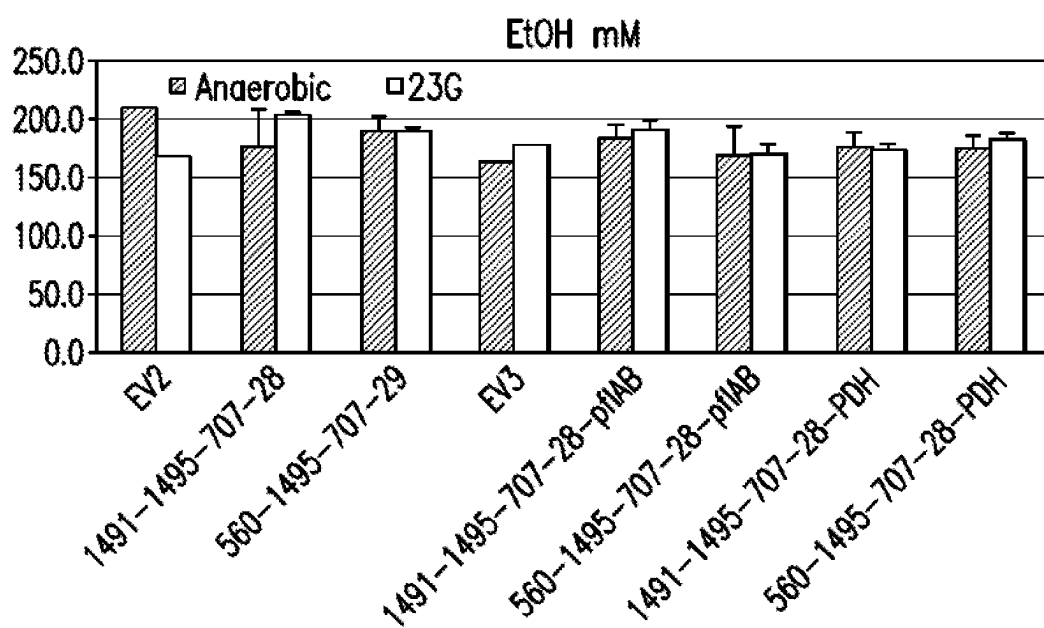
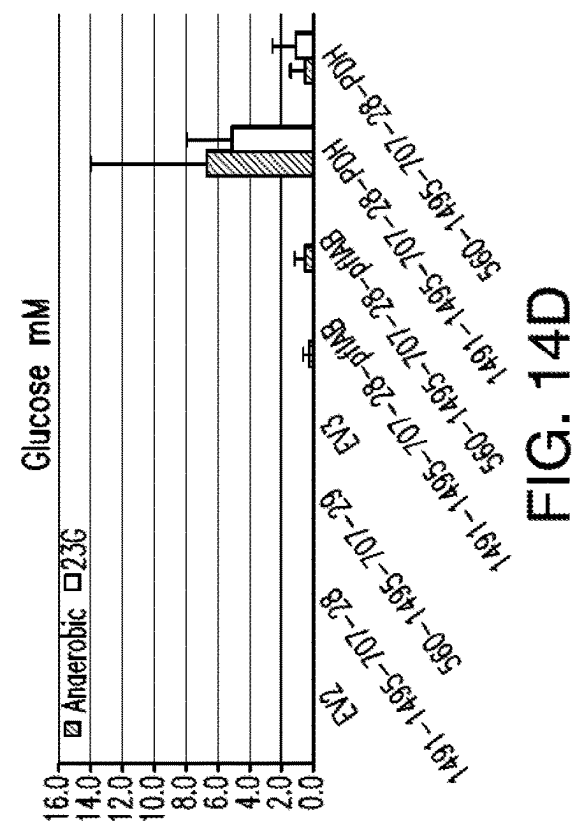
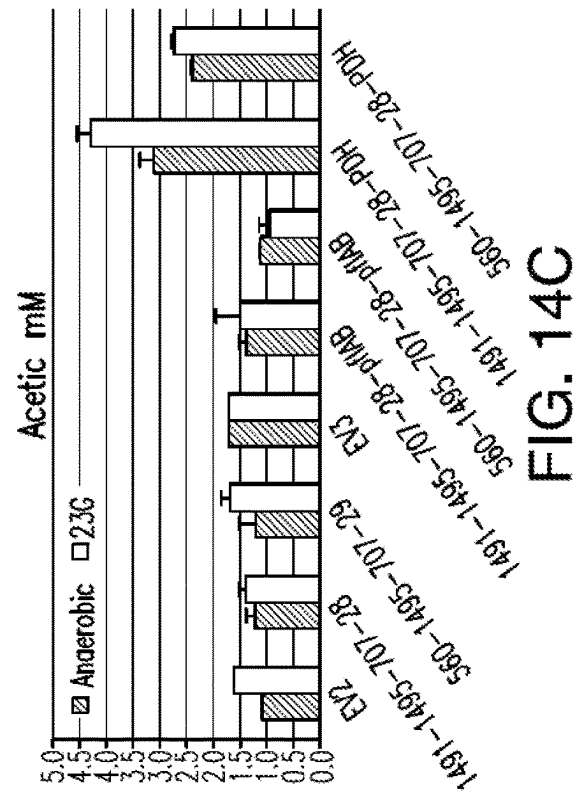
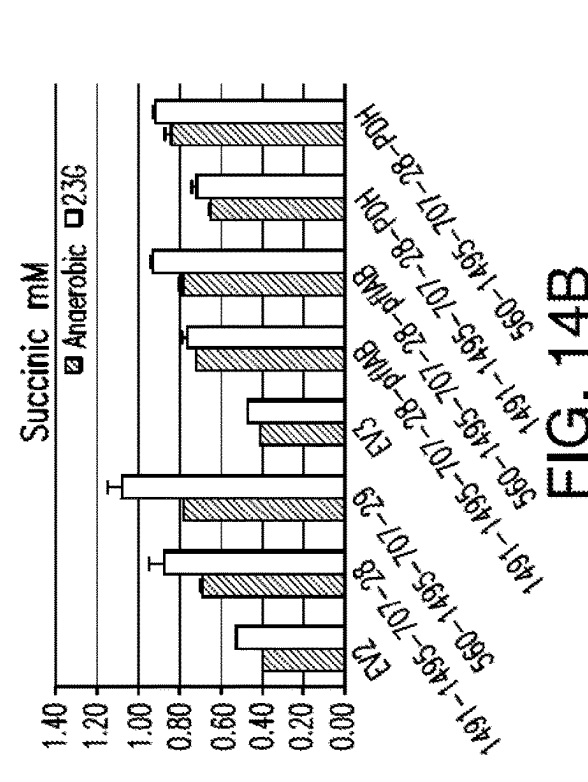
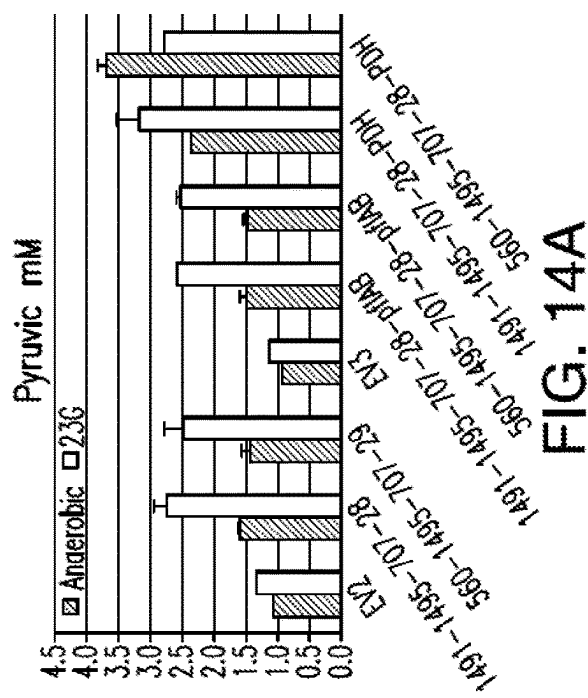


FIG. 13B



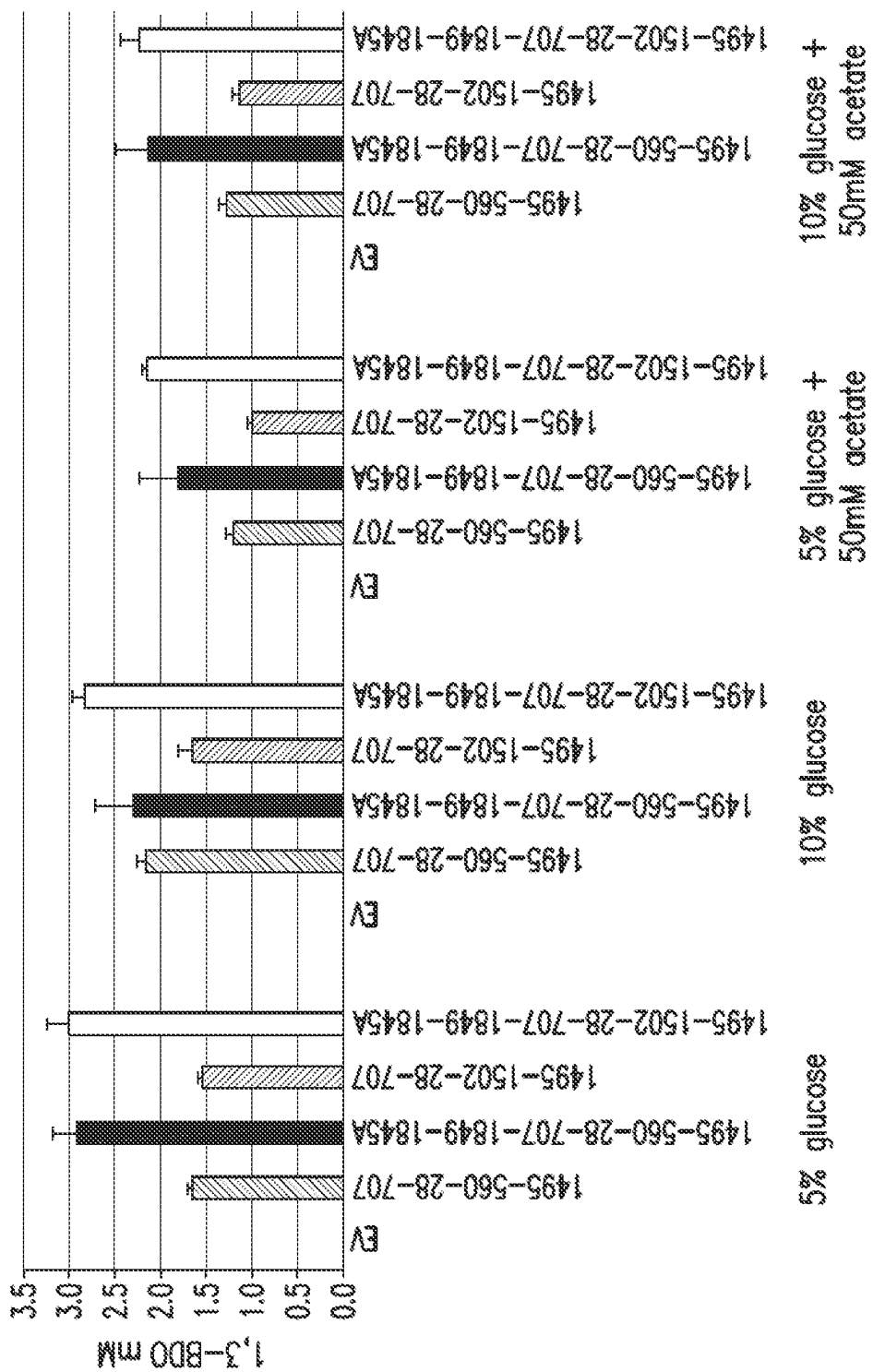


FIG. 15

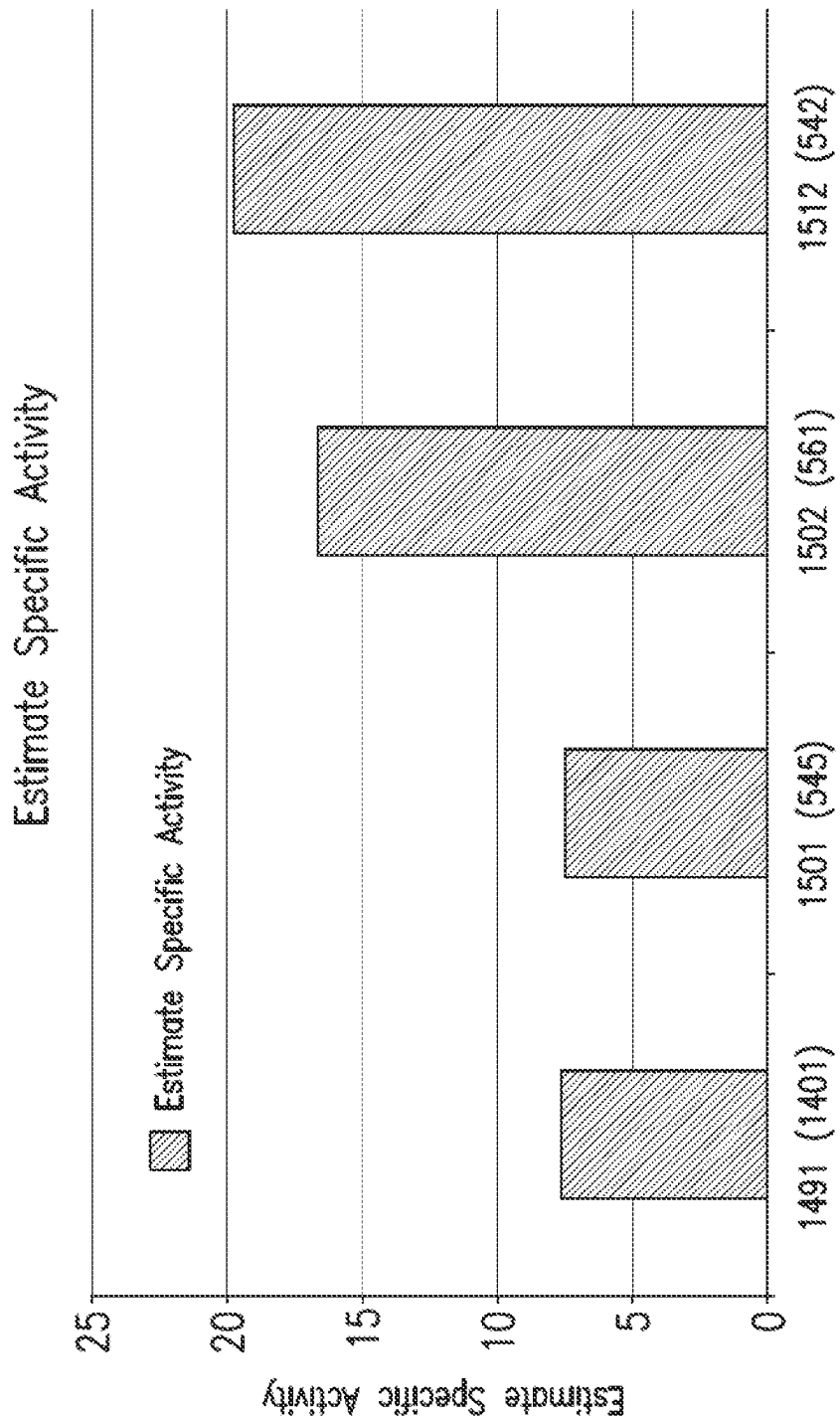


FIG. 16

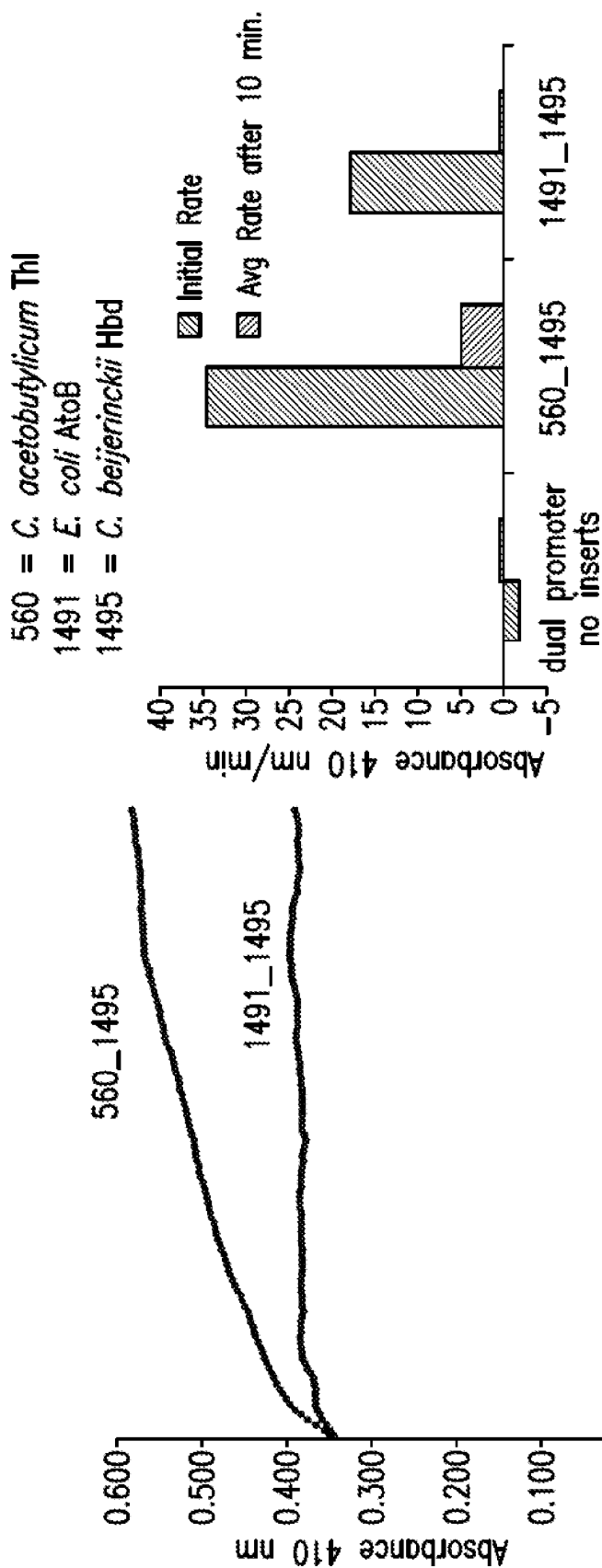


FIG. 17B

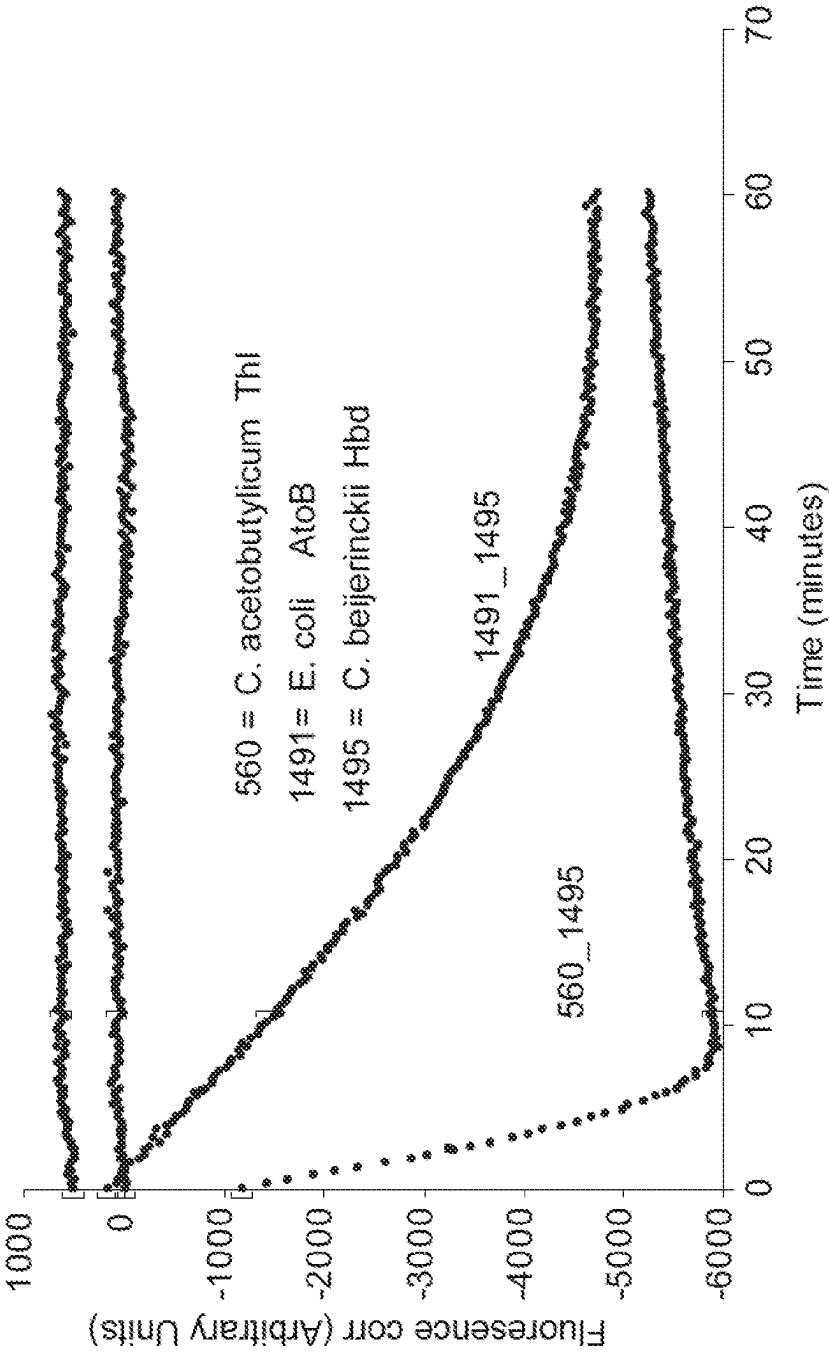


FIG. 18

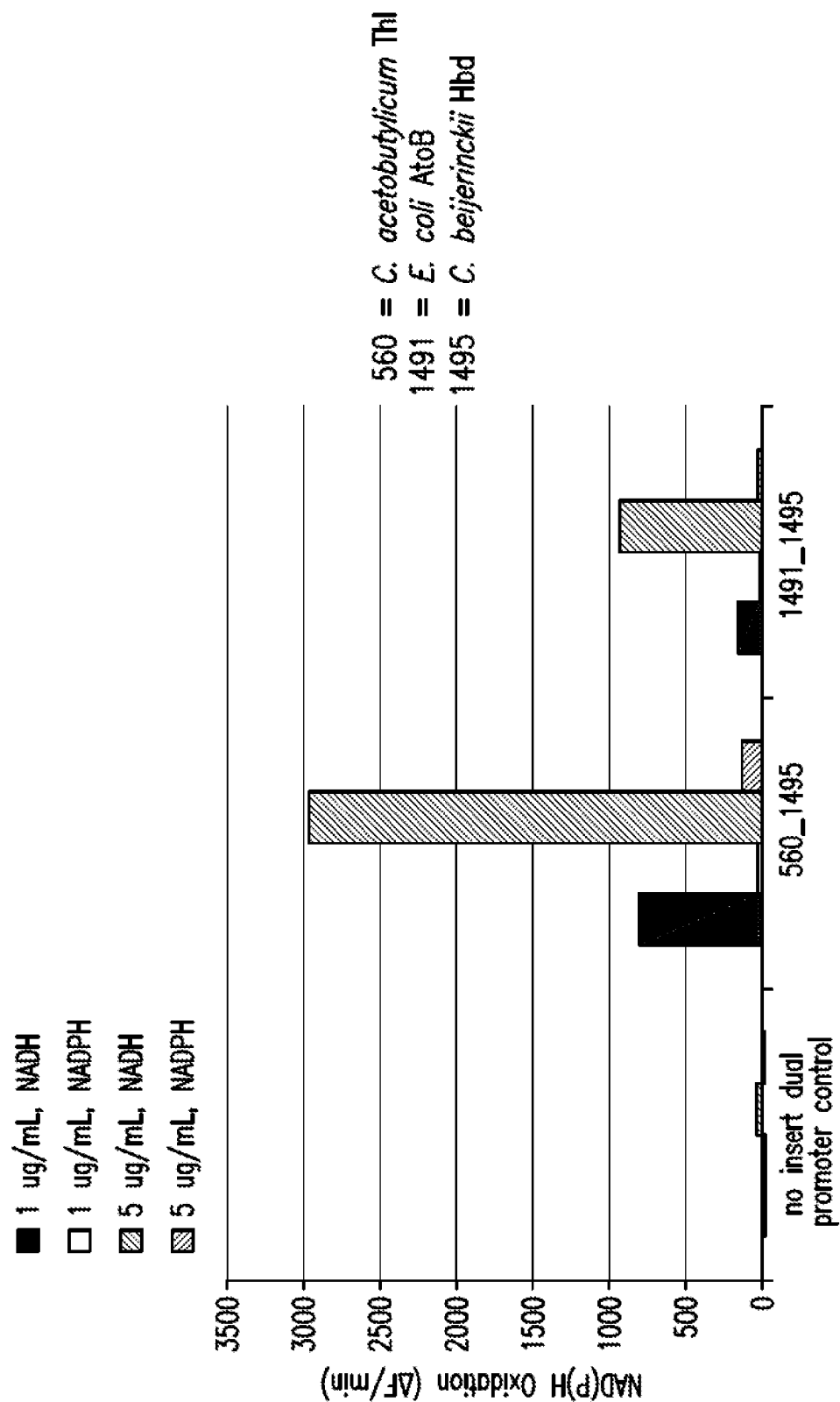
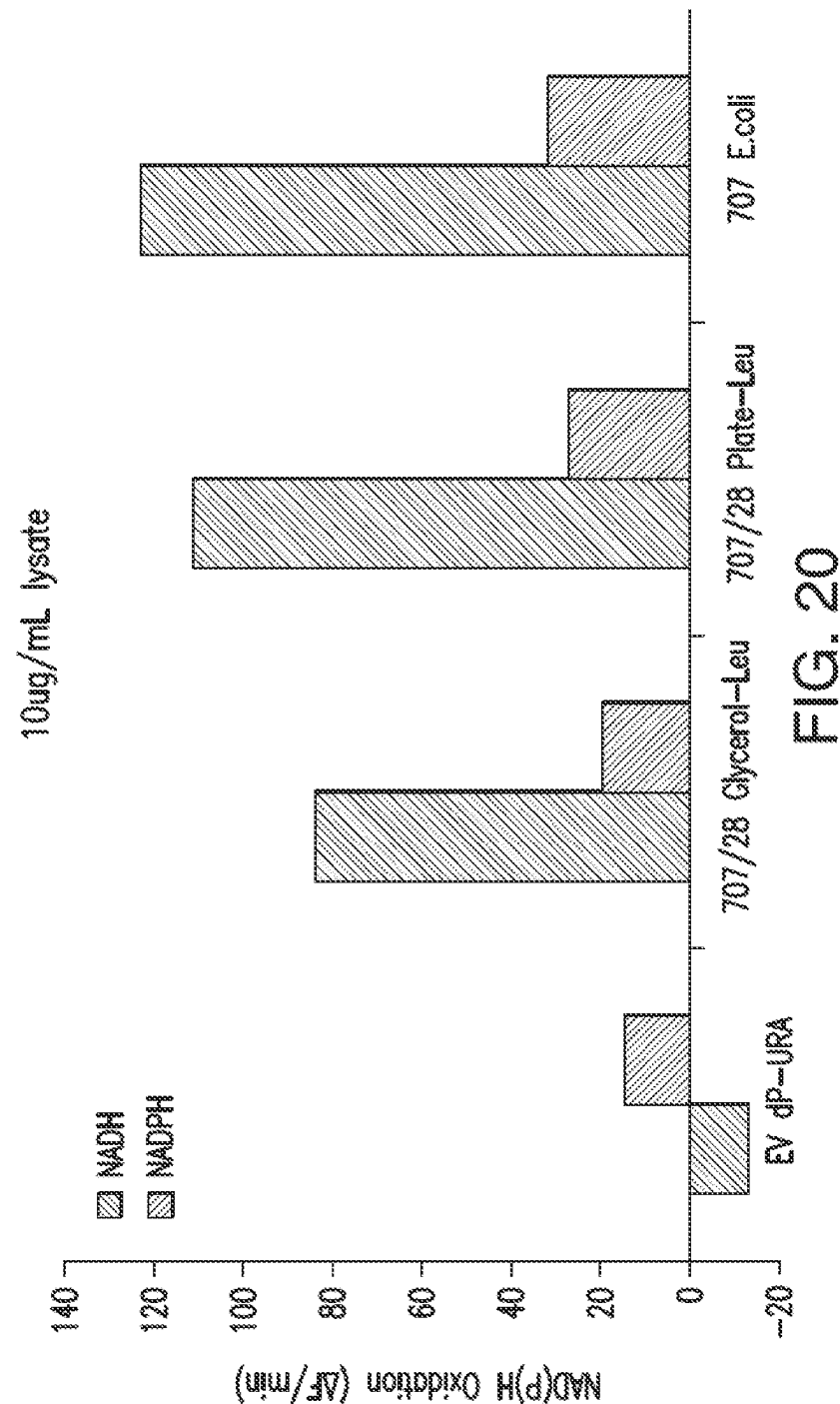


FIG. 19



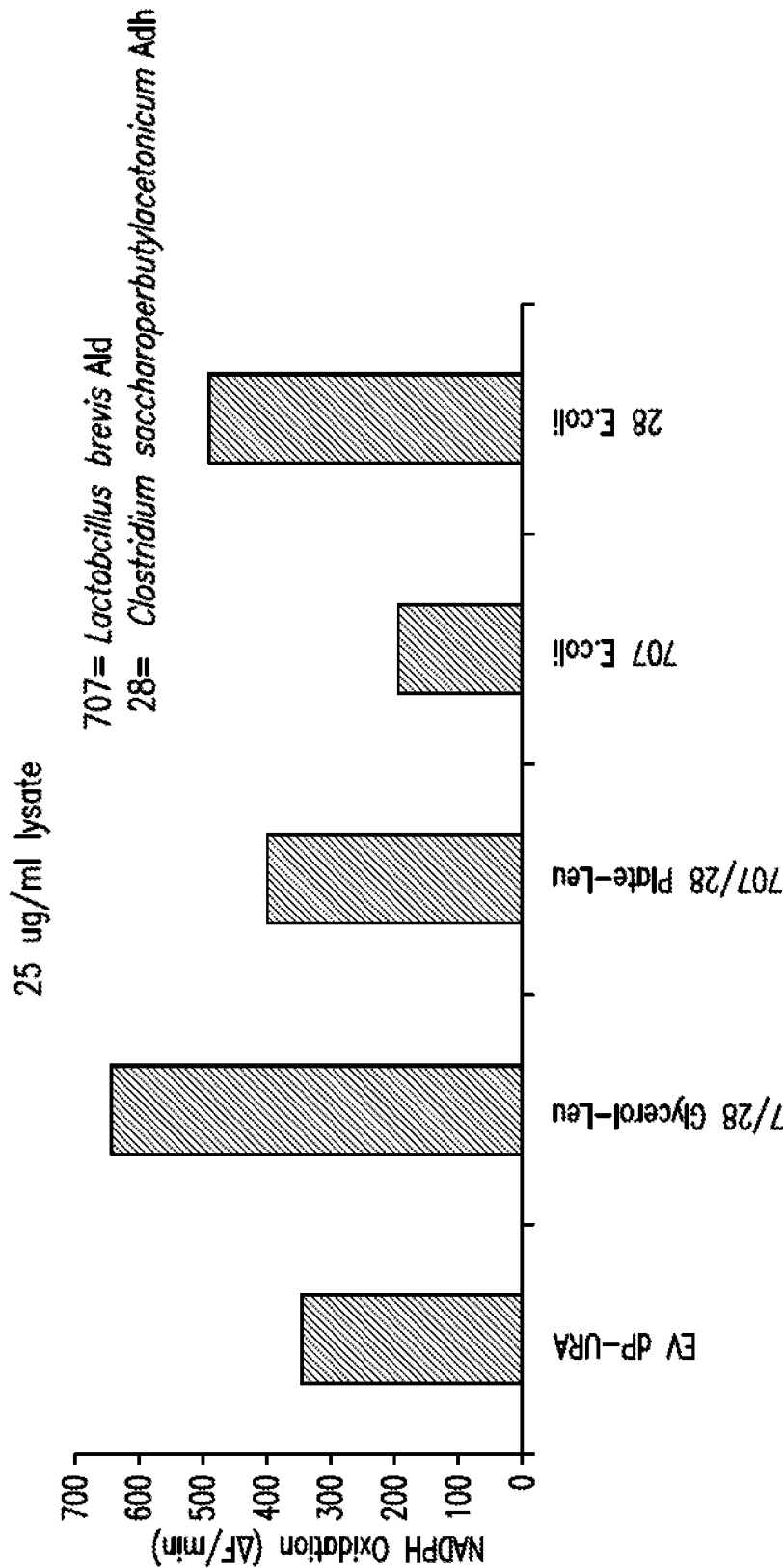


FIG. 21

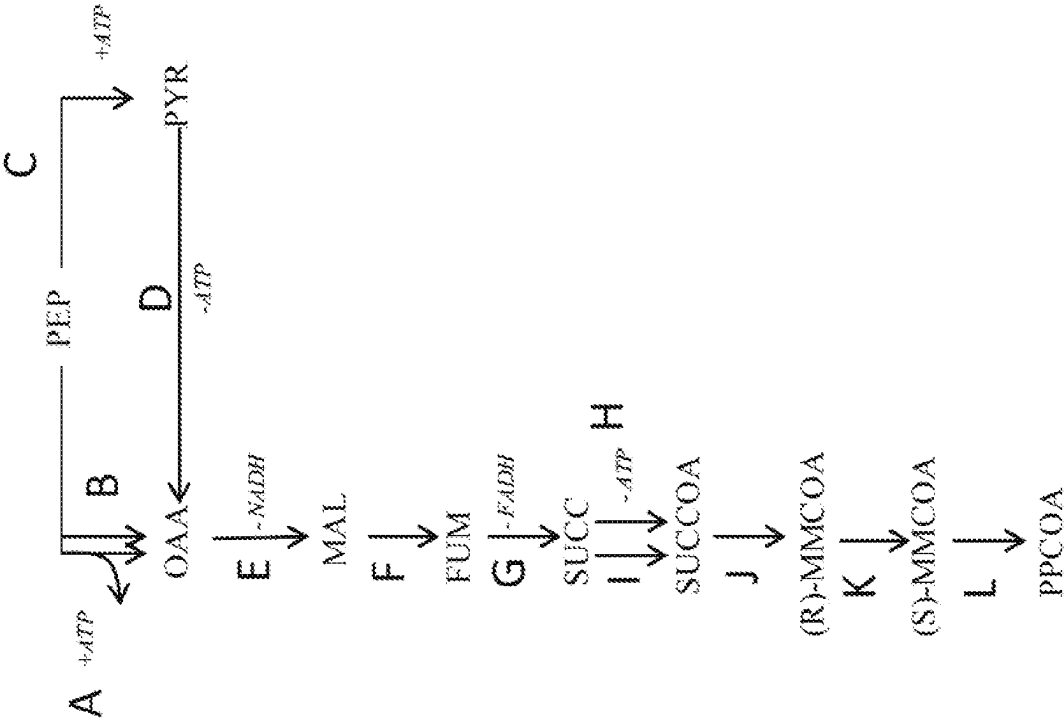


FIG. 22

1

METHODS FOR ENHANCING MICROBIAL PRODUCTION OF SPECIFIC LENGTH FATTY ALCOHOLS IN THE PRESENCE OF METHANOL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of United States application Ser. No. 15/038,922, ABN, which is a United States National Stage Application under 35 U.S.C. § 371 of International Patent Application No. PCT/US2014/067282, filed Nov. 25, 2014, which claims the benefit of priority of U.S. Provisional Application Nos. 61/945,003, filed Feb. 26, 2014, 61/911,374, filed Dec. 3, 2013, and 61/908,652, filed Nov. 25, 2013, the entire contents of which are each incorporated herein by reference.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jan. 5, 2022, is named 199683-999336_US_SL.txt and is 23,022 bytes in size.

BACKGROUND OF THE INVENTION

The present invention relates generally to biosynthetic processes, and more specifically to organisms having specific length fatty alcohol, fatty aldehyde or fatty acid biosynthetic capacity or having isopropanol biosynthetic capacity.

Primary alcohols are a product class of compounds having a variety of industrial applications which include a variety of biofuels and specialty chemicals. Primary alcohols also can be used to make a large number of additional industrial products including polymers and surfactants. For example, higher primary alcohols, also known as fatty alcohols (C₄-C₂₄) and their ethoxylates are used as surfactants in many consumer detergents, cleaning products and personal care products worldwide such as laundry powders and liquids, dishwashing liquid and hard surface cleaners. They are also used in the manufacture of a variety of industrial chemicals and in lubricating oil additives. Specific length fatty alcohols, such as octanol and hexanol, have useful organoleptic properties and have long been employed as fragrance and flavor materials. Smaller chain length C₄-C₈ alcohols (e.g., butanol) are used as chemical intermediates for production of derivatives such as acrylates used in paints, coatings, and adhesives applications.

Fatty alcohols are currently produced from, for example, hydrogenation of fatty acids, hydroformylation of terminal olefins, partial oxidation of n-paraffins and the Al-catalyzed polymerization of ethylene. Unfortunately, it is not commercially viable to produce fatty alcohols directly from the oxidation of petroleum-based linear hydrocarbons (n-paraffins). This impracticality is because the oxidation of n-paraffins produces primarily secondary alcohols, tertiary alcohols or ketones, or a mixture of these compounds, but does not produce high yields of fatty alcohols. Additionally, currently known methods for producing fatty alcohols suffer from the disadvantage that they are restricted to feedstock which is relatively expensive, notably ethylene, which is produced via the thermal cracking of petroleum. In addition, current methods require several steps, and several catalyst types.

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Fatty alcohol production by microorganisms involves fatty acid synthesis followed by acyl-reduction steps. The universal fatty acid biosynthesis pathway found in most cells has been investigated for production of fatty alcohols and other fatty acid derivatives. There is currently a great deal of improvement that can be achieved to provide more efficient biosynthesis pathways for fatty alcohol production with significantly higher theoretical product and energy yields.

Isopropanol (IPA) is a colorless, flammable liquid that mixes completely with most solvents, including water. The largest use for IPA is as a solvent, including its well known yet small use as "rubbing alcohol," which is a mixture of IPA and water. As a solvent, IPA is found in many everyday products such as paints, lacquers, thinners, inks, adhesives, general-purpose cleaners, disinfectants, cosmetics, toiletries, de-icers, and pharmaceuticals. Low-grade IPA is also used in motor oils. The second largest use is as a chemical intermediate for the production of isopropylamines, isopropylethers, and isopropyl esters. Isopropanol can potentially be dehydrated to form propylene, a polymer precursor with an annual market of more than 2 million metric tons.

Current global production capacity of IPA is approximately 6 B lb/yr, with approximately 74% of global IPA capacity concentrated in the US, Europe, and Japan. Isopropanol is manufactured by two petrochemical routes. The predominant process entails the hydration of propylene either with or without sulfuric acid catalysis. Secondly, IPA is produced via hydrogenation of acetone, which is a by-product formed in the production of phenol and propylene oxide. High-priced propylene is currently driving costs up and margins down throughout the chemical industry motivating the need for an expanded range of low cost feedstocks.

Thus, there exists a need for alternative means for effectively producing commercial quantities of fatty alcohols, isopropanol and related compounds. The present invention satisfies this need and provides related advantages as well.

SUMMARY OF INVENTION

The invention provides non-naturally occurring microbial organisms containing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway. For production of a fatty alcohol, fatty aldehyde, or fatty acid, in some embodiments, the non-naturally occurring microbial organism of the invention has: a formaldehyde fixation pathway, a formate assimilation pathway, and/or a methanol metabolic pathway; and a malonyl-CoA independent fatty acyl-CoA elongation (MI-FAE) cycle and/or a malonyl-CoA dependent fatty acyl-CoA elongation (MD-FAE) cycle in combination with a termination pathway, as depicted in FIGS. 1, 2, 7, 8 and 10. Alternatively, in some embodiments, the non-naturally occurring microbial organism of the invention has: a formaldehyde fixation pathway, a formate assimilation pathway, and/or a methanol metabolic pathway; and a fatty acyl-ACP elongation (FAACPE) cycle in combination with a termination pathway, as depicted in FIGS. 1, 10 and 12.

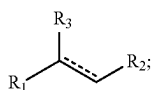
For production of isopropanol, in some embodiments, the non-naturally occurring microbial organism of the invention has: a formaldehyde fixation pathway, a formate assimilation pathway, and/or a methanol metabolic pathway; and an isopropanol pathway, as depicted in FIGS. 1, 10 and 11.

In one aspect, the formaldehyde fixation pathway, formate assimilation pathway, and/or a methanol metabolic pathway present in the microbial organisms of the invention enhances the availability of substrates and/or pathway intermediates, such as acetyl-CoA and malonyl-CoA, and/or reducing

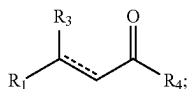
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equivalents, which can be utilized for fatty alcohol, fatty aldehyde, fatty acid, or isopropanol production through one or more fatty alcohol, fatty aldehyde, fatty acid, or isopropanol pathways of the invention. For example, in some embodiments, a non-naturally occurring microbial organism of the invention that includes a methanol metabolic pathway can enhance the availability of reducing equivalents in the presence of methanol and/or convert methanol to formaldehyde, a substrate for the formaldehyde fixation pathway. Likewise, a non-naturally occurring microbial organism of the invention having a formate assimilation pathway can reutilize formate to generate substrates and pathway intermediates such as formaldehyde, pyruvate and/or acetyl-CoA. Such substrates, intermediates and reducing equivalents can be used to increase the yield of a fatty alcohol, a fatty aldehyde, a fatty acid, or isopropanol produced by the microbial organism.

In some embodiments, the microbial organisms of the invention advantageously enhance the production of substrates and/or pathway intermediates for the production of a chain length specific fatty alcohol, fatty aldehyde, fatty acid. Accordingly, some embodiments, one or more enzymes of the formaldehyde fixation pathway, formate assimilation pathway, methanol metabolic pathway, MI-FAE cycle, MD-FAE cycle, FAACPE cycle or termination pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce a fatty alcohol, fatty aldehyde or fatty acid of Formula (I):



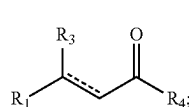
wherein R_1 is C_{1-24} linear alkyl; R_2 is CH_2OH , CHO , or $COOH$; R_3 is H , OH , or oxo ($=O$); and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four. In order to be able to produce a chain length specific compound, the enzymes of the MI-FAE cycle, the MD-FAE cycle, the FAACPE cycle and/or the termination pathway are selective for a particular substrate. Accordingly, in some embodiments, the substrate of each of the enzymes of the MI-FAE cycle, the MD-FAE cycle and/or the termination pathway are independently selected from a compound of Formula (II), malonyl-CoA, propionyl-CoA or acetyl-CoA:



wherein R_1 is C_{1-24} linear alkyl; R_3 is H , OH , or oxo ($=O$); R_4 is $S-CoA$, ACP , OH or H ; and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four, wherein said one or more enzymes of the MI-FAE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said fatty alcohol, fatty aldehyde or fatty acid of Formula (I), wherein said one or more enzymes of the MD-FAE

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cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said fatty alcohol, fatty aldehyde or fatty acid of Formula (I), and wherein said one or more enzymes of the termination pathway are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no less than the number of carbon atoms at R_1 of said fatty alcohol, fatty aldehyde or fatty acid of Formula (I). Alternatively, in some embodiments, the substrate of each of the enzymes of the FAACPE cycle and/or the termination pathway are independently selected from a compound of Formula (II) or malonyl-ACP:



wherein R_1 is C_{1-24} linear alkyl; R_3 is H , OH , or oxo ($=O$); R_4 is $S-CoA$, ACP , OH or H ; and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four, wherein the one or more enzymes of the FAACPE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said compound of Formula (I), and wherein the one or more enzymes of the termination pathway are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no less than the number of carbon atoms at R_1 of said compound of Formula (I).

In some embodiments, the invention provides a non-naturally occurring microbial organism containing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway further having an acetyl-CoA pathway, a methanol oxidation pathway, a hydrogenase and/or a carbon monoxide dehydrogenase. Accordingly, in some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway, wherein the microbial organism further includes an acetyl-CoA pathway and at least one exogenous nucleic acid encoding an acetyl-CoA pathway enzyme expressed in a sufficient amount to produce or enhance carbon flux through acetyl-CoA, wherein the acetyl-CoA pathway includes a pathway shown in FIG. 1, 3, 4, 5 or 6. In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway, wherein the microbial organism further includes a methanol oxidation pathway enzyme expressed in a sufficient amount to produce formaldehyde in the presence of methanol. An exemplary methanol oxidation pathway enzyme is a methanol dehydrogenase as depicted in FIG. 1, Step A. In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway, wherein the microbial organism further includes a hydrogenase and/or a carbon monoxide dehydrogenase for generating reducing equivalents as depicted in FIG. 10.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the microbial organism has one or more gene disruptions, wherein the one or more gene disruptions occur in endog-

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enous genes encoding proteins or enzymes involved in: native production of ethanol, glycerol, pyruvate, acetate, formate, lactate, CO₂, fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native degradation of a MI-FAE cycle intermediate, MD-FAE cycle intermediate, FAACPE cycle intermediate or a termination pathway intermediate by the microbial organism, the one or more gene disruptions confer increased production of a fatty alcohol, fatty aldehyde or fatty acid in the microbial organism.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein one or more enzymes of the MI-FAE cycle, MD-FAE cycle, FAACPE cycle or the termination pathway preferentially react with an NADH cofactor or have reduced preference for reacting with an NAD(P)H cofactor.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the microbial organism has one or more gene disruptions in genes encoding proteins or enzymes that result in an increased ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism following the disruptions.

In some embodiments, the non-naturally occurring microbial organism of the invention is Crabtree positive and is in culture medium comprising excess glucose. In such conditions, as described herein, the microbial organism can result in increasing the ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the microbial organism has at least one exogenous nucleic acid encoding an extracellular transporter or an extracellular transport system for a fatty alcohol, fatty aldehyde or fatty acid of the invention.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the microbial organism one or more endogenous enzymes involved in: native production of ethanol, glycerol, pyruvate, acetate, formate, lactate, CO₂, fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native degradation of a MI-FAE cycle intermediate, a MD-FAE cycle intermediate, FAACPE cycle intermediate or a termination pathway intermediate by said microbial organism, has attenuated enzyme activity or expression levels.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the microbial organism has attenuated enzyme activity or expression levels for one or more endogenous enzymes involved in the oxidation of NAD(P)H or NADH.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the microbial organism further includes attenuation of one or more endogenous enzymes, which enhances carbon flux through acetyl-CoA, or a gene disruption of one or more endogenous nucleic acids encoding such enzymes. For example, in some aspects, the endogenous enzyme can be selected from DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase, DHA synthase or any combination thereof.

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The invention further provides non-naturally occurring microbial organisms that have elevated or enhanced synthesis or yields of acetyl-CoA (e.g. intracellular) or biosynthetic products such as a fatty alcohol, fatty aldehyde, fatty acid or isopropanol and methods of using those non-naturally occurring organisms to produce such biosynthetic products. The enhanced synthesis of intracellular acetyl-CoA enables enhanced production of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol from which acetyl-CoA is an intermediate and further, may have been rate limiting.

In some embodiments, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the microbial organism further includes attenuation of one or more endogenous enzymes of a competing formaldehyde assimilation or dissimilation pathway or a gene disruption of one or more endogenous nucleic acids encoding enzymes of a competing formaldehyde assimilation or dissimilation pathway. Examples of these endogenous enzymes are described herein.

The invention additionally provides methods of using the above microbial organisms to produce a fatty alcohol, a fatty aldehyde, a fatty acid or isopropanol by culturing a non-naturally occurring microbial organism containing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway as described herein under conditions and for a sufficient period of time to produce a fatty alcohol, fatty aldehyde, fatty acid or isopropanol.

The invention still further provides a bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol produced by a microbial organism of the invention, culture medium having the bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol of the invention, compositions having the bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol of the invention, a biobased product comprising the bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol of the invention, and a process for producing a bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows exemplary metabolic pathways enabling the conversion of CO₂, formate, formaldehyde (Fald), methanol (MeOH), glycerol, xylose (XYL) and glucose (GLC) to acetyl-CoA (ACCOA) and exemplary endogenous enzyme targets for optional attenuation or disruption. The exemplary pathways and endogenous enzyme targets can be combined with the cycles and pathways depicted herein that utilize ACCOA, such as those depicted in FIGS. 1, 11 and 12. The enzyme targets are indicated by arrows having "X" markings. The endogenous enzyme targets include DHA kinase, methanol oxidase (AOX), PQQ-dependent methanol dehydrogenase (PQQ) and/or DHA synthase. The enzymatic transformations shown are carried out by the following enzymes: A) methanol dehydrogenase, B) 3-hexulose-6-phosphate synthase, C) 6-phospho-3-hexuloisomerase, D) dihydroxyacetone synthase, E) formate reductase, F) formate ligase, formate transferase, or formate synthetase, G) formyl-CoA reductase, H) formyltetrahydrofolate synthetase, I) methenyltetrahydrofolate cyclohydrolase, J) methylenetetrahydrofolate dehydrogenase, K) spontaneous or formaldehyde-forming enzyme, L) glycine cleavage system, M) serine hydroxymethyltransferase, N) serine deaminase, O) methylenetetrahydrofolate reductase, P) acetyl-CoA synthase, Q) pyruvate formate lyase, R) pyruvate dehydrogenase, pyruvate ferredoxin oxidoreductase, or

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pyruvate:NADP⁺ oxidoreductase, S) formate dehydrogenase, T) fructose-6-phosphate phosphoketolase, U) xylulose-5-phosphate phosphoketolase, V) phosphotransacetylase, W) acetate kinase, X) acetyl-CoA transferase, synthetase, or ligase, Y) lower glycolysis including glyceraldehyde-3-phosphate dehydrogenase, Z) fructose-6-phosphate aldolase. See abbreviation list below for compound names.

FIG. 2 shows an exemplary MI-FAE cycle and/or MD-FAE cycle in combination with termination pathways for production of fatty alcohols, aldehydes, or acids from the acyl-CoA intermediate of the MI-FAE cycle or MD-FAE cycle. Enzymes are: A. Thiolase; B. 3-Oxoacyl-CoA reductase; C. 3-Hydroxyacyl-CoA dehydratase; D. Enoyl-CoA reductase; E. Acyl-CoA reductase (aldehyde forming); F. Alcohol dehydrogenase; G. Acyl-CoA reductase (alcohol forming); H. acyl-CoA hydrolase, transferase or synthase; J. Acyl-ACP reductase; K. Acyl-CoA:ACP acyltransferase; L. Thioesterase; N. Aldehyde dehydrogenase (acid forming) or carboxylic acid reductase; O. Elongase; and P. acyl-ACP reductase (alcohol forming).

FIG. 3 shows exemplary pathways for production of cytosolic acetyl-CoA from pyruvate or threonine. Enzymes are: A. pyruvate oxidase (acetate-forming); B. acetyl-CoA synthetase, ligase or transferase; C. acetate kinase; D. phosphotransacetylase; E. pyruvate decarboxylase; F. acetaldehyde dehydrogenase; G. pyruvate oxidase (acetyl-phosphate forming); H. pyruvate dehydrogenase, pyruvate:fenedoxin oxidoreductase, pyruvate:NAD(P)H oxidoreductase or pyruvate formate lyase; I. acetaldehyde dehydrogenase (acylating); and J. threonine aldolase.

FIG. 4 shows exemplary pathways for production of acetyl-CoA from phosphoenolpyruvate (PEP). Enzymes are: A. PEP carboxylase or PEP carboxykinase; B. oxaloacetate decarboxylase; C. malonate semialdehyde dehydrogenase (acylating); D. acetyl-CoA carboxylase or malonyl-CoA decarboxylase; E. oxaloacetate dehydrogenase or oxaloacetate oxidoreductase; G. malonate semialdehyde dehydrogenase (acylating); H. pyruvate carboxylase; J. malonate semialdehyde dehydrogenase; K. malonyl-CoA synthetase or transferase; L. malic enzyme; M. malate dehydrogenase or oxidoreductase; and N. pyruvate kinase or PEP phosphatase.

FIG. 5 shows exemplary pathways for production of cytosolic acetyl-CoA from mitochondrial acetyl-CoA using citrate and malate transporters. Enzymes are: A. citrate synthase; B. citrate transporter; C. citrate/malate transporter; D. ATP citrate lyase; E. citrate lyase; F. acetyl-CoA synthetase or transferase; H. cytosolic malate dehydrogenase; I. malate transporter; J. mitochondrial malate dehydrogenase; K. acetate kinase; and L. phosphotransacetylase.

FIG. 6 shows exemplary pathways for production of cytosolic acetyl-CoA from mitochondrial acetyl-CoA using citrate and oxaloacetate transporters. Enzymes are: A. citrate synthase; B. citrate transporter; C. citrate/oxaloacetate transporter; D. ATP citrate lyase; E. citrate lyase; F. acetyl-CoA synthetase or transferase; G) oxaloacetate transporter; K) acetate kinase; and L) phosphotransacetylase.

FIG. 7 shows an exemplary MI-FAE cycle and/or MD-FAE cycle for elongating the linear alkyl of R₁. Enzymes are: A. Thiolase; B. 3-Ketoacyl-CoA reductase; C. 3-Hydroxyacyl-CoA dehydratase; D. Enoyl-CoA reductase; and E. Elongase.

FIG. 8 shows an exemplary termination cycle for generating a fatty alcohol, fatty aldehyde or fatty acid from any of the MI-FAE cycle intermediates or MD-FAE cycle intermediates of FIG. 7. Enzymes are: E. MI-FAE/MD-FAE inter-

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mediate-CoA reductase (aldehyde forming); F. Alcohol dehydrogenase; G. MI-FAE/MD-FAE intermediate-CoA reductase (alcohol forming); H. MI-FAE/MD-FAE intermediate-CoA hydrolase, transferase or synthase; J. MI-FAE/MD-FAE intermediate-ACP reductase; K. MI-FAE/MD-FAE intermediate-CoA:ACP acyltransferase; L. Thioesterase; N. Aldehyde dehydrogenase (acid forming) or carboxylic acid reductase; and P. acyl-ACP reductase (alcohol forming). R₁ is C1-24 linear alkyl; R₃ is H, OH, or oxo (=O) and represents a single or double bond with the proviso that the valency of the carbon atom to which R₃ is attached is four.

FIG. 9 shows exemplary compounds that can be produced from the four MI-FAE or MD-FAE cycle intermediates using the cycles depicted in FIG. 6 and the termination pathways depicted in FIG. 7. R is C₁₋₂₄ linear alkyl.

FIG. 10 shows exemplary metabolic pathways that provide the extraction of reducing equivalents from methanol, hydrogen, or carbon monoxide. Enzymes are: A) methanol methyltransferase, B) methylenetetrahydrofolate reductase, C) methylenetetrahydrofolate dehydrogenase, D) methylenetetrahydrofolate cyclohydrolase, E) formyltetrahydrofolate deformylase, F) formyltetrahydrofolate synthetase, G) formate hydrogen lyase, H) hydrogenase, I) formate dehydrogenase, J) methanol dehydrogenase, K) spontaneous or formaldehyde activating enzyme, L) formaldehyde dehydrogenase, M) spontaneous or S-(hydroxymethyl)glutathione synthase, N) Glutathione-Dependent Formaldehyde Dehydrogenase, O) S-formylglutathione hydrolase, P) carbon monoxide dehydrogenase. See abbreviation list below for compound names.

FIG. 11 shows exemplary metabolic pathways enabling the conversion of acetyl-CoA to isopropanol. Enzymes are: T) acetyl-CoA carboxylase, U) acetoacetyl-CoA synthase, V) acetyl-CoA:acetyl-CoA acyltransferase, W) acetoacetyl-CoA hydrolase, acetoacetyl-CoA transferase, acetoacetyl-CoA ligase, phosphotransacetoacetylase/acetoacetate kinase, X) acetoacetate decarboxylase, Y) acetone reductase (or isopropanol dehydrogenase). See abbreviation list below for compound names.

FIG. 12 shows an exemplary β -ketoacyl-ACP pathway, a FAACPE cycle in combination with termination pathways for production of fatty alcohols, aldehydes, or acids from the acyl-ACP intermediate of the FAACPE cycle. Enzymes are: A) Acetyl-CoA carboxylase, B) Malonyl-CoA ACP transacylase, C) Acetoacetyl-ACP synthase, D) β -Ketoacyl-ACP synthase, E) β -Ketoacyl-ACP reductase, F) β -Hydroxyacyl-ACP reductase, G) Enoyl ACP-reductase, H) β -Ketoacyl-ACP synthase, I) Thioesterase, J) Fatty acyl-ACP reductase, K) Acyl-CoA synthase, L) Acyl-CoA reductase, M) Fatty aldehyde reductase, N) Fatty alcohol forming acyl-CoA reductase (FAR), O) Carboxylic acid reductase (CAR), and P) acyl-ACP reductase (alcohol forming).

FIGS. 13A and 13B depict the production of 1,3-butanediol (FIG. 13A) or ethanol (FIG. 13B) in *S. cerevisiae* transformed with plasmids comprising genes encoding various MI-FAE cycle and termination pathway enzymes, either with or without pflAV or PDH bypass, as provided in Example XIII.

FIGS. 14A-14D depict the production of pyruvic acid (FIG. 14A), succinic acid (FIG. 14B), acetic acid (FIG. 14C) or glucose (FIG. 14D) in *S. cerevisiae* transformed with plasmids comprising genes encoding various MI-FAE cycle and termination pathway enzymes, either with or without pflAV or PDH bypass, as provided in Example XIII.

FIG. 15 depicts the production of 1,3-butanediol in *S. cerevisiae* transformed with plasmids comprising genes encoding various MI-FAE cycle and termination pathway enzymes, either with or without pflAV or PDH bypass, as provided in Example XIII.

FIG. 16 depicts the estimated specific activity of five thiolases for acetyl-CoA condensation activity in *E. coli* as provided in Example XIV.

FIGS. 17A and 17B depict the estimated specific activity of two thiolases (1491 and 560) cloned in dual promoter yeast vectors with 1495 (a 3-hydroxybutyryl-CoA dehydrogenase) for acetyl-CoA condensation activity in *E. coli* as provided in Example XIV.

FIG. 18 depicts the time course of fluorescence detection of oxidation of NADH, which is used to measure the metabolism of acetoacetyl-CoA to 3-hydroxybutyryl-CoA by 3-hydroxybutyryl-CoA dehydrogenase, as provided in Example XIV. Acetoacetyl-CoA is metabolized to 3-hydroxybutyryl-CoA by 3-hydroxybutyryl-CoA dehydrogenase. The reaction requires oxidation of NADH, which can be monitored by fluorescence at an excitation wavelength at 340 nm and an emission at 460 nm. The oxidized form, NAD⁺, does not fluoresce. 1495, the Hbd from *Clostridium beijerinckii*, was assayed in the dual promoter yeast vectors that contained either 1491 (vector id=pY3Hd17) or 560 (vector id=pY3Hd16).

FIG. 19 depicts levels of NAD(P)H oxidation in the presence of 1 or 5 ug/ml NADH or 1 or 5 ug/ml NADPH, and shows that the Hbd prefers NADH over NADPH, as provided in Example XIV.

FIG. 20 depicts the activity data for crude lysates of an aldehyde reductase that converts 3-hydroxybutyryl-CoA to 3-hydroxybutyraldehyde and requires NAD(P)H oxidation, which can be used to monitor enzyme activity, as provided in Example XIV. The Ald from *Lactobacillus brevis* (Gene ID 707) was cloned in a dual vector that contained the alcohol dehydrogenase from *Clostridium saccharoperbutylacetonicum* (Gene ID 28). These two enzymes were cloned in another dual promoter yeast vector containing a Leu marker. A 707 lysate from *E. coli* was used as a standard.

FIG. 21 depicts the evaluation of ADH (Gene 28) in the dual promoter vector with ALD (Gene 707) with butyraldehyde, a surrogate substrate for 3-hydroxybutyraldehyde. 1,3-BDO is formed by an alcohol dehydrogenase (Adh), which reduces 3-hydroxybutyraldehyde in the presence of NAD(P)H, and the oxidation of NAD(P)H is used to monitor the reaction.

FIG. 22 depicts exemplary pathways for production of propionyl-CoA. Enzymes are: A) PEP carboxykinase, B) PEP carboxylase, C) Pyruvate kinase, D) Pyruvate carboxylase, E) Malate dehydrogenase, F) Fumarase, G) Fumarate reductase, H) Succinyl-CoA synthetase, I) Succinyl-CoA:3-ketoacid-CoA transferase, J) Methylmalonyl-CoA mutase, K) Methylmalonyl-CoA epimerase, L) Methylmalonyl-CoA decarboxylase. See abbreviation list below for compound names.

DETAILED DESCRIPTION OF THE INVENTION

The present invention is directed to metabolic and biosynthetic processes and microbial organisms capable of producing fatty alcohols, fatty aldehydes, fatty acids or isopropanol. The invention disclosed herein is based, at least in part, on non-naturally occurring microbial organisms capable of synthesizing fatty alcohols, fatty aldehydes, or fatty acids using a formaldehyde fixation pathway, a formate

assimilation pathway and/or a methanol metabolic pathway with a malonyl-CoA-independent fatty acid elongation (MI-FAE) cycle and/or malonyl-CoA dependent fatty acid elongation cycle (MD-FAE) cycle in combination with a termination pathway, or in some embodiments a fatty acyl-ACP elongation (FAACPE) cycle in combination with a termination pathway. The invention disclosed herein is also based, at least in part, on non-naturally occurring microbial organisms capable of synthesizing isopropanol using a formaldehyde fixation pathway, a formate assimilation pathway and/or a methanol metabolic pathway in combination with an isopropanol pathway. Additionally, in some embodiments, the non-naturally occurring microbial organisms can further include a methanol oxidation pathway, an acetyl-CoA pathway, a hydrogenase and/or a carbon monoxide dehydrogenase.

The following is a list of abbreviations and their corresponding compound or composition names. These abbreviations, which are used throughout the disclosure and the figures. It is understood that one of ordinary skill in the art can readily identify these compounds/compositions by such nomenclature. MeOH or MEOH=methanol; Fald=formaldehyde; GLC=glucose; G6P=glucose-6-phosphate; H6P=hexulose-6-phosphate; F6P=fructose-6-phosphate; FDP=fructose diphosphate or fructose-1,6-diphosphate; DHA=dihydroxyacetone; DHAP=dihydroxyacetone phosphate; G3P=and glyceraldehyde-3-phosphate; PYR=pyruvate; ACTP=acetyl-phosphate; ACCOA=acetyl-CoA; AACOA=acetoacetyl-CoA; MALCOA=malonyl-CoA; FTHF=formyltetrahydrofolate; THF=tetrahydrofolate; E4P=erythrose-4-phosphate; Xu5P=xylose-5-phosphate; Ru5P=ribulose-5-phosphate; 57P=sedoheptulose-7-phosphate; R₅P=ribose-5-phosphate; TCA=tricarboxylic acid; PEP=Phosphoenolpyruvate; OAA=Oxaloacetate; MAL=malate; FUM=Fumarate; SUCC=Succinate; SUCCOA=Succinyl-CoA; (R)-MMCOA=R-Methylmalonyl-CoA; (S)-MMCOA=S-Methylmalonyl-CoA; PPCOA=Propionyl-CoA.

It is also understood that association of multiple steps in a pathway can be indicated by linking their step identifiers with or without spaces or punctuation; for example, the following are equivalent to describe the 4-step pathway comprising Step W, Step X, Step Y and Step Z: steps WXYZ or W,X,Y,Z or W;X;Y;Z or W-X-Y-Z. One of ordinary skill can readily distinguish a single step designator of "AA" or "AB" or "AD" from a multiple step pathway description based on context and use in the description and figures herein.

As used herein, the term "non-naturally occurring" when used in reference to a microbial organism or microorganism of the invention is intended to mean that the microbial organism has at least one genetic alteration not normally found in a naturally occurring strain of the referenced species, including wild-type strains of the referenced species. Genetic alterations include, for example, modifications introducing expressible nucleic acids encoding metabolic polypeptides, other nucleic acid additions, nucleic acid deletions and/or other functional disruption of the microbial organism's genetic material. Such modifications include, for example, coding regions and functional fragments thereof, for heterologous, homologous or both heterologous and homologous polypeptides for the referenced species. Additional modifications include, for example, non-coding regulatory regions in which the modifications alter expression of a gene or operon. Exemplary metabolic polypeptides include enzymes or proteins within a fatty alcohol, fatty aldehyde or fatty alcohol biosynthetic pathway.

A metabolic modification refers to a biochemical reaction that is altered from its naturally occurring state. Therefore, non-naturally occurring microorganisms can have genetic modifications to nucleic acids encoding metabolic polypeptides, or functional fragments thereof. Exemplary metabolic modifications are disclosed herein.

As used herein, the term “isolated” when used in reference to a microbial organism is intended to mean an organism that is substantially free of at least one component as the referenced microbial organism is found in nature. The term includes a microbial organism that is removed from some or all components as it is found in its natural environment. The term also includes a microbial organism that is removed from some or all components as the microbial organism is found in non-naturally occurring environments. Therefore, an isolated microbial organism is partly or completely separated from other substances as it is found in nature or as it is grown, stored or subsisted in non-naturally occurring environments. Specific examples of isolated microbial organisms include partially pure microbes, substantially pure microbes and microbes cultured in a medium that is non-naturally occurring.

As used herein, the terms “microbial,” “microbial organism” or “microorganism” are intended to mean any organism that exists as a microscopic cell that is included within the domains of archaea, bacteria or eukarya. Therefore, the term is intended to encompass prokaryotic or eukaryotic cells or organisms having a microscopic size and includes bacteria, archaea and eubacteria of all species as well as eukaryotic microorganisms such as yeast and fungi. The term also includes cell cultures of any species that can be cultured for the production of a biochemical.

As used herein, the term “CoA” or “coenzyme A” is intended to mean an organic cofactor or prosthetic group (nonprotein portion of an enzyme) whose presence is required for the activity of many enzymes (the apoenzyme) to form an active enzyme system. Coenzyme A functions in certain condensing enzymes, acts in acetyl or other acyl group transfer and in fatty acid synthesis and oxidation, pyruvate oxidation and in other acetylation.

As used herein, the term “ACP” or “acyl carrier protein” refers to any of the relatively small acidic proteins that are associated with the fatty acid synthase system of many organisms, from bacteria to plants. ACPs can contain one 4'-phosphopantetheine prosthetic group bound covalently by a phosphate ester bond to the hydroxyl group of a serine residue. The sulfhydryl group of the 4'-phosphopantetheine moiety serves as an anchor to which acyl intermediates are (thio)esterified during fatty-acid synthesis. An example of an ACP is *Escherichia coli* ACP, a separate single protein, containing 77 amino-acid residues (8.85 kDa), wherein the phosphopantetheine group is linked to serine 36.

As used herein, the term “substantially anaerobic” when used in reference to a culture or growth condition is intended to mean that the amount of oxygen is less than about 10% of saturation for dissolved oxygen in liquid media. The term also is intended to include sealed chambers of liquid or solid medium maintained with an atmosphere of less than about 1% oxygen.

“Exogenous” as it is used herein is intended to mean that the referenced molecule or the referenced activity is introduced into the host microbial organism. The molecule can be introduced, for example, by introduction of an encoding nucleic acid into the host genetic material such as by integration into a host chromosome or as non-chromosomal genetic material such as a plasmid. Therefore, the term as it is used in reference to expression of an encoding nucleic

acid refers to introduction of the encoding nucleic acid in an expressible form into the microbial organism. When used in reference to a biosynthetic activity, the term refers to an activity that is introduced into the host reference organism. The source can be, for example, a homologous or heterologous encoding nucleic acid that expresses the referenced activity following introduction into the host microbial organism. Therefore, the term “endogenous” refers to a referenced molecule or activity that is present in the host. Similarly, the term when used in reference to expression of an encoding nucleic acid refers to expression of an encoding nucleic acid contained within the microbial organism. The term “heterologous” refers to a molecule or activity derived from a source other than the referenced species whereas “homologous” refers to a molecule or activity derived from the host microbial organism. Accordingly, exogenous expression of an encoding nucleic acid of the invention can utilize either or both a heterologous or homologous encoding nucleic acid.

It is understood that when more than one exogenous nucleic acid is included in a microbial organism that the more than one exogenous nucleic acids refers to the referenced encoding nucleic acid or biosynthetic activity, as discussed above. It is further understood, as disclosed herein, that such more than one exogenous nucleic acids can be introduced into the host microbial organism on separate nucleic acid molecules, on polycistronic nucleic acid molecules, or a combination thereof, and still be considered as more than one exogenous nucleic acid. For example, as disclosed herein a microbial organism can be engineered to express two or more exogenous nucleic acids encoding a desired pathway enzyme or protein. In the case where two exogenous nucleic acids encoding a desired activity are introduced into a host microbial organism, it is understood that the two exogenous nucleic acids can be introduced as a single nucleic acid, for example, on a single plasmid, on separate plasmids, can be integrated into the host chromosome at a single site or multiple sites, and still be considered as two exogenous nucleic acids. Similarly, it is understood that more than two exogenous nucleic acids can be introduced into a host organism in any desired combination, for example, on a single plasmid, on separate plasmids, can be integrated into the host chromosome at a single site or multiple sites, and still be considered as two or more exogenous nucleic acids, for example three exogenous nucleic acids. Thus, the number of referenced exogenous nucleic acids or biosynthetic activities refers to the number of encoding nucleic acids or the number of biosynthetic activities, not the number of separate nucleic acids introduced into the host organism.

As used herein, the term “gene disruption,” or grammatical equivalents thereof, is intended to mean a genetic alteration that renders the encoded gene product inactive or attenuated. The genetic alteration can be, for example, deletion of the entire gene, deletion of a regulatory sequence required for transcription or translation, deletion of a portion of the gene which results in a truncated gene product, or by any of various mutation strategies that inactivate or attenuate the encoded gene product, for example, replacement of a gene's promoter with a weaker promoter, replacement or insertion of one or more amino acid of the encoded protein to reduce its activity, stability or concentration, or inactivation of a gene's transactivating factor such as a regulatory protein. One particularly useful method of gene disruption is complete gene deletion because it reduces or eliminates the occurrence of genetic reversions in the non-naturally occurring microorganisms of the invention. A gene disruption also

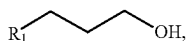
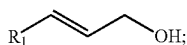
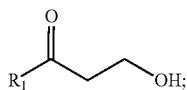
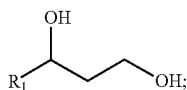
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includes a null mutation, which refers to a mutation within a gene or a region containing a gene that results in the gene not being transcribed into RNA and/or translated into a functional gene product. Such a null mutation can arise from many types of mutations including, for example, inactivating point mutations, deletion of a portion of a gene, entire gene deletions, or deletion of chromosomal segments.

As used herein, the term “growth-coupled” when used in reference to the production of a biochemical product is intended to mean that the biosynthesis of the referenced biochemical product is produced during the growth phase of a microorganism. In a particular embodiment, the growth-coupled production can be obligatory, meaning that the biosynthesis of the referenced biochemical is an obligatory product produced during the growth phase of a microorganism.

As used herein, the term “attenuate,” or grammatical equivalents thereof, is intended to mean to weaken, reduce or diminish the activity or amount of an enzyme or protein. Attenuation of the activity or amount of an enzyme or protein can mimic complete disruption if the attenuation causes the activity or amount to fall below a critical level required for a given pathway to function. However, the attenuation of the activity or amount of an enzyme or protein that mimics complete disruption for one pathway, can still be sufficient for a separate pathway to continue to function. For example, attenuation of an endogenous enzyme or protein can be sufficient to mimic the complete disruption of the same enzyme or protein for production of a fatty alcohol, fatty aldehyde or fatty acid product of the invention, but the remaining activity or amount of enzyme or protein can still be sufficient to maintain other pathways, such as a pathway that is critical for the host microbial organism to survive, reproduce or grow. Attenuation of an enzyme or protein can also be weakening, reducing or diminishing the activity or amount of the enzyme or protein in an amount that is sufficient to increase yield of a fatty alcohol, fatty aldehyde or fatty acid product of the invention, but does not necessarily mimic complete disruption of the enzyme or protein.

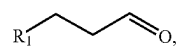
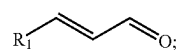
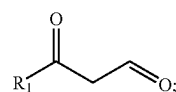
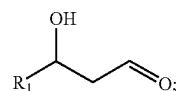
The term “fatty alcohol,” as used herein, is intended to mean an aliphatic compound that contains one or more hydroxyl groups and contains a chain of 4 or more carbon atoms. The fatty alcohol possesses the group $-\text{CH}_2\text{OH}$ that can be oxidized so as to form a corresponding aldehyde or acid having the same number of carbon atoms. A fatty alcohol can also be a saturated fatty alcohol, an unsaturated fatty alcohol, a 1,3-diol, or a 3-oxo-alkan-1-ol. Exemplary fatty alcohols include a compound of Formula (III)—(VI):



wherein R_1 is a C_{1-24} linear alkyl.

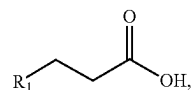
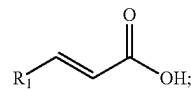
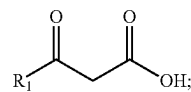
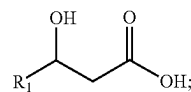
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The term “fatty aldehyde,” as used herein, is intended to mean an aliphatic compound that contains an aldehyde (CHO) group and contains a chain of 4 or more carbon atoms. The fatty aldehyde can be reduced to form the corresponding alcohol or oxidized to form the carboxylic acid having the same number of carbon atoms. A fatty aldehyde can also be a saturated fatty aldehyde, an unsaturated fatty aldehyde, a 3-hydroxyaldehyde or 3-oxoaldehyde. Exemplary fatty aldehydes include a compound of Formula (VII)—(X):



wherein R_1 is a C_{1-24} linear alkyl.

The term “fatty acid,” as used herein, is intended to mean an aliphatic compound that contains a carboxylic acid group and contains a chain of 4 or more carbon atoms. The fatty acid can be reduced to form the corresponding alcohol or aldehyde having the same number of carbon atoms. A fatty acid can also be a saturated fatty acid, an unsaturated fatty acid, a 3-hydroxyacid or a 3-oxoacids. Exemplary fatty acids include a compound of Formula (XI)—(XIV):



wherein R_1 is a C_{1-24} linear alkyl.

The term “alkyl” refers to a linear saturated monovalent hydrocarbon. The alkyl can be a linear saturated monovalent hydrocarbon that has 1 to 24 (C_{1-24}), 1 to 17 (C_{1-17}), or 9 to 13 (C_{9-13}) carbon atoms. Examples of alkyl groups include, but are not limited to, methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl, octyl, nonyl, decyl, undecyl and dodecyl. For example, C_{9-13} alkyl refers to a linear saturated monovalent hydrocarbon of 9 to 13 carbon atoms.

As used herein, “isopropanol” is intended to mean a secondary alcohol, with the molecular formula of $\text{C}_3\text{H}_8\text{O}$

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and a molecular mass of 60.1 g/mol, wherein the alcohol carbon is attached to two other carbons. This attachment is sometimes shown as $(\text{CH}_3)_2\text{CHOH}$. Isopropanol is also known in the art as propan-2-ol, 2-propanol or the abbreviation IPA. Isopropanol is an isomer of n-propanol.

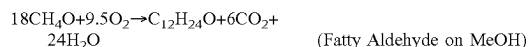
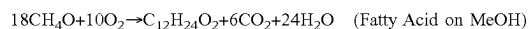
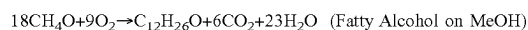
As used herein, the phrase “enhance carbon flux” is intended to mean to intensify, increase, or further improve the extent or flow of metabolic carbon through or to a desired pathway, pathway product, intermediate, or compound. The intensity, increase or improvement can be relative to a predetermined baseline of a pathway product, intermediate or compound. For example, an increased yield of acetyl-CoA can be achieved per mole of methanol with a phosphoketolase enzyme described herein (see, e.g., FIG. 1) than in the absence of a phosphoketolase enzyme. Similarly, an increased yield of acetyl-CoA can be achieved per mole of methanol with the formate assimilation enzymes (see, e.g., FIG. 1) than in the absence of the enzymes. Since an increased yield of acetyl-CoA can be achieved, a higher yield of acetyl-CoA derived products, such as fatty alcohols, fatty acids, fatty aldehydes or isopropanol of the invention, can also be achieved.

Provided herein are methanol metabolic pathways and a methanol oxidation pathway to improve that availability of reducing equivalents and/or substrates for production of a compound of the invention. Because methanol is a relatively inexpensive organic feedstock that can be used as a redox, energy, and carbon source for the production of chemicals such as fatty alcohols, fatty acids, fatty aldehydes or isopropanol, and their intermediates, it is a desirable substrate for the non-naturally occurring microbial organisms of the invention. Employing one or more methanol metabolic enzymes as described herein, for example as shown in FIGS. 1 and 10, methanol can enter central metabolism in most production hosts by employing methanol dehydrogenase (FIG. 1, step A) along with a pathway for formaldehyde assimilation. One exemplary formaldehyde assimilation pathway that can utilize formaldehyde produced from the oxidation of methanol is shown in FIG. 1, which involves condensation of formaldehyde and D-ribulose-5-phosphate to form hexulose-6-phosphate (H6P) by hexulose-6-phosphate synthase (FIG. 1, step B). The enzyme can use Mg^{2+} or Mn^{2+} for maximal activity, although other metal ions are useful, and even non-metal-ion-dependent mechanisms are contemplated. H6P is converted into fructose-6-phosphate by 6-phospho-3-hexuloisomerase (FIG. 1, step C). Another exemplary pathway that involves the detoxification and assimilation of formaldehyde produced from the oxidation of methanol proceeds through dihydroxyacetone. Dihydroxyacetone synthase (FIG. 1, step D) is a transketolase that first transfers a glycoaldehyde group from xylulose-5-phosphate to formaldehyde, resulting in the formation of dihydroxyacetone (DHA) and glyceraldehyde-3-phosphate (G3P), which is an intermediate in glycolysis. The DHA obtained from DHA synthase can be then further phosphorylated to form DHA phosphate by a DHA kinase DHAP can be assimilated into glycolysis, e.g. via isomerization to G3P, and several other pathways. Alternatively, DHA and G3P can be converted by fructose-6-phosphate aldolase to form fructose-6-phosphate (F6P).

By combining the pathways for methanol oxidation (FIG. 1, step A) and formaldehyde fixation (FIG. 1, Steps B and C or Step D), molar yields of 0.333 mol acetyl-CoA/mol methanol can be achieved for production of a fatty alcohol, a fatty acid, a fatty aldehyde, isopropanol, and their intermediates. The following maximum theoretical yield stoichiometries for a fatty alcohol (e.g., a C12), a fatty acid (e.g.,

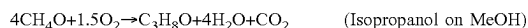
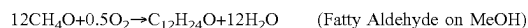
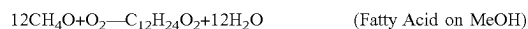
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a C12), a fatty aldehyde (e.g., a C12), isopropanol are thus made possible by combining the steps for methanol oxidation, formaldehyde fixation, and product synthesis.



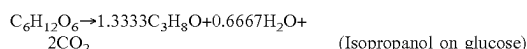
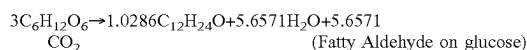
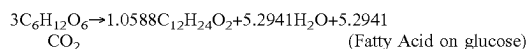
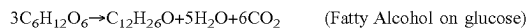
The yield on several substrates, including methanol, can be further increased by capturing some of the carbon lost from the conversion of pathway intermediates, e.g. pyruvate to acetyl-CoA, using one of the formate reutilization pathways shown in FIG. 1. For example, the CO_2 generated by conversion of pyruvate to acetyl-CoA (FIG. 1, step R) can be converted to formate via formate dehydrogenase (FIG. 1, step S). Alternatively, pyruvate formate lyase, which forms formate directly instead of CO_2 , can be used to convert pyruvate to acetyl-CoA (FIG. 1, step Q). Formate can be converted to formaldehyde by using: 1) formate reductase (FIG. 1, step E), 2) a formyl-CoA synthetase, transferase, or ligase along with formyl-CoA reductase (FIG. 1, steps F-G), or 3) formyltetrahydrofolate synthetase, methenyltetrahydrofolate cyclohydrolase, methylenetetrahydrofolate dehydrogenase, and formaldehyde-forming enzyme (FIG. 1, steps H—I-J-K). Conversion of methylene-THF to formaldehyde alternatively will occur spontaneously. Alternatively, formate can be reutilized by converting it to pyruvate or acetyl-CoA using FIG. 1, steps H—I-J-L-M-N or FIG. 1, steps H—I-J-O—P, respectively. Formate reutilization is also useful when formate is an external carbon source. For example, formate can be obtained from organocatalytic, electrochemical, or photoelectrochemical conversion of CO_2 to formate. An alternative source of methanol for use in the present methods is organocatalytic, electrochemical, or photoelectrochemical conversion of CO_2 to methanol.

By combining the pathways for methanol oxidation (FIG. 1, step A), formaldehyde fixation (FIG. 1, Steps B and C or Step D), and formate reutilization, molar yields as high as 0.500 mol acetyl-CoA/mol methanol can be achieved for production of a fatty alcohol, a fatty acid, a fatty aldehyde, isopropanol, and their intermediates. Thus, for example, the following maximum theoretical yield stoichiometries for a fatty alcohol (e.g., a C12), a fatty acid (e.g., a C12), a fatty aldehyde (e.g., a C12), and isopropanol are thus made possible by combining the steps for methanol oxidation, formaldehyde fixation, formate reutilization, and product synthesis.

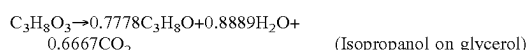


By combining pathways for formaldehyde fixation and formate reutilization, yield increases on additional substrates are also available including but not limited to glucose, glycerol, sucrose, fructose, xylose, arabinose and galactose. For example, the following maximum theoretical yield stoichiometries for a fatty alcohol (e.g., a C12), a fatty acid (e.g., a C12), a fatty aldehyde (e.g., a C12), and isopropanol on glucose are made possible by combining the steps for formaldehyde fixation, formate reutilization, and product synthesis.

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Similarly, the maximum theoretical yield of a fatty alcohol, a fatty acid, a fatty aldehyde, or isopropanol from glycerol can be increased by enabling fixation of formaldehyde from generation and utilization of formate. The following maximum theoretical yield stoichiometries for a fatty alcohol (e.g., a C12), a fatty acid (e.g., a C12), a fatty aldehyde (e.g., a C12), and isopropanol on glycerol are thus made possible by combining the steps for formaldehyde fixation, formate reutilization, and product synthesis.



In numerous engineered pathways, product yields based on carbohydrate feedstock are hampered by insufficient reducing equivalents or by loss of reducing equivalents to byproducts. Methanol is a relatively inexpensive organic feedstock that can be used to generate reducing equivalents by employing one or more methanol metabolic enzymes as shown in FIG. 10. Reducing equivalents can also be extracted from hydrogen and carbon monoxide by employing hydrogenase and carbon monoxide dehydrogenase enzymes, respectively, as shown in FIG. 10. The reducing equivalents are then passed to acceptors such as oxidized ferredoxins, oxidized quinones, oxidized cytochromes, NAD(P)⁺, water, or hydrogen peroxide to form reduced ferredoxin, reduced quinones, reduced cytochromes, NAD(P)H, H₂, or water, respectively. Reduced ferredoxin, reduced quinones and NAD(P)H are particularly useful as they can serve as redox carriers for various Wood-Ljungdahl pathway, reductive TCA cycle, or product pathway enzymes.

The reducing equivalents produced by the metabolism of methanol, hydrogen, and carbon monoxide can be used to power several fatty alcohol, fatty acid, fatty aldehyde, and isopropanol production pathways. For example, the maximum theoretical yield of a fatty alcohol, a fatty acid, a fatty aldehyde, or isopropanol from glucose and glycerol can be increased by enabling fixation of formaldehyde, formate reutilization, and extraction of reducing equivalents from an external source such as hydrogen. In fact, by combining pathways for formaldehyde fixation, formate reutilization, reducing equivalent extraction, and product synthesis, the following maximum theoretical yield stoichiometries for fatty alcohol, a fatty acid, a fatty aldehyde, and isopropanol on glucose and glycerol are made possible.



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In most instances, achieving such maximum yield stoichiometries may require some oxidation of reducing equivalents (e.g., $\text{H}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{H}_2\text{O}$, $\text{CO} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}_2$, $\text{CH}_4\text{O} + 1.5\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O}$) to provide sufficient energy for the substrate to product pathways to operate. Nevertheless, if sufficient reducing equivalents are available, enabling pathways for fixation of formaldehyde, formate reutilization, extraction of reducing equivalents, and product synthesis can even lead to production of a fatty alcohol, a fatty acid, a fatty aldehyde, isopropanol, and their intermediates, directly from CO₂.

Pathways identified herein, and particularly pathways exemplified in specific combinations presented herein, are superior over other pathways based in part on the applicant's ranking of pathways based on attributes including maximum theoretical compound yield, maximal carbon flux, maximal production of reducing equivalents, minimal production of CO₂, pathway length, number of non-native steps, thermodynamic feasibility, number of enzymes active on pathway substrates or structurally similar substrates, and having steps with currently characterized enzymes, and furthermore, the latter pathways are even more favored by having in addition at least the fewest number of non-native steps required, the most enzymes known active on pathway substrates or structurally similar substrates, and the fewest total number of steps from central metabolism.

In some embodiments, the microorganisms of the invention can utilize a heterologous MI-FAE cycle and/or a MD-FAE cycle coupled with an acyl-CoA termination pathway to form fatty alcohols, fatty aldehydes, or fatty acids. The MI-FAE cycle can include a thiolase, a 3-oxoacyl-CoA reductase, a 3-hydroxyacyl-CoA dehydratase and an enoyl-CoA reductase. The MD-FAE cycle can include an elongase, a 3-oxoacyl-CoA reductase, a 3-hydroxyacyl-CoA dehydratase and an enoyl-CoA reductase. Each passage through the MI-FAE cycle and/or the MD-FAE cycle results in the formation of an acyl-CoA elongated by a single two carbon unit compared to the acyl-CoA substrate entering the elongation cycle. Products can be even or odd chain length, depending on the initial substrate entering the acyl-CoA elongation pathway, i.e. two acetyl-CoA substrates, malonyl-CoA or one acetyl-CoA substrate combined with a propionyl-CoA substrate. Elongation of the two acetyl-CoA substrates or malonyl-CoA produces an even chain length product, whereas elongation with the propionyl-CoA substrate produces an odd chain length product. A termination pathway catalyzes the conversion of a MI-FAE intermediate and/or a MD-FAE intermediate, such as the acyl-CoA, to its corresponding fatty alcohol, fatty aldehyde, or fatty acid product. MI-FAE cycle, MD-FAE cycle and termination pathway enzymes can be expressed in one or more com-

partments of the microorganism. For example, in one embodiment, all MI-FAE cycle and termination pathway enzymes are expressed in the cytosol. In another embodiment, all MD-FAE cycle and termination pathway enzymes are expressed in the cytosol. Additionally, the microorganisms of the invention can be engineered to optionally secrete the desired product into the culture media or fermentation broth for further manipulation or isolation.

In some embodiments, the microorganisms of the invention can utilize a heterologous FAACPE cycle coupled with an acyl-ACP termination pathway to form fatty alcohols, fatty aldehydes, or fatty acids. The FAACPE cycle can include a β -ketoacyl-ACP synthase, a β -ketoacyl-ACP reductase, a β -hydroxyacyl-ACP reductase, and an enoyl ACP-reductase. Each passage through the FAACPE cycle results in the formation of an acyl-ACP elongated by a single two carbon unit compared to the acyl-ACP substrate entering the elongation cycle. Products can be even or odd chain length, depending on the initial substrate entering the FAACPE pathway, i.e. acetoacetyl-ACP or 3-oxovaleryl-ACP. Elongation of the acetoacetyl-ACP substrates produces an even chain length product, whereas elongation with the 3-oxovaleryl-ACP substrate produces an odd chain length product. A termination pathway catalyzes the conversion of a FAACPE intermediate, such as the acyl-ACP, to its corresponding fatty alcohol, fatty aldehyde, or fatty acid product. FAACPE cycle and termination pathway enzymes can be expressed in one or more compartments of the microorganism. For example, in one embodiment, all FAACPE cycle and termination pathway enzymes are expressed in the cytosol. Additionally, the microorganisms of the invention can be engineered to optionally secrete the desired product into the culture media or fermentation broth for further manipulation or isolation.

Products of the invention include fatty alcohols, fatty aldehydes, or fatty acids derived from intermediates of the MI-FAE cycle, MD-FAE cycle, and/or FAACPE cycle. For example, alcohol products can include saturated fatty alcohols, unsaturated fatty alcohols, 1,3-diols, and 3-oxo-alkan-1-ols. Aldehyde products can include saturated fatty aldehydes, unsaturated fatty aldehydes, 3-hydroxyaldehydes and 3-oxoaldehydes. Acid products can include saturated fatty acids, unsaturated fatty acids, 3-hydroxyacids and 3-oxoacids. These products can further be converted to derivatives such as fatty esters, either by chemical or enzymatic means. Methods for converting fatty alcohols to esters are well known in the art. Another product of the invention is isopropanol.

The invention also encompasses fatty alcohol, fatty aldehyde, and fatty acid chain-length control strategies in conjunction with host strain engineering strategies, such that the non-naturally occurring microorganism of the invention efficiently directs carbon and reducing equivalents toward fermentation products of a specific chain length.

Recombinant microorganisms of the invention can produce commercial quantities of a fatty alcohol, fatty aldehyde, or fatty acid ranging in chain length from four carbon atoms (C_4) to twenty-four carbon atoms (C_{24}) or more carbon atoms. The microorganism of the invention can produce a desired product that is at least 50%, 60%, 70%, 75%, 85%, 90%, 95% or more selective for a particular chain length. The carbon chain-length of the product can be controlled by one or more enzymes of the MI-FAE cycle (steps A/B/C/D of FIG. 7) and/or one or more enzymes of the MD-FAE cycle (steps E/B/C/D of FIG. 7) in combination with one or more termination pathway enzymes (steps E-N of FIG. 8). Chain length can be capped during the

elongation cycle by one or more MI-FAE cycle enzymes (thiolase, 3-oxoacyl-CoA reductase, 3-hydroxyacyl-CoA dehydratase and/or enoyl-CoA reductase) exhibiting selectivity for MI-FAE cycle substrates having a number of carbon atoms that are no greater than the desired product size. Alternatively, or in addition, chain length can be capped during the elongation cycle by one or more MD-FAE cycle enzymes (elongase, 3-oxoacyl-CoA reductase, 3-hydroxyacyl-CoA dehydratase and/or enoyl-CoA reductase). Chain length can be further constrained by one or more enzymes catalyzing the conversion of the MI-FAE cycle intermediate to the fatty alcohol, fatty aldehyde or fatty acid product such that the one or more termination enzymes only reacts with substrates having a number of carbon atoms that are no less than the desired fatty alcohol, fatty aldehyde or fatty acid product.

The termination pathway enzymes catalyzing conversion of a MI-FAE-CoA intermediate or MD-FAE-CoA intermediate to a fatty alcohol can include enzyme combinations of a fatty acyl-CoA reductase (alcohol or aldehyde forming), a fatty aldehyde reductase, an acyl-ACP reductase, an acyl-CoA:ACP acyltransferase, a thioesterase, an acyl-CoA hydrolase and/or a carboxylic acid reductase (see, e.g., pathways G; E/F; K/J/F; H/N/F; or K/LN/F of FIG. 8). Termination pathway enzymes for converting a MI-FAE-CoA intermediate or MD-FAE-CoA intermediate to a fatty acid can include enzyme combinations of a thioesterase, a CoA hydrolase, an acyl-CoA:ACP acyltransferase, an aldehyde dehydrogenase and/or an acyl-ACP reductase (see, e.g., pathways H; K/L; E/N; K/J/N of FIG. 8). For production of a fatty aldehyde, the termination pathway enzymes can include enzyme combinations of a fatty acyl-CoA reductase (aldehyde forming), an acyl-ACP reductase, an acyl-CoA:ACP acyltransferase, a thioesterase, an acyl-CoA hydrolase and/or a carboxylic acid reductase (see, e.g., pathways E; K/J; H/N; or K/LN of FIG. 8).

The carbon chain-length of the product can also be controlled by one or more enzymes of the FAACPE cycle (steps H/E/F/G of FIG. 12) in combination with one or more termination pathway enzymes (steps I-O of FIG. 12). Chain length can be capped during the elongation cycle by one or more FAACPE cycle enzymes (β -ketoacyl-ACP synthase, β -ketoacyl-ACP reductase, β -hydroxyacyl-ACP reductase, and/or enoyl ACP-reductase) exhibiting selectivity for FAACPE cycle substrates having a number of carbon atoms that are no greater than the desired product size. Chain length can be further constrained by one or more enzymes catalyzing the conversion of the FAACPE cycle intermediate to the fatty alcohol, fatty aldehyde or fatty acid product such that the one or more termination enzymes only reacts with substrates having a number of carbon atoms that are no less than the desired fatty alcohol, fatty aldehyde or fatty acid product.

The termination pathway enzymes catalyzing conversion of a FAACPE cycle intermediate to a fatty alcohol can include enzyme combinations of a thioesterase, a fatty acyl-ACP reductase, an acyl-CoA synthase, an acyl-CoA reductase, a fatty aldehyde reductase, a fatty alcohol forming acyl-CoA reductase (FAR), and/or a carboxylic acid reductase (CAR) (see, e.g., pathways J/M; I/K/L/M; I/O/M; and I/K/N of FIG. 12). Termination pathway enzyme for converting a FAACPE intermediate to a fatty acid can include a thioesterase (see, e.g., pathways I of FIG. 12). For production of a fatty aldehyde, the termination pathway enzymes can include combinations of a thioesterase, a fatty acyl-ACP reductase, an acyl-CoA synthase, an acyl-CoA

reductase, a fatty aldehyde reductase, and/or a carboxylic acid reductase (CAR), (see, e.g., pathways J; I/K/L; and I/O of FIG. 12).

The non-naturally occurring microbial organisms of the invention can also efficiently direct cellular resources, including carbon, energy and reducing equivalents, to the production of fatty alcohols, fatty aldehydes and fatty acids, thereby resulting in improved yield, productivity and/or titer relative to a naturally occurring organism. In one embodiment, the microorganism is modified to increase cytosolic acetyl-CoA levels. In another embodiment, the microorganism is modified to efficiently direct cytosolic acyl-CoA into fatty alcohols, fatty aldehydes or fatty acids rather than other byproducts or cellular processes. Enzymes or pathways that lead to the formation of byproducts can be attenuated or deleted. Exemplary byproducts include, but are not limited to, ethanol, glycerol, lactate, acetate, esters and carbon dioxide. Additional byproducts can include fatty-acyl-CoA derivatives such as alcohols, alkenes, alkanes, esters, acids and aldehydes. Accordingly, a byproduct can include any fermentation product diverting carbon and/or reducing equivalents from the product of interest.

In another embodiment, the availability of reducing equivalents or redox ratio is increased. In yet another embodiment, the cofactor requirements of the microorganism are balanced such that the same reduced cofactors generated during carbon assimilation and central metabolism are utilized by MI-FAE cycle, MD-FAE cycle and/or termination pathway enzymes. In yet another embodiment, the fatty alcohol, fatty aldehyde or fatty acid producing organism expresses a transporter which exports the fatty alcohol, fatty aldehyde or fatty acid from the cell.

Microbial organisms capable of fatty alcohol production are exemplified herein with reference to the *Saccharomyces cerevisiae* genetic background. However, with the complete genome sequence available now for thousands of species (with more than half of these available on public databases such as the NCBI), the identification of an alternate species homolog for one or more genes, including for example, orthologs, paralogs and nonorthologous gene displacements, and the interchange of genetic alterations between eukaryotic organisms is routine and well known in the art. Accordingly, the metabolic alterations enabling production of fatty alcohols described herein with reference to a particular organism such as *Saccharomyces cerevisiae* can be readily applied to other microorganisms. Given the teachings and guidance provided herein, those skilled in the art understand that a metabolic alteration exemplified in one organism can be applied equally to other organisms.

The methods of the invention are applicable to various prokaryotic and eukaryotic organisms such as bacteria, yeast and fungus. For example, the yeast can include *Saccharomyces cerevisiae* and *Rhizopus arrhizus*. Exemplary eukaryotic organisms can also include Crabtree positive and negative yeasts, and yeasts in the genera *Saccharomyces*, *Kluyveromyces*, *Candida* or *Pichia*. Further exemplary eukaryotic species include those selected from *Schizosaccharomyces pombe*, *Kluyveromyces lactis*, *Kluyveromyces marxianus*, *Aspergillus terreus*, *Aspergillus niger*, *Rhizopus arrhizus*, *Rhizopus oryzae*, *Candida albicans*, *Candida boidinii*, *Candida sonorensis*, *Candida tropicalis*, *Yarrowia lipolytica* and *Pichia pastoris*. Additionally, select cells from larger eukaryotic organisms are also applicable to methods of the present invention. Exemplary bacteria include species selected from *Escherichia coli*, *Klebsiella oxytoca*, *Anaerobiospinillum succiniciproducens*, *Actinobacillus succinogenes*, *Mannheimia succiniciproducens*, *Rhizobium etli*,

Bacillus subtilis, *Corynebacterium glutamicum*, *Gluconobacter oxydans*, *Zymomonas mobilis*, *Lactococcus lactis*, *Lactobacillus plantarum*, *Streptomyces coelicolor*, *Clostridium acetobutylicum*, *Pseudomonas fluorescens*, and *Pseudomonas putida*.

In some aspects of the invention, production of fatty alcohols, fatty aldehydes and fatty acids through the MI-FAE cycle and termination pathways disclosed herein are particularly useful because the cycle and pathways result in higher product and ATP yields than through naturally occurring biosynthetic pathways such as the well-known malonyl-CoA dependent fatty acid synthesis pathway, or in some aspects the malonyl-ACP dependent fatty acid synthesis pathway. For example, using acetyl-CoA as a C2 extension unit (e.g. step A, FIG. 2) instead of malonyl-acyl carrier protein (malonyl-ACP) saves one ATP molecule per unit flux of acetyl-CoA entering the MI-FAE cycle. The MI-FAE cycle results in acyl-CoA instead of acyl-ACP, and can preclude the need of the ATP-consuming acyl-CoA synthase reactions for the production of octanol and other fatty alcohols, fatty aldehydes or fatty acids if acetyl-CoA is used as the extender unit. The fatty alcohol, fatty aldehyde and fatty acid producing organisms of the invention can additionally allow the use of biosynthetic processes to convert low cost renewable feedstock for the manufacture of chemical products.

The eukaryotic organism of the invention can be further engineered to metabolize and/or co-utilize a variety of feedstocks including glucose, xylose, fructose, syngas, methanol, and the like.

Chain length control can be achieved using a combination of highly active enzymes with suitable substrate ranges appropriate for biosynthesis of the desired fatty alcohol, fatty aldehyde, or fatty acid. Chain length of the product can be controlled using one or more enzymes of MI-FAE cycle, MD-FAE cycle, FAACPE cycle or termination pathway. As described herein, chain length can be capped during the MI-FAE cycle by one or more MI-FAE cycle enzymes (thiolase, 3-oxoacyl-CoA reductase, 3-hydroxyacyl-CoA dehydratase and/or enoyl-CoA reductase), in the case of the MD-FAE cycle, one or more MD-FAE cycle enzymes (elongase, 3-oxoacyl-CoA reductase, 3-hydroxyacyl-CoA dehydratase and/or enoyl-CoA reductase), and in the case of the FAACPE cycle, one or more enzymes (β -ketoacyl-ACP synthase, β -ketoacyl-ACP reductase, β -hydroxyacyl-ACP reductase and/or enoyl ACP-reductase), exhibiting selectivity for MI-FAE cycle, MD-FAE cycle and/or FAACPE cycle substrates having a number of carbon atoms that are no greater than the desired product size. Since enzymes are reversible, any of the elongation pathway enzymes can serve in this capacity. Selecting enzymes with broad substrate ranges but defined chain-length boundaries enables the use of a single enzyme to catalyze multiple cycles of elongation, while conferring product specificity. To further hone specificity and prevent the accumulation of shorter byproducts, selectivity is further constrained by product-forming termination enzymes, such that one or more enzymes are selective for acyl-CoA, acyl-ACP or other termination pathway substrates having a number of carbon atoms that are no less than the desired chain length. The deletion or attenuation of endogenous pathway enzymes that produce different chain length products can further hone product specificity.

Using the approaches outlined herein, one skilled in the art can select enzymes from the literature with characterized substrate ranges that selectively produce a fatty alcohol, fatty aldehyde or fatty acid product of a specific chain length. To selectively produce fatty alcohols, fatty aldehydes

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or fatty acids of a desired length, one can utilize combinations of known enzymes in the literature with different selectivity ranges as described above. For example, a non-naturally occurring microbial organism that produces C₁₆ fatty alcohol can express enzymes such as the *Rattus norvegicus* Acaal thiolase and the enoyl-CoA reductase of *Mycobacterium smegmatis*, which only accept substrates up to length C₁₆. Coupling one or both chain elongation enzymes with a C₁₆-C₁₈ fatty acyl-CoA reductase (alcohol or aldehyde forming) such as FAR of *Simmondsia chinensis* further increases product specificity by reducing the synthesis of shorter alcohol products. As another example, a non-naturally occurring microbial organism of the invention can selectively produce alcohols of length C₁₄ by combining the 3-hydroxyacyl-CoA dehydratase of *Arabidopsis thaliana* with the acyl-CoA reductase Ac1 of *Acinetobacter* sp. Strain M-1. To produce 3-oxoacids of length C₁₄, one can, for example, combine the rat thiolase with the 3-oxoacyl-CoA hydrolase of *Solanum lycopersicum*. As still a further example, to produce C₁₈ fatty acids, one can combine the *Salmonella enterica* fadE enoyl-CoA reductase with the tesB thioesterase of *E. coli*. In yet another example, selective production of C₆ alcohols are formed by combining the paaH1 thiolase from *Ralstonia eutropha* with the *Leifsonia* sp. S749 alcohol dehydrogenase lsadh.

Exemplary MI-FAE cycle, MD-FAE cycle and termination pathway enzymes are described in detail in Example IV. The biosynthetic enzymes described herein exhibit varying degrees of substrate specificity. Exemplary substrate ranges of enzymes characterized in the literature are shown in the table below and described in further detail in Example IV.

Pathway step	Chain length	Gene	Organism
2A	C4	AtoB	<i>Escherichia coli</i>
2A	C6	PhaD	<i>Pseudomonas putida</i>
2A	C6-C8	BktB	<i>Ralstonia eutropha</i>
2A	C10-C16	Acaal1a	<i>Rattus norvegicus</i>
2B	C4	Hbd	<i>Clostridium acetobutylicum</i>
2B	C4-C6	paaH1	<i>Ralstonia eutropha</i>
2B	C4-C10	HADH	<i>Sus scrofa</i>
2B/C	C4-C18	FadB	<i>Escherichia coli</i>
2B/C	C4-C18	Fox2	<i>Candida tropicalis</i>
2B/C	C4-C18	Fox2	<i>Saccharomyces cerevisiae</i>
2C	C4-C6	crt	<i>Clostridium acetobutylicum</i>
2C	C4-C7	pimF	<i>Rhodospirillum rubrum</i>
2C	C4-C14	MFP2	<i>Arabidopsis thaliana</i>
2D	C4-C6	ECR1	<i>Euglena gracilis</i>
2D	C6-C8	ECR3	<i>Euglena gracilis</i>
2D	C8-10	ECR2	<i>Euglena gracilis</i>
2D	C8-C16	ECR	<i>Rattus norvegicus</i>
2D	C10-C16	ECR	<i>Mycobacterium smegmatis</i>
2D	C2-C18	fadE	<i>Salmonella enterica</i>
2E	C2-C4	bphG	<i>Pseudomonas sp</i>
2E	C4	Bld	<i>Clostridium saccharoperbutylacetonicum</i>

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Pathway step	Chain length	Gene	Organism
5 2E	C12-C20	ACR	<i>Acinetobacter calcoaceticus</i>
2E	C14-C18	Acrl	<i>Acinetobacter</i> sp. Strain M-1
2E	C16-C18	Rv1543, Rv3391	<i>Mycobacterium tuberculosis</i>
10 2E	C18	FAR1, FAR2	<i>Mus musculus</i>
2E	C12-C20	orf1594	<i>Synechococcus elongatus</i> PCC7942
2E	C6-C18	Maqu_2220	<i>Marinobacter aquaeolei</i>
15 2F	C6-C7	lsadh	<i>Leifsonia</i> sp. S749
2F	C2-C8	yqhD	<i>Escherichia coli</i>
2F	C3-C10	Adh	<i>Pseudomonas putida</i>
20 2F	C2-C14	alrA	<i>Acinetobacter</i> sp. strain M-1
2F	C2-C30	ADH1	<i>Geobacillus thermophilus</i>
2F	C3-C8	ADH6	<i>Saccharomyces cerevisiae</i> s288c
25 2G	C2	adhE	<i>Escherichia coli</i>
2G	C2-C8	adhe2	<i>Clostridium acetobutylicum</i>
2G	C14-C16	At3g11980	<i>Arabidopsis thaliana</i>
2G	C16	At3g44560	<i>Arabidopsis thaliana</i>
30 2G	C16-C18	FAR	<i>Simmondsia chinensis</i>
2H	C4	Cat2	<i>Clostridium kluyveri</i>
2H	C4-C6	Acot12	<i>Rattus norvegicus</i>
2H	C14	MKS2	<i>Solanum lycopersicum</i>
2L	C8-C10	fatB2	<i>Cuphea hookeriana</i>
2L	C12	fatB	<i>Umbellularia californica</i>
40 2L	C14-C16	fatB3	<i>Cuphea hookeriana</i>
2L	C18	tesA	<i>Escherichia coli</i>
2N	C12-C18	Car	<i>Nocardia iowensis</i>
2N	C12-C16	Car	<i>Mycobacterium</i> sp. (strain JLS)
20	C4-C8	ELO1	<i>Trypanosoma brucei</i>
20	C10-C12	ELO2	<i>Trypanosoma brucei</i>
50 20	C14-C16	ELO3	<i>Trypanosoma brucei</i>
20	C14-C16	ELO1	<i>Saccharomyces cerevisiae</i>
20	C18-C20	ELO2	<i>Saccharomyces cerevisiae</i>
55 20	C22-C24	ELO3	<i>Saccharomyces cerevisiae</i>

Taking into account the differences in chain-length specificities of each enzyme in the MI-FAE cycle, MD-FAE cycle or FAACPE cycle, one skilled in the art can select one or more enzymes for catalyzing each elongation cycle reaction step (e.g., steps A-D or steps E/B/C/D of FIG. 6, or H/E/F/G of FIG. 12). For example, for the thiolase step of the MI-FAE cycle, some thiolase enzymes such as bktB of *Ralstonia eutropha* catalyze the elongation of short- and medium-chain acyl-CoA intermediates (C₆-C₈), whereas others such as Acaal of *R. norvegicus* are active on longer-

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chain substrates (C₁₀-C₁₆). Thus, a microbial organism producing a fatty alcohol, fatty aldehyde or fatty acid can comprise one, two, three, four or more variants of a thiolase, elongase, 3-oxoacyl-CoA reductase, 3-hydroxyacyl-CoA dehydratase and/or enoyl-CoA reductase.

Chain length specificity of enzymes can be assayed by methods well known in the art (eg. Wrensford et al, *Anal Biochem* 192:49-54 (1991)). The substrate ranges of fatty alcohol, fatty aldehyde, or fatty acid producing enzymes can be further extended or narrowed by methods well known in the art. Variants of biologically-occurring enzymes can be generated, for example, by rational and directed evolution, mutagenesis and enzyme shuffling as described herein. As one example, a rational engineering approach for altering chain length specificity was taken by Denic and Weissman (Denic and Weissman, *Cell* 130:663-77 (2008)). Denic and Weissman mapped the region of the yeast elongase protein ELOp responsible for chain length, and introduced mutations to vary the length of fatty acid products. In this instance, the geometry of the hydrophobic substrate pocket set an upper boundary on chain length. A similar approach can be useful for altering the chain length specificities of enzymes of the MI-FAE cycle, MD-FAE cycle and/or termination pathways.

Enzyme mutagenesis, expression in a host, and screening for fatty alcohol production is another useful approach for generating enzyme variants with improved properties for the desired application. For example, US patent application 2012/0009640 lists hundreds of variants of *Mannobacter algicola* and *Marinobacter aquaeolei* FAR enzymes with improved activity over the wild type enzyme, and varying product profiles.

Enzyme mutagenesis (random or directed) in conjunction with a selection platform is another useful approach. For example, Machado and coworkers developed a selection platform aimed at increasing the activity of acyl-CoA elongation cycle enzymes on longer chain length substrates (Machado et al., *Met Eng* 14(5):504-511(2012)). Machado et al. identified the chain-length limiting step of their pathway (a 3-hydroxyacyl-CoA dehydrogenase) and evolved it for improved activity on C₆-C₈ substrates using an anaerobic growth rescue platform. Additional variants of enzymes useful for producing fatty alcohols are listed in the table below

Enzyme	Protein/ GenBankID/ GI number	Organism	Variant (s)	Reference
3-Ketoacyl-CoA thiolase	Acaa2 NP_569117.1 GI: 18426866	<i>Rattus norvegicus</i>	H352A, H352E, H352K, H352Y	Zeng et al., <i>Prot. Expr. Purif.</i> 35: 320-326 (2004)
3-Hydroxyacyl-CoA dehydrogenase	Hadh NP_476534.1 GI: 17105336	<i>Rattus norvegicus</i>	S137A, S137C, S137T	Liu et al., <i>Prot. Expr. Purif.</i> 37: 344-351 (2004).
Enoyl-CoA hydratase	Ech1 NP_072116.1 GI: 12018256	<i>Rattus norvegicus</i>	E144A, E144A/ Q162L, E164A, Q162A, Q162L, Q162M	Kiema et al., <i>Biochem.</i> 38: 2991-2999 (1999)
Enoyl-CoA reductase	InhA AAY54545.1 GI: 66737267	<i>Mycobacterium tuberculosis</i>	K165A, K165Q, Y158F	Poletto, S. et al., <i>Prot. Expr. Purif.</i> 34: 118-125 (2004).

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Enzyme	Protein/ GenBankID/ GI number	Organism	Variant (s)	Reference
Acyl-CoA reductase	LuxC AAT00788.1 GI:46561111	<i>Photobacterium phosphoreum</i>	C171S, C279S, C286S	Lee, C. et al., <i>Biochim. Biophys. Acta.</i> 1338: 215-222 (1997).
Alcohol dehydrogenase	YADH-1 P00330.4 GI: 1168350	<i>Saccharomyces cerevisiae</i>	D223G, D49N, E68Q, G204A, G224I, H47R, H51E, L203A	Leskovac et al., <i>FEMS Yeast Res.</i> 2(4): 481-94 (2002).
Fatty alcohol forming acyl-CoA reductase (FAR)	AdhE NP_415757.1 GI: 16129202	<i>Escherichia coli</i>	A267T/ E568K, A267T	Membrillo et al., <i>J. Biol. Chem.</i> 275(43): 333869-75 (2000).

Those skilled in the art will understand that the genetic alterations, including metabolic modifications exemplified herein, are described with reference to a suitable host organism such as *E. coli* or *S. cerevisiae* and their corresponding metabolic reactions or a suitable source organism for desired genetic material such as genes for a desired metabolic pathway. However, given the complete genome sequencing of a wide variety of organisms and the high level of skill in the area of genomics, those skilled in the art will readily be able to apply the teachings and guidance provided herein to essentially all other organisms. For example, the metabolic alterations exemplified herein can readily be applied to other species by incorporating the same or analogous encoding nucleic acid from species other than the referenced species. Such genetic alterations include, for example, genetic alterations of species homologs, in general, and in particular, orthologs, paralogs or nonorthologous gene displacements.

An ortholog is a gene or genes that are related by vertical descent and are responsible for substantially the same or identical functions in different organisms. For example, mouse epoxide hydrolase and human epoxide hydrolase can be considered orthologs for the biological function of hydrolysis of epoxides. Genes are related by vertical descent when, for example, they share sequence similarity of sufficient amount to indicate they are homologous, or related by evolution from a common ancestor. Genes can also be considered orthologs if they share three-dimensional structure but not necessarily sequence similarity, of a sufficient amount to indicate that they have evolved from a common ancestor to the extent that the primary sequence similarity is not identifiable. Genes that are orthologous can encode proteins with sequence similarity of about 25% to 100% amino acid sequence identity. Genes encoding proteins sharing an amino acid similarity less than 25% can also be considered to have arisen by vertical descent if their three-dimensional structure also shows similarities. Members of the serine protease family of enzymes, including tissue plasminogen activator and elastase, are considered to have arisen by vertical descent from a common ancestor.

Orthologs include genes or their encoded gene products that through, for example, evolution, have diverged in structure or overall activity. For example, where one species encodes a gene product exhibiting two functions and where

such functions have been separated into distinct genes in a second species, the three genes and their corresponding products are considered to be orthologs. For the production of a biochemical product, those skilled in the art will understand that the orthologous gene harboring the metabolic activity to be introduced or disrupted is to be chosen for construction of the non-naturally occurring microorganism. An example of orthologs exhibiting separable activities is where distinct activities have been separated into distinct gene products between two or more species or within a single species. A specific example is the separation of elastase proteolysis and plasminogen proteolysis, two types of serine protease activity, into distinct molecules as plasminogen activator and elastase. A second example is the separation of mycoplasma 5'-3' exonuclease and *Drosophila* DNA polymerase III activity. The DNA polymerase from the first species can be considered an ortholog to either or both of the exonuclease or the polymerase from the second species and vice versa.

In contrast, paralogs are homologs related by, for example, duplication followed by evolutionary divergence and have similar or common, but not identical functions. Paralogs can originate or derive from, for example, the same species or from a different species. For example, microsomal epoxide hydrolase (epoxide hydrolase I) and soluble epoxide hydrolase (epoxide hydrolase II) can be considered paralogs because they represent two distinct enzymes, co-evolved from a common ancestor, that catalyze distinct reactions and have distinct functions in the same species. Paralogs are proteins from the same species with significant sequence similarity to each other suggesting that they are homologous, or related through co-evolution from a common ancestor. Groups of paralogous protein families include HipA homologs, luciferase genes, peptidases, and others.

A nonorthologous gene displacement is a nonorthologous gene from one species that can substitute for a referenced gene function in a different species. Substitution includes, for example, being able to perform substantially the same or a similar function in the species of origin compared to the referenced function in the different species. Although generally, a nonorthologous gene displacement will be identifiable as structurally related to a known gene encoding the referenced function, less structurally related but functionally similar genes and their corresponding gene products nevertheless will still fall within the meaning of the term as it is used herein. Functional similarity requires, for example, at least some structural similarity in the active site or binding region of a nonorthologous gene product compared to a gene encoding the function sought to be substituted. Therefore, a nonorthologous gene includes, for example, a paralog or an unrelated gene.

Therefore, in identifying and constructing the non-naturally occurring microbial organisms of the invention having fatty alcohol, fatty aldehyde or fatty acid biosynthetic capability, those skilled in the art will understand with applying the teaching and guidance provided herein to a particular species that the identification of metabolic modifications can include identification and inclusion or inactivation of orthologs. To the extent that paralogs and/or nonorthologous gene displacements are present in the referenced microorganism that encode an enzyme catalyzing a similar or substantially similar metabolic reaction, those skilled in the art also can utilize these evolutionally related genes. Similarly for a gene disruption, evolutionally related genes can also be disrupted or deleted in a host microbial organism to reduce or eliminate functional redundancy of enzymatic activities targeted for disruption.

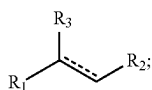
Orthologs, paralogs and nonorthologous gene displacements can be determined by methods well known to those skilled in the art. For example, inspection of nucleic acid or amino acid sequences for two polypeptides will reveal sequence identity and similarities between the compared sequences. Based on such similarities, one skilled in the art can determine if the similarity is sufficiently high to indicate the proteins are related through evolution from a common ancestor. Algorithms well known to those skilled in the art, such as Align, BLAST, Clustal W and others compare and determine a raw sequence similarity or identity, and also determine the presence or significance of gaps in the sequence which can be assigned a weight or score. Such algorithms also are known in the art and are similarly applicable for determining nucleotide sequence similarity or identity. Parameters for sufficient similarity to determine relatedness are computed based on well known methods for calculating statistical similarity, or the chance of finding a similar match in a random polypeptide, and the significance of the match determined. A computer comparison of two or more sequences can, if desired, also be optimized visually by those skilled in the art. Related gene products or proteins can be expected to have a high similarity, for example, 25% to 100% sequence identity. Proteins that are unrelated can have an identity which is essentially the same as would be expected to occur by chance, if a database of sufficient size is scanned (about 5%). Sequences between 5% and 24% may or may not represent sufficient homology to conclude that the compared sequences are related. Additional statistical analysis to determine the significance of such matches given the size of the data set can be carried out to determine the relevance of these sequences.

Exemplary parameters for determining relatedness of two or more sequences using the BLAST algorithm, for example, can be as set forth below. Briefly, amino acid sequence alignments can be performed using BLASTP version 2.0.8 (Jan. 5, 1999) and the following parameters: Matrix: 0 BLOSUM62; gap open: 11; gap extension: 1; x_dropoff: 50; expect: 10.0; wordsize: 3; filter: on. Nucleic acid sequence alignments can be performed using BLASTN version 2.0.6 (Sep. 16, 1998) and the following parameters: Match: 1; mismatch: -2; gap open: 5; gap extension: 2; x_dropoff: 50; expect: 10.0; wordsize: 11; filter: off. Those skilled in the art will know what modifications can be made to the above parameters to either increase or decrease the stringency of the comparison, for example, and determine the relatedness of two or more sequences.

In some embodiments, the invention provides a non-naturally occurring microbial organism having: (i) a formaldehyde fixation pathway; (ii) a formate assimilation pathway; and/or (iii) a methanol metabolic pathway as depicted in FIGS. 1 and 10, and a MI-FAE cycle or a MD-FAE cycle in combination with a termination pathway as depicted in FIGS. 2, 7 and 8, wherein said formaldehyde fixation pathway comprises: (1) 1B and 1C; (2) 1D; or (3) 1D and 1Z, wherein 1B is a 3-hexulose-6-phosphate synthase, wherein 1C is a 6-phospho-3-hexuloisomerase, wherein 1D is a dihydroxyacetone synthase, wherein 1Z is a fructose-6-phosphate aldolase, wherein said formate assimilation pathway comprises a pathway selected from: (4) 1E; (5) 1F, and 1G; (6) 1H, 1I, 1J, and 1K; (7) 1H, 1I, 1J, 1L, 1M, and 1N; (8) 1E, 1H, 1I, 1J, 1L, 1M, and 1N; (9) 1F, 1G, 1H, 1I, 1J, 1L, 1M, and 1N; (10) 1K, 1H, 1I, 1J, 1L, 1M, and 1N; and (11) 1H, 1I, 1J, 1O, and 1P, wherein 1E is a formate reductase, 1F is a formate ligase, a formate transferase, or a formate synthetase, wherein 1G is a formyl-CoA reductase, wherein 1H is a formyltetrahydrofolate synthetase, wherein

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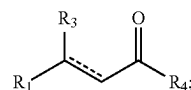
11 is a methenyltetrahydrofolate cyclohydrolase, wherein 1J is a methylenetetrahydrofolate dehydrogenase, wherein 1K is a formaldehyde-forming enzyme or spontaneous, wherein 1L is a glycine cleavage system, wherein 1M is a serine hydroxymethyltransferase, wherein 1N is a serine deaminase, wherein 1O is a methylenetetrahydrofolate reductase, wherein 1P is an acetyl-CoA synthase, wherein said methanol metabolic pathway comprises a pathway selected from: (12) 10J; (13) 10A, (14) 10A and 10B; (15) 10A, 10B and 10C; (16) 10J, 10K and 10C; (17) 10J, 10M, and 10N; (18) 10J and 10L; (19) 10J, 10L and 10G; (20) 10J, 10L, and 10I; (21) 10A, 10B, 10C, 10D, and 10E; (22) 10A, 10B, 10C, 10D, and 10F; (23) 10J, 10K, 10C, 10D, and 10E; (24) 10J, 10K, 10C, 10D, and 10F; (25) 10J, 10M, 10N, and 10O; (26) 10A, 10B, 10C, 10D, 10E, and 10G; (27) 10A, 10B, 10C, 10D, 10F, and 10G; (28) 10J, 10K, 10C, 10D, 10E, and 10G; (29) 10J, 10K, 10C, 10D, 10F, and 10G; (30) 10J, 10M, 10N, 10O, and 10G; (31) 10A, 10B, 10C, 10D, 10E, and 10I; (32) 10A, 10B, 10C, 10D, 10F, and 10I; (33) 10J, 10K, 10C, 10D, 10E, and 10I; (34) 10J, 10K, 10C, 10D, 10F, and 10I; and (35) 10J, 10M, 10N, 10O, and 10I, wherein 10A is a methanol methyltransferase, wherein 10B is a methylenetetrahydrofolate reductase, wherein 10C is a methylenetetrahydrofolate dehydrogenase, wherein 10D is a methenyltetrahydrofolate cyclohydrolase, wherein 10E is a formyltetrahydrofolate deformylase, wherein 10F is a formyltetrahydrofolate synthetase, wherein 10G is a formate hydrogen lyase, wherein 10I is a formate dehydrogenase, wherein 10J is a methanol dehydrogenase, wherein 10K is a formaldehyde activating enzyme or spontaneous, wherein 10L is a formaldehyde dehydrogenase, wherein 10M is a S-(hydroxymethyl)glutathione synthase or spontaneous, wherein 10N is a glutathione-dependent formaldehyde dehydrogenase, wherein 10O is a S-formylglutathione hydrolase, wherein the MI-FAE cycle includes one or more thiolase, one or more 3-oxoacyl-CoA reductase, one or more 3-hydroxyacyl-CoA dehydratase, and one or more enoyl-CoA reductase, wherein the MD-FAE cycle includes one or more elongase, one or more 3-oxoacyl-CoA reductase, one or more 3-hydroxyacyl-CoA dehydratase, and one or more enoyl-CoA reductase, wherein the termination pathway includes a pathway selected from: (36) 2H; (37) 2K and 2L; (38) 2E and 2N; (39) 2K, 2J, and 2N; (40) 2E; (41) 2K and 2J; (42) 2H and 2N; (43) 2K, 2L, and 2N; (44) 2E and 2F; (45) 2K, 2J, and 2F; (46) 2H, 2N, and 2F; (47) 2K, 2L, 2N, and 2F; (48) 2G; (49) 2P, wherein 2E is an acyl-CoA reductase (aldehyde forming), wherein 2F is an alcohol dehydrogenase, wherein 2G is an acyl-CoA reductase (alcohol forming), wherein 2H is an acyl-CoA hydrolase, acyl-CoA transferase or acyl-CoA synthase, wherein 2J is an acyl-ACP reductase, wherein 2K is an acyl-CoA:ACP acyltransferase, wherein 2L is a thioesterase, wherein 2N is an aldehyde dehydrogenase (acid forming) or a carboxylic acid reductase, wherein 2P is an acyl-ACP reductase (alcohol forming) wherein an enzyme of the formaldehyde fixation pathway, the formate assimilation pathway, the methanol metabolic pathway, the MI-FAE cycle, MD-FAE cycle or termination pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce a compound of Formula (I):



(I)

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wherein R_1 is C_{1-24} linear alkyl; R_2 is CH_2OH , CHO , or $COOH$; R_3 is H , OH , or $oxo (=O)$; and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four, wherein the substrate of each of said enzymes of the MI-FAE cycle, the MD-FAE cycle and the termination pathway are independently selected from a compound of Formula (II), malonyl-CoA, propionyl-CoA or acetyl-CoA:

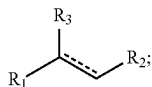


(II)

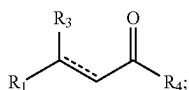
wherein R_1 is C_{1-24} linear alkyl; R_3 is H , OH , or $oxo (=O)$; R_4 is $S-CoA$, ACP , OH or H ; and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four; wherein said one or more enzymes of the MI-FAE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said compound of Formula (I), wherein said one or more enzymes of the MD-FAE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said compound of Formula (I), and wherein said one or more enzymes of the termination pathway are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no less than the number of carbon atoms at R_1 of said compound of Formula (I).

In some embodiments, the invention provides a non-naturally occurring microbial organism having: (i) a formaldehyde fixation pathway; (ii) a formate assimilation pathway; and/or (iii) a methanol metabolic pathway as depicted in FIGS. 1 and 10, and a FAACPE cycle in combination with a termination pathway as depicted in FIG. 12, wherein said formaldehyde fixation pathway comprises: (1) 1B and 1C; (2) 1D; (3) 1D and 1Z, wherein 1B is a 3-hexulose-6-phosphate synthase, wherein 1C is a 6-phospho-3-hexuloisomerase, wherein 1D is a dihydroxyacetone synthase, wherein 1Z is a fructose-6-phosphate aldolase, wherein said formate assimilation pathway comprises a pathway selected from: (4) 1E; (5) 1F, and 1G; (6) 1H, 1I, 1J, and 1K; (7) 1H, 1I, 1J, 1L, 1M, and 1N; (8) 1E, 1H, 1I, 1J, 1L, 1M, and 1N; (9) 1F, 1G, 1H, 1I, 1J, 1L, 1M, and 1N; (10) 1K, 1H, 1I, 1J, 1L, 1M, and 1N; and (11) 1H, 1I, 1J, 1O, and 1P, wherein 1E is a formate reductase, 1F is a formate ligase, a formate transferase, or a formate synthetase, wherein 1G is a formyl-CoA reductase, wherein 1H is a formyltetrahydrofolate synthetase, wherein 1I is a methenyltetrahydrofolate cyclohydrolase, wherein 1J is a methylenetetrahydrofolate dehydrogenase, wherein 1K is a formaldehyde-forming enzyme or spontaneous, wherein 1L is a glycine cleavage system, wherein 1M is a serine hydroxymethyltransferase, wherein 1N is a serine deaminase, wherein 1O is a methylenetetrahydrofolate reductase, wherein 1P is an acetyl-CoA synthase, wherein said methanol metabolic pathway comprises a pathway selected from: (12) 10J; (13) 10A, (14) 10A and 10B; (15) 10A, 10B and 10C; (16) 10J, 10K and 10C; (17) 10J, 10M, and 10N; (18) 10J and 10L; (19) 10J, 10L and 10G; (20) 10J, 10L, and 10I; (21) 10A, 10B, 10C, 10D, and

10E; (22) 10A, 10B, 10C, 10D, and 10F; (23) 10J, 10K, 10C, 10D, and 10E; (24) 10J, 10K, 10C, 10D, and 10F; (25) 10J, 10M, 10N, and 10O; (26) 10A, 10B, 10C, 10D, 10E, and 10G; (27) 10A, 10B, 10C, 10D, 10F, and 10G; (28) 10J, 10K, 10C, 10D, 10E, and 10G; (29) 10J, 10K, 10C, 10D, 10F, and 10G; (30) 10J, 10M, 10N, 10O, and 10G; (31) 10A, 10B, 10C, 10D, 10E, and 10I; (32) 10A, 10B, 10C, 10D, 10F, and 10I; (33) 10J, 10K, 10C, 10D, 10E, and 10I; (34) 10J, 10K, 10C, 10D, 10F, and 10I; and (35) 10J, 10M, 10N, 10O, and 10I, wherein 10A is a methanol methyltransferase, wherein 10B is a methylenetetrahydrofolate reductase, wherein 10C is a methylenetetrahydrofolate dehydrogenase, wherein 10D is a methenyltetrahydrofolate cyclohydrolase, wherein 10E is a formyltetrahydrofolate deformylase, wherein 10F is a formyltetrahydrofolate synthetase, wherein 10G is a formate hydrogen lyase, wherein 10I is a formate dehydrogenase, wherein 10J is a methanol dehydrogenase, wherein 10K is a formaldehyde activating enzyme or spontaneous, wherein 10L is a formaldehyde dehydrogenase, wherein 10M is a S-(hydroxymethyl)glutathione synthase or spontaneous, wherein 10N is a glutathione-dependent formaldehyde dehydrogenase, wherein 10O is a S-formylglutathione hydrolase, wherein said FAACPE cycle comprises one or more β -ketoacyl-ACP synthase, one or more β -ketoacyl-ACP reductase, one or more β -hydroxyacyl-ACP reductase, and one or more enoyl ACP-reductase, wherein said termination pathway comprises a pathway selected from: (36) 12I; (37) 12J; (38) 12I, 12K, and 12L; (39) 12I and 12O; (40) 12J and 12M; (41) 12I, 12K, 12L, and 12M; (42) 12I, 12O, and 12M; (43) 12I, 12K and 12N; (44) 12P, wherein 12I is a thioesterase, wherein 12J is a fatty acyl-ACP reductase, wherein 12K is an acyl-CoA synthase, wherein 12L is an acyl-CoA reductase, wherein 12M is a fatty aldehyde reductase, wherein 12N is a fatty alcohol forming acyl-CoA reductase (FAR), wherein 12O is a carboxylic acid reductase (CAR), wherein 12P is an acyl-ACP reductase (alcohol forming), wherein an enzyme of the formaldehyde fixation pathway, the formate assimilation pathway, the methanol metabolic pathway, the FAACPE cycle or the termination pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce a compound of Formula (I):



wherein R₁ is C₁₋₂₄ linear alkyl; R₂ is CH₂OH, CHO, or COOH; R₃ is H, OH, or oxo (=O); and $\overline{\hspace{1cm}}$ represents a single or double bond with the proviso that the valency of the carbon atom to which R₃ is attached is four, wherein the substrate of each of said enzymes of the FAACPE cycle and the termination pathway are independently selected from a compound of Formula (II) or malonyl-ACP:



wherein R₁ is C₁₋₂₄ linear alkyl; R₃ is H, OH, or oxo (=O); R₄ is S-CoA, ACP, OH or H; and ----- represents a single or double bond with the proviso that the valency of the carbon atom to which R₃ is attached is four; wherein said one or more enzymes of the FAACPE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R₁ that is no greater than the number of carbon atoms at R₁ of said compound of Formula (I), and wherein said one or more enzymes of the termination pathway are each selective for a compound of Formula (II) having a number of carbon atoms at R₁ that is no less than the number of carbon atoms at R₁ of said compound of Formula (I).

In some embodiments, the non-naturally occurring microbial organism of the invention has a combination of one or more pathways for generating substrates, intermediates and/or reducing equivalents that can be used with elongation cycles and termination pathways described herein for producing a fatty alcohol, fatty acid or fatty aldehyde of the invention. Accordingly, in some embodiments, the microbial organism has a formaldehyde fixation pathway and a MI-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formate assimilation pathway and a MI-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, and a MI-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway and a MD-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formate assimilation pathway and a MD-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, and a MD-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a methanol metabolic pathway and a MI-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a methanol metabolic pathway and a MD-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a methanol metabolic pathway and a MI-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formate assimilation pathway, a methanol metabolic pathway and a MI-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and a MI-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and a MD-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and a MD-FAE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and a FAACPE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formate assimilation pathway and an FAACPE cycle in

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combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, and an FAACPE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a methanol metabolic pathway and an FAACPE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a methanol metabolic pathway and an FAACPE cycle in combination with a termination pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and an FAACPE cycle in combination with a termination pathway.

In some embodiment, the non-naturally occurring microbial organism of the invention having FAACPE cycle in combination with a termination pathway as described herein, can further include a pathway for production of substrants for the FAACPE cycle, such as acetoacetyl-ACP or 3-oxovalery-ACP. Accordingly, in some embodiments, the microbial organism further comprises an acetoacetyl-ACP pathway of: (1) 12A, 12B, and 12C; or (2) 12A, 12B, and 12D, wherein 12A is an acetyl-CoA carboxylase, wherein 12B is malonyl-CoA ACP transacylase, wherein 12C is an acetoacetyl-ACP synthase, and wherein 12D is a β -ketoacyl-ACP synthase. In some embodiments, the microbial organism further comprises a 3-oxovalery-ACP pathway comprising an acetyl-CoA carboxylase, a malonyl-CoA ACP transacylase, and a β -ketoacyl-ACP synthase. In some aspects of the invention, an enzyme of the acetoacetyl-ACP pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce acetoacetyl-ACP wherein the acetoacetyl-ACP is a β -ketoacyl-ACP of the FAACPE cycle. In some aspects of the invention, an enzyme of the 3-oxovalery-ACP pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce 3-oxovalery-ACP, wherein the 3-oxovalery-ACP is a β -ketoacyl-ACP of the FAACPE cycle.

In some aspects of the invention, non-naturally occurring microbial organism of the invention can produce a compound of Formula (I) wherein R_1 is C_{1-17} linear alkyl. In another aspect of the invention, the R_1 of the compound of Formula (I) is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} linear alkyl, C_{12} linear alkyl or C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some aspects of the invention, the microbial organism includes two, three, or four exogenous nucleic acids each encoding an enzyme of the MI-FAE cycle, the MD-FAE cycle, or the FAACPE cycle. In some aspects of the invention, the microbial organism includes two, three, or four exogenous nucleic acids each encoding an enzyme of the termination pathway. In some aspects of the invention, the microbial organism includes one, two, three, four, five, six, seven, or eight exogenous nucleic acids each encoding a formaldehyde fixation pathway enzyme, a formate assimilation pathway enzyme, or a methanol metabolic pathway enzyme. In some aspects of the invention, the microbial organism includes exogenous nucleic acids encoding each of the enzymes of at least one of the pathways selected from

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(1)-(49) for a microbial organism having a MI-FAE cycle or a MD-FAE cycle in combination with a termination pathway as depicted in FIGS. 1, 2, 7, 8 and 10. In some aspects of the invention, the microbial organism includes exogenous nucleic acids encoding each of the enzymes of at least one of the pathways selected from (1)-(44) for a microbial organism having a fatty acyl-ACP elongation (FAACPE) cycle in combination with a termination pathway as depicted in FIGS. 1, 10 and 12.

In some embodiments, the invention provides a non naturally occurring microbial organism, wherein the one or more enzymes of the MI-FAE cycle, MD-FAE cycle, FAACPE cycle or termination pathway is expressed in a sufficient amount to produce a fatty alcohol selected from the Formulas (III)–(VI):



wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} linear alkyl, C_{12} linear alkyl, C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a non naturally occurring microbial organism, wherein the one or more enzymes of the MI-FAE cycle, MD-FAE cycle, FAACPE cycle or termination pathway is expressed in a sufficient amount to produce a fatty aldehyde selected from the Formula (VII)–(X):

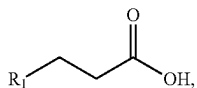
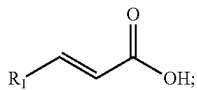
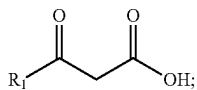
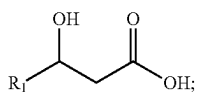


wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear

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alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} , linear alkyl, C_{12} linear alkyl, C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a non naturally occurring microbial organism, wherein the one or more enzymes of the MI-FAE cycle, MD-FAE cycle, FAACPE cycle or termination pathway is expressed in a sufficient amount to produce a fatty acid selected from the Formula (XI)—(XIV):



wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} , linear alkyl, C_{12} linear alkyl, C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a non naturally occurring microbial organism, wherein one or more enzymes of the MI-FAE cycle and/or MD-FAE cycle are each selective for a compound of Formula (II) wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} , linear alkyl, C_{12} linear alkyl, C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a non naturally occurring microbial organism, wherein one or more enzymes of the FAACPE cycle are each selective for a compound of Formula (II) wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear

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alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} , linear alkyl, C_{12} linear alkyl, C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a non naturally occurring microbial organism, wherein one or more enzymes of the termination pathway are each selective for a compound of Formula (II) wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} , linear alkyl, C_{12} linear alkyl, C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a non-naturally occurring microbial organism having: (i) a formaldehyde fixation pathway; (ii) a formate assimilation pathway; and/or (iii) a methanol metabolic pathway as depicted in FIGS. 1 and 10, and an isopropanol pathway as depicted in FIG. 11, wherein said formaldehyde fixation pathway comprises: (1) 1B and 1C; (2) 1D; or (3) 1D and 1Z, wherein 1B is a 3-hexulose-6-phosphate synthase, wherein 1C is a 6-phospho-3-hexuloisomerase, wherein 1D is a dihydroxyacetone synthase, wherein 1Z is a fructose-6-phosphate aldolase, wherein said formate assimilation pathway comprises a pathway selected from: (4) 1E; (5) 1F, and 1G; (6) 1H, 1I, 1J, and 1K; (7) 1H, 1I, 1J, 1L, 1M, and 1N; (8) 1E, 1H, 1I, 1J, 1L, 1M, and 1N; (9) 1F, 1G, 1H, 1I, 1J, 1L, 1M, and 1N; (10) 1K, 1H, 1I, 1J, 1L, 1M, and 1N; and (11) 1H, 1I, 1J, 1O, and 1P, wherein 1E is a formate reductase, 1F is a formate ligase, a formate transferase, or a formate synthetase, wherein 1G is a formyl-CoA reductase, wherein 1H is a formyltetrahydrofolate synthetase, wherein 1I is a methenyltetrahydrofolate cyclohydrolase, wherein 1J is a methylenetetrahydrofolate dehydrogenase, wherein 1K is a formaldehyde-forming enzyme or spontaneous, wherein 1L is a glycine cleavage system, wherein 1M is a serine hydroxymethyltransferase, wherein 1N is a serine deaminase, wherein 1O is a methylenetetrahydrofolate reductase, wherein 1P is an acetyl-CoA synthase, wherein said methanol metabolic pathway comprises a pathway selected from: (12) 1OJ; (13) 10A, (14) 10A and 10B; (15) 10A, 10B and 10C; (16) 10J, 10K and 10C; (17) 10J, 10M, and 10N; (18) 10J and 10L; (19) 10J, 10L and 10G; (20) 10J, 10L, and 10I; (21) 10A, 10B, 10C, 10D, and 10E; (22) 10A, 10B, 10C, 10D, and 10F; (23) 10J, 10K, 10C, 10D, and 10E; (24) 10J, 10K, 10C, 10D, and 10F; (25) 10J, 10M, 10N, and 10O; (26) 10A, 10B, 10C, 10D, 10E, and 10G; (27) 10A, 10B, 10C, 10D, 10F, and 10G; (28) 10J, 10K, 10C, 10D, 10E, and 10G; (29) 10J, 10K, 10C, 10D, 10F, and 10G; (30) 10J, 10M, 10N, 10O, and 10G; (31) 10A, 10B, 10C, 10D, 10E, and 10I; (32) 10A, 10B, 10C, 10D, 10F, and 10I; (33) 10J, 10K, 10C, 10D, 10E, and 10I; (34) 10J, 10K, 10C, 10D, 10F, and 10I; and (35) 10J, 10M, 10N, 10O, and 10I, wherein 10A is a methanol methyltransferase, wherein 10B is a methylenetetrahydrofolate reductase, wherein 10C is a methylenetetrahydrofolate dehydrogenase, wherein 10D is a methenyltetrahydrofolate cyclohydrolase, wherein 10E is a formyltetrahydrofolate deformylase, wherein 10F is a formyltetrahydrofolate synthetase, wherein 10G is a formate hydrogen lyase, wherein 10I is a formate dehydrogenase, wherein 10J is a methanol dehydrogenase, wherein 10K is a formaldehyde activating enzyme or spontaneous, wherein

10L is a formaldehyde dehydrogenase, wherein 10M is a S-(hydroxymethyl)glutathione synthase or spontaneous, wherein 10N is a glutathione-dependent formaldehyde dehydrogenase, wherein 10O is a S-formylglutathione hydrolase, wherein said isopropanol pathway comprises: (36) 11V, 11W, 11X, and 11Y; or (37) 11T, 11U, 11W, 11X, and 11Y, wherein 11T is an acetyl-CoA carboxylase, wherein 11U is an acetoacetyl-CoA synthase, wherein 11V is an acetyl-CoA:acetyl-CoA acyltransferase, wherein 11W is an acetoacetyl-CoA hydrolase, an acetoacetyl-CoA transferase, an acetoacetyl-CoA ligase, or a phosphotransacetylase/acetoacetate kinase, wherein 11X is an acetoacetate decarboxylase, wherein 11Y is an acetone reductase or isopropanol dehydrogenase, wherein an enzyme of the formaldehyde fixation pathway, formate assimilation pathway, methanol metabolic pathway, or isopropanol pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce isopropanol.

In some embodiments, the non-naturally occurring microbial organism of the invention has a combination of one or more pathways for generating substrates, intermediates and/or reducing equivalents that can be used with isopropanol pathways described herein for producing isopropanol of the invention. Accordingly, in some embodiments, the microbial organism has a formaldehyde fixation pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formate assimilation pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, and an isopropanol pathway. In some embodiments, the microbial organism has a methanol metabolic pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a methanol metabolic pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formate assimilation pathway, a methanol metabolic pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and an isopropanol pathway.

In some aspects of the invention, the microbial organism includes two, three, four, five or six exogenous nucleic acids each encoding an enzyme of the isopropanol pathway. In some aspects of the invention, the microbial organism includes one, two, three, four, five, six, seven, or eight exogenous nucleic acids each encoding a formaldehyde fixation pathway enzyme, a formate assimilation pathway enzyme, or a methanol metabolic pathway enzyme. In some aspects of the invention, the microbial organism includes exogenous nucleic acids encoding each of the enzymes of at least one of the pathways selected from (1)-(37) for a microbial organism having an isopropanol pathway as depicted in FIGS. 1, 10 and 11.

In some embodiments, a non-naturally occurring microbial organism of the invention having a formate assimilation pathway further includes wherein the formate assimilation pathway comprises: (1) 1Q; (2) 1R, and 1S; (3) 1Y and 1Q; (4) 1Y, 1R and 1S, wherein 1Q is a pyruvate formate lyase, wherein 1R is a pyruvate dehydrogenase, a pyruvate ferredoxin oxidoreductase, or a pyruvate:NADP⁺ oxidoreductase, wherein 1S is a formate dehydrogenase, wherein 1Y is a glyceraldehydes-3-phosphate dehydrogenase or an enzyme of lower glycolysis. In addition to a glyceraldehyde-3-phosphate dehydrogenase, lower glycolysis includes a phosphoglycerate kinase, a phosphoglyceromutase, an enolase, a pyruvate kinase or a PTS-dependant substrate import. Accordingly, in some embodiments, the

formate assimilation pathway comprising 1Y includes an enzyme selected from a phosphoglycerate kinase, a phosphoglyceromutase, an enolase, a pyruvate kinase and a PTS-dependant substrate import.

In some embodiments, a non-naturally occurring microbial organism of the invention includes a methanol oxidation pathway. Such a pathway can include at least one exogenous nucleic acid encoding a methanol oxidation pathway enzyme expressed in a sufficient amount to produce formaldehyde in the presence of methanol. An exemplary methanol oxidation pathway enzyme is a methanol dehydrogenase. Accordingly, in some embodiments, a non-naturally occurring microbial organism of the invention includes at least one exogenous nucleic acid encoding a methanol dehydrogenase expressed in a sufficient amount to produce formaldehyde in the presence of methanol.

In some embodiments, the exogenous nucleic acid encoding a methanol dehydrogenase is expressed in a sufficient amount to produce an amount of formaldehyde greater than or equal to 1 μ M, 10 μ M, 20 μ M, or 50 μ M, or a range thereof, in culture medium or intracellularly. In other embodiments, the exogenous nucleic acid encoding a methanol dehydrogenase is capable of producing an amount of formaldehyde greater than or equal to 1 μ M, 10 μ M, 20 μ M, or 50 μ M, or a range thereof, in culture medium or intracellularly. In some embodiments, the range is from 1 μ M to 50 μ M or greater. In other embodiments, the range is from 10 μ M to 50 μ M or greater. In other embodiments, the range is from 20 μ M to 50 μ M or greater. In other embodiments, the amount of formaldehyde production is 50 μ M or greater. In specific embodiments, the amount of formaldehyde production is in excess of, or as compared to, that of a negative control, e.g., the same species of organism that does not comprise the exogenous nucleic acid, such as a wild-type microbial organism or a control microbial organism thereof. In certain embodiments, the methanol dehydrogenase is selected from those provided herein, e.g., as exemplified in Example II (see FIG. 1, Step A, or FIG. 10, Step J). In certain embodiments, the amount of formaldehyde production is determined by a whole cell assay, such as that provided in Example II (see FIG. 1, Step A, or FIG. 10, Step J), or by another assay provided herein or otherwise known in the art. In certain embodiments, formaldehyde utilization activity is absent in the whole cell.

In certain embodiments, the exogenous nucleic acid encoding a methanol dehydrogenase is expressed in a sufficient amount to produce at least 1 $\times$, 2 $\times$, 3 $\times$, 4 $\times$, 5 $\times$, 6 $\times$, 7 $\times$, 8 $\times$, 9 $\times$, 10 $\times$, 15 $\times$, 20 $\times$, 30 $\times$, 40 $\times$, 50 $\times$, 100 $\times$ or more formaldehyde in culture medium or intracellularly. In other embodiments, the exogenous nucleic acid encoding a methanol dehydrogenase is capable of producing an amount of formaldehyde at least 1 $\times$, 2 $\times$, 3 $\times$, 4 $\times$, 5 $\times$, 6 $\times$, 7 $\times$, 8 $\times$, 9 $\times$, 10 $\times$, 15 $\times$, 20 $\times$, 30 $\times$, 40 $\times$, 50 $\times$, 100 $\times$, or a range thereof, in culture medium or intracellularly. In some embodiments, the range is from 1 $\times$ to 100 $\times$. In other embodiments, the range is from 2 $\times$ to 100 $\times$. In other embodiments, the range is from 5 $\times$ to 100 $\times$. In other embodiments, the range is from 10 $\times$ to 100 $\times$. In other embodiments, the range is from 50 $\times$ to 100 $\times$. In some embodiments, the amount of formaldehyde production is at least 20 $\times$. In other embodiments, the amount of formaldehyde production is at least 50 $\times$. In specific embodiments, the amount of formaldehyde production is in excess of, or as compared to, that of a negative control, e.g., the same species of organism that does not comprise the exogenous nucleic acid, such as a wild-type microbial organism or a control microbial organism thereof. In certain embodiments, the methanol dehydrogenase is selected from those

provided herein, e.g., as exemplified in Example II (see FIG. 1, Step A, or FIG. 10, Step J). In certain embodiments, the amount of formaldehyde production is determined by a whole cell assay, such as that provided in Example II (see FIG. 1, Step A, or FIG. 10, Step J), or by another assay provided herein or otherwise known in the art. In certain embodiments, formaldehyde utilization activity is absent in the whole cell.

In some embodiments, a non-naturally occurring microbial organism of the invention includes one or more enzymes for generating reducing equivalents. For example, the microbial organism can further include a hydrogenase and/or a carbon monoxide dehydrogenase. In some aspects, the organism comprises an exogenous nucleic acid encoding the hydrogenase or the carbon monoxide dehydrogenase.

A reducing equivalent can also be readily obtained from a glycolysis intermediate by any of several central metabolic reactions including glyceraldehyde-3-phosphate dehydrogenase, pyruvate dehydrogenase, pyruvate formate lyase and NAD(P)-dependant formate dehydrogenase, isocitrate dehydrogenase, alpha-ketoglutarate dehydrogenase, succinate dehydrogenase, and malate dehydrogenase. Additionally, reducing equivalents can be generated from glucose 6-phosphate-1-dehydrogenase and 6-phosphogluconate dehydrogenase of the pentose phosphate pathway. Overall, at most twelve reducing equivalents can be obtained from a C₆ glycolysis intermediate (e.g., glucose-6-phosphate, fructose-6-phosphate, fructose-1,6-diphosphate) and at most six reducing equivalents can be generated from a C₃ glycolysis intermediate (e.g., dihydroxyacetone phosphate, glyceraldehyde-3-phosphate).

In some embodiments, the at least one exogenous nucleic acid included in the non-naturally occurring microbial organism of the invention is a heterologous nucleic acid. Accordingly, in some embodiments, the at least one exogenous nucleic acid encoding a formaldehyde fixation pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a formate assimilation pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a methanol metabolic pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a MI-FAE cycle enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a MD-FAE cycle enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a FAACPE cycle enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a termination pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding an acetoacetyl-ACP pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a 3-oxovaleryl-ACP pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding an isopropanol pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a methanol oxidation pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a hydrogenase or a carbon monoxide dehydrogenase is a heterologous nucleic acid.

In some embodiments, the non-naturally occurring microbial organism of the invention is in a substantially anaerobic culture medium.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes an acetyl-CoA pathway and at least one exogenous nucleic acid encoding an acetyl-CoA pathway enzyme expressed in a sufficient amount to produce or enhance carbon flux through acetyl-CoA, wherein the acetyl-CoA pathway includes a pathway shown in FIG. 1, 3, 4, 5 or 6 selected from: (1) 3A and 3B; (2) 3A, 3C, and 3D; (3) 3H; (4) 3G and 3D; (5) 3E, 3F and 3B; (6) 3E and 3I; (7) 3J, 3F and 3B; (8) 3J and 3I; (9) 4A, 4B, and 4C; (10) 4A, 4B, 4J, 4K, and 4D; (11) 4A, 4B, 4G, and 4D; (12) 4A, 4F, and 4D; (13) 4N, 4H, 4B and 4C; (14) 4N, 4H, 4B, 4J, 4K, and 4D; (15) 4N, 4H, 4B, 4G, and 4D; (16) 4N, 4H, 4F, and 4D; (17) 4L, 4M, 4B and 4C; (18) 4L, 4M, 4B, 4J, 4K, and 4D; (19) 4L, 4M, 4B, 4G, and 4D; (20) 4L, 4M, 4F, and 4D; (21) 5A, 5B, 5D, 5H, 5I, and 5J; (22) 5A, 5B, 5E, 5F, 5H, 5I, and 5J; (23) 5A, 5B, 5E, 5K, 5L, 5H, 5I, and 5J; (24) 5A, 5C, 5D, 5H, and 5J; (25) 5A, 5C, 5E, 5F, 5H, and 5J; (26) 5A, 5C, 5E, 5K, 5L, 5H, and 5J; (27) 6A, 6B, 6D, and 6G; (28) 6A, 6B, 6E, 6F, and 6G; (29) 6A, 6B, 6E, 6K, 6L, and 6G; (30) 6A, 6C, and 6D; (31) 6A, 6C, 6E, and 6F; (32) 6A, 6C, 6E, 6K, and 6L; (33) 1T and 1V; (34) 1T, 1W, and 1X; (35) 1U and 1V; and (36) 1U, 1W, and 1X, wherein 3A is a pyruvate oxidase (acetate-forming), wherein 3B is an acetyl-CoA synthetase, an acetyl-CoA ligase or an acetyl-CoA transferase, wherein 3C is an acetate kinase, wherein 3D is a phosphotransacetylase, wherein 3E is a pyruvate decarboxylase, wherein 3F is an acetaldehyde dehydrogenase, wherein 3G is a pyruvate oxidase (acetyl-phosphate forming), wherein 3H is a pyruvate dehydrogenase, a pyruvate:ferredoxin oxidoreductase, a pyruvate:NAD(P)H oxidoreductase or a pyruvate formate lyase, wherein 3I is an acetaldehyde dehydrogenase (acylating), wherein 3J is a threonine aldolase, wherein 4A is a phosphoenolpyruvate (PEP) carboxylase or a PEP carboxykinase, wherein 4B is an oxaloacetate decarboxylase, wherein 4C is a malonate semialdehyde dehydrogenase (acetylating), wherein 4D is an acetyl-CoA carboxylase or a malonyl-CoA decarboxylase, wherein 4F is an oxaloacetate dehydrogenase or an oxaloacetate oxidoreductase, wherein 4G is a malonate semialdehyde dehydrogenase (acylating), wherein 4H is a pyruvate carboxylase, wherein 4J is a malonate semialdehyde dehydrogenase, wherein 4K is a malonyl-CoA synthetase or a malonyl-CoA transferase, wherein 4L is a malic enzyme, wherein 4M is a malate dehydrogenase or a malate oxidoreductase, wherein 4N is a pyruvate kinase or a PEP phosphatase, wherein 5A is a citrate synthase, wherein 5B is a citrate transporter, wherein 5C is a citrate/malate transporter, wherein 5D is an ATP citrate lyase, wherein 5E is a citrate lyase, wherein 5F is an acetyl-CoA synthetase or an acetyl-CoA transferase, wherein 5H is a cytosolic malate dehydrogenase, wherein 5I is a malate transporter, wherein 5J is a mitochondrial malate dehydrogenase, wherein 5K is an acetate kinase, wherein 5L is a phosphotransacetylase, wherein 6A is a citrate synthase, wherein 6B is a citrate transporter, wherein 6C is a citrate/oxaloacetate transporter, wherein 6D is an ATP citrate lyase, wherein 6E is a citrate lyase, wherein 6F is an acetyl-CoA synthetase or an acetyl-CoA transferase, wherein 6G is an oxaloacetate transporter, wherein 6K is an acetate kinase, wherein 6L is a phosphotransacetylase, wherein 1T is a fructose-6-phosphate phosphoketolase, wherein 1U is a xylulose-5-phosphate phosphoketolase, wherein 1V is a phosphotransacetylase, wherein 1W is an acetate kinase,

wherein 1X is an acetyl-CoA transferase, an acetyl-CoA synthetase, or an acetyl-CoA ligase.

In some aspects, the microbial organism of the invention can include two, three, four, five, six, seven or eight exogenous nucleic acids each encoding an acetyl-CoA pathway enzyme. In some aspects, the microbial organism includes exogenous nucleic acids encoding each of the acetyl-CoA pathway enzymes of at least one of the pathways selected from (1)-(36).

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes a propionyl-CoA pathway and at least one exogenous nucleic acid encoding a propionyl-CoA pathway enzyme expressed in a sufficient amount to produce propionyl-CoA, wherein the propionyl-CoA pathway includes a pathway shown in FIG. 22. For example, in some embodiments, the propionyl-CoA pathway comprises a pathway selected from: (1) 22A, 22E, 22F, 22G, 22I, 22J, 22K and 22L; (2) 22A, 22E, 22F, 22G, 22H, 22J, 22K and 22L; (3) 22B, 22E, 22F, 22G, 22I, 22J, 22K and 22L; (4) 22B, 22E, 22F, 22G, 22H, 22J, 22K and 22L; (5) 22C, 22D, 22E, 22F, 22G, 22I, 22J, 22K and 22L; and (6) 22C, 22D, 22E, 22F, 22G, 22H, 22J, 22K and 22L, wherein 22A is a PEP carboxykinase, wherein 22B is a PEP carboxylase, wherein 22C is a Pyruvate kinase, wherein 22D is a Pyruvate carboxylase, wherein 22E is a Malate dehydrogenase, wherein 22F is a Fumarase, wherein 22G is a Fumarate reductase, wherein 22H is a Succinyl-CoA synthetase, wherein 22I is a Succinyl-CoA:3-ketoacid-CoA transferase, wherein 22J is a Methylmalonyl-CoA mutase, wherein 22K is a Methyl-malonyl-CoA epimerase, and wherein 22L is a Methylmalonyl-CoA decarboxylase.

In an additional embodiment, the invention provides a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde or fatty acid pathway, wherein the non-naturally occurring microbial organism comprises at least one exogenous nucleic acid encoding an enzyme or protein that converts a substrate to a product selected from the group consisting of MeOH to Fald, Fald to H6P, H6P to F6P, Fald to DHA and G3P, DHA and G3P to F6P, F6P to ACTP and E4P, ACTP to ACCOA, ACTP to acetate, acetate to ACCOA, Xu5P to ACTP and G3P, G3P to PYR, PYR to formate and ACCOA, PYR to CO₂ and ACCOA, CO₂ to formate, formate to Fald, formate to Formyl-CoA, Formyl-CoA to Fald, Formate to FTHF, FTHF to methenyl-THF, methenyl-THF to methylene-THF, methylene-THF to Fald, methylene-THF to glycine, glycine to serine, serine to PYR, methylene-THF to methyl-THF, methyl-THF to ACCOA, two acetyl-CoA molecules to a 3-ketoacyl-CoA, acetyl-CoA plus propionyl-CoA to a ketoacyl-CoA, malonyl-CoA to 3-ketoacyl-CoA, a 3-ketoacyl-CoA to a 3-hydroxyacyl-CoA, a 3-hydroxyacyl-CoA to an enoyl-CoA, an enoyl-CoA to an acyl-CoA, an acyl-CoA plus an acetyl-CoA to a 3-ketoacyl-CoA, an acyl-CoA plus malonyl-CoA to a 3-ketoacyl-CoA, an acyl-CoA to a fatty aldehyde, a fatty aldehyde to a fatty alcohol, an acyl-CoA to a fatty alcohol, an acyl-CoA to an acyl-ACP, an acyl-ACP to a fatty acid, an acyl-CoA to a fatty acid, an acyl-ACP to a fatty aldehyde, a fatty acid to a fatty aldehyde, a fatty aldehyde to a fatty acid, formaldehyde to S-hydroxymethylglutathione, S-hydroxymethylglutathione to S-formylglutathione to formate, formaldehyde to formate, MeOH to methyl-THF, methyl-THF to methylene-THF, formaldehyde to methylene-THF, methylene-THF to methenyl-THF, methenyl-THF to formyl-THF, formyl-THF to formate, formaldehyde to formate, ACCOA to MALCOA, ACCOA to AACOA, MALCOA to AACOA, AACOA to acetoacetate, acetoacetate to

acetone, acetone to isopropanol, malonyl-CoA to malonyl-ACP, malonyl-ACP and acetyl-CoA to acetoacetyl-ACP, malonyl-ACP and acetyl-ACP to acetoacetyl-ACP, malonyl-ACP and propionyl-CoA to 3-oxovalery-ACP, malonyl-ACP and an acyl-ACP to a β -ketoacyl-ACP, a β -ketoacyl-ACP to a β -hydroxyacyl-ACP, a β -hydroxyacyl-ACP to a trans-2-enoyl-ACP, a trans-2-enoyl-ACP to an acyl-ACP, an acyl-ACP to a fatty acid, an acyl-ACP to a fatty aldehyde, a fatty acid to a fatty aldehyde, a fatty acid to an acyl-CoA, an acyl-CoA to a fatty aldehyde, a fatty aldehyde to a fatty alcohol, a fatty aldehyde to a fatty alcohol, PEP to OAA, OAA to MAL, MAL to FUM, FUM to SUCC, SUCCOA to (R)-MMCOA, (R)-MMCOA to (S)-MMCOA, MMCOA to PPCOA, PEP to PYR, pyruvate to acetate, acetate to acetyl-CoA, pyruvate to acetyl-CoA, pyruvate to acetaldehyde, threonin to acetaldehyde, acetaldehyde to acetate, acetaldehyde to acetyl-CoA, pyruvate to acetyl-phosphate, acetate to acetyl-phosphate, acetyl-phosphate to acetyl-CoA, phosphoenolpyruvate (PEP) to pyruvate, pyruvate to malate, malate to oxaloacetate, pyruvate to oxaloacetate, PEP to oxaloacetate, oxaloacetate to malonate semialdehyde, oxaloacetate to malonyl-CoA, malonate semialdehyde to malonate, malonate to malonyl-CoA, malonate semialdehyde to malonyl-CoA, malonyl-CoA to acetyl-CoA, malonate semialdehyde to acetyl-CoA, oxaloacetate plus acetyl-CoA to citrate, citrate to oxaloacetate plus acetyl-CoA, citrate to oxaloacetate plus acetate, and oxaloacetate to malate. One skilled in the art will understand that these are merely exemplary and that any of the substrate-product pairs disclosed herein suitable to produce a desired product and for which an appropriate activity is available for the conversion of the substrate to the product can be readily determined by one skilled in the art based on the teachings herein. Thus, the invention provides a non-naturally occurring microbial organism containing at least one exogenous nucleic acid encoding an enzyme or protein, where the enzyme or protein converts the substrates and products of a fatty alcohol, fatty aldehyde, fatty acid, or isopropanol pathway, such as that shown in FIGS. 1-12 and 22.

While generally described herein as a microbial organism that contains a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway, it is understood that the invention additionally provides a non-naturally occurring microbial organism comprising at least one exogenous nucleic acid encoding a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein expressed in a sufficient amount to produce an intermediate of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway. For example, as disclosed herein, a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway is exemplified in FIGS. 1-12 and 22. Therefore, in addition to a microbial organism containing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway that produces fatty alcohol, fatty aldehyde, fatty acid or isopropanol, the invention additionally provides a non-naturally occurring microbial organism comprising at least one exogenous nucleic acid encoding a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme, where the microbial organism produces a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate, for example, pyruvate, formate, formaldehyde, H6P, DHA, G3P, F6P, ACTP, E4P, formyl-CoA, FTHF, methenyl-THF, methylene-THF, glycine, serine, methyl-THF, CO₂, a 3-ketoacyl-CoA, a 3-hydroxyacyl-CoA, an enoyl-CoA, a β -ketoacyl-ACP, a β -hydroxyacyl-ACP, a trans-2-enoyl-ACP, an acyl-CoA, an acyl-ACP, acetoacetate, acetone, acetate, acetaldehyde, acetyl-phosphate, oxaloac-

etate, matate, malonate semialdehyde, malonate, malonyl-ACP, propionyl-CoA, malonyl-CoA, acetyl-CoA, or citrate.

It is understood that any of the pathways disclosed herein, as described in the Examples and exemplified in the Figures, including the pathways of FIGS. 1-12 and 22, can be utilized to generate a non-naturally occurring microbial organism that produces any pathway intermediate or product, as desired. As disclosed herein, such a microbial organism that produces an intermediate can be used in combination with another microbial organism expressing downstream pathway enzymes to produce a desired product. However, it is understood that a non-naturally occurring microbial organism that produces a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate can be utilized to produce the intermediate as a desired product.

In some embodiments, the invention provides a non-naturally occurring microbial organism having an acetyl-CoA pathway, wherein said acetyl-CoA pathway comprises a pathway selected from: (1) 1T and 1V; (2) 1T, 1W, and 1X; (3) 1U and 1V; (4) 1U, 1W, and 1X; wherein 1T is a fructose-6-phosphate phosphoketolase, wherein 1U is a xylulose-5-phosphate phosphoketolase, wherein 1V is a phosphotransacetylase, wherein 1W is an acetate kinase, wherein 1X is an acetyl-CoA transferase, an acetyl-CoA synthetase, or an acetyl-CoA ligase, wherein said non-naturally occurring microbial organism further comprises a pathway capable of producing isopropanol and an exogenous nucleic acid encoding an isopropanol pathway enzyme expressed in a sufficient amount to produce isopropanol, wherein said isopropanol pathway comprises a pathway selected from: (1) 11V, 11W, 11X, and 11Y; or (2) 11T, 11U, 11W, 11X, and 11Y, wherein 11T is an acetyl-CoA carboxylase, wherein 11U is an acetoacetyl-CoA synthase, wherein 11V is an acetyl-CoA:acetyl-CoA acyltransferase, wherein 11W is an acetoacetyl-CoA hydrolase, an acetoacetyl-CoA transferase, an acetoacetyl-CoA ligase, or a phosphotransacetoacetylase/acetoacetate kinase, wherein 11X is an acetoacetate decarboxylase, wherein 11Y is an acetone reductase or isopropanol dehydrogenase.

The invention further provides non-naturally occurring microbial organisms that have elevated or enhanced synthesis or yields of acetyl-CoA (e.g. intracellular) or biosynthetic products such as a fatty alcohol, fatty aldehyde, fatty acid or isopropanol and methods of using those non-naturally occurring organisms to produce such biosynthetic products. The enhanced synthesis of intracellular acetyl-CoA enables enhanced production of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol from which acetyl-CoA is an intermediate and further, may have been rate limiting.

The non-naturally occurring microbial organisms having enhanced yields of a biosynthetic product include one or more of the various pathway configurations employing a methanol dehydrogenase for methanol oxidation, a formaldehyde fixation pathway and/or an acetyl-CoA enhancing pathway, e.g. phosphoketolase, for directing the carbon from methanol into acetyl-CoA and other desired products via formaldehyde fixation. The various different methanol oxidation and formaldehyde fixation configurations exemplified below can be engineered in conjunction with any or each of the various methanol oxidation, formaldehyde fixation, formate reutilization, fatty alcohol, fatty aldehyde, fatty acid and/or isopropanol pathways exemplified previously and herein. The metabolic modifications exemplified below increase biosynthetic product yields over, for example, endogenous methanol utilization pathways because they further focus methanol derived carbon into the assimilation pathways described herein, decrease inefficient use of

methanol carbon through competing methanol utilization and/or formaldehyde fixation pathways and/or increase the production of reducing equivalents.

In this regard, methylotrophs microbial organisms utilize methanol as the sole source of carbon and energy. In such methylotrophic organisms, the oxidation of methanol to formaldehyde is catalyzed by one of three different enzymes: NADH dependent methanol dehydrogenase (MeDH), PQQ-dependent methanol dehydrogenase (MeDH-PQQ) and alcohol oxidase (AOX). Methanol oxidase is a specific type of AOX with activity on methanol. Gram positive bacterial methylotrophs such as *Bacillus methanolicus* utilize a cytosolic MeDH which generates reducing equivalents in the form of NADH. Gram negative bacterial methylotrophs utilize periplasmic PQQ-containing methanol dehydrogenase enzymes which transfer electrons from methanol to specialized cytochromes CL, and subsequently to a cytochrome oxidase (Afolabi et al, *Biochem* 40:9799-9809 (2001)). Eukaryotic methylotrophs employ a peroxisomal oxygen-consuming and hydrogen-peroxide producing alcohol oxidase.

Bacterial methylotrophs are found in the genera *Bacillus*, *Methylobacterium*, *Methyloversatilis*, *Methylococcus*, *Methylocystis* and *Hyphomicrobium*. These organisms utilize either the serine cycle (type II) or the RuMP cycle (type I) to further assimilate formaldehyde into central metabolism (Hanson and Hanson, *Microbiol Rev* 60:439-471 (1996)). As described previously, the RuMP pathway combines formaldehyde with ribulose monophosphate to form hexulose-6-phosphate, which is further converted to fructose-6-phosphate (see FIG. 1, step C). In the serine cycle formaldehyde is initially converted to 5,10-methylene-THF, which is combined with glycine to form serine. Overall, the reactions of the serine cycle produce one equivalent of acetyl-CoA from three equivalents of methanol (Anthony, *Science Prog* 94:109-37 (2011)). The RUMP cycle also yields one equivalent of acetyl-CoA from three equivalents methanol in the absence of phosphoketolase activity or a formate assimilation pathway. Genetic tools are available for numerous prokaryotic methylotrophs and methanotrophs.

Eukaryotic methylotrophs are found in the genera *Candida*, *Pichia*, *Ogataea*, *Kuraishia* and *Komagataella*. Particularly useful methylotrophic host organisms are those with well-characterized genetic tools and gene expression systems such as *Hansenula polymorpha*, *Pichia pastoris*, *Candida boidinii* and *Pichia methanolica* (for review see Yurimoto et al, *Int J Microbiol* (2011)). The initial step of methanol assimilation in eukaryotic methylotrophs occurs in the peroxisomes, where methanol and oxygen are oxidized to formaldehyde and hydrogen peroxide by alcohol oxidase (AOX). Formaldehyde assimilation with xylulose-5-phosphate via DHA synthase also occurs in the peroxisomes. During growth on methanol, the two enzymes DHA synthase and AOX together comprise 80% of the total cell protein (Horiguchi et al, *J Bacteriol* 183:6372-83 (2001)). DHA synthase products, DHA and glyceraldehyde-3-phosphate, are secreted into the cytosol where they undergo a series of rearrangements catalyzed by pentose phosphate pathway enzymes, and are ultimately converted to cellular constituents and xylulose-5-phosphate, which is transported back into the peroxisomes. The initial step of formaldehyde dissimilation, catalyzed by S-(hydroxymethyl)-glutathione synthase, also occurs in the peroxisomes. Like the bacterial methylotrophic pathways described above, eukaryotic methylotrophic pathways convert three equivalents of methanol to at most one equivalent of acetyl-CoA because they lack phosphoketolase activity or a formate assimilation pathway.

As exemplified further below, the various configurations of metabolic modifications disclosed herein for enhancing product yields via methanol derived carbon include enhancing methanol oxidation and production of reducing equivalents using either an endogenous NADH dependent methanol dehydrogenase, an exogenous NADH dependent methanol dehydrogenase, both an endogenous NADH dependent methanol dehydrogenase and exogenous NADH dependent methanol dehydrogenase alone or in combination with one or more metabolic modifications that attenuate, for example, DHA synthase and/or AOX. In addition, other metabolic modifications as exemplified below that reduce carbon flux away from methanol oxidation and formaldehyde fixation also can be included, alone or in combination, with the methanol oxidation and formaldehyde fixation pathway configurations disclosed herein that enhance carbon flux into product precursors such as acetyl-CoA and, therefore, enhance product yields.

Accordingly, the microbial organisms of the invention having one or more of any of the above and/or below metabolic modifications to a methanol utilization pathway and/or formaldehyde assimilation pathway configurations for enhancing product yields can be combined with any one or more, including all of the previously described methanol oxidation, formaldehyde fixation, formate reutilization, fatty alcohol, fatty aldehyde, acid and/or isopropanol pathways to enhance the yield and/or production of a product such as any of the fatty alcohol, fatty aldehyde, fatty acids and/or isopropanol described herein.

Given the teachings and guidance provided herein, the methanol oxidation and formaldehyde fixation pathway configurations can be equally engineered into both prokaryotic and eukaryotic organisms. In prokaryotic microbial organisms, for example, one skilled in the art will understand that utilization of an endogenous methanol oxidation pathway enzyme or expression of an exogenous nucleic acid encoding a methanol oxidation pathway enzyme will naturally occur cytosolically because prokaryotic organisms lack peroxisomes. In eukaryotic microbial organisms one skilled in the art will understand that certain methanol oxidation pathways occur in the peroxisome as described above and that cytosolic expression of the methanol oxidation pathway or pathways described herein to enhance product yields can be beneficial. The peroxisome located pathways and competing pathways remain or, alternatively, attenuated as described below to further enhance methanol oxidation and formaldehyde fixation.

With respect to eukaryotic microbial host organisms, those skilled in the art will know that yeasts and other eukaryotic microorganisms exhibit certain characteristics distinct from prokaryotic microbial organisms. When such characteristics are desirable, one skilled in the art can choose to use such eukaryotic microbial organisms as a host for engineering the various different methanol oxidation and formaldehyde fixation configurations exemplified herein for enhancing product yields. For example, yeast are robust organisms, able to grow over a wide pH range and able to tolerate more impurities in the feedstock. Yeast also ferment under low growth conditions and are not susceptible to infection by phage. Less stringent aseptic design requirements can also reduce production costs. Cell removal, disposal and propagation are also cheaper, with the added potential for by-product value for animal feed applications. The potential for cell recycle and semi-continuous fermentation offers benefits in increased overall yields and rates. Other benefits include: potential for extended fermentation times under low growth conditions, lower viscosity broth (vs

E. coli) with insoluble hydrophobic products, the ability to employ large fermenters with external loop heat exchangers.

Eukaryotic host microbial organisms suitable for engineering carbon efficient methanol utilization capability can be selected from, and the non-naturally occurring microbial organisms generated in, for example, yeast, fungus or any of a variety of other microorganisms applicable to fermentation processes. As described previously, exemplary yeasts or fungi include species selected from the genera *Saccharomyces*, *Schizosaccharomyces*, *Schizochytrium*, *Rhodotorula*, *Thraustochytrium*, *Aspergillus*, *Kluyveromyces*, *Issatchenkia*, *Yarrowia*, *Candida*, *Pichia*, *Ogataea*, *Kuraishia*, *Hansenula* and *Komagataella*. Useful host organisms include *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Hansenula polymorpha*, *Pichia methanolica*, *Candida boidinii*, *Kluyveromyces lactis*, *Kluyveromyces marxianus*, *Aspergillus terreus*, *Aspergillus niger*, *Pichia pastoris*, *Rhizopus arrhizus*, *Rhizobus oryzae*, *Yarrowia lipolytica*, *Issatchenkia orientalis* and the like.

The methanol oxidation and/or formaldehyde assimilation pathway configurations described herein for enhancing product yields include, for example, a NADH-dependent methanol dehydrogenase (MeDH), one or more formaldehyde assimilation pathways and/or one or more phosphoketolases. Such engineered pathways provide a yield advantage over endogenous pathways found in methylotrophic organisms. For example, methanol assimilation via methanol dehydrogenase provides reducing equivalents in the useful form of NADH, whereas alcohol oxidase and PQQ-dependent methanol dehydrogenase do not. Several product pathways described herein have several NADH-dependant enzymatic steps. In addition, deletion of redox-inefficient methanol oxidation enzymes as described further below, combined with increased cytosolic or peroxisomal expression of an NADH-dependent methanol dehydrogenase, improves the ability of the organism to extract useful reducing equivalents from methanol. In some aspects, if NADH-dependent methanol dehydrogenase is engineered into the peroxisome, an efficient means of shuttling redox in the form of NADH out of the peroxisome and into the cytosol can be included. Further employment of a formaldehyde assimilation pathway in combination with a phosphoketolase or formate assimilation pathway enables high yield conversion of methanol to acetyl-CoA, and subsequently to acetyl-CoA derived products.

For example, in a eukaryotic organism such as *Pichia pastoris*, deleting the endogenous alcohol oxidase and peroxisomal formaldehyde assimilation and dissimilation pathways, and expressing redox and carbon-efficient cytosolic methanol utilization pathways significantly improves the yield of dodecanol, an acetyl-CoA derived product. The maximum dodecanol yield of *Pichia pastoris* from methanol using endogenous methanol oxidase and formaldehyde assimilation enzymes is 0.256 g dodecanol/g methanol. Adding one or more heterologous cytosolic phosphoketolase enzymes, in combination with a formaldehyde assimilation pathway such as the DHA pathway or the RUMP pathway, boosts the dodecanol yield to 0.306 g dodecanol/g methanol. Deletion of peroxisomal methanol oxidase and formaldehyde assimilation pathway enzymes (alcohol oxidase, DHA synthase), and replacement with cytosolic methanol dehydrogenase (NADH dependent) and formaldehyde assimilation pathways, together with a phosphoketolase, provides a significant boost of yield to 0.422 g/g.

Strain design (assumes DHA pathway)	Max FA yield (g dodecanol/g MeOH)
Pichia + AOX + fatty acid pathway	0.256
Pichia + AOX + PK	0.306
Pichia + MeDH + PK	0.422

Metabolic modifications for enabling redox- and carbon-efficient cytosolic methanol utilization in a eukaryotic or prokaryotic organism are exemplified in further detail below.

In one embodiment, the invention provides cytosolic expression of one or more methanol oxidation and/or formaldehyde assimilation pathways. Engineering into a host microbial organism carbon- and redox-efficient cytosolic formaldehyde assimilation can be achieved by expression of one or more endogenous or exogenous methanol oxidation pathways and/or one or more endogenous or exogenous formaldehyde assimilation pathway enzymes in the cytosol. An exemplary pathway for methanol oxidation includes NADH dependent methanol dehydrogenase as shown in FIG. 1. Exemplary pathways for converting cytosolic formaldehyde into glycolytic intermediates also are shown in FIG. 1. Such pathways include methanol oxidation via expression of an cytosolic NADH dependent methanol dehydrogenase, formaldehyde fixation via expression of cytosolic DHA synthase, both methanol oxidation via expression of an cytosolic NADH dependent methanol dehydrogenase and formaldehyde fixation via expression of cytosolic DHA synthase alone or together with the metabolic modifications exemplified below that attenuate less beneficial methanol oxidation and/or formaldehyde fixation pathways. Such attenuating metabolic modifications include, for example, attenuation of alcohol oxidase, attenuation of DHA kinase and/or when utilization of ribulose-5-phosphate (Ru5P) pathway for formaldehyde fixation attenuation of DHA synthase.

For example, in the carbon-efficient DHA pathway of formaldehyde assimilation shown in FIG. 1, step D, formaldehyde is converted to dihydroxyacetone (DHA) and glyceraldehyde-3-phosphate (GAP) by DHA synthase (FIG. 1D). DHA and G3P are then converted to fructose-6-phosphate in one step by F6P aldolase (FIG. 1C) or in three steps by DHA kinase, FBP aldolase and fructose-1,6-bisphosphatase (not shown). Formation of F6P from DHA and G3P by F6P aldolase is more ATP-efficient than using DHA kinase, FBP aldolase and fructose-1,6-bisphosphatase. Rearrangement of F6P and E4P by enzymes of the pentose phosphate pathway (transaldolase, transketolase, R₅P epimerase and Ru5P epimerase) regenerates xylulose-5-phosphate, the DHA synthase substrate. Conversion of F6P to acetyl-phosphate and E4P (FIG. 11), or Xu5P to G3P and acetyl-phosphate (FIGS. 1T and 1U) by one or more phosphoketolase enzymes results in the carbon-efficient generation of cytosolic acetyl-CoA. Exemplary enzymes catalyzing each step of the carbon efficient DHA pathway are described elsewhere herein.

An alternate carbon efficient pathway for formaldehyde assimilation proceeding through ribulose-5-phosphate (Ru5P) is shown in FIG. 1, step B. The formaldehyde assimilation enzyme of this pathway is 3-hexulose-6-phosphate synthase, which combines ru5p and formaldehyde to form hexulose-6-phosphate (FIG. 1B). 6-Phospho-3-hexuloisomerase converts H6P to F6P (FIG. 1C). Regeneration of Ru5P from F6P proceeds by pentose phosphate pathway enzymes. Carbon-efficient phosphoketolase enzymes catalyze the conversion of F6P and/or Xu5P to acetyl-phosphate

and pentose phosphate intermediates. Exemplary enzymes catalyzing each step of the carbon efficient RUMP pathway are described elsewhere herein.

Thus, in this embodiment, conversion of cytosolic formaldehyde into glycolytic intermediates can occur via expression of a cytosolic 3-hexulose-6-phosphate (3-Hu6P) synthase and 6-phospho-3-hexuloisomerase. Thus, exemplary pathways that can be engineered into a microbial organism of the invention can include methanol oxidation via expression of a cytosolic NADH dependent methanol dehydrogenase, formaldehyde fixation via expression of cytosolic 3-Hu6P synthase and 6-phospho-3-hexuloisomerase, both methanol oxidation via expression of an cytosolic NADH dependent dehydrogenase and formaldehyde fixation via expression of cytosolic 3-Hu6P synthase and 6-phospho-3-hexuloisomerase alone or together with the metabolic modifications exemplified below that attenuate less beneficial methanol oxidation and/or formaldehyde fixation pathways. Such attenuating metabolic modifications include, for example, attenuation of alcohol oxidase, attenuation of DHA kinase and/or when utilization of ribulose-5-phosphate (Ru5P) pathway for formaldehyde fixation attenuation of DHA synthase.

In yet another embodiment increased product yields can be accomplished by engineering into the host microbial organism of the invention both the RUMP and DHA pathways as shown in FIG. 1. In this embodiment, the microbial organisms can have cytosolic expression of one or more methanol oxidation and/or formaldehyde assimilation pathways. The formaldehyde assimilation pathways can include both assimilation through cytosolic DHA synthase and 3-Hu6P synthase. Such pathways include methanol oxidation via expression of a cytosolic NADH dependent methanol dehydrogenase, formaldehyde fixation via expression of cytosolic DHA synthase and 3-Hu6P synthase, both methanol oxidation via expression of an cytosolic NADH dependent dehydrogenase and formaldehyde fixation via expression of cytosolic DHA synthase and 3-Hu6P synthase alone or together with the metabolic modifications exemplified previously and also below that attenuate less beneficial methanol oxidation and/or formaldehyde fixation pathways. Such attenuating metabolic modifications include, for example, attenuation of alcohol oxidase, attenuation of DHA kinase and/or attenuation of DHA synthase (e.g. when ribulose-5-phosphate (Ru5P) pathway for formaldehyde fixation is utilized).

Increasing the expression and/or activity of one or more formaldehyde assimilation pathway enzymes in the cytosol can be utilized to assimilate formaldehyde at a high rate. Increased activity can be achieved by increased expression, altering the ribosome binding site, altering the enzyme activity, or altering the sequence of the gene to ensure, for example, that codon usage is balanced with the needs of the host organism, or that the enzyme is targeted to the cytosol as disclosed herein.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes attenuation of one or more endogenous enzymes, which enhances carbon flux through acetyl-CoA. For example, in some aspects, the endogenous enzyme can be selected from DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase, DHA synthase or any combination thereof. Accordingly, in some aspects, the attenuation is of the endogenous enzyme DHA kinase. In some aspects, the attenuation is of the endogenous enzyme methanol oxidase. In some aspects, the attenuation is of the endogenous enzyme PQQ-depen-

dent methanol dehydrogenase. In some aspects, the attenuation is of the endogenous enzyme DHA synthase. The invention also provides a microbial organism wherein attenuation is of any combination of two or three endogenous enzymes described herein. For example, a microbial organism of the invention can include attenuation of DHA kinase and DHA synthase, or alternatively methanol oxidase and PQQ-dependent methanol dehydrogenase, or alternatively DHA kinase, methanol oxidase, and PQQ-dependent methanol dehydrogenase, or alternatively DHA kinase, methanol oxidase, and DHA synthase. The invention also provides a microbial organism wherein attenuation is of all endogenous enzymes described herein. For example, in some aspects, a microbial organism described herein includes attenuation of DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase and DHA synthase.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes attenuation of one or more endogenous enzymes of a competing formaldehyde assimilation or dissimilation pathway. Examples of these endogenous enzymes are disclosed in FIG. 1 and described in Example XXIII. It is understood that a person skilled in the art would be able to readily identify enzymes of such competing pathways. Competing pathways can be dependent upon the host microbial organism and/or the exogenous nucleic acid introduced into the microbial organism as described herein. Accordingly, in some aspects of the invention, the microbial organism includes attenuation of one, two, three, four, five, six, seven, eight, nine, ten or more endogenous enzymes of a competing formaldehyde assimilation or dissimilation pathway.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes a gene disruption of one or more endogenous nucleic acids encoding enzymes, which enhances carbon flux through acetyl-CoA. For example, in some aspects, the endogenous enzyme can be selected from DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase, DHA synthase or any combination thereof. According, in some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme DHA kinase. In some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme methanol oxidase. In some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme PQQ-dependent methanol dehydrogenase. In some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme DHA synthase. The invention also provides a microbial organism wherein the gene disruption is of any combination of two or three nucleic acids encoding endogenous enzymes described herein. For example, a microbial organism of the invention can include a gene disruption of DHA kinase and DHA synthase, or alternatively methanol oxidase and PQQ-dependent methanol dehydrogenase, or alternatively DHA kinase, methanol oxidase, and PQQ-dependent methanol dehydrogenase, or alternatively DHA kinase, methanol oxidase, and DHA synthase. The invention also provides a microbial organism wherein all endogenous nucleic acids encoding enzymes described herein are disrupted. For example, in some aspects, a microbial organism described herein includes disruption of DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase and DHA synthase.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein,

wherein the microbial organism further includes a gene disruption of one or more endogenous enzymes of a competing formaldehyde assimilation or dissimilation pathway. Examples of these endogenous enzymes are disclosed in FIG. 1 and described in Example XXIII. It is understood that a person skilled in the art would be able to readily identify enzymes of such competing pathways. Competing pathways can be dependent upon the host microbial organism and/or the exogenous nucleic acid introduced into the microbial organism as described herein. Accordingly, in some aspects of the invention, the microbial organism includes a gene disruption of one, two, three, four, five, six, seven, eight, nine, ten or more endogenous nucleic acids encoding enzymes of a competing formaldehyde assimilation or dissimilation pathway.

The invention is described herein with general reference to the metabolic reaction, reactant or product thereof, or with specific reference to one or more nucleic acids or genes encoding an enzyme associated with or catalyzing, or a protein associated with, the referenced metabolic reaction, reactant or product. Unless otherwise expressly stated herein, those skilled in the art will understand that reference to a reaction also constitutes reference to the reactants and products of the reaction. Similarly, unless otherwise expressly stated herein, reference to a reactant or product also references the reaction, and reference to any of these metabolic constituents also references the gene or genes encoding the enzymes that catalyze or proteins involved in the referenced reaction, reactant or product. Likewise, given the well known fields of metabolic biochemistry, enzymology and genomics, reference herein to a gene or encoding nucleic acid also constitutes a reference to the corresponding encoded enzyme and the reaction it catalyzes or a protein associated with the reaction as well as the reactants and products of the reaction.

The non-naturally occurring microbial organisms of the invention can be produced by introducing expressible nucleic acids encoding one or more of the enzymes or proteins participating in one or more fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathways. Depending on the host microbial organism chosen for biosynthesis, nucleic acids for some or all of a particular fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway can be expressed. For example, if a chosen host is deficient in one or more enzymes or proteins for a desired biosynthetic pathway, then expressible nucleic acids for the deficient enzyme(s) or protein(s) are introduced into the host for subsequent exogenous expression. Alternatively, if the chosen host exhibits endogenous expression of some pathway genes, but is deficient in others, then an encoding nucleic acid is needed for the deficient enzyme(s) or protein(s) to achieve fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthesis. Thus, a non-naturally occurring microbial organism of the invention can be produced by introducing exogenous enzyme or protein activities to obtain a desired biosynthetic pathway or a desired biosynthetic pathway can be obtained by introducing one or more exogenous enzyme or protein activities that, together with one or more endogenous enzymes or proteins, produces a desired product such as fatty alcohol, fatty aldehyde, fatty acid or isopropanol.

Host microbial organisms can be selected from, and the non-naturally occurring microbial organisms generated in, for example, bacteria, yeast, fungus or any of a variety of other microorganisms applicable or suitable to fermentation processes. Exemplary bacteria include any species selected from the order Enterobacteriales, family Enterobacteriaceae,

including the genera *Escherichia* and *Klebsiella*; the order Aeromonadales, family Succinivibrionaceae, including the genus *Anaerobiospirillum*; the order Pasteurellales, family Pasteurellaceae, including the genera *Actinobacillus* and *Mannheimia*; the order Rhizobiales, family Bradyrhizobiaceae, including the genus *Rhizobium*; the order Bacillales, family Bacillaceae, including the genus *Bacillus*; the order Actinomycetales, families Corynebacteriaceae and Streptomycetaceae, including the genus *Corynebacterium* and the genus *Streptomyces*, respectively; order Rhodospirillales, family Acetobacteraceae, including the genus *Gluconobacter*; the order Sphingomonadales, family Sphingomonadaceae, including the genus *Zymomonas*; the order Lactobacillales, families Lactobacillaceae and Streptococcaceae, including the genus *Lactobacillus* and the genus *Lactococcus*, respectively; the order Clostridiales, family Clostridiaceae, genus *Clostridium*; and the order Pseudomonadales, family Pseudomonadaceae, including the genus *Pseudomonas*. Non-limiting species of host bacteria include *Escherichia coli*, *Klebsiella oxytoca*, *Anaerobiospirillum succiniciproducens*, *Actinobacillus succinogenes*, *Mannheimia succiniciproducens*, *Rhizobium elli*, *Bacillus subtilis*, *Corynebacterium glutamicum*, *Gluconobacter oxydans*, *Zymomonas mobilis*, *Lactococcus lactis*, *Lactobacillus plantarum*, *Streptomyces coelicolor*, *Clostridium acetobutylicum*, *Pseudomonas fluorescens*, and *Pseudomonas putida*. Exemplarily bacterial methylotrophs include, for example, *Bacillus*, *Methylobacterium*, *Methyloversatilis*, *Methylococcus*, *Methylocystis* and *Hyphomicrobium*.

Similarly, exemplary species of yeast or fungi species include any species selected from the order Saccharomycetales, family Saccaromycetaceae, including the genera *Saccharomyces*, *Kluyveromyces* and *Pichia*; the order Saccharomycetales, family Dipodascaceae, including the genus *Yarrowia*; the order Schizosaccharomycetales, family Schizosaccharomycetaceae, including the genus *Schizosaccharomyces*; the order Eurotiales, family Trichocomaceae, including the genus *Aspergillus*; and the order Mucorales, family Mucoraceae, including the genus *Rhizopus*. Non-limiting species of host yeast or fungi include *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Kluyveromyces lactis*, *Kluyveromyces marxianus*, *Aspergillus terreus*, *Aspergillus niger*, *Pichia pastoris*, *Rhizopus arrhizus*, *Rhizobus oryzae*, *Yarrowia lipolytica*, and the like. *E. coli* is a particularly useful host organism since it is a well characterized microbial organism suitable for genetic engineering. Other particularly useful host organisms include yeast such as *Saccharomyces cerevisiae* and yeasts or fungi selected from the genera *Saccharomyces*, *Schizosaccharomyces*, *Schizochytrium*, *Rhodotorula*, *Thraustochytrium*, *Aspergillus*, *Kluyveromyces*, *Issatchenkia*, *Yarrowia*, *Candida*, *Pichia*, *Ogataea*, *Kuraishia*, *Hansenula* and *Komagataella*. Useful host organisms include *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, *Hansenula polymorpha*, *Pichia methanolica*, *Candida boidinii*, *Kluyveromyces lactis*, *Kluyveromyces marxianus*, *Aspergillus terreus*, *Aspergillus niger*, *Pichia pastoris*, *Rhizopus arrhizus*, *Rhizobus oryzae*, *Yarrowia lipolytica*, *Issatchenkia orientalis* and the like. Exemplarily eukaryotic methylotrophs include, for example, eukaryotic methylotrophs found in the genera *Candida*, *Pichia*, *Ogataea*, *Kuraishia* and *Komagataella*. Particularly useful methylotrophic host organisms include, for example, *Hansenula polymorpha*, *Pichia pastoris*, *Candida bochmii* and *Pichia methanolica*. It is understood that any suitable microbial host organism can be used to introduce metabolic and/or genetic modifications to produce a desired product.

Depending on the fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway constituents of a selected host microbial organism, the non-naturally occurring microbial organisms of the invention will include at least one exogenously expressed fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway-encoding nucleic acid and up to all encoding nucleic acids for one or more fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathways. For example, fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthesis can be established in a host deficient in a pathway enzyme or protein through exogenous expression of the corresponding encoding nucleic acid. In a host deficient in all enzymes or proteins of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway, exogenous expression of all enzyme or proteins in the pathway can be included, although it is understood that all enzymes or proteins of a pathway can be expressed even if the host contains at least one of the pathway enzymes or proteins. For example, exogenous expression of all enzymes or proteins in a pathway for production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be included, such as a thiolase, a 3-oxoacyl-CoA reductase, a 3-hydroxyacyl-CoA dehydratase, an enoyl-CoA reductase, an acyl-CoA reductase (aldehyde forming) and an alcohol dehydrogenase, for production of a fatty alcohol.

Given the teachings and guidance provided herein, those skilled in the art will understand that the number of encoding nucleic acids to introduce in an expressible form will, at least, parallel the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway deficiencies of the selected host microbial organism. Therefore, a non-naturally occurring microbial organism of the invention can have one, two, three, four, five, six, seven or eight up to all nucleic acids encoding the enzymes or proteins constituting a fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway disclosed herein. In some embodiments, the non-naturally occurring microbial organisms also can include other genetic modifications that facilitate or optimize fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthesis or that confer other useful functions onto the host microbial organism. One such other functionality can include, for example, augmentation of the synthesis of one or more of the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway precursors such as acetyl-CoA, malonyl-ACP, malonyl-CoA or propionyl-CoA.

Generally, a host microbial organism is selected such that it produces the precursor of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway, either as a naturally produced molecule or as an engineered product that either provides de novo production of a desired precursor or increased production of a precursor naturally produced by the host microbial organism. For example, acetyl-CoA is produced naturally in a host organism such as *E. coli*. A host organism can be engineered to increase production of a precursor, as disclosed herein. In addition, a microbial organism that has been engineered to produce a desired precursor can be used as a host organism and further engineered to express enzymes or proteins of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway.

In some embodiments, a non-naturally occurring microbial organism of the invention is generated from a host that contains the enzymatic capability to synthesize fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In this specific embodiment it can be useful to increase the synthesis or accumulation of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway product to, for example, drive fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway

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reactions toward fatty alcohol, fatty aldehyde, fatty acid or isopropanol production. Increased synthesis or accumulation can be accomplished by, for example, overexpression of nucleic acids encoding one or more of the above-described fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzymes or proteins. Overexpression of the enzyme or enzymes and/or protein or proteins of the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway can occur, for example, through exogenous expression of the endogenous gene or genes, or through exogenous expression of the heterologous gene or genes. Therefore, naturally occurring organisms can be readily generated to be non-naturally occurring microbial organisms of the invention, for example, producing fatty alcohol, fatty aldehyde, fatty acid or isopropanol, through overexpression of one, two, three, four, five, six, seven, or eight, that is, up to all nucleic acids encoding fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway enzymes or proteins. In addition, a non-naturally occurring organism can be generated by mutagenesis of an endogenous gene that results in an increase in activity of an enzyme in the fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway.

In particularly useful embodiments, exogenous expression of the encoding nucleic acids is employed. Exogenous expression confers the ability to custom tailor the expression and/or regulatory elements to the host and application to achieve a desired expression level that is controlled by the user. However, endogenous expression also can be utilized in other embodiments such as by removing a negative regulatory effector or induction of the gene's promoter when linked to an inducible promoter or other regulatory element. Thus, an endogenous gene having a naturally occurring inducible promoter can be up-regulated by providing the appropriate inducing agent, or the regulatory region of an endogenous gene can be engineered to incorporate an inducible regulatory element, thereby allowing the regulation of increased expression of an endogenous gene at a desired time. Similarly, an inducible promoter can be included as a regulatory element for an exogenous gene introduced into a non-naturally occurring microbial organism.

It is understood that, in methods of the invention, any of the one or more exogenous nucleic acids can be introduced into a microbial organism to produce a non-naturally occurring microbial organism of the invention. The nucleic acids can be introduced so as to confer, for example, a fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway onto the microbial organism. Alternatively, encoding nucleic acids can be introduced to produce an intermediate microbial organism having the biosynthetic capability to catalyze some of the required reactions to confer fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic capability. For example, a non-naturally occurring microbial organism having a fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway can comprise at least two exogenous nucleic acids encoding desired enzymes or proteins, such as the combination of a thiolase and an acyl-CoA reductase (alcohol forming), or alternatively a 2-oxoacyl-CoA reductase and an acyl-CoA hydrolase, or alternatively a enoyl-CoA reductase and an acyl-CoA reductase (aldehyde forming), or alternatively a methanol methyltransferase and an acetone reductase, or alternatively a 3-hexulose-6-phosphate synthase and an enoyl ACP-reductase, and the like. Thus, it is understood that any combination of two or more enzymes or proteins of a biosynthetic pathway can be included in a non-naturally occurring microbial organism of the invention. Similarly, it is understood that any combination of three or more

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enzymes or proteins of a biosynthetic pathway can be included in a non-naturally occurring microbial organism of the invention, for example, a thiolase, an enoyl-CoA reductase and a aldehyde dehydrogenase (acid forming), or alternatively a 3-hydroxyacyl-coA dehydratase, an acyl-CoA:ACP acyltransferase and a thioesterase, or alternatively a 3-oxoacyl-CoA reductase, an acyl-CoA hydrolase and a carboxylic acid reductase, or alternatively a dihydroxyacetone synthase, a S-formylglutathione hydrolase and an acetoacetyl-CoA ligase, or alternatively a 6-phospho-3-hexulose isomerase, a β -hydroxyacyl-ACP reductase and a fatty alcohol forming acyl-CoA reductase, and so forth, as desired, so long as the combination of enzymes and/or proteins of the desired biosynthetic pathway results in production of the corresponding desired product. Similarly, any combination of four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, sixteen, seventeen, eighteen, nineteen, twenty or more enzymes or proteins of a biosynthetic pathway as disclosed herein can be included in a non-naturally occurring microbial organism of the invention, as desired, so long as the combination of enzymes and/or proteins of the desired biosynthetic pathway results in production of the corresponding desired product.

In addition to the biosynthesis of fatty alcohol, fatty aldehyde, fatty acid or isopropanol as described herein, the non-naturally occurring microbial organisms and methods of the invention also can be utilized in various combinations with each other and/or with other microbial organisms and methods well known in the art to achieve product biosynthesis by other routes. For example, one alternative to produce fatty alcohol, fatty aldehyde, fatty acid or isopropanol other than use of the fatty alcohol, fatty aldehyde, fatty acid or isopropanol producers is through addition of another microbial organism capable of converting a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate to fatty alcohol, fatty aldehyde, fatty acid or isopropanol. One such procedure includes, for example, the fermentation of a microbial organism that produces a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate. The fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate can then be used as a substrate for a second microbial organism that converts the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate to fatty alcohol, fatty aldehyde, fatty acid or isopropanol. The fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate can be added directly to another culture of the second organism or the original culture of the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate producers can be depleted of these microbial organisms by, for example, cell separation, and then subsequent addition of the second organism to the fermentation broth can be utilized to produce the final product without intermediate purification steps.

In other embodiments, the non-naturally occurring microbial organisms and methods of the invention can be assembled in a wide variety of subpathways to achieve biosynthesis of, for example, fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In these embodiments, biosynthetic pathways for a desired product of the invention can be segregated into different microbial organisms, and the different microbial organisms can be co-cultured to produce the final product. In such a biosynthetic scheme, the product of one microbial organism is the substrate for a second microbial organism until the final product is synthesized. For example, the biosynthesis of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be accomplished by constructing a microbial organism that contains biosynthetic path-

ways for conversion of one pathway intermediate to another pathway intermediate or the product. Alternatively, fatty alcohol, fatty aldehyde, fatty acid or isopropanol also can be biosynthetically produced from microbial organisms through co-culture or co-fermentation using two organisms in the same vessel, where the first microbial organism produces a fatty alcohol, fatty aldehyde, fatty acid or isopropanol intermediate and the second microbial organism converts the intermediate to fatty alcohol, fatty aldehyde, fatty acid or isopropanol.

Given the teachings and guidance provided herein, those skilled in the art will understand that a wide variety of combinations and permutations exist for the non-naturally occurring microbial organisms and methods of the invention together with other microbial organisms, with the co-culture of other non-naturally occurring microbial organisms having subpathways and with combinations of other chemical and/or biochemical procedures well known in the art to produce fatty alcohol, fatty aldehyde, fatty acid or isopropanol.

Similarly, it is understood by those skilled in the art that a host organism can be selected based on desired characteristics for introduction of one or more gene disruptions to increase production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. Thus, it is understood that, if a genetic modification is to be introduced into a host organism to disrupt a gene, any homologs, orthologs or paralogs that catalyze similar, yet non-identical metabolic reactions can similarly be disrupted to ensure that a desired metabolic reaction is sufficiently disrupted. Because certain differences exist among metabolic networks between different organisms, those skilled in the art will understand that the actual genes disrupted in a given organism may differ between organisms. However, given the teachings and guidance provided herein, those skilled in the art also will understand that the methods of the invention can be applied to any suitable host microorganism to identify the cognate metabolic alterations needed to construct an organism in a species of interest that will increase fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthesis. In a particular embodiment, the increased production couples biosynthesis of fatty alcohol, fatty aldehyde, fatty acid or isopropanol to growth of the organism, and can obligatorily couple production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol to growth of the organism if desired and as disclosed herein.

Sources of encoding nucleic acids for a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein can include, for example, any species where the encoded gene product is capable of catalyzing the referenced reaction. Such species include both prokaryotic and eukaryotic organisms including, but not limited to, bacteria, including archaea and eubacteria, and eukaryotes, including yeast, plant, insect, animal, and mammal, including human. Exemplary species for such sources include, for example, *Escherichia coli* 255956237, *Penicillium chrysogenum* Wisconsin 54-1255, *Acetobacter pasteurians*, *Acidaminococcus fermentans*, *Acinetobacter baumannii* Naval-82 *Acinetobacter baylyi*, *Acinetobacter calcoaceticus*, *Acinetobacter* sp. ADP1, *Acinetobacter* sp. Strain M-1, *Actinobacillus succinogenes*, *Actinobacillus succinogenes* 130Z *Aedes aegypti*, *Allochromatium vinosum* DWI 180, *Aminomonas aminovorans*, *Anabaena variabilis* ATCC 29413, *Anaerobiospirillum succiniciproducens*, *Aquifex aeolicus*, *Arabidopsis thaliana*, *Archaeoglobus fulgidus*, *Archaeoglobus fulgidus* DSM 4304, *Arthrobacter globiformis*, *Ascaris suum*, *Aspergillus fumigatus*, *Aspergillus nidulans*, *Aspergillus niger*, *Aspergillus niger* CBS 513.88, *Aspergillus terreus* NIH2624, *Asper-*

gillus Synechococcus elongatus PCC 6301, *Azotobacter vinelandii* DT B. subtilis 168, *Bacillus alcalophilus* ATCC 27647, *Bacillus anthracis*, *Bacillus azotoformans* LMG 9581, *Bacillus cereus*, *Bacillus cereus* ATCC 14579, *Bacillus coagulans* 36D1, *Bacillus megaterium*, *Bacillus methanolicus* MGA3, *Bacillus methanolicus* PB1, *Bacillus selenitireducens* MLS10, *Bacillus* sp. SG-1, *Bacillus sphaericus*, *Bacillus subtilis*, *Bacteroides fragilis*, *Bifidobacterium bifidum*, *Bifidobacterium longum* NCC2705, *Bombyx mori*, *Bos taurus*, *Bradyrhizobium japonicum*, *Bradyrhizobium japonicum* USDA110, *Brassica juncea*, *Brassica napus*, *Burkholderia ambifaria* AMMD, *Burkholderia cenocepacia*, *Burkholderia cepacia*, *Burkholderia multivorans*, *Burkholderia phymatum*, *Burkholderia pyrrocinia*, *Burkholderia stabilis*, *Burkholderia thailandensis* E264, *Burkholderiales bacterium* Joshi_001, butyrate-producing bacterium L2-50, *Caenorhabditis elegans*, *Campylobacter curvus* 525.92, *Campylobacter jejuni*, *Candida albicans*, *Candida boidinii*, *Candida methylica*, *Candida parapsilosis*, *Candida tropicalis*, *Candida tropicalis* MYA-3404, *Candida tropicalis* MYA-3404, *Candida tropicalis*, *Carboxydothermus hydrogenoformans*, *Carboxydothermus hydrogenoformans* Z-2901, *Carthamus tinctorius*, *Caulobacter* sp. AP07, *Chlamydomonas reinhardtii*, *Chlorobium limicola*, *Chlorobium phaeobacteroides* DSM266, *Chlorobium tepidum*, *Chloroflexus aurantiacus*, *Cinnamomum camphorum*, *Citrobacter koseri* ATCC BAA-895, *Citrus junos*, *Clostridium acetobutylicum*, *Clostridium acetobutylicum* ATCC 824, *Clostridium aminoburicum*, *Clostridium beijerinckii*, *Clostridium beijerinckii* NCIMB 8052, *Clostridium carboxidivorans* P7, *Clostridium cellulolyticum* H10, *Clostridium cellulovorans* 743B, *Clostridium kluyveri*, *Clostridium kluyveri* DSM 555, *Clostridium ljungdahlii*, *Clostridium ljungdahlii* DSM 13528, *Clostridium pasteurianum*, *Clostridium pasteurianum* DSM525, *Clostridium perfringens*, *Clostridium perfringens* ATCC 13124, *Clostridium perfringens* str. 13, *Clostridium phytofermentans* ISDg, *Clostridium saccharoperbutylacetonicum*, *Clostridium symbiosum*, *Corynebacterium glutamicum*, *Corynebacterium glutamicum* ATCC 14067, *Corynebacterium glutamicum* R, *Corynebacterium* sp., *Corynebacterium* sp. U-96, *Corynebacterium ulcerans*, *Corynebacterium variabile*, *Cryptosporidium parvum* Iowa II, *Cuphea hookeriana*, *Cuphea palustris*, *Cupriavidus necator*, *Cupriavidus necator* N-1, *Cupriavidus taiwanensis*, *Cyanobium* PCC7001, *Cyanotheca* sp. PCC 7425, *Danio rerio*, *Desulfatibacillum alkenivorans* AK-01, *Desulfobacterium hafsiense*, *Desulfobacterium metallireducens* DSM 15288, *Desulfococcus oleovorans* Hxd3, *Desulfotomaculum reducens* MI-1, *Desulfovibrio africanus*, *Desulfovibrio africanus* str. Walvis Bay, *Desulfovibrio alaskensis*, *Desulfovibrio desulfuricans* subsp. *desulfuricans* str. ATCC 27774, *Desulfovibrio fructosovorans* JJ, *Desulfovibrio vulgaris* str. 'Miyazaki F' *Desulfovibrio vulgaris* str. Hildenborough, *Dictyostelium discoideum* AX4, *E. coli*, *Erythrobacter* sp. NAP1, *Escherichia coli* DH1, *Escherichia coli* K-12, *Escherichia coli* K-12 MG1655, *Escherichia coli* K-12 MG1655 niger CBS 513.88, *Escherichia coli* LW1655F+, *Escherichia coli* MG1655, *Escherichia coli* str. K-12 substr. MG1655, *Euglena gracilis*, *Flavobacterium frigidus*, *Fusobacterium nucleatum*, *Geobacillus* sp. GHH01, *Geobacillus* sp. M10EXG, *Geobacillus* sp. Y4.1MC1, *Geobacillus themodenitrificans* NG80-2, *Geobacillus themodenitrificans*, *Geobacter bemidiensis* Bem, *Geobacter metallireducens* GS-15, *Geobacter sulfurreducens*, *Geobacter sulfurreducens* PCA, *Haemophilus influenza*, *Haloarcula marismortui*, *Haloarcula marismortui* ATCC 43049, *Halomonas* sp.

HTNK1, *Helianthus annuus*, *Helicobacter pylori*, *Helicobacter pylori* 26695, *Homo sapiens*, human gut metagenome, *Hydrogenobacter thermophilus*, *Hyphomicrobium denitrificans* ATCC 51888, *Hyphomicrobium zavarzinii*, *Kineococcus radiotolerans*, *Klebsiella pneumoniae*, *Klebsiella pneumoniae*, *Klebsiella pneumoniae* subsp. *pneumoniae* MGH 78578, *Kluyveromyces lactis*, *Kluyveromyces lactis* NRRL Y-1140, *Lactobacillus acidophilus*, *Lactobacillus brevis* ATCC 367, *Lactobacillus casei*, *Lactobacillus plantarum*, *Lactobacillus reuteri*, *Lactococcus lactis*, *Lactococcus lactis* subsp. *lactis*, *Leifsonia* sp. S749, *Leuconostoc mesenteroides*, *Listeria monocytogenes*, *Lyngbya* sp. PCC 8106, *Lysinibacillus fusiformis*, *Lysinibacillus sphaericus*, *Mannheimia succiniciproducens*, marine gamma proteobacterium HTCC2080, *Marinobacter aquaeolei*, *Megathyrus maximus*, *Mesorhizobium loti*, *Mesorhizobium loti* MAFF303099, *Metallosphaera sedula*, *Metallosphaera sedula*, *Metarhizium acridum* CQMa 102, *Methanosarcina acetivorans*, *Methanosarcina acetivorans* C2A, *Methanosarcina barkeri*, *Methanosarcina mazei* Tuc01, *Methanosarcina thermophila*, *Methanothermobacter thermautotrophicus*, *Methylobacillus flagellatus*, *Methylobacillus flagellatus* KT, *Methylobacter marinus*, *Methylobacterium extorquens*, *Methylobacterium extorquens* AM1, *Methylococcus capsulatus*, *Methylococcus capsulatus* BG8, *Methylobacterium aminofaciens*, *Methylovorus glucosetrophus* SIP3-4, *Methylovorus* sp. MP688, *Moorella thermoacetica*, *Moorella thermoacetica* ATCC 39073, *Mus musculus*; *Mycobacter* sp. strain JC1 DSM3803, *Mycobacterium avium* subsp. *paratuberculosis* K-10, *Mycobacterium bovis* BCG, *Mycobacterium gastri*, *Mycobacterium kansasii* ATCC 12478, *Mycobacterium marinum* M, *Mycobacterium smegmatis*, *Mycobacterium smegmatis* MC2 155, *Mycobacterium smegmatis* str. MC2 155, *Mycobacterium* sp. strain JLS, *Mycobacterium tuberculosis*, *Mycobacterium tuberculosis* H37Rv, *Neurospora crassa* OR 74A, *Nicotiana tabacum*, *Nitrosopumilus salaria* BD31, *Nitrososphaera gargensis* Ga9.2, *Nocardia brasiliensis*, *Nocardia farcinica* IFM10152, *Nocardia iowensis*, *Nocardia iowensis* (sp. NRRL 5646), *Nodularia spumigena* CCY9414, *Nostoc azollae*, *Nostoc* sp. PCC 7120, *Ogataea parapolymorpha* DL-1 (*Hansenula polymorpha* DL-1), *Oxalobacter formigenes*, *Paenibacillus peoriae* KCTC 3763, *Paracoccus denitrificans*, *Pelobacter carbinolicus* DSM 2380, *Penicillium chrysogenum*, *Perkinsus marinus* ATCC 50983, *Photobacterium leiognathi* PL741, *Photobacterium phosphoreum*, *Photobacterium profundum* 3TCK *Phytomonas* sp., *Pichia pastoris*, *Pichia pastoris* GS115, *Picrophilus torridus* DSM9790, *Plasmodium falciparum*, *Porphyromonas gingivalis*, *Porphyromonas gingivalis* W83, *Prochlorococcus marinus* MIT 9312, *Propionibacterium acnes*, *Propionibacterium freudenreichii* sp. *shennanii*, *Propionibacterium freudenreichii*, *Propionibacterium freudenreichii* subsp. *Shermanii*, *Propionigenium modestum*, *Pseudomonas aeruginosa*, *Pseudomonas aeruginosa* PA01, *Pseudomonas fluorescens*, *Pseudomonas fluorescens* Pf0-1, *Pseudomonas knackmussii*, *Pseudomonas knackmussii* (B13), *Pseudomonas mendocina*, *Pseudomonas putida*, *Pseudomonas putida* GB-1, *Pseudomonas putida* GB-1 *Trypanosoma brucei*, *Pseudomonas* sp., *Pseudomonas* sp. CF600, *Pseudomonas stutzeri*, *Pseudomonas syringae*, *Pseudomonas syringae* pv. *syringae* B728a, *Pyrobaculum aerophilum* str. IM2, *Pyrococcus abyssi*, *Pyrococcus furiosus*, *Pyrococcus horikoshii* OT3, *Ralstonia eutropha*, *Ralstonia eutropha* H16, *Ralstonia metallidurans*, *Rattus norvegicus*, *Rhizobium leguminosarum*, *Rhizopus oryzae*, *Rhodobacter capsulatus*, *Rhodobacter sphaeroides*, *Rhodobacter sphaeroides* ATCC 17025, *Rhodococcus eryth-*

ropolis SK121, *Rhodococcus opacus* B4, *Rhodopseudomonas palustris*, *Rhodopseudomonas palustris* CGA009, *Rhodopseudomonas palustris* DX-1, *Rhodospirillum rubrum*, *Roseiflexus castenholzii*, *Saccharomyces cerevisiae*, *Saccharomyces cerevisiae* S288c, *Salmonella enteric*, *Salmonella enterica*, *Salmonella enterica* LT2, *Salmonella enterica* subsp. *enterica* serovar *Typhimurium* str. LT2, *Salmonella enterica* *Typhimurium*, *Salmonella typhimurium*, *Salmonella typhimurium* LT2, *Schizosaccharomyces pombe*, *Seibaldella termitidis* ATCC 33386, *Shewanella oneidensis* MR-1, *Shigella dysenteriae*, *Shigella flexneri*, *Shigella sonnei*, *Simmondsia chinensis*, *Sinorhizobium meliloti* 1021, *Solanum lycopersicum*, *Sordaria macrospora*, *Staphylococcus aureus*, *Staphylococcus aureus* MW2, *Streptococcus mutans*, *Streptococcus mutans* UA159, *Streptococcus pneumoniae*, *Streptococcus sanguinis*, *Streptomyces anulatus*, *Streptomyces avermitilis* MA-4680, *Streptomyces avermitilis*, *Streptomyces cinnamonensis*, *Streptomyces coelicolor*, *Streptomyces glaucescens*, *Streptomyces griseus* subsp. *griseus* NBRC 13350, *Streptomyces luridus*, *Streptomyces* sp. CL190, *Streptomyces* sp. CL190, *Streptomyces* sp. KO-3988, *Streptomyces viridochromogenes*, *Streptomyces wedmorensis*, *Sulfolobus acidocaldarius*, *Sulfolobus solfataricus*, *Sulfolobus solfataricus* P-2, *Sulfolobus tokodaii*, *Sulfurihydrogenibium subterraneum*, *Sulfurimonas denitrificans*, *Sus scrofa*, *Synechococcus elongatus* PCC 6301, *Synechococcus elongatus* PCC7942, *Synechococcus* sp. PCC 7002, *Synechocystis* str. PCC 6803, *Syntrophobacter fumaroxidans*, *Syntrophus aciditrophicus*, *Thauera aromatic*, *Thermoanaerobacter ethanolicus* JW 200, *Thermoanaerobacter pseudethanolicus* ATCC 33223, *Thermoanaerobacter* sp. X514, *Thermoanaerobacter tengcongensis* MB4, *Thermoanaerobacter brockii*, *Thermococcus kodakaraensis*, *Thermococcus litoralis*, *Thermomyces lanuginosus*, *Thermoplasma acidophilum*, *Thermoproteus neutrophilus*, *Thermotoga maritime*, *Thiocapsa roseopersicina*, *Treponema denticola*, *Trichomonas vaginalis* G3, *Triticum aestivum*, *Trypanosoma brucei*, *Trypanosoma cruzi* strain CL Brener, *Tsukamurella paurometabola* DSM 20162, *Umbellularia californica*, uncultured organism, *Veillonella parvula*, *Vibrio harveyi* ATCC BAA-1116, *Xanthobacter autotrophicus* Py2, *Xenopus tropicalis*, *Yarrowia lipolytica*, *Yersinia frederiksenii*, *Zea mays*, *Zoogloea ramigera*, *Zymomonas mobilis*, *Zymomonas mobilis* subsp. *mobilis* ZM4, *Clostridium beijerinckii*, *Deinococcus radiodurans* R1, *Aquifex aeolicus* VF5, *Methanocaldococcus janaschii*, *Yersinia pestis*, *Bifidobacterium animalis lactis*, *Bifidobacterium dentium* ATCC 27678, *Bifidobacterium pseudolongum* subsp. *globosum*, *Bifidobacterium breve*, *Lactobacillus paraplantarum*, *Corynebacterium glutamicum* ATCC 13032, as well as other exemplary species disclosed herein or available as source organisms for corresponding genes. However, with the complete genome sequence available for now more than 550 species (with more than half of these available on public databases such as the NCBI), including 395 microorganism genomes and a variety of yeast, fungi, plant, and mammalian genomes, the identification of genes encoding the requisite fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic activity for one or more genes in related or distant species, including for example, homologues, orthologs, paralogs and nonorthologous gene displacements of known genes, and the interchange of genetic alterations between organisms is routine and well known in the art. Accordingly, the metabolic alterations allowing biosynthesis of fatty alcohol, fatty aldehyde, fatty acid or isopropanol described herein with reference to a particular organism such as *E. coli* can be readily applied to

other microorganisms, including prokaryotic and eukaryotic organisms alike. Given the teachings and guidance provided herein, those skilled in the art will know that a metabolic alteration exemplified in one organism can be applied equally to other organisms.

In some instances, such as when an alternative fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway exists in an unrelated species, fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthesis can be conferred onto the host species by, for example, exogenous expression of a paralog or paralogs from the unrelated species that catalyzes a similar, yet non-identical metabolic reaction to replace the referenced reaction. Because certain differences among metabolic networks exist between different organisms, those skilled in the art will understand that the actual gene usage between different organisms may differ. However, given the teachings and guidance provided herein, those skilled in the art also will understand that the teachings and methods of the invention can be applied to all microbial organisms using the cognate metabolic alterations to those exemplified herein to construct a microbial organism in a species of interest that will synthesize fatty alcohol, fatty aldehyde, fatty acid or isopropanol. A nucleic acid molecule encoding a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein of the invention can also include a nucleic acid molecule that hybridizes to a nucleic acid disclosed herein by SEQ ID NO, GenBank and/or GI number or a nucleic acid molecule that hybridizes to a nucleic acid molecule that encodes an amino acid sequence disclosed herein by SEQ ID NO, GenBank and/or GI number. Hybridization conditions can include highly stringent, moderately stringent, or low stringency hybridization conditions that are well known to one of skill in the art such as those described herein. Similarly, a nucleic acid molecule that can be used in the invention can be described as having a certain percent sequence identity to a nucleic acid disclosed herein by SEQ ID NO, GenBank and/or GI number or a nucleic acid molecule that hybridizes to a nucleic acid molecule that encodes an amino acid sequence disclosed herein by SEQ ID NO, GenBank and/or GI number. For example, the nucleic acid molecule can have at least 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98% or 99% sequence identity to a nucleic acid described herein.

Stringent hybridization refers to conditions under which hybridized polynucleotides are stable. As known to those of skill in the art, the stability of hybridized polynucleotides is reflected in the melting temperature (T_m) of the hybrids. In general, the stability of hybridized polynucleotides is a function of the salt concentration, for example, the sodium ion concentration and temperature. A hybridization reaction can be performed under conditions of lower stringency, followed by washes of varying, but higher, stringency. Reference to hybridization stringency relates to such washing conditions. Highly stringent hybridization includes conditions that permit hybridization of only those nucleic acid sequences that form stable hybridized polynucleotides in 0.018M NaCl at 65° C., for example, if a hybrid is not stable in 0.018M NaCl at 65° C., it will not be stable under high stringency conditions, as contemplated herein. High stringency conditions can be provided, for example, by hybridization in 50% formamide, 5× Denhart's solution, 5× SSPE, 0.2% SDS at 42° C., followed by washing in 0.1× SSPE, and 0.1% SDS at 65° C. Hybridization conditions other than highly stringent hybridization conditions can also be used to describe the nucleic acid sequences disclosed herein. For example, the phrase moderately stringent hybridization

refers to conditions equivalent to hybridization in 50% formamide, 5× Denhart's solution, 5× SSPE, 0.2% SDS at 42° C., followed by washing in 0.2× SSPE, 0.2% SDS, at 42° C. The phrase low stringency hybridization refers to conditions equivalent to hybridization in 10% formamide, 5× Denhart's solution, 6× SSPE, 0.2% SDS at 22° C., followed by washing in 1× SSPE, 0.2% SDS, at 37° C. Denhart's solution contains 1% Ficoll, 1% polyvinylpyrrolidone, and 1% bovine serum albumin (BSA). 20× SSPE (sodium chloride, sodium phosphate, ethylene diamine tetraacetic acid (EDTA)) contains 3M sodium chloride, 0.2M sodium phosphate, and 0.025 M (EDTA). Other suitable low, moderate and high stringency hybridization buffers and conditions are well known to those of skill in the art and are described, for example, in Sambrook et al., *Molecular Cloning: A Laboratory Manual*, Third Ed., Cold Spring Harbor Laboratory, New York (2001); and Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley and Sons, Baltimore, MD (1999).

A nucleic acid molecule encoding a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein of the invention can have at least a certain sequence identity to a nucleotide sequence disclosed herein. According, in some aspects of the invention, a nucleic acid molecule encoding a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein has a nucleotide sequence of at least 65% identity, at least 70% identity, at least 75% identity, at least 80% identity, at least 85% identity, at least 90% identity, at least 91% identity, at least 92% identity, at least 93% identity, at least 94% identity, at least 95% identity, at least 96% identity, at least 97% identity, at least 98% identity, or at least 99% identity to a nucleic acid disclosed herein by SEQ ID NO, GenBank and/or GI number or a nucleic acid molecule that hybridizes to a nucleic acid molecule that encodes an amino acid sequence disclosed herein by SEQ ID NO, GenBank and/or GI number.

Sequence identity (also known as homology or similarity) refers to sequence similarity between two nucleic acid molecules or between two polypeptides. Identity can be determined by comparing a position in each sequence, which may be aligned for purposes of comparison. When a position in the compared sequence is occupied by the same base or amino acid, then the molecules are identical at that position. A degree of identity between sequences is a function of the number of matching or homologous positions shared by the sequences. The alignment of two sequences to determine their percent sequence identity can be done using software programs known in the art, such as, for example, those described in Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley and Sons, Baltimore, MD. (1999). Preferably, default parameters are used for the alignment. One alignment program well known in the art that can be used is BLAST set to default parameters. In particular, programs are BLASTN and BLASTP, using the following default parameters: Genetic code=standard; filter=none; strand=both; cutoff=60; expect=10; Matrix=BLOSUM62; Descriptions=50 sequences; sort by=HIGH SCORE; Databases=non-redundant, GenBank+EMBL+DDBJ+PDB+GenBank CDS translations+SwissProtein+SPupdate+PIR. Details of these programs can be found at the National Center for Biotechnology Information.

Methods for constructing and testing the expression levels of a non-naturally occurring fatty alcohol, fatty aldehyde, fatty acid or isopropanol-producing host can be performed, for example, by recombinant and detection methods well

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known in the art. Such methods can be found described in, for example, Sambrook et al., *Molecular Cloning: A Laboratory Manual*, Third Ed., Cold Spring Harbor Laboratory, New York (2001); and Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley and Sons, Baltimore, MD. (1999).

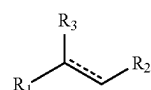
Exogenous nucleic acid sequences involved in a pathway for production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be introduced stably or transiently into a host cell using techniques well known in the art including, but not limited to, conjugation, electroporation, chemical transformation, transduction, transfection, and ultrasound transformation. For exogenous expression in *E. coli* or other prokaryotic cells, some nucleic acid sequences in the genes or cDNAs of eukaryotic nucleic acids can encode targeting signals such as an N-terminal mitochondrial or other targeting signal, which can be removed before transformation into prokaryotic host cells, if desired. For example, removal of a mitochondrial leader sequence led to increased expression in *E. coli* (Hoffmeister et al., *J. Biol. Chem.* 280:4329-4338 (2005)). For exogenous expression in yeast or other eukaryotic cells, genes can be expressed in the cytosol without the addition of leader sequence, or can be targeted to mitochondrion or other organelles, or targeted for secretion, by the addition of a suitable targeting sequence such as a mitochondrial targeting or secretion signal suitable for the host cells. Thus, it is understood that appropriate modifications to a nucleic acid sequence to remove or include a targeting sequence can be incorporated into an exogenous nucleic acid sequence to impart desirable properties. Furthermore, genes can be subjected to codon optimization with techniques well known in the art to achieve optimized expression of the proteins.

An expression vector or vectors can be constructed to include one or more fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathway encoding nucleic acids as exemplified herein operably linked to expression control sequences functional in the host organism. Expression vectors applicable for use in the microbial host organisms of the invention include, for example, plasmids, phage vectors, viral vectors, episomes and artificial chromosomes, including vectors and selection sequences or markers operable for stable integration into a host chromosome. Additionally, the expression vectors can include one or more selectable marker genes and appropriate expression control sequences. Selectable marker genes also can be included that, for example, provide resistance to antibiotics or toxins, complement auxotrophic deficiencies, or supply critical nutrients not in the culture media. Expression control sequences can include constitutive and inducible promoters, transcription enhancers, transcription terminators, and the like which are well known in the art. When two or more exogenous encoding nucleic acids are to be co-expressed, both nucleic acids can be inserted, for example, into a single expression vector or in separate expression vectors. For single vector expression, the encoding nucleic acids can be operationally linked to one common expression control sequence or linked to different expression control sequences, such as one inducible promoter and one constitutive promoter. The transformation of exogenous nucleic acid sequences involved in a metabolic or synthetic pathway can be confirmed using methods well known in the art. Such methods include, for example, nucleic acid analysis such as Northern blots or polymerase chain reaction (PCR) amplification of mRNA, or immunoblotting for expression of gene products, or other suitable analytical methods to test the expression of an introduced nucleic acid sequence or its corresponding gene

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product. It is understood by those skilled in the art that the exogenous nucleic acid is expressed in a sufficient amount to produce the desired product, and it is further understood that expression levels can be optimized to obtain sufficient expression using methods well known in the art and as disclosed herein.

In some embodiments, the invention provides a method for producing a compound of Formula (I):



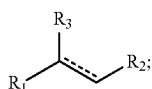
(I)

wherein R_1 is C_{1-24} linear alkyl; R_2 is CH_2OH , CHO , or $COOH$; R_3 is H , OH , or oxo ($=O$); and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four, comprising culturing a non-naturally occurring microbial organism of the invention under conditions and for a sufficient period of time to produce the compound of Formula (I).

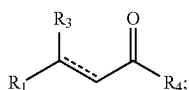
In some aspects of the invention, the microbial organism used in a method of the invention includes a non-naturally occurring having: (i) a formaldehyde fixation pathway; (ii) a formate assimilation pathway; and/or (iii) a methanol metabolic pathway as depicted in FIGS. 1 and 10, and a MI-FAE cycle or a MD-FAE cycle in combination with a termination pathway as depicted in FIGS. 2, 7 and 8, wherein said formaldehyde fixation pathway comprises: (1) 1B and 1C; (2) 1D; or (3) 1D and 1Z, wherein 1B is a 3-hexulose-6-phosphate synthase, wherein 1C is a 6-phospho-3-hexuloisomerase, wherein 1D is a dihydroxyacetone synthase, wherein 1Z is a fructose-6-phosphate aldolase, wherein said formate assimilation pathway comprises a pathway selected from: (4) 1E; (5) 1F, and 1G; (6) 1H, 1I, 1J, and 1K; (7) 1H, 1I, 1J, 1L, 1M, and 1N; (8) 1E, 1H, 1I, 1J, 1L, 1M, and 1N; (9) 1F, 1G, 1H, 1I, 1J, 1L, 1M, and 1N; (10) 1K, 1H, 1I, 1J, 1L, 1M, and 1N; and (11) 1H, 1I, 1J, 1O, and 1P, wherein 1E is a formate reductase, 1F is a formate ligase, a formate transferase, or a formate synthetase, wherein 1G is a formyl-CoA reductase, wherein 1H is a formyltetrahydrofolate synthetase, wherein 1I is a methylenetetrahydrofolate cyclohydrolase, wherein 1J is a methylenetetrahydrofolate dehydrogenase, wherein 1K is a formaldehyde-forming enzyme or spontaneous, wherein 1L is a glycine cleavage system, wherein 1M is a serine hydroxymethyltransferase, wherein 1N is a serine deaminase, wherein 1O is a methylenetetrahydrofolate reductase, wherein 1P is an acetyl-CoA synthase, wherein said methanol metabolic pathway comprises a pathway selected from: (12) 10J; (13) 10A, (14) 10A and 10B; (15) 10A, 10B and 10C; (16) 10J, 10K and 10C; (17) 10J, 10M, and 10N; (18) 10J and 10L; (19) 10J, 10L and 10G; (20) 10J, 10L, and 10I; (21) 10A, 10B, 10C, 10D, and 10E; (22) 10A, 10B, 10C, 10D, and 10F; (23) 10J, 10K, 10C, 10D, and 10E; (24) 10J, 10K, 10C, 10D, and 10F; (25) 10J, 10M, 10N, and 10O; (26) 10A, 10B, 10C, 10D, 10E, and 10G; (27) 10A, 10B, 10C, 10D, 10F, and 10G; (28) 10J, 10K, 10C, 10D, 10E, and 10G; (29) 10J, 10K, 10C, 10D, 10F, and 10G; (30) 10J, 10M, 10N, 10O, and 10G; (31) 10A, 10B, 10C, 10D, 10E, and 10I; (32) 10A, 10B, 10C, 10D, 10F, and 10I; (33) 10J, 10K, 10C, 10D, 10E, and 10I; (34) 10J, 10K, 10C, 10D, 10F, and 10I; and (35) 10J, 10M, 10N, 10O, and 10I, wherein 10A is a methanol methyltransferase, wherein 10B is a methylenetetrahydro-

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folate reductase, wherein 10C is a methylenetetrahydrofolate dehydrogenase, wherein 10D is a methenyltetrahydrofolate cyclohydrolase, wherein 10E is a formyltetrahydrofolate deformylase, wherein 10F is a formyltetrahydrofolate synthetase, wherein 10G is a formate hydrogen lyase, wherein 10I is a formate dehydrogenase, wherein 10J is a methanol dehydrogenase, wherein 10K is a formaldehyde activating enzyme or spontaneous, wherein 10L is a formaldehyde dehydrogenase, wherein 10M is a S-(hydroxymethyl)glutathione synthase or spontaneous, wherein 10N is a glutathione-dependent formaldehyde dehydrogenase, wherein 10O is a S-formylglutathione hydrolase, wherein the MI-FAE cycle includes one or more thiolase, one or more 3-oxoacyl-CoA reductase, one or more 3-hydroxyacyl-CoA dehydratase, and one or more enoyl-CoA reductase, wherein the MD-FAE cycle includes one or more elongase, one or more 3-oxoacyl-CoA reductase, one or more 3-hydroxyacyl-CoA dehydratase, and one or more enoyl-CoA reductase, wherein the termination pathway includes a pathway selected from: (36) 2H; (37) 2K and 2L; (38) 2E and 2N; (39) 2K, 2J, and 2N; (40) 2E; (41) 2K and 2J; (42) 2H and 2N; (43) 2K, 2L, and 2N; (44) 2E and 2F; (45) 2K, 2J, and 2F; (46) 2H, 2N, and 2F; (47) 2K, 2L, 2N, and 2F; (48) 2G; and (49) 2P, wherein 2E is an acyl-CoA reductase (aldehyde forming), wherein 2F is an alcohol dehydrogenase, wherein 2G is an acyl-CoA reductase (alcohol forming), wherein 2H is an acyl-CoA hydrolase, acyl-CoA transferase or acyl-CoA synthase, wherein 2J is an acyl-ACP reductase, wherein 2K is an acyl-CoA:ACP acyl-transferase, wherein 2L is a thioesterase, wherein 2N is an aldehyde dehydrogenase (acid forming) or a carboxylic acid reductase, wherein 2P is an acyl-ACP reductase (alcohol forming) wherein an enzyme of the formaldehyde fixation pathway, the formate assimilation pathway, the methanol metabolic pathway, the MI-FAE cycle, MD-FAE cycle or termination pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce a compound of Formula (I):



wherein R_1 is C_{1-24} linear alkyl; R_2 is CH_2OH , CHO , or $COOH$; R_3 is H , OH , or oxo ($=O$); and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four, wherein the substrate of each of said enzymes of the MI-FAE cycle, the MD-FAE cycle and the termination pathway are independently selected from a compound of Formula (II), malonyl-CoA, propionyl-CoA or acetyl-CoA:



wherein R_1 is C_{1-24} linear alkyl; R_3 is H , OH , or oxo ($=O$); R_4 is S-CoA, ACP, OH or H ; and $-----$ represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is

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four; wherein said one or more enzymes of the MI-FAE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said compound of Formula (I), wherein said one or more enzymes of the MD-FAE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said compound of Formula (I), and wherein said one or more enzymes of the termination pathway are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no less than the number of carbon atoms at R_1 of said compound of Formula (I).

In some aspects of the invention, the microbial organism used in a method of the invention includes a non-naturally occurring having: (i) a formaldehyde fixation pathway; (ii) a formate assimilation pathway; and/or (iii) a methanol metabolic pathway as depicted in FIGS. 1 and 10, and a FAACPE cycle in combination with a termination pathway as depicted in FIG. 12, wherein said formaldehyde fixation pathway comprises: (1) 1B and 1C; (2) 1D; (3) 1D and 1Z, wherein 1B is a 3-hexulose-6-phosphate synthase, wherein 1C is a 6-phospho-3-hexuloisomerase, wherein 1D is a dihydroxyacetone synthase, wherein 1Z is a fructose-6-phosphate aldolase, wherein said formate assimilation pathway comprises a pathway selected from: (4) 1E; (5) 1F, and 1G; (6) 1H, 1I, 1J, and 1K; (7) 1H, 1I, 1J, 1L, 1M, and 1N; (8) 1E, 1H, 1I, 1J, 1L, 1M, and 1N; (9) 1F, 1G, 1H, 1I, 1J, 1L, 1M, and 1N; (10) 1K, 1H, 1I, 1J, 1L, 1M, and 1N; and (11) 1H, 1I, 1J, 1O, and 1P, wherein 1E is a formate reductase, 1F is a formate ligase, a formate transferase, or a formate synthetase, wherein 1G is a formyl-CoA reductase, wherein 1H is a formyltetrahydrofolate synthetase, wherein 1I is a methenyltetrahydrofolate cyclohydrolase, wherein 1J is a methylenetetrahydrofolate dehydrogenase, wherein 1K is a formaldehyde-forming enzyme or spontaneous, wherein 1L is a glycine cleavage system, wherein 1M is a serine hydroxymethyltransferase, wherein 1N is a serine deaminase, wherein 1O is a methylenetetrahydrofolate reductase, wherein 1P is an acetyl-CoA synthase, wherein said methanol metabolic pathway comprises a pathway selected from: (12) 10J; (13) 10A, (14) 10A and 10B; (15) 10A, 10B and 10C; (16) 10J, 10K and 10C; (17) 10J, 10M, and 10N; (18) 10J and 10L; (19) 10J, 10L and 10G; (20) 10J, 10L, and 10I; (21) 10A, 10B, 10C, 10D, and 10E; (22) 10A, 10B, 10C, 10D, and 10F; (23) 10J, 10K, 10C, 10D, and 10E; (24) 10J, 10K, 10C, 10D, and 10F; (25) 10J, 10M, 10N, and 10O; (26) 10A, 10B, 10C, 10D, 10E, and 10G; (27) 10A, 10B, 10C, 10D, 10F, and 10G; (28) 10J, 10K, 10C, 10D, 10E, and 10G; (29) 10J, 10K, 10C, 10D, 10F, and 10G; (30) 10J, 10M, 10N, 10O, and 10G; (31) 10A, 10B, 10C, 10D, 10E, and 10I; (32) 10A, 10B, 10C, 10D, 10F, and 10I; (33) 10J, 10K, 10C, 10D, 10E, and 10I; (34) 10J, 10K, 10C, 10D, 10F, and 10I; and (35) 10J, 10M, 10N, 10O, and 10I, wherein 10A is a methanol methyltransferase, wherein 10B is a methylenetetrahydrofolate reductase, wherein 10C is a methylenetetrahydrofolate dehydrogenase, wherein 10D is a methenyltetrahydrofolate cyclohydrolase, wherein 10E is a formyltetrahydrofolate deformylase, wherein 10F is a formyltetrahydrofolate synthetase, wherein 10G is a formate hydrogen lyase, wherein 10I is a formate dehydrogenase, wherein 10J is a methanol dehydrogenase, wherein 10K is a formaldehyde activating enzyme or spontaneous, wherein 10L is a formaldehyde dehydrogenase, wherein 10M is a S-(hydroxymethyl)glutathione synthase or spontaneous, wherein 10N is a glutathione-dependent formaldehyde

In some aspects of the invention, the microbial organism used in a method of the invention that includes a FAACPE cycle in combination with a termination pathway as described herein, can further include a pathway for production of substrates for the FAACPE cycle, such as acetoacetyl-ACP or 3-oxovaleryl-ACP. Accordingly, in some

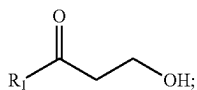
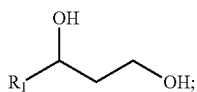
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embodiments, the microbial organism further comprises an acetoacetyl-ACP pathway of: (1) 12A, 12B, and 12C; or (2) 12A, 12B, and 12D, wherein 12A is an acetyl-CoA carboxylase, wherein 12B is malonyl-CoA ACP transacylase, wherein 12C is an acetoacetyl-ACP synthase, and wherein 12D is a β -ketoacyl-ACP synthase. In some embodiments, the microbial organism further comprises a 3-oxovaleryl-ACP pathway comprising an acetyl-CoA carboxylase, a malonyl-CoA ACP transacylase, and a β -ketoacyl-ACP synthase. In some aspects of the invention, an enzyme of the acetoacetyl-ACP pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce acetoacetyl-ACP wherein the acetoacetyl-ACP is a β -ketoacyl-ACP of the FAACPE cycle. In some aspects of the invention, an enzyme of the 3-oxovaleryl-ACP pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce 3-oxovaleryl-ACP, wherein the 3-oxovaleryl-ACP is a β -ketoacyl-ACP of the FAACPE cycle.

In some embodiments, the invention provides a method for producing a compound of Formula (I) wherein R_1 is C_{1-17} linear alkyl. In another aspect of the invention, the R_1 of the compound of Formula (I) is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} linear alkyl, C_{12} linear alkyl or C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

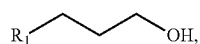
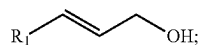
In some aspects of the invention, the microbial organism used in a method of the invention includes two, three, or four exogenous nucleic acids each encoding an enzyme of the MI-FAE cycle, the MD-FAE cycle, or the FAACPE cycle. In some aspects of the invention, the microbial organism includes two, three, or four exogenous nucleic acids each encoding an enzyme of the termination pathway. In some aspects of the invention, the microbial organism includes one, two, three, four, five, six, seven, or eight exogenous nucleic acids each encoding a formaldehyde fixation pathway enzyme, a formate assimilation pathway enzyme, or a methanol metabolic pathway enzyme. In some aspects of the invention, the microbial organism includes exogenous nucleic acids encoding each of the enzymes of at least one of the pathways selected from (1)-(49) for a microbial organism having a MI-FAE cycle or a MD-FAE cycle in combination with a termination pathway as depicted in FIGS. 1, 2, 7, 8 and 10. In some aspects of the invention, the microbial organism includes exogenous nucleic acids encoding each of the enzymes of at least one of the pathways selected from (1)-(44) for a microbial organism having a fatty acyl-ACP elongation (FAACPE) cycle in combination with a termination pathway as depicted in FIGS. 1, 10 and 12.

In some embodiments, the invention provides a method for producing a fatty alcohol selected from the Formulas (III)—(VI):



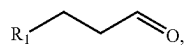
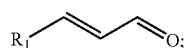
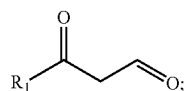
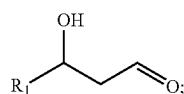
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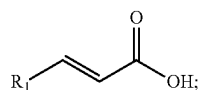
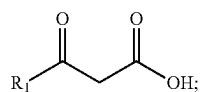
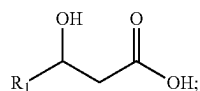
wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} linear alkyl, C_{12} linear alkyl or C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a method for producing a fatty aldehyde selected from the Formulas (VII)—(X):



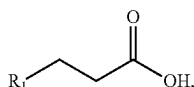
wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} linear alkyl, C_{12} linear alkyl or C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a method for producing a fatty acid selected from the Formulas (XI)—(XIV):



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(XIV)

wherein R_1 is C_{1-24} linear alkyl, or alternatively R_1 is C_{1-17} linear alkyl, or alternatively R_1 is C_{9-13} linear alkyl. In some aspects of the invention, R_1 is C_1 linear alkyl, C_2 linear alkyl, C_3 linear alkyl, C_4 linear alkyl, C_5 linear alkyl, C_6 linear alkyl, C_7 linear alkyl, C_8 linear alkyl, C_9 linear alkyl, C_{10} linear alkyl, C_{11} linear alkyl, C_{12} linear alkyl or C_{13} linear alkyl, C_{14} linear alkyl, C_{15} linear alkyl, C_{16} linear alkyl, C_{17} linear alkyl, C_{18} linear alkyl, C_{19} linear alkyl, C_{20} linear alkyl, C_{21} linear alkyl, C_{22} linear alkyl, C_{23} linear alkyl, or C_{24} linear alkyl.

In some embodiments, the invention provides a method for producing isopropanol comprising culturing the non-naturally occurring a microbial organism of the invention under conditions for a sufficient period of time to produce isopropanol.

In some aspects of the invention, the microbial organism used in a method of the invention includes a non-naturally occurring having: (i) a formaldehyde fixation pathway; (ii) a formate assimilation pathway; and/or (iii) a methanol metabolic pathway as depicted in FIGS. 1 and 10, and an isopropanol pathway as depicted in FIG. 11, wherein said formaldehyde fixation pathway comprises: (1) 1B and 1C; (2) 1D; or (3) 1D and 1Z, wherein 1B is a 3-hexulose-6-phosphate synthase, wherein 1C is a 6-phospho-3-hexulose isomerase, wherein 1D is a dihydroxyacetone synthase, wherein 1Z is a fructose-6-phosphate aldolase, wherein said formate assimilation pathway comprises a pathway selected from: (4) 1E; (5) 1F, and 1G; (6) 1H, 1I, 1J, and 1K; (7) 1H, 1I, 1J, 1L, 1M, and 1N; (8) 1E, 1H, 1I, 1J, 1L, 1M, and 1N; (9) 1F, 1G, 1H, 1I, 1J, 1L, 1M, and 1N; (10) 1K, 1H, 1I, 1J, 1L, 1M, and 1N; and (11) 1H, 1I, 1J, 1O, and 1P, wherein 1E is a formate reductase, 1F is a formate ligase, a formate transferase, or a formate synthetase, wherein 1G is a formyl-CoA reductase, wherein 1H is a formyltetrahydrofolate synthetase, wherein 1I is a methenyltetrahydrofolate cyclohydrolase, wherein 1J is a methylenetetrahydrofolate dehydrogenase, wherein 1K is a formaldehyde-forming enzyme or spontaneous, wherein 1L is a glycine cleavage system, wherein 1M is a serine hydroxymethyltransferase, wherein 1N is a serine deaminase, wherein 1O is a methylenetetrahydrofolate reductase, wherein 1P is an acetyl-CoA synthase, wherein said methanol metabolic pathway comprises a pathway selected from: (12) 10J; (13) 10A, (14) 10A and 10B; (15) 10A, 10B and 10C; (16) 10J, 10K and 10C; (17) 10J, 10M, and 10N; (18) 10J and 10L; (19) 10J, 10L and 10G; (20) 10J, 10L, and 10I; (21) 10A, 10B, 10C, 10D, and 10E; (22) 10A, 10B, 10C, 10D, and 10F; (23) 10J, 10K, 10C, 10D, and 10E; (24) 10J, 10K, 10C, 10D, and 10F; (25) 10J, 10M, 10N, and 10O; (26) 10A, 10B, 10C, 10D, 10E, and 10G; (27) 10A, 10B, 10C, 10D, 10F, and 10G; (28) 10J, 10K, 10C, 10D, 10E, and 10G; (29) 10J, 10K, 10C, 10D, 10F, and 10G; (30) 10J, 10M, 10N, 10O, and 10G; (31) 10A, 10B, 10C, 10D, 10E, and 10I; (32) 10A, 10B, 10C, 10D, 10F, and 10I; (33) 10J, 10K, 10C, 10D, 10E, and 10I; (34) 10J, 10K, 10C, 10D, 10F, and 10I; and (35) 10J, 10M, 10N, 10O, and 10I, wherein 10A is a methanol methyltransferase, wherein 10B is a methylenetetrahydrofolate reductase, wherein 10C is a methylenetetrahydrofolate dehydrogenase, wherein 10D is a methenyltetrahydrofolate cyclohydrolase, wherein 10E is a formyltetrahydrofolate deformylase,

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wherein 10F is a formyltetrahydrofolate synthetase, wherein 10G is a formate hydrogen lyase, wherein 10I is a formate dehydrogenase, wherein 10J is a methanol dehydrogenase, wherein 10K is a formaldehyde activating enzyme or spontaneous, wherein 10L is a formaldehyde dehydrogenase, wherein 10M is a S-(hydroxymethyl)glutathione synthase or spontaneous, wherein 10N is a glutathione-dependent formaldehyde dehydrogenase, wherein 10O is a S-formylglutathione hydrolase, wherein said isopropanol pathway comprises: (36) 11V, 11W, 11X, and 11Y; or (37) 11T, 11U, 11W, 11X, and 11Y, wherein 11T is an acetyl-CoA carboxylase, wherein 11U is an acetoacetyl-CoA synthase, wherein 11V is an acetyl-CoA:acetyl-CoA acyltransferase, wherein 11W is an acetoacetyl-CoA hydrolase, an acetoacetyl-CoA transferase, an acetoacetyl-CoA ligase, or a phosphotransacetoacetylase/acetoacetate kinase, wherein 11X is an acetoacetate decarboxylase, wherein 11Y is an acetone reductase or isopropanol dehydrogenase, wherein an enzyme of the formaldehyde fixation pathway, formate assimilation pathway, methanol metabolic pathway, or isopropanol pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce isopropanol. In some embodiments, the non-naturally occurring microbial organism described herein comprises an acetyl-CoA pathway that comprises 1T and 1V and a formaldehyde fixation pathway that comprises 1D and 1Z. In some embodiments, the non-naturally occurring microbial organism described herein comprises an acetyl-CoA pathway that comprises 1T and 1V and a formaldehyde fixation pathway comprises 1B and 1C.

In some aspects of the invention, the microbial organism used in a method of the invention has a combination of one or more pathways for generating substrates, intermediates and/or reducing equivalents that can be used with isopropanol pathways described herein for producing isopropanol of the invention. Accordingly, in some embodiments, the microbial organism has a formaldehyde fixation pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formate assimilation pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, and an isopropanol pathway. In some embodiments, the microbial organism has a methanol metabolic pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a methanol metabolic pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formate assimilation pathway, a methanol metabolic pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and an isopropanol pathway. In some embodiments, the microbial organism has a formaldehyde fixation pathway, a formate assimilation pathway, a methanol metabolic pathway and an isopropanol pathway.

In some aspects of the invention, the microbial organism used in a method of the invention includes two, three, four, five or six exogenous nucleic acids each encoding an enzyme of the isopropanol pathway. In some aspects of the invention, the microbial organism includes one, two, three, four, five, six, seven, or eight exogenous nucleic acids each encoding a formaldehyde fixation pathway enzyme, a formate assimilation pathway enzyme, or a methanol metabolic pathway enzyme. In some aspects of the invention, the microbial organism includes exogenous nucleic acids encoding each of the enzymes of at least one of the pathways selected from (1)-(37) for a microbial organism having an isopropanol pathway as depicted in FIGS. 1, 10 and 11.

In some aspects of the invention, the microbial organism used in a method of the invention having a formate assimilation

lation pathway further includes wherein the formate assimilation pathway comprises: (1) 1Q; (2) 1R, and 1S; (3) 1Y and 1Q; or (4) 1Y, 1R, and 1S, wherein 1Q is a pyruvate formate lyase, wherein 1R is a pyruvate dehydrogenase, a pyruvate ferredoxin oxidoreductase, or a pyruvate:NADP+ oxidoreductase, wherein 1S is a formate dehydrogenase wherein 10Y is a glyceraldehydes-3-phosphate dehydrogenase or an enzyme of lower glycolysis. In addition to a glyceraldehyde-3-phosphate dehydrogenase, lower glycolysis includes a phosphoglycerate kinase, a phosphoglyceromutase, an enolase, a pyruvate kinase or a PTS-dependant substrate import. Accordingly, in some embodiments, the formate assimilation pathway comprising 1Y includes an enzyme selected from a phosphoglycerate kinase, a phosphoglyceromutase, an enolase, a pyruvate kinase and a PTS-dependant substrate import.

In some aspects of the invention, the microbial organism used in a method of the invention includes a methanol oxidation pathway. Such a pathway can include at least one exogenous nucleic acid encoding a methanol oxidation pathway enzyme expressed in a sufficient amount to produce formaldehyde in the presence of methanol. An exemplary methanol oxidation pathway enzyme is a methanol dehydrogenase. Accordingly, in some aspects, the microbial organism used in the method of the invention includes a non-naturally occurring having at least one exogenous nucleic acid encoding a methanol dehydrogenase expressed in a sufficient amount to produce formaldehyde in the presence of methanol.

In some aspects of the invention, the microbial organism used in a method of the invention includes one or more enzymes for generating reducing equivalents. For example, the microbial organism can further include a hydrogenase and/or a carbon monoxide dehydrogenase. In some aspects, the microbial organism used in the method of the invention includes a non-naturally occurring having an exogenous nucleic acid encoding the hydrogenase or the carbon monoxide dehydrogenase.

In some aspects of the invention, the microbial organism used in a method of the invention includes a non-naturally occurring having at least one exogenous nucleic acid that is a heterologous nucleic acid. Accordingly, in some embodiments, the at least one exogenous nucleic acid encoding a formaldehyde fixation pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a formate assimilation pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a methanol metabolic pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a MI-FAE cycle enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a MD-FAE cycle enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a FAACPE cycle enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a termination pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding an acetoacetyl-ACP pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a 3-oxovalery-ACP pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding an isopropanol path-

way enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a methanol oxidation pathway enzyme described herein is a heterologous nucleic acid. In some embodiments, the at least one exogenous nucleic acid encoding a hydrogenase or a carbon monoxide dehydrogenase is a heterologous nucleic acid.

In some embodiments, the method for producing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol described herein includes using a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes an acetyl-CoA pathway and at least one exogenous nucleic acid encoding an acetyl-CoA pathway enzyme expressed in a sufficient amount to produce or enhance carbon flux through acetyl-CoA, wherein the acetyl-CoA pathway includes a pathway shown in FIG. 1, 3, 4, 5 or 6 selected from: (1) 3A and 3B; (2) 3A, 3C, and 3D; (3) 3H; (4) 3G and 3D; (5) 3E, 3F and 3B; (6) 3E and 3I; (7) 3J, 3F and 3B; (8) 3J and 3I; (9) 4A, 4B, and 4C; (10) 4A, 4B, 4J, 4K, and 4D; (11) 4A, 4B, 4G, and 4D; (12) 4A, 4F, and 4D; (13) 4N, 4H, 4B and 4C; (14) 4N, 4H, 4B, 4J, 4K, and 4D; (15) 4N, 4H, 4B, 4G, and 4D; (16) 4N, 4H, 4F, and 4D; (17) 4L, 4M, 4B and 4C; (18) 4L, 4M, 4B, 4J, 4K, and 4D; (19) 4L, 4M, 4B, 4G, and 4D; (20) 4L, 4M, 4F, and 4D; (21) 5A, 5B, 5D, 5H, 5I, and 5J; (22) 5A, 5B, 5E, 5F, 5H, 5I, and 5J; (23) 5A, 5B, 5E, 5K, 5L, 5H, 5I, and 5J; (24) 5A, 5C, 5D, 5H, and 5J; (25) 5A, 5C, 5E, 5F, 5H, and 5J; (26) 5A, 5C, 5E, 5K, 5L, 5H, and 5J; (27) 6A, 6B, 6D, and 6G; (28) 6A, 6B, 6E, 6F, and 6G; (29) 6A, 6B, 6E, 6K, 6L, and 6G; (30) 6A, 6C, and 6D; (31) 6A, 6C, 6E, and 6F; (32) 6A, 6C, 6E, 6K, and 6L; (33) 1T and 1V; (34) 1T, 1W, and 1X; (35) 1U and 1V; and (36) 1U, 1W, and 1X, wherein 3A is a pyruvate oxidase (acetate-forming), wherein 3B is an acetyl-CoA synthetase, an acetyl-CoA ligase or an acetyl-CoA transference, wherein 3C is an acetate kinase, wherein 3D is a phosphotransacetylase, wherein 3E is a pyruvate decarboxylase, wherein 3F is an acetaldehyde dehydrogenase, wherein 3G is a pyruvate oxidase (acetyl-phosphate forming), wherein 3H is a pyruvate dehydrogenase, a pyruvate:ferredoxin oxidoreductase, a pyruvate:NAD(P)H oxidoreductase or a pyruvate formate lyase, wherein 3I is an acetaldehyde dehydrogenase (acylating), wherein 3J is a threonine aldolase, wherein 4A is a phosphoenolpyruvate (PEP) carboxylase or a PEP carboxykinase, wherein 4B is an oxaloacetate decarboxylase, wherein 4C is a malonate semialdehyde dehydrogenase (acetylating), wherein 4D is an acetyl-CoA carboxylase or a malonyl-CoA decarboxylase, wherein 4F is an oxaloacetate dehydrogenase or an oxaloacetate oxidoreductase, wherein 4G is a malonate semialdehyde dehydrogenase (acylating), wherein 4H is a pyruvate carboxylase, wherein 4J is a malonate semialdehyde dehydrogenase, wherein 4K is a malonyl-CoA synthetase or a malonyl-CoA transference, wherein 4L is a malic enzyme, wherein 4M is a malate dehydrogenase or a malate oxidoreductase, wherein 4N is a pyruvate kinase or a PEP phosphatase, wherein 5A is a citrate synthase, wherein 5B is a citrate transporter, wherein 5C is a citrate/malate transporter, wherein 5D is an ATP citrate lyase, wherein 5E is a citrate lyase, wherein 5F is an acetyl-CoA synthetase or an acetyl-CoA transferase, wherein 5H is a cytosolic malate dehydrogenase, wherein 5I is a malate transporter, wherein 5J is a mitochondrial malate dehydrogenase, wherein 5K is an acetate kinase, wherein 5L is a phosphotransacetylase, wherein 6A is a citrate synthase, wherein 6B is a citrate transporter, wherein 6C is a citrate/oxaloacetate transporter, wherein 6D is an ATP citrate lyase, wherein 6E is a citrate lyase, wherein 6F is an acetyl-CoA synthetase or an acetyl-

CoA transference, wherein 6G is an oxaloacetate transporter, wherein 6K is an acetate kinase, and wherein 6L is a phosphotransacetylase, wherein 1T is a fructose-6-phosphate phosphoketolase, wherein 1U is a xylulose-5-phosphate phosphoketolase, wherein 1V is a phosphotransacetylase, wherein 1W is an acetate kinase, wherein 1X is an acetyl-CoA transference, an acetyl-CoA synthetase, or an acetyl-CoA ligase.

In some aspects, the microbial organism used in a method of the invention includes two, three, four, five, six, seven or eight exogenous nucleic acids each encoding an acetyl-CoA pathway enzyme. In some aspects, the microbial organism used in a method of the invention includes exogenous nucleic acids encoding each of the acetyl-CoA pathway enzymes of at least one of the pathways selected from (1)-(36).

In some aspects, the microbial organism used in a method of the invention includes further includes a propionyl-CoA pathway and at least one exogenous nucleic acid encoding a propionyl-CoA pathway enzyme expressed in a sufficient amount to produce propionyl-CoA, wherein the propionyl-CoA pathway includes a pathway shown in FIG. 22. For example, in some embodiments, the propionyl-CoA pathway comprises a pathway selected from: (1) 22A, 22E, 22F, 22G, 22I, 22J, 22K and 22L; (2) 22A, 22E, 22F, 22G, 22H, 22J, 22K and 22L; (3) 22B, 22E, 22F, 22G, 22I, 22J, 22K and 22L; (4) 22B, 22E, 22F, 22G, 22H, 22J, 22K and 22L; (5) 22C, 22D, 22E, 22F, 22G, 22I, 22J, 22K and 22L; and (6) 22C, 22D, 22E, 22F, 22G, 22H, 22J, 22K and 22L, wherein 22A is a PEP carboxykinase, wherein 22B is a PEP carboxylase, wherein 22C is a Pyruvate kinase, wherein 22D is a Pyruvate carboxylase, wherein 22E is a Malate dehydrogenase, wherein 22F is a Fumarase, wherein 22G is a Fumarate reductase, wherein 22H is a Succinyl-CoA synthetase, wherein 22I is a Succinyl-CoA:3-ketoacid-CoA transferase, wherein 22J is a Methylmalonyl-CoA mutase, wherein 22K is a Methylmalonyl-CoA epimerase, and wherein 22L is a Methylmalonyl-CoA decarboxylase.

In some embodiments, the invention provides a method for producing isopropanol, wherein the method includes culturing a non-naturally occurring microbial organism described herein under conditions and for a sufficient period of time to produce isopropanol, wherein the microbial organism comprises an acetyl-CoA pathway, wherein said acetyl-CoA pathway comprises a pathway selected from: (1) 1T and 1V; (2) 1T, 1W, and 1X; (3) 1U and 1V; (4) 1U, 1W, and 1X; wherein 1T is a fructose-6-phosphate phosphoketolase, wherein 1U is a xylulose-5-phosphate phosphoketolase, wherein 1V is a phosphotransacetylase, wherein 1W is an acetate kinase, wherein 1X is an acetyl-CoA transferase, an acetyl-CoA synthetase, or an acetyl-CoA ligase, wherein said non-naturally occurring microbial organism further comprises a pathway capable of producing isopropanol and an exogenous nucleic acid encoding an isopropanol pathway enzyme expressed in a sufficient amount to produce isopropanol, wherein said isopropanol pathway comprises a pathway selected from: (1) 11V, 11W, 11X, and 11Y; or (2) 11T, 11U, 11W, 11X, and 11Y, wherein 11T is an acetyl-CoA carboxylase, wherein 11U is an acetoacetyl-CoA synthase, wherein 11V is an acetyl-CoA:acetyl-CoA acyltransferase, wherein 11W is an acetoacetyl-CoA hydrolase, an acetoacetyl-CoA transferase, an acetoacetyl-CoA ligase, or a phosphotransacetoacetylase/acetoacetate kinase, wherein 11X is an acetoacetate decarboxylase, wherein 11Y is an acetone reductase or isopropanol dehydrogenase.

In other aspects, the invention further provides methods for producing elevated or enhanced yields of biosynthetic products such as a fatty alcohol, fatty aldehyde, fatty acid and/or isopropanol.

The methods for producing enhanced yields of a fatty alcohol, fatty aldehyde, fatty acid and/or isopropanol described herein include using a non-naturally occurring microbial organisms having one or more of the various pathway configurations employing a methanol dehydrogenase for methanol oxidation, a formaldehyde fixation pathway, and/or a phosphoketolase for directing the carbon from methanol into acetyl-CoA and other desired products via formaldehyde fixation as described previously. The methods include using a non-naturally occurring microbial organism of the invention having one or more of the various different methanol oxidation and formaldehyde fixation configurations exemplified previously and below engineered in conjunction with any or each of the various methanol oxidation, formaldehyde fixation, formate reutilization, fatty alcohol, fatty aldehyde, fatty acid and/or isopropanol pathway exemplified previously. Accordingly, the methods of the invention can use a microbial organism having one or more of the metabolic modifications exemplified previously and also below that increase biosynthetic product yields over, for example, endogenous methanol utilization pathways because they further focus methanol derived carbon into the assimilation pathways described herein, decrease inefficient use of methanol carbon through competing methanol utilization and/or formaldehyde fixation pathways and/or increase the production of reducing equivalents.

In some aspects, the methods of the invention can use microbial organisms containing or engineered to contain one or more of the various configurations of metabolic modifications disclosed herein for enhancing product yields via methanol derived carbon include enhancing methanol oxidation and production of reducing equivalents using either an endogenous NADH dependent methanol dehydrogenase, an exogenous NADH dependent methanol dehydrogenase, both an endogenous NADH dependent methanol dehydrogenase and exogenous NADH dependent methanol dehydrogenase alone or in combination with one or more metabolic modifications that attenuate, for example, DHA synthase and/or AOX. In addition, other metabolic modifications as exemplified previously and further below that reduce carbon flux away from methanol oxidation and formaldehyde fixation also can be included, alone or in combination, with the methanol oxidation and formaldehyde fixation pathway configurations disclosed herein that enhance carbon flux into product precursors such as acetyl-CoA and, therefore, enhance product yields.

Accordingly, in some embodiments, the microbial organisms used in a method of the invention can include one or more of any of the above and/or below metabolic modifications to a methanol utilization pathway and/or formaldehyde assimilation pathway configurations for enhancing product yields can be combined with any one or more, including all of the previously described methanol oxidation, formaldehyde fixation, formate reutilization, fatty alcohol, fatty aldehyde, fatty acid and/or isopropanol pathway to enhance the yield and/or production of a product such as any of the fatty alcohol, fatty aldehyde, fatty acids and/or isopropanol described herein.

Given the teachings and guidance provided herein, both prokaryotic and eukaryotic microbial organisms engineered to have methanol oxidation and/or formaldehyde fixation pathway configurations for enhancing product yields can be used in the methods of the invention. As exemplified herein

and well known in the art, those skilled in the art will know which organism to select for a particular application. For example, with respect to eukaryotic microbial host organisms, those skilled in the art will know that yeasts and other eukaryotic microorganisms exhibit certain characteristics distinct from prokaryotic microbial organisms. When such characteristics are desirable, one skilled in the art can choose to use such eukaryotic microbial organisms having one or more of the various different methanol oxidation and formaldehyde fixation configurations exemplified herein for enhancing product yields in a method of the invention. Such characteristics have been described previously.

In some embodiments, the microbial organism used in a method of the invention and having a methanol oxidation and/or formaldehyde assimilation pathway configurations described herein for enhancing product yields can include, for example, a NADH-dependent methanol dehydrogenase (MeDH), one or more formaldehyde assimilation pathways and/or one or more phosphoketolases.

In one embodiment, the methods of the invention use microbial organisms that have cytosolic expression of one or more methanol oxidation and/or formaldehyde assimilation pathways. As described previously, exemplary pathways for converting cytosolic formaldehyde into glycolytic intermediates are shown in FIG. 1. Such pathways include methanol oxidation via expression of a cytosolic NADH dependent methanol dehydrogenase, formaldehyde fixation via expression of cytosolic DHA synthase, both methanol oxidation via expression of an cytosolic NADH dependent methanol dehydrogenase and formaldehyde fixation via expression of cytosolic DHA synthase alone or together with the metabolic modifications exemplified previously and also below that attenuate less beneficial methanol oxidation and/or formaldehyde fixation pathways. Such attenuating metabolic modifications include, for example, attenuation of alcohol oxidase, attenuation of DHA kinase and/or attenuation of DHA synthase (e.g. when ribulose-5-phosphate (Ru5P) pathway for formaldehyde fixation is utilized).

In another embodiment, conversion of cytosolic formaldehyde into glycolytic intermediates can occur via expression of a cytosolic 3-hexulose-6-phosphate (3-Hu6P) synthase. Thus, exemplary pathways that can be engineered into a microbial organism used in a method of the invention can include methanol oxidation via expression of a cytosolic NADH dependent methanol dehydrogenase, formaldehyde fixation via expression of cytosolic 3-Hu6P synthase, both methanol oxidation via expression of an cytosolic NADH dependent dehydrogenase and formaldehyde fixation via expression of cytosolic 3-Hu6P synthase alone or together with the metabolic modifications exemplified previously and also below that attenuate less beneficial methanol oxidation and/or formaldehyde fixation pathways. Such attenuating metabolic modifications include, for example, attenuation of alcohol oxidase, attenuation of DHA kinase and/or attenuation of DHA synthase (e.g. when ribulose-5-phosphate (Ru5P) pathway for formaldehyde fixation is utilized).

In yet another embodiment, the methods of the invention use microbial organisms that have cytosolic expression of one or more methanol oxidation and/or formaldehyde assimilation pathways. The formaldehyde assimilation pathways can include both assimilation through cytosolic DHA synthase and 3-Hu6P synthase. In this specific embodiment, such pathways include methanol oxidation via expression of a cytosolic NADH dependent methanol dehydrogenase, formaldehyde fixation via expression of cytosolic DHA synthase and 3-Hu6P synthase, both methanol oxidation via expression of an cytosolic NADH dependent dehydrogenase

and formaldehyde fixation via expression of cytosolic DHA synthase and 3-Hu6P synthase alone or together with the metabolic modifications exemplified previously and also below that attenuate less beneficial methanol oxidation and/or formaldehyde fixation pathways. Such attenuating metabolic modifications include, for example, attenuation of alcohol oxidase, attenuation of DHA kinase and/or attenuation of DHA synthase (e.g. when ribulose-5-phosphate (Ru5P) pathway for formaldehyde fixation is utilized).

In some embodiments, the method for producing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol described herein includes using a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes attenuation of one or more endogenous enzymes, which enhances carbon flux through acetyl-CoA. For example, in some aspects, the endogenous enzyme can be selected from DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase, DHA synthase or any combination thereof. Accordingly, in some aspects, the attenuation is of the endogenous enzyme DHA kinase. In some aspects, the attenuation is of the endogenous enzyme methanol oxidase. In some aspects, the attenuation is of the endogenous enzyme PQQ-dependent methanol dehydrogenase. In some aspects, the attenuation is of the endogenous enzyme DHA synthase. The invention also provides a method wherein the microbial organism used includes attenuation of any combination of two or three endogenous enzymes described herein. For example, a microbial organism can include attenuation of DHA kinase and DHA synthase, or alternatively methanol oxidase and PQQ-dependent methanol dehydrogenase, or alternatively DHA kinase, methanol oxidase, and PQQ-dependent methanol dehydrogenase, or alternatively DHA kinase, methanol oxidase, and DHA synthase. The invention also provides a method wherein the microbial organism used includes attenuation of all endogenous enzymes described herein. For example, in some aspects, a microbial organism includes attenuation of DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase and DHA synthase.

In some embodiments, the method for producing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol described herein includes using a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes attenuation of one or more endogenous enzymes of a competing formaldehyde assimilation or dissimilation pathway. Examples of these endogenous enzymes are disclosed in FIG. 1 and described in Example XXIII. It is understood that a person skilled in the art would be able to readily identify enzymes of such competing pathways. Competing pathways can be dependent upon the host microbial organism and/or the exogenous nucleic acid introduced into the microbial organism as described herein. Accordingly, in some aspects of the invention, the method includes a microbial organism having attenuation of one, two, three, four, five, six, seven, eight, nine, ten or more endogenous enzymes of a competing formaldehyde assimilation or dissimilation pathway.

In some embodiments, the method for producing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol described herein includes using a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes a gene disruption of one or more endogenous nucleic acids encoding enzymes, which enhances carbon flux through acetyl-CoA. For example, in some aspects, the endogenous enzyme can be selected from DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase, DHA synthase or any combination thereof.

According, in some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme DHA kinase. In some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme methanol oxidase. In some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme PQQ-dependent methanol dehydrogenase. In some aspects, the gene disruption is of an endogenous nucleic acid encoding the enzyme DHA synthase. The invention also provides a method wherein the microbial organism used includes the gene disruption of any combination of two or three nucleic acids encoding endogenous enzymes described herein. For example, a microbial organism of the invention can include a gene disruption of DHA kinase and DHA synthase, or alternatively methanol oxidase and PQQ-dependent methanol dehydrogenase, or alternatively DHA kinase, methanol oxidase, and DHA synthase. The invention also provides a method wherein the microbial organism used includes wherein all endogenous nucleic acids encoding enzymes described herein are disrupted. For example, in some aspects, a microbial organism described herein includes disruption of DHA kinase, methanol oxidase, PQQ-dependent methanol dehydrogenase and DHA synthase.

In some embodiments, the method for producing a fatty alcohol, fatty aldehyde, fatty acid or isopropanol described herein includes using a non-naturally occurring microbial organism as described herein, wherein the microbial organism further includes a gene disruption of one or more endogenous enzymes of a competing formaldehyde assimilation or dissimilation pathway. Examples of these endogenous enzymes are disclosed in FIG. 1 and described in Example XXIII. It is understood that a person skilled in the art would be able to readily identify enzymes of such competing pathways. Competing pathways can be dependent upon the host microbial organism and/or the exogenous nucleic acid introduced into the microbial organism as described herein. Accordingly, in some aspects of the invention, the microbial organism used in the method includes a gene disruption of one, two, three, four, five, six, seven, eight, nine, ten or more endogenous nucleic acids encoding enzymes of a competing formaldehyde assimilation or dissimilation pathway.

Suitable purification and/or assays to test for the production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be performed using well known methods. Suitable replicates such as triplicate cultures can be grown for each engineered strain to be tested. For example, product and byproduct formation in the engineered production host can be monitored. The final product and intermediates, and other organic compounds, can be analyzed by methods such as HPLC (High Performance Liquid Chromatography), GC-MS (Gas Chromatography-Mass Spectroscopy) and LC-MS (Liquid Chromatography-Mass Spectroscopy) or other suitable analytical methods using routine procedures well known in the art. The release of product in the fermentation broth can also be tested with the culture supernatant. Byproducts and residual glucose can be quantified by HPLC using, for example, a refractive index detector for glucose and alcohols, and a UV detector for organic acids (Lin et al., *Biotechnol. Bioeng.* 90:775-779 (2005)), or other suitable assay and detection methods well known in the art. The individual enzyme or protein activities from the exogenous DNA sequences can also be assayed using methods well known in the art.

The fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be separated from other components in the culture using

a variety of methods well known in the art. Such separation methods include, for example, extraction procedures as well as methods that include continuous liquid-liquid extraction, pervaporation, membrane filtration, membrane separation, reverse osmosis, electrodialysis, distillation, crystallization, centrifugation, extractive filtration, ion exchange chromatography, size exclusion chromatography, adsorption chromatography, and ultrafiltration. All of the above methods are well known in the art.

Any of the non-naturally occurring microbial organisms described herein can be cultured to produce and/or secrete the biosynthetic products of the invention. For example, the fatty alcohol, fatty aldehyde, fatty acid or isopropanol producers can be cultured for the biosynthetic production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. Accordingly, in some embodiments, the invention provides culture medium having the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate described herein. In some aspects, the culture medium can also be separated from the non-naturally occurring microbial organisms of the invention that produced the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate. Methods for separating a microbial organism from culture medium are well known in the art. Exemplary methods include filtration, flocculation, precipitation, centrifugation, sedimentation, and the like.

For the production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, the recombinant strains are cultured in a medium with carbon source and other essential nutrients. It is sometimes desirable and can be highly desirable to maintain anaerobic conditions in the fermenter to reduce the cost of the overall process. Such conditions can be obtained, for example, by first sparging the medium with nitrogen and then sealing the flasks with a septum and crimp-cap. For strains where growth is not observed anaerobically, microaerobic or substantially anaerobic conditions can be applied by perforating the septum with a small hole for limited aeration. Exemplary anaerobic conditions have been described previously and are well-known in the art. Exemplary aerobic and anaerobic conditions are described, for example, in United State publication 2009/0047719, filed Aug. 10, 2007. Fermentations can be performed in a batch, fed-batch or continuous manner, as disclosed herein. Fermentations can also be conducted in two phases, if desired. The first phase can be aerobic to allow for high growth and therefore high productivity, followed by an anaerobic phase of high fatty alcohol, fatty aldehyde, fatty acid or isopropanol yields.

If desired, the pH of the medium can be maintained at a desired pH, in particular neutral pH, such as a pH of around 7 by addition of a base, such as NaOH or other bases, or acid, as needed to maintain the culture medium at a desirable pH. The growth rate can be determined by measuring optical density using a spectrophotometer (600 nm), and the glucose uptake rate by monitoring carbon source depletion over time.

The growth medium, can include, for example, any carbohydrate source which can supply a source of carbon to the non-naturally occurring microbial organism of the invention. Such sources include, for example, sugars such as glucose, xylose, arabinose, galactose, mannose, fructose, sucrose and starch; or glycerol, alone as the sole source of carbon or in combination with other carbon sources described herein or known in the art. In one embodiment, the carbon source is a sugar. In one embodiment, the carbon source is a sugar-containing biomass. In some embodiments, the sugar is glucose. In one embodiment, the sugar is xylose.

In another embodiment, the sugar is arabinose. In one embodiment, the sugar is galactose. In another embodiment, the sugar is fructose. In other embodiments, the sugar is sucrose. In one embodiment, the sugar is starch. In certain embodiments, the carbon source is glycerol. In some embodiments, the carbon source is crude glycerol. In one embodiment, the carbon source is crude glycerol without treatment. In other embodiments, the carbon source is glycerol and glucose. In another embodiment, the carbon source is methanol and glycerol. In one embodiment, the carbon source is carbon dioxide. In one embodiment, the carbon source is formate. In one embodiment, the carbon source is methane. In one embodiment, the carbon source is methanol. In certain embodiments, methanol is used alone as the sole source of carbon or in combination with other carbon sources described herein or known in the art. In a specific embodiment, the methanol is the only (sole) carbon source. In one embodiment, the carbon source is chemoelectro-generated carbon (see, e.g., Liao et al. (2012) *Science* 335:1596). In one embodiment, the chemoelectro-generated carbon is methanol. In one embodiment, the chemoelectro-generated carbon is formate. In one embodiment, the chemoelectro-generated carbon is formate and methanol. In one embodiment, the carbon source is a carbohydrate and methanol. In one embodiment, the carbon source is a sugar and methanol. In another embodiment, the carbon source is a sugar and glycerol. In other embodiments, the carbon source is a sugar and crude glycerol. In yet other embodiments, the carbon source is a sugar and crude glycerol without treatment. In one embodiment, the carbon source is a sugar-containing biomass and methanol. In another embodiment, the carbon source is a sugar-containing biomass and glycerol. In other embodiments, the carbon source is a sugar-containing biomass and crude glycerol. In yet other embodiments, the carbon source is a sugar-containing biomass and crude glycerol without treatment. In some embodiments, the carbon source is a sugar-containing biomass, methanol and a carbohydrate. Other sources of carbohydrate include, for example, renewable feedstocks and biomass. Exemplary types of biomasses that can be used as feedstocks in the methods provided herein include cellulosic biomass, hemicellulosic biomass and lignin feedstocks or portions of feedstocks. Such biomass feedstocks contain, for example, carbohydrate substrates useful as carbon sources such as glucose, xylose, arabinose, galactose, mannose, fructose and starch. Given the teachings and guidance provided herein, those skilled in the art will understand that renewable feedstocks and biomass other than those exemplified above also can be used for culturing the microbial organisms provided herein for the production of succinate and other pathway intermediates.

In one embodiment, the carbon source is glycerol. In certain embodiments, the glycerol carbon source is crude glycerol or crude glycerol without further treatment. In a further embodiment, the carbon source comprises glycerol or crude glycerol, and also sugar or a sugar-containing biomass, such as glucose. In a specific embodiment, the concentration of glycerol in the fermentation broth is maintained by feeding crude glycerol, or a mixture of crude glycerol and sugar (e.g., glucose). In certain embodiments, sugar is provided for sufficient strain growth. In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of from 200:1 to 1:200. In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of from 100:1 to 1:100. In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of

glycerol to sugar of from 100:1 to 5:1. In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of from 50:1 to 5:1. In certain embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 100:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 90:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 80:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 70:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 60:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 50:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 40:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 30:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 20:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 10:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 5:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 2:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:1. In certain embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:100. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:90. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:80. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:70. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:60. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:50. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:40. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:30. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:20. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:10. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:5. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of glycerol to sugar of 1:2. In certain embodiments of the ratios provided above, the sugar is a sugar-containing biomass. In certain other embodiments of the ratios provided above, the glycerol is a crude glycerol or a crude glycerol without further treatment. In other embodiments of the ratios provided above, the sugar is a sugar-containing biomass, and the glycerol is a crude glycerol or a crude glycerol without further treatment.

Crude glycerol can be a by-product produced in the production of biodiesel, and can be used for fermentation without any further treatment. Biodiesel production methods include (1) a chemical method wherein the glycerol-group of vegetable oils or animal oils is substituted by low-carbon alcohols such as methanol or ethanol to produce a corresponding fatty acid methyl esters or fatty acid ethyl esters by

transesterification in the presence of acidic or basic catalysts; (2) a biological method where biological enzymes or cells are used to catalyze transesterification reaction and the corresponding fatty acid methyl esters or fatty acid ethyl esters are produced; and (3) a supercritical method, wherein transesterification reaction is carried out in a supercritical solvent system without any catalysts. The chemical composition of crude glycerol can vary with the process used to produce biodiesel, the transesterification efficiency, recovery efficiency of the biodiesel, other impurities in the feedstock, and whether methanol and catalysts were recovered. For example, the chemical compositions of eleven crude glycerol collected from seven Australian biodiesel producers reported that glycerol content ranged between 38% and 96%, with some samples including more than 14% methanol and 29% ash. In certain embodiments, the crude glycerol comprises from 5% to 99% glycerol. In some embodiments, the crude glycerol comprises from 10% to 90% glycerol. In some embodiments, the crude glycerol comprises from 10% to 80% glycerol. In some embodiments, the crude glycerol comprises from 10% to 70% glycerol. In some embodiments, the crude glycerol comprises from 10% to 60% glycerol. In some embodiments, the crude glycerol comprises from 10% to 50% glycerol. In some embodiments, the crude glycerol comprises from 10% to 40% glycerol. In some embodiments, the crude glycerol comprises from 10% to 30% glycerol. In some embodiments, the crude glycerol comprises from 10% to 20% glycerol. In some embodiments, the crude glycerol comprises from 80% to 90% glycerol. In some embodiments, the crude glycerol comprises from 70% to 90% glycerol. In some embodiments, the crude glycerol comprises from 60% to 90% glycerol. In some embodiments, the crude glycerol comprises from 50% to 90% glycerol. In some embodiments, the crude glycerol comprises from 40% to 90% glycerol. In some embodiments, the crude glycerol comprises from 30% to 90% glycerol. In some embodiments, the crude glycerol comprises from 20% to 90% glycerol. In some embodiments, the crude glycerol comprises from 20% to 40% glycerol. In some embodiments, the crude glycerol comprises from 40% to 60% glycerol. In some embodiments, the crude glycerol comprises from 60% to 80% glycerol. In some embodiments, the crude glycerol comprises from 50% to 70% glycerol. In one embodiment, the glycerol comprises 5% glycerol. In one embodiment, the glycerol comprises 10% glycerol. In one embodiment, the glycerol comprises 15% glycerol. In one embodiment, the glycerol comprises 20% glycerol. In one embodiment, the glycerol comprises 25% glycerol. In one embodiment, the glycerol comprises 30% glycerol. In one embodiment, the glycerol comprises 35% glycerol. In one embodiment, the glycerol comprises 40% glycerol. In one embodiment, the glycerol comprises 45% glycerol. In one embodiment, the glycerol comprises 50% glycerol. In one embodiment, the glycerol comprises 55% glycerol. In one embodiment, the glycerol comprises 60% glycerol. In one embodiment, the glycerol comprises 65% glycerol. In one embodiment, the glycerol comprises 70% glycerol. In one embodiment, the glycerol comprises 75% glycerol. In one embodiment, the glycerol comprises 80% glycerol. In one embodiment, the glycerol comprises 85% glycerol. In one embodiment, the glycerol comprises 90% glycerol. In one embodiment, the glycerol comprises 95% glycerol. In one embodiment, the glycerol comprises 99% glycerol.

In one embodiment, the carbon source is methanol or formate. In certain embodiments, methanol is used as a carbon source in a formaldehyde fixation pathway provided

herein. In one embodiment, the carbon source is methanol or formate. In other embodiments, formate is used as a carbon source in a formaldehyde fixation pathway provided herein. In specific embodiments, methanol is used as a carbon source in a methanol oxidation pathway provided herein, either alone or in combination with the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathways provided herein. In one embodiment, the carbon source is methanol. In another embodiment, the carbon source is formate.

In one embodiment, the carbon source comprises methanol, and sugar (e.g., glucose) or a sugar-containing biomass. In another embodiment, the carbon source comprises formate, and sugar (e.g., glucose) or a sugar-containing biomass. In one embodiment, the carbon source comprises methanol, formate, and sugar (e.g., glucose) or a sugar-containing biomass. In specific embodiments, the methanol or formate, or both, in the fermentation feed is provided as a mixture with sugar (e.g., glucose) or sugar-comprising biomass. In certain embodiments, sugar is provided for sufficient strain growth.

In certain embodiments, the carbon source comprises methanol and a sugar (e.g., glucose). In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of from 200:1 to 1:200. In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of from 100:1 to 1:100. In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of from 100:1 to 5:1. In some embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of from 50:1 to 5:1. In certain embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 100:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 90:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 80:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 70:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 60:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 50:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 40:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 30:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 20:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 10:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 5:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 2:1. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 1:1. In certain embodiments, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 1:100. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 1:90. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 1:80. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 1:70. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol to sugar of 1:60. In one

of 1:50. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol and formate to sugar of 1:40. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol and formate to sugar of 1:30. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol and formate to sugar of 1:20. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol and formate to sugar of 1:10. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol and formate to sugar of 1:5. In one embodiment, the sugar (e.g., glucose) is provided at a molar concentration ratio of methanol and formate to sugar of 1:2. In certain embodiments of the ratios provided above, the sugar is a sugar-containing biomass.

In addition to renewable feedstocks such as those exemplified above, the fatty alcohol, fatty aldehyde, fatty acid or isopropanol producing microbial organisms of the invention also can be modified for growth on syngas as its source of carbon. In this specific embodiment, one or more proteins or enzymes are expressed in the fatty alcohol, fatty aldehyde, fatty acid or isopropanol producing organisms to provide a metabolic pathway for utilization of syngas or other gaseous carbon source.

Synthesis gas, also known as syngas or producer gas, is the major product of gasification of coal and of carbonaceous materials such as biomass materials, including agricultural crops and residues. Syngas is a mixture primarily of H₂ and CO and can be obtained from the gasification of any organic feedstock, including but not limited to coal, coal oil, natural gas, biomass, and waste organic matter. Gasification is generally carried out under a high fuel to oxygen ratio. Although largely H₂ and CO, syngas can also include CO₂ and other gases in smaller quantities. Thus, synthesis gas provides a cost effective source of gaseous carbon such as CO and, additionally, CO₂.

Accordingly, given the teachings and guidance provided herein, those skilled in the art will understand that a non-naturally occurring microbial organism can be produced that secretes the biosynthesized compounds of the invention when grown on a carbon source such as a carbohydrate. Such compounds include, for example, fatty alcohol, fatty aldehyde, fatty acid or isopropanol and any of the intermediate metabolites in the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway. All that is required is to engineer in one or more of the required enzyme or protein activities to achieve biosynthesis of the desired compound or intermediate including, for example, inclusion of some or all of the fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthetic pathways. Accordingly, the invention provides a non-naturally occurring microbial organism that produces and/or secretes fatty alcohol, fatty aldehyde, fatty acid or isopropanol when grown on a carbohydrate or other carbon source and produces and/or secretes any of the intermediate metabolites shown in the fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway when grown on a carbohydrate or other carbon source. The fatty alcohol, fatty aldehyde, fatty acid or isopropanol producing microbial organisms of the invention can initiate synthesis from an intermediate, for example, a 3-ketoacyl-CoA, a 3-hydroxyacyl-CoA, an enoyl-CoA, an acyl-CoA, an acyl-ACP, acetate, acetaldehyde, acetyl-phosphate, oxaloacetate, malate, malonate semialdehyde, malonate, malonyl-CoA, acetyl-CoA, or citrate.

The non-naturally occurring microbial organisms of the invention are constructed using methods well known in the art as exemplified herein to exogenously express at least one

nucleic acid encoding a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein in sufficient amounts to produce fatty alcohol, fatty aldehyde, fatty acid or isopropanol. It is understood that the microbial organisms of the invention are cultured under conditions sufficient to produce fatty alcohol, fatty aldehyde, fatty acid or isopropanol. Following the teachings and guidance provided herein, the non-naturally occurring microbial organisms of the invention can achieve biosynthesis of fatty alcohol, fatty aldehyde, fatty acid or isopropanol resulting in intracellular concentrations between about 0.1-200 mM or more. Generally, the intracellular concentration of fatty alcohol, fatty aldehyde, fatty acid or isopropanol is between about 3-150 mM, particularly between about 5-125 mM and more particularly between about 8-100 mM, including about 10 mM, 20 mM, 50 mM, 80 mM, or more. Intracellular concentrations between and above each of these exemplary ranges also can be achieved from the non-naturally occurring microbial organisms of the invention.

In some embodiments, culture conditions include anaerobic or substantially anaerobic growth or maintenance conditions. Exemplary anaerobic conditions have been described previously and are well known in the art. Exemplary anaerobic conditions for fermentation processes are described herein and are described, for example, in U.S. publication 2009/0047719, filed Aug. 10, 2007. Any of these conditions can be employed with the non-naturally occurring microbial organisms as well as other anaerobic conditions well known in the art. Under such anaerobic or substantially anaerobic conditions, the fatty alcohol, fatty aldehyde, fatty acid or isopropanol producers can synthesize fatty alcohol, fatty aldehyde, fatty acid or isopropanol at intracellular concentrations of 5-10 mM or more as well as all other concentrations exemplified herein. It is understood that, even though the above description refers to intracellular concentrations, fatty alcohol, fatty aldehyde, fatty acid or isopropanol producing microbial organisms can produce fatty alcohol, fatty aldehyde, fatty acid or isopropanol intracellularly and/or secrete the product into the culture medium.

Exemplary fermentation processes include, but are not limited to, fed-batch fermentation and batch separation; fed-batch fermentation and continuous separation; and continuous fermentation and continuous separation. In an exemplary batch fermentation protocol, the production organism is grown in a suitably sized bioreactor sparged with an appropriate gas. Under anaerobic conditions, the culture is sparged with an inert gas or combination of gases, for example, nitrogen, N₂/CO₂ mixture, argon, helium, and the like. As the cells grow and utilize the carbon source, additional carbon source(s) and/or other nutrients are fed into the bioreactor at a rate approximately balancing consumption of the carbon source and/or nutrients. The temperature of the bioreactor is maintained at a desired temperature, generally in the range of 22-37 degrees C., but the temperature can be maintained at a higher or lower temperature depending on the growth characteristics of the production organism and/or desired conditions for the fermentation process. Growth continues for a desired period of time to achieve desired characteristics of the culture in the fermenter, for example, cell density, product concentration, and the like. In a batch fermentation process, the time period for the fermentation is generally in the range of several hours to several days, for example, 8 to 24 hours, or 1, 2, 3, 4 or 5 days, or up to a week, depending on the desired culture conditions. The pH can be controlled or not, as desired, in which case a culture in which pH is not controlled will

typically decrease to pH 3-6 by the end of the run. Upon completion of the cultivation period, the fermenter contents can be passed through a cell separation unit, for example, a centrifuge, filtration unit, and the like, to remove cells and cell debris. In the case where the desired product is expressed intracellularly, the cells can be lysed or disrupted enzymatically or chemically prior to or after separation of cells from the fermentation broth, as desired, in order to release additional product. The fermentation broth can be transferred to a product separations unit. Isolation of product occurs by standard separations procedures employed in the art to separate a desired product from dilute aqueous solutions. Such methods include, but are not limited to, liquid-liquid extraction using a water immiscible organic solvent (e.g., toluene or other suitable solvents, including but not limited to diethyl ether, ethyl acetate, tetrahydrofuran (THF), methylene chloride, chloroform, benzene, pentane, hexane, heptane, petroleum ether, methyl tertiary butyl ether (MTBE), dioxane, dimethylformamide (DMF), dimethyl sulfoxide (DMSO), and the like) to provide an organic solution of the product, if appropriate, standard distillation methods, and the like, depending on the chemical characteristics of the product of the fermentation process.

In an exemplary fully continuous fermentation protocol, the production organism is generally first grown up in batch mode in order to achieve a desired cell density. When the carbon source and/or other nutrients are exhausted, feed medium of the same composition is supplied continuously at a desired rate, and fermentation liquid is withdrawn at the same rate. Under such conditions, the product concentration in the bioreactor generally remains constant, as well as the cell density. The temperature of the fermenter is maintained at a desired temperature, as discussed above. During the continuous fermentation phase, it is generally desirable to maintain a suitable pH range for optimized production. The pH can be monitored and maintained using routine methods, including the addition of suitable acids or bases to maintain a desired pH range. The bioreactor is operated continuously for extended periods of time, generally at least one week to several weeks and up to one month, or longer, as appropriate and desired. The fermentation liquid and/or culture is monitored periodically, including sampling up to every day, as desired, to assure consistency of product concentration and/or cell density. In continuous mode, fermenter contents are constantly removed as new feed medium is supplied. The exit stream, containing cells, medium, and product, are generally subjected to a continuous product separations procedure, with or without removing cells and cell debris, as desired. Continuous separations methods employed in the art can be used to separate the product from dilute aqueous solutions, including but not limited to continuous liquid-liquid extraction using a water immiscible organic solvent (e.g., toluene or other suitable solvents, including but not limited to diethyl ether, ethyl acetate, tetrahydrofuran (THF), methylene chloride, chloroform, benzene, pentane, hexane, heptane, petroleum ether, methyl tertiary butyl ether (MTBE), dioxane, dimethylformamide (DMF), dimethyl sulfoxide (DMSO), and the like), standard continuous distillation methods, and the like, or other methods well known in the art.

In addition to the culturing and fermentation conditions disclosed herein, growth condition for achieving biosynthesis of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can include the addition of an osmoprotectant to the culturing conditions. In certain embodiments, the non-naturally occurring microbial organisms of the invention can be sustained, cultured or fermented as described herein in the

presence of an osmoprotectant. Briefly, an osmoprotectant refers to a compound that acts as an osmolyte and helps a microbial organism as described herein survive osmotic stress. Osmoprotectants include, but are not limited to, betaines, amino acids, and the sugar trehalose. Non-limiting examples of such are glycine betaine, praline betaine, dimethylthetin, dimethylsulfoniopropionate, 3-dimethylsulfonio-2-methylpropionate, pipercolic acid, dimethylsulfonioacetate, choline, L-carnitine and ectoine. In one aspect, the osmoprotectant is glycine betaine. It is understood to one of ordinary skill in the art that the amount and type of osmoprotectant suitable for protecting a microbial organism described herein from osmotic stress will depend on the microbial organism used. The amount of osmoprotectant in the culturing conditions can be, for example, no more than about 0.1 mM, no more than about 0.5 mM, no more than about 1.0 mM, no more than about 1.5 mM, no more than about 2.0 mM, no more than about 2.5 mM, no more than about 3.0 mM, no more than about 5.0 mM, no more than about 7.0 mM, no more than about 10 mM, no more than about 50 mM, no more than about 100 mM or no more than about 500 mM.

In some embodiments, the carbon feedstock and other cellular uptake sources such as phosphate, ammonia, sulfate, chloride and other halogens can be chosen to alter the isotopic distribution of the atoms present in fatty alcohol, fatty aldehyde, fatty acid or isopropanol or any fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate. The various carbon feedstock and other uptake sources enumerated above will be referred to herein, collectively, as "uptake sources." Uptake sources can provide isotopic enrichment for any atom present in the product fatty alcohol, fatty aldehyde, fatty acid or isopropanol or fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate, or for side products generated in reactions diverging away from a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway. Isotopic enrichment can be achieved for any target atom including, for example, carbon, hydrogen, oxygen, nitrogen, sulfur, phosphorus, chloride or other halogens.

In some embodiments, the uptake sources can be selected to alter the carbon-12, carbon-13, and carbon-14 ratios. In some embodiments, the uptake sources can be selected to alter the oxygen-16, oxygen-17, and oxygen-18 ratios. In some embodiments, the uptake sources can be selected to alter the hydrogen, deuterium, and tritium ratios. In some embodiments, the uptake sources can be selected to alter the nitrogen-14 and nitrogen-15 ratios. In some embodiments, the uptake sources can be selected to alter the sulfur-32, sulfur-33, sulfur-34, and sulfur-35 ratios. In some embodiments, the uptake sources can be selected to alter the phosphorus-31, phosphorus-32, and phosphorus-33 ratios. In some embodiments, the uptake sources can be selected to alter the chlorine-35, chlorine-36, and chlorine-37 ratios.

In some embodiments, the isotopic ratio of a target atom can be varied to a desired ratio by selecting one or more uptake sources. An uptake source can be derived from a natural source, as found in nature, or from a man-made source, and one skilled in the art can select a natural source, a man-made source, or a combination thereof, to achieve a desired isotopic ratio of a target atom. An example of a man-made uptake source includes, for example, an uptake source that is at least partially derived from a chemical synthetic reaction. Such isotopically enriched uptake sources can be purchased commercially or prepared in the laboratory and/or optionally mixed with a natural source of the uptake source to achieve a desired isotopic ratio. In some

embodiments, a target atom isotopic ratio of an uptake source can be achieved by selecting a desired origin of the uptake source as found in nature. For example, as discussed herein, a natural source can be a biobased derived from or synthesized by a biological organism or a source such as petroleum-based products or the atmosphere. In some such embodiments, a source of carbon, for example, can be selected from a fossil fuel-derived carbon source, which can be relatively depleted of carbon-14, or an environmental or atmospheric carbon source, such as CO₂, which can possess a larger amount of carbon-14 than its petroleum-derived counterpart.

The unstable carbon isotope carbon-14 or radiocarbon makes up for roughly 1 in 10¹² carbon atoms in the earth's atmosphere and has a half-life of about 5700 years. The stock of carbon is replenished in the upper atmosphere by a nuclear reaction involving cosmic rays and ordinary nitrogen (¹⁴N). Fossil fuels contain no carbon-14, as it decayed long ago. Burning of fossil fuels lowers the atmospheric carbon-14 fraction, the so-called "Suess effect".

Methods of determining the isotopic ratios of atoms in a compound are well known to those skilled in the art. Isotopic enrichment is readily assessed by mass spectrometry using techniques known in the art such as accelerated mass spectrometry (AMS), Stable Isotope Ratio Mass Spectrometry (SIRMS) and Site-Specific Natural Isotopic Fractionation by Nuclear Magnetic Resonance (SNIF-NMR). Such mass spectral techniques can be integrated with separation techniques such as liquid chromatography (LC), high performance liquid chromatography (HPLC) and/or gas chromatography, and the like.

In the case of carbon, ASTM D6866 was developed in the United States as a standardized analytical method for determining the biobased content of solid, liquid, and gaseous samples using radiocarbon dating by the American Society for Testing and Materials (ASTM) International. The standard is based on the use of radiocarbon dating for the determination of a product's biobased content. ASTM D6866 was first published in 2004, and the current active version of the standard is ASTM D6866-11 (effective Apr. 1, 2011). Radiocarbon dating techniques are well known to those skilled in the art, including those described herein.

The biobased content of a compound is estimated by the ratio of carbon-14 (¹⁴C) to carbon-12 (¹²C). Specifically, the Fraction Modern (Fm) is computed from the expression: $Fm = (S-B)/(M-B)$, where B, S and M represent the ¹⁴C/¹²C ratios of the blank, the sample and the modern reference, respectively. Fraction Modern is a measurement of the deviation of the ¹⁴C/¹²C ratio of a sample from "Modern." Modern is defined as 95% of the radiocarbon concentration (in AD 1950) of National Bureau of Standards (NBS) Oxalic Acid I (i.e., standard reference materials (SRM) 4990b) normalized to $\delta^{13}C_{VPDB} = -19$ per mil (Olsson, *The use of Oxalic acid as a Standard*, in, *Radiocarbon Variations and Absolute Chronology*, Nobel Symposium, 12th Proc., John Wiley & Sons, New York (1970)). Mass spectrometry results, for example, measured by ASM, are calculated using the internationally agreed upon definition of 0.95 times the specific activity of NBS Oxalic Acid I (SRM 4990b) normalized to $\delta^{13}C_{VPDB} = -19$ per mil. This is equivalent to an absolute (AD 1950) ¹⁴C/¹²C ratio of $1.176 \pm 0.010 \times 10^{-12}$ (Karlen et al., *Arkiv Geofysik*, 4:465-471 (1968)). The standard calculations take into account the differential uptake of one isotope with respect to another, for example, the preferential uptake in biological systems of C¹² over C¹³ over C¹⁴, and these corrections are reflected as a Fm corrected for $\delta^{13}C$.

An oxalic acid standard (SRM 4990b or HOx 1) was made from a crop of 1955 sugar beet. Although there were 1000 lbs made, this oxalic acid standard is no longer commercially available. The Oxalic Acid II standard (HOx 2; N.I.S.T designation SRM 4990 C) was made from a crop of 1977 French beet molasses. In the early 1980's, a group of 12 laboratories measured the ratios of the two standards. The ratio of the activity of Oxalic acid II to 1 is 1.2933 ± 0.001 (the weighted mean). The isotopic ratio of HOx II is -17.8 per mil. ASTM D6866-11 suggests use of the available Oxalic Acid II standard SRM 4990 C (Hox2) for the modern standard (see discussion of original vs. currently available oxalic acid standards in Mann, *Radiocarbon*, 25(2):519-527 (1983)). A Fm=0% represents the entire lack of carbon-14 atoms in a material, thus indicating a fossil (for example, petroleum based) carbon source. A Fm=100%, after correction for the post-1950 injection of carbon-14 into the atmosphere from nuclear bomb testing, indicates an entirely modern carbon source. As described herein, such a "modern" source includes biobased sources.

As described in ASTM D6866, the percent modern carbon (@MC) can be greater than 100% because of the continuing but diminishing effects of the 1950s nuclear testing programs, which resulted in a considerable enrichment of carbon-14 in the atmosphere as described in ASTM D6866-11. Because all sample carbon-14 activities are referenced to a "pre-bomb" standard, and because nearly all new biobased products are produced in a post-bomb environment, all pMC values (after correction for isotopic fraction) must be multiplied by 0.95 (as of 2010) to better reflect the true biobased content of the sample. A biobased content that is greater than 103% suggests that either an analytical error has occurred, or that the source of biobased carbon is more than several years old.

ASTM D6866 quantifies the biobased content relative to the material's total organic content and does not consider the inorganic carbon and other non-carbon containing substances present. For example, a product that is 50% starch-based material and 50% water would be considered to have a Biobased Content=100% (50% organic content that is 100% biobased) based on ASTM D6866. In another example, a product that is 50% starch-based material, 25% petroleum-based, and 25% water would have a Biobased Content=66.7% (75% organic content but only 50% of the product is biobased). In another example, a product that is 50% organic carbon and is a petroleum-based product would be considered to have a Biobased Content=0% (50% organic carbon but from fossil sources). Thus, based on the well known methods and known standards for determining the biobased content of a compound or material, one skilled in the art can readily determine the biobased content and/or prepared downstream products that utilize of the invention having a desired biobased content.

Applications of carbon-14 dating techniques to quantify bio-based content of materials are known in the art (Currie et al., *Nuclear Instruments and Methods in Physics Research B*, 172:281-287 (2000)). For example, carbon-14 dating has been used to quantify bio-based content in terephthalate-containing materials (Colonna et al., *Green Chemistry*, 13:2543-2548 (2011)). Notably, polypropylene terephthalate (PPT) polymers derived from renewable 1,3-propanediol and petroleum-derived terephthalic acid resulted in Fm values near 30% (i.e., since 3/11 of the polymeric carbon derives from renewable 1,3-propanediol and 8/11 from the fossil end member terephthalic acid) (Currie et al., *supra*, 2000). In contrast, polybutylene terephthalate polymer derived from both renewable 1,4-butanediol and renewable

terephthalic acid resulted in bio-based content exceeding 90% (Colonna et al., supra, 2011).

Accordingly, in some embodiments, the present invention provides fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate that has a carbon-12, carbon-13, and carbon-14 ratio that reflects an atmospheric carbon, also referred to as environmental carbon, uptake source. For example, in some aspects the fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate can have an Fm value of at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, at least 98% or as much as 100%. In some such embodiments, the uptake source is CO₂. In some embodiments, the present invention provides fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate that has a carbon-12, carbon-13, and carbon-14 ratio that reflects petroleum-based carbon uptake source. In this aspect, the fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate can have an Fm value of less than 95%, less than 90%, less than 85%, less than 80%, less than 75%, less than 70%, less than 65%, less than 60%, less than 55%, less than 50%, less than 45%, less than 40%, less than 35%, less than 30%, less than 25%, less than 20%, less than 15%, less than 10%, less than 5%, less than 2% or less than 1%. In some embodiments, the present invention provides fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate that has a carbon-12, carbon-13, and carbon-14 ratio that is obtained by a combination of an atmospheric carbon uptake source with a petroleum-based uptake source. Using such a combination of uptake sources is one way by which the carbon-12, carbon-13, and carbon-14 ratio can be varied, and the respective ratios would reflect the proportions of the uptake sources.

Further, the present invention relates to the biologically produced fatty alcohol, fatty aldehyde, fatty acid or isopropanol or fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate as disclosed herein, and to the products derived therefrom, wherein the fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate has a carbon-12, carbon-13, and carbon-14 isotope ratio of about the same value as the CO₂ that occurs in the environment. For example, in some aspects the invention provides bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol intermediate having a carbon-12 versus carbon-13 versus carbon-14 isotope ratio of about the same value as the CO₂ that occurs in the environment, or any of the other ratios disclosed herein. It is understood, as disclosed herein, that a product can have a carbon-12 versus carbon-13 versus carbon-14 isotope ratio of about the same value as the CO₂ that occurs in the environment, or any of the ratios disclosed herein, wherein the product is generated from bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate as disclosed herein, wherein the bioderived product is chemically modified to generate a final product. Methods of chemically modifying a bioderived product of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, or an intermediate thereof,

to generate a desired product are well known to those skilled in the art, as described herein. The invention further provides biofuels, chemicals, polymers, surfactants, soaps, detergents, shampoos, lubricating oil additives, fragrances, flavor materials or acrylates having a carbon-12 versus carbon-13 versus carbon-14 isotope ratio of about the same value as the CO₂ that occurs in the environment, wherein the biofuels, chemicals, polymers, surfactants, soaps, detergents, shampoos, lubricating oil additives, fragrances, flavor materials or acrylates are generated directly from or in combination with bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol or a bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway intermediate as disclosed herein.

Fatty alcohol, fatty aldehyde or fatty acid is a chemical used in commercial and industrial applications. Non-limiting examples of such applications include production of biofuels, chemicals, polymers, surfactants, soaps, detergents, shampoos, lubricating oil additives, fragrances, flavor materials and acrylates. Accordingly, in some embodiments, the invention provides biobased biofuels, chemicals, polymers, surfactants, soaps, detergents, shampoos, lubricating oil additives, fragrances, flavor materials and acrylates comprising one or more bioderived fatty alcohol, fatty aldehyde or fatty acid or bioderived fatty alcohol, fatty aldehyde or fatty acid pathway intermediate produced by a non-naturally occurring microorganism of the invention or produced using a method disclosed herein.

Isopropanol is a chemical used in commercial and industrial applications. Non-limiting examples of such applications include production of solvents, including rubbing alcohol. As a solvent, isopropanol is found in products such as paints, lacquers, thinners, inks, adhesives, general-purpose cleaners, disinfectants, cosmetics, toiletries, de-icers, and pharmaceuticals. Low-grade isopropanol is also used in motor oils. Isopropanol is also used as a chemical intermediate for the production of isopropylamines, isopropylethers, and isopropyl esters. Isopropanol can potentially be dehydrated to form propylene, a polymer precursor. Accordingly, in some embodiments, the invention provides biobased solvents, rubbing alcohol, paints, lacquers, thinners, inks, adhesives, general-purpose cleaners, disinfectants, cosmetics, toiletries, de-icers, pharmaceuticals, motor oils, isopropylamines, isopropylethers, isopropyl esters, propylene and polymers, comprising bioderived isopropanol produced by a non-naturally occurring microorganism of the invention or produced using a method disclosed herein.

As used herein, the term "bioderived" means derived from or synthesized by a biological organism and can be considered a renewable resource since it can be generated by a biological organism. Such a biological organism, in particular the microbial organisms of the invention disclosed herein, can utilize feedstock or biomass, such as, sugars or carbohydrates obtained from an agricultural, plant, bacterial, or animal source. Alternatively, the biological organism can utilize atmospheric carbon. As used herein, the term "biobased" means a product as described above that is composed, in whole or in part, of a bioderived compound of the invention. A biobased or bioderived product is in contrast to a petroleum derived product, wherein such a product is derived from or synthesized from petroleum or a petrochemical feedstock.

In some embodiments, the invention provides a biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate comprising bioderived fatty alcohol, fatty aldehyde or fatty acid or bioderived fatty alcohol, fatty aldehyde or fatty acid

pathway intermediate, wherein the bioderived fatty alcohol, fatty aldehyde or fatty acid or bioderived fatty alcohol, fatty aldehyde or fatty acid pathway intermediate includes all or part of the fatty alcohol, fatty aldehyde or fatty acid or fatty alcohol, fatty aldehyde or fatty acid pathway intermediate used in the production of a biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate. For example, the final biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate can contain the bioderived fatty alcohol, fatty aldehyde or fatty acid, fatty alcohol, fatty aldehyde or fatty acid pathway intermediate, or a portion thereof that is the result of the manufacturing of the biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate. Such manufacturing can include chemically reacting the bioderived fatty alcohol, fatty aldehyde or fatty acid, or bioderived fatty alcohol, fatty aldehyde or fatty acid pathway intermediate (e.g. chemical conversion, chemical functionalization, chemical coupling, oxidation, reduction, polymerization, copolymerization and the like) with itself or another compound in a reaction that produces the final biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate. Thus, in some aspects, the invention provides a biobased biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate comprising at least 2%, at least 3%, at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or 100% bioderived fatty alcohol, fatty aldehyde or fatty acid or bioderived fatty alcohol, fatty aldehyde or fatty acid pathway intermediate as disclosed herein. In some aspects, when the product is a biobased polymer that includes or is obtained from a bioderived fatty alcohol, fatty aldehyde or fatty acid, or or fatty alcohol, fatty aldehyde or fatty acid pathway intermediate described herein, the biobased polymer can be molded using methods well known in the art. Accordingly, in some embodiments, provided herein is a molded product comprising the biobased polymer described herein.

Additionally, in some embodiments, the invention provides a composition having a bioderived fatty alcohol, fatty aldehyde or fatty acid, or fatty alcohol, fatty aldehyde or fatty acid pathway intermediate disclosed herein and a compound other than the bioderived fatty alcohol, fatty aldehyde or fatty acid or fatty alcohol, fatty aldehyde or fatty acid pathway intermediate. For example, in some aspects, the invention provides a biobased biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate wherein the fatty alcohol, fatty aldehyde or fatty acid or fatty alcohol, fatty aldehyde or fatty acid pathway intermediate used in its production is a combination of bioderived and petroleum derived fatty alcohol, fatty aldehyde or fatty acid or fatty alcohol, fatty aldehyde or fatty acid pathway intermediate. For example, a biobased biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate can be produced using 50% bioderived fatty alcohol, fatty aldehyde or fatty acid and 50% petroleum derived fatty alcohol, fatty aldehyde or fatty acid or other desired ratios such as 60%/40%, 70%/30%, 80%/20%, 90%/10%, 95%/5%, 100%/0%, 40%/60%, 30%/70%, 20%/80%, 10%/90% of bioderived/petroleum derived precursors, so long as at least a portion of the

product comprises a bioderived product produced by the microbial organisms disclosed herein. It is understood that methods for producing a biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating oil additive, fragrance, flavor material or acrylate using the bioderived fatty alcohol, fatty aldehyde or fatty acid or bioderived fatty alcohol, fatty aldehyde or fatty acid pathway intermediate of the invention are well known in the art.

In some embodiments, the invention provides a solvent, a paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or a polymer comprising bioderived isopropanol or bioderived isopropanol pathway intermediate, wherein the bioderived isopropanol or bioderived isopropanol pathway intermediate includes all or part of the isopropanol or isopropanol pathway intermediate used in the production of a solvent, a paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or a polymer. For example, the final solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer can contain the bioderived isopropanol, isopropanol pathway intermediate, or a portion thereof that is the result of the manufacturing of a solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer. Such manufacturing can include chemically reacting the bioderived isopropanol or bioderived isopropanol pathway intermediate (e.g. chemical conversion, chemical functionalization, chemical coupling, oxidation, reduction, polymerization, copolymerization and the like) into the final solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer. Thus, in some aspects, the invention provides a biobased solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer comprising at least 2%, at least 3%, at least 5%, at least 10%, at least 15%, at least 20%, at least 25%, at least 30%, at least 35%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 98% or 100% bioderived isopropanol or bioderived isopropanol pathway intermediate as disclosed herein.

Additionally, in some embodiments, the invention provides a composition having a bioderived isopropanol or isopropanol pathway intermediate disclosed herein and a compound other than the bioderived isopropanol or isopropanol pathway intermediate. For example, in some aspects, the invention provides a biobased solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer wherein the isopropanol or isopropanol pathway intermediate used in its production is a combination of bioderived and petroleum derived isopropanol or isopropanol pathway intermediate. For example, a biobased solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer can be produced using 50% bioderived isopropanol and 50% petroleum derived isopropanol or other desired ratios such as 60%/40%, 70%/30%, 80%/20%, 90%/10%, 95%/5%,

100%/0%, 40%/60%, 30%/70%, 20%/80%, 10%/90% of bioderived/petroleum derived precursors, so long as at least a portion of the product comprises a bioderived product produced by the microbial organisms disclosed herein. It is understood that methods for producing a solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer using the bioderived isopropanol or bioderived isopropanol pathway intermediate of the invention are well known in the art.

The invention further provides a composition comprising bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol, and a compound other than the bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. The compound other than the bioderived product can be a cellular portion, for example, a trace amount of a cellular portion of, or can be fermentation broth or culture medium, or a purified or partially purified fraction thereof produced in the presence of, a non-naturally occurring microbial organism of the invention having a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway. The composition can comprise, for example, a reduced level of a byproduct when produced by an organism having reduced byproduct formation, as disclosed herein. The composition can comprise, for example, bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol, or a cell lysate or culture supernatant of a microbial organism of the invention.

In certain embodiments, provided herein is a composition comprising a bioderived fatty alcohol, fatty aldehyde or fatty acid provided herein, for example, a bioderived fatty alcohol, fatty aldehyde or fatty acid produced by culturing a non-naturally occurring microbial organism having a formaldehyde fixation pathway, a formate assimilation pathway and/or a methanol metabolic pathway, and a MI-FAE cycle, a MD-FAE cycle, and/or a FAACPE cycle in combination with a termination pathway, as provided herein. In some embodiments, the composition further comprises a compound other than said bioderived fatty alcohol, fatty aldehyde or fatty acid. In certain embodiments, the compound other than said bioderived fatty alcohol, fatty aldehyde or fatty acid is a trace amount of a cellular portion of a non-naturally occurring microbial organism having a formaldehyde fixation pathway, a formate assimilation pathway and/or a methanol metabolic pathway, and a MI-FAE cycle, a MD-FAE cycle, and/or a FAACPE cycle in combination with a termination pathway, as provided herein.

In certain embodiments, provided herein is a composition comprising bioderived isopropanol provided herein, for example, bioderived isopropanol produced by culturing a non-naturally occurring microbial organism having a formaldehyde fixation pathway, a formate assimilation pathway and/or a methanol metabolic pathway, and an isopropanol pathway, as provided herein. In some embodiments, the composition further comprises a compound other than said bioderived isopropanol. In certain embodiments, the compound other than said bioderived isopropanol is a trace amount of a cellular portion of a non-naturally occurring microbial organism having a formaldehyde fixation pathway, a formate assimilation pathway and/or a methanol metabolic pathway, and an isopropanol pathway, as provided herein.

In some embodiments, provided herein is a biobased product comprising a bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol provided herein. In certain embodiments, the biobased product is a biofuel, chemical, polymer, surfactant, soap, detergent, shampoo, lubricating

oil additive, fragrance, flavor material or acrylate. In certain embodiments, the biobased product is a solvent, paint, lacquer, thinner, ink, adhesive, cleaner, disinfectant, cosmetic, toiletry, de-icer, pharmaceutical, motor oil, isopropylamine, isopropylether, isopropyl ester, propylene or polymer. In certain embodiments, the biobased product comprises at least 5% bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In certain embodiments, the biobased product comprises at least 10% bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In some embodiments, the biobased product comprises at least 20% bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In other embodiments, the biobased product comprises at least 30% bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In some embodiments, the biobased product comprises at least 40% bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In other embodiments, the biobased product comprises at least 50% bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In one embodiment, the biobased product comprises a portion of said bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol as a repeating unit. In another embodiment, provided herein is a molded product obtained by molding the biobased product provided herein. In other embodiments, provided herein is a process for producing a biobased product provided herein, comprising chemically reacting said bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol with itself or another compound in a reaction that produces said biobased product. In certain embodiments, provided herein is a polymer comprising or obtained by converting the bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In other embodiments, provided herein is a method for producing a polymer, comprising chemically or enzymatically converting the bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol to the polymer. In yet other embodiments, provided herein is a composition comprising the bioderived fatty alcohol, fatty aldehyde, fatty acid or isopropanol, or a cell lysate or culture supernatant thereof.

The culture conditions can include, for example, liquid culture procedures as well as fermentation and other large scale culture procedures. As described herein, particularly useful yields of the biosynthetic products of the invention can be obtained under anaerobic or substantially anaerobic culture conditions.

As described herein, one exemplary growth condition for achieving biosynthesis of fatty alcohol, fatty aldehyde, fatty acid or isopropanol includes anaerobic culture or fermentation conditions. In certain embodiments, the non-naturally occurring microbial organisms of the invention can be sustained, cultured or fermented under anaerobic or substantially anaerobic conditions. Briefly, an anaerobic condition refers to an environment devoid of oxygen. Substantially anaerobic conditions include, for example, a culture, batch fermentation or continuous fermentation such that the dissolved oxygen concentration in the medium remains between 0 and 10% of saturation. Substantially anaerobic conditions also includes growing or resting cells in liquid medium or on solid agar inside a sealed chamber maintained with an atmosphere of less than 1% oxygen. The percent of oxygen can be maintained by, for example, sparging the culture with an N₂/CO₂ mixture or other suitable non-oxygen gas or gases.

The culture conditions described herein can be scaled up and grown continuously for manufacturing of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. Exemplary growth procedures include, for example, fed-batch fermentation and

batch separation; fed-batch fermentation and continuous separation, or continuous fermentation and continuous separation. All of these processes are well known in the art. Fermentation procedures are particularly useful for the biosynthetic production of commercial quantities of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. Generally, and as with non-continuous culture procedures, the continuous and/or near-continuous production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol will include culturing a non-naturally occurring fatty alcohol, fatty aldehyde, fatty acid or isopropanol producing organism of the invention in sufficient nutrients and medium to sustain and/or nearly sustain growth in an exponential phase. Continuous culture under such conditions can include, for example, growth or culturing for 1 day, 2, 3, 4, 5, 6 or 7 days or more. Additionally, continuous culture can include longer time periods of 1 week, 2, 3, 4 or 5 or more weeks and up to several months. Alternatively, organisms of the invention can be cultured for hours, if suitable for a particular application. It is to be understood that the continuous and/or near-continuous culture conditions also can include all time intervals in between these exemplary periods. It is further understood that the time of culturing the microbial organism of the invention is for a sufficient period of time to produce a sufficient amount of product for a desired purpose.

Fermentation procedures are well known in the art. Briefly, fermentation for the biosynthetic production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be utilized in, for example, fed-batch fermentation and batch separation; fed-batch fermentation and continuous separation, or continuous fermentation and continuous separation. Examples of batch and continuous fermentation procedures are well known in the art.

In addition to the above fermentation procedures using the fatty alcohol, fatty aldehyde, fatty acid or isopropanol producers of the invention for continuous production of substantial quantities of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, the fatty alcohol, fatty aldehyde, fatty acid or isopropanol producers also can be, for example, simultaneously subjected to chemical synthesis and/or enzymatic procedures to convert the product to other compounds or the product can be separated from the fermentation culture and sequentially subjected to chemical and/or enzymatic conversion to convert the product to other compounds, if desired.

To generate better producers, metabolic modeling can be utilized to optimize growth conditions. Modeling can also be used to design gene knockouts that additionally optimize utilization of the pathway (see, for example, U.S. patent publications US 2002/0012939, US 2003/0224363, US 2004/0029149, US 2004/0072723, US 2003/0059792, US 2002/0168654 and US 2004/0009466, and U.S. Pat. No. 7,127,379). Modeling analysis allows reliable predictions of the effects on cell growth of shifting the metabolism towards more efficient production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol.

In addition to active and selective enzymes producing fatty alcohols, fatty aldehydes, fatty acid or isopropanols at high yield, titer and productivity, a robust host organism that can efficiently direct carbon and reducing equivalents to fatty alcohol, fatty aldehyde and fatty acid biosynthesis can be beneficial. Host modifications described herein are particularly useful in combination with selective enzymes described herein that favor formation of the desired fatty alcohol, fatty aldehyde, fatty acid or isopropanol product. Several host modifications described herein entail introducing heterologous enzyme activities into the host organism.

Other modifications involve overexpressing or elevating enzyme activity relative to wild type levels. Yet other modifications include disrupting endogenous genes or attenuating endogenous enzyme activities.

In one embodiment of the invention, the microbial organisms efficiently directs carbon and energy sources into production of acetyl-CoA, which is used as both a primer and extension unit in the MI-FAE cycle. In one embodiment of the invention, the microbial organisms efficiently directs carbon and energy sources into production of malonyl-CoA, which is used as both a primer and extension unit in the MD-FAE cycle. In unmodified microbial organism, fatty alcohol, fatty aldehyde and fatty acid production in the cytosol relies on the native cell machinery to provide the necessary precursors. Thus, high concentrations of cytosolic acetyl-CoA and/or malonyl-CoA are desirable for facilitating deployment of a cytosolic fatty alcohol, fatty aldehyde or fatty acid production pathway that originates from acetyl-CoA or malonyl-CoA. Metabolic engineering strategies for increasing cytosolic acetyl-CoA and malonyl-CoA are disclosed herein.

Since many eukaryotic organisms synthesize most of their acetyl-CoA in the mitochondria during growth on glucose, increasing the availability of acetyl-CoA in the cytosol can be obtained by introduction of a cytosolic acetyl-CoA biosynthesis pathway. Accordingly, acetyl-CoA biosynthesis pathways are described herein. In one embodiment, utilizing the pathways shown in FIG. 3, acetyl-CoA can be synthesized in the cytosol from a pyruvate or threonine precursor. In other embodiment, acetyl-CoA can be synthesized in the cytosol from phosphoenolpyruvate (PEP) or pyruvate (FIG. 4). In yet another embodiment acetyl-CoA can be synthesized in cellular compartments and transported to the cytosol. For example, one mechanism involves converting mitochondrial acetyl-CoA to a metabolic intermediate such as citrate or citramalate, transporting those intermediates to the cytosol, and then regenerating the acetyl-CoA (see FIGS. 5 and 6). Exemplary acetyl-CoA pathways and corresponding enzymes are further described in Examples V-VII.

In another embodiment, increasing cytosolic acetyl-CoA availability for fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthesis is to disrupt or attenuate competing enzymes and pathways that utilize acetyl-CoA or its precursors. Exemplary competing enzyme activities include, but are not limited to, pyruvate decarboxylase, lactate dehydrogenase, short-chain aldehyde and alcohol dehydrogenases, acetate kinase, phosphotransacetylase, glyceraldehyde-3-phosphate dehydrogenases, pyruvate oxidase and acetyl-CoA carboxylase. Exemplary acetyl-CoA consuming pathways whose disruption or attenuation can improve fatty alcohol, fatty aldehyde, fatty acid or isopropanol production include the mitochondrial TCA cycle, fatty acid biosynthesis, ethanol production and amino acid biosynthesis. These enzymes and pathways are further described herein.

Yet another strategy for increasing cytosolic acetyl-CoA production is to increase the pool of CoA available in the cytoplasm. This can be accomplished by overexpression of CoA biosynthetic enzymes in the cytosol. In particular, expression of pantothenate kinase (EC 2.7.1.33) can be used. This enzyme catalyzes the first step and rate-limiting enzyme of CoA biosynthesis. Exemplary pantothenate kinase variants resistant to feedback inhibition by CoA are well known in the art (Rock et al, *J Bacteriol* 185: 3410-5 (2003)) and are described in the below table.

Protein	Accession #	GI number	Organism
coaA	AAC76952	1790409	<i>Escherichia coli</i>
CAB1	NP_010820.3	398366683	<i>Saccharomyces cerevisiae</i>
KLLA0C00869g	XP_452233.1	50304555	<i>Kluyveromyces lactis</i>
YALI0D25476g	XP_503275.1	50551601	<i>Yarrowia lipolytica</i>
ANI_1_3272024	XP_001400486.2	317028058	<i>Aspergillus niger</i>

Competing enzymes and pathways that divert acyl-CoA substrates from production of fatty alcohols, fatty aldehydes or fatty acids of the invention can also be attenuated or disrupted. Exemplary enzymes for attenuation include acyltransferases, carnitine shuttle enzymes and negative regulators of MI-FAE cycle, MD-FAE cycle, FAACPE cycle or termination pathway enzymes.

Disruption or attenuation of acyltransferases that transfer acyl moieties from CoA to other acceptors such as ACP, glycerol, ethanol and others, can increase the availability of acyl-CoA for fatty alcohol, fatty aldehyde or fatty acid production. For example, Acyl-CoA:ACP transacylase (EC 2.3.1.38; 2.3.1.39) enzymes such as fabH (KASIII) of *E. coli* transfer acyl moieties from CoA to ACP. FabH is active on acetyl-CoA and butyryl-CoA (Prescott et al, *Adv. Enzymol. Relat. Areas Mol.* 36:269-311(1972)). Acetyl-CoA:ACP transacylase enzymes from *Plasmodium falciparum* and *Streptomyces avermitilis* have been heterologously expressed in *E. coli* (Lobo et al, *Biochem* 40:11955-64 (2001)). A synthetic KASIII (FabH) from *P. falciparum* expressed in a fabH-deficient *Lactococcus lactis* host was able to complement the native fadH activity (Du et al, *AEM* 76:3959-66 (2010)). The acetyl-CoA:ACP transacylase enzyme from *Spinacia oleracea* accepts other acyl-ACP molecules as substrates, including butyryl-ACP (Shimakata et al, *Methods Enzym* 122:53-9 (1986)). Malonyl-CoA:ACP transacylase enzymes include FabD of *E. coli* and *Brassica napus* (Verwoert et al, *J Bacteriol.* 174:2851-7 (1992); Simon et al, *FEBS Lett* 435:204-6 (1998)). FabD of *B. napus* was able to complement fabD-deficient *E. coli*. The multifunctional eukaryotic fatty acid synthase enzyme complexes (described herein) also catalyze this activity. Other exemplary acyltransferases include diacylglycerol acyltransferases such as LRO1 and DGA1 of *S. cerevisiae* and DGA1 and DGA2 of *Yarrowia lipolytica*, glycerolipid acyltransferase enzymes such as plsB of *E. coli* (GenBank: AAC77011.2, GI:87082362; Heath and Rock, *J Bacteriol* 180:1425-30 (1998)), sterol acyltransferases such as ARE1 and ARE2 of *S. cerevisiae*, ethanol acyltransferases (EEB1, EHT1), putative acyltransferases (YMR210W) and others.

Protein	GenBank ID	GI Number	Organism
fabH	AAC74175.1	1787333	<i>Escherichia coli</i>
fadA	NP_824032.1	29829398	<i>Streptomyces avermitilis</i>
fabH	AAC63960.1	3746429	<i>Plasmodium falciparum</i>
Synthetic construct	ACX34097.1	260178848	<i>Plasmodium falciparum</i>
fabH	CAL98359.1	124493385	<i>Lactococcus lactis</i>
fabD	AAC74176.1	1787334	<i>Escherichia coli</i>
fabD	CAB45522.1	5139348	<i>Brassica napus</i>
LRO1	NP_014405.1	6324335	<i>Saccharomyces cerevisiae</i>
DGA1	NP_014888.1	6324819	<i>Saccharomyces cerevisiae</i>
DGA1	CAG79269.1	49649549	<i>Yarrowia lipolytica</i>
DGA2	XP_504700.1	50554583	<i>Yarrowia lipolytica</i>
ARE1	NP_009978.1	6319896	<i>Saccharomyces cerevisiae</i>
ARE2	NP_014416.1	6324346	<i>Saccharomyces cerevisiae</i>
EEB1	NP_015230.1	6325162	<i>Saccharomyces cerevisiae</i>
EHT1	NP_009736.3	398365307	<i>Saccharomyces cerevisiae</i>

-continued

Protein	GenBank ID	GI Number	Organism
YMR210W	NP_013937.1	6323866	<i>Saccharomyces cerevisiae</i>
ALE1	NP_014818.1	6324749	<i>Saccharomyces cerevisiae</i>

Increasing production of fatty alcohols, fatty aldehydes or fatty acids may necessitate disruption or attenuation of enzymes involved in the trafficking of acetyl-CoA and acyl-CoA molecules from the cytosol to other compartments of the organism such as mitochondria, endoplasmic reticulum, proteoliposomes and peroxisomes. In these compartments, the acyl-CoA intermediate can be degraded or used as building blocks to synthesize fatty acids, cofactors and other byproducts.

Acetyl-CoA and acyl-CoA molecules localized in the cytosol can be transported into other cellular compartments with the aid of the carrier molecule carnitine via carnitine shuttles (van Roermund et al., *EMBO J* 14:3480-86 (1995)). Acyl-carnitine shuttles between cellular compartments have been characterized in yeasts such as *Candida albicans* (Strijbis et al, *J Biol Chem* 285:24335-46 (2010)). In these shuttles, the acyl moiety of acyl-CoA is reversibly transferred to carnitine by acylcarnitine transferase enzymes. Acetylcarnitine can then be transported across the membrane by organelle-specific acylcarnitine/carnitine translocase enzymes. After translocation, the acyl-CoA is regenerated by acetylcarnitine transferase. Enzymes suitable for disruption or attenuation include carnitine acyltransferase enzymes, acylcarnitine translocases, acylcarnitine carrier proteins and enzymes involved in carnitine biosynthesis.

Carnitine acetyltransferase (CAT, EC 2.3.1.7) reversibly links acetyl units from acetyl-CoA to the carrier molecule, carnitine. *Candida albicans* encodes three CAT isozymes: Cat2, Yat1 and Yat2 (Strijbis et al., *J Biol Chem* 285:24335-46 (2010)). Cat2 is expressed in both the mitochondrion and the peroxisomes, whereas Yat1 and Yat2 are cytosolic. The Cat2 transcript contains two start codons that are regulated under different carbon source conditions. The longer transcript contains a mitochondrial targeting sequence whereas the shorter transcript is targeted to peroxisomes. Cat2 of *Saccharomyces cerevisiae* and AcuJ of *Aspergillus nidulans* employ similar mechanisms of dual localization (Elgersma et al., *EMBO J* 14:3472-9 (1995); Hynes et al., *Euk Cell* 10:547-55 (2011)). The cytosolic CAT of *A. nidulans* is encoded by facC. Other exemplary CAT enzymes are found in *Rattus norvegicus* and *Homo sapiens* (Cordente et al., *Biochem* 45:6133-41(2006)). Exemplary carnitine acyltransferase enzymes (EC 2.3.1.21) are the Cpt1 and Cpt2 gene products of *Rattus norvegicus* (de Vries et al., *Biochem* 36:5285-92 (1997)).

Protein	Accession #	GI number	Organism
Cat2	AAN31660.1	23394954	<i>Candida albicans</i>
Yat1	AAN31659.1	23394952	<i>Candida albicans</i>
Yat2	XP_711005.1	68490355	<i>Candida albicans</i>
Cat2	CAA88327.1	683665	<i>Saccharomyces cerevisiae</i>
Yat1	AAC09495.1	456138	<i>Saccharomyces cerevisiae</i>
Yat2	NP_010941.1	6320862	<i>Saccharomyces cerevisiae</i>
AcuJ	CBF69795.1	259479509	<i>Aspergillus nidulans</i>
FacC	AAC82487.1	2511761	<i>Aspergillus nidulans</i>
Crat	AAH83616.1	53733439	<i>Rattus norvegicus</i>
Crat	P43155.5	215274265	<i>Homo sapiens</i>
Cpt1	AAB48046.1	1850590	<i>Rattus norvegicus</i>
Cpt2	AAB02339.1	1374784	<i>Rattus norvegicus</i>

Carnitine-acylcarnitine translocases can catalyze the bidirectional transport of carnitine and carnitine-fatty acid complexes. The Cact gene product provides a mechanism for transporting acyl-carnitine substrates across the mitochondrial membrane (Ramsay et al *Biochim Biophys Acta* 1546: 21-42 (2001)). A similar protein has been studied in humans (Sekoguchi et al., *J Biol Chem* 278:38796-38802 (2003)). The *Saccharomyces cerevisiae* mitochondrial carnitine carrier is Crc1 (van Roermund et al., supra; Palmieri et al., *Biochimica et Biophys Acta* 1757:1249-62 (2006)). The human carnitine translocase was able to complement a Crc1-deficient strain of *S. cerevisiae* (van Roermund et al., supra). Two additional carnitine translocases found in *Drosophila melanogaster* and *Caenorhabditis elegans* were also able to complement Crc1-deficient yeast (Oey et al., *Mol Genet Metab* 85:121-24 (2005)). Four mitochondrial carnitine/acetylcarnitine carriers were identified in *Trypanosoma brucei* based on sequence homology to the yeast and human transporters (Colasante et al., *Mol Biochem Parasit* 167:104-117 (2009)). The carnitine transporter of *Candida albicans* was also identified by sequence homology. An additional mitochondrial carnitine transporter is the acuH gene product of *Aspergillus nidulans*, which is exclusively localized to the mitochondrial membrane (Lucas et al., *FEMS Microbiol Lett* 201:193-8 (2006)).

Protein	GenBank ID	GI Number	Organism
Cact	P97521.1	2497984	<i>Rattus norvegicus</i>
Cactl	NP_001034444.1	86198310	<i>Homo sapiens</i>
CaO19.2851	XP_715782.1	68480576	<i>Candida albicans</i>
Crc1	NP_014743.1	6324674	<i>Saccharomyces cerevisiae</i>
Dif-1	CAA88283.1	829102	<i>Caenorhabditis elegans</i>
colt	CAA73099.1	1944534	<i>Drosophila melanogaster</i>
Tb11.02.2960	EAN79492.1	70833990	<i>Trypanosoma brucei</i>
Tb11.03.0870	EAN79007.1	70833505	<i>Trypanosoma brucei</i>
Tb11.01.5040	EAN80288.1	70834786	<i>Trypanosoma brucei</i>
Tb927.8.5810	AAX69329.1	62175181	<i>Trypanosoma brucei</i>
acuH	CAB44434.1	5019305	<i>Aspergillus nidulans</i>

Transport of carnitine and acylcarnitine across the peroxisomal membrane has not been well-characterized. Specific peroxisomal acylcarnitine carrier proteins in yeasts have not been identified to date. However, mitochondrial carnitine translocases can also function in the peroxisomal transport of carnitine and acetylcarnitine. Experimental evidence suggests that the OCTN3 protein of *Mus musculus* is a peroxisomal carnitine/acetylcarnitine translocase.

Yet another possibility is that acyl-CoA or acyl-carnitine are transported across the peroxisomal or mitochondrial membranes by an acyl-CoA transporter such as the Pxa1 and Pxa2 ABC transporter of *Saccharomyces cerevisiae* or the ALDP ABC transporter of *Homo sapiens* (van Roermund et al., *FASEB J* 22:4201-8 (2008)). Pxa1 and Pxa2 (Pat1 and Pat2) form a heterodimeric complex in the peroxisomal membrane and catalyze the ATP-dependent transport of fatty acyl-CoA esters into the peroxisome (Verleur et al., *Eur J Biochem* 249: 657-61(1997)). The mutant phenotype of a pxa1/pxa2 deficient yeast can be rescued by heterologous expression of ALDP, which was shown to transport a range of acyl-CoA substrates (van Roermund et al., *FASEB J* 22:4201-8 (2008)). Deletion of the Pxa12 transport system, in tandem with deletion of the peroxisomal fatty acyl-CoA synthetase (Faa7) abolished peroxisomal beta-oxidation in *S. cerevisiae*. Yet another strategy for reducing transport of pathway intermediates or products into the peroxisome is to attenuate or eliminate peroxisomal function, by interfering

with systems involved in peroxisomal biogenesis. An exemplary target is Pex10 of *Yarrowia lipolytica* and homologs.

Protein	Accession #	GI number	Organism
OCTN3	BAA78343.1	4996131	<i>Mus musculus</i>
Pxa1	AAC49009.1	619668	<i>Saccharomyces cerevisiae</i>
Pxa2	AAB51597.1	1931633	<i>Saccharomyces cerevisiae</i>
Faa2	NP_010931.3	398364331	<i>Saccharomyces cerevisiae</i>
ALDP	NP_000024.2	7262393	<i>Homo sapiens</i>
Pex10	BAA99413.1	9049374	<i>Yarrowia lipolytica</i>

Carnitine biosynthetic pathway enzymes are also suitable candidates for disruption or attenuation. In *Candida albicans*, for example, carnitine is synthesized from trimethyl-L-lysine in four enzymatic steps (Strijbis et al., *FASEB J* 23:2349-59 (2009)). The carnitine pathway precursor, trimethyllysine (TML), is produced during protein degradation. TML dioxygenase (CaO13.4316) hydroxylates TML to form 3-hydroxy-6-N-trimethyllysine. A pyridoxal-5'-phosphate dependent aldolase (CaO19.6305) then cleaves HTML into 4-trimethylaminobutyraldehyde. The 4-trimethylaminobutyraldehyde is subsequently oxidized to 4-trimethylaminobutyrate by a dehydrogenase (CaO19.6306). In the final step, 4-trimethylaminobutyrate is hydroxylated to form carnitine by the gene product of CaO19.7131. Flux through the carnitine biosynthesis pathway is limited by the availability of the pathway substrate and very low levels of carnitine seem to be sufficient for normal carnitine shuttle activity (Strejbs et al., *IUBMB Life* 62:357-62 (2010)).

Protein	Accession #	GI number	Organism
CaO19.4316	XP_720623.1	68470755	<i>Candida albicans</i>
CaO19.6305	XP_711090.1	68490151	<i>Candida albicans</i>
CaO19.6306	XP_711091.1	68490153	<i>Candida albicans</i>
CaO19.7131	XP_715182.1	68481628	<i>Candida albicans</i>

Carbon flux towards production of fatty alcohols, fatty aldehydes or fatty acids can be improved by deleting or attenuating competing pathways. Typical fermentation products of yeast include ethanol, glycerol and CO₂. The elimination or reduction of these byproducts can be accomplished by approaches described herein. For example, carbon loss due to respiration can be reduced. Other potential byproducts include lactate, acetate, formate, fatty acids and amino acids.

The conversion of acetyl-CoA into ethanol can be detrimental to the production of fatty alcohols, fatty aldehydes, fatty acids or isopropanol because the conversion process can draw away both carbon and reducing equivalents from the MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway or isopropanol pathway. Ethanol can be formed from pyruvate in two enzymatic steps catalyzed by pyruvate decarboxylase and ethanol dehydrogenase. *Saccharomyces cerevisiae* has three pyruvate decarboxylases (PDC1, PDC5 and PDC6). PDC1 is the major isozyme and is strongly expressed in actively fermenting cells. PDC5 also functions during glycolytic fermentation, but is expressed only in the absence of PDC1 or under thiamine limiting conditions. PDC6 functions during growth on nonfermentable carbon sources. Deleting PDC1 and PDC5 can reduce ethanol production significantly; however these deletions can lead to mutants with increased PDC6 expression. Deletion of all three eliminates ethanol formation completely but also can cause a growth defect because of inability of the cells to form sufficient acetyl-CoA for biomass formation.

This, however, can be overcome by evolving cells in the presence of reducing amounts of C2 carbon source (ethanol or acetate) (van Mans et al, *AEM* 69:2094-9 (2003)). It has also been reported that deletion of the positive regulator PDC2 of pyruvate decarboxylases PDC1 and PDC5, reduced ethanol formation to ~10% of that made by wild-type (Hohmann et al, *Mol Gen Genet* 241:657-66 (1993)). Protein sequences and identifiers of PDC enzymes are listed in Example V.

Alternatively, alcohol dehydrogenases that convert acetaldehyde into ethanol and/or other short chain alcohol dehydrogenases can be disrupted or attenuated to provide carbon and reducing equivalents for the MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway or isopropanol pathway. To date, seven alcohol dehydrogenases, ADH1-ADHVII, have been reported in *S. cerevisiae* (de Smidt et al, *FEMS Yeast Res* 8:967-78 (2008)). ADH1 (GI:1419926) is the key enzyme responsible for reducing acetaldehyde to ethanol in the cytosol under anaerobic conditions. It has been reported that a yeast strain deficient in ADH1 cannot grow anaerobically because an active respiratory chain is the only alternative path to regenerate NADH and lead to a net gain of ATP (Drewke et al, *J Bacteriol* 172:3909-17 (1990)). This enzyme is an ideal candidate for downregulation to limit ethanol production. ADH2 is severely repressed in the presence of glucose. In *K. lactis*, two NAD-dependent cytosolic alcohol dehydrogenases have been identified and characterized. These genes also show activity for other aliphatic alcohols. The genes ADH1 (GI:113358) and ADHII (GI:51704293) are preferentially expressed in glucose-grown cells (Bozzi et al, *Biochim Biophys Acta* 1339:133-142 (1997)). Cytosolic alcohol dehydrogenases are encoded by ADH1 (GI:608690) in *C. albicans*, ADH1 (GI:3810864) in *S. pombe*, ADH1 (GI:5802617) in *Y. lipolytica*, ADH1 (GI:2114038) and ADHII (GI:2143328) in *Pichia stipitis* or *Scheffersomyces stipitis* (Passoth et al, *Yeast* 14:1311-23 (1998)). Candidate alcohol dehydrogenases are shown the table below.

Protein	GenBank ID	GI number	Organism
SADH	BAA24528.1	2815409	<i>Candida parapsilosis</i>
ADH1	NP_014555.1	6324486	<i>Saccharomyces cerevisiae</i> s288c
ADH2	NP_014032.1	6323961	<i>Saccharomyces cerevisiae</i> s288c
ADH3	NP_013800.1	6323729	<i>Saccharomyces cerevisiae</i> s288c
ADH4	NP_011258.2	269970305	<i>Saccharomyces cerevisiae</i> s288c
ADH5 (SFA1)	NP_010113.1	6320033	<i>Saccharomyces cerevisiae</i> s288c
ADH6	NP_014051.1	6323980	<i>Saccharomyces cerevisiae</i> s288c
ADH7	NP_010030.1	6319949	<i>Saccharomyces cerevisiae</i> s288c
adhP	CAA44614.1	2810	<i>Kluyveromyces lactis</i>
ADH1	P20369.1	113358	<i>Kluyveromyces lactis</i>
ADH2	CAA45739.1	2833	<i>Kluyveromyces lactis</i>
ADH3	P49384.2	51704294	<i>Kluyveromyces lactis</i>
ADH1	CAA57342.1	608690	<i>Candida albicans</i>
ADH2	CAA21988.1	3859714	<i>Candida albicans</i>
SAD	XP_712899.1	68486457	<i>Candida albicans</i>
ADH1	CAA21782.1	3810864	<i>Schizosaccharomyces pombe</i>
ADH1	AAD51737.1	5802617	<i>Yarrowia lipolytica</i>
ADH2	AAD51738.1	5802619	<i>Yarrowia lipolytica</i>
ADH3	AAD51739.1	5802621	<i>Yarrowia lipolytica</i>
AlcB	AAX53105.1	61696864	<i>Aspergillus niger</i>
ANL1_282024	XP_001399347.1	145231748	<i>Aspergillus niger</i>

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Protein	GenBank ID	GI number	Organism
ANL1_126164	XP_001398574.2	317037131	<i>Aspergillus niger</i>
ANL1_1756104	XP_001395505.2	317033815	<i>Aspergillus niger</i>
ADH2	CAA73827.1	2143328	<i>Scheffersomyces stipitis</i>

Attenuation or disruption of one or more glycerol-3-phosphatase or glycerol-3-phosphate (G3P) dehydrogenase enzymes can eliminate or reduce the formation of glycerol, and thereby conserving carbon and reducing equivalents for production of fatty alcohols, fatty aldehydes, fatty acids or isopropanol.

G3P phosphatase catalyzes the hydrolysis of G3P to glycerol. Enzymes with this activity include the glycerol-1-phosphatase (EC 3.1.3.21) enzymes of *Saccharomyces cerevisiae* (GPP1 and GPP2), *Candida albicans* and *Dunalella parva* (Popp et al, *Biotechnol Bioeng* 100:497-505 (2008); Fan et al, *FEMS Microbiol Lett* 245:107-16 (2005)). The *D. parva* gene has not been identified to date. These and additional G3P phosphatase enzymes are shown in the table below.

Protein	GenBank ID	GI Number	Organism
GPP1	DAA08494.1	285812595	<i>Saccharomyces cerevisiae</i>
GPP2	NP_010984.1	6320905	<i>Saccharomyces cerevisiae</i>
GPP1	XP_717809.1	68476319	<i>Candida albicans</i>
KLLA0C08217g	XP_452565.1	50305213	<i>Kluyveromyces lactis</i>
KLLA0C11143g	XP_452697.1	50305475	<i>Kluyveromyces lactis</i>
ANL1_380074	XP_001392369.1	145239445	<i>Aspergillus niger</i>
ANL1_444054	XP_001390913.2	317029125	<i>Aspergillus niger</i>

S. cerevisiae has three G3P dehydrogenase enzymes encoded by GPD1 and GPD2 in the cytosol and GUT2 in the mitochondrion. GPD2 is known to encode the enzyme responsible for the majority of the glycerol formation and is responsible for maintaining the redox balance under anaerobic conditions. GPD1 is primarily responsible for adaptation of *S. cerevisiae* to osmotic stress (Bakker et al., *FEMS Microbiol Rev* 24:15-37 (2001)). Attenuation of GPD1, GPD2 and/or GUT2 will reduce glycerol formation. GPD1 and GUT2 encode G3P dehydrogenases in *Yarrowia lipolytica* (Beopoulos et al, *AEM* 74:7779-89 (2008)). GPD1 and GPD2 encode for G3P dehydrogenases in *S. pombe*. Similarly, G3P dehydrogenase is encoded by CTRL_02011 in *Candida tropicalis* and a gene represented by GI:20522022 in *Candida albicans*.

Protein	GenBank ID	GI number	Organism
GPD1	CAA98582.1	1430995	<i>Saccharomyces cerevisiae</i>
GPD2	NP_014582.1	6324513	<i>Saccharomyces cerevisiae</i>
GUT2	NP_012111.1	6322036	<i>Saccharomyces cerevisiae</i>
GPD1	CAA22119.1	6066826	<i>Yarrowia lipolytica</i>
GUT2	CAG83113.1	49646728	<i>Yarrowia lipolytica</i>
GPD1	CAA22119.1	3873542	<i>Schizosaccharomyces pombe</i>

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Protein	GenBank ID	GI number	Organism
GPD2	CAA91239.1	1039342	<i>Schizosaccharomyces pombe</i>
ANI_1_786014	XP_001389035.2	317025419	<i>Aspergillus niger</i>
ANI_1_1768134	XP_001397265.1	145251503	<i>Aspergillus niger</i>
KLLA0C04004g	XP_452375.1	50304839	<i>Kluyveromyces lactis</i>
CTRG_02011	XP_002547704.1	255725550	<i>Candida tropicalis</i>
GPD1	XP_714362.1	68483412	<i>Candida albicans</i>
GPD2	XP_713824.1	68484586	<i>Candida albicans</i>

Enzymes that form acid byproducts such as acetate, formate and lactate can also be attenuated or disrupted. Such enzymes include acetate kinase, phosphotransacetylase and pyruvate oxidase. Disruption or attenuation of pyruvate formate lyase and formate dehydrogenase could limit formation of formate and carbon dioxide. These enzymes are described in further detail in Example V.

Alcohol dehydrogenases that convert pyruvate to lactate are also candidates for disruption or attenuation. Lactate dehydrogenase enzymes include LdhA of *E. coli* and Ldh from *Ralstonia eutropha* (Steinbuechel and Schlegel, *Eur. J. Biochem.* 130:329-334 (1983)). Other alcohol dehydrogenases listed above may also exhibit LDH activity.

Protein	GenBank ID	GI number	Organism
LdhA	NP_415898.1	16129341	<i>Escherichia coli</i>
Ldh	YP_725182.1	113866693	<i>Ralstonia eutropha</i>

Tuning down activity of the mitochondrial pyruvate dehydrogenase complex will limit flux into the mitochondrial TCA cycle. Under anaerobic conditions and in conditions where glucose concentrations are high in the medium, the capacity of this mitochondrial enzyme is very limited and there is no significant flux through it. However, in some embodiments, this enzyme can be disrupted or attenuated to increase fatty alcohol, fatty aldehyde or fatty acid production. Exemplary pyruvate dehydrogenase genes include PDB1, PDA1, LAT1 and LPD1. Accession numbers and homologs are listed in Example V.

Another strategy for reducing flux into the TCA cycle is to limit transport of pyruvate into the mitochondria by tuning down or deleting the mitochondrial pyruvate carrier. Transport of pyruvate into the mitochondria in *S. cerevisiae* is catalyzed by a heterocomplex encoded by MPC1 and MPC2 (Herzig et al, *Science* 337:93-6 (2012); Bricker et al, *Science* 337:96-100 (2012)). *S. cerevisiae* encodes five other putative monocarboxylate transporters (MCH1-5), several of which may be localized to the mitochondrial membrane (Makuc et al, *Yeast* 18:1131-43 (2001)). NDT1 is another putative pyruvate transporter, although the role of this protein is disputed in the literature (Todisco et al, *J Biol Chem* 20:1524-31(2006)). Exemplary pyruvate and monocarboxylate transporters are shown in the table below:

Protein	GenBank ID	GI number	Organism
MPC1	NP_011435.1	6321358	<i>Saccharomyces cerevisiae</i>
MPC2	NP_012032.1	6321956	<i>Saccharomyces cerevisiae</i>
MPC1	XP_504811.1	50554805	<i>Yarrowia lipolytica</i>
MPC2	XP_501390.1	50547841	<i>Yarrowia lipolytica</i>
MPC1	XP_719951.1	68471816	<i>Candida albicans</i>

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Protein	GenBank ID	GI number	Organism
MPC2	XP_716190.1	68479656	<i>Candida albicans</i>
MCH1	NP_010229.1	6320149	<i>Saccharomyces cerevisiae</i>
MCH2	NP_012701.2	330443640	<i>Saccharomyces cerevisiae</i>
MCH3	NP_014274.1	6324204	<i>Saccharomyces cerevisiae</i>
MCH5	NP_014951.2	330443742	<i>Saccharomyces cerevisiae</i>
NDT1	NP_012260.1	6322185	<i>Saccharomyces cerevisiae</i>
ANI_1_1592184	XP_001401484.2	317038471	<i>Aspergillus niger</i>
CaJ7_0216	XP_888808.1	77022728	<i>Candida albicans</i>
YAL10E16478g	XP_504023.1	50553226	<i>Yarrowia lipolytica</i>
KLLA0D14036g	XP_453688.1	50307419	<i>Kluyveromyces lactis</i>

Disruption or attenuation of enzymes that synthesize malonyl-CoA and fatty acids can increase the supply of carbon available for fatty alcohol, fatty aldehyde or fatty acid biosynthesis from acetyl-CoA. Exemplary enzymes for disruption or attenuation include fatty acid synthase, acetyl-CoA carboxylase, biotin:apoenzyme ligase, acyl carrier protein, thioesterase, acyltransferases, ACP malonyltransferase, fatty acid elongase, acyl-CoA synthetase, acyl-CoA transferase and acyl-CoA hydrolase.

Another strategy to reduce fatty acid biosynthesis is expression or overexpression of regulatory proteins which repress fatty acid forming genes. Acetyl-CoA carboxylase (EC 6.4.1.2) catalyzes the first step of fatty acid biosynthesis in many organisms: the ATP-dependent carboxylation of acetyl-CoA to malonyl-CoA. This enzyme utilizes biotin as a cofactor. Exemplary ACC enzymes are encoded by accABCD of *E. coli* (Davis et al, *J Biol Chem* 275:28593-8 (2000)), ACC1 of *Saccharomyces cerevisiae* and homologs (Sumper et al, *Methods Enzym* 71:34-7 (1981)). The mitochondrial acetyl-CoA carboxylase of *S. cerevisiae* is encoded by HFA1. Acetyl-CoA carboxylase holoenzyme formation requires attachment of biotin by a biotin:apoprotein ligase such as BPL1 of *S. cerevisiae*.

Protein	GenBank ID	GI Number	Organism
ACC1	CAA96294.1	1302498	<i>Saccharomyces cerevisiae</i>
KLLA0F06072g	XP_455355.1	50310667	<i>Kluyveromyces lactis</i>
ACC1	XP_718624.1	68474502	<i>Candida albicans</i>
YAL10C11407p	XP_501721.1	50548503	<i>Yarrowia lipolytica</i>
ANI_1_1724104	XP_001395476.1	145246454	<i>Aspergillus niger</i>
accA	AAC73296.1	1786382	<i>Escherichia coli</i>
accB	AAC76287.1	1789653	<i>Escherichia coli</i>
accC	AAC76288.1	1789654	<i>Escherichia coli</i>
accD	AAC75376.1	1788655	<i>Escherichia coli</i>
HFA1	NP_013934.1	6323863	<i>Saccharomyces cerevisiae</i>
BPL1	NP_010140.1	6320060	<i>Saccharomyces cerevisiae</i>

Proteins participating in the synthesis of fatty acids are shown below. The fatty acid synthase enzyme complex of yeast is composed of two multifunctional subunits, FAS1 and FAS2, which together catalyze the net conversion of acetyl-CoA and malonyl-CoA to fatty acids (Lomakin et al, *Cell* 129: 319-32 (2007)). Additional proteins associated with mitochondrial fatty acid synthesis include OAR1, Mct1, ETR1, ACP1 and PPT2. ACP1 is the mitochondrial acyl carrier protein and PPT2 encodes a phosphopantetheine transferase, which pantetheinylates mitochondrial ACP and is required for fatty acid biosynthesis in the mitochondria

(Stuible et al, *J Biol Chem*: 273: 22334-9 (1998)). A non-genetic strategy for reducing activity of fatty acid synthases is to add an inhibitor such as cerulenin. Global regulators of lipid biosynthesis can also be altered to tune down endogenous fatty acid biosynthesis pathways during production of long chain alcohols or related products. An exemplary global regulator is SNF1 of *Yarrowia lipolytica* and *Saccharomyces cerevisiae*.

Protein	GenBank ID	GI Number	Organism
FAS1	NP_012739.1	6322666	<i>Saccharomyces cerevisiae</i>
FAS2	NP_015093.1	6325025	<i>Saccharomyces cerevisiae</i>
FAS1	XP_451653.1	50303423	<i>Kluyveromyces lactis</i>
FAS2	XP_452914.1	50305907	<i>Kluyveromyces lactis</i>
FAS1	XP_716817.1	68478392	<i>Candida albicans</i>
FAS2	XP_723014.1	68465892	<i>Candida albicans</i>
FAS1	XP_500912.1	50546885	<i>Yarrowia lipolytica</i>
FAS2	XP_501096.1	50547253	<i>Yarrowia lipolytica</i>
FAS1	XP_001393490.2	317031809	<i>Aspergillus niger</i>
FAS2	XP_001388458.1	145228299	<i>Aspergillus niger</i>
OAR1	NP_012868.1	6322795	<i>Saccharomyces cerevisiae</i>
MCT1	NP_014864.4	398365823	<i>Saccharomyces cerevisiae</i>
ETR1	NP_009582.1	6319500	<i>Saccharomyces cerevisiae</i>
ACP1	NP_012729.1	6322656	<i>Saccharomyces cerevisiae</i>
PPT2	NP_015177.2	37362701	<i>Saccharomyces cerevisiae</i>
SNF1	CAG80498.1	49648180	<i>Yarrowia lipolytica</i>
SNF1	P06782.1	134588	<i>Saccharomyces cerevisiae</i>

Disruption or attenuation of elongase enzymes which convert acyl-CoA substrates to longer-chain length fatty acid derivatives longer than the product of interest can also be used to increase fatty alcohol, fatty aldehyde or fatty acid production. Elongase enzymes are found in compartments such as the mitochondria, endoplasmic reticulum, proteoliposomes and peroxisomes. For example, some yeast such as *S. cerevisiae* are able to synthesize long-chain fatty acids of chain length C16 and higher via a mitochondrial elongase which accepts exogenous or endogenous acyl-CoA substrates (Bessoule et al, *FEBS Lett* 214: 158-162 (1987)). This system requires ATP for activity. The endoplasmic reticulum also has an elongase system for synthesizing very long chain fatty acids (C18+) from acyl-CoA substrates of varying lengths (Kohlwein et al, *Mol Cell Biol* 21:109-25 (2001)). Genes involved in this system include TSC13, ELO2 and ELO3. ELO1 catalyzes the elongation of C12 acyl-CoAs to C16-C18 fatty acids.

Protein	Accession #	GI number	Organism
ELO2	NP_009963.1	6319882	<i>Saccharomyces cerevisiae</i>
ELO3	NP_013476.3	398366027	<i>Saccharomyces cerevisiae</i>
TSC13	NP_010269.1	6320189	<i>Saccharomyces cerevisiae</i>
ELO1	NP_012339.1	6322265	<i>Saccharomyces cerevisiae</i>

Native enzymes converting acyl-CoA pathway intermediates to acid byproducts can also reduce fatty alcohol, fatty aldehyde or fatty acid yield. For example, CoA hydrolases, transferases and synthetases can act on acyl-CoA intermediates to form short-, medium- or long chain acids. Disruption or attenuation of endogenous CoA hydrolases, CoA transferases and/or reversible CoA synthetases can be used to increase fatty alcohol, fatty aldehyde or fatty acid yield. Exemplary enzymes are shown in the table below.

Protein	GenBank ID	GI number	Organism
Tes1	NP_012553.1	6322480	<i>Saccharomyces cerevisiae</i> s288c

-continued

Protein	GenBank ID	GI number	Organism
ACH1	NP_009538.1	6319456	<i>Saccharomyces cerevisiae</i> s288c
EHD3	NP_010321.1	6320241	<i>Saccharomyces cerevisiae</i> s288c
YALI0F14729p	XP_505426.1	50556036	<i>Yarrowia lipolytica</i>
YALI0E30965p	XP_504613.1	50554409	<i>Yarrowia lipolytica</i>
KLLA0E16523g	XP_454694.1	50309373	<i>Kluyveromyces lactis</i>
KLLA0E10561g	XP_454427.1	50308845	<i>Kluyveromyces lactis</i>
ACH1	P83773.2	229462795	<i>Candida albicans</i>
CaO19.10681	XP_714720.1	68482646	<i>Candida albicans</i>
ANL1_318184	XP_001401512.1	145256774	<i>Aspergillus niger</i>
ANL1_1594124	XP_001401252.2	317035188	<i>Aspergillus niger</i>
tesB	NP_414986.1	16128437	<i>Escherichia coli</i>
tesB	NP_355686.2	159185364	<i>Agrobacterium tumefaciens</i>
atoA	2492994	P76459.1	<i>Escherichia coli</i>
atoD	2492990	P76458.1	<i>Escherichia coli</i>

Enzymes that favor the degradation of products, MI-FAE cycle intermediates, MD-FAE cycle intermediates, FAACPE cycle intermediates, termination pathway intermediates, or isopropanol pathway intermediates can also be disrupted or attenuated. Examples include aldehyde dehydrogenases, aldehyde decarbonylases, oxidative alcohol dehydrogenases, and irreversible fatty acyl-CoA degrading enzymes.

For production of fatty alcohols, fatty aldehydes, fatty acids or isopropanol of the invention, deletion or attenuation of non-specific aldehyde dehydrogenases can improve yield. For production of fatty acids, expression of such an enzyme may improve product formation. Such enzymes can, for example, convert acetyl-CoA into acetaldehyde, fatty aldehydes to fatty acids, or fatty alcohols to fatty acids. Acylating aldehyde dehydrogenase enzymes are described in Example IV. Acid-forming aldehyde dehydrogenase are described in Examples VI and XII.

The pathway enzymes that favor the reverse direction can also be disrupted or attenuated, if they are detrimental to fatty alcohol, fatty aldehyde, fatty acid or isopropanol production. An example is long chain alcohol dehydrogenases (EC 1.1.1.192) that favor the oxidative direction. Exemplary long chain alcohol dehydrogenases are ADH1 and ADH2 of *Geobacillus thermodenitrificans*, which oxidize alcohols up to a chain length of C30 (Liu et al, *Physiol Biochem* 155:2078-85 (2009)). These and other exemplary fatty alcohol dehydrogenase enzymes are listed in Examples IV and V. If an alcohol-forming acyl-CoA reductase is utilized for fatty alcohol, fatty aldehyde or fatty acid biosynthesis, deletion of endogenous fatty alcohol dehydrogenases will substantially reduce back lux.

Beta-oxidation enzymes may be reversible and operate in the direction of acyl-CoA synthesis. However, if they are irreversible or strongly favored in the degradation direction they are candidates for disruption or attenuation. An enzyme that fall into this category includes FOX2 of *S. cerevisiae*, a multifunctional enzyme with 3-hydroxyacyl-CoA dehydrogenase and enoyl-CoA hydratase activity (Hiltunen et al, *J Biol Chem* 267: 6646-6653 (1992)). Additional genes include degradative thiolases such as POT1 and acyl-CoA dehydrogenases that utilize cofactors other than NAD(P)H (EG. EC 1.3.8.) such as fadE of *E. coli*.

Protein	GenBank ID	GI Number	Organism
POT1	NP_012106.1	6322031	<i>Saccharomyces cerevisiae</i>
FOX2	NP_012934.1	6322861	<i>Saccharomyces cerevisiae</i>
fadE	AAC73325.2	87081702	<i>Escherichia coli</i>

Fatty acyl-CoA oxidase enzymes such as PDX1 of *S. cerevisiae* catalyze the oxygen-dependent oxidation of fatty acyl-CoA substrates. Enzymes with this activity can be disrupted or attenuated, if they are expressed under fatty alcohol, fatty aldehyde or fatty acid producing conditions. PDX1 (EC 1.3.3.6) genes and homologs are shown in the table below. PDX1 is subject to regulation by OAF1, which also activates genes involved in peroxisomal beta-oxidation, organization and biogenesis (Luo et al, *J Biol Chem* 271: 12068-75 (1996)). Regulators with functions similar to OAF1 and peroxisomal fatty acid transporters PXA1 and PXA2 are also candidates for deletion.

Protein	GenBank ID	GI Number	Organism
POX1	NP_011310.1	6321233	<i>Saccharomyces cerevisiae</i>
OAF1	NP_009349.3	330443370	<i>Saccharomyces cerevisiae</i>
PXA1	NP_015178.1	6325110	<i>Saccharomyces cerevisiae</i>
PXA2	NP_012733.1	6322660	<i>Saccharomyces cerevisiae</i>
YAL10F10857g	XP_505264.1	50555712	<i>Yarrowia lipolytica</i>
YAL10D24750p	XP_503244.1	50551539	<i>Yarrowia lipolytica</i>
YAL10E32835p	XP_504703.1	50554589	<i>Yarrowia lipolytica</i>
YAL10E06567p	XP_503632.1	50552444	<i>Yarrowia lipolytica</i>
YAL10E27654p	XP_504475.1	50554133	<i>Yarrowia lipolytica</i>
YAL10C23859p	XP_502199.1	50549457	<i>Yarrowia lipolytica</i>
POX	XP_455532.1	50311017	<i>Kluyveromyces lactis</i>
POX104	XP_721610.1	68468582	<i>Candida albicans</i>
POX105	XP_717995.1	68475844	<i>Candida albicans</i>
POX102	XP_721613.1	68468588	<i>Candida albicans</i>

Another candidate for disruption or attenuation is an acyl-CoA binding protein. The acyl binding protein ACB1 of *S. cerevisiae*, for example, binds acyl-CoA esters and shuttles them to acyl-CoA utilizing processes (Schjerling et al, *J Biol Chem* 271: 22514-21(1996)). Deletion of this protein did not impact growth rate and lead to increased accumulation of longer-chain acyl-CoA molecules. Acyl-CoA esters are involved in diverse cellular processes including lipid biosynthesis and homeostasis, signal transduction, growth regulation and cell differentiation (Rose et al, *PNAS USA* 89: 11287-11291(1992)).

Protein	GenBank ID	GI Number	Organism
ACB1	P31787.3	398991	<i>Saccharomyces cerevisiae</i>
KLLA0B05643g	XP_451787.2	302309983	<i>Kluyveromyces lactis</i>
YAL10E23185g	XP_002143080.1	210076210	<i>Yarrowia lipolytica</i>
ANI_1_1084034	XP_001390082.1	145234867	<i>Aspergillus niger</i>

To achieve high yields of fatty alcohols, fatty aldehydes fatty acids or isopropanol, it is desirable that the host organism can supply the cofactors required by the MI-FAE cycle, MD-FAE cycle, FAACPE cycle, the termination pathway and/or isopropanol pathway in sufficient quantities. In several organisms, in particular eukaryotic organisms, such as several *Saccharomyces*, *Kluyveromyces*, *Candida*, *Aspergillus*, and *Yarrowia* species, NADH is more abundant than NADPH in the cytosol as it is produced in large quantities by glycolysis. NADH can be made even more abundant by converting pyruvate to acetyl-CoA by means of heterologous or native NAD-dependant enzymes such as

NAD-dependant pyruvate dehydrogenase, NAD-dependant formate dehydrogenase, NADH:ferredoxin oxidoreductase, or NAD-dependant acylating acetylaldehyde dehydrogenase in the cytosol. Given the abundance of NADH in the cytosol of most organisms, it can be beneficial for all reduction steps of the MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway and/or isopropanol pathway to accept NADH as the reducing agent preferentially over other reducing agents such as NADPH. High yields of fatty alcohols, fatty aldehydes or fatty acids can thus be accomplished by, for example: 1) identifying and implementing endogenous or exogenous MI-FAE cycle, MD-FAE cycle and/or termination pathway enzymes with a stronger preference for NADH than other reducing equivalents such as NADPH; 2) attenuating one or more endogenous MI-FAE cycle, MD-FAE cycle or termination pathway enzymes that contribute NADPH-dependant reduction activity; 3) altering the cofactor specificity of endogenous or exogenous MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway or isopropanol pathway enzymes so that they have a stronger preference for NADH than their natural versions; or 4) altering the cofactor specificity of endogenous or exogenous MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway or isopropanol pathway enzymes so that they have a weaker preference for NADPH than their natural versions.

Strategies for engineering NADH-favoring MI-FAE cycle, MD-FAE cycle, FAACPE cycle termination pathways and/or isopropanol pathways are described in further detail in Example VIII. Methods for changing the cofactor specificity of an enzyme are well known in the art, and an example is described in Example IX.

If one or more of the MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway and/or isopropanol pathway enzymes utilizes NADPH as the cofactor, it can be beneficial to increase the production of NADPH in the host organism. In particular, if the MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway and/or isopropanol pathway is present in the cytosol of the host organism, methods for increasing NADPH production in the cytosol can be beneficial. Several approaches for increasing cytosolic production of NADPH can be implemented including channeling an increased amount of flux through the oxidative branch of the pentose phosphate pathway relative to wild-type, channeling an increased amount of flux through the Entner Doudoroff pathway relative to wild-type, introducing a soluble or membrane-bound transhydrogenase to convert NADH to NADPH, or employing NADP-dependant versions of the following enzymes: phosphorylating or non-phosphorylating glyceraldehyde-3-phosphate dehydrogenase, pyruvate dehydrogenase, formate dehydrogenase, or acylating acetylaldehyde dehydrogenase. These activities can be augmented by disrupting or attenuating native NAD-dependant enzymes including glyceraldehyde-3-phosphate dehydrogenase, pyruvate dehydrogenase, formate dehydrogenase, or acylating acetylaldehyde dehydrogenase. Strategies for engineering increased NADPH availability are described in Example X.

Synthesis of fatty alcohols, fatty aldehydes, fatty acids or isopropanol in the cytosol can be dependent upon the availability of sufficient carbon and reducing equivalents. Therefore, without being bound to any particular theory of operation, increasing the redox ratio of NAD(P)H to NAD(P) can help drive the MI-FAE cycle, MD-FAE cycle, FAACPE cycle, termination pathway and/or isopropanol pathway in the forward direction. Methods for increasing the redox ratio of NAD(P)H to NAD(P) include limiting respi-

ration, attenuating or disrupting competing pathways that produce reduced byproducts such as ethanol and glycerol, attenuating or eliminating the use of NADH by NADH dehydrogenases, and attenuating or eliminating redox shuttles between compartments.

One exemplary method to provide an increased number of reducing equivalents, such as NAD(P)H, for enabling the formation of fatty alcohols, fatty aldehydes, fatty acids or isopropanol is to constrain the use of such reducing equivalents during respiration. Respiration can be limited by: reducing the availability of oxygen, attenuating NADH dehydrogenases and/or cytochrome oxidase activity, attenuating G3P dehydrogenase, and/or providing excess glucose to Crabtree positive organisms.

Restricting oxygen availability by culturing the non-naturally occurring eukaryotic organisms in a fermenter is one example for limiting respiration and thereby increasing the ratio of NAD(P)H to NAD(P). The ratio of NAD(P)H/NAD(P) increases as culture conditions become more anaerobic, with completely anaerobic conditions providing the highest ratios of the reduced cofactors to the oxidized ones. For example, it has been reported that the ratio of NADH/NAD=0.02 in aerobic conditions and 0.75 in anaerobic conditions in *E. coli* (de Graes et al, J Bacteriol 181: 2351-57 (1999)).

Respiration can also be limited by reducing expression or activity of NADH dehydrogenases and/or cytochrome oxidases in the cell under aerobic conditions. In this case, respiration can be limited by the capacity of the electron transport chain. Such an approach has been used to enable anaerobic metabolism of *E. coli* under completely aerobic conditions (Portnoy et al, AEM 74:7561-9 (2008)). *S. cerevisiae* can oxidize cytosolic NADH directly using external NADH dehydrogenases, encoded by NDE1 and NDE2. One such NADH dehydrogenase in *Yarrowia lipolytica* is encoded by NDH2 (Kerscher et al, J Cell Sci 112:2347-54 (1999)). These and other NADH dehydrogenase enzymes are listed in the table below.

Protein	GenBank ID	GI number	Organism
NDE1	NP_013865.1	6323794	<i>Saccharomyces cerevisiae</i> s288c
NDE2	NP_010198.1	6320118	<i>Saccharomyces cerevisiae</i> s288c
NDH2	AJ006852.1	3718004	<i>Yarrowia lipolytica</i>
ANI_1_610074	XP_001392541.2	317030427	<i>Aspergillus niger</i>
ANI_1_2462094	XP_001394893.2	317033119	<i>Aspergillus niger</i>
KLLA0E21891g	XP_454942.1	50309857	<i>Kluyveromyces lactis</i>
KLLA0C06336g	XP_452480.1	50305045	<i>Kluyveromyces lactis</i>
NDE1	XP_720034.1	68471982	<i>Candida albicans</i>
NDE2	XP_717986.1	68475826	<i>Candida albicans</i>

Cytochrome oxidases of *Saccharomyces cerevisiae* include the COX gene products. COX1-3 are the three core subunits encoded by the mitochondrial genome, whereas COX4-13 are encoded by nuclear genes. Attenuation or disruption of any of the cytochrome genes results in a decrease or block in respiratory growth (Hermann and Funes, Gene 354:43-52 (2005)). Cytochrome oxidase genes in other organisms can be inferred by sequence homology.

Protein	GenBank ID	GI number	Organism
COX1	CAA09824.1	4160366	<i>Saccharomyces cerevisiae</i> s288c
COX2	CAA09845.1	4160387	<i>Saccharomyces cerevisiae</i> s288c
COX3	CAA09846.1	4160389	<i>Saccharomyces cerevisiae</i> s288c

-continued

Protein	GenBank ID	GI number	Organism
COX4	NP_011328.1	6321251	<i>Saccharomyces cerevisiae</i> s288c
COX5A	NP_014346.1	6324276	<i>Saccharomyces cerevisiae</i> s288c
COX5B	NP_012155.1	6322080	<i>Saccharomyces cerevisiae</i> s288c
COX6	NP_011918.1	6321842	<i>Saccharomyces cerevisiae</i> s288c
COX7	NP_013983.1	6323912	<i>Saccharomyces cerevisiae</i> s288c
COX8	NP_013499.1	6323427	<i>Saccharomyces cerevisiae</i> s288c
COX9	NP_010216.1	6320136	<i>Saccharomyces cerevisiae</i> s288c
COX12	NP_013139.1	6323067	<i>Saccharomyces cerevisiae</i> s288c
COX13	NP_011324.1	6321247	<i>Saccharomyces cerevisiae</i> s288c

Cytosolic NADH can also be oxidized by the respiratory chain via the G3P dehydrogenase shuttle, consisting of cytosolic NADH-linked G3P dehydrogenase and a membrane-bound G3P:ubiquinone oxidoreductase. The deletion or attenuation of G3P dehydrogenase enzymes will also prevent the oxidation of NADH for respiration. Enzyme candidates encoding these enzymes are described herein.

Additionally, in Crabtree positive organisms, fermentative metabolism can be achieved in the presence of excess of glucose. For example, *S. cerevisiae* makes ethanol even under aerobic conditions. The formation of ethanol and glycerol can be reduced/eliminated and replaced by the production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol in a Crabtree positive organism by feeding excess glucose to the Crabtree positive organism. In another aspect, provided herein is a method for producing fatty alcohols, fatty aldehydes, fatty acids, isopropanol comprising culturing a non-naturally occurring eukaryotic organism under conditions and for a sufficient period of time to produce fatty alcohol, fatty aldehyde, fatty acid or isopropanol, wherein the eukaryotic organism is a Crabtree positive organism, and wherein the eukaryotic organism is in a culture medium comprising excess glucose.

Preventing formation of reduced fermentation byproducts will increase the availability of both carbon and reducing equivalents for fatty alcohol, fatty aldehyde, fatty acid or isopropanol production. The two key reduced byproducts under anaerobic and microaerobic conditions are ethanol and glycerol. Ethanol is typically formed from pyruvate in two enzymatic steps catalyzed by pyruvate decarboxylase and ethanol dehydrogenase. Glycerol is formed from the glycolytic intermediate dihydroxyacetone phosphate by the enzymes glycerol-3-phosphate dehydrogenase and glycerol-3-phosphate phosphatase. Attenuation of one or more of these enzyme activities will increase the yield of fatty alcohols, fatty aldehydes, fatty acids or isopropanol. Strain engineering strategies for reducing or eliminating ethanol and glycerol formation are described herein.

Yeast such as *S. cerevisiae* can produce glycerol to allow for regeneration of NAD(P) under anaerobic conditions. Another way to reduce or eliminate glycerol production is by oxygen-limited cultivation (Bakker et al, supra). Glycerol formation only sets in when the specific oxygen uptake rates of the cells decrease below the rate that is required to reoxidize the NADH formed in biosynthesis.

In addition to the redox sinks listed above, malate dehydrogenase can potentially draw away reducing equivalents when it functions in the reductive direction. Several redox shuttles believed to be functional in *S. cerevisiae* utilize this enzyme to transfer reducing equivalents between the cytosol and the mitochondria. This transfer of redox can be prevented by attenuating malate dehydrogenase and/or malic enzyme activity. The redox shuttles that can be blocked by the attenuation of mdh include (i) malate-aspartate shuttle,

(ii) malate-oxaloacetate shuttle, and (iii) malate-pyruvate shuttle. Genes encoding malate dehydrogenase and malic enzymes are listed in the table below.

Protein	GenBank ID	GI Number	Organism
MDH1	NP_012838.1	6322765	<i>Saccharomyces cerevisiae</i>
MDH2	NP_014515.2	116006499	<i>Saccharomyces cerevisiae</i>
MDH3	NP_010205.1	6320125	<i>Saccharomyces cerevisiae</i>
MAE1	NP_012896.1	6322823	<i>Saccharomyces cerevisiae</i>
MDH1	XP_722674.1	68466384	<i>Candida albicans</i>
MDH2	XP_718638.1	68474530	<i>Candida albicans</i>
MAE1	XP_716669.1	68478574	<i>Candida albicans</i>
KLLA0F25960g	XP_456236.1	50312405	<i>Kluyveromyces lactis</i>
KLLA0E18635g	XP_454793.1	50309563	<i>Kluyveromyces lactis</i>
KLLA0E07525g	XP_454288.1	50308571	<i>Kluyveromyces lactis</i>
YALI0D16753p	XP_502909.1	50550873	<i>Yarrowia lipolytica</i>
YALI0E18634p	XP_504112.1	50553402	<i>Yarrowia lipolytica</i>
ANI_1_268064	XP_001391302.1	145237310	<i>Aspergillus niger</i>
ANI_1_12134	XP_001396546.1	145250065	<i>Aspergillus niger</i>
ANI_1_22104	XP_001395105.2	317033225	<i>Aspergillus niger</i>

Overall, disruption or attenuation of the aforementioned sinks for redox either individually or in combination with the other redox sinks can eliminate or lower the use of reducing power for respiration or byproduct formation. It has been reported that the deletion of the external NADH dehydrogenases (NDE1 and NDE2) and the mitochondrial G3P dehydrogenase (GUT2) almost completely eliminates cytosolic NAD⁺ regeneration in *S. cerevisiae* (Overkamp et al, *J Bacterial* 182:2823-30 (2000)).

Microorganisms of the invention produce fatty alcohols, fatty aldehydes fatty acids or isopropanol and optionally secrete the fatty alcohols, fatty aldehydes fatty acids or isopropanol into the culture medium. *S. cerevisiae*, *Yarrowia lipolytica* and *E. coli* harboring heterologous fatty alcohol forming activities accumulated fatty alcohols intracellularly; however fatty alcohols were not detected in the culture medium (Behrouzian et al, United States Patent Application 20100298612). The introduction of fatty acyl-CoA reductase enzymes with improved activity resulted in higher levels of fatty alcohol secreted into the culture media. Alternately, introduction of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol transporter or transport system can improve extracellular accumulation of fatty alcohols, fatty aldehydes or fatty acids. Exemplary transporters are listed in the table below.

Protein	GenBank ID	GI Number	Organism
Fatp	NP_524723.2	24583463	<i>Drosophila melanogaster</i>
AY161280.1:45..1757	AAN73268.1	34776949	<i>Rhodococcus erythropolis</i>
acrA	CAF23274.1	46399825	<i>Candidatus Protochlamydia amoebophila</i>
acrB	CAF23275.1	46399826	<i>Candidatus Protochlamydia amoebophila</i>
CER5	AY734542.1	52354013	<i>Arabidopsis thaliana</i>
AmiS2	JC5491	7449112	<i>Rhodococcus sp.</i>
ANI_1_1160064	XP_001391993.1	145238692	<i>Aspergillus niger</i>
YALI0E1601g	XP_504004.1	50553188	<i>Yarrowia lipolytica</i>

Thus, in some embodiments, the invention provides a non-naturally occurring microbial organism as disclosed

herein having one or more gene disruptions, wherein the one or more gene disruptions occur in endogenous genes encoding proteins or enzymes involved in: native production of ethanol, glycerol, acetate, formate, lactate, CO₂, fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native degradation of a MI-FAE cycle intermediate, a MD-FAE cycle intermediate, a FAACPE cycle intermediate or a termination pathway intermediate by the microbial organism, the one or more gene disruptions confer increased production of a fatty alcohol, fatty aldehyde or fatty acid in the microbial organism. Accordingly, the protein or enzyme can be a fatty acid synthase, an acetyl-CoA carboxylase, a biotin:apoenzyme ligase, an acyl carrier protein, a thioesterase, an acyltransferase, an ACP malonyl-transfemse, a fatty acid elongase, an acyl-CoA synthetase, an acyl-CoA transferase, an acyl-CoA hydrolase, a pyruvate decarboxylase, a lactate dehydrogenase, an alcohol dehydrogenase, an acid-forming aldehyde dehydrogenases, an acetate kinase, a phosphotransacetylase, a pyruvate oxidase, a glycerol-3-phosphate dehydrogenase, a glycerol-3-phosphate phosphatase, a mitochondrial pyruvate carrier, a peroxisomal fatty acid transporter, a peroxisomal acyl-CoA transporter, a peroxisomal carnitine/acylcarnitine transferase, an acyl-CoA oxidase, or an acyl-CoA binding protein. In some aspects, the one or more gene disruptions include a deletion of the one or more genes.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein one or more enzymes of the MI-FAE cycle, the MD-FAE cycle, the FAACPE cycle or the termination pathway preferentially react with an NADH cofactor or have reduced preference for reacting with an NAD(P)H cofactor. For example, the one or more enzymes of the MI-FAE cycle can be a 3-ketoacyl-CoA reductase or an enoyl-CoA reductase. As another example, the one or more enzymes of the FAACPE cycle can be a 3-ketoacyl-ACP reductase or an enoyl-ACP reductase. For the termination pathway, the one or more enzymes can be an acyl-CoA reductase (aldehyde forming), an alcohol dehydrogenase, an acyl-CoA reductase (alcohol forming), an aldehyde decarbonylase, an acyl-ACP reductase, an aldehyde dehydrogenase (acid forming) or a carboxylic acid reductase.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein having one or more gene disruptions in genes encoding proteins or enzymes that result in an increased ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism following the disruptions. Accordingly, the gene encoding a protein or enzyme that results in an increased ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism following the disruptions can be an NADH dehydrogenase, a cytochrome oxidase, a G3P dehydrogenase, G3P phosphatase, an alcohol dehydrogenase, a pyruvate decarboxylase, an aldehyde dehydrogenase (acid forming), a lactate dehydrogenase, a glycerol-3-phosphate dehydrogenase, a glycerol-3-phosphate:quinone oxidoreductase, a malic enzyme and a malate dehydrogenase. In some aspects, the one or more gene disruptions include a deletion of the one or more genes.

In some embodiments, the non-naturally occurring microbial organism of the invention is Crabtree positive and is in culture medium comprising excess glucose. In such conditions, as described herein, the microbial organism can result in increasing the ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein having at least one exogenous nucleic acid encoding an extracellular transporter or an extracellular transport system for a fatty alcohol, fatty aldehyde or fatty acid of the invention.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein one or more endogenous enzymes involved in: native production of ethanol, glycerol, acetate, formate, lactate, CO₂, fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native degradation of a MI-FAE cycle intermediate, a MD-FAE cycle intermediate, a FAACPE cycle intermediate or a termination pathway intermediate by said microbial organism, has attenuated enzyme activity or expression levels. Accordingly, the endogenous enzyme can be a fatty acid synthase, an acetyl-CoA carboxylase, a biotin:apoenzyme ligase, an acyl carrier protein, a thioesterase, an acyltransferase, an ACP malonyl-transferase, a fatty acid elongase, an acyl-CoA synthetase, an acyl-CoA transferase, an acyl-CoA hydrolase, a pyruvate decarboxylase, a lactate dehydrogenase, an alcohol dehydrogenase, an acid-forming aldehyde dehydrogenases, an acetate kinase, a phosphotransacetylase, a pyruvate oxidase, a glycerol-3-phosphate dehydrogenase, a glycerol-3-phosphate phosphatase, a mitochondrial pyruvate carrier, a peroxisomal fatty acid transporter, a peroxisomal acyl-CoA transporter, a peroxisomal carnitine/acylcarnitine transferase, an acyl-CoA oxidase, or an acyl-CoA binding protein.

In some embodiments, the invention provides a non-naturally occurring microbial organism as described herein, wherein one or more endogenous enzymes involved in the oxidation of NAD(P)H or NADH, has attenuated enzyme activity or expression levels. Accordingly, the one or more endogenous enzymes can be a NADH dehydrogenase, a cytochrome oxidase, a G3P dehydrogenase, G3P phosphatase, an alcohol dehydrogenase, a pyruvate decarboxylase, an aldehyde dehydrogenase (acid forming), a lactate dehydrogenase, a glycerol-3-phosphate dehydrogenase, a glycerol-3-phosphate:quinone oxidoreductase, a malic enzyme and a malate dehydrogenase.

The non-naturally occurring microbial organisms of the invention can contain stable genetic alterations, which refers to microorganisms that can be cultured for greater than five generations without loss of the alteration. Generally, stable genetic alterations include modifications that persist greater than 10 generations, particularly stable modifications will persist more than about 25 generations, and more particularly, stable genetic modifications will be greater than 50 generations, including indefinitely.

In the case of gene disruptions, a particularly useful stable genetic alteration is a gene deletion. The use of a gene deletion to introduce a stable genetic alteration is particularly useful to reduce the likelihood of a reversion to a phenotype prior to the genetic alteration. For example, stable growth-coupled production of a biochemical can be achieved, for example, by deletion of a gene encoding an enzyme catalyzing one or more reactions within a set of metabolic modifications. The stability of growth-coupled production of a biochemical can be further enhanced through multiple deletions, significantly reducing the likelihood of multiple compensatory reversions occurring for each disrupted activity.

Also provided is a method of producing a non-naturally occurring microbial organisms having stable growth-coupled production of fatty alcohol, fatty aldehyde, fatty

acid or isopropanol. The method can include identifying in silico a set of metabolic modifications that increase production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, for example, increase production during exponential growth; genetically modifying an organism to contain the set of metabolic modifications that increase production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, and culturing the genetically modified organism. If desired, culturing can include adaptively evolving the genetically modified organism under conditions requiring production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. The methods of the invention are applicable to bacterium, yeast and fungus as well as a variety of other cells and microorganism, as disclosed herein.

Thus, the invention provides a non-naturally occurring microbial organism comprising one or more gene disruptions that confer increased production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In one embodiment, the one or more gene disruptions confer growth-coupled production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, and can, for example, confer stable growth-coupled production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. In another embodiment, the one or more gene disruptions can confer obligatory coupling of fatty alcohol, fatty aldehyde, fatty acid or isopropanol production to growth of the microbial organism. Such one or more gene disruptions reduce the activity of the respective one or more encoded enzymes.

The non-naturally occurring microbial organism can have one or more gene disruptions included in a gene encoding an enzyme or protein disclosed herein. As disclosed herein, the one or more gene disruptions can be a deletion. Such non-naturally occurring microbial organisms of the invention include bacteria, yeast, fungus, or any of a variety of other microorganisms applicable to fermentation processes, as disclosed herein.

Thus, the invention provides a non-naturally occurring microbial organism, comprising one or more gene disruptions, where the one or more gene disruptions occur in genes encoding proteins or enzymes where the one or more gene disruptions confer increased production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol in the organism. The production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be growth-coupled or not growth-coupled. In a particular embodiment, the production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol can be obligatorily coupled to growth of the organism, as disclosed herein.

The invention provides non naturally occurring microbial organisms having genetic alterations such as gene disruptions that increase production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, for example, growth-coupled production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. Product production can be, for example, obligatorily linked to the exponential growth phase of the microorganism by genetically altering the metabolic pathways of the cell, as disclosed herein. The genetic alterations can increase the production of the desired product or even make the desired product an obligatory product during the growth phase. Metabolic alterations or transformations that result in increased production and elevated levels of fatty alcohol, fatty aldehyde, fatty acid or isopropanol biosynthesis are exemplified herein. Each alteration corresponds to the requisite metabolic reaction that should be functionally disrupted. Functional disruption of all reactions within one or more of the pathways can result in the increased production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol by the engineered strain during the growth phase.

Each of these non-naturally occurring alterations result in increased production and an enhanced level of fatty alcohol, fatty aldehyde, fatty acid or isopropanol production, for example, during the exponential growth phase of the microbial organism, compared to a strain that does not contain such metabolic alterations, under appropriate culture conditions. Appropriate conditions include, for example, those disclosed herein, including conditions such as particular carbon sources or reactant availabilities and/or adaptive evolution.

Given the teachings and guidance provided herein, those skilled in the art will understand that to introduce a metabolic alteration such as attenuation of an enzyme, it can be necessary to disrupt the catalytic activity of the one or more enzymes involved in the reaction. Alternatively, a metabolic alteration can include disrupting expression of a regulatory protein or cofactor necessary for enzyme activity or maximal activity. Furthermore, genetic loss of a cofactor necessary for an enzymatic reaction can also have the same effect as a disruption of the gene encoding the enzyme. Disruption can occur by a variety of methods including, for example, deletion of an encoding gene or incorporation of a genetic alteration in one or more of the encoding gene sequences. The encoding genes targeted for disruption can be one, some, or all of the genes encoding enzymes involved in the catalytic activity. For example, where a single enzyme is involved in a targeted catalytic activity, disruption can occur by a genetic alteration that reduces or eliminates the catalytic activity of the encoded gene product. Similarly, where the single enzyme is multimeric, including heteromeric, disruption can occur by a genetic alteration that reduces or destroys the function of one or all subunits of the encoded gene products. Destruction of activity can be accomplished by loss of the binding activity of one or more subunits required to form an active complex, by destruction of the catalytic subunit of the multimeric complex or by both. Other functions of multimeric protein association and activity also can be targeted in order to disrupt a metabolic reaction of the invention. Such other functions are well known to those skilled in the art. Similarly, a target enzyme activity can be reduced or eliminated by disrupting expression of a protein or enzyme that modifies and/or activates the target enzyme, for example, a molecule required to convert an apoenzyme to a holoenzyme. Further, some or all of the functions of a single polypeptide or multimeric complex can be disrupted according to the invention in order to reduce or abolish the catalytic activity of one or more enzymes involved in a reaction or metabolic modification of the invention. Similarly, some or all of enzymes involved in a reaction or metabolic modification of the invention can be disrupted so long as the targeted reaction is reduced or eliminated.

Given the teachings and guidance provided herein, those skilled in the art also will understand that an enzymatic reaction can be disrupted by reducing or eliminating reactions encoded by a common gene and/or by one or more orthologs of that gene exhibiting similar or substantially the same activity. Reduction of both the common gene and all orthologs can lead to complete abolishment of any catalytic activity of a targeted reaction. However, disruption of either the common gene or one or more orthologs can lead to a reduction in the catalytic activity of the targeted reaction sufficient to promote coupling of growth to product biosynthesis. Exemplified herein are both the common genes encoding catalytic activities for a variety of metabolic modifications as well as their orthologs. Those skilled in the art will understand that disruption of some or all of the genes

encoding a enzyme of a targeted metabolic reaction can be practiced in the methods of the invention and incorporated into the non-naturally occurring microbial organisms of the invention in order to achieve the increased production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol or growth-coupled product production.

Given the teachings and guidance provided herein, those skilled in the art also will understand that enzymatic activity or expression can be attenuated using well known methods. Reduction of the activity or amount of an enzyme can mimic complete disruption of a gene if the reduction causes activity of the enzyme to fall below a critical level that is normally required for a pathway to function. Reduction of enzymatic activity by various techniques rather than use of a gene disruption can be important for an organism's viability. Methods of reducing enzymatic activity that result in similar or identical effects of a gene disruption include, but are not limited to: reducing gene transcription or translation; destabilizing mRNA, protein or catalytic RNA; and mutating a gene that affects enzyme activity or kinetics (See, Sambrook et al., *Molecular Cloning: A Laboratory Manual*, Third Ed., Cold Spring Harbor Laboratory, New York (2001); and Ausubel et al., *Current Protocols in Molecular Biology*, John Wiley and Sons, Baltimore, MD (1999). Natural or imposed regulatory controls can also accomplish enzyme attenuation including: promoter replacement (See, Wang et al., *Mol. Biotechnol.* 52(2):300-308 (2012)); loss or alteration of transcription factors (Dietrick et al., *Annu. Rev. Biochem.* 79:563-590 (2010); and Simicevic et al., *Mol. Biosyst.* 6(3):462-468 (2010)); introduction of inhibitory RNAs or peptides such as siRNA, antisense RNA, RNA or peptide/small-molecule binding aptamers, ribozymes, aptazymes and riboswitches (Wieland et al., *Methods* 56(3): 351-357 (2012); O'Sullivan, *Anal. Bioanal. Chem.* 372(1): 44-48 (2002); and Lee et al., *Curr. Opin. Biotechnol.* 14(5): 505-511(2003)); and addition of drugs or other chemicals that reduce or disrupt enzymatic activity such as an enzyme inhibitor, an antibiotic or a target-specific drug.

One skilled in the art will also understand and recognize that attenuation of an enzyme can be done at various levels. For example, at the gene level, a mutation causing a partial or complete null phenotype, such as a gene disruption, or a mutation causing epistatic genetic effects that mask the activity of a gene product (Miko, *Nature Education* 1(1) (2008)), can be used to attenuate an enzyme. At the gene expression level, methods for attenuation include: coupling transcription to an endogenous or exogenous inducer, such as isopropylthio- β -galactoside (IPTG), then adding low amounts of inducer or no inducer during the production phase (Donovan et al., *J. Ind. Microbiol.* 16(3):145-154 (1996); and Hansen et al., *Curr. Microbiol.* 36(6):341-347 (1998)); introducing or modifying a positive or a negative regulator of a gene; modify histone acetylation/deacetylation in a eukaryotic chromosomal region where a gene is integrated (Yang et al., *Curr. Opin. Genet. Dev.* 13(2):143-153 (2003) and Kurdistani et al., *Nat. Rev. Mol. Cell Biol.* 4(4):276-284 (2003)); introducing a transposition to disrupt a promoter or a regulatory gene (Bleykasten-Broshans et al., *C. R. Biol.* 33(8-9):679-686 (2011); and McCue et al., *PLoS Genet.* 8(2):e1002474 (2012)); flipping the orientation of a transposable element or promoter region so as to modulate gene expression of an adjacent gene (Wang et al., *Genetics* 120(4):875-885 (1988); Hayes, *Annu. Rev. Genet.* 37:3-29 (2003); in a diploid organism, deleting one allele resulting in loss of heterozygosity (Daigaku et al., *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis* 600(1-2):177-183 (2006)); introducing nucleic

acids that increase RNA degradation (Houseley et al., *Cell*, 136(4):763-776 (2009); or in bacteria, for example, introduction of a transfer-messenger RNA (tmRNA) tag, which can lead to RNA degradation and ribosomal stalling (Sunohara et al., *RNA* 10(3):378-386 (2004); and Sunohara et al., *J. Biol. Chem.* 279:15368-15375 (2004)). At the translational level, attenuation can include: introducing rare codons to limit translation (Angov, *Biotechnol. J.* 6(6):650-659 (2011)); introducing RNA interference molecules that block translation (Castel et al., *Nat. Rev. Genet.* 14(2):100-112 (2013); and Kawasaki et al., *Curr. Opin. Mol. Ther.* 7(2): 125-131(2005); modifying regions outside the coding sequence, such as introducing secondary structure into an untranslated region (UTR) to block translation or reduce efficiency of translation (Ringner et al., *PLoS Comput. Biol.* 1(7):e72 (2005)); adding RNAase sites for rapid transcript degradation (Pasquinelli, *Nat. Rev. Genet.* 13(4):271-282 (2012); and Arraiano et al., *FEMS Microbiol. Rev.* 34(5): 883-932 (2010); introducing antisense RNA oligomers or antisense transcripts (Nashizawa et al., *Front. Biosci.* 17:938-958 (2012)); introducing RNA or peptide aptamers, ribozymes, aptazymes, riboswitches (Wieland et al., *Methods* 56(3):351-357 (2012); O'Sullivan, *Anal. Bioanal. Chem.* 372(1):44-48 (2002); and Lee et al., *Curr. Opin. Biotechnol.* 14(5):505-511(2003)); or introducing translational regulatory elements involving RNA structure that can prevent or reduce translation that can be controlled by the presence or absence of small molecules (Araujo et al., *Comparative and Functional Genomics*, Article ID 475731, 8 pages (2012)). At the level of enzyme localization and/or longevity, enzyme attenuation can include: adding a degradation tag for faster protein turnover (Hochstrasser, *Annual Rev. Genet.* 30:405-439 (1996); and Yuan et al., *PLoS One* 8(4):e62529 (2013)); or adding a localization tag that results in the enzyme being secreted or localized to a subcellular compartment in a eukaryotic cell, where the enzyme would not be able to react with its normal substrate (Nakai et al., *Genomics* 14(4):897-911 (1992); and Russell et al., *J. Bact.* 189(21):7581-7585 (2007)). At the level of post-translational regulation, enzyme attenuation can include: increasing intracellular concentration of known inhibitors; or modifying post-translational modified sites (Mann et al., *Nature Biotech.* 21:255-261 (2003)). At the level of enzyme activity, enzyme attenuation can include: adding an endogenous or an exogenous inhibitor, such as an enzyme inhibitor, an antibiotic or a target-specific drug, to reduce enzyme activity; limiting availability of essential cofactors, such as vitamin B12, for an enzyme that requires the cofactor; chelating a metal ion that is required for enzyme activity; or introducing a dominant negative mutation. The applicability of a technique for attenuation described above can depend upon whether a given host microbial organism is prokaryotic or eukaryotic, and it is understood that a determination of what is the appropriate technique for a given host can be readily made by one skilled in the art.

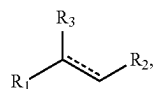
In some embodiments, microaerobic designs can be used based on the growth-coupled formation of the desired product. To examine this, production cones can be constructed for each strategy by first maximizing and, subsequently minimizing the product yields at different rates of biomass formation feasible in the network. If the rightmost boundary of all possible phenotypes of the mutant network is a single point, it implies that there is a unique optimum yield of the product at the maximum biomass formation rate possible in the network. In other cases, the rightmost boundary of the feasible phenotypes is a vertical line, indicating that at the point of maximum biomass the network can make any

amount of the product in the calculated range, including the lowest amount at the bottommost point of the vertical line. Such designs are given a low priority.

The fatty alcohol, fatty aldehyde, fatty acid or isopropanol-production strategies identified in the various tables disclosed herein can be disrupted to increase production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. Accordingly, the invention also provides a non-naturally occurring microbial organism having metabolic modifications coupling fatty alcohol, fatty aldehyde, fatty acid or isopropanol production to growth of the organism, where the metabolic modifications includes disruption of one or more genes selected from the genes encoding proteins and/or enzymes shown in the various tables disclosed herein.

Each of the strains can be supplemented with additional deletions if it is determined that the strain designs do not sufficiently increase the production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol and/or couple the formation of the product with biomass formation. Alternatively, some other enzymes not known to possess significant activity under the growth conditions can become active due to adaptive evolution or random mutagenesis. Such activities can also be knocked out. However, the list of gene deletion disclosed herein allows the construction of strains exhibiting high-yield production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol, including growth-coupled production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol.

In some embodiments, the invention provides a method for producing a compound of Formula (I):



(I)

wherein R_1 is C_{1-24} linear alkyl; R_2 is CH_2OH , CHO , or COOH ; R_3 is H , OH , or $\text{oxo} (=O)$; and --- represents a single or double bond with the proviso that the valency of the carbon atom to which R_3 is attached is four, comprising culturing a non-naturally occurring microbial organism described herein under conditions and for a sufficient period of time to produce the compound of Formula (I), wherein the non-naturally occurring microbial organism has one or more gene disruptions, wherein the one or more gene disruptions occur in endogenous genes encoding proteins or enzymes involved in: native production of ethanol, glycerol, acetate, formate, lactate, CO_2 , fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native degradation of a MI-FAE cycle intermediate, a MD-FAE cycle intermediate or a termination pathway intermediate by the microbial organism, the one or more gene disruptions confer increased production of a fatty alcohol, fatty aldehyde or fatty acid in the microbial organism. Accordingly, the protein or enzyme can be a fatty acid synthase, an acetyl-CoA carboxylase, a biotin:apoenzyme ligase, an acyl carrier protein, a thioesterase, an acyltransferases, an ACP malonyltransferase, a fatty acid elongase, an acyl-CoA synthetase, an acyl-CoA transferase, an acyl-CoA hydrolase, a pyruvate decarboxylase, a lactate dehydrogenase, an alcohol dehydrogenase, an acid-forming aldehyde dehydrogenases, an acetate kinase, a phosphotransacetylase, a pyruvate oxidase, a glycerol-

3-phosphate dehydrogenase, a glycerol-3-phosphate phosphatase, a mitochondrial pyruvate carrier, a peroxisomal fatty acid transporters, a peroxisomal acyl-CoA transporters, a peroxisomal carnitine/acylcarnitine transferases, an acyl-CoA oxidase, or an acyl-CoA binding protein. In some aspects, the one or more gene disruptions include a deletion of the one or more genes.

In some embodiments, the invention provides a method for producing a fatty alcohol, fatty aldehyde or fatty acid using a non-naturally occurring microbial organism as described herein, wherein one or more enzymes of the MI-FAE cycle, MD-FAE cycle, FAACPE cycle or the termination pathway preferentially react with an NADH cofactor or have reduced preference for reacting with an NAD(P)H cofactor. For example, the one or more enzymes of the MI-FAE cycle or MD-FAE cycle can be a 3-ketoacyl-CoA reductase or an enoyl-CoA reductase. As another example, the one or more enzymes of the FAACPE cycle can be a 3-ketoacyl-ACP reductase or an enoyl-ACP reductase. For the termination pathway, the one or more enzymes can be an acyl-CoA reductase (aldehyde forming), an alcohol dehydrogenase, an acyl-CoA reductase (alcohol forming), an aldehyde decarbonylase, an acyl-ACP reductase, an aldehyde dehydrogenase (acid forming) or a carboxylic acid reductase.

In some embodiments, the invention provides a method for producing a fatty alcohol, fatty aldehyde or fatty acid using a non-naturally occurring microbial organism as described herein having one or more gene disruptions in genes encoding proteins or enzymes that result in an increased ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism following the disruptions. Accordingly, the gene encoding a protein or enzyme that results in an increased ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism following the disruptions can be an NADH dehydrogenase, a cytochrome oxidase, a glycerol-3-phosphate (G3P) dehydrogenase, a glycerol-3-phosphate (G3P) phosphatase, an alcohol dehydrogenase, a pyruvate decarboxylase, an aldehyde dehydrogenase (acid forming), a lactate dehydrogenase, a glycerol-3-phosphate dehydrogenase, a glycerol-3-phosphate:quinone oxidoreductase, a malic enzyme and a malate dehydrogenase. In some aspects, the one or more gene disruptions include a deletion of the one or more genes.

In some embodiments, the invention provides a method for producing a fatty alcohol, fatty aldehyde or fatty acid using a non-naturally occurring microbial organism of the invention that is Crabtree positive and is in culture medium comprising excess glucose. In such conditions, as described herein, the microbial organism can result in increasing the ratio of NAD(P)H to NAD(P) present in the cytosol of the microbial organism.

In some embodiments, the invention provides a method for producing a fatty alcohol, fatty aldehyde or fatty acid using a non-naturally occurring microbial organism as described herein having at least one exogenous nucleic acid encoding an extracellular transporter or an extracellular transport system for a fatty alcohol, fatty aldehyde or fatty acid of the invention.

In some embodiments, the invention provides a method for producing a fatty alcohol, fatty aldehyde or fatty acid using a non-naturally occurring microbial organism as described herein, wherein one or more endogenous enzymes involved in: native production of ethanol, glycerol, acetate, formate, lactate, CO₂, fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native

degradation of a MI-FAE cycle intermediate, a MD-FAE cycle intermediate, a FAACPE cycle intermediate or a termination pathway intermediate by said microbial organism, has attenuated enzyme activity or expression levels. Accordingly, the endogenous enzyme can be a fatty acid synthase, an acetyl-CoA carboxylase, a biotin:apoenzyme ligase, an acyl carrier protein, a thioesterase, an acyltransferase, an ACP malonyltransferase, a fatty acid elongase, an acyl-CoA synthetase, an acyl-CoA transferase, an acyl-CoA hydrolase, a pyruvate decarboxylase, a lactate dehydrogenase, an alcohol dehydrogenase, an acid-forming aldehyde dehydrogenases, an acetate kinase, a phosphotransacetylase, a pyruvate oxidase, a glycerol-3-phosphate dehydrogenase, a glycerol-3-phosphate phosphatase, a mitochondrial pyruvate carrier, a peroxisomal fatty acid transporter, a peroxisomal acyl-CoA transporter, a peroxisomal carnitine/acylcarnitine transferase, an acyl-CoA oxidase, and an acyl-CoA binding protein.

In some embodiments, the invention provides a method for producing a fatty alcohol, fatty aldehyde or fatty acid using a non-naturally occurring microbial organism as described herein, wherein one or more endogenous enzymes involved in the oxidation of NAD(P)H or NADH, has attenuated enzyme activity or expression levels. Accordingly, the one or more endogenous enzymes can be NADH dehydrogenase, a cytochrome oxidase, a glycerol-3-phosphate dehydrogenase, glycerol-3-phosphate phosphatase, an alcohol dehydrogenase, a pyruvate decarboxylase, an aldehyde dehydrogenase (acid forming), a lactate dehydrogenase, a glycerol-3-phosphate dehydrogenase, a glycerol-3-phosphate:quinone oxidoreductase, a malic enzyme and a malate dehydrogenase.

A fatty alcohol, fatty aldehyde or fatty acid can be harvested or isolated at any time point during the culturing of the microbial organism, for example, in a continuous and/or near-continuous culture period, as disclosed herein. Generally, the longer the microorganisms are maintained in a continuous and/or near-continuous growth phase, the proportionally greater amount of fatty alcohol, fatty aldehyde or fatty acid can be produced.

Therefore, the invention additionally provides a method for producing fatty alcohol, fatty aldehyde or fatty acid that includes culturing a non-naturally occurring microbial organism having one or more gene disruptions, as disclosed herein. The disruptions can occur in one or more genes encoding an enzyme that increases production of fatty alcohol, fatty aldehyde or fatty acid, including optionally coupling fatty alcohol, fatty aldehyde or fatty acid production to growth of the microorganism when the gene disruption reduces or eliminates an activity of the enzyme. For example, the disruptions can confer stable growth-coupled production of fatty alcohol, fatty aldehyde or fatty acid onto the non-naturally microbial organism.

In some embodiments, the gene disruption can include a complete gene deletion. In some embodiments other methods to disrupt a gene include, for example, frameshifting by omission or addition of oligonucleotides or by mutations that render the gene inoperable. One skilled in the art will recognize the advantages of gene deletions, however, because of the stability it confers to the non-naturally occurring organism from reverting to a parental phenotype in which the gene disruption has not occurred. In particular, the gene disruptions are selected from the gene sets as disclosed herein.

Once computational predictions are made of gene sets for disruption to increase production of fatty alcohol, fatty aldehyde or fatty acid, the strains can be constructed,

evolved, and tested. Gene disruptions, including gene deletions, are introduced into host organism by methods well known in the art. A particularly useful method for gene disruption is by homologous recombination, as disclosed herein.

The engineered strains can be characterized by measuring the growth mte, the substrate uptake mte, and/or the product/byproduct secretion mte. Cultures can be grown and used as inoculum for a fresh batch culture for which measurements are taken during exponential growth. The growth mte can be determined by measuring optical density using a spectrophotometer (A600). Concentrations of glucose and other organic acid byproducts in the culture supernatant can be determined by well known methods such as HPLC, GC-MS or other well known analytical methods suitable for the analysis of the desired product, as disclosed herein, and used to calculate uptake and secretion rates.

Strains containing gene disruptions can exhibit suboptimal growth mtes until their metabolic networks have adjusted to their missing functionalities. To assist in this adjustment, the strains can be adaptively evolved. By subjecting the strains to adaptive evolution, cellular growth rate becomes the primary selection pressure and the mutant cells are compelled to reallocate their metabolic fluxes in order to enhance their rates of growth. This reprogramming of metabolism has been recently demonstrated for several *E. coli* mutants that had been adaptively evolved on various substrates to reach the growth rates predicted a priori by an in silico model (Fong and Palsson, *Nat. Genet.* 36:1056-1058 (2004)). The growth improvements brought about by adaptive evolution can be accompanied by enhanced rates of fatty alcohol, fatty aldehyde or fatty acid production. The strains are generally adaptively evolved in replicate, running in parallel, to account for differences in the evolutionary patterns that can be exhibited by a host organism (Fong and Palsson, *Nat. Genet.* 36:1056-1058 (2004); Fong et al., *J. Bacteriol.* 185:6400-6408 (2003); Ibarra et al., *Nature* 420:186-189 (2002)) that could potentially result in one strain having superior production qualities over the others. Evolutions can be run for a period of time, typically 2-6 weeks, depending upon the rate of growth improvement attained. In general, evolutions are stopped once a stable phenotype is obtained.

Following the adaptive evolution process, the new strains are characterized again by measuring the growth rate, the substrate uptake rate, and the product/byproduct secretion rate. These results are compared to the theoretical predictions by plotting actual growth and production yields along side the production envelopes from metabolic modeling. The most successful design/evolution combinations are chosen to pursue further, and are characterized in lab-scale batch and continuous fermentations. The growth-coupled biochemical production concept behind the methods disclosed herein such as OptKnock approach should also result in the generation of genetically stable overproducers. Thus, the cultures are maintained in continuous mode for an extended period of time, for example, one month or more, to evaluate long-term stability. Periodic samples can be taken to ensure that yield and productivity are maintained.

Adaptive evolution is a powerful technique that can be used to increase growth rates of mutant or engineered microbial strains, or of wild-type strains growing under unnatural environmental conditions. It is especially useful for strains designed via methods such as OptKnock, which results in growth-coupled product formation. Therefore, evolution toward optimal growing strains will indirectly optimize production as well. Unique strains of *E. coli* K-12

MG1655 were created through gene knockouts and adaptive evolution. (Fong and Palsson, *Nat. Genet.* 36:1056-1058 (2004)). In this work, all adaptive evolutionary cultures were maintained in prolonged exponential growth by serial passage of batch cultures into fresh medium before the stationary phase was reached, thus rendering growth rate as the primary selection pressure. Knockout strains were constructed and evolved on minimal medium supplemented with different carbon substrates (four for each knockout strain). Evolution cultures were carried out in duplicate or triplicate, giving a total of 50 evolution knockout strains. The evolution cultures were maintained in exponential growth until a stable growth rate was reached. The computational predictions were accurate (within 10%) at predicting the post-evolution growth rate of the knockout strains in 38 out of the 50 cases examined. Furthermore, a combination of OptKnock design with adaptive evolution has led to improved lactic acid production strains. (Fong et al., *Bio-technol. Bioeng.* 91:643-648 (2005)). Similar methods can be applied to the strains disclosed herein and applied to various host strains.

There are a number of developed technologies for carrying out adaptive evolution. Exemplary methods are disclosed herein. In some embodiments, optimization of a non-naturally occurring organism of the present invention includes utilizing adaptive evolution techniques to increase fatty alcohol, fatty aldehyde, fatty acid or isopropanol production and/or stability of the producing strain.

Serial culture involves repetitive transfer of a small volume of grown culture to a much larger vessel containing fresh growth medium. When the cultured organisms have grown to saturation in the new vessel, the process is repeated. This method has been used to achieve the longest demonstrations of sustained culture in the literature (Lenski and Travisano, *Proc. Natl. Acad. Sci. USA* 91:6808-6814 (1994)) in experiments which clearly demonstrated consistent improvement in reproductive rate over a period of years. Typically, transfer of cultures is usually performed during exponential phase, so each day the transfer volume is precisely calculated to maintain exponential growth through the next 24 hour period. Manual serial dilution is inexpensive and easy to parallelize.

In continuous culture the growth of cells in a chemostat represents an extreme case of dilution in which a very high fraction of the cell population remains. As a culture grows and becomes saturated, a small proportion of the grown culture is replaced with fresh media, allowing the culture to continually grow at close to its maximum population size. Chemostats have been used to demonstrate short periods of rapid improvement in reproductive rate (Dykhuizen, *Methods Enzymol.* 613-631 (1993)). The potential usefulness of these devices was recognized, but traditional chemostats were unable to sustain long periods of selection for increased reproduction rate, due to the unintended selection of dilution-resistant (static) variants. These variants are able to resist dilution by adhering to the surface of the chemostat, and by doing so, outcompete less adherent individuals, including those that have higher reproductive rates, thus obviating the intended purpose of the device (Chao and Ramsdell, *J. Gen. Microbiol.* 20:132-138 (1985)). One possible way to overcome this drawback is the implementation of a device with two growth chambers, which periodically undergo transient phases of sterilization, as described previously (Marliere and Mutzel, U.S. Pat. No. 6,686,194).

Evolugator™ is a continuous culture device developed by Evolugate, LLC (GainesvilleFLa.) and exhibits significant time and effort savings over traditional evolution techniques

(de Crecy et al., *Appl. Microbiol. Biotechnol.* 77:489-496 (2007)). The cells are maintained in prolonged exponential growth by the serial passage of batch cultures into fresh medium before the stationary phase is attained. By automating optical density measurement and liquid handling, the Evolugator™ can perform serial transfer at high rates using large culture volumes, thus approaching the efficiency of a chemostat in evolution of cell fitness. For example, a mutant of *Acinetobacter* sp ADP1 deficient in a component of the translation apparatus, and having severely hampered growth, was evolved in 200 generations to 80% of the wild-type growth rate. However, in contrast to the chemostat which maintains cells in a single vessel, the machine operates by moving from one "reactor" to the next in subdivided regions of a spool of tubing, thus eliminating any selection for wall-growth. The transfer volume is adjustable, and normally set to about 50%. A drawback to this device is that it is large and costly, thus running large numbers of evolutions in parallel is not practical. Furthermore, gas addition is not well regulated, and strict anaerobic conditions are not maintained with the current device configuration. Nevertheless, this is an alternative method to adaptively evolve a production strain.

As disclosed herein, a nucleic acid encoding a desired activity of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway can be introduced into a host organism. In some cases, it can be desirable to modify an activity of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein to increase production of fatty alcohol, fatty aldehyde, fatty acid or isopropanol. For example, known mutations that increase the activity of a protein or enzyme can be introduced into an encoding nucleic acid molecule. Additionally, optimization methods can be applied to increase the activity of an enzyme or protein and/or decrease an inhibitory activity, for example, decrease the activity of a negative regulator.

One such optimization method is directed evolution. Directed evolution is a powerful approach that involves the introduction of mutations targeted to a specific gene in order to improve and/or alter the properties of an enzyme. Improved and/or altered enzymes can be identified through the development and implementation of sensitive high-throughput screening assays that allow the automated screening of many enzyme variants (for example, $>10^4$). Iterative rounds of mutagenesis and screening typically are performed to afford an enzyme with optimized properties. Computational algorithms that can help to identify areas of the gene for mutagenesis also have been developed and can significantly reduce the number of enzyme variants that need to be generated and screened. Numerous directed evolution technologies have been developed (for reviews, see Hibbert et al., *Biomol. Eng* 22:11-19 (2005); Huisman and Lalonde, In *Biocatalysis in the pharmaceutical and biotechnology industries* pgs. 717-742 (2007), Patel (ed.), CRC Press; Often and Quax, *Biomol. Eng* 22:1-9 (2005); and Sen et al., *Appl Biochem. Biotechnol* 143:212-223 (2007)) to be effective at creating diverse variant libraries, and these methods have been successfully applied to the improvement of a wide range of properties across many enzyme classes. Enzyme characteristics that have been improved and/or altered by directed evolution technologies include, for example: selectivity/specificity, for conversion of non-natural substrates; temperature stability, for robust high temperature processing; pH stability, for bioprocessing under lower or higher pH conditions; substrate or product tolerance, so that high product titers can be achieved; binding (K_m), including broadening substrate binding to include non-natural sub-

strates; inhibition (K_i), to remove inhibition by products, substrates, or key intermediates; activity (kcat), to increase enzymatic reaction rates to achieve desired flux; expression levels, to increase protein yields and overall pathway flux; oxygen stability, for operation of air sensitive enzymes under aerobic conditions; and anaerobic activity, for operation of an aerobic enzyme in the absence of oxygen.

Described below in more detail are exemplary methods that have been developed for the mutagenesis and diversification of genes to target desired properties of specific enzymes. Such methods are well known to those skilled in the art. Any of these can be used to alter and/or optimize the activity of a fatty alcohol, fatty aldehyde, fatty acid or isopropanol pathway enzyme or protein.

EpPCR (Pritchard et al., *J Theor. Biol.* 234:497-509 (2005)) introduces random point mutations by reducing the fidelity of DNA polymerase in PCR reactions by the addition of Mn^{2+} ions, by biasing dNTP concentrations, or by other conditional variations. The five step cloning process to confine the mutagenesis to the target gene of interest involves: 1) error-prone PCR amplification of the gene of interest; 2) restriction enzyme digestion; 3) gel purification of the desired DNA fragment 4) ligation into a vector; 5) transformation of the gene variants into a suitable host and screening of the library for improved performance. This method can generate multiple mutations in a single gene simultaneously, which can be useful to screen a larger number of potential variants having a desired activity. A high number of mutants can be generated by EpPCR, so a high-throughput screening assay or a selection method, for example, using robotics, is useful to identify those with desirable characteristics.

Error-prone Rolling Circle Amplification (epRCA) (Fujii et al., *Nucleic Acids Res.* 32:e145 (2004); and Fujii et al., *Nat. Protoc.* 1:2493-2497 (2006)) has many of the same elements as epPCR except a whole circular plasmid is used as the template and random 6-mers with exonuclease resistant thiophosphate linkages on the last 2 nucleotides are used to amplify the plasmid followed by transformation into cells in which the plasmid is re-circularized at tandem repeats. Adjusting the Mn^{2+} concentration can vary the mutation rate somewhat. This technique uses a simple error-prone, single-step method to create a full copy of the plasmid with 3-4 mutations/kbp. No restriction enzyme digestion or specific primers are required. Additionally, this method is typically available as a commercially available kit.

DNA or Family Shuffling (Stemmer, *Proc Natl Acad Sci USA* 91:10747-10751(1994)); and Stemmer, *Nature* 370: 389-391(1994)) typically involves digestion of two or more variant genes with nucleases such as Dnase I or EndoV to generate a pool of random fragments that are reassembled by cycles of annealing and extension in the presence of DNA polymerase to create a library of chimeric genes. Fragments prime each other and recombination occurs when one copy primes another copy (template switch). This method can be used with >1 kbp DNA sequences. In addition to mutational recombinants created by fragment reassembly, this method introduces point mutations in the extension steps at a rate similar to error-prone PCR. The method can be used to remove deleterious, random and neutral mutations.

Staggered Extension (StEP) (Zhao et al., *Nat. Biotechnol.* 16:258-261(1998)) entails template priming followed by repeated cycles of 2 step PCR with denaturation and very short duration of annealing/extension (as short as 5 sec). Growing fragments anneal to different templates and extend further, which is repeated until full-length sequences are made. Template switching means most resulting fragments

have multiple parents. Combinations of low-fidelity polymerases (Taq and Mutazyme) reduce error-prone biases because of opposite mutational spectra.

In Random Priming Recombination (RPR) random sequence primers are used to generate many short DNA fragments complementary to different segments of the template (Shao et al., *Nucleic Acids Res* 26:681-683 (1998)). Base misincorporation and mispriming via epPCR give point mutations. Short DNA fragments prime one another based on homology and are recombined and reassembled into full-length by repeated thermocycling. Removal of templates prior to this step assures low parental recombinants. This method, like most others, can be performed over multiple iterations to evolve distinct properties. This technology avoids sequence bias, is independent of gene length, and requires very little parent DNA for the application.

In Heteroduplex Recombination linearized plasmid DNA is used to form heteroduplexes that are repaired by mismatch repair (Volkov et al., *Nucleic Acids Res.* 27:e18 (1999); and Volkov et al., *Methods Enzymol.* 328:456-463 (2000)). The mismatch repair step is at least somewhat mutagenic. Heteroduplexes transform more efficiently than linear homoduplexes. This method is suitable for large genes and whole operons.

Random Chimeragenesis on Transient Templates (RACHITT) (Coco et al., *Nat. Biotechnol.* 19:354-359 (2001)) employs Dnase I fragmentation and size fractionation of single stranded DNA (ssDNA). Homologous fragments are hybridized in the absence of polymerase to a complementary ssDNA scaffold. Any overlapping unhybridized fragment ends are trimmed down by an exonuclease. Gaps between fragments are filled in and then ligated to give a pool of full-length diverse strands hybridized to the scaffold, which contains U to preclude amplification. The scaffold then is destroyed and is replaced by a new strand complementary to the diverse strand by PCR amplification. The method involves one strand (scaffold) that is from only one parent while the priming fragments derive from other genes, and the parent scaffold is selected against. Thus, no reannealing with parental fragments occurs. Overlapping fragments are trimmed with an exonuclease. Otherwise, this is conceptually similar to DNA shuffling and StEP. Therefore, there should be no siblings, few inactives, and no unshuffled parentals. This technique has advantages in that few or no parental genes are created and many more crossovers can result relative to standard DNA shuffling.

Recombined Extension on Truncated templates (RETT) entails template switching of unidirectionally growing strands from primers in the presence of unidirectional ssDNA fragments used as a pool of templates (Lee et al., *J. Molec. Catalysis* 26:119-129 (2003)). No DNA endonucleases are used. Unidirectional ssDNA is made by DNA polymerase with random primers or serial deletion with exonuclease. Unidirectional ssDNA are only templates and not primers Random priming and exonucleases do not introduce sequence bias as true of enzymatic cleavage of DNA shuffling/RACHITT. RETT can be easier to optimize than StEP because it uses normal PCR conditions instead of very short extensions. Recombination occurs as a component of the PCR steps, that is, no direct shuffling. This method can also be more random than StEP due to the absence of pauses.

In Degenerate Oligonucleotide Gene Shuffling (DOGS) degenerate primers are used to control recombination between molecules; (Bergquist and Gibbs, *Methods Mol. Biol.* 352:191-204 (2007); Bergquist et al., *Biomol. Eng* 22:63-72 (2005); Gibbs et al., *Gene* 271:13-20 (2001)) this

can be used to control the tendency of other methods such as DNA shuffling to regenerate parental genes. This method can be combined with random mutagenesis (epPCR) of selected gene segments. This can be a good method to block the reformation of parental sequences. No endonucleases are needed. By adjusting input concentrations of segments made, one can bias towards a desired backbone. This method allows DNA shuffling from unrelated parents without restriction enzyme digests and allows a choice of random mutagenesis methods.

Incremental Truncation for the Creation of Hybrid Enzymes (ITCHY) creates a combinatorial library with 1 base pair deletions of a gene or gene fragment of interest (Ostermeier et al., *Proc. Natl. Acad. Sci. USA* 96:3562-3567 (1999); and Ostermeier et al., *Nat. Biotechnol.* 17:1205-1209 (1999)). Truncations are introduced in opposite direction on pieces of 2 different genes. These are ligated together and the fusions are cloned. This technique does not require homology between the 2 parental genes. When ITCHY is combined with DNA shuffling, the system is called SCRATCHY (see below). A major advantage of both is no need for homology between parental genes; for example, functional fusions between an *E. coli* and a human gene were created via ITCHY. When ITCHY libraries are made, all possible crossovers are captured.

Thio-Incremental Truncation for the Creation of Hybrid Enzymes (THIO-ITCHY) is similar to ITCHY except that phosphothioate dNTPs are used to generate truncations (Lutz et al., *Nucleic Acids Res* 29:E16 (2001)). Relative to ITCHY, THIO-ITCHY can be easier to optimize, provide more reproducibility, and adjustability.

SCRATCHY combines two methods for recombining genes, ITCHY and DNA shuffling (Lutz et al., *Proc. Natl. Acad. Sci. USA* 98:11248-11253 (2001)). SCRATCHY combines the best features of ITCHY and DNA shuffling. First, ITCHY is used to create a comprehensive set of fusions between fragments of genes in a DNA homology-independent fashion. This artificial family is then subjected to a DNA-shuffling step to augment the number of crossovers. Computational predictions can be used in optimization. SCRATCHY is more effective than DNA shuffling when sequence identity is below 80%.

In Random Drift Mutagenesis (RNDM) mutations are made via epPCR followed by screening/selection for those retaining usable activity (Bergquist et al., *Biomol. Eng* 22:63-72 (2005)). Then, these are used in DOGS to generate recombinants with fusions between multiple active mutants or between active mutants and some other desirable parent. Designed to promote isolation of neutral mutations; its purpose is to screen for retained catalytic activity whether or not this activity is higher or lower than in the original gene. RNDM is usable in high throughput assays when screening is capable of detecting activity above background. RNDM has been used as a front end to DOGS in generating diversity. The technique imposes a requirement for activity prior to shuffling or other subsequent steps; neutral drill libraries are indicated to result in higher/quicker improvements in activity from smaller libraries. Though published using epPCR, this could be applied to other large-scale mutagenesis methods.

Sequence Saturation Mutagenesis (SeSaM) is a random mutagenesis method that: 1) generates a pool of random length fragments using random incorporation of a phosphothioate nucleotide and cleavage; this pool is used as a template to 2) extend in the presence of "universal" bases such as inosine; 3) replication of an inosine-containing complement gives random base incorporation and, conse-

quently, mutagenesis (Wong et al., *Biotechnol. J.* 3:74-82 (2008); Wong et al., *Nucleic Acids Res.* 32:e26 (2004); and Wong et al., *Anal. Biochem.* 341:187-189 (2005)). Using this technique it can be possible to generate a large library of mutants within 2 to 3 days using simple methods. This technique is non-directed in comparison to the mutational bias of DNA polymerases. Differences in this approach makes this technique complementary (or an alternative) to epPCR.

In Synthetic Shuffling, overlapping oligonucleotides are designed to encode "all genetic diversity in targets" and allow a very high diversity for the shuffled progeny (Ness et al., *Nat. Biotechnol.* 20:1251-1255 (2002)). In this technique, one can design the fragments to be shuffled. This aids in increasing the resulting diversity of the progeny. One can design sequence/codon biases to make more distantly related sequences recombine at rates approaching those observed with more closely related sequences. Additionally, the technique does not require physically possessing the template genes.

Nucleotide Exchange and Excision Technology NextT exploits a combination of dUTP incorporation followed by treatment with uracil DNA glycosylase and then piperidine to perform endpoint DNA fragmentation (Muller et al., *Nucleic Acids Res.* 33:e117 (2005)). The gene is reassembled using internal PCR primer extension with proof-reading polymerase. The sizes for shuffling are directly controllable using varying dUPT::dTTP ratios. This is an end point reaction using simple methods for uracil incorporation and cleavage. Other nucleotide analogs, such as 8-oxo-guanine, can be used with this method. Additionally, the technique works well with very short fragments (86 bp) and has a low error rate. The chemical cleavage of DNA used in this technique results in very few unshuffled clones.

In Sequence Homology-Independent Protein Recombination (SHIPREC), a linker is used to facilitate fusion between two distantly related or unrelated genes. Nuclease treatment is used to generate a range of chimeras between the two genes. These fusions result in libraries of single-crossover hybrids (Sieber et al., *Nat. Biotechnol.* 19:456-460 (2001)). This produces a limited type of shuffling and a separate process is required for mutagenesis. In addition, since no homology is needed, this technique can create a library of chimeras with varying fractions of each of the two unrelated parent genes. SHIPREC was tested with a heme-binding domain of a bacterial CP450 fused to N-terminal regions of a mammalian CP450; this produced mammalian activity in a more soluble enzyme.

In Gene Site Saturation Mutagenesis™ (GSSM™) the starting materials are a supercoiled dsDNA plasmid containing an insert and two primers which are degenerate at the desired site of mutations (Kretz et al., *Methods Enzymol.* 388:3-11 (2004)). Primers carrying the mutation of interest, anneal to the same sequence on opposite strands of DNA. The mutation is typically in the middle of the primer and flanked on each side by approximately 20 nucleotides of correct sequence. The sequence in the primer is NNN or NNK (coding) and MNN (noncoding) (N=all 4, K=G, T, M=A, C). After extension, DpnI is used to digest dam-methylated DNA to eliminate the wild-type template. This technique explores all possible amino acid substitutions at a given locus (that is, one codon). The technique facilitates the generation of all possible replacements at a single-site with no nonsense codons and results in equal to near-equal representation of most possible alleles. This technique does not require prior knowledge of the structure, mechanism, or domains of the target enzyme. If followed by shuffling or

Gene Reassembly, this technology creates a diverse library of recombinants containing all possible combinations of single-site up-mutations. The usefulness of this technology combination has been demonstrated for the successful evolution of over 50 different enzymes, and also for more than one property in a given enzyme.

Combinatorial Cassette Mutagenesis (CCM) involves the use of short oligonucleotide cassettes to replace limited regions with a large number of possible amino acid sequence alterations (Reidhaar-Olson et al. *Methods Enzymol.* 208: 564-586 (1991); and Reidhaar-Olson et al. *Science* 241:53-57 (1988)). Simultaneous substitutions at two or three sites are possible using this technique. Additionally, the method tests a large multiplicity of possible sequence changes at a limited range of sites. This technique has been used to explore the information content of the lambda repressor DNA-binding domain.

Combinatorial Multiple Cassette Mutagenesis (CMCM) is essentially similar to CCM except it is employed as part of a larger program: 1) use of epPCR at high mutation rate to 2) identify hot spots and hot regions and then 3) extension by CMCM to cover a defined region of protein sequence space (Reetz et al., *Angew. Chem. Int. Ed Engl.* 40:3589-3591 (2001)). As with CCM, this method can test virtually all possible alterations over a target region. If used along with methods to create random mutations and shuffled genes, it provides an excellent means of generating diverse, shuffled proteins. This approach was successful in increasing, by 51-fold, the enantioselectivity of an enzyme.

In the Mutator Strains technique, conditional ts mutator plasmids allow increases of 20 to 4000-X in random and natural mutation frequency during selection and block accumulation of deleterious mutations when selection is not required (Selifonova et al., *Appl. Environ. Microbiol.* 67:3645-3649 (2001)). This technology is based on a plasmid-derived mutD5 gene, which encodes a mutant subunit of DNA polymerase III. This subunit binds to endogenous DNA polymerase III and compromises the proofreading ability of polymerase III in any strain that harbors the plasmid. A broad-spectrum of base substitutions and frameshift mutations occur. In order for effective use, the mutator plasmid should be removed once the desired phenotype is achieved; this is accomplished through a temperature sensitive (ts) origin of replication, which allows for plasmid curing at 41° C. It should be noted that mutator strains have been explored for quite some time (see Low et al., *J. Mol. Biol.* 260:359-3680 (1996)). In this technique, very high spontaneous mutation rates are observed. The conditional property minimizes non-desired background mutations. This technology could be combined with adaptive evolution to enhance mutagenesis rates and more rapidly achieve desired phenotypes.

Look-Through Mutagenesis (LTM) is a multidimensional mutagenesis method that assesses and optimizes combinatorial mutations of selected amino acids (Rajpal et al., *Proc. Natl. Acad. Sci. USA* 102:8466-8471 (2005)). Rather than saturating each site with all possible amino acid changes, a set of nine is chosen to cover the range of amino acid R-group chemistry. Fewer changes per site allows multiple sites to be subjected to this type of mutagenesis. A>800-fold increase in binding affinity for an antibody from low nanomolar to picomolar has been achieved through this method. This is a rational approach to minimize the number of random combinations and can increase the ability to find improved traits by greatly decreasing the numbers of clones to be screened. This has been applied to antibody engineer-

ing, specifically to increase the binding affinity and/or reduce dissociation. The technique can be combined with either screens or selections.

Gene Reassembly is a DNA shuffling method that can be applied to multiple genes at one time or to create a large library of chimeras (multiple mutations) of a single gene (Tunable GeneReassembly™ (TGR™) Technology supplied by Verenum Corporation). Typically this technology is used in combination with ultra-high-throughput screening to query the represented sequence space for desired improvements. This technique allows multiple gene recombination independent of homology. The exact number and position of cross-over events can be pre-determined using fragments designed via bioinformatic analysis. This technology leads to a very high level of diversity with virtually no parental gene reformation and a low level of inactive genes. Combined with GSSM™, a large range of mutations can be tested for improved activity. The method allows “blending” and “fine tuning” of DNA shuffling, for example, codon usage can be optimized.

In Silico Protein Design Automation (PDA) is an optimization algorithm that anchors the structurally defined protein backbone possessing a particular fold, and searches sequence space for amino acid substitutions that can stabilize the fold and overall protein energetics (Hayes et al., *Proc. Natl. Acad. Sci. USA* 99:15926-15931(2002)). This technology uses in silico structure-based entropy predictions in order to search for structural tolerance toward protein amino acid variations. Statistical mechanics is applied to calculate coupling interactions at each position. Structural tolerance toward amino acid substitution is a measure of coupling. Ultimately, this technology is designed to yield desired modifications of protein properties while maintaining the integrity of structural characteristics. The method computationally assesses and allows filtering of a very large number of possible sequence variants (10^{50}). The choice of sequence variants to test is related to predictions based on the most favorable thermodynamics. Ostensibly only stability or properties that are linked to stability can be effectively addressed with this technology. The method has been successfully used in some therapeutic proteins, especially in engineering immunoglobulins. In silico predictions avoid testing extraordinarily large numbers of potential variants. Predictions based on existing three-dimensional structures are more likely to succeed than predictions based on hypothetical structures. This technology can readily predict and allow targeted screening of multiple simultaneous mutations, something not possible with purely experimental technologies due to exponential increases in numbers.

Iterative Saturation Mutagenesis (ISM) involves: 1) using knowledge of structure/function to choose a likely site for enzyme improvement; 2) performing saturation mutagenesis at chosen site using a mutagenesis method such as Strata-gene QuikChange (Stratagene; San DiegoCA.); 3) screening/ selecting for desired properties; and 4) using improved clone(s), start over at another site and continue repeating until a desired activity is achieved (Reetz et al., *Nat. Protoc.* 2:891-903 (2007); and Reetz et al., *Angew. Chem. Int. Ed Engl.* 45:7745-7751 (2006)). This is a proven methodology, which assures all possible replacements at a given position are made for screening/selection.

Any of the aforementioned methods for mutagenesis can be used alone or in any combination. Additionally, any one or combination of the directed evolution methods can be used in conjunction with adaptive evolution techniques, as described herein.

It is understood that modifications which do not substantially affect the activity of the various embodiments of this invention are also provided within the definition of the invention provided herein. Accordingly, the following examples are intended to illustrate but not limit the present invention.

Example I

Formate Assimilation Pathways

This example describes enzymatic pathways for converting pyruvate to formaldehyde, and optionally in combination with producing acetyl-CoA and/or reproducing pyruvate.

Step E, FIG. 1: Formate Reductase

The conversion of formate to formaldehyde can be carried out by a formate reductase (step E, FIG. 1). A suitable enzyme for these transformations is the aryl-aldehyde dehydrogenase, or equivalently a carboxylic acid reductase, from *Nocardia iowensis*. Carboxylic acid reductase catalyzes the magnesium, ATP and NADPH-dependent reduction of carboxylic acids to their corresponding aldehydes (Venkitasubramanian et al., *J. Biol. Chem.* 282:478-485 (2007)). This enzyme, encoded by car, was cloned and functionally expressed in *E. coli* (Venkitasubramanian et al., *J. Biol. Chem.* 282:478-485 (2007)). Expression of the npt gene product improved activity of the enzyme via post-transcriptional modification. The npt gene encodes a specific phosphopantetheine transferase (PPTase) that converts the inactive apo-enzyme to the active holo-enzyme. The natural substrate of this enzyme is vanillic acid, and the enzyme exhibits broad acceptance of aromatic and aliphatic substrates (Venkitasubramanian et al., in *Biocatalysis in the Pharmaceutical and Biotechnology Industries*, ed. R. N. Patel, Chapter 15, pp. 425-440, CRC Press LLC, Boca RatonFLa. (2006)). Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
Car	AAR91681.1	40796035	<i>Nocardia iowensis</i> (sp. NRRL 5646)
Npt	ABI83656.1	114848891	<i>Nocardia iowensis</i> (sp. NRRL 5646)

Additional car and npt genes can be identified based on sequence homology.

Protein	GenBank ID	GI number	Organism
fadD9	YP_978699.1	121638475	<i>Mycobacterium bovis</i> BCG
BCG_2812c	YP_978898.1	121638674	<i>Mycobacterium bovis</i> BCG
nfa20150	YP_118225.1	54023983	<i>Nocardia farcinica</i> IFM 10152
nfa40540	YP_120266.1	54026024	<i>Nocardia farcinica</i> IFM 10152
SGR_6790	YP_001828302.1	182440583	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350
SGR_665	YP_001822177.1	182434458	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350
MSMEG 2956	YP_887275.1	118473501	<i>Mycobacterium smegmatis</i> MC2155

-continued

Protein	GenBank ID	GI number	Organism
MSMEG_5739	YP_889972.1	118469671	<i>Mycobacterium smegmatis</i> MC2155
MSMEG_2648	YP_886985.1	118471293	<i>Mycobacterium smegmatis</i> MC2155
MAP1040c	NP_959974.1	41407138	<i>Mycobacterium avium</i> subsp. <i>paratuberculosis</i> K-10
MAP2899c	NP_961833.1	41408997	<i>Mycobacterium avium</i> subsp. <i>paratuberculosis</i> K-10
MMAR_2117	YP_001850422.1	183982131	<i>Mycobacterium marinum</i> M
MM4R_2936	YP_001851230.1	183982939	<i>Mycobacterium marinum</i> M
MMAR_1916	YP_001850220.1	183981929	<i>Mycobacterium marinum</i> M
TpauDRAFT_33060	ZP_04027864.1	227980601	<i>Tsukamwella paurometabola</i> DSM 20162
TpauDRAFT_20920	ZP_04026660.1	227979396	<i>Tsukamwella paurometabola</i> DSM 20162
CPCC7001_1320	ZP_05045132.1	254431429	<i>Cyanobium PCC7001</i>
DDBDRAFT_0187729	XP_636931.1	66806417	<i>Dictyostelium discoideum</i> AX4

An additional enzyme candidate found in *Streptomyces griseus* is encoded by the griC and griD genes. This enzyme is believed to convert 3-amino-4-hydroxybenzoic acid to 3-amino-4-hydroxybenzaldehyde as deletion of either griC or griD led to accumulation of extracellular 3-acetylamin-4-hydroxybenzoic acid, a shunt product of 3-amino-4-hydroxybenzoic acid metabolism (Suzuki, et al., *J. Antibiot.* 60(6):380-387 (2007)). Co-expression of griC and griD with SGR 665, an enzyme similar in sequence to the *Nocardia iowensis* npt, can be beneficial. Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
griC	YP_001825755.1	182438036	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350
griD	YP_001825756.1	182438037	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350

An enzyme with similar characteristics, alpha-aminoadipate reductase (AAR, EC 1.2.1.31), participates in lysine biosynthesis pathways in some fungal species. This enzyme naturally reduces alpha-aminoadipate to alpha-aminoadipate semialdehyde. The carboxyl group is first activated through the ATP-dependent formation of an adenylate that is then reduced by NAD(P)H to yield the aldehyde and AMP. Like CAR, this enzyme utilizes magnesium and requires activation by a PPTase. Enzyme candidates for AAR and its corresponding PPTase are found in *Saccharomyces cerevisiae* (Morris et al., *Gene* 98:141-145 (1991)), *Candida albicans* (Guo et al., *Mol. Genet. Genomics* 269:271-279 (2003)), and *Schizosaccharomyces pombe* (Ford et al., *Curr. Genet.* 28:131-137 (1995)). The AAR from *S. pombe* exhibited significant activity when expressed in *E. coli* (Guo et al., *Yeast* 21:1279-1288 (2004)). The AAR from *Penicillium chrysogenum* accepts S-carboxymethyl-L-cysteine as an alternate substrate, but did not react with adipate, L-glutamate or diaminopimelate (Hijarrubia et al., *J. Biol. Chem.* 278:8250-8256 (2003)). The gene encoding the *P. chryso-*

genum PPTase has not been identified to date. Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
LYS2	AAA34747.1	171867	<i>Saccharomyces cerevisiae</i>
LYS5	P50113.1	1708896	<i>Saccharomyces cerevisiae</i>
LYS2	AAC02241.1	2853226	<i>Candida albicans</i>
LYS5	AAO26020.1	28136195	<i>Candida albicans</i>
Lys1p	P40976.3	13124791	<i>Schizosaccharomyces pombe</i>
Lys7p	Q10474.1	1723561	<i>Schizosaccharomyces pombe</i>
Lys2	CAA74300.1	3282044	<i>Penicillium chrysogenum</i>

Tani et al (Agric Biol Chem, 1978, 42: 63-68; Agric Biol Chem, 1974, 38: 2057-2058) showed that purified enzymes from *Escherichia coli* strain B could reduce the sodium salts of different organic acids (e.g. formate, glycolate, acetate, etc.) to their respective aldehydes (e.g. formaldehyde, glycoaldehyde, acetaldehyde, etc.). Of three purified enzymes examined by Tani et al (1978), only the "A" isozyme was shown to reduce formate to formaldehyde. Collectively, this group of enzymes was originally termed glycoaldehyde dehydrogenase; however, their novel reductase activity led the authors to propose the name glycolate reductase as being more appropriate (Morita et al, Agric Biol Chem, 1979, 43: 185-186). Morita et al (Agric Biol Chem, 1979, 43: 185-186) subsequently showed that glycolate reductase activity is relatively widespread among microorganisms, being found for example in: *Pseudomonas*, *Agrobacterium*, *Escherichia*, *Flavobacterium*, *Micrococcus*, *Staphylococcus*, *Bacillus*, and others. Without wishing to be bound by any particular theory, it is believed that some of these glycolate reductase enzymes are able to reduce formate to formaldehyde.

Any of these CAR or CAR-like enzymes can exhibit formate reductase activity or can be engineered to do so.

Step F, FIG. 1 Formate Ligase, Formate Transferase, Formate Synthetase

The acylation of formate to formyl-CoA is catalyzed by enzymes with formate transferase, synthetase, or ligase activity (Step F, FIG. 1). Formate transferase enzymes have been identified in several organisms including *Escherichia coli* (Toyota, et al., *J Bacteriol.* 2008 April; 190(7):2556-64), *Oxalobacter formigenes* (Toyota, et al., *J Bacteriol.* 2008 April; 190(7):2556-64; Baetz et al., *J Bacteriol.* 1990 July; 172(7):3537-40; Ricagno, et al., *EMBO J.* 2003 Jul. 1; 22(13):3210-9)), and *Lactobacillus acidophilus* (Azcarate-Peril, et al., *Appl. Environ. Microbiol.* 2006 72(3) 1891-1899). Homologs exist in several other organisms. Enzymes acting on the CoA-donor for formate transferase may also be expressed to ensure efficient regeneration of the CoA-donor. For example, if oxalyl-CoA is the CoA donor substrate for formate transferase, an additional transferase, synthetase, or ligase may be required to enable efficient regeneration of oxalyl-CoA from oxalate. Similarly, if succinyl-CoA or acetyl-CoA is the CoA donor substrate for formate transferase, an additional transferase, synthetase, or ligase may be required to enable efficient regeneration of succinyl-CoA from succinate or acetyl-CoA from acetate, respectively.

Protein	GenBank ID	GI number	Organism
YfdW	NP_416875.1	16130306	<i>Escherichia coli</i>
fre	O06644.3	21542067	<i>Oxalobacter formigenes</i>
fre	ZP_04021099.1	227903294	<i>Lactobacillus acidophilus</i>

Suitable CoA-donor regeneration or formate transferase enzymes are encoded by the gene products of cat1, cat2, and cats of *Clostridium kluyveri*. These enzymes have been shown to exhibit succinyl-CoA, 4-hydroxybutyryl-CoA, and butyryl-CoA acetyltransferase activity, respectively (Seedorf et al., *Proc. Natl. Acad. Sci. USA* 105:2128-2133 (2008); Sohling and Gottschalk, *J. Bacteriol.* 178:871-880 (1996)). Similar CoA transferase activities are also present in *Trichomonas vaginalis* (van Grinsven et al., *J. Biol. Chem.* 283:1411-1418 (2008)) and *Trypanosoma brucei* (Riviere et al., *J. Biol. Chem.* 279:45337-45346 (2004)). Yet another transferase capable of the desired conversions is butyryl-CoA:acetoacetate CoA-transferase. Exemplary enzymes can be found in *Fusobacterium nucleatum* (Barker et al., *J. Bacteriol.* 152(1):201-7 (1982)), *Clostridium* SB4 (Barker et al., *J. Biol. Chem.* 253(4):1219-25 (1978)), and *Clostridium acetobutylicum* (Wiesenbom et al., *Appl. Environ. Microbiol.* 55(2):323-9 (1989)). Although specific gene sequences were not provided for butyryl-CoA:acetoacetate CoA-transferase in these references, the genes FN0272 and FN0273 have been annotated as a butyrate-acetoacetate CoA-transferase (Kapatral et al., *J. Bact.* 184(7) 2005-2018 (2002)). Homologs in *Fusobacterium nucleatum* such as FN1857 and FN1856 also likely have the desired acetoacetyl-CoA transferase activity. FN1857 and FN1856 are located adjacent to many other genes involved in lysine fermentation and are thus very likely to encode an acetoacetate:butyrate CoA transferase (Kreimeyer, et al., *J. Biol. Chem.* 282 (10) 7191-7197 (2007)). Additional candidates from *Porphyromonas gingivalis* and *Thermoanaerobacter tengcongensis* can be identified in a similar fashion (Kreimeyer, et al., *J. Biol. Chem.* 282 (10) 7191-7197 (2007)). Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
Cat1	P38946.1	729048	<i>Clostridium kluyveri</i>
Cat2	P38942.2	1705614	<i>Clostridium kluyveri</i>
Cat3	EDK35586.1	146349050	<i>Clostridium kluyveri</i>
TVAG_395550	XP_001330176	123975034	<i>Trichomonas vaginalis</i> G3
Tb11.02.0290	XP_828352	71754875	<i>Trypanosoma brucei</i>
FN0272	NP_603179.1	19703617	<i>Fusobacterium nucleatum</i>
FN0273	NP_603180.1	19703618	<i>Fusobacterium nucleatum</i>
FN1857	NP_602657.1	19705162	<i>Fusobacterium nucleatum</i>
FN1856	NP_602656.1	19705161	<i>Fusobacterium nucleatum</i>
PG1066	NP_905281.1	34540802	<i>Porphyromonas gingivalis</i> W83
PG1075	NP_905290.1	34540811	<i>Porphyromonas gingivalis</i> W83
TTE0720	NP_622378.1	20807207	<i>Thermoanaerobacter tengcongensis</i> MB4
TTE0721	NP_622379.1	20807208	<i>Thermoanaerobacter tengcongensis</i> MB4

Additional transferase enzymes of interest include the gene products of atoAD from *E. coli* (Hanai et al., *Appl Environ Microbiol* 73:7814-7818 (2007)), cffAB from *C. acetobutylicum* (Jojima et al., *Appl Microbiol Biotechnol* 77:1219-1224 (2008)), and cffAB from *Clostridium saccharoperbutylacetonicum* (Kosaka et al., *Biosci. Biotechnol Biochem.* 71:58-68 (2007)). Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
AtoA	P76459.1	2492994	<i>Escherichia coli</i>
AtoD	P76458.1	2492990	<i>Escherichia coli</i>
CtfA	NP_149326.1	15004866	<i>Clostridium acetobutylicum</i>
CtfB	NP_149327.1	15004867	<i>Clostridium acetobutylicum</i>
CtfA	AAP42564.1	31075384	<i>Clostridium saccharoperbutylacetonicum</i>
CtfB	AAP42565.1	31075385	<i>Clostridium saccharoperbutylacetonicum</i>

Succinyl-CoA:3-ketoacid-CoA transferase naturally converts succinate to succinyl-CoA while converting a 3-ketoacyl-CoA to a 3-ketoacid. Exemplary succinyl-CoA:3-ketoacid-CoA transferases are present in *Helicobacter pylori* (Corthesy-Theulaz et al., *J. Biol. Chem.* 272:25659-25667 (1997)), *Bacillus subtilis* (Stols et al., *Protein. Expr. Purif.* 53:396-403 (2007)), and *Homo sapiens* (Fukao et al., *Genomics* 68:144-151(2000); Tanaka et al., *Mol. Hum. Reprod* 8:16-23 (2002)). Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
HPAG1_0676	YP_627417	108563101	<i>Helicobacter pylori</i>
HPAG1_0677	YP_627418	108563102	<i>Helicobacter pylori</i>
ScoA	NP_391778	16080950	<i>Bacillus subtilis</i>
ScoB	NP_391777	16080949	<i>Bacillus subtilis</i>
OXCT1	NP_000427	4557817	<i>Homo sapiens</i>
OXCT2	NP_071403	11545841	<i>Homo sapiens</i>

Two additional enzymes that catalyze the activation of formate to formyl-CoA reaction are AMP-forming formyl-CoA synthetase and ADP-forming formyl-CoA synthetase. Exemplary enzymes, known to function on acetate, are found in *E. coli* (Brown et Gen. *Microbiol.* 102:327-336 (1977)), *Ralstonia eutropha* (Priefert and Steinbuchel, *J. Bacteriol.* 174:6590-6599 (1992)), *Methanothermobacter thermautotrophicus* (Ingram-Smith and Smith, *Archaea* 2:95-107 (2007)), *Salmonella enterica* (Gulick et al., *Biochemistry* 42:2866-2873 (2003)) and *Saccharomyces cerevisiae* (Jogl and Tong, *Biochemistry* 43:1425-1431(2004)). Such enzymes may also acylate formate naturally or can be engineered to do so.

Protein	Gen Bank ID	GI Number	Organism
acs	AAC77039.1	1790505	<i>Escherichia coli</i>
acoE	AAA21945.1	141890	<i>Ralstonia eutropha</i>
acs1	ABC87079.1	86169671	<i>Methanothermobacter thermautotrophicus</i>
acs1	AAL23099.1	16422835	<i>Salmonella enterica</i>
ACS1	Q01574.2	257050994	<i>Saccharomyces cerevisiae</i>

ADP forming acetyl-CoA synthetase (ACD, EC 6.2.1.13) is another candidate enzyme that couples the conversion of acyl-CoA esters to their corresponding acids with the concurrent synthesis of ATP. Several enzymes with broad substrate specificities have been described in the literature. ACD I from *Archaeoglobus fulgidus*, encoded by AF1211, was shown to operate on a variety of linear and branched-chain substrates including acetyl-CoA, propionyl-CoA, butyryl-CoA, acetate, propionate, butyrate, isobutyryate, isovakrate, succinate, fumarate, phenylacetate, indoleacetate (Musfeldt et *J. Bacteriol.* 184:636-644 (2002)). The enzyme from *Haloarcula marismortui* (annotated as a succinyl-CoA synthetase) accepts propionate, butyrate, and branched-chain acids (isovalerate and isobutyrate) as substrates, and was

shown to operate in the forward and reverse directions (Brasen et al., *Arch. Microbiol.* 182:277-287 (2004)). The ACD encoded by PAE3250 from hyperthermophilic crenarchaeon *Pyrobaculum aerophilum* showed the broadest substrate range of all characterized ACDs, reacting with acetyl-CoA, isobutyryl-CoA (preferred substrate) and phenylacetyl-CoA (Brasen et al., supra (2004)). The enzymes from *A. filgidus*, *H. marismortui* and *P. aerophilum* have all been cloned, functionally expressed, and characterized in *E. coli* (Musfeldt et al., supra; Brasen et al., supra (2004)). Additional candidates include the succinyl-CoA synthetase encoded by sucCD in *E. coli* (Buck et al., *Biochemistry* 24:6245-6252 (1985)) and the acyl-CoA ligase from *Pseudomonas putida* (Fernandez-Valverde et al., *Appl. Environ. Microbiol.* 59:1149-1154 (1993)). Such enzymes may also acylate formate naturally or can be engineered to do so. Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
AF1211	NP_070039.1	11498810	<i>Archaeoglobus fulgidus</i> DSM4304
AF1983	NP_070807.1	11499565	<i>Archaeoglobus fulgidus</i> DSM4304
scs	YP_135572.1	55377722	<i>Haloarcula marismortui</i> ATCC 43049
PAE3250	NP_560604.1	18313937	<i>Pyrobaculum aerophilum</i> str. IM2
sucC	NP_415256.1	16128703	<i>Escherichia coli</i>
sueD	AAC73823.1	1786949	<i>Escherichia coli</i>
paaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>

An alternative method for adding the CoA moiety to formate is to apply a pair of enzymes such as a phosphate-transferring acyltransferase and a kinase. These activities enable the net formation of formyl-CoA with the simultaneous consumption of ATP. An exemplary phosphate-transferring acyltransferase is phosphotransacetylase, encoded by pta. The pta gene from *E. coli* encodes an enzyme that can convert acetyl-CoA into acetyl-phosphate, and vice versa (Suzuki, T. *Biochim. Biophys. Acta* 191:559-569 (1969)). This enzyme can also utilize propionyl-CoA instead of acetyl-CoA forming propionate in the process (Hesslinger et al. *Mol. Microbiol.* 27:477-492 (1998)). Homologs exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*. Such enzymes may also phosphorylate formate naturally or can be engineered to do so.

Protein	GenBank ID	GI number	Organism
Pta	NP_416800.1	16130232	<i>Escherichia coli</i>
Pta	NP_461280.1	16765665	<i>Salmonella enterica</i> subsp. <i>enterica</i> serovar <i>Typhimurium</i> str. LT2
PAT2	XP_001694504.1	159472743	<i>Chlamydomonas reinhardtii</i>
PAT1	XP_001691787.1	159467202	<i>Chlamydomonas reinhardtii</i>

An exemplary acetate kinase is the *E. coli* acetate kinase, encoded by ackA (Skarstedt and Silverstein *J. Biol. Chem.* 251:6775-6783 (1976)). Homologs exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*. It is likely that such enzymes naturally possess formate kinase activity or can be engineered to have this activity. Information related to these proteins and genes is shown below:

Protein	GenBank ID	GI number	Organism
AckA	NP_416799.1	16130231	<i>Escherichia coli</i>
AckA	NP_461279.1	16765664	<i>Salmonella enterica</i> subsp. <i>enterica</i> serovar <i>Typhimurium</i> str. LT2
ACK1	XP_001694505.1	159472745	<i>Chlamydomonas reinhardtii</i>
ACK2	XP_001691682.1	159466992	<i>Chlamydomonas reinhardtii</i>

The acylation of formate to formyl-CoA can also be carried out by a formate ligase. For example, the product of the LSC1 and LSC2 genes of *S. cerevisiae* and the sucC and sucD genes of *E. coli* naturally form a succinyl-CoA ligase complex that catalyzes the formation of succinyl-CoA from succinate with the concomitant consumption of one ATP, a reaction which is reversible in vivo (Grays et al., U.S. Pat. No. 5,958,745, filed Sep. 28, 1999). Such enzymes may also acylate formate naturally or can be engineered to do so. Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
SucC	NP_415256.1	16128703	<i>Escherichia coli</i>
SucD	AAC73823.1	1786949	<i>Escherichia coli</i>
LSC1	NP_014785	6324716	<i>Saccharomyces cerevisiae</i>
LSC2	NP_011760	6321683	<i>Saccharomyces cerevisiae</i>

Additional exemplary CoA-ligases include the rat dicarboxylate-CoA ligase for which the sequence is yet uncharacterized (Vamecq et al., *Biochemical J.* 230:683-693 (1985)), either of the two characterized phenylacetate-CoA ligases from *P. chrysogenum* (Lamas-Maceiras et al., *Biochem. J.* 395:147-155 (2005); Wang et al., *Biochem Biophys Res Commun* 360(2):453-458 (2007)), the phenylacetate-CoA ligase from *Pseudomonas putida* (Martinez-Blanco et al., *J. Biol. Chem.* 265:7084-7090 (1990)), and the 6-carboxyhexanoate-CoA ligase from *Bacillus subtilis* (Bower et al., *J. Bacteriol.* 178(14):4122-4130 (1996)). Additional candidate enzymes are acetoacetyl-CoA synthetases from *Mus musculus* (Hasegawa et al., *Biochim. Biophys. Acta* 1779:414-419 (2008)) and *Homo sapiens* (Ohgami et al., *Biochem. Pharmacol.* 65:989-994 (2003)), which naturally catalyze the ATP-dependant conversion of acetoacetate into acetoacetyl-CoA. 4-Hydroxybutyryl-CoA synthetase activity has been demonstrated in *Metallosphaera sedula* (Berg et al., *Science* 318:1782-1786 (2007)). This function has been tentatively assigned to the Msed_1422 gene. Such enzymes may also acylate formate naturally or can be engineered to do so. Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
Phl	CAJ15517.1	77019264	<i>Penicillium chrysogenum</i>
PhIB	ABS19624.1	152002983	<i>Penicillium chrysogenum</i>
PaaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>
BioW	NP_390902.2	50812281	<i>Bacillus subtilis</i>
AACS	NP_084486.1	21313520	<i>Mus musculus</i>
AACS	NP_076417.2	31982927	<i>Homo sapiens</i>
Msed 1422	YP_001191504	146304188	<i>Metallosphaera sedula</i>

Step G, FIG. 1: Formyl-CoA reductase

Several acyl-CoA dehydrogenases are capable of reducing an acyl-CoA (e.g., formyl-CoA) to its corresponding

aldehyde (e.g., formaldehyde) (Steps F, FIG. 1). Exemplary genes that encode such enzymes include the *Acinetobacter calcoaceticus* *acr1* encoding a fatty acyl-CoA reductase (Reiser and Somerville, *J. Bacteriol.* 179:2969-2975 (1997)), the *Acinetobacter* sp. M-1 fatty acyl-CoA reductase (Ishige et al., *Appl. Environ. Microbiol.* 68:1192-1195 (2002)), and a CoA- and NADP-dependent succinate semialdehyde dehydrogenase encoded by the *sucD* gene in *Clostridium kluyveri* (Sohling and Gottschalk, *J. Bacteriol.* 178:871-880 (1996); Sohling and Gottschalk, *J. Bacteriol.* 177:871-880 (1996)). *SucD* of *P. gingivalis* is another succinate semialdehyde dehydrogenase (Takahashi et al., *J. Bacteriol.* 182:4704-4710 (2000)). The enzyme acylating acetaldehyde dehydrogenase in *Pseudomonas* sp. encoded by *bphG*, is yet another candidate as it has been demonstrated to oxidize and acylate acetaldehyde, propionaldehyde, butyraldehyde, isobutyraldehyde and formaldehyde (Powlowski et al., *J. Bacteriol.* 175:377-385 (1993)). In addition to reducing acetyl-CoA to ethanol, the enzyme encoded by *adhE* in *Leuconostoc mesenteroides* has been shown to oxidize the branched chain compound isobutyraldehyde to isobutyryl-CoA (Kazahaya et al., *J. Gen. Appl. Microbiol.* 18:45-55 (1972); Koo et al., *Biotechnol. Lett.* 27:505-510 (2005)). Butyraldehyde dehydrogenase catalyzes a similar reaction, conversion of butyryl-CoA to butyraldehyde, in solventogenic organisms such as *Clostridium saccharoperbutylacetonicum* (Kosaka et al., *Biosci. Biotechnol. Biochem.* 71:58-68 (2007)). Additional aldehyde dehydrogenase enzyme candidates are found in *Desulfatibacillum alkenivorans*, *Citrobacter koseri*, *Salmonella enterica*, *Lactobacillus brevis* and *Bacillus selenitireducens*. Such enzymes may be capable of naturally converting formyl-CoA to formaldehyde or can be engineered to do so.

Protein	GenBank ID	GI number	Organism
acr1	YP_047869.1	50086355	<i>Acinetobacter calcoaceticus</i>
acr1	AAC45217	1684886	<i>Acinetobacter baylyi</i>
acr1	BAB85476.1	18857901	<i>Acinetobacter</i> sp. Strain M-1
sucD	P38947.1	172046062	<i>Clostridium kluyveri</i>
sucD	NP_904963.1	34540484	<i>Porphyromonas gingivalis</i>
bphG	BAA03892.1	425213	<i>Pseudomonas</i> sp
adhE	AAV66076.1	55818563	<i>Leuconostoc mesenteroides</i>
Bld	AAP42563.1	31075383	<i>Clostridium saccharoperbutylacetonicum</i>
Ald	ACL06658.1	218764192	<i>Desulfatibacillum alkenivorans</i> AK-01
Ald	YP_001452373	157145054	<i>Citrobacter koseri</i> ATCC BAA-895
pduP	NP_460996.1	16765381	<i>Salmonella enterica</i> Typhimurium
pduP	ABJ64680.1	116099531	<i>Lactobacillus brevis</i> ATCC 367
BseIDRAFT_1651	ZP_02169447	163762382	<i>Bacillus selenitireducens</i> MLS10

An additional enzyme type that converts an acyl-CoA to its corresponding aldehyde is malonyl-CoA reductase which transforms malonyl-CoA to malonic semialdehyde. Malonyl-CoA reductase is a key enzyme in autotrophic carbon fixation via the 3-hydroxypropionate cycle in thermoacidophilic archaeal bacteria (Berg et al., *Science* 318:1782-1786 (2007); Thauer, *Science* 318:1732-1733 (2007)). The enzyme utilizes NADPH as a cofactor and has been characterized in *Metallosphaera* and *Sulfolobus* spp (Alber et *Bacteriol.* 188:8551-8559 (2006); Hugler et *J. Bacteriol.* 184:2404-2410 (2002)). The enzyme is encoded by *Msd_0709* in *Metallosphaera sedula* (Alber et al., supra (2006); Berg et al., *Science* 318:1782-1786 (2007)). A gene

encoding a malonyl-CoA reductase from *Sulfolobus tokodaii* was cloned and heterologously expressed in *E. coli* (Alber et *J. Bacteriol.* 188:8551-8559 (2006)). This enzyme has also been shown to catalyze the conversion of methylmalonyl-CoA to its corresponding aldehyde (WO 2007/141208 (2007)). Although the aldehyde dehydrogenase functionality of these enzymes is similar to the bifunctional dehydrogenase from *Chloroflexus aurantiacus*, there is little sequence similarity. Both malonyl-CoA reductase enzyme candidates have high sequence similarity to aspartate-semialdehyde dehydrogenase, an enzyme catalyzing the reduction and concurrent dephosphorylation of aspartyl-4-phosphate to aspartate semialdehyde. Additional gene candidates can be found by sequence homology to proteins in other organisms including *Sulfolobus solfataricus* and *Sulfolobus acidocaldarius* and have been listed below. Yet another candidate for CoA-acylating aldehyde dehydrogenase is the *ald* gene from *Clostridium beijerinckii* (Toth et al., *Appl. Environ. Microbiol.* 65:4973-4980 (1999)). This enzyme has been reported to reduce acetyl-CoA and butyryl-CoA to their corresponding aldehydes. This gene is very similar to *eutE* that encodes acetaldehyde dehydrogenase of *Salmonella typhimurium* and *E. coli* (Toth et al., supra). Such enzymes may be capable of naturally converting formyl-CoA to formaldehyde or can be engineered to do so.

Protein	GenBank ID	GI number	Organism
Msd_0709	YP_001190808.1	146303492	<i>Metallosphaera sedula</i>
Mc	NP_378167.1	15922498	<i>Sulfolobus tokodaii</i>
asd-2	NP_343563.1	15898958	<i>Sulfolobus solfataricus</i>
Saci 2370	YP_256941.1	70608071	<i>Sulfolobus acidocaldarius</i>
Ald	AAT66436	9473535	<i>Clostridium beijerinckii</i>
eutE	AAA80209	687645	<i>Salmonella typhimurium</i>
eutE	P77445	2498347	<i>Escherichia coli</i>

Step H, FIG. 1: Formyltetrahydrofolate Synthetase

Formyltetrahydrofolate synthetase ligates formate to tetrahydrofolate at the expense of one ATP. This reaction is catalyzed by the gene product of *Moth_0109* in *M. thermoacetica* (O'Brien et al., *Experientia Suppl.* 26:249-262 (1976); Lovell et al., *Arch. Microbiol.* 149:280-285 (1988); Lovell et al., *Biochemistry* 29:5687-5694 (1990)), FHS in *Clostridium acidurici* (Whitehead and Rabinowitz, *J. Bacteriol.* 167:203-209 (1986); Whitehead and Rabinowitz, *J. Bacteriol.* 170:3255-3261 (1988)), and *CHY_2385* in *C. hydrogenoformans* (Wu et al., *PLoS Genet.* 1:e65 (2005)). Homologs exist in *C. carboxivorans* P7. This enzyme is found in several other organisms as listed below.

Protein	Gen Bank ID	GI number	Organism
Moth_0109	YP_428991.1	83588982	<i>Moorella thermoacetica</i>
CHY_2385	YP_361182.1	78045024	<i>Carboxydotherrmus hydrogenoformans</i>
FHS	P13419.1	120562	<i>Clostridium acidurici</i>
CcarbDRAFT_1913	ZP_05391913.1	255524966	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2946	ZP_05392946.1	255526022	<i>Clostridium carboxidivorans</i> P7
Dhaf_0555	ACL18622.1	219536883	<i>Desulfotobacterium hafniense</i>
fhs	YP_001393842.1	153953077	<i>Clostridium kluyveri</i> DSM 555

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-continued

Protein	Gen Bank ID	GI number	Organism
fhs	YP_003781893.1	300856909	<i>Clostridium ljundahlia</i> DSM 13528
MGA3_08300	EIJ83208.1	387590889	<i>Bacillus methanolicus</i> MGA3
PB1_13509	ZP_10132113.1	387929436	<i>Bacillus methanolicus</i> PB1

Steps I and J, FIG. 1: Formyltetrahydrofolate Synthetase and Methylenetetrahydrofolate Dehydrogenase

In *M. thermoacetica*, *E. coli*, and *C. hydrogenoformans*, methylenetetrahydrofolate cyclohydrolase and methylenetet-

Protein	GenBank ID	GI number	Organism
dmgo	ZP_09278452.1	359775109	<i>Arthrobacter globiformis</i>
dmgo	YP_002778684.1	226360906	<i>Rhodococcus opacus</i> B4
dmgo	EFY87157.1	322695347	<i>Metarhizium acridum</i> CQMa 102
shd	AAD53398.2	5902974	<i>Homo sapiens</i>
shd	NP_446116.1	GI: 25742657	<i>Rattus norvegicus</i>
dmgdh	NP_037523.2	24797151	<i>Homo sapiens</i>
dmgdh	Q63342.1	2498527	<i>Rattus norvegicus</i>

rahydrofolate dehydrogenase are carried out by the bi-functional gene products of Moth_1516, folD, and CHY_1878, respectively (Pierce et al., *Environ. Microbiol* 10:2550-2573 (2008); Wu et al., *PLoS Genet.* 1:e65 (2005); D'Ari and Rabinowitz, *J. Biol. Chem.* 266:23953-23958 (1991)). A homolog exists in *C. carboxidivorans* P7. Several other organisms also encode for this bifunctional protein as tabulated below.

Protein	Gen Bank ID	GI number	Organism
Moth_1516	YP_430368.1	83590359	<i>Moorella thermoacetica</i>
folD	NP_415062.1	16128513	<i>Escherichia coli</i>
CHY_1878	YP_360698.1	78044829	<i>Carboxydotherrmus hydrogenoformans</i>
CcarbDRAFT_2948	ZP_05392948.1	255526024	<i>Clostridium carboxidivorans</i> P7
folD	ADK16789.1	300437022	<i>Clostridium ljundahlia</i> DSM 13528
folD-2	NP_951919.1	39995968	<i>Geobacter sulfurreducens</i> PCA
folD	YP_725874.1	113867385	<i>Ralstonia eutropha</i> H16
folD	NP_348702.1	15895353	<i>Clostridium acetobutylicum</i> ATCC 824
folD	YP_696506.1	110800457	<i>Clostridium perfringens</i>
MGA3_09460	EIJ83438.1	387591119	<i>Bacillus methanolicus</i> MGA3
PB1_14689	ZP_10132349.1	387929672	<i>Bacillus methanolicus</i> PB1

Steps K, FIG. 1: Formaldehyde-Forming Enzyme or Spontaneous

Methylene-THF, or active formaldehyde, will spontaneously decompose to formaldehyde and THF (Thomdike and Beck, *Cancer Res.* 1977, 37(4) 1125-32; Ordonez and Caraballo, *Psychopharmacol Commun.* 1975 1(3) 253-60; Kallen and Jencks, 1966, *J Biol Chem* 241(24) 5851-63). To achieve higher rates, a formaldehyde-forming enzyme can be applied. Such an activity can be obtained by engineering

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an enzyme that reversibly forms methylene-THF from THF and a formaldehyde donor, to release free formaldehyde. Such enzymes include glycine cleavage system enzymes which naturally transfer a formaldehyde group from methylene-THF to glycine (see Step L, FIG. 1 for candidate enzymes). Additional enzymes include serine hydroxymethyltransferase (see Step M, FIG. 1 for candidate enzymes), dimethylglycine dehydrogenase (Porter, et al., *Arch Biochem Biophys.* 1985, 243(2) 396-407; Brizio et al., 2004, (37) 2, 434-442), sarcosine dehydrogenase (Porter, et al., *Arch Biochem Biophys.* 1985, 243(2) 396-407), and dimethylglycine oxidase (Leys, et al., 2003, *The EMBO Journal* 22(16) 4038-4048).

Step L, FIG. 1: Glycine Cleavage System

The reversible NAD(P)H-dependent conversion of 5,10-methylenetetrahydrofolate and CO₂ to glycine is catalyzed by the glycine cleavage complex, also called glycine cleavage system, composed of four protein components; P, H, T and L. The glycine cleavage complex is involved in glycine catabolism in organisms such as *E. coli* and glycine biosynthesis in eukaryotes (Kikuchi et al, *Proc Jpn Acad Ser* 84:246 (2008)). The glycine cleavage system of *E. coli* is encoded by four genes: gcvPHT and lpdA (Okamura et al, *Eur J Biochem* 216:539-48 (1993); Heil et al, *Microbiol* 148:2203-14 (2002)). Activity of the glycine cleavage system in the direction of glycine biosynthesis has been demonstrated in vivo in *Saccharomyces cerevisiae* (Maaheimo et al, *Eur J Biochem* 268:2464-79 (2001)). The yeast GCV is encoded by GCV1, GCV2, GCV3 and LPD1.

Protein	GenBank ID	GI Number	Organism
gcvP	AAC75941.1	1789269	<i>Escherichia coli</i>
gcvT	AAC75943.1	1789272	<i>Escherichia coli</i>
gcvH	AAC75942.1	1789271	<i>Escherichia coli</i>
lpdA	AAC73227.1	1786307	<i>Escherichia coli</i>
GCV1	NP_010302.1	6320222	<i>Saccharomyces cerevisiae</i>
GCV2	NP_013914.1	6323843	<i>Saccharomyces cerevisiae</i>
GCV3	NP_009355.3	269970294	<i>Saccharomyces cerevisiae</i>
LPD1	NP_116635.1	14318501	<i>Saccharomyces cerevisiae</i>

Step M, FIG. 1: Serine hydroxymethyltransferase

Conversion of glycine to serine is catalyzed by serine hydroxymethyltransferase, also called glycine hydroxymethyltransferase. This enzyme reversibly converts glycine and 5,10-methylenetetrahydrofolate to serine and THF. Serine methyltransferase has several side reactions including the reversible cleavage of 3-hydroxyacids to glycine and an aldehyde, and the hydrolysis of 5,10-methenyl-THF to 5-formyl-THF. This enzyme is encoded by glyA of *E. coli* (Plamann et al, *Gene* 22:9-18 (1983)). Serine hydroxymeth-

yltransferase enzymes of *S. cerevisiae* include SHM1 (mitochondrial) and SHM2 (cytosolic) (McNeil et al, *J Biol Chem* 269:9155-65 (1994)). Similar enzymes have been studied in *Corynebacterium glutamicum* and *Methylobacterium extorquens* (Chistoserdova et al, *J Bacteriol* 176:6759-62 (1994); Schweitzer et al, *J Biotechnol* 139:214-21 (2009)).

Protein	GenBank ID	GI Number	Organism
glyA	AAC75604.1	1788902	<i>Escherichia coli</i>
SHM1	NP_009822.2	37362622	<i>Saccharomyces cerevisiae</i>
SHM2	NP_013159.1	6323087	<i>Saccharomyces cerevisiae</i>
glyA	AAA64456.1	496116	<i>Methylobacterium extorquens</i>
gly	AAK60516.1	14334055	<i>Corynebacterium glutamicum</i>

Step N, FIG. 1: Serine Deaminase

Serine can be deaminated to pyruvate by serine deaminase. Serine deaminase enzymes are present in several organisms including *Clostridium acidurici* (Carter, et al., 1972, *J Bacteriol.*, 109(2) 757-763), *Escherichia coli* (Cicchillo et al., 2004, *J Biol Chem.*, 279(31) 32418-25), and *Cornebacterium* sp. (Netzer et al., *Appl Environ Microbiol.* 2004 December; 70(12):7148-55).

Protein	GenBank ID	GI Number	Organism
sdaA	YP_490075.1	388477887	<i>Escherichia coli</i>
sdaB	YP_491005.1	388478813	<i>Escherichia coli</i>
tdcG	YP_491301.1	388479109	<i>Escherichia coli</i>
tdcB	YP_491307.1	388479115	<i>Escherichia coli</i>
sdaA	YP_225930.1	62390528	<i>Corynebacterium</i> sp.

Step O, FIG. 1: Methylenetetrahydrofolate Reductase

In *M. thermoacetica*, this enzyme is oxygen-sensitive and contains an iron-sulfur cluster (Clark and Ljungdahl, *J. Biol. Chem.* 259:10845-10849 (1984)). This enzyme is encoded by metF in *E. coli* (Sheppard et al., *J. Bacteriol.* 181:718-725 (1999) and CHY_1233 in *C. hydrogenoformans* (Wu et al., *PLoS Genet.* 1:e65 (2005)). The *M. thermoacetica* genes, and its *C. hydrogenoformans* counterpart, are located near the CODH/ACS gene cluster, separated by putative hydrogenase and heterodisulfide reductase genes. Some additional gene candidates found bioinformatically are listed below. In *Acetobacterium woodii* metF is coupled to the Rnf complex through RnfC2 (Poehlein et al, *PLoS One.* 7:e33439). Homologs of RnfC are found in other organisms by blast search. The Rnf complex is known to be a reversible complex (Fuchs (2011) *Annu. Rev. Microbiol.* 65:631-658).

Protein	GenBank ID	GI number	Organism
Moth_1191	YP_430048.1	83590039	<i>Moorella thermoacetica</i>
Moth_1192	YP_430049.1	83590040	<i>Moorella thermoacetica</i>
metF	NP_418376.1	16131779	<i>Escherichia coli</i>
CH_1233	YP_360071.1	78044792	<i>Carboxydotherrnus hydrogenoformans</i>
CLJU_c37610	YP_003781889.1	300856905	<i>Clostridium ljungdahlii</i> DSM 13528
DesfrDRAFT_3717	ZP_07335241.1	303248996	<i>Desulfovibrio fructosovorans</i> JJ
CcarbDRAFT_2950	ZP_05392950.1	255526026	<i>Clostridium carboxidivorans</i> P7
Ccel74_010100023124	ZP_07633513.1	307691067	<i>Clostridium cellulovorans</i> 743B
Cphy_3110	YP_001560205.1	160881237	<i>Clostridium phytofermentans</i> ISDg

Step P, FIG. 1: Acetyl-CoA Synthase

Acetyl-CoA synthase is the central enzyme of the carbonyl branch of the Wood-Ljungdahl pathway. It catalyzes the synthesis of acetyl-CoA from carbon monoxide, coenzyme A, and the methyl group from a methylated corrinoid-iron-sulfur protein. The corrinoid-iron-sulfur-protein is methylated by methyltetrahydrofolate via a methyltransferase. Expression in a foreign host entails introducing one or more of the following proteins and their corresponding activities: Methyltetrahydrofolate:corrinoid protein methyltransferase (AcsE), Corrinoid iron-sulfur protein (AcsD), Nickel-protein assembly protein (AcsF), Ferredoxin (Orf7), Acetyl-CoA synthase (AcsB and AcsC), Carbon monoxide dehydrogenase (AcsA), and Nickel-protein assembly protein (CooC).

The genes used for carbon-monoxide dehydrogenase/acetyl-CoA synthase activity typically reside in a limited region of the native genome that can be an extended operon (Ragsdale, S. W., *Crit. Rev. Biochem. Mol. Biol.* 39:165-195 (2004); Morton et al., *J. Biol. Chem.* 266:23824-23828 (1991); Roberts et al., *Proc. Natl. Acad. Sci. U.S.A.* 86:32-36 (1989). Each of the genes in this operon from the acetogen, *M. thermoacetica*, has already been cloned and expressed actively in *E. coli* (Morton et al. supra; Roberts et al. supra; Lu et al., *J. Biol. Chem.* 268:5605-5614 (1993)). The protein sequences of these genes can be identified by the following GenBank accession numbers.

Protein	GenBank ID	GI number	Organism
AcsE	YP_430054	83590045	<i>Moorella thermoacetica</i>
AcsD	YP_430055	83590046	<i>Moorella thermoacetica</i>
AcsF	YP_430056	83590047	<i>Moorella thermoacetica</i>
Orf7	YP_430057	83590048	<i>Moorella thermoacetica</i>
AcsC	YP_430058	83590049	<i>Moorella thermoacetica</i>
AcsB	YP_430059	83590050	<i>Moorella thermoacetica</i>
AcsA	YP_430060	83590051	<i>Moorella thermoacetica</i>
CooC	YP_430061	83590052	<i>Moorella thermoacetica</i>

The hydrogenic bacterium, *Carboxydotherrnus hydrogenoformans*, can utilize carbon monoxide as a growth substrate by means of acetyl-CoA synthase (Wu et al., *PLoS Genet.* 1:e65 (2005)). In strain Z-2901, the acetyl-CoA synthase enzyme complex lacks carbon monoxide dehydrogenase due to a frameshift mutation (Wu et al. supra (2005)), whereas in strain DSM 6008, a functional unframeshifted full-length version of this protein has been purified (Svetlitchnyi et al., *Proc. Natl. Acad. Sci. U.S.A.* 101:446-451 (2004)). The protein sequences of the *C. hydrogenoformans* genes from strain Z-2901 can be identified by the following GenBank accession numbers.

Protein	GenBank ID	GI number	Organism
AcsE	YP_360065	78044202	<i>Carboxydotherrnus hydrogenoformans</i>
AcsD	YP_360064	78042962	<i>Carboxydotherrnus hydrogenoformans</i>
AcsF	YP_360063	78044060	<i>Carboxydotherrnus hydrogenoformans</i>
Orf7	YP_360062	78044449	<i>Carboxydotherrnus hydrogenoformans</i>
AcsC	YP_360061	78043584	<i>Carboxydotherrnus hydrogenoformans</i>
AcsB	YP_360060	78042742	<i>Carboxydotherrnus hydrogenoformans</i>
CooC	YP_360059	78044249	<i>Carboxydotherrnus hydrogenoformans</i>

Homologous ACS/CODH genes can also be found in the draft genome assembly of *Clostridium carboxidivorans* P7.

two active CODH/ACS operons inM acetivorans include catalytic properties (i.e., K_m , V_{max} , k_{cat}) that favor carboxi-

Protein	GenBank ID	GI Number	Organism
AcsA	ZP_05392944.1	255526020	<i>Clostridium carboxidivorans</i> P7
CooC	ZP_05392945.1	255526021	<i>Clostridium carboxidivorans</i> P7
AcsF	ZP_05392952.1	255526028	<i>Clostridium carboxidivorans</i> P7
AcsD	ZP_05392953.1	255526029	<i>Clostridium carboxidivorans</i> P7
AcsC	ZP_05392954.1	255526030	<i>Clostridium carboxidivorans</i> P7
AcsE	ZP_05392955.1	255526031	<i>Clostridium carboxidivorans</i> P7
AcsB	ZP_05392956.1	255526032	<i>Clostridium carboxidivorans</i> P7
Orf7	ZP_05392958.1	255526034	<i>Clostridium carboxidivorans</i> P7

The methanogenic archaeon, *Methanosarcina acetiv-* orans, can also grow on carbon monoxide, exhibits acetyl-CoA synthase/carbon monoxide dehydrogenase activity, and produces both acetate and formate (Lessner et al., *Proc. Natl. Acad. Sci. U.S.A.* 103:17921-17926 (2006)). This organism contains two sets of genes that encode ACS/ CODH activity (Rother and Metcalf, *Proc. Natl. Acad. Sci. U.S.A.* 101:16929-16934 (2004)). The protein sequences of both sets ofM acetivorans genes are identified by the following GenBank accession numbers.

Protein	GenBank ID	GI number	Organism
AcsC	NP_618736	20092661	<i>Methanosarcina acetivorans</i>
AcsD	NP_618735	20092660	<i>Methanosarcina acetivorans</i>
AcsF, CooC	NP_618734	20092659	<i>Methanosarcina acetivorans</i>
AcsB	NP_618733	20092658	<i>Methanosarcina acetivorans</i>
AcsEps	NP_618732	20092657	<i>Methanosarcina acetivorans</i>
AcsA	NP_618731	20092656	<i>Methanosarcina acetivorans</i>
AcsC	NP_615961	20089886	<i>Methanosarcina acetivorans</i>
AcsD	NP_615962	20089887	<i>Methanosarcina acetivorans</i>
AcsF, CooC	NP_615963	20089888	<i>Methanosarcina acetivorans</i>
AcsB	NP_615964	20089889	<i>Methanosarcina acetivorans</i>
AcsEps	NP_615965	20089890	<i>Methanosarcina acetivorans</i>
AcsA	NP_615966	20089891	<i>Methanosarcina acetivorans</i>

The AcsC, AcsD, AcsB, AcsEps, and AcsA proteins are commonly referred to as the gamma, delta, beta, epsilon, and alpha subunits of the methanogenic CODH/ACS. Homologs to the epsilon encoding genes are not present in acetogens such as *M. thermoacetica* or hydrogenogenic bacteria such as *C. hydrogenoformans*. Hypotheses for the existence of

drotrophic or acetoclastic growth or differential gene regulation enabling various stimuli to induce CODH/ACS expression (Rother et al., *Arch. Microbiol.* 188:463-472 (2007)).

Step Y, FIG. 1: Glyceraldehydes-3-phosphate Dehydrogenase and Enzymes of Lower Glycolysis

Enzymes comprising Step Y, G3P to PYR include: Glyceraldehyde-3-phosphate dehydrogenase; Phosphoglycerate

kinase; Phosphoglyceromutase; Enolase; Pyruvate kinase or PTS-dependant substrate import.

Glyceraldehyde-3-phosphate dehydrogenase enzymes include:

NADP-dependant glyceraldehyde-3-phosphate dehydrogenase, exemplary enzymes are:

Protein	GenBank ID	GI Number	Organism
gapN	AAA91091.1	642667	<i>Streptococcus mutans</i>
NP-GAPDH	AEC07555.1	330252461	<i>Arabidopsis thaliana</i>
GAPN	AAM77679.2	82469904	<i>Triticum aestivum</i>
gapN	CAI56300.1	87298962	<i>Clostridium acetobutylicum</i>
NADP-GAPDH	2D2I_A	112490271	<i>Synechococcus elongatus</i> PCC 7942

-continued

Protein	GenBank ID	GI Number	Organism
NADP-GAPDH	CAA62619.1	4741714	<i>Synechococcus elongatus</i> PCC 7942
GDP1	XP_455496.1	50310947	<i>Kluyveromyces lactis</i> NRRL Y-1140
HP1346	NP_208138.1	15645959	<i>Helicobacter pylori</i> 26695

and NAD-dependant glyceraldehyde-3-phosphate dehydrogenase, exemplary enzymes are:

Pyruvate kinase (also known as phosphoenolpyruvate kinase and phosphoenolpyruvate kinase) or PTS-dependant

Protein	GenBank ID	GI Number	Organism
TDH1	NP_012483.1	6322409	<i>Saccharomyces cerevisiae</i> s288c
TDH2	NP_012542.1	6322468	<i>Saccharomyces cerevisiae</i> s288c
TDH3	NP_011708.1	632163	<i>Saccharomyces cerevisiae</i> s288c
KLLA0A11858g	XP_451516.1	50303157	<i>Kluyveromyces lactis</i> NRRL Y-1140
KLLA0F20988g	XP_456022.1	50311981	<i>Kluyveromyces lactis</i> NRRL Y-1140
ANI_1_256144	XP_001397496.1	145251966	<i>Aspergillus niger</i> CBS 513.88
YALI0C06369g	XP_501515.1	50548091	<i>Yarrowia lipolytica</i>
CTRG_05666	XP_002551368.1	255732890	<i>Candida tropicalis</i> MYA-3404
HPODL 1089	EFW97311.1	320583095	<i>Hansenula polymorpha</i> DL-1
gapA	YP_490040.1	388477852	<i>Escherichia coli</i>

Phosphoglycemte kinase enzymes include:

Protein	GenBank ID	GI Number	Organism
PGK1	NP_009938.2	10383781	<i>Saccharomyces cerevisiae</i> s288c
PGK	BAD83658.1	57157302	<i>Candida boidinii</i>
PGK	EFW98395.1	320584184	<i>Hansenula polymorpha</i> DL-1
pgk	EIJ77825.1	387585500	<i>Bacillus methanolicus</i> MGA3
pgk	YP_491126.1	388478934	<i>Escherichia coli</i>

Phosphoglyceromutase (aka phosphoglycemte mutase) enzymes include;

Protein	GenBank ID	GI Number	Organism
GPM1	NP_012770.1	6322697	<i>Saccharomyces cerevisiae</i> s288c
GPM2	NP_010263.1	6320183	<i>Saccharomyces cerevisiae</i> s288c
GPM3	NP_014585.1	6324516	<i>Saccharomyces cerevisiae</i> s288c
HPODL_1391	EFW96681.1	320582464	<i>Hansenula polymorpha</i> DL-1
HPODL_0376	EFW97746.1	320583533	<i>Hansenula polymorpha</i> DL-1
gpmI	EIJ77827.1	387585502	<i>Bacillus methanolicus</i> MGA3
gpmA	YP_489028.1	388476840	<i>Escherichia coli</i>
gpmM	AAC76636.1	1790041	<i>Escherichia coli</i>

Enolase (also known as phosphopyruvate hydratase and 2-phosphoglycemte dehydratase) enzymes include:

Protein	GenBank ID	GI Number	Organism
ENO1	NP_011770.3	398366315	<i>Saccharomyces cerevisiae</i> s288c
ENO2	AAB68019.1	458897	<i>Saccharomyces cerevisiae</i> s288c
HPODL 2596	EFW95743.1	320581523	<i>Hansenula polymorpha</i> DL-1
eno	EIJ77828.1	387585503	<i>Bacillus methanolicus</i> MGA3
eno	AAC75821.1	1789141	<i>Escherichia coli</i>

substrate import enzymes include those below. Pyruvate kinase, also known as phosphoenolpyruvate synthase (EC 2.7.9.2), converts pyruvate and ATP to PEP and AMP. This enzyme is encoded by the PYK1 (Burke et al., J. Biol. Chem. 258:2193-2201 (1983)) and PYK2 (Boles et al., J. Bacteriol. 179:2987-2993 (1997)) genes in *S. cerevisiae*. In *E. coli*, this activity is catalyzed by the gene products of *pykF* and *pykA*. Note that *pykA* and *pykF* are genes encoding separate enzymes potentially capable of carrying out the PYK reaction. Selected homologs of the *S. cerevisiae* enzymes are also shown in the table below.

Protein	GenBank ID	GI Number	Organism
PYK1	NP_009362	6319279	<i>Saccharomyces cerevisiae</i>
PYK2	NP_014992	6324923	<i>Saccharomyces cerevisiae</i>
pykF	NP_416191.1	16129632	<i>Escherichia coli</i>
pykA	NP_416368.1	16129807	<i>Escherichia coli</i>
KLLA0F23397g	XP_456122.1	50312181	<i>Kluyveromyces lactis</i>
CaO19.3575	XP_714934.1	68482353	<i>Candida albicans</i>
CaO19.11059	XP_714997.1	68482226	<i>Candida albicans</i>
YAL10F09185p	XP_505195	210075987	<i>Yarrowia lipolytica</i>
ANI_1_1126064	XP_001391973	145238652	<i>Aspergillus niger</i>
MGA3_03005	EIJ84220.1	387591903	<i>Bacillus methanolicus</i> MGA3
HPODL_1539	EFW96829.1	320582612	<i>Hansenula polymorpha</i> DL-1

Alternatively, Phosphoenolpyruvate phosphatase (EC 3.1.3.60) catalyzes the hydrolysis of PEP to pyruvate and phosphate. Numerous phosphatase enzymes catalyze this activity, including alkaline phosphatase (EC 3.1.3.1), acid phosphatase (EC 3.1.3.2), phosphoglycerate phosphatase (EC 3.1.3.20) and PEP phosphatase (EC 3.1.3.60). PEP phosphatase enzymes have been characterized in plants such as *Vigna radiate*, *Bruguiera sexangula* and *Brassica nigra*. The phytase from *Aspergillus fumigatus*, the acid phosphatase from *Homo sapiens* and the alkaline phosphatase of *E. coli* also catalyze the hydrolysis of PEP to pyruvate (Brugger et al, Appl Microbiol Biotech 63:383-9 (2004); Hayman et al, Biochem J 261:601-9 (1989); et al, The Enzymes 3rd Ed. 4:373-415 (1971)). Similar enzymes have been characterized in *Campylobacter jejuni* (van Mourik et al., Microbiol. 154:584-92 (2008)), *Saccharomyces cerevisiae* (Oshima et al., Gene 179:171-7 (1996)) and *Staphylococcus aureus* (Shah and Blobel, J. Bacteriol. 94:780-1 (1967)). Enzyme engineering and/or removal of targeting sequences may be required for alkaline phosphatase enzymes to function in the cytoplasm.

Protein	GenBank ID	GI Number	Organism
phyA	O00092.1	41017447	<i>Aspergillus fumigatus</i>
Acp5	P13686.3	56757583	<i>Homo sapiens</i>
phoA	NP_414917.2	49176017	<i>Escherichia coli</i>
phoX	ZP_01072054.1	86153851	<i>Campylobacter jejuni</i>
PHO8	AAA34871.1	172164	<i>Saccharomyces cerevisiae</i>
SaurJH1_2706	YP_001317815.1	150395140	<i>Staphylococcus aureus</i>

Step Q, FIG. 1: Pyruvate Formate Lyase

Pyruvate formate-lyase (PFL, EC 2.3.1.54), encoded by pflB in *E. coli*, can convert pyruvate into acetyl-CoA and formate. The activity of PFL can be enhanced by an activating enzyme encoded by pflA (Knappe et al., *Proc. Natl. Acad. Sci USA* 81:1332-1335 (1984); Wong et al., *Biochemistry* 32:14102-14110 (1993)). Keto-acid formate-lyase (EC 2.3.1.-), also known as 2-ketobutyrate formate-lyase (KFL) and pyruvate formate-lyase 4, is the gene product of tdcE in *E. coli*. This enzyme catalyzes the conversion of 2-ketobutyrate to propionyl-CoA and formate during anaerobic threonine degradation, and can also substitute for pyruvate formate-lyase in anaerobic catabolism (Simanshu et al., *J Biosci.* 32:1195-1206 (2007)). The enzyme is oxygen-sensitive and, like PflB, can require post-translational modification by PFL-AE to activate a glycyl radical in the active site (Hesslinger et al., *Mol. Microbiol* 27:477-492 (1998)). A pyruvate formate-lyase from *Archaeoglobus fulgidus* encoded by pflD has been cloned, expressed in *E. coli* and characterized (Lehtio et al., *Protein Eng Des Sel* 17:545-552

(2004)). The crystal structures of the *A. fulgidus* and *E. coli* enzymes have been resolved (Lehtio et al., *J Mol. Biol.* 357:221-235 (2006); Leppanen et al., *Structure.* 7:733-744 (1999)). Additional PFL and PFL-AE candidates are found in *Lactococcus lactis* (Melchiorson et al., *Appl Microbiol Biotechnol* 58:338-344 (2002)), and *Streptococcus mutans* (Takahashi-Abbe et al., *Oral. Microbiol Immunol.* 18:293-297 (2003)), *Chlamydomonas reinhardtii* (Hemschemeier et al., *Eukaryot. Cell* 7:518-526 (2008b); Atteia et al., *J. Biol. Chem.* 281:9909-9918 (2006)) and *Clostridium pasteurianum* (Weidner et al., *J Bacteriol.* 178:2440-2444 (1996)).

Protein	GenBank ID	GI Number	Organism
pflB	NP_415423	16128870	<i>Escherichia coli</i>
pflA	NP_415422.1	16128869	<i>Escherichia coli</i>
tdcE	AAT48170.1	48994926	<i>Escherichia coli</i>
pflD	NP_070278.1	11499044	<i>Archaeoglobus fulgidus</i>
Pfl	CAA03993	2407931	<i>Lactococcus lactis</i>
Pfl	BAA09085	1129082	<i>Streptococcus mutans</i>
PFL1	XP_001689719.1	159462978	<i>Chlamydomonas reinhardtii</i>

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-continued

Protein	GenBank ID	GI Number	Organism
pflA1	XP_001700657.1	159485246	<i>Chlamydomonas reinhardtii</i>
Pfl	Q46266.1	2500058	<i>Clostridium pasteurianum</i>
Act	CAA63749.1	1072362	<i>Clostridium pasteurianum</i>

Step R, FIG. 1: Pyruvate Dehydrogenase, Pyruvate Ferredoxin Oxidoreductase, Pyruvate:nadp+Oxidoreductase

The pyruvate dehydrogenase (PDH) complex catalyzes the conversion of pyruvate to acetyl-CoA (FIG. 3H). The *E. coli* PDH complex is encoded by the genes aceEF and lpdA. Enzyme engineering efforts have improved the *E. coli* PDH enzyme activity under anaerobic conditions (Kim et al., *J. Bacteriol.* 190:3851-3858 (2008); Kim et al., *Appl. Environ. Microbiol.* 73:1766-1771 (2007); Zhou et al., *Biotechnol. Lett.* 30:335-342 (2008)). In contrast to the *E. coli* PDH, the *B. subtilis* complex is active and required for growth

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under anaerobic conditions (Nakano et al., 179:6749-6755 (1997)). The *Klebsiella pneumoniae* PDH, characterized during growth on glycerol, is also active under anaerobic conditions (Menzel et al., 56:135-142 (1997)). Crystal structures of the enzyme complex from bovine kidney (Zhou et al., 98:14802-14807 (2001)) and the E2 catalytic domain from *Azotobacter vinelandii* are available (Mattevi et al., *Science*. 255:1544-1550 (1992)). Some mammalian PDH enzymes complexes can react on alternate substrates such as 2-oxobutanoate. Comparative kinetics of *Rattus norvegicus* PDH and BCKAD indicate that BCKAD has higher activity on 2-oxobutanoate as a substrate (Paxton et al., *Biochem.J.* 234:295-303 (1986)). The *S. cerevisiae* PDH complex can consist of an E2 (LAT1) core that binds E1 (PDA1, PDB1), E3 (LPD1), and Protein X (PDX1) components (Pronk et al., *Yeast* 12:1607-1633 (1996)). The PDH complex of *S. cerevisiae* is regulated by phosphorylation of E1 involving PKP1 (PDH kinase I), PTCS (PDH phosphatase I), PKP2 and PTC6. Modification of these regulators may also enhance PDH activity. Coexpression of lipoyl ligase (LpiA of *E. coli* and AIM22 in *S. cerevisiae*) with PDH in the cytosol may be necessary for activating the PDH enzyme complex. Increasing the supply of cytosolic lipoate, either by modifying a metabolic pathway or media supplementation with lipoate, may also improve PDH activity.

Gene	Accession No.	GI Number	Organism
aceE	NP_414656.1	16128107	<i>Escherichia coli</i>
aceF	NP_414657.1	16128108	<i>Escherichia coli</i>
lpd	NP_414658.1	16128109	<i>Escherichia coli</i>
lplA	NP_418803.1	16132203	<i>Escherichia coli</i>
pdhA	P21881.1	3123238	<i>Bacillus subtilis</i>
pdhB	P21882.1	129068	<i>Bacillus subtilis</i>
pdhC	P21883.2	129054	<i>Bacillus subtilis</i>
pdhD	P21880.1	118672	<i>Bacillus subtilis</i>
aceE	YP_001333808.1	152968699	<i>Klebsiella pneumoniae</i>
aceF	YP_001333809.1	152968700	<i>Klebsiella pneumoniae</i>
lpdA	YP_001333810.1	152968701	<i>Klebsiella pneumoniae</i>
Pdha1	NP_001004072.2	124430510	<i>Rattus norvegicus</i>
Pdha2	NP_446446.1	16758900	<i>Rattus norvegicus</i>
Dlat	NP_112287.1	78365255	<i>Rattus norvegicus</i>
Dld	NP_955417.1	40786469	<i>Rattus norvegicus</i>
LAT1	NP_014328	6324258	<i>Saccharomyces cerevisiae</i>
PDA1	NP_011105	37362644	<i>Saccharomyces cerevisiae</i>
PDB1	NP_009780	6319698	<i>Saccharomyces cerevisiae</i>
LPD1	NP_116635	14318501	<i>Saccharomyces cerevisiae</i>
PDX1	NP_011709	6321632	<i>Saccharomyces cerevisiae</i>
AIM22	NP_012489.2	83578101	<i>Saccharomyces cerevisiae</i>

As an alternative to the large multienzyme PDH complexes described above, some organisms utilize enzymes in

oxygen (Pieulle et al., *J. Bacteriol.* 179:5684-5692 (1997)). Oxygen stability is relatively uncommon in PFORs and is believed to be conferred by a 60 residue extension in the polypeptide chain of the *D. africanus* enzyme. The *M. thermoacetica* PFOR is also well characterized (Menon et al., *Biochemistry* 36:8484-8494 (1997)) and was even shown to have high activity in the direction of pyruvate synthesis during autotrophic growth (Furdui et al., *J. Biol. Chem.* 275:28494-28499 (2000)). Further, *E. coli* possesses an uncharacterized open reading frame, ydbK that encodes a protein that is 51% identical to the *M. thermoacetica* PFOR. Evidence for pyruvate oxidoreductase activity in *E. coli* has been described (Blaschkowski et al., *Eur. J. Biochem.* 123: 563-569 (1982)). Several additional PFOR enzymes are described in Ragsdale, *Chem. Rev.* 103:2333-2346 (2003). Finally, flavodoxin reductases (e.g., fqrB from *Helicobacter pylori* or *Campylobacter jejuni* (St Maurice et al., *J. Bacteriol.* 189:4764-4773 (2007))) or Rnf-type proteins (Seedorf et al., *Proc. Natl. Acad. Sci. USA.* 105:2128-2133 (2008); Herrmann et al., *J. Bacteriol.* 190:784-791 (2008)) provide a means to generate NADH or NADPH from the reduced ferredoxin generated by PFOR. These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
Por	CAA70873.1	1770208	<i>Desulfovibrio africanus</i>
Por	YP_428946.1	83588937	<i>Moorella thermoacetica</i>
ydbK	NP_415896.1	16129339	<i>Escherichia coli</i>
fqrB	NP_207955.1	15645778	<i>Helicobacter pylori</i>
fqrB	YP_001482096.1	157414840	<i>Campylobacter jejuni</i>
RnfC	EDK33306.1	146346770	<i>Clostridium kluyveri</i>
RnfD	EDK33307.1	146346771	<i>Clostridium kluyveri</i>
RnfG	EDK33308.1	146346772	<i>Clostridium kluyveri</i>
RnfE	EDK33309.1	146346773	<i>Clostridium kluyveri</i>
RnfA	EDK33310.1	146346774	<i>Clostridium kluyveri</i>
RnfB	EDK33311.1	146346775	<i>Clostridium kluyveri</i>

Pyruvate:NADP oxidoreductase (PNO) catalyzes the conversion of pyruvate to acetyl-CoA. This enzyme is encoded by a single gene and the active enzyme is a homodimer, in contrast to the multi-subunit PDH enzyme complexes described above. The enzyme from *Euglena gracilis* is stabilized by its cofactor, thiamin pyrophosphate (Nakazawa et al., *Arch. Biochem. Biophys.* 411:183-8 (2003)). The mitochondrial targeting sequence of this enzyme should be removed for expression in the cytosol. The PNO protein of *E. gracilis* and other NADP-dependant pyruvate:NADP+ oxidoreductase enzymes are listed in the table below.

Protein	GenBank ID	GI Number	Organism
PNO	Q94IN5.1	33112418	<i>Euglena gracilis</i>
cgd4_690	XP_625673.1	66356990	<i>Cryptosporidium parvum</i> Iowa II
TPP_PFOR_PNO	XP_002765111.11	294867463	<i>Perkinsus marinus</i> ATCC 50983

the 2-ketoacid oxidoreductase family (OFOR) to catalyze acylating oxidative decarboxylation of 2-keto-acids. Unlike the PDH complexes, PFOR enzymes contain iron-sulfur clusters, utilize different cofactors and use ferredoxin or flavodoxin as electron acceptors in lieu of NAD(P)H. Pyruvate ferredoxin oxidoreductase (PFOR) can catalyze the oxidation of pyruvate to form acetyl-CoA (FIG. 3H). The PFOR from *Desulfovibrio africanus* has been cloned and expressed in *E. coli* resulting in an active recombinant enzyme that was stable for several days in the presence of

Step S, FIG. 1: Formate Dehydrogenase

Formate dehydrogenase (FDH) catalyzes the reversible transfer of electrons from formate to an acceptor. Enzymes with FDH activity utilize various electron carriers such as, for example, NADH (EC 1.2.1.2), NADPH (EC 1.2.1.43), quinols (EC 1.1.5.6), cytochromes (EC 1.2.2.3) and hydrogenases (EC 1.1.99.33). FDH enzymes have been characterized from *Moorella thermoacetica* (Andresen and Ljungdahl, *J. Bacteriol.* 116:867-873 (1973); Li et al., *J.*

Bacteriol 92:405-412 (1966); Yamamoto et al., *J Biol Chem.* 258:1826-1832 (1983). The loci, Moth_2312 is responsible for encoding the alpha subunit of formate dehydrogenase while the beta subunit is encoded by Moth_2314 (Pierce et al., *Environ Microbiol* (2008)). Another set of genes encoding formate dehydrogenase activity with a propensity for CO₂ reduction is encoded by Sfum_2703 through Sfum_2706 in *Syntrophobacter fumaroxidans* (de Bok et al., *Eur J Biochem.* 270:2476-2485 (2003)); Reda et al., *PNAS* 105:10654-10658 (2008)). A similar set of genes presumed to carry out the same function are encoded by CHY_0731, CHY_0732, and CHY_0733 in *C. hydrogenoformans* (Wu et al., *PLoS Genet* 1:e65 (2005)). Formate dehydrogenases are also found many additional organisms including *C. carboxidivorans* P7, *Bacillus methanolicus*, *Burkholderia stabilis*, *Moorella thermoacetica* ATCC 39073, *Candida boidinii*, *Candida methylca*, and *Saccharomyces cerevisiae* S288c. The soluble formate dehydrogenase from *Ralstonia eutropha* reduces NAD⁺ (fdsG, —B, —A, —C, —D) (Oh and Bowien, 1998)

Several EM8 enzymes have been identified that have higher specificity for NADP as the cofactor as compared to NAD. This enzyme has been deemed as the NADP-dependent formate dehydrogenase and has been reported from 5 species of the *Burkholderia cepacia* complex. It was tested and verified in multiple strains of *Burkholderia multivorans*, *Burkholderia stabilis*, *Burkholderia pyrrocinia*, and *Burkholderia cenocepacia* (Hatrongjit et al., *Enzyme and Microbial Tech.*, 46: 557-561 (2010)). The enzyme from *Burkholderia stabilis* has been characterized and the apparent K_m of the enzyme were reported to be 55.5 mM, 0.16 mM and 1.43 mM for formate, NADP, and NAD respectively. More gene candidates can be identified using sequence homology of proteins deposited in Public databases such as NCBI, JGI and the metagenomic databases.

Protein	GenBank ID	GI Number	Organism
Moth_2312	YP_431142	148283121	<i>Moorella thermoacetica</i>
Moth_2314	YP_431144	83591135	<i>Moorella thermoacetica</i>
Sfum_2703	YP_846816.1	116750129	<i>Syntrophobacter fumaroxidans</i>
Sfum_2704	YP_846817.1	116750130	<i>Syntrophobacter fumaroxidans</i>
Sfum_2705	YP_846818.1	116750131	<i>Syntrophobacter fumaroxidans</i>
Sfum_2706	YP_846819.1	116750132	<i>Syntrophobacter fumaroxidans</i>
CHY_0731	YP_359585.1	78044572	<i>Carboxydotherrmus hydrogenoformans</i>
CHY_0732	YP_359586.1	78044500	<i>Carboxydotherrmus hydrogenoformans</i>
CHY_0733	YP_359587.1	78044647	<i>Carboxydotherrmus hydrogenoformans</i>
CcarbDRAFT_0901	ZP_05390901.1	255523938	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_4380	ZP_05394380.1	255527512	<i>Clostridium carboxidivorans</i> P7
fdhA, MGA3_06625	EIJ82879.1	387590560	<i>Bacillus methanolicus</i> MGA3
fdhA, PB1_11719	ZP_10131761.1	387929084	<i>Bacillus methanolicus</i> PB1
fdhD, MGA3_06630	EIJ82880.1	387590561	<i>Bacillus methanolicus</i> MGA3
fdhD, PB1_11724	ZP_10131762.1	387929085	<i>Bacillus methanolicus</i> PB1
fdh	ACF35003.1	194220249	<i>Burkholderia stabilis</i>
fdh	ACF35004.1	194220251	<i>Burkholderia pyrrocinia</i>
fdh	ACF35002.1	194220247	<i>Burkholderia cenocepacia</i>
fdh	ACF35001.1	194220245	<i>Burkholderia multivorans</i>
fdh	ACF35000.1	194220243	<i>Burkholderia cepacia</i>
FDH1	AAC49766.1	2276465	<i>Candida boidinii</i>
fdh	CAA57036.1	1181204	<i>Candida methylca</i>
FDH2	P0CF35.1	294956522	<i>Saccharomyces cerevisiae</i> S288c
FDH1	NP_015033.1	6324964	<i>Saccharomyces cerevisiae</i> S288c
fdsG	YP_725156.1	113866667	<i>Ralstonia eutropha</i>
fdsB	YP_725157.1	113866668	<i>Ralstonia eutropha</i>
fdsA	YP_725158.1	113866669	<i>Ralstonia eutropha</i>
fdsC	YP_725159.1	113866670	<i>Ralstonia eutropha</i>
fdsD	YP_725160.1	113866671	<i>Ralstonia eutropha</i>

Production of Reducing Equivalents

This example describes methanol metabolic pathways and other additional enzymes generating reducing equivalents as shown in FIG. 10.

FIG. 10, Step A—Methanol Methyltransferase

A complex of 3-methyltransferase proteins, denoted MtaA, MtaB, and MtaC, perform the desired methanol methyltransferase activity (Sauer et al., *Eur. Biochem.* 243: 670-677 (1997); Naidu and Ragsdale, *J. Bacteriol.* 183: 3276-3281 (2001); Tallant and Krzycki, *J. Biol. Chem.* 276:4485-4493 (2001); Tallant and Krzycki, *J. Bacteriol.* 179:6902-6911 (1997); Tallant and Krzycki, *J. Bacteriol.* 178:1295-1301 (1996); Ragsdale, S. W., *Crit. Rev. Biochem. Mol. Biol.* 39:165-195 (2004)).

MtaB is a zinc protein that can catalyze the transfer of a methyl group from methanol to MtaC, a corrinoid protein. Exemplary genes encoding MtaB and MtaC can be found in methanogenic archaea such as *Methanosarcina barkeri* (Maeder et al., *J. Bacteriol.* 188:7922-7931 (2006) and *Methanosarcina acetivorans* (Galagan et al., *Genome Res.* 12:532-542 (2002), as well as the acetogen, *Moorella thermoacetica* (Das et al., *Proteins* 67:167-176 (2007)). In general, the MtaB and MtaC genes are adjacent to one another on the chromosome as their activities are tightly interdependent. The protein sequences of various MtaB and MtaC encoding genes in *M. barkeri*, *M. acetivorans*, and *M. thermoacetica* can be identified by their following GenBank accession numbers.

Protein	GenBank ID	GI number	Organism
MtaB1	YP_304299	73668284	<i>Methanosarcina barkeri</i>
MtaC1	YP_304298	73668283	<i>Methanosarcina barkeri</i>
MtaB2	YP_307082	73671067	<i>Methanosarcina barkeri</i>
MtaC2	YP_307081	73671066	<i>Methanosarcina barkeri</i>
MtaB3	YP_304612	73668597	<i>Methanosarcina barkeri</i>
MtaC3	YP_304611	73668596	<i>Methanosarcina barkeri</i>
MtaB1	NP_615421	20089346	<i>Methanosarcina acetivorans</i>
MtaB1	NP_615422	20089347	<i>Methanosarcina acetivorans</i>
MtaB2	NP_619254	20093179	<i>Methanosarcina acetivorans</i>
MtaC2	NP_619253	20093178	<i>Methanosarcina acetivorans</i>
MtaB3	NP_616549	20090474	<i>Methanosarcina acetivorans</i>
MtaC3	NP_616550	20090475	<i>Methanosarcina acetivorans</i>
MtaB	YP_430066	83590057	<i>Moorella thermoacetica</i>
MtaC	YP_430065	83590056	<i>Moorella thermoacetica</i>
MtaA	YP_430064	83590056	<i>Moorella thermoacetica</i>

The MtaB1 and MtaC1 genes, YP_304299 and YP_304298, from *M. barkeri* were cloned into *E. coli* and sequenced (Sauer et al., *Eur. J. Biochem.* 243:670-677 (1997)). The crystal structure of this methanol-cobalamin methyltransferase complex is also available (Hagemeier et al., *Proc. Natl. Acad. Sci. U.S.A.* 103:18917-18922 (2006)). The MtaB genes, YP_307082 and YP_304612, in *M. barkeri* were identified by sequence homology to YP_304299. In general, homology searches are an effective means of identifying methanol methyltransferases because MtaB encoding genes show little or no similarity to methyltransferases that act on alternative substrates such as trimethylamine, dimethylamine, monomethylamine, or dimethylsulfide. The MtaC genes, YP_307081 and YP_304611 were identified based on their proximity to the MtaR genes and also their homology to YP_304298. The three sets of MtaB and MtaC genes from *M. acetivorans* have been genetically, physiologically, and biochemically characterized (Pritchett and Metcalf, *Mol. Microbiol.* 56:1183-1194 (2005)). Mutant strains lacking two of the sets were able to grow on methanol, whereas a strain lacking all three sets of MtaB and MtaC genes sets could not grow on methanol. This suggests that each set of genes plays a role in methanol utilization. The *M. thermoacetica* MtaB gene was identified based on homology to the methanogenic MtaB genes and also by its adjacent chromosomal proximity to the methanol-induced corrinoid protein, MtaC, which has been crystallized (Zhou et al., *Acta Crystallogr. Sect. F. Struct. Biol. Cryst. Commun.* 61:537-540 (2005) and further characterized by Northern hybridization and Western Blotting ((Das et al., *Proteins* 67:167-176 (2007)).

MtaA is zinc protein that catalyzes the transfer of the methyl group from MtaC to either Coenzyme M in methanogens or methyltetrahydrofolate in acetogens. MtaA can also utilize methylcobalamin as the methyl donor. Exemplary genes encoding MtaA can be found in methanogenic archaea such as *Methanosarcina barkeri* (Maeder et al., *J. Bacteriol.* 188:7922-7931 (2006) and *Methanosarcina acetivorans* (Galagan et al., *Genome Res.* 12:532-542 (2002), as well as the acetogen, *Moorella thermoacetica* ((Das et al., *Proteins* 67:167-176 (2007)). In general, MtaA proteins that catalyze the transfer of the methyl group from CH₃-MtaC are difficult to identify bioinformatically as they share similarity to other corrinoid protein methyltransferases and

are not oriented adjacent to the MtaB and MtaC genes on the chromosomes. Nevertheless, a number of MtaA encoding genes have been characterized. The protein sequences of these genes in *M. barkeri* and *M. acetivorans* can be identified by the following GenBank accession numbers.

Protein	GenBank ID	GI number	Organism
MtaA	YP_304602	73668587	<i>Methanosarcina barkeri</i>
MtaA1	NP_619241	20093166	<i>Methanosarcina acetivorans</i>
MtaA2	NP_616548	20090473	<i>Methanosarcina acetivorans</i>

The MtaA gene, YP_304602, from *M. barkeri* was cloned, sequenced, and functionally overexpressed in *E. coli* (Harms and Thauer, *Eur. J. Biochem.* 235:653-659 (1996)). In *M. acetivorans*, MtaA1 is required for growth on methanol, whereas MtaA2 is dispensable even though methane production from methanol is reduced in MtaA2 mutants (Bose et al., *J. Bacteriol.* 190:4017-4026 (2008)). There are multiple additional MtaA homologs in *M. barkeri* and *M. acetivorans* that are as yet uncharacterized, but may also catalyze corrinoid protein methyltransferase activity.

Putative MtaA encoding genes in *M. thermoacetica* were identified by their sequence similarity to the characterized methanogenic MtaA genes. Specifically, three *M. thermoacetica* genes show high homology (>30% sequence identity) to YP_304602 from *M. barkeri*. Unlike methanogenic MtaA proteins that naturally catalyze the transfer of the methyl group from CH₃-MtaC to Coenzyme M, anM thermoacetica MtaA is likely to transfer the methyl group to methyltetrahydrofolate given the similar roles of methyltetrahydrofolate and Coenzyme M in methanogens and acetogens, respectively. The protein sequences of putative MtaA encoding genes from *M. thermoacetica* can be identified by the following GenBank accession numbers.

Protein	GenBank ID	GI number	Organism
MtaA	YP_430937	83590928	<i>Moorella thermoacetica</i>
MtaA	YP_431175	83591166	<i>Moorella thermoacetica</i>
MtaA	YP_430935	83590926	<i>Moorella thermoacetica</i>
MtaA	YP_430064	83590056	<i>Moorella thermoacetica</i>

FIG. 10, Step B—Methylenetetrahydrofolate Reductase

The conversion of methyl-THF to methylenetetrahydrofolate is catalyzed by methylenetetrahydrofolate reductase. In *M. thermoacetica*, this enzyme is oxygen-sensitive and contains an iron-sulfur cluster (Clark and Ljungdahl, *J. Biol. Chem.* 259:10845-10849 (1984). This enzyme is encoded by metF in *E. coli* (Sheppard et al., *J. Bacteriol.* 181:718-725 (1999) and CHY_1233 in *C. hydrogenoformans* (Wu et al., *PLoS Genet.* 1:e65 (2005). The *M. thermoacetica* genes, and its *C. hydrogenoformans* counterpart, are located near the CODH/ACS gene cluster, separated by putative hydrogenase and heterodisulfide reductase genes. Some additional gene candidates found bioinformatically are listed below. In *Acetobacterium woodii* metF is coupled to the Rnf complex through RnfC2 (Poehlein et al, *PLoS One.* 7:e33439). Homologs of RnfC are found in other organisms by blast search. The Rnf complex is known to be a reversible complex (Fuchs (2011) *Annu. Rev. Microbiol.* 65:631-658).

Protein	GenBank ID	GI number	Organism
Moth_1191	YP_430048.1	83590039	<i>Moorella thermoacetica</i>
Moth_1192	YP_430049.1	83590040	<i>Moorella thermoacetica</i>
metF	NP_418376.1	16131779	<i>Escherichia coli</i>
CHY_1233	YP_360071.1	78044792	<i>Carboxydotherrmus hydrogenoformans</i>
CLJU_c37610	YP_003781889.1	300856905	<i>Clostridium ljunydhlii</i> DSM 13528
DesfrDRAFT_3717	ZP_07335241.1	303248996	<i>Desulfovibrio fructosovorans</i> JJ
CcarbDRAFT_2950	ZP_05392950.1	255526026	<i>Clostridium carboxidivorans</i> P7
Ccel74_010100023124	ZP_07633513.1	307691067	<i>Clostridium cellulovorans</i> 743B
Cphy_3110	YP_001560205.1	160881237	<i>Clostridium phytofermentans</i> ISDg

FIG. 10, Steps C and D—Methylenetetrahydrofolate Dehydrogenase, Methenyltetrahydrofolate Cyclohydrolase

In *M. thermoacetica*, *E. coli*, and *C. hydrogenoformans*,¹⁵ methenyltetrahydrofolate cyclohydrolase and methylenetetrahydrofolate dehydrogenase are carried out by the bifunctional gene products of Moth_1516, folD, and CHY_1878, respectively (Pierce et al., *Environ. Microbiol.* 10:2550-2573 (2008); Wu et al., *PLoS Genet.* 1:e65 (2005); D'Ari and Rabinowitz, *J. Biol. Chem.* 266:23953-23958 (1991)). A homolog exists in *C. carboxidivorans* P7. Several other organisms also encode for this bifunctional protein as²⁵ tabulated below.

Protein	GenBank ID	GI number	Organism
Moth_1516	YP_430368.1	83590359	<i>Moorella thermoacetica</i>
folD	NP_415062.1	16128513	<i>Escherichia coli</i>
CHY_1878	YP_360698.1	78044829	<i>Carboxydotherrmus hydrogenoformans</i>
CcarbDRAFT_2948	ZP_05392948.1	255526024	<i>Clostridium carboxidivorans</i> P7
folD	ADK16789.1	300437022	<i>Clostridium ljunydhlii</i> DSM 13528
folD-2	NP_951919.1	39995968	<i>Geobacter sulfurreducens</i> PCA
folD	YP_725874.1	113867385	<i>Ralstonia eutropha</i> H16
folD	NP_348702.1	15895353	<i>Clostridium acetobutylicum</i> ATCC 824
folD	YP_696506.1	110800457	<i>Clostridium perfringens</i>
MGA3_09460	EIJ83438.1	387591119	<i>Bacillus methanolicus</i> MGA3
PB1_14689	ZP_10132349.1	387929672	<i>Bacillus methanolicus</i> PB1

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FIG. 10, Step E—Formyltetrahydrofolate Deformylase

This enzyme catalyzes the hydrolysis of 10-formyltetrahydrofolate (formyl-THF) to THF and formate. In *E. coli*, this enzyme is encoded by purU and has been overproduced, purified, and characterized (Nagy, et al., *J. Bacteriol.* 3:1292-1298 (1995)). Homologs exist in *Corynebacterium* sp. U-96 (Suzuki, et al., *Biosci. Biotechnol. Biochem.* 69(5): 952-956 (2005)), *Corynebacterium glutamicum* ATCC 14067, *Salmonella enterica*, and several additional organisms.⁵⁵

Protein	GenBank ID	GI number	Organism
purU	AAC74314.1	1787483	<i>Escherichia coli</i> K-12 MG1655
purU	BAD97821.1	63002616	<i>Corynebacterium</i> sp. U-96
purU	EHE84645.1	354511740	<i>Corynebacterium glutamicum</i> ATCC 14067
purU	NP_460715.1	16765100	<i>Salmonella enterica</i> subsp. <i>enterica</i> serovar <i>Typhimurium</i> str. LT2

FIG. 10, Step F—Formyltetrahydrofolate Synthetase

Formyltetrahydrofolate synthetase ligates formate to tetrahydrofolate at the expense of one ATP. This reaction is catalyzed by the gene product of Moth_0109 in *M. thermoacetica* (O'Brien et al., *Experientia Suppl.* 26:249-262 (1976); Lovell et al., *Arch. Microbiol.* 149:280-285 (1988); Lovell et al., *Biochemistry* 29:5687-5694 (1990)), FHS in *Clostridium acidurici* (Whitehead and Rabinowitz, *J. Bacteriol.* 167:203-209 (1986); Whitehead and Rabinowitz, *J. Bacteriol.* 170:3255-3261 (1988), and CHY_2385 in *C. hydrogenoformans* (Wu et al., *PLoS Genet.* 1:e65 (2005)). Homologs exist in *C. carboxidivorans* P7. This enzyme is found in several other organisms as listed below.

Protein	GenBank ID	GI number	Organism
Moth_0109	YP_428991.1	83588982	<i>Moorella thermoacetica</i>
CHY_2385	YP_361182.1	78045024	<i>Carboxydotherrus hydrogenoformans</i>
FHS	P13419.1	120562	<i>Clostridium acidurici</i>
CcarbDRAFT_1913	ZP_05391913.1	255524966	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2946	ZP_05392946.1	255526022	<i>Clostridium carboxidivorans</i> P7
Dhaf_0555	ACL18622.1	219536883	<i>Desulfitobacterium hafniense</i>
fhs	YP_001393842.1	153953077	<i>Clostridium kluyveri</i> DSM 555
fhs	YP_003781893.1	300856909	<i>Clostridium ljumydahlil</i> DSM 13528
MGA3_08300	EIJ83208.1	387590889	<i>Bacillus methanolicus</i> MGA3
PB1_13509	ZP_10132113.1	387929436	<i>Bacillus methanolicus</i> PB1

FIG. 10, Step G—Formate Hydrogen Lyase

A formate hydrogen lyase enzyme can be employed to convert formate to carbon dioxide and hydrogen. An exemplary formate hydrogen lyase enzyme can be found in *Escherichia coli*. The *E. coli* formate hydrogen lyase consists of hydrogenase 3 and formate dehydrogenase-H (Maeda et al., *Appl Microbiol Biotechnol* 77:879-890 (2007)). It is activated by the gene product of fhfA. (Maeda et al., *Appl Microbiol Biotechnol* 77:879-890 (2007)). The addition of the trace elements, selenium, nickel and molybdenum, to a fermentation broth has been shown to enhance formate hydrogen lyase activity (Soini et al., *Microb. Cell Fact.* 7:26 (2008)). Various hydrogenase 3, formate dehydrogenase and transcriptional activator genes are shown below.

Protein	GenBank ID	GI number	Organism
hycA	NP_417205	16130632	<i>Escherichia coli</i> K-12 MG1655
hycB	NP_417204	16130631	<i>Escherichia coli</i> K-12 MG1655
hycC	NP_417203	16130630	<i>Escherichia coli</i> K-12 MG1655
hycD	NP_417202	16130629	<i>Escherichia coli</i> K-12 MG1655
hycE	NP_417201	16130628	<i>Escherichia coli</i> K-12 MG1655
hycF	NP_417200	16130627	<i>Escherichia coli</i> K-12 MG1655
hycG	NP_417199	16130626	<i>Escherichia coli</i> K-12 MG1655
hycH	NP_417198	16130625	<i>Escherichia coli</i> K-12 MG1655
hycI	NP_417197	16130624	<i>Escherichia coli</i> K-12 MG1655
fdhF	NP_418503	16131905	<i>Escherichia coli</i> K-12 MG1655
fhfA	NP_417211	16130638	<i>Escherichia coli</i> K-12 MG1655

A formate hydrogen lyase enzyme also exists in the hyperthermophilic archaeon, *Thermococcus litoralis* (Takacs et al., *BMC Microbiol* 8:88 (2008)).

Protein	GenBank ID	GI number	Organism
mhyC	ABW05543	157954626	<i>Thermococcus litoralis</i>
mhyD	ABW05544	157954627	<i>Thermococcus litoralis</i>
mhyE	ABW05545	157954628	<i>Thermococcus litoralis</i>
myhF	ABW05546	157954629	<i>Thermococcus litoralis</i>
myhG	ABW05547	157954630	<i>Thermococcus litoralis</i>

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Protein	GenBank ID	GI number	Organism
myhH	ABW05548	157954631	<i>Thermococcus litoralis</i>
fdhA	AAB94932	2146136	<i>Thermococcus litoralis</i>
fdhB	AAB94931	157954625	<i>Thermococcus litoralis</i>

Additional formate hydrogen lyase systems have been found in *Salmonella typhimurium*, *Klebsiella pneumoniae*, *Rhodospirillum rubrum*, *Methanobacterium formicicum* (Vardar-Schara et al., *Microbial Biotechnology* 1:107-125 (2008)).

FIG. 10, Step H—Hydrogenase

Hydrogenase enzymes can convert hydrogen gas to protons and transfer electrons to acceptors such as ferredoxins, NAD⁺, or NADP⁺. *Ralstonia eutropha* H16 uses hydrogen as an energy source with oxygen as a terminal electron acceptor. Its membrane-bound uptake [NiFe]-hydrogenase is an "O₂-tolerant" hydrogenase (Cracknell, et al. *Proc Natl Acad Sci*, 106(49) 20681-20686 (2009)) that is periplasmically-oriented and connected to the respiratory chain via a b-type cytochrome (Schink and Schlegel, *Biochim. Biophys. Acta*, 567, 315-324 (1979); Bernhard et al., *Eur. J. Biochem.* 248, 179-186 (1997)). *R. eutropha* also contains an O₂-tolerant soluble hydrogenase encoded by the Hox operon which is cytoplasmic and directly reduces NAD⁺ at the expense of hydrogen (Schneider and Schlegel, *Biochim. Biophys. Acta* 452, 66-80 (1976); Burgdorf, *J. Bact.* 187(9) 3122-3132(2005)). Soluble hydrogenase enzymes are additionally present in several other organisms including *Geobacter sulfurreducens* (Coppi, *Microbiology* 151, 1239-1254 (2005)), *Synechocystis* str. PCC 6803 (Germer, *J. Biol. Chem.*, 284(52), 36462-36472 (2009)), and *Thiocapsa roseopersicina* (Rakhely, *Appl. Environ. Microbiol.* 70(2) 722-728 (2004)). The *Synechocystis* enzyme is capable of generating NADPH from hydrogen. Overexpression of both the Hox operon from *Synechocystis* str. PCC 6803 and the accessory genes encoded by the Hyp operon from *Nostoc* sp. PCC 7120 led to increased hydrogenase activity compared to expression of the Hox genes alone (Germer, *J. Biol. Chem.* 284(52), 36462-36472 (2009)).

Protein	Gen Bank ID	GI Number	Organism
HoxF	NP_942727.1	38637753	<i>Ralstonia eutropha</i> HI 6
HoxU	NP_942728.1	38637754	<i>Ralstonia eutropha</i> HI 6
HoxY	NP_942729.1	38637755	<i>Ralstonia eutropha</i> HI 6
HoxH	NP_942730.1	38637756	<i>Ralstonia eutropha</i> HI 6
HoxW	NP_942731.1	38637757	<i>Ralstonia eutropha</i> HI 6
HoxI	NP_942732.1	38637758	<i>Ralstonia eutropha</i> HI 6
HoxE	NP_953767.1	39997816	<i>Geobacter sulfurreducens</i>
HoxF	NP_953766.1	39997815	<i>Geobacter sulfurreducens</i>

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Protein	Gen Bank ID	GI Number	Organism
HoxU	NP_953765.1	39997814	<i>Geobacter sulfurreducens</i>
HoxY	NP_953764.1	39997813	<i>Geobacter sulfurreducens</i>
HoxH	NP_953763.1	39997812	<i>Geobacter sulfurreducens</i>
GSU2717	NP_953762.1	39997811	<i>Geobacter sulfurreducens</i>
HoxE	NP_441418.1	16330690	<i>Synechocystis</i> str. PCC 6803
HoxF	NP_441417.1	16330689	<i>Synechocystis</i> str. PCC 6803
Unknown function	NP_441416.1	16330688	<i>Synechocystis</i> str. PCC 6803
HoxU	NP_441415.1	16330687	<i>Synechocystis</i> str. PCC 6803
HoxY	NP_441414.1	16330686	<i>Synechocystis</i> str. PCC 6803
Unknown function	NP_441413.1	16330685	<i>Synechocystis</i> str. PCC 6803
Unknown function	NP_441412.1	16330684	<i>Synechocystis</i> str. PCC 6803
HoxH	NP_441411.1	16330683	<i>Synechocystis</i> str. PCC 6803
HypF	NP_484737.1	17228189	<i>Nostoc</i> sp. PCC 7120
HypC	NP_484738.1	17228190	<i>Nostoc</i> sp. PCC 7120
HypD	NP_484739.1	17228191	<i>Nostoc</i> sp. PCC 7120
Unknown function	NP_484740.1	17228192	<i>Nostoc</i> sp. PCC 7120
HypE	NP_484741.1	17228193	<i>Nostoc</i> sp. PCC 7120
HypA	NP_484742.1	17228194	<i>Nostoc</i> sp. PCC 7120
HypB	NP_484743.1	17228195	<i>Nostoc</i> sp. PCC 7120
Hox1E	AAP50519.1	37787351	<i>Thiocapsa roseopersicina</i>
Hox1F	AAP50520.1	37787352	<i>Thiocapsa roseopersicina</i>
Hox1U	AAP50521.1	37787353	<i>Thiocapsa roseopersicina</i>
Hox1Y	AAP50522.1	37787354	<i>Thiocapsa roseopersicina</i>
Hox1H	AAP50523.1	37787355	<i>Thiocapsa roseopersicina</i>

The genomes of *E. coli* and other enteric bacteria encode up to four hydrogenase enzymes (Sawers, G., *Antonie Van Leeuwenhoek* 66:57-88 (1994); Sawers et al., *J Bacteriol.* 164:1324-1331(1985); Sawers and Boxer, *Eur J Biochem.* 156:265-275 (1986); Sawers et al., *J Bacteriol.* 168:398-404 (1986)). Given the multiplicity of enzyme activities *E. coli* or another host organism can provide sufficient hydrogenase activity to split incoming molecular hydrogen and reduce the corresponding acceptor. Endogenous hydrogen-lyase enzymes of *E. coli* include hydrogenase 3, a membrane-bound enzyme complex using ferredoxin as an acceptor, and hydrogenase 4 that also uses a ferredoxin acceptor. Hydrogenase 3 and 4 are encoded by the hyc and hyf gene clusters, respectively. Hydrogenase activity in *E. coli* is also dependent upon the expression of the hyp genes whose corresponding proteins are involved in the assembly of the hydrogenase complexes (Jacobi et al., *Arch.Microbiol* 158: 444-451(1992); Rangarajan et al., *J Bacteriol.* 190:1447-1458 (2008)). The *M. thermoacetica* and *Clostridium ljungdahli* hydrogenases are suitable for a host that lacks sufficient endogenous hydrogenase activity. *M. thermoacetica* and *C. ljungdahli* can grow with CO₂ as the exclusive carbon source indicating that reducing equivalents are extracted from H₂ to enable acetyl-CoA synthesis via the Wood-Ljungdahl pathway (Drake, H. L., *J Bacteriol.* 150: 702-709 (1982); Drake and Daniel, *Res Microbiol* 155:869-883 (2004); Kellum and Drake, *J Bacteriol.* 160:466-469 (1984)). *M. thermoacetica* has homologs to several hyp, hyc, and hyf genes from *E. coli*. These protein sequences encoded for by these genes are identified by the following GenBank accession numbers. In addition, several gene clusters encoding hydrogenase functionality are present in *M. thermoacetica* and *C. ljungdahli* (see for example US 2012/0003652).

Protein	GenBank ID	GI Number	Organism
HypA	NP_417206	16130633	<i>Escherichia coli</i>
HypB	NP_417207	16130634	<i>Escherichia coli</i>
HypC	NP_417208	16130635	<i>Escherichia coli</i>
HypD	NP_417209	16130636	<i>Escherichia coli</i>
HypE	NP_417210	226524740	<i>Escherichia coli</i>

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Protein	GenBank ID	GI Number	Organism
HypF	NP_417192	16130619	<i>Escherichia coli</i>
HycA	NP_417205	16130632	<i>Escherichia coli</i>
HycB	NP_417204	16130631	<i>Escherichia coli</i>
HycC	NP_417203	16130630	<i>Escherichia coli</i>
HycD	NP_417202	16130629	<i>Escherichia coli</i>
HycE	NP_417201	16130628	<i>Escherichia coli</i>
HycF	NP_417200	16130627	<i>Escherichia coli</i>
HycG	NP_417199	16130626	<i>Escherichia coli</i>
HycH	NP_417198	16130625	<i>Escherichia coli</i>
Hycl	NP_417197	16130624	<i>Escherichia coli</i>
HyfA	NP_416976	90111444	<i>Escherichia coli</i>
HyfB	NP_416977	16130407	<i>Escherichia coli</i>
HyfC	NP_416978	90111445	<i>Escherichia coli</i>
HyfD	NP_416979	16130409	<i>Escherichia coli</i>
HyfE	NP_416980	16130410	<i>Escherichia coli</i>
HyfF	NP_416981	16130411	<i>Escherichia coli</i>
HyfG	NP_416982	16130412	<i>Escherichia coli</i>
HyfH	NP_416983	16130413	<i>Escherichia coli</i>
HyfI	NP_416984	16130414	<i>Escherichia coli</i>
HyfJ	NP_416985	90111446	<i>Escherichia coli</i>
HyfR	NP_416986	90111447	<i>Escherichia coli</i>

Proteins in *M. thermoacetica* whose genes are homologous to the *E. coli* hydrogenase genes are shown below.

Protein	GenBank ID	GI Number	Organism
Moth_2175	YP_431007	83590998	<i>Moorella thermoacetica</i>
Moth_2176	YP_431008	83590999	<i>Moorella thermoacetica</i>
Moth_2177	YP_431009	83591000	<i>Moorella thermoacetica</i>
Moth_2178	YP_431010	83591001	<i>Moorella thermoacetica</i>
Moth_2179	YP_431011	83591002	<i>Moorella thermoacetica</i>
Moth_2180	YP_431012	83591003	<i>Moorella thermoacetica</i>
Moth_2181	YP_431013	83591004	<i>Moorella thermoacetica</i>
Moth_2182	YP_431014	83591005	<i>Moorella thermoacetica</i>
Moth_2183	YP_431015	83591006	<i>Moorella thermoacetica</i>
Moth_2184	YP_431016	83591007	<i>Moorella thermoacetica</i>
Moth_2185	YP_431017	83591008	<i>Moorella thermoacetica</i>
Moth_2186	YP_431018	83591009	<i>Moorella thermoacetica</i>
Moth_2187	YP_431019	83591010	<i>Moorella thermoacetica</i>
Moth_2188	YP_431020	83591011	<i>Moorella thermoacetica</i>
Moth_2189	YP_431021	83591012	<i>Moorella thermoacetica</i>
Moth_2190	YP_431022	83591013	<i>Moorella thermoacetica</i>
Moth_2191	YP_431023	83591014	<i>Moorella thermoacetica</i>
Moth_2192	YP_431024	83591015	<i>Moorella thermoacetica</i>
Moth_0439	YP_429313	83589304	<i>Moorella thermoacetica</i>
Moth_0440	YP_429314	83589305	<i>Moorella thermoacetica</i>
Moth_0441	YP_429315	83589306	<i>Moorella thermoacetica</i>
Moth_0442	YP_429316	83589307	<i>Moorella thermoacetica</i>
Moth_0809	YP_429670	83589661	<i>Moorella thermoacetica</i>
Moth_0810	YP_429671	83589662	<i>Moorella thermoacetica</i>
Moth_0811	YP_429672	83589663	<i>Moorella thermoacetica</i>
Moth_0812	YP_429673	83589664	<i>Moorella thermoacetica</i>
Moth_0814	YP_429674	83589665	<i>Moorella thermoacetica</i>
Moth_0815	YP_429675	83589666	<i>Moorella thermoacetica</i>
Moth_0816	YP_429676	83589667	<i>Moorella thermoacetica</i>
Moth_1193	YP_430050	83590041	<i>Moorella thermoacetica</i>
Moth_1194	YP_430051	83590042	<i>Moorella thermoacetica</i>
Moth_1195	YP_430052	83590043	<i>Moorella thermoacetica</i>
Moth_1196	YP_430053	83590044	<i>Moorella thermoacetica</i>
Moth_1717	YP_430562	83590553	<i>Moorella thermoacetica</i>
Moth_1718	YP_430563	83590554	<i>Moorella thermoacetica</i>
Moth_1719	YP_430564	83590555	<i>Moorella thermoacetica</i>
Moth_1883	YP_430726	83590717	<i>Moorella thermoacetica</i>
Moth_1884	YP_430727	83590718	<i>Moorella thermoacetica</i>

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Protein	GenBank ID	GI Number	Organism
Moth_1885	YP_430728	83590719	<i>Moorella thermoacetica</i>
Moth_1886	YP_430729	83590720	<i>Moorella thermoacetica</i>
Moth_1887	YP_430730	83590721	<i>Moorella thermoacetica</i>
Moth_1888	YP_430731	83590722	<i>Moorella thermoacetica</i>
Moth_1452	YP_430305	83590296	<i>Moorella thermoacetica</i>
Moth_1453	YP_430306	83590297	<i>Moorella thermoacetica</i>
Moth_1454	YP_430307	83590298	<i>Moorella thermoacetica</i>

Genes encoding hydrogenase enzymes from *C. ljungdahli* are shown below.

Protein	GenBank ID	GI Number	Organism
CLJU_c02090	ADK15091.1	300435324	<i>Clostridium ljungdahli</i>
CLJU_c07030	ADK13773.1	300434006	<i>Clostridium ljungdahli</i>
CLJU_c07040	ADK13774.1	300434007	<i>Clostridium ljungdahli</i>
CLJU_c07050	ADK13775.1	300434008	<i>Clostridium ljungdahli</i>
CLJU_c07060	ADK13776.1	300434009	<i>Clostridium ljungdahli</i>
CLJU_c07070	ADK13777.1	300434010	<i>Clostridium ljungdahli</i>
CLJU_c07080	ADK13778.1	300434011	<i>Clostridium ljungdahli</i>
CLJU_c14730	ADK14541.1	300434774	<i>Clostridium ljungdahli</i>
CLJU_c14720	ADK14540.1	300434773	<i>Clostridium ljungdahli</i>
CLJU_c14710	ADK14539.1	300434772	<i>Clostridium ljungdahli</i>
CLJU_c14700	ADK14538.1	300434771	<i>Clostridium ljungdahli</i>
CLJU_c28670	ADK15915.1	300436148	<i>Clostridium ljungdahli</i>
CLJU_c28660	ADK15914.1	300436147	<i>Clostridium ljungdahli</i>
CLJU_c28650	ADK15913.1	300436146	<i>Clostridium ljungdahli</i>
CLJU_c28640	ADK15912.1	300436145	<i>Clostridium ljungdahli</i>

In some cases, hydrogenase encoding genes are located adjacent to a CODH. In *Rhodospirillum rubrum*, the encoded CODH/hydrogenase proteins form a membrane-bound enzyme complex that has been indicated to be a site where energy, in the form of a proton gradient, is generated from the conversion of CO and H₂O to CO₂ and H₂ (Fox et al., *J. Bacteriol.* 178:6200-6208 (1996)). The CODH-I of *C. hydrogenoformans* and its adjacent genes have been proposed to catalyze a similar functional role based on their similarity to the *R. rubrum* CODH/hydrogenase gene cluster (Wu et al., *PLoS Genet.* 1:e65 (2005)). The *C. hydrogenoformans* CODH-I was also shown to exhibit intense CO oxidation and CO₂ reduction activities when linked to an electrode (Parkin et al., *J. Am. Chem. Soc.* 129:10328-10329 (2007)).

Protein	GenBank ID	GI Number	Organism
CooL	AAC45118	1515468	<i>Rhodospirillum rubrum</i>
CooX	AAC45119	1515469	<i>Rhodospirillum rubrum</i>
CooU	AAC45120	1515470	<i>Rhodospirillum rubrum</i>
CooH	AAC45121	1498746	<i>Rhodospirillum rubrum</i>
CooF	AAC45122	1498747	<i>Rhodospirillum rubrum</i>
CODH(CooS)	AAC45123	1498748	<i>Rhodospirillum rubrum</i>
CooC	AAC45124	1498749	<i>Rhodospirillum rubrum</i>
CooT	AAC45125	1498750	<i>Rhodospirillum rubrum</i>
CooJ	AAC45126	1498751	<i>Rhodospirillum rubrum</i>
CODH-I (CooS-I)	YP_360644	78043418	<i>Carboxydotherrmus hydrogenoformans</i>
CooF	YP_360645	78044791	<i>Carboxydotherrmus hydrogenoformans</i>
HypA	YP_360646	78044340	<i>Carboxydotherrmus hydrogenoformans</i>
CooH	YP_360647	78043871	<i>Carboxydotherrmus hydrogenoformans</i>
CooU	YP_360648	78044023	<i>Carboxydotherrmus hydrogenoformans</i>
CooX	YP_360649	78043124	<i>Carboxydotherrmus hydrogenoformans</i>
CooL	YP_360650	78043938	<i>Carboxydotherrmus hydrogenoformans</i>
CooK	YP_360651	78044700	<i>Carboxydotherrmus hydrogenoformans</i>
CooM	YP_360652	78043942	<i>Carboxydotherrmus hydrogenoformans</i>
CooC	YP_360654.1	78043296	<i>Carboxydotherrmus hydrogenoformans</i>
CooA-1	YP_360655.1	78044021	<i>Carboxydotherrmus hydrogenoformans</i>

Some hydrogenase and CODH enzymes transfer electrons to ferredoxins. Ferredoxins are small acidic proteins containing one or more iron-sulfur clusters that function as intracellular electron carriers with a low reduction potential. Reduced ferredoxins donate electrons to Fe-dependent enzymes such as ferredoxin-NADP⁺ oxidoreductase, pyruvate:ferredoxin oxidoreductase (PFOR) and 2-oxoglutarate:ferredoxin oxidoreductase (OFOR). The *H. thermophilus* gene *fdx1* encodes a [4Fe-4S]-type ferredoxin that is required for the reversible carboxylation of 2-oxoglutarate and pyruvate by OFOR and PFOR, respectively (Yamamoto et al., *Extremophiles* 14:79-85 (2010)). The ferredoxin associated with the *Sulfolobus solfataricus* 2-oxoacid:ferredoxin reductase is a monomeric dimer [3Fe-4S][4Fe-4S] type ferredoxin (Park et al. 2006). While the gene associated with this protein has not been fully sequenced, the N-terminal domain shares 93% homology with the *zfx* ferredoxin from *S. acidocaldarius*. The *E. coli* genome encodes a soluble ferredoxin of unknown physiological function, *fdx*. Some evidence indicates that this protein can function in iron-sulfur cluster assembly (Takahashi and Nakamura, 1999). Additional ferredoxin proteins have been characterized in *Helicobacter pylori* (Mukhopadhyay et al. 2003) and *Campylobacter jejuni* (van Vliet et al. 2001). A 2Fe-2S ferredoxin from *Clostridium pasteurianum* has been cloned and expressed in *E. coli* (Fujinaga and Meyer, Biochemical and Biophysical Research Communications, 192(3): (1993)). Acetogenic bacteria such as *Moorella thermoacetica*, *Clostridium carboxidivorans* P7, *Clostridium ljungdahli* and *Rhodospirillum rubrum* are predicted to encode several ferredoxins, listed below.

Protein	GenBank ID	GI Number	Organism
fdx1	BAE02673.1	68163284	<i>Hydrogenobacter thermophilus</i>
M11214.1	AAA83524.1	144806	<i>Clostridium pasteurianum</i>
Zfx	AAAY79867.1	68566938	<i>Sulfolobus acidocalarius</i>
Fdx	AAC75578.1	1788874	<i>Escherichia coli</i>
hlp_0277	AAD07340.1	2313367	<i>Helicobacter pylori</i>
fdxA	CAL34484.1	112359698	<i>Campylobacter jejuni</i>
Moth_0061	ABC18400.1	83571848	<i>Moorella thermoacetica</i>
Moth_1200	ABC19514.1	83572962	<i>Moorella thermoacetica</i>
Moth_1888	ABC20188.1	83573636	<i>Moorella thermoacetica</i>
Moth_2112	ABC20404.1	83573852	<i>Moorella thermoacetica</i>
Moth_1037	ABC19351.1	83572799	<i>Moorella thermoacetica</i>
CcarbDRAFT_4383	ZP_05394383.1	255527515	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2958	ZP_05392958.1	255526034	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2281	ZP_05392281.1	255525342	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_5296	ZP_05395295.1	255528511	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_1615	ZP_05391615.1	255524662	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_1304	ZP_05391304.1	255524347	<i>Clostridium carboxidivorans</i> P7
cooF	AAG29808.1	11095245	<i>Carboxydotherrnus hydrogenoformans</i>
fdxN	CAA35699.1	46143	<i>Rhodobacter capsulatus</i>
Rru_A2264	ABC23064.1	83576513	<i>Rhodospirillum rubrum</i>
Rru_A1916	ABC22716.1	83576165	<i>Rhodospirillum rubrum</i>
Rru_A2026	ABC22826.1	83576275	<i>Rhodospirillum rubrum</i>
cooF	AAC45122.1	1498747	<i>Rhodospirillum rubrum</i>
fdxN	AAA26460.1	152605	<i>Rhodospirillum rubrum</i>
Alvin_2884	ADC63789.1	288897953	<i>Allochrochromatium vinosum</i> DSM 180
Fdx	YP_002801146.1	226946073	<i>Azotobacter vinelandii</i> DJ
CKL_3790	YP_001397146.1	153956381	<i>Clostridium kluyveri</i> DSM 555
fer1	NP_949965.1	39937689	<i>Rhodopseudomonas palustris</i> CGA009
Fdx	CAA12251.1	3724172	<i>Thauera aromatica</i>
CHY_2405	YP_361202.1	78044690	<i>Carboxydotherrnus hydrogenoformans</i>
Fer	YP_359966.1	78045103	<i>Carboxydotherrnus hydrogenoformans</i>
Fer	AAC83945.1	1146198	<i>Bacillus subtilis</i>
fdx1	NP_249053.1	15595559	<i>Pseudomonas aeruginosa</i> PA01
yfhL	AP_003148.1	89109368	<i>Escherichia coli</i> K-12
CLJU_c00930	ADK13195.1	300433428	<i>Clostridium ljungdahli</i>
CLJU_c00010	ADK13115.1	300433348	<i>Clostridium ljungdahli</i>
CLJU_c01820	ADK13272.1	300433505	<i>Clostridium ljungdahli</i>
CLJU_cl 7980	ADK14861.1	300435094	<i>Clostridium ljungdahli</i>
CLJU_c17970	ADK14860.1	300435093	<i>Clostridium ljungdahli</i>
CLJU_c22510	ADK15311.1	300435544	<i>Clostridium ljungdahli</i>
CLJU_c26680	ADK15726.1	300435959	<i>Clostridium ljungdahli</i>
CLJU_c29400	ADK15988.1	300436221	<i>Clostridium ljungdahli</i>

Ferredoxin oxidoreductase enzymes transfer electrons from ferredoxins or flavodoxins to NAD(P)H. Two enzymes catalyzing the reversible transfer of electrons from reduced ferredoxins to NAD(P)⁺ are ferredoxin:NAD⁺ oxidoreductase (EC 1.18.1.3) and ferredoxin:NADP⁺ oxidoreductase (FNR, EC 1.18.1.2). Ferredoxin:NADP⁺ oxidoreductase (FNR, EC 1.18.1.2) has a noncovalently bound FAD cofactor that facilitates the reversible transfer of electrons from NADPH to low-potential acceptors such as ferredoxins or flavodoxins (Blaschkowski et al., *Eur. J. Biochem.* 123:563-569 (1982); Fujii et al., 1977). The *Helicobacter pylori* FNR, encoded by HP1164 (fqrB), is coupled to the activity of pyruvate:ferredoxin oxidoreductase (PFOR) resulting in the pyruvate-dependent production of NADPH (St et al. 2007). An analogous enzyme is found in *Campylobacter jejuni* (St Maurice et al., *J. Bacteriol.* 189: 4764-4773 (2007)). A ferredoxin:NADP⁺ oxidoreductase enzyme is encoded in the *E. coli* genome by fpr (Bianchi et al. 1993). Ferredoxin:NAD⁺ oxidoreductase utilizes

reduced ferredoxin to generate NADH from NAD⁺. In several organisms, including *E. coli*, this enzyme is a component of multifunctional dioxygenase enzyme complexes. The ferredoxin:NAD⁺ oxidoreductase of *E. coli*, encoded by hcaD, is a component of the 3-phenylpropionate dioxygenase system involved in aromatic acid utilization (Diaz et al. 1998). NADH:ferredoxin reductase activity was detected in cell extracts of *Hydrogenobacter thermophilus*, although a gene with this activity has not yet been indicated (Yoon et al. 2006). Additional ferredoxin:NAD(P)⁺ oxidoreductases have been annotated in *Clostridium carboxidivorans* P7. The NADH-dependent reduced ferredoxin: NADP oxidoreductase of *C. kluyveri*, encoded by nfnAB, catalyzes the concomitant reduction of ferredoxin and NAD⁺ with two equivalents of NADPH (Wang et al, *J Bacteriol* 192: 5115-5123 (2010)). Finally, the energy-conserving membrane-associated Rnf-type proteins (Seedorf et al, *PNAS* 105:2128-2133 (2008); and Hermann, *J. Bacteriol* 190:784-791 (2008)) provide a means to generate NADH or NADPH from reduced ferredoxin.

Protein	GenBank ID	GI Number	Organism
fqrB	NP_207955.1	15645778	<i>Helicobacter pylori</i>
fqrB	YP_001482096.1	157414840	<i>Campylobacter jejuni</i>
RPA3954	CAE29395.1	39650872	<i>Rhodopseudomonas palustris</i>
Fpr	BAH29712.1	225320633	<i>Hydrogenobacter thermophilus</i>
yumC	NP_391091.2	255767736	<i>Bacillus subtilis</i>

Protein	GenBank ID	GI Number	Organism
Fpr	P28861.4	399486	<i>Escherichia coli</i>
hcaD	AAC75595.1	1788892	<i>Escherichia coli</i>
LOC100282643	NP_001149023.1	226497434	<i>Zea mays</i>
NfnA	YP_001393861.1	153953096	<i>Clostridium kluyveri</i>
NfnB	YP_001393862.1	153953097	<i>Clostridium kluyveri</i>
CcarbDRAFT_2639	ZP_05392639.1	255525707	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2638	ZP_05392638.1	255525706	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2636	ZP_05392636.1	255525704	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_5060	ZP_05395060.1	255528241	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2450	ZP_05392450.1	255525514	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_1084	ZP_05391084.1	255524124	<i>Clostridium carboxidivorans</i> P7
RnfC	EDK33306.1	146346770	<i>Clostridium kluyveri</i>
RnfD	EDK33307.1	146346771	<i>Clostridium kluyveri</i>
RnfG	EDK33308.1	146346772	<i>Clostridium kluyveri</i>
RnfE	EDK33309.1	146346773	<i>Clostridium kluyveri</i>
RnfA	EDK33310.1	146346774	<i>Clostridium kluyveri</i>
RnfB	EDK33311.1	146346775	<i>Clostridium kluyveri</i>
CLJU_c11410 (RnfB)	ADK14209.1	300434442	<i>Clostridium ljungdahli</i>
CLJU_c11400 (RnfA)	ADK14208.1	300434441	<i>Clostridium ljungdahli</i>
CLJU_c11390 (RnfE)	ADK14207.1	300434440	<i>Clostridium ljungdahli</i>
CLJU_c11380 (RnfG)	ADK14206.1	300434439	<i>Clostridium ljungdahli</i>
CLJU_c11370 (RnfD)	ADK14205.1	300434438	<i>Clostridium ljungdahli</i>
CLJU_c11360 (RnfC)	ADK14204.1	300434437	<i>Clostridium ljungdahli</i>
MOTH_1518 (NfnA)	YP_430370.1	83590361	<i>Moorella thermoacetica</i>
MOTH_1517 (NfnB)	YP_430369.1	83590360	<i>Moorella thermoacetica</i>
CHY 1992 (NfnA)	YP_360811.1	78045020	<i>Carboxydotherrmus hydrogenoformans</i>
CHY 1993 (NfnB)	YP_360812.1	78044266	<i>Carboxydotherrmus hydrogenoformans</i>
CLJU_c37220 (NfnAB)	YP_003781850.1	300856866	<i>Clostridium ljungdahli</i>

FIG. 10, Step I—Formate Dehydrogenase

Formate dehydrogenase (FDH) catalyzes the reversible transfer of electrons from formate to an acceptor. Enzymes with FDH activity utilize various electron carriers such as, for example, NADH (EC 1.2.1.2), NADPH (EC 1.2.1.43), quinols (EC 1.1.5.6), cytochromes (EC 1.2.2.3) and hydrogenases (EC 1.1.99.33). FDH enzymes have been characterized from *Moorella thermoacetica* (Andreesen and Ljungdahl, *J Bacteriol* 116:867-873 (1973); Li et al., *J Bacteriol* 92:405-412 (1966); Yamamoto et al., *J Biol Chem.* 258:1826-1832 (1983). The loci, Moth_2312 is responsible for encoding the alpha subunit of formate dehydrogenase while the beta subunit is encoded by Moth_2314 (Pierce et al., *Environ Microbiol* (2008)). Another set of genes encoding formate dehydrogenase activity with a propensity for

CO₂ reduction is encoded by Sfum_2703 through Sfum_2706 in *Syntrophobacter fumaroxidans* (de Bok et al., *Eur J Biochem.* 270:2476-2485 (2003)); Reda et al., *PNAS* 105:10654-10658 (2008)). A similar set of genes presumed to carry out the same function are encoded by CHY_0731, CHY 0732, and CHY 0733 in *C. hydrogenoformans* (Wu et al., *PLoS Genet* 1:e65 (2005)). Formate dehydrogenases are also found many additional organisms including *C. carboxidivorans* P7, *Bacillus methanolicus*, *Burkholderia stabilis*, *Moorella thermoacetica* ATCC 39073, *Candida boidinii*, *Candida methylica*, and *Saccharomyces cerevisiae* S288c. The soluble formate dehydrogenase from *Ralstonia eutropha* reduces NAD⁺ (fdsG, —B, -A, —C, -D) (Oh and Bowien, 1998)

Protein	GenBank ID	GI Number	Organism
Moth_2312	YP_431142	148283121	<i>Moorella thermoacetica</i>
Moth_2314	YP_431144	83591135	<i>Moorella thermoacetica</i>
Sfum_2703	YP_846816.1	116750129	<i>Syntrophobacter fumaroxidans</i>
Sfum_2704	YP_846817.1	116750130	<i>Syntrophobacter fumaroxidans</i>
Sfum_2705	YP_846818.1	116750131	<i>Syntrophobacter fumaroxidans</i>
Sfum_2706	YP_846819.1	116750132	<i>Syntrophobacter fumaroxidans</i>
CHY_0731	YP_359585.1	78044572	<i>Carboxydotherrmus hydrogenoformans</i>
CHY_0732	YP_359586.1	78044500	<i>Carboxydotherrmus hydrogenoformans</i>
CHY_0733	YP_359587.1	78044647	<i>Carboxydotherrmus hydrogenoformans</i>
CcarbDRAFT_0901	ZP_05390901.1	255523938	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_4380	ZP_05394380.1	255527512	<i>Clostridium carboxidivorans</i> P7
fdhA,MGA3_06625	EIJ82879.1	387590560	<i>Bacillus methanolicus</i> MGA3
fdhA,PB1_11719	ZP_10131761.1	387929084	<i>Bacillus methanolicus</i> PBI
fdhD, MGA3_06630	EIJ82880.1	387590561	<i>Bacillus methanolicus</i> MGA3
fdhD, PB1_11724	ZP_10131762.1	387929085	<i>Bacillus methanolicus</i> PBI
fdh	ACF35003.	194220249	<i>Burkholderia stabilis</i>
FDH1	AAC49766.1	2276465	<i>Candida boidinii</i>
fdh	CAA57036.1	1181204	<i>Candida methylica</i>
FDH2	P0CF35.1	294956522	<i>Saccharomyces cerevisiae</i> S288c
FDH1	NP_015033.1	6324964	<i>Saccharomyces cerevisiae</i> S288c
fdsG	YP_725156.1	113866667	<i>Ralstonia eutropha</i>
fdsB	YP_725157.1	113866668	<i>Ralstonia eutropha</i>

Protein	GenBank ID	GI Number	Organism
fdsA	YP_725158.1	113866669	<i>Ralstonia eutropha</i>
fdsC	YP_725159.1	113866670	<i>Ralstonia eutropha</i>
fdsD	YP_725160.1	113866671	<i>Ralstonia eutropha</i>

FIG. 10, Step J—Methanol Dehydrogenase

NAD⁺ dependent methanol dehydrogenase enzymes (EC 1.1.1.244) catalyze the conversion of methanol and NAD⁺ to formaldehyde and NADH. An enzyme with this activity was first characterized in *Bacillus methanolicus* (Heggeset, et al., *Applied and Environmental Microbiology*, 78(15): 5170-5181 (2012)). This enzyme is zinc and magnesium dependent, and activity of the enzyme is enhanced by the activating enzyme encoded by act (Kloosterman et al, *J Biol Chem* 277:34785-92 (2002)). The act is a Nudix hydrolase. Several of these candidates have been identified and shown to have activity on methanol. Additional NAD(P)⁺ depen-

dent enzymes can be identified by sequence homology. Methanol dehydrogenase enzymes utilizing different electron acceptors are also known in the art. Examples include cytochrome dependent enzymes such as mxalF of the methylophilic *Methylobacterium extorquens* (Nunn et al, *Nucl Acid Res* 16:7722 (1988)). Methanol dehydrogenase enzymes of methanotrophs such as *Methylococcus capsulatus* function in a complex with methane monooxygenase (MMO) (Myronova et al, *Biochem* 45:11905-14 (2006)). Methanol can also be oxidized to formaldehyde by alcohol oxidase enzymes such as methanol oxidase (EC 1.1.3.13) of *Candida boidinii* (Sakai et al, *Gene* 114: 67-73 (1992)).

Protein	GenBank ID	GI Number	Organism
mdh, MGA3_17392	ELJ77596.1	387585261	<i>Bacillus methanolicus</i> MGA3
mdh2, MGA3_07340	ELJ83020.1	387590701	<i>Bacillus methanolicus</i> MGA3
mdh3, MGA3_10725	ELJ80770.1	387588449	<i>Bacillus methanolicus</i> MGA3
act, MGA3_09170	ELJ83380.1	387591061	<i>Bacillus methanolicus</i> MGA3
mdh, PB1_17533	ZP_10132907.1	387930234	<i>Bacillus methanolicus</i> PB1
mdh1, PB1_14569	ZP_10132325.1	387929648	<i>Bacillus methanolicus</i> PB1
mdh2, PB1_12584	ZP_10131932.1	387929255	<i>Bacillus methanolicus</i> PB1
act, PB1_14394	ZP_10132290.1	387929613	<i>Bacillus methanolicus</i> PB1
BFZC1_05383	ZP_07048751.1	299535429	<i>Lysinibacillus fusiformis</i>
BFZC1_20163	ZP_07051637.1	299538354	<i>Lysinibacillus fusiformis</i>
Bsph_4187	YP_001699778.1	169829620	<i>Lysinibacillus sphaericus</i>
Bsph_1706	YP_001697432.1	169827274	<i>Lysinibacillus sphaericus</i>
mdh2	YP_004681552.1	339322658	<i>Cupriavidus necator</i> N-1
nudF1	YP_004684845.1	339325152	<i>Cupriavidus necator</i> N-1
BthaA_010200007655	ZP_05587334.1	257139072	<i>Burkholderia thailandensis</i> E264
BTH_11076 (MutT/NUDIX NTP pyrophosphatase)	YP_441629.1	83721454	<i>Burkholderia thailandensis</i> E264
BalcAV_11743	ZP_10819291.1	402299711	<i>Bacillus alcalophilus</i> ATCC 27647
BalcAV_05251	ZP_10818002.1	402298299	<i>Bacillus alcalophilus</i> ATCC 27647
alcohol dehydrogenase P3TCK_27679	YP_001447544	156976638	<i>Vibrio harveyi</i> ATCC BAA-1116
alcohol dehydrogenase adhB	ZP_01220157.1	90412151	<i>Photobacterium profundum</i> 3TCK
alcohol dehydrogenase	YP_694908	110799824	<i>Clostridium perfringens</i> ATCC 13124
alcohol dehydrogenase	NP_717107	24373064	<i>Shewanella oneidensis</i> MR-1
alcohol dehydrogenase	YP_237055	66047214	<i>Pseudomonas syringae</i> pv. <i>syringae</i> B728a
alcohol dehydrogenase	YP_359772	78043360	<i>Carboxydotherrmus hydrogenoformans</i> Z-2901
alcohol dehydrogenase PpeoK3_010100018471	YP_003990729	312112413	<i>Geobacillus</i> sp. Y4.1MC1
OBE_12016	ZP_10241531.1	390456003	<i>Paenibacillus peoriae</i> KCTC 3763
alcohol dehydrogenase dhaT	EKC54576	406526935	human gut metaypnome
alcohol dehydrogenase BAZO_10081	YP_001343716	152978087	<i>Actinobacillus succinopenes</i> 130Z
alcohol dehydrogenase	AAC45651	2393887	<i>Clostridium pastewianum</i> DSM 525
alcohol dehydrogenase	NP_561852	18309918	<i>Clostridium perfringens</i> str. 13
alcohol dehydrogenase	ZP_11313277.1	410459529	<i>Bacillus azotoformans</i> LMG 9581
alcohol dehydrogenase	YP_007491369	452211255	<i>Methanosarcina mazei</i> Tuc01
alcohol dehydrogenase	YP_004860127	347752562	<i>Bacillus coagulans</i> 36D1
alcohol dehydrogenase	YP_002138168	197117741	<i>Geobacter bemidiensis</i> Bem
DesmeDRAFT_1354	ZP_08977641.1	354558386	<i>Desulfotobacterium metallireducens</i> DSM 15288
alcohol dehydrogenase	YP_001337153	152972007	<i>Klebsiella pneumoniae</i> subsp. <i>pneumoniae</i> MGH 78578
alcohol dehydrogenase	YP_001113612	134300116	<i>Desulfotomaculum reducens</i> MI-1
alcohol dehydrogenase	YP_001663549	167040564	<i>Thermoanaerobacter</i> sp. X514
ACINNAV82_2382	ZP_16224338.1	421788018	<i>Acinetobacter baumannii</i> Naval-82
alcohol dehydrogenase	YP_005052855	374301216	<i>Desulfovibrio africanus</i> str. Walvis Bay
alcohol dehydrogenase	AGF87161	451936849	uncultured organism
DesfrDRAFT_3929	ZP_07335453.1	303249216	<i>Desulfovibrio fructosovorans</i> JJ
alcohol dehydrogenase	NP_617528	20091453	<i>Methanosarcina acetivorans</i> C2A
alcohol dehydrogenase	NP_343875.1	15899270	<i>Sulfolobus solfataricus</i> P-2
adh4	YP_006863258	408405275	<i>Nitrososphaera parvipes</i> Ga9.2
Ta0841	NP_394301.1	16081897	<i>Thermoplasma acidophilum</i>
PTO1151	YP_023929.1	48478223	<i>Picrophilus torridus</i> DSM9790

-continued

Protein	GenBank ID	GI Number	Organism
alcohol dehydrogenase	ZP_10129817.1	387927138	<i>Bacillus methanolicus</i> PB-1
cgR_2695	YP_001139613.1	145296792	<i>Corynebacterium glutamicum</i> R
alcohol dehydrogenase	YP_004758576.1	340793113	<i>Corynebacterium variabile</i>
HMPREF1015_01790	ZP_09352758.1	365156443	<i>Bacillus smithii</i>
ADH1	NP_014555.1	6324486	<i>Saccharomyces cerevisiae</i>
NADH-dependent butanol dehydrogenase A	YP_001126968.1	138896515	<i>Geobacillus thermodenitrificans</i> NG80-2
alcohol dehydrogenase	WP_007139094.1	494231392	<i>Flavobacterium frigidis</i>
methanol dehydrogenase	WP_003897664.1	489994607	<i>Mycobacterium smegmatis</i>
ADH1B	NP_000659.2	34577061	<i>Homo sapiens</i>
PMI01_01199	ZP_10750164.1	399072070	<i>Caulobacter</i> sp. AP07
YiaY	YP_026233.1	49176377	<i>Escherichia coli</i>
MCA0299	YP_112833.1	53802410	<i>Methylococcus capsulatus</i>
MCA0782	YP_113284.1	53804880	<i>Methylococcus capsulatus</i>
mxal	YP_002965443.1	240140963	<i>Methylobacterium extorquens</i>
mxalF	YP_002965446.1	240140966	<i>Methylobacterium extorquens</i>
AOD1	AAA34321.1	170820	<i>Candida boidinii</i>
hypothetical protein	EDA87976.1	142827286	Marine metapenome
GOS_1920437		JCVI_SCAF_1096627185304	
alcohol dehydrogenase	CAA80989.1	580823	<i>Geobacillus stearothermophilus</i>

An in vivo assay was developed to determine the activity of methanol dehydrogenases. This assay relies on the detection of formaldehyde (HCHO), thus measuring the forward activity of the enzyme (oxidation of methanol). To this end, a strain comprising a BDOP and lacking frmA, frmB, frmR was created using Lambda Red recombinase technology (Datsenko and Wanner, *Proc. Natl. Acad. Sci. USA*, 6 97(12): 6640-5 (2000)). Plasmids expressing methanol dehydrogenases were transformed into the strain, then grown to saturation in LB medium+antibiotic at 37° C. with shaking. Transformation of the strain with an empty vector served as a negative control. Cultures were adjusted by O.D. and then diluted 1:10 into M9 medium+0.5% glucose+antibiotic and cultured at 37° C. with shaking for 6-8 hours until late log phase. Methanol was added to 2% v/v and the cultures were further incubated for 30 min. with shaking at 37° C. Cultures were spun down and the supernatant was assayed for formaldehyde produced using DEFECTX Formaldehyde Detection kit (Arbor Assays; Ann Arbor MIh.) according to manufacturer's instructions. The frmA, frmB, frmR deletions resulted in the native formaldehyde utilization pathway to be deleted, which enables the formation of formaldehyde that can be used to detect methanol dehydrogenase activity in the NNOMO.

The activity of several enzymes was measured using the assay described above. The results of four independent experiments are provided in Table 5 below.

TABLE 5

Results of in vivo assays showing formaldehyde (HCHO) production by various NNOMO comprising a plasmid expressing a methanol dehydrogenase.	
Accession number	HCHO (μM)
Experiment 1	
ELJ177596.1	>50
ELJ83020.1	>20
ELJ80770.1	>50
ZP_10132907.1	>20
ZP_10132325.1	>20
ZP_10131932.1	>50

TABLE 5-continued

Results of in vivo assays showing formaldehyde (HCHO) production by various NNOMO comprising a plasmid expressing a methanol dehydrogenase.	
Accession number	HCHO (μM)
ZP_07048751.1	>50
YP_001699778.1	>50
YP_004681552.1	>10
ZP_10819291.1	<1
Empty vector Experiment 2	2.33
ELJ77596.1	>50
NP_00659.2	>50
YP_004758576.1	>20
ZP_09352758.1	>50
ZP_10129817.1	>20
YP_001139613.1	>20
NP_014555.1	>10
WP_007139094.1	>10
NP_343875.1	>1
YP_006863258	>1
NP_394301.1	>1
ZP_10750164.1	>1
YP_023929.1	>1
ZP_08977641.1	<1
ZP_10117398.1	<1
YP_004108045.1	<1
ZP_09753449.1	<1
Empty vector Experiment 3	0.17
ELJ77596.1	>50
NP_561852	>50
YP_002138168	>50
YP_026233.1	>50
YP_001447544	>50
Metalibrary	>50
YP_359772	>50
ZP_01220157.1	>50
ZP_07335453.1	>20
YP_001337153	>20
YP_694908	>20
NP_717107	>20
AAC45651	>10
ZP_11313277.1	>10
ZP_16224338.1	>10
YP_001113612	>10

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TABLE 5-continued

Results of in vivo assays showing formaldehyde (HCHO) production by various NNOMO comprising a plasmid expressing a methanol dehydrogenase.	
Accession number	HCHO (μ M)
YP_004860127	>10
YP_003310546	>10
YP_001343716	>10
NP_717107	>10
YP_002434746	>10
Empty vector Experiment 4	0.11
ELJ77596.1	>20
ZP_11313277.1	>50
YP_001113612	>50
YP_001447544	>20
AGF87161	>50
EDA87976.1	>20
Empty vector	-0.8

FIG. 10, Step K—Spontaneous or Formaldehyde Activating Enzyme

The conversion of formaldehyde and THF to methyl-enetetrahydrofolate can occur spontaneously. It is also possible that the rate of this reaction can be enhanced by a formaldehyde activating enzyme. A formaldehyde activating enzyme (Fae) has been identified in *Methylobacterium extorquens* AM1 which catalyzes the condensation of formaldehyde and tetrahydromethanopterin to methylene tetrahydromethanopterin (Vorholt, et al., J. Bacteriol., 182(23), 6645-6650 (2000)). It is possible that a similar enzyme exists or can be engineered to catalyze the condensation of formaldehyde and tetrahydrofolate to methylenetetrahydro-

Protein	GenBank ID	GI Number	Organism
MexAM1_META1p1766	Q9FA38.3	17366061	<i>Methylobacterium extorquens</i> AM1
Xaut_0032	YP_001414948.1	154243990	<i>Xanthobacter autotrophicus</i> Py2
Hden_1474	YP_003755607.1	300022996	<i>Hyphomicrobium denitrificans</i> ATCC 51888

FIG. 10, Step L—Formaldehyde Dehydrogenase

Oxidation of formaldehyde to formate is catalyzed by formaldehyde dehydrogenase. An NAD⁺ dependent formaldehyde dehydrogenase enzyme is encoded by fdhA of *Pseudomonas putida* (Ito et al, J Bacteriol 176: 2483-2491 (1994)). Additional formaldehyde dehydrogenase enzymes include the NAD⁺ and glutathione independent formaldehyde dehydrogenase from *Hyphomicrobium zavarzinii* (Jerome et al, Appl Microbiol Biotechnol 77:779-88 (2007)), the glutathione dependent formaldehyde dehydrogenase of *Pichia pastoris* (Sunga et al, Gene 330:39-47 (2004)) and the NAD(P)⁺ dependent formaldehyde dehydrogenase of *Methylobacter marinus* (Speer et al, FEMS Microbiol Lett, 121(3):349-55 (1994)).

Protein	GenBank ID	GI Number	Organism
fdhA	P46154.3	1169603	<i>Pseudomonas putida</i>
faoA	CAC85637.1	19912992	<i>Hyphomicrobium zavarzinii</i>

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Protein	GenBank ID	GI Number	Organism
Fld1	CCA39112.1	328352714	<i>Pichia pastoris</i>
fdh	P47734.2	221222447	<i>Methylobacter marinus</i>

In addition to the formaldehyde dehydrogenase enzymes listed above, alternate enzymes and pathways for converting formaldehyde to formate are known in the art. For example, many organisms employ glutathione-dependent formaldehyde oxidation pathways, in which formaldehyde is converted to formate in three steps via the intermediates S-hydroxymethylglutathione and S-formylglutathione (Vorholt et al, J Bacteriol 182:6645-50 (2000)). The enzymes of this pathway are S-(hydroxymethyl)glutathione synthase (EC 4.4.1.22), glutathione-dependent formaldehyde dehydrogenase (EC 1.1.1.284) and S-formylglutathione hydrolase (EC 3.1.2.12).

FIG. 10, Step M—Spontaneous or S-(hydroxymethyl)glutathione Synthase

While conversion of formaldehyde to S-hydroxymethylglutathione can occur spontaneously in the presence of glutathione, it has been shown by Goenrich et al (Goenrich, et al., J Biol Chem 277(5); 3069-72 (2002)) that an enzyme from *Paracoccus denitrificans* can accelerate this spontaneous condensation reaction. The enzyme catalyzing the conversion of formaldehyde and glutathione was purified and named glutathione-dependent formaldehyde-activating enzyme (Gfa). The gene encoding it, which was named gfa, is located directly upstream of the gene for glutathione-dependent formaldehyde dehydrogenase, which catalyzes the subsequent oxidation of S-hydroxymethylglutathione. Putative proteins with sequence identity to Gfa from *P. denitrificans* are present also in *Rhodobacter sphaeroides*, *Sinorhizobium meliloti*, and *Mesorhizobium loti*.

Protein	GenBank ID	GI Number	Organism
Gfa	Q51669.3	38257308	<i>Paracoccus denitrificans</i>
Gfa	ABP71667.1	145557054	<i>Rhodobacter sphaeroides</i> ATCC 17025
Gfa	Q92WX6.1	38257348	<i>Sinorhizobium meliloti</i> 1021
Gfa	Q98LU4.2	38257349	<i>Mesorhizobium loti</i> MAFF303099

FIG. 10, Step N—Glutathione-Dependent Formaldehyde Dehydrogenase

Glutathione-dependent formaldehyde dehydrogenase (GS-FDH) belongs to the family of class III alcohol dehydrogenases. Glutathione and formaldehyde combine non-enzymatically to form hydroxymethylglutathione, the true substrate of the GS-FDH catalyzed reaction. The product, S-formylglutathione, is further metabolized to formic acid.

Protein	GenBank ID	GI Number	Organism
fimA	YP_488650.1	388476464	<i>Escherichia coli</i> K-12MG1655
SFA1	NP_010113.1	6320033	<i>Saccharomyces cerevisiae</i> S288c
fthA	AAC44551.1	1002865	<i>Paracoccus denitrificans</i>
adhI	AAB09774.1	986949	<i>Rhodobacter sphaeroides</i>

FIG. 10, Step O—S-formylglutathione Hydrolase

S-formylglutathione hydrolase is a glutathione thiol esterase found in bacteria, plants and animals. It catalyzes conversion of S-formylglutathione to formate and glutathione. The fghA gene of *P. denitrificans* is located in the same operon with gfa and fthA, two genes involved in the oxidation of formaldehyde to formate in this organism. In *E. coli*, FrmB is encoded in an operon with FrmR and FnnA, which are proteins involved in the oxidation of formaldehyde. YeiG of *E. coli* is a promiscuous serine hydrolase; its highest specific activity is with the substrate S-formylglutathione.

Protein	GenBankID	GI Number	Organism
fimB	NP_414889.1	16128340	<i>Escherichia coli</i> K-12MG1655
yeiG	AAC75215.1	1788477	<i>Escherichia coli</i> K-12MG1655
fghA	AAC44554.1	1002868	<i>Paracoccus denitrificans</i>

Protein	Gen Bank ID	GI Number	Organism
CODH (putative)	YP_430813	83590804	<i>Moorella thermoacetica</i>
CODH-II (CooS-II)	YP_358957	78044574	<i>Carboxydotherrnus hydrogenoformans</i>
CooF	YP_358958	78045112	<i>Carboxydotherrnus hydrogenoformans</i>
CODH (putative)	ZP_05390164.1	255523193	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_0341	ZP_05390341.1	255523371	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_1756	ZP_05391756.1	255524806	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_2944	ZP_05392944.1	255526020	<i>Clostridium carboxidivorans</i> P7
CODH	YP_384856.1	78223109	<i>Geobacter metallireducens</i> GS-15
Cpha266_0148 (cytochrome c)	YP_910642.1	119355998	<i>Chlorobium phaeobacteroides</i> DSM 266
Cpha266_0149 (CODH)	YP_910643.1	119355999	<i>Chlorobium phaeobacteroides</i> DSM 266
Ccel_0438	YP_002504800.1	220927891	<i>Clostridium cellulolyticum</i> H10
Ddes_0382 (CODH)	YP_002478973.1	220903661	<i>Desulfovibrio desulfuricans</i> subsp. desulfuricans str. ATCC 27774
Ddes_0381 (CooC)	YP_002478972.1	220903660	<i>Desulfovibrio desulfuricans</i> subsp. desulfuricans str. ATCC 27774
Pcar_0057 (CODH)	YP_355490.1	7791767	<i>Pelobacter carbinolicus</i> DSM2380
Pcar_0058 (CooC)	YP_355491.1	7791766	<i>Pelobacter carbinolicus</i> DSM2380
Pcar_0058 (HypA)	YP_355492.1	7791765	<i>Pelobacter carbinolicus</i> DSM2380
CooS(CODH)	YP_001407343.1	154175407	<i>Campylobacter curvus</i> 525.92
CLJU_c09110	ADK13979.1	300434212	<i>Clostridium ljungdahli</i>
CLJU_C09100	ADK13978.1	300434211	<i>Clostridium ljungdahli</i>
CLJU_C09090	ADK13977.1	300434210	<i>Clostridium ljungdahli</i>

FIG. 10, Step P—Carbon Monoxide Dehydrogenase (CODH)

CODH is a reversible enzyme that interconverts CO and CO₂ at the expense or gain of electrons. The natural physiological role of the CODH in ACS/CODH complexes is to convert CO₂ to CO for incorporation into acetyl-CoA by acetyl-CoA synthase. Nevertheless, such CODH enzymes are suitable for the extraction of reducing equivalents from CO due to the reversible nature of such enzymes. Expressing such CODH enzymes in the absence of ACS allows them to operate in the direction opposite to their natural physiological role (i.e., CO oxidation).

In *M. thermoacetica*, *C. hydrogenoformans*, *C. carboxidivorans* P7, and several other organisms, additional CODH

encoding genes are located outside of the ACS/CODH operons. These enzymes provide a means for extracting electrons (or reducing equivalents) from the conversion of carbon monoxide to carbon dioxide. The *M. thermoacetica* gene (Genbank Accession Number: YP_430813) is expressed by itself in an operon and is believed to transfer electrons from CO to an external mediator like ferredoxin in a “Ping-pong” reaction. The reduced mediator then couples to other reduced nicotinamide adenine dinucleotide phosphate (NAD(P)H) carriers or ferredoxin-dependent cellular processes (Ragsdale, *Annals of the New York Academy of Sciences* 1125: 129-136 (2008)). The genes encoding the *C. hydrogenoformans* CODH-II and CooF, a neighboring protein, were cloned and sequenced (Gonzalez and Robb, *FEMS Microbiol Lett.* 191:243-247 (2000)). The resulting complex was membrane-bound, although cytoplasmic fractions of CODH-II were shown to catalyze the formation of NADPH suggesting an anabolic role (Svetlitchnyi et al., *J Bacteriol.* 183:5134-5144 (2001)). The crystal structure of the CODH-II is also available (Dobbek et al., *Science* 293:1281-1285 (2001)). Similar ACS-free CODH enzymes can be found in a diverse array of organisms including *Geobacter metallireducens* GS-15, *Chlorobium phaeobacteroides* DSM 266, *Clostridium cellulolyticum* H10, *Desulfovibrio desulfuricans* subsp. *desulfuricans* str. ATCC 27774, *Pelobacter carbinolicus* DSM 2380, *C. ljungdahli* and *Campylobacter curvus* 525.92.

Example III

Methods for Formaldehyde Fixation

Provided herein are exemplary pathways, which utilize formaldehyde produced from the oxidation of methanol (see, e.g., FIG. 1, step A, or FIG. 10, step J) or from formate assimilation pathways described in Example I (see, e.g., FIG. 1) in the formation of intermediates of certain central metabolic pathways that can be used for the production of compounds disclosed herein.

One exemplary pathway that can utilize formaldehyde produced from the oxidation of methanol is shown in FIG. 1, which involves condensation of formaldehyde and

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D-ribulose-5-phosphate to form hexulose-6-phosphate (h6p) by hexulose-6-phosphate synthase (FIG. 1, step B). The enzyme can use Mg^{2+} or Mn^{2+} for maximal activity, although other metal ions are useful, and even non-metal-ion-dependent mechanisms are contemplated. H6p is converted into fructose-6-phosphate by 6-phospho-3-hexuloisomerase (FIG. 1, step C).

Another exemplary pathway that involves the detoxification and assimilation of formaldehyde produced from the oxidation of methanol is shown in FIG. 1 and proceeds through dihydroxyacetone. Dihydroxyacetone synthase is a special transketolase that first transfers a glycoaldehyde group from xylulose-5-phosphate to formaldehyde, resulting in the formation of dihydroxyacetone (DHA) and glyceraldehyde-3-phosphate (G3P), which is an intermediate in glycolysis (FIG. 1). The DHA obtained from DHA synthase can be further phosphorylated to form DHA phosphate and assimilated into glycolysis and several other pathways (FIG. 1). Alternatively, or in addition, a fructose-6-phosphate aldolase can be used to catalyze the conversion of DHA and G3P to fructose-6-phosphate (FIG. 1, step Z).

FIG. 1, Steps B and C—Hexulose-6-phosphate synthase (Step B) and 6-phospho-3-hexuloisomerase (Step C)

Both of the hexulose-6-phosphate synthase and 6-phospho-3-hexuloisomerase enzymes are found in several organisms, including methanotrophs and methylotrophs where they have been purified (Kato et al., 2006, BioSci Biotechnol

Biochem. 70(1):10-21. In addition, these enzymes have been reported in heterotrophs such as *Bacillus subtilis* also where they are reported to be involved in formaldehyde detoxification (Mitsui et al., 2003, AEM 69(10):6128-32, Yasueda et al., 1999, J Bac 181(23):7154-60. Genes for these two enzymes from the methylotrophic bacterium *Mycobacterium gastri* MB19 have been fused and *E. coli* strains harboring the hps-phi construct showed more efficient utilization of formaldehyde (Orita et al. 2007, Appl Microbiol Biotechnol. 76:439-445). In some organisms, these two enzymes naturally exist as a fused version that is bifunctional.

Exemplary candidate genes for hexulose-6-phosphate synthase are:

Protein	GenBank ID	GI number	Organism
Hps	AAR39392.1	40074227	<i>Bacillus methanolicus</i> MGA3
Hps	EIJ81375.1	387589055	<i>Bacillus methanolicus</i> PB1
RmpA	BAA83096.1	5706381	<i>Methylobacillus aminofaciens</i>
RmpA	BAA90546.1	6899861	<i>Mycobacterium gastri</i>

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Protein	GenBank ID	GI number	Organism
YckG	BAA08980.1	1805418	<i>Bacillus subtilis</i>
Hps	YP_544362.1	91774606	<i>Methylobacillus flagellatus</i>
Hps	YP_545763.1	91776007	<i>Methylobacillus flagellatus</i>
Hps	AAG29505.1	11093955	<i>Aminomonas aminovorans</i>
SgbH	YP_004038706.1	313200048	<i>Methylovorus</i> sp. MP688
Hps	YP_003050044.1	253997981	<i>Methylovorus glucosetrophus</i> SIP3-4
Hps	YP_003990382.1	312112066	<i>Geobacillus</i> sp. Y4.1MC1
Hps	gb AAR91478.1	40795504	<i>Geobacillus</i> sp. M10EXG
Hps	YP_007402409.1	448238351	<i>Geobacillus</i> sp. GHH01

Exemplary gene candidates for 6-phospho-3-hexuloisomerase are:

Protein	GenBank ID	GI number	Organism
Phi	AAR39393.1	40074228	<i>Bacillus methanolicus</i> MGA3
Phi	EIJ81376.1	387589056	<i>Bacillus methanolicus</i> PB1
Phi	BAA83098.1	5706383	<i>Methylobacillus aminofaciens</i>
RmpB	BAA90545.1	6899860	<i>Mycobacterium gastri</i>
Phi	YP_545762.1	91776006	<i>Methylobacillus flagellatus</i> KT
Phi	YP_003051269.1	253999206	<i>Methylovorus glucosetrophus</i> SIP3-4
Phi	YP_003990383.1	312112067	<i>Geobacillus</i> sp. Y4.1MC1
Phi	YP_007402408.1	448238350	<i>Geobacillus</i> sp. GHH01

Candidates for enzymes where both of these functions have been fused into a single open reading frame include the following.

Protein	GenBank ID	GI number	Organism
PH1938	NP_143767.1	14591680	<i>Pyrococcus horikoshii</i> OT3
PF0220	NP_577949.1	18976592	<i>Pyrococcus furiosus</i>
TK0475	YP_182888.1	57640410	<i>Thermococcus kodakaraensis</i>
PAB1222	NP_127388.1	14521911	<i>Pyrococcus abyssi</i>
MCA2738	YP_115138.1	53803128	<i>Methylococcus capsulatus</i>
Metal_3152	EIC30826.1	380884949	<i>Methylobacillus album</i> BG8

FIG. 1, Step D—Dihydroxyacetone Synthase

The dihydroxyacetone synthase enzyme in *Candida boidinii* uses thiamine pyrophosphate and Mg^{2+} as cofactors and is localized in the peroxisome. The enzyme from the methanol-growing carboxydobacterium, *Mycobacter* sp. strain JC1 DSM 3803, was also found to have DHA synthase and kinase activities (Ro et al., 1997, JBac 179(19):6041-7). DHA synthase from this organism also has similar cofactor requirements as the enzyme from *C. boidinii*. The K_m s for formaldehyde and xylulose 5-phosphate were reported to be 1.86 mM and 33.3 microM, respectively. Several other mycobacteria, excluding only *Mycobacterium tuberculosis*, can use methanol as the sole source of carbon and energy and are reported to use dihydroxyacetone synthase (Part et al., 2003, JBac 185(1):142-7).

Protein	GenBank ID	GI number	Organism
DAS1	AAC83349.1	3978466	<i>Candida boidinii</i>
HPODL_2613	EFW95760.1	320581540	<i>Ogataea parapolymorpha</i> DL-1 (<i>Hansenula polymorpha</i> DL-1)
	AAG12171.2	18497328	<i>Mycobacter</i> sp. strain JC1 DSM 3803

FIG. 1, Step Z—Fructose-6-phosphate aldolase

Fructose-6-phosphate aldolase (F6P aldolase) can catalyze the combination of dihydroxyacetone (DHA) and glyceraldehyde-3-phosphate (G3P) to form fructose-6-phosphate. This activity was recently discovered in *E. coli* and the corresponding gene candidate has been termed *fsa* (Schunnann and Sprenger, *J. Biol. Chem.*, 2001, 276(14), 11055-11061). The enzyme has narrow substrate specificity and cannot utilize fructose, fructose 1-phosphate, fructose 1,6-bisphosphate, or dihydroxyacetone phosphate. It can however use hydroxybutanone and acetol instead of DHA. The purified enzyme displayed a V_{max} of 7 units/mg of protein for fructose 6-phosphate cleavage (at 30 degrees C., pH 8.5 in 50 mM glycylglycine buffer). For the aldolization reaction a V_{max} of 45 units/mg of protein was found; K_m values for the substrates were 9 mM for fructose 6-phosphate, 35 mM for dihydroxyacetone, and 0.8 mM for glyceraldehyde 3-phosphate. The enzyme prefers the aldol formation over the cleavage reaction.

The selectivity of the *E. coli* enzyme towards DHA can be improved by introducing point mutations. For example, the mutation A129S improved reactivity towards DHA by over 17 fold in terms of K_{cat}/K_m (Gutierrez et al., *Chem Commun (Carob)*, 2011, 47(20), 5762-5764). The same mutation reduced the catalytic efficiency on hydroxyacetone by more than 3 fold and reduced the affinity for glycoaldehyde by more than 3 fold compared to that of the wild type enzyme (Castillo et al., *Advanced Synthesis & Catalysis*, 352(6), 1039-1046). Genes similar to *fsa* have been found in other genomes by sequence homology. Some exemplary gene candidates have been listed below.

Gene	Protein accession number	GI number	Organism
<i>fsa</i>	AAC73912.2	87081788	<i>Escherichia coli</i> K12
<i>talC</i>	AAC76928.1	1790382	<i>Escherichia coli</i> K12
<i>fsa</i>	WP_017209835.1	515777235	<i>Clostridium beijerinickii</i>
DR_1337	AAF10909.1	6459090	<i>Deinococcus radiodurans</i> R1
<i>talC</i>	NP_213080.1	15605703	<i>Aquifex aeolicus</i> VF5
MJ_0960	NP_247955.1	15669150	<i>Methanocaldococcus janaschii</i>
<i>mipB</i>	NP_993370.2	161511381	<i>Yersinia pestis</i>

As described below, there is an energetic advantage to using F6P aldolase in the DHA pathway.

The assimilation of formaldehyde formed by the oxidation of methanol can proceed either via the dihydroxyacetone (DHA) pathway (step D, FIG. 1) or the Ribulose monophosphate (RuMP) pathway (steps B and C, FIG. 1). In the RUMP pathway, formaldehyde combines with ribulose-5-phosphate to form F6P. F6P is then either metabolized via glycolysis or used for regeneration of ribulose-5-phosphate to enable further formaldehyde assimilation. Notably, ATP hydrolysis is not required to form F6P from formaldehyde and ribulose-5-phosphate via the RuMP pathway.

In contrast, in the DHA pathway, formaldehyde combines with xylulose-5-phosphate (X5P) to form dihydroxyacetone (DHA) and glyceraldehyde-3-phosphate (G3P). Some of the DHA and G3P must be metabolized to F6P to enable

regeneration of xylulose-5-phosphate. In the standard DHA pathway, DHA and G3P are converted to F6P by three enzymes: DHA kinase, fructose bisphosphate aldolase, and fructose bisphosphatase. The net conversion of DHA and G3P to F6P requires ATP hydrolysis as described below. First, DHA is phosphorylated to form DHA phosphate (DHAP) by DHA kinase at the expense of an ATP. DHAP and G3P are then combined by fructose bisphosphate aldolase to form fructose-1,6-diphosphate (FDP). FDP is converted to F6P by fructose bisphosphatase, thus wasting a high energy phosphate bond.

A more ATP efficient sequence of reactions is enabled if DHA synthase functions in combination with F6P aldolase as opposed to in combination with DHA kinase, fructose bisphosphate aldolase, and fructose bisphosphatase. F6P aldolase enables direct conversion of DHA and G3P to F6P, bypassing the need for ATP hydrolysis. Overall, DHA synthase when combined with F6P aldolase is identical in energy demand to the RuMP pathway. Both of these formaldehyde assimilation options (i.e., RuMP pathway, DHA synthase+F6P aldolase) are superior to DHA synthase combined with DHA kinase, fructose bisphosphate aldolase, and fructose bisphosphatase in terms of ATP demand.

Example IV

Production of Fatty Alcohols and Fatty Aldehydes by MI-FAE Cycle, MD-FAE Cycle and Acyl-CoA Termination Pathways

Encoding nucleic acids and species that can be used as sources for conferring fatty alcohol and fatty aldehyde production capability onto a host microbial organism are exemplified further below.

Multienzyme Complexes

In one exemplary embodiment, the genes *fadA* and *fadB* encode a multienzyme complex that exhibits three constituent activities of the malonyl-CoA independent FAS pathway, namely, ketoacyl-CoA thiolase, 3-hydroxyacyl-CoA dehydrogenase, and enoyl-CoA hydratase activities (Nakahi-

gashi, K. and H. Inokuchi, *Nucleic Acids Research* 18:4937 (1990); Yang et al., *Journal of Bacteriology* 173:7405-7406 (1991); Yang et al., *Journal of Biological Chemistry* 265: 10424-10429 (1990); Yang et al., *Biochemistry* 30:6788-6795 (1990)). The fadI and fadJ genes encode similar activities which can substitute for the above malonyl-CoA

eutropha (Jenkins et al, *Journal of Bacteriology* 169:42-52 (1987)). The acetoacetyl-CoA thiolase from *Zoogloea ramigera* is irreversible in the biosynthetic direction and a crystal structure is available (Merilainen et al, *Biochem* 48: 11011-25 (2009)). Accession numbers for these thiolases and homologs are included in the table below.

Protein	GenBank ID	GI Number	Organism
atoB	NP_416728	16130161	<i>Escherichia coli</i>
yqeF	NP_417321.2	90111494	<i>Escherichia coli</i>
thlA	NP_349476.1	15896127	<i>Clostridium acetobutylicum</i>
thlB	NP_149242.1	15004782	<i>Clostridium acetobutylicum</i>
ERG10	NP_015297	6325229	<i>Saccharomyces cerevisiae</i>
POT1	NP_012106.1	6322031	<i>Saccharomyces cerevisiae</i>
phaA	YP_725941	113867452	<i>Ralstonia eutropha</i>
phbA	P07097.4	135759	<i>Zoogloea ramigera</i>
h16_A1713	YP_726205.1	113867716	<i>Ralstonia eutropha</i>
pcaF	YP_728366.1	116694155	<i>Ralstonia eutropha</i>
h16_B1369	YP_840888.1	116695312	<i>Ralstonia eutropha</i>
h16_A0170	YP_724690.1	113866201	<i>Ralstonia eutropha</i>
h16_A0462	YP_724980.1	113866491	<i>Ralstonia eutropha</i>
h16_A1528	YP_726028.1	113867539	<i>Ralstonia eutropha</i>
h16_B0381	YP_728545.1	116694334	<i>Ralstonia eutropha</i>
h16_B0662	YP_728824.1	116694613	<i>Ralstonia eutropha</i>
h16_B0759	YP_728921.1	116694710	<i>Ralstonia eutropha</i>
h16_B0668	YP_728830.1	116694619	<i>Ralstonia eutropha</i>
h16_A1720	YP_726212.1	113867723	<i>Ralstonia eutropha</i>
h16_A1887	YP_726356.1	113867867	<i>Ralstonia eutropha</i>
bktB	YP_002005382.1	194289475	<i>Cupriavidus taiwanensis</i>
Rmet_1362	YP_583514.1	94310304	<i>Ralstonia metallidurans</i>
Bphy_0975	YP_001857210.1	186475740	<i>Burkholderia phymatum</i>

independent FAS conferring genes fadA and fadB. The acyl-CoA dehydrogenase of *E. coli* is encoded by fadE (Campbell et al, *J Bacteriol* 184: 3759-64)). This enzyme catalyzes the rate-limiting step of beta-oxidation (O'Brien et al, *J Bacteriol* 132:532-40 (1977)). The nucleic acid sequences for each of the above fad genes are well known in the art and can be accessed in the public databases such as Genbank using the following accession numbers.

Protein	GenBank ID	GI Number	Organism
fadA	YP_026272.1	49176430	<i>Escherichia coli</i>
fadB	NP_418288.1	16131692	<i>Escherichia coli</i>
fadI	NP_416844.1	16130275	<i>Escherichia coli</i>
fadJ	NP_416843.1	16130274	<i>Escherichia coli</i>
fadR	NP_415705.1	16129150	<i>Escherichia coli</i>
fadE	AAC73325.2	87081702	<i>Escherichia coli</i>

Step A. Thiolase

Thiolase enzymes, also known as beta-keto thiolase, acyl-CoA C-acetyltransferase, acyl-CoA:acetyl-CoA C-acyltransferase, 3-oxoacyl-CoA thiolase, 3-ketoacyl-CoA thiolase, beta-ketoacyl-CoA thiolase, and acyl-CoA thiolase, that are suitable for fatty alcohol, fatty aldehyde or fatty acid production are described herein (FIGS. 2A and 7A). Exemplary acetoacetyl-CoA thiolase enzymes include the gene products of atoB and homolog yqeF from *E. coli* (Martin et al., *Nat.Biotechnol* 21:796-802 (2003)), thlA and thlB from *C. acetobutylicum* (Hanai et al., *Appl Environ Microbiol* 73:7814-7818 (2007); Winzer et al., *J.Mol.Microbiol Biotechnol* 2:531-541(2000)), and ERG10 from *S. cerevisiae* (Hiser et al., *J.Biol.Chem.* 269:31383-31389 (1994)). A degradative thiolase of *S. cerevisiae* is encoded by POT1. Another candidate thiolase is the phaA gene product of *R.*

Many thiolase enzymes catalyze the formation of longer-chain acyl-CoA products. Exemplary thiolases include, for example, 3-oxoadipyl-CoA thiolase (EC 2.3.1.174) and acyl-CoA thiolase (EC 2.3.1.16). 3-Oxoadipyl-CoA thiolase converts succinyl-CoA and acetyl-CoA to 3-oxoadipyl-CoA, and is a key enzyme of the beta-ketoadipate pathway for aromatic compound degradation. The enzyme is widespread in soil bacteria and fungi including *Pseudomonas putida* (Harwood et al., *J Bacteriol.* 176:6479-6488 (1994)) and *Acinetobacter calcoaceticus* (Doten et al., *J Bacteriol.* 169: 3168-3174 (1987)). The gene products encoded by pcaF in *Pseudomonas* strain B13 (Kaschabek et al., *J Bacteriol.* 184:207-215 (2002)), phaD in *Pseudomonas putida* U (Olivera et al., *Proc.Natl.AcadSci USA* 95:6419-6424 (1998)), paaE in *Pseudomonas fluorescens* ST (Di et al., *Arch.Microbiol* 188:117-125 (2007)), and paaJ from *E. coli* (Nogales et al., *Microbiology* 153:357-365 (2007)) also catalyze this transformation. Several beta-ketothiolases exhibit significant and selective activities in the oxoadipyl-CoA forming direction including bkt from *Pseudomonas putida*, pcaF and bkt from *Pseudomonas aeruginosa* PAO1, bkt from *Burkholderia ambifaria* AMMD, paaJ from *E. coli*, and phaD from *P. putida*. Two gene products of *Ralstonia eutropha* (formerly known as *Alcaligenes eutrophus*), encoded by genes bktB and bktC, catalyze the formation of 3-oxopimeloyl-CoA (Slater et al., *J.Bacteriol.* 180:1979-1987 (1998); Haywood et al., *FEMS Microbiology Letters* 52:91-96 (1988)). The sequence of the BktB protein is known; however, the sequence of the BktC protein has not been reported. BktB is also active on substrates of length C6 and C8 (Machado et al, Met Eng in press (2012)). The pim operon of *Rhodospirillum rubrum* also encodes a beta-ketothiolase, encoded by pimB, predicted to catalyze this transformation in the degradative direction during benzoyl-CoA degradation (Harrison et al., *Microbiology* 151: 727-736 (2005)). A beta-ketothiolase enzyme candidate in *S. aciditrophicus* was identified by sequence homology to bktB (43% identity, value=1e-93).

Gene name	GI#	GenBank Accession #	Organism
paaJ	16129358	NP_415915.1	<i>Escherichia coli</i>
pcaF	17736947	AAL02407	<i>Pseudomonas knackmussii</i> (B13)
phaD	3253200	AAC24332.1	<i>Pseudomonas putida</i>
pcaF	506695	AAA85138.1	<i>Pseudomonas putida</i>
pcaF	141777	AAC37148.1	<i>Acinetobacter calcoaceticus</i>
paaE	106636097	ABF82237.1	<i>Pseudomonas fluorescens</i>
bkt	115360515	YP_777652.1	<i>Burkholderia ambifaria</i> AMMD
bkt	9949744	AAG06977.1	<i>Pseudomonas aeruginosa</i> PAO1
pcaF	9946065	AAG03617.1	<i>Pseudomonas aeruginosa</i> PAO1
bktB	YP_725948	11386745	<i>Ralstonia eutropha</i>
pimB	CAE29156	39650633	<i>Rhodopseudomonas palustris</i>
syn_02642	YP_462685.1	85860483	<i>Syntrophus aciditrophicus</i>

Acyl-CoA thiolase (EC 2.3.1.16) enzymes involved in the beta-oxidation cycle of fatty acid degradation exhibit activity on a broad range of acyl-CoA substrates of varying chain length. Exemplary acyl-CoA thiolases are found in *Arabidopsis thaliana* (Cruz et al, Plant Physiol 135:85-94 (2004)), *Homo sapiens* (Mannaerts et al, Cell Biochem Biophys 32:73-87 (2000)), *Helianthus annuus* (Schiedel et al, Prot Expr Purif 33:25-33 (2004)). The chain length specificity of thiolase enzymes can be assayed by methods well known in the art (Wrensford et al, Anal Biochem 192:49-54 (1991)). A peroxisomal thiolase found in rat liver catalyze the acetyl-CoA dependent formation of longer chain acyl-CoA products from octanoyl-CoA (Horie et al, Arch Biochem Biophys 274: 64-73 (1989); Hijikata et al, J Biol Chem 265, 4600-4606 (1990)).

Protein	GenBank ID	GI Number	Organism
AY308827.1:1 . . . 1350	AAQ77242.1	34597334	<i>Helianthus annuus</i>
KAT2	Q56WD9.2	73919871	<i>Arabidopsis thaliana</i>
KAT1	Q8LF48.2	73919870	<i>Arabidopsis thaliana</i>
KAT5	Q570C8.2	73919872	<i>Arabidopsis thaliana</i>
ACAA1	P09110.2	135751	<i>Homo sapiens</i>
LCTHIO	AAF04612.1	6165556	<i>Sus scrofa</i>
Acaa1a	NP_036621.1	6978429	<i>Rattus norvegicus</i>
Acaa1b	NP_001035108.1	90968642	<i>Rattus norvegicus</i>
Acaa2	NP_569117.1	18426866	<i>Rattus norvegicus</i>

Acetoacetyl-CoA can also be synthesized from acetyl-CoA and malonyl-CoA by acetoacetyl-CoA synthase (EC 2.3.1.194). This enzyme (FhsA) has been characterized in the soil bacterium *Streptomyces* sp. CL190 where it participates in mevalonate biosynthesis (Okamura et al, PNAS USA 107:11265-70 (2010)). As this enzyme catalyzes an essentially irreversible reaction, it is particularly useful for metabolic engineering applications for overproducing metabolites, fuels or chemicals derived from acetoacetyl-CoA such as long chain alcohols. Other acetoacetyl-CoA synthase genes can be identified by sequence homology to fhsA. Acyl-CoA synthase enzymes such as fhsA and homologs can be engineered or evolved to accept longer acyl-CoA substrates by methods known in the art.

Protein	GenBank ID	GI Number	Organism
fhsA	BAJ83474.1	325302227	<i>Streptomyces</i> sp CL190
AB183750.1:11991 . . . 12971	BAD86806.1	57753876	<i>Streptomyces</i> sp. KO-3988
epzT	ADQ43379.1	312190954	<i>Streptomyces cinnamonensis</i>
ppzT	CAX48662.1	238623523	<i>Streptomyces anulatus</i>
O3I_22085	ZP_09840373.1	378817444	<i>Nocardia brasiliensis</i>

Chain length selectivity of selected thiolase enzymes described above is summarized in the table below.

Chain length	Gene	Organism
C4	atoB	<i>Escherichia coli</i>
C6	phaD	<i>Pseudomonas putida</i>
C6-C8	bktB	<i>Ralstonia eutropha</i>
C10-C16	Acaa1a	<i>Rattus norvegicus</i>

Step B. 3-Oxoacyl-CoA Reductase

3-Oxoacyl-CoA reductases (also known as 3-hydroxyacyl-CoA dehydrogenases, 3-ketoacyl-CoA reductases, beta-

ketoacyl-CoA reductases, beta-hydroxyacyl-CoA dehydrogenases, hydroxyacyl-CoA dehydrogenases, and ketoacyl-CoA reductases) catalyze the reduction of 3-oxoacyl-CoA substrates to 3-hydroxyacyl-CoA products (FIG. 2B and FIG. 7B). These enzymes are often involved in fatty acid beta-oxidation and aromatic degradation pathways. For example, subunits of two fatty acid oxidation complexes in *E. coli*, encoded by fadB and fadJ, function as 3-hydroxyacyl-CoA dehydrogenases (Binstock et al., Methods Enzymol. 71 Pt C:403-411 (1981)). Knocking out a negative regulator encoded by fadR can be utilized to activate the fadB gene product (Sato et al., J Biosci.Bioeng 103:38-44 (2007)). Another 3-hydroxyacyl-CoA dehydrogenase from *E. coli* is paaH (Ismail et al., European Journal of Biochemistry 270:3047-3054 (2003)). Additional 3-oxoacyl-CoA

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enzymes include the gene products of phaC in *Pseudomonas putida* (Olives et al., *Proc.Natl.Acad.Sci USA* 95:6419-6424 (1998)) and paaC in *Pseudomonas fluorescens* (Di et al., 188:117-125 (2007)). These enzymes catalyze the reversible oxidation of 3-hydroxyadipyl-CoA to 3-oxoadipyl-CoA during the catabolism of phenylacetate or styrene. Other suit-

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preferentially active on short and medium chain acyl-CoA substrates whereas the heart enzyme is less selective (He et al, *Biochim Biophys Acta* 1392:119-26 (1998)). The *S. cerevisiae* enzyme FOX2 is active in beta-degradation pathways and also has enoyl-CoA hydratase activity (Hiltunen et al, *J Biol Chem* 267: 6646-6653 (1992)).

Protein	Genbank ID	GI number	Organism
fadB	P21177.2	119811	<i>Escherichia coli</i>
fadJ	P77399.1	3334437	<i>Escherichia coli</i>
paaH	NP_415913.1	16129356	<i>Escherichia coli</i>
Hbd2	EDK34807.1	146348271	<i>Clostridium kluyveri</i>
Hbd1	EDK32512.1	146345976	<i>Clostridium kluyveri</i>
phaC	NP_745425.1	26990000	<i>Pseudomonas putida</i>
paaC	ABF82235.1	106636095	<i>Pseudomonas fluorescens</i>
HSD17B10	O02691.3	3183024	<i>Bos taurus</i>
phbB	P23238.1	130017	<i>Zoogloea ramigera</i>
phaB	YP_353825.1	77464321	<i>Rhodobacter sphaeroides</i>
paaH1	CAJ91433.1	113525088	<i>Ralstonia eutropha</i>
phaB	BAA08358	675524	<i>Paracoccus denitrificans</i>
Hbd	NP_349314.1	15895965	<i>Clostridium acetobutylicum</i>
Hbd	AAM14586.1	20162442	<i>Clostridium beijerinckii</i>
Msed_1423	YP_001191505	146304189	<i>Metallosphaera sedula</i>
Msed_0399	YP_001190500	146303184	<i>Metallosphaera sedula</i>
Msed_0389	YP_001190490	146303174	<i>Metallosphaera sedula</i>
Msed_1993	YP_001192057	146304741	<i>Metallosphaera sedula</i>
Fox2	Q02207	399508	<i>Candida tropicalis</i>
HSD17B10	O02691.3	3183024	<i>Bos taurus</i>
HADH	NP_999496.1	47523722	<i>Bos taurus</i>
3HCDH	AAO72312.1	29293591	<i>Euglena gracilis</i>
FOX2	NP_012934.1	6322861	<i>Saccharomyces cerevisiae</i>

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able enzyme candidates include AAO72312.1 from *E. gracilis* (Winkler et al., *Plant Physiology* 131:753-762 (2003)) and paaC from *Pseudomonas putida* (Olivera et al., *PNAS USA* 95:6419-6424 (1998)). Enzymes catalyzing the reduction of acetoacetyl-CoA to 3-hydroxybutyryl-CoA include hbd of *Clostridium acetobutylicum* (Youngleson et al., *J Bacteriol.* 171:6800-6807 (1989)), phbB from *Zoogloea ramigera* (Ploux et al., *Eur.JBiochem.* 174:177-182 (1988)), phaB from *Rhodobacter sphaeroides* (Alber et al., *Mol.Microbiol.* 61:297-309 (2006)) and paaH1 of *Ralstonia eutropha* (Machado et al, *Met Eng*, In Press (2012)). The *Z. ramigera* enzyme is NADPH-dependent and also accepts 3-oxopropionyl-CoA as a substrate (Ploux et al., *Eur.J Biochem.* 174:177-182 (1988)). Additional genes include phaB in *Paracoccus denitrificans*, Hbd1 (C-terminal domain) and Hbd2 (N-terminal domain) in *Clostridium kluyveri* (Hilimer and Gottschalk, *Biochim. Biophys. Acta* 3334:12-23 (1974)) and HSD17B10 in *Bos taurus* (Wakil et al., *J Biol.Chem.* 207:631-638 (1954)). The enzyme from *Paracoccus denitrificans* has been functionally expressed and characterized in *E. coli* (Yabutani et al., *FEMS Microbiol Lett.* 133:85-90 (1995)). A number of similar enzymes have been found in other species of *Clostridia* and in *Metallosphaera sedula* (Berg et al., *Science.* 318:1782-1786 (2007)). The enzyme from *Candida tropicalis* is a component of the peroxisomal fatty acid beta-oxidation multifunctional enzyme type 2 (MFE-2). The dehydrogenase B domain of this protein is catalytically active on acetoacetyl-CoA. The domain has been functionally expressed in *E. coli*, a crystal structure is available, and the catalytic mechanism is well-understood (Ylianttila et al., *Biochem Biophys Res Commun* 324:25-30 (2004); Ylianttila et al., *J Mol Biol* 358:1286-1295 (2006)). 3-Hydroxyacyl-CoA dehydrogenases that accept longer acyl-CoA substrates (eg. EC 1.1.1.35) are typically involved in beta-oxidation. An example is HSD17B10 in *Bos taurus* (Wakil et al., *J Biol.Chem.* 207:631-638 (1954)). The pig liver enzyme is

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Chain length specificity of selected hydroxyacyl-CoA dehydrogenase enzymes is shown below. Directed evolution can enhance selectivity of enzymes for longer-chain substrates. For example, Machado and coworkers developed a selection platform for directed evolution of chain elongation enzymes that favor longer acyl-CoA substrates. This group evolved paaH1 of *Ralstonia eutropha* for improved activity on 3-oxo-hexanoyl-CoA (Machado et al, *Met Eng*, In Press (2012)).

Chain length	Gene	Organism
C4	hbd	<i>Clostridium acetobutylicum</i>
C5	phbB	<i>Zoogloea ramigera</i>
C4-C6	paaH1	<i>Ralstonia eutropha</i>
C4-C10	HADH	<i>Sus scrofa</i>
C4-C18	fadB	<i>Escherichia coli</i>

Step C. 3-Hydroxyacyl-CoA Dehydratase

3-Hydroxyacyl-CoA dehydratases (eg. EC 4.2.1.17, also known as enoyl-CoA hydratases) catalyze the dehydration of a range of 3-hydroxyacyl-CoA substrates (Roberts et al., *Arch.Microbiol.* 117:99-108 (1978); Agnihotri et al., *Bioorg.Med.Chem.* 11:9-20 (2003); Conrad et al., *J Bacteriol.* 118:103-111(1974)) and can be used in the conversion of 3-hydroxyacyl-CoA to enoyl-CoA (FIGS. 2C and 7C). The ech gene product of *Pseudomonas putida* catalyzes the conversion of 3-hydroxybutyryl-CoA to crotonyl-CoA (Roberts et al., *Arch.Microbiol.* 117:99-108 (1978)). This transformation is also catalyzed by the crt gene product of *Clostridium acetobutylicum*, the crt1 gene product of *C. kluyveri*, and other clostridial organisms Atsumi et al., *Metab Eng* 10:305-311(2008); Boynton et al., *J Bacteriol.* 178:3015-3024 (1996); Hillmer et al., *FEBS Lett.* 21:351-354 (1972)). Additional enoyl-CoA hydratase candidates are

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phaA and phaB, of *P. putida*, and paaA and paaB from *P. fluorescens* (Olives et al., *Proc.Natl.Acad.Sci U.S.A* 95:6419-6424 (1998)). The gene product of pimF in *Rhodopseudomonas palustris* is predicted to encode an enoyl-CoA hydratase that participates in pimeloyl-CoA degradation (Harrison et al., *Microbiology* 151:727-736 (2005)). Lastly, a number of *Escherichia coli* genes have been shown to demonstrate enoyl-CoA hydratase functionality including maoC (Park et al., *J Bacteriol.* 185:5391-5397 (2003)), paaF (Ismail et al., *Eur.J Biochem.* 270:3047-3054 (2003); Park et al., *Appl.Biochem.Biotechnol* 113-116:335-346 (2004); Park et al., *Biotechnol Bioeng* 86:681-686 (2004)) and paaG (Ismail et al., *Eur.J Biochem.* 270:3047-3054 (2003); Park and Lee, *Appl.Biochem.Biotechnol* 113-116:335-346 (2004); Park and Yup, *Biotechnol Bioeng* 86:681-686 (2004)). Enzymes with 3-hydroxyacyl-CoA dehydratase activity in *S. cerevisiae* include PHS1 and FOX2.

Gene	GenBank Accession No.	GI No.	Organism
ech	NP_745498.1	26990073	<i>Pseudomonas putida</i>
crt	NP_349318.1	15895969	<i>Clostridium acetobutylicum</i>
crt1	YP_001393856	153953091	<i>Clostridium kluyveri</i>
phaA	ABF82233.1	26990002	<i>Pseudomonas putida</i>
phaB	ABF82234.1	26990001	<i>Pseudomonas putida</i>
paaA	NP_745427.1	106636093	<i>Pseudomonas fluorescens</i>
paaB	NP_745426.1	106636094	<i>Pseudomonas fluorescens</i>
pimF	CAE29158.1	39650635	<i>Rhodopseudomonas palustris</i>
maoC	NP_415905.1	16129348	<i>Escherichia coli</i>
paaF	NP_415911.1	16129354	<i>Escherichia coli</i>
paaG	NP_415912.1	16129355	<i>Escherichia coli</i>
FOX2	NP_012934.1	6322861	<i>Saccharomyces cerevisiae</i>
PHS1	NP_012438.1	6322364	<i>Saccharomyces cerevisiae</i>

Enoyl-CoA hydratases involved in beta-oxidation can also be used in an fatty alcohol, fatty aldehyde and fatty acid biosynthetic pathway. For example, the multifunctional MFP2 gene product of *Arabidopsis thaliana* exhibits an enoyl-CoA reductase activity selective for chain lengths less than or equal to C14 (Arent et al, *J Biol Chem* 285:24066-77 (2010)). Alternatively, the *E. coli* gene products of fadA and fadB encode a multienzyme complex involved in fatty acid oxidation that exhibits enoyl-CoA hydratase activity (Yang et al., *Biochemistry* 30:6788-6795 (1991); Yang, *J Bacteriol.* 173:7405-7406 (1991); Nakahigashi et al., *Nucleic Acids Res.* 18:4937 (1990)). The fadI and fadJ genes encode

Protein	GenBank ID	GI Number	Organism
MFP2	AAD18042.1	4337027	<i>Arabidopsis thaliana</i>
fadA	YP_026272.1	49176430	<i>Escherichia coli</i>

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Protein	GenBank ID	GI Number	Organism
fadB	NP_418288.1	16131692	<i>Escherichia coli</i>
fadI	NP_416844.1	16130275	<i>Escherichia coli</i>
fadJ	NP_416843.1	16130274	<i>Escherichia coli</i>
fadR	NP_415705.1	16129150	<i>Escherichia coli</i>

Chain length specificity of selected 3-hydroxyacyl-CoA dehydratase enzymes is shown below.

Chain length	Gene	Organism
C4-C6	crt	<i>Clostridium acetobutylicum</i>
C4-C7	pimF	<i>Rhodopseudomonas palustris</i>
C4-C14	MFP2	<i>Arabidopsis thaliana</i>

Step D. Enoyl-CoA Reductase

Enoyl-CoA reductases (also known as acyl-CoA dehydrogenases, trans-2-enoyl-CoA reductases, or acyl-CoA oxidoreductases) catalyze the conversion of an enoyl-CoA to an acyl-CoA (step D of FIGS. 2 and 7). Exemplary acyl-CoA dehydrogenase or enoyl-CoA reductase (ECR) enzymes are the gene products of fadE of *E. coli* and *Salmonella enterica* (Tram et al, *J Bacteriol* 188:599-608 (2006)). YdiO of *E. coli* encodes a ferridoxin-dependent enoyl-CoA reductase (Delomonaco et al *Nature* 476:355 (2011)). The bcd gene product from *Clostridium acetobutylicum* (Atsumi et al., 10:305-311(2008); Boynton et al., *J Bacteriol.* 178:3015-3024 (1996)) catalyzes the reduction of crotonyl-CoA to butyryl-CoA (EC 1.3.99.2). This enzyme participates in the acetyl-CoA fermentation pathway to butyrate in *Clostridial* species (Jones et al., *Microbiol Rev.* 50:484-524 (1986)). Activity of butyryl-CoA reductase can be enhanced by expressing bcd in conjunction with expression of the *C. acetobutylicum* effAB genes, which encode an electron transfer flavoprotein. An additional candidate for the enoyl-CoA reductase step is the enoyl-CoA reductase (EC 1.3.1.44) TER from *E. gracilis* (Hoffmeister et al., *J Biol.Chem* 280:4329-4338 (2005)). A construct derived from this sequence following the removal of its mitochondrial targeting leader sequence was cloned in *E. coli* resulting in an active enzyme. A close homolog of the ECR protein from the prokaryote *Treponema denticola*, encoded by TDE0597, has also been cloned and expressed in *E. coli* (Tucci et al., *FEBS Lett*, 581:1561-1566 (2007)). Six genes in *Syntrophus aciditrophicus* were identified by sequence homology to the *C. acetobutylicum* bcd gene product. The *S. aciditrophicus* genes syn_02637 and syn_02636 bear high sequence homology to the effAB genes of *C. acetobutylicum*, and are predicted to encode the alpha and beta subunits of an electron transfer flavoprotein.

Protein	GenBank ID	GI Number	Organism
fadE	AAC73325.2	87081702	<i>Escherichia coli</i>
ydiO	YP_489957.1	4E+08	<i>Escherichia coli</i>
fadE	YP_005241256.1	379699528	<i>Salmonella enterica</i>
bcd	NP_349317.1	15895968	<i>Clostridium acetobutylicum</i>
etfA	NP_349315.1	15895966	<i>Clostridium acetobutylicum</i>
etfB	NP_349316.1	15895967	<i>Clostridium acetobutylicum</i>
TER	Q5EU90.1	62287512	<i>Euglena gracilis</i>
TER	NP_612558.1	19924091	<i>Rattus norvegicus</i>
TDE0597	NP_971211.1	42526113	<i>Treponema denticola</i>
syn_02587	ABC76101	85721158	<i>Syntrophus aciditrophicus</i>

Protein	GenBank ID	GI Number	Organism
syn_02586	ABC76100	85721157	<i>Syntrophus aciditrophicus</i>
syn_01146	ABC76260	85721317	<i>Syntrophus aciditrophicus</i>
syn_00480	ABC77899	85722956	<i>Syntrophus aciditrophicus</i>
syn_02128	ABC76949	85722006	<i>Syntrophus aciditrophicus</i>
syn_01699	ABC78863	85723920	<i>Syntrophus aciditrophicus</i>
syn_02637	ABC78522.1	85723579	<i>Syntrophus aciditrophicus</i>
syn_02636	ABC78523.1	85723580	<i>Syntrophus aciditrophicus</i>

Additional enoyl-CoA reductase enzyme candidates are found in organisms that degrade aromatic compounds. *Rhodopseudomonas palustris*, a model organism for benzoate degradation, has the enzymatic capability to degrade pimelate via beta-oxidation of pimeloyl-CoA. Adjacent genes in the pim operon, pimC and pimD, bear sequence homology to *C. acetobutylicum* bcd and are predicted to encode a flavin-containing pimeloyl-CoA dehydrogenase (Harrison et al., 151:727-736 (2005)). The genome of nitrogen-fixing soybean symbiont *Bradyrhizobium japonicum* also contains a pim operon composed of genes with high sequence similarity to pimC and pimD of *R. palustris* (Harrison and Harwood, *Microbiology* 151:727-736 (2005)).

Protein	GenBank ID	GI Number	Organism
pimC	CAE29155	39650632	<i>Rhodopseudomonas palustris</i>
pimD	CAE29154	39650631	<i>Rhodopseudomonas palustris</i>
pimC	BAC53083	27356102	<i>Bradyrhizobium japonicum</i>
pimD	BAC53082	27356101	<i>Bradyrhizobium japonicum</i>

An additional candidate is 2-methyl-branched chain enoyl-CoA reductase (EC 1.3.1.52 and EC 1.3.99.12), an enzyme catalyzing the reduction of sterically hindered trans-enoyl-CoA substrates. This enzyme participates in branched-chain fatty acid synthesis in the nematode *Ascaris suum* and is capable of reducing a variety of linear and branched chain substrates including 2-methylvaleryl-CoA, 2-methylbutanoyl-CoA, 2-methylpentanoyl-CoA, octanoyl-CoA and pentanoyl-CoA (Duran et al., 268:22391-22396 (1993)). Two isoforms of the enzyme, encoded by genes acad1 and acad, have been characterized.

Protein	GenBank ID	GI Number	Organism
acad1	AAC48316.1	2407655	<i>Ascaris suum</i>
acad	AAA16096.1	347404	<i>Ascaris suum</i>

At least three mitochondrial enoyl-CoA reductase enzymes exist in *E. gracilis* and are applicable for use in the invention. Three mitochondrial enoyl-CoA reductase enzymes of *E. gracilis* (ECR1-3) exhibit different chain length preferences (Inui et al., *European Journal of Biochemistry* 142:121-126 (1984)), which is particularly useful for dictating the chain length of the desired fatty alcohol, fatty aldehyde or fatty acid products. EST's ELL00002199, ELL00002335, and ELL00002648, which are all annotated as mitochondrial trans-2-enoyl-CoA reductases, can be used to isolate these additional enoyl-CoA reductase genes by methods known in the art. Two ECR enzymes from rat liver microsomes also exhibit different substrate specificities (Nagi et al, *Arch Biochem Biophys* 226:50-64 (1983)). The sequences of these enzymes have not been identified to date. The *Mycobacterium smegmatis* enoyl-CoA reductase

accepts acyl-CoA substrates of chain lengths between C10-C16 (Shimakata et al, *J Biochem* 89:1075-80 (1981)).

Protein	GenBank ID	GI Number	Organism
acad	WP_015308343.1	5E+08	<i>Mycobacterium smegmatis</i>
caiA	WP_015308454.1	5E+08	<i>Mycobacterium smegmatis</i>

Enoyl-CoA reductases and their chain length specificities are shown in the table below.

Chain length	Gene	Organism
C4-C6	ECR1	<i>Euglena gracilis</i>
C6-C8	ECR3	<i>Euglena gracilis</i>
C8-10	ECR2	<i>Euglena gracilis</i>
C8-C16	Long chain ECR	<i>Rattus norvegicus</i>
C10-C16	ECR (acad; caiA)	<i>Mycobacterium smegmatis</i>
C2-C18	fadE	<i>Salmonella enterica</i>

Step E. Acyl-CoA Reductase (Aldehyde Forming)

Reduction of an acyl-CoA to a fatty alcohol is catalyzed by either a single enzyme or pair of enzymes that exhibit acyl-CoA reductase and alcohol dehydrogenase activities. Acyl-CoA dehydrogenases that reduce an acyl-CoA to its corresponding aldehyde include fatty acyl-CoA reductase (EC 1.2.1.42, 1.2.1.50), succinyl-CoA reductase (EC 1.2.1.76), acetyl-CoA reductase, butyryl-CoA reductase and propionyl-CoA reductase (EC 1.2.1.3). Aldehyde forming acyl-CoA reductase enzymes with demonstrated activity on acyl-CoA, 3-hydroxyacyl-CoA and 3-oxoacyl-CoA substrates are known in the literature. Several acyl-CoA reductase enzymes are active on 3-hydroxyacyl-CoA substrates. For example, some butyryl-CoA reductases from Clostridial organisms, are active on 3-hydroxybutyryl-CoA and propionyl-CoA reductase of *L. reuteri* is active on 3-hydroxypropionyl-CoA. An enzyme for converting 3-oxoacyl-CoA substrates to their corresponding aldehydes is malonyl-CoA reductase. Enzymes in this class that demonstrate activity on enoyl-CoA substrates have not been identified to date. Specificity for a particular substrate can be refined using evolution or enzyme engineering methods known in the art.

Exemplary fatty acyl-CoA reductases enzymes are encoded by acrl of *Acinetobacter calcoaceticus* (Reiser, *Journal of Bacteriology* 179:2969-2975 (1997)) and *Acinetobacter* sp. M-1 (Ishige et al., *Appl. Environ. Microbiol.* 68:1192-1195 (2002)). Two gene products from *Mycobacterium tuberculosis* accept longer chain fatty acyl-CoA substrates of length C16-C18 (Harmander Singh, U. Central Florida (2007)). Yet another fatty acyl-CoA reductase is LuxC of *Photobacterium phosphoreum* (Lee et al, *Biochim Biophys Acta* 1388:215-22 (1997)). Enzymes with succinyl-

CoA reductase activity are encoded by *sucD* of *Clostridium kluyveri* (Sohling, *J. Bacteriol.* 178:871-880 (1996)) and *sucD* of *P. gingivalis* (Takahashi, *J. Bacteriol.* 182:4704-

forming acyl-CoA reductase activity (Schirmer et al, *Science*, 329: 559-62 (2010)). Acyl-ACP reductase enzymes and homologs are described in further detail in Example XII.

Protein	GenBank ID	GI Number	Organism
acr1	YP_047869.1	50086359	<i>Acinetobacter calcoaceticus</i>
acr1	AAC45217	1684886	<i>Acinetobacter baylyi</i>
acr1	BAB85476.1	18857901	<i>Acinetobacter</i> sp. Strain M-1
Rv1543	NP_216059.1	15608681	<i>Mycobacterium tuberculosis</i>
Rv3391	NP_217908.1	15610527	<i>Mycobacterium tuberculosis</i>
LuxC	Q03324	547874	<i>Photobacterium leiognathi</i> PL741
LuxC	AAT00788.1	46561111	<i>Photobacterium phosphoreum</i>
Msd_0709	YP_001190808.1	146303492	<i>Metallosphaera sedula</i>
Tneu_0421	ACB39369.1	170934108	<i>Thermoproteus neutrophilus</i>
sucD	P38947.1	172046062	<i>Clostridium kluyveri</i>
sucD	NP_904963.1	34540484	<i>Porphyromonas gingivalis</i>
bphG	BAA03892.1	425213	<i>Pseudomonas</i> sp
adhE	AAV66076.1	55818563	<i>Leuconostoc mesenteroides</i>
bld	AAP42563.1	31075383	<i>Clostridium saccharoperbutylacetonicum</i>
pduP	NP_460996	16765381	<i>Salmonella typhimurium</i> LT2
eutE	NP_416950	16130380	<i>Escherichia coli</i>
pduP	CCC03595.1	337728491	<i>Lactobacillus reuteri</i>
wax2	AAN06975.1	22900949	<i>Arabidopsis thaliana</i>
FAR1	AAH07178.1	13938126	<i>Mus musculus</i>
FAR2	AAH55759	33416982	<i>Mus musculus</i>

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4710 (2000)). Additional succinyl-CoA reductase enzymes participate in the 3-hydroxypropionate/4-hydroxybutyrate cycle of thermophilic archaea including *Metallosphaera sedula* (Berg et al., *Science* 318:1782-1786 (2007)) and *Thermoproteus neutrophilus* (Ramos-Vera et al., *J. Bacteriol.* 191:4286-4297 (2009)). The *M. sedula* enzyme, encoded by Msd_0709, is strictly NADPH-dependent and also has malonyl-CoA reductase activity. The *T. neutrophilus* enzyme is active with both NADPH and NADH. The enzyme acylating acetaldehyde dehydrogenase in *Pseudomonas* sp, encoded by bphG, is yet another as it has been demonstrated to oxidize and acylate acetaldehyde, propionaldehyde, butyraldehyde, isobutyraldehyde and formaldehyde (Powlowski, *J. Bacteriol.* 175:377-385 (1993)). In addition to reducing acetyl-CoA to ethanol, the enzyme encoded by adhE in *Leuconostoc mesenteroides* has been shown to oxidize the branched chain compound isobutyraldehyde to isobutyryl-CoA (Kazahaya, *J. Gen. Appl. Microbiol.* 18:43-55 (1972); and Koo et al., *Biotechnol. Lett.* 27:505-510 (2005)). Butyraldehyde dehydrogenase catalyzes a similar reaction, conversion of butyryl-CoA to butyraldehyde, in solventogenic organisms such as *Clostridium saccharoperbutylacetonicum* (Kosaka et al., *Biosci. Biotechnol. Biochem.*, 71:58-68 (2007)). Exemplary propionyl-CoA reductase enzymes include pduP of *Salmonella typhimurium* LT2 (Leal, *Arch. Microbiol.* 180:353-361 (2003)) and eutE from *E. coli* (Skmlly, WO Patent No. 2004/024876). The propionyl-CoA reductase of *Salmonella typhimurium* LT2, which naturally converts propionyl-CoA to propionaldehyde, also catalyzes the reduction of 5-hydroxyvaleryl-CoA to 5-hydroxypentanal (WO 2010/068953A2). The propionaldehyde dehydrogenase of *Lactobacillus reuteri*, PduP, has a broad substrate range that includes butyraldehyde, valeraldehyde and 3-hydroxypropionaldehyde (Luo et al, *Appl. Microbiol. Biotech.*, 89: 697-703 (2011)). Additional FAR enzymes are encoded by wax2 of *Arabidopsis thaliana* and FAR₁ and FAR₂ of *Mus musculus* (Chen et al, *Plant Cell* 15:1170-85 (2003); Cheng and Russel, *J. Biol. Chem.* 279:37789-97 (2004)). Both mouse FAR enzymes accept substrates with a chain length of C16-18. Additionally, some acyl-ACP reductase enzymes such as the orf1594 gene product of *Synechococcus elongatus* PCC7942 also exhibit aldehyde-

An additional enzyme type that converts an acyl-CoA to its corresponding aldehyde is malonyl-CoA reductase which transforms malonyl-CoA to malonic semialdehyde. Malonyl-CoA reductase is a key enzyme in autotrophic carbon fixation via the 3-hydroxypropionate cycle in thermoacidophilic archaeal bacteria (Berg, *Science* 318:1782-1786 (2007); and Thauer, *Science* 318:1732-1733 (2007)). The enzyme utilizes NADPH as a cofactor and has been characterized in *Metallosphaera* and *Sulfolobus* sp. (Alber et al., *J. Bacteriol.* 188:8551-8559 (2006); and Hugler, *J. Bacteriol.* 184:2404-2410 (2002)). The enzyme is encoded by Msd_0709 in *Metallosphaera sedula* (Alber et al., *J. Bacteriol.* 188:8551-8559 (2006); and Berg, *Science* 318:1782-1786 (2007)). A gene encoding a malonyl-CoA reductase from *Sulfolobus tokodaii* was cloned and heterologously expressed in *E. coli* (Alber et al., *J. Bacteriol.* 188:8551-8559 (2006)). This enzyme has also been shown to catalyze the conversion of methylmalonyl-CoA to its corresponding aldehyde (WO2007141208 (2007)). Although the aldehyde dehydrogenase functionality of these enzymes is similar to the bifunctional dehydrogenase from *Chlorollexus aurantiacus*, there is little sequence similarity. Both malonyl-CoA reductase enzyme candidates have high sequence similarity to aspartate-semialdehyde dehydrogenase, an enzyme catalyzing the reduction and concurrent dephosphorylation of aspartyl-4-phosphate to aspartate semialdehyde. Additional gene candidates can be found by sequence homology to proteins in other organisms including *Sulfolobus solfataricus* and *Sulfolobus acidocaldarius* and have been listed below. Yet another candidate for CoA-acylating aldehyde dehydrogenase is the ald gene from *Clostridium beijerinckii* (Toth, *Appl. Environ. Microbiol.* 65:4973-4980 (1999)). This enzyme has been reported to reduce acetyl-CoA and butyryl-CoA to their corresponding aldehydes. This gene is very similar to eutE that encodes acetaldehyde dehydrogenase of *Salmonella typhimurium* and *E. coli* (Toth, *Appl. Environ. Microbiol.* 65:4973-4980 (1999)).

Gene	GenBank ID	GI Number	Organism
Msed_0709	YP_001190808.1	146303492	<i>Metallosphaera sedula</i>
mcr	NP_378167.1	15922498	<i>Sulfolobus tokodaii</i>
asd-2	NP_343563.1	15898958	<i>Sulfolobus solfataricus</i>
Saci_2370	YP_256941.1	70608071	<i>Sulfolobus acidocaldarius</i>
Ald	AAT66436	49473535	<i>Clostridium beijerinckii</i>
eutE	AAA80209	687645	<i>Salmonella typhimurium</i>
eutE	NP_416950	16130380	<i>Escherichia coli</i>

Chain length specificity ranges of selected aldehyde-forming acyl-CoA reductase enzymes are shown in the table below.

Chain length	Gene	Organism
C2-C4	bphG	<i>Pseudomonas</i> sp
C4	Bld	<i>Clostridium saccharoperbutylacetonicum</i>
C12-C20	ACR	<i>Acinetobacter calcoaceticus</i>
C14-C18	Acr1	<i>Acinetobacter</i> sp. Strain M-1
C16-C18	Rv1543, Rv3391	<i>Mycobacterium tuberculosis</i>
C16-C18	FAR1, FAR2	<i>Mus musculus</i>
C18	Wax2	<i>Arabidopsis thaliana</i>

Step G. Acyl-CoA Reductase (Alcohol Forming)

Bifunctional alcohol-forming acyl-CoA reductase enzymes catalyze step G (i.e. step E and F) of FIGS. 2 and 7. Enzymes with this activity include adhE of *E. coli* (Kessler et al., *FEBS Lett.* 281:59-63 (1991)) and adhE2 of *Clostridium acetobutylicum* (Fontaine et al., *J. Bacteriol.* 184:821-830 (2002)). The *E. coli* enzyme is active on C2 substrates, whereas the *C. acetobutylicum* enzyme has a broad substrate range that spans C2-C8 (Dekishima et al., *J. Am Chem Soc* 133:11399-11401 (2011)). The *C. acetobutyli-*

Its overexpression in *E. coli* resulted in FAR activity and the accumulation of C16-C18 fatty alcohols (Metz et al., *Plant Physiol.* 122:635-644 (2000)). FAR enzymes in *Arabidopsis*

thaliana include the gene products of At3g11980, At3g44560 and CER4 (Doan et al, *J Plant Physiol* 166 (2006); Rowland et al, *Plant Physiol* 142:866-77 (2006)). Bifunctional prokaryotic FAR enzymes are found in *Marinobacter aquaeolei* VT8 (Hofvander et al, *FEBS Lett* 3538-43 (2011)), *Marinobacter algicola* and *Oceanobacter* strain RED65 (US Pat Appl 20110000125). Other suitable enzymes include bfar from *Bombyx mori*, mfar1 and mfar2 from *Mus musculus*; mfar2 from *Mus musculus*; acrM1 from *Acinetobacter* sp. M1; and hfar from *H. sapiens*.

Protein	GenBank ID	GI Number	Organism
adhE	NP_415757.1	16129202	<i>Escherichia coli</i>
adhE2	AAK09379.1	12958626	<i>Clostridium acetobutylicum</i>
bdh I	NP_349892.1	15896543	<i>Clostridium acetobutylicum</i>
bdh II	NP_349891.1	15896542	<i>Clostridium acetobutylicum</i>
adhE	AAV66076.1	55818563	<i>Leucorhynchus mesenteroides</i>
mcr	AAS20429.1	42561982	<i>Chloroflexus aurantiacus</i>
Rcas_2929	YP_001433009.1	156742880	<i>Roseiflexus castenholzii</i>
NAP1_02720	ZP_01039179.1	85708113	<i>Erythrobacter</i> sp. NAP1
MGP2080_00535	ZP_01626393.1	119504313	<i>marine gamma proteobacterium</i> HTCC2080
FAR	AAD38039.1	5020215	<i>Simmondsia chinensis</i>
At3g11980	NP_191229.1	15228993	<i>Arabidopsis thaliana</i>
At3g44560	NP_190042.2	145339120	<i>Arabidopsis thaliana</i>
CER4	AEE86278.1	332660878	<i>Arabidopsis thaliana</i>
FAR	YP_959486.1	120555135	<i>Marinobacter aquaeolei</i>
bfar	Q8R079	81901336	<i>Bombyx mori</i>

cum enzymes encoded by bdh I and bdh II (Walter, et *J. Bacteriol.* 174:7149-7158 (1992)), reduce acetyl-CoA and butyryl-CoA to ethanol and butanol, respectively. The adhE gene product from *Leuconostoc mesenteroides* is active on acetyl-CoA and isobutyryl-CoA (Kazahaya et al., *J. Gen. Appl. Microbiol.* 18:43-55 (1972); Koo et al., *Biotechnol Lett*, 27:505-510 (2005)). Enzyme candidates in other organisms including *Roseiflexus castenholzii*, *Erythrobacter* sp. NAP1 and marine gamma proteobacterium HTCC2080 can be inferred by sequence similarity. Longer chain acyl-CoA molecules can be reduced to their corresponding alcohols by enzymes such as the jojoba (*Simmondsia chinensis*) FAR which encodes an alcohol-forming fatty acyl-CoA reductase.

Chain length specificity ranges of selected alcohol-forming acyl-CoA reductase enzymes are shown in the table below.

Chain length	Gene	Organism
C2	adhE	<i>Escherichia coli</i>
C2-C8	adhE2	<i>Clostridium acetobutylicum</i>
C14-C16	At3g11980	<i>Arabidopsis thaliana</i>
C16	At3g44560	<i>Arabidopsis thaliana</i>

-continued

Chain length	Gene	Organism
C16-C18	FAR	<i>Simmondsia chinensis</i>
C14-C18	FAR	<i>Marinobacter aquaeolei</i>
C24-C26	CER4	<i>Arabidopsis thaliana</i>

Step F. Fatty Aldehyde Reductase

Exemplary genes encoding enzymes that catalyze the conversion of an aldehyde to alcohol (i.e., alcohol dehydrogenase or equivalently aldehyde reductase) include alrA encoding a medium-chain alcohol dehydrogenase for C2-C14 (Tani et al., *Appl. Environ. Microbiol.* 66:5231-5235 (2000)), yqhD and fucO from *E. coli* (Sulzenbacher et al., 342:489-502 (2004)), and bdh I and bdh II from *C. acetobutylicum* which converts butyrylaldehyde into butanol (Walter et al., *J Bacteriol* 174:7149-7158 (1992)). The alrA gene product showed no activity on aldehydes longer than C14, and favored the reductive direction (Tani et al, supra). YqhD catalyzes the reduction of a wide range of aldehydes using

NADPH as the cofactor, with a preference for chain lengths longer than C(3) (Sulzenbacher et al, *J Mol Biol* 342:489-502 (2004); Perez et al., *J Biol.Chem.* 283:7346-7353 (2008)). The adhA gene product from *Zymomonas mobilis* has been demonstrated to have activity on a number of aldehydes including formaldehyde, acetaldehyde, propionaldehyde, butyraldehyde, and acrolein (Kinoshita et al., *Appl Microbiol Biotechnol* 22:249-254 (1985)). Additional aldehyde reductase candidates are encoded by bdh in *C. saccharoperbulaceticum* and Cbei_1722, Cbei_2181 and Cbei_2421 in *C. beijerinckii*. The alcohol dehydrogenase from *Leifsonia* sp. S749 shows maximal activity on medium chain-length substrates of length C6-C7 (Inoue et al, *AEM* 71: 3633-3641 (2005)). The adh gene product of *Pseudomonas putida* is active on substrates of length C3-C10 (Nagashima et al, *J Ferment Bioeng* 82:328-33(1996)). The alcohol dehydrogenase enzymes ADH1 and ADH2 of *Geobacillus thermodenitrificans* oxidize alcohols up to a chain length of C30 (Liu et al, *Physiol Biochem* 155:2078-85 (2009)). Three additional alcohol dehydrogenase enzymes from *Geobacillus thermodenitrificans* are active on C2-C14 substrates (Liu et al, supra).

Protein	GenBank ID	GI Number	Organism
alrA	BAB12273.1	9967138	<i>Acinetobacter</i> sp. strain M-1
ADH2	NP_014032.1	6323961	<i>Saccharomyces cerevisiae</i>
yqhD	NP_417484.1	16130909	<i>Escherichia coli</i>
fucO	NP_417279.1	16130706	<i>Escherichia coli</i>
bdh I	NP_349892.1	15896543	<i>Clostridium acetobutylicum</i>
bdh II	NP_349891.1	15896542	<i>Clostridium acetobutylicum</i>
adhA	YP_162971.1	56552132	<i>Zymomonas mobilis</i>
bdh	BAF45463.1	124221917	<i>Clostridium saccharoperbutylacetonicum</i>
Cbei_1722	YP_001308850	150016596	<i>Clostridium beijerinckii</i>
Cbei_2181	YP_001309304	150017050	<i>Clostridium beijerinckii</i>
Cbei_2421	YP_001309535	150017281	<i>Clostridium beijerinckii</i>
lsadh	BAD99642.1	67625613	<i>Leifsonia</i> sp. S749
adh			<i>Pseudomonas putida</i>
ADH1	YP_001126968.1	138896515	<i>Geobacillus thermodenitrificans</i>
ADH2	YP_001125863.1	138895410	<i>Geobacillus thermodenitrificans</i>
GTNG_0872	YP_001124995.1	138894542	<i>Geobacillus thermodenitrificans</i>
GTNG_1287	YP_001125402.1	138894949	<i>Geobacillus thermodenitrificans</i>
GTNG_1851	YP_001125956.1	138895503	<i>Geobacillus thermodenitrificans</i>

Native alcohol dehydrogenases also convert aldehyde substrates to alcohol products. To date, seven alcohol dehydrogenases, ADH1-ADHVII, have been reported in *S. cerevisiae* (de Smidt et al, *FEMS Yeast Res* 8:967-78 (2008)). ADH1 (GI:1419926) is the key enzyme responsible for reducing acetaldehyde to ethanol in the cytosol under anaerobic conditions. In *K. lactis*, two NAD-dependent cytosolic alcohol dehydrogenases have been identified and characterized. These genes also show activity for other aliphatic alcohols. The genes ADH1 (GI:113358) and ADHIII (GI:51704293) are preferentially expressed in glucose-grown cells (Bozzi et al, *Biochim Biophys Acta* 1339: 133-142 (1997)). Cytosolic alcohol dehydrogenases are encoded by ADH1 (GI:608690) in *C. albicans*, ADH1 (GI:3810864) in *S. pombe*, ADH1 (GI:5802617) in *Y. lipolytica*, ADH1 (GI:2114038) and ADHIII (GI:2143328) in *Pichia stipitis* or *Scheffersomyces stipitis* (Passoth et al, *Yeast* 14:1311-23 (1998)). Candidate alcohol dehydrogenases are shown the table below.

Protein	GenBank ID	GI number	Organism
SADH	BAA24528.1	2815409	<i>Candida parapsilosis</i>
ADH1	NP_014555.1	6324486	<i>Saccharomyces cerevisiae</i> s288c
ADH2	NP_014032.1	6323961	<i>Saccharomyces cerevisiae</i> s288c
ADH3	NP_013800.1	6323729	<i>Saccharomyces cerevisiae</i> s288c

Protein	GenBank ID	GI number	Organism
ADH4	NP_011258.2	269970305	<i>Saccharomyces cerevisiae</i> s288c
ADH5 (SFA1)	NP_010113.1	6320033	<i>Saccharomyces cerevisiae</i> s288c
ADH6	NP_014051.1	6323980	<i>Saccharomyces cerevisiae</i> s288c
ADH7	NP_010030.1	6319949	<i>Saccharomyces cerevisiae</i> s288c
adhP	CAA44614.1	2810	<i>Kluyveromyces lactis</i>
ADH1	P20369.1	113358	<i>Kluyveromyces lactis</i>
ADH2	CAA45739.1	2833	<i>Kluyveromyces lactis</i>
ADH3	P49384.2	51704294	<i>Kluyveromyces lactis</i>

Substrate specificity ranges of selected alcohol dehydrogenase enzymes are shown in the table below.

Chain length	Gene	Organism
C6-C7	lsadh	<i>Leifsonia</i> sp. S749
C2-C8	yqhD	<i>Escherichia coli</i>
C3-C10	Adh	<i>Pseudomonas putida</i>
C2-C14	alrA	<i>Acinetobacter</i> sp. strain M-1
C2-C14	ADH	<i>Geobacillus thermodenitrificans</i>
C2-C30	ADH1	<i>Geobacillus thermodenitrificans</i>

Step O. Elongase

Elongase (ELO) enzymes utilize malonyl-CoA to add a C2 unit to a growing acyl-CoA chain. This process also involves decarboxylation and is thus largely irreversible. *Trypanosoma brucei*, a eukaryotic human parasite, is known to produce long chain fatty acids using an elongase system. The process is initiated by butyryl-CoA. In particular, the ELO system esterifies the growing fatty acid chain to CoA intermediates rather than ACP intermediates like the bacterial and other microbial counterparts (Lee et al, Cell 126, 691-699, 2006; Cronan, Cell, 126, 2006). This is in contrast to typical bacterial fatty acid elongation which is initiated following the formation of acetoacetyl acyl-ACP from malonyl-ACP. So far, four ELOs (encoded by ELO1-4) that are homologous to their animal counterparts have been found in *T. brucei* (Lee et al, Nature Reviews Microbiology, Vol 5, 287-297, 2007). ELO1-3 together account for synthesis of saturated fatty acids up to a chain length of C18. ELO1 converts C4 to C10, ELO2 extends the chain length from C10 to myristate (C14), and ELO3 extends myristate to C18. There is some overlap in ELO specificity; for example, ELO1 can extend a C10 primer to C12, albeit with low activity. ELO4 is an example of an ELO that is specific for poly unsaturated fatty acids (PUFAs). It extends arachidonate (C20:4) by two carbon atoms. Several additional ELO enzymes can be found by sequence homology (see Lee et al, Nature Reviews Microbiology, Vol 5, 287-297, 2007).

Elongase enzymes are found in several compartments including the mitochondria, endoplasmic reticulum, proteoliposomes and peroxisomes. For example, some yeast such as *S. cerevisiae* are able to synthesize long-chain fatty acids of chain length C16 and higher via a mitochondrial elongase which accepts exogenous or endogenous acyl-CoA substrates (Bessoule et al, *FEBS Lett* 214: 158-162 (1987)). This system requires ATP for activity. The endoplasmic reticulum also has an elongase system for synthesizing very long chain fatty acids (C18+) from acyl-CoA substrates of varying lengths (Kohlwein et al, *Mol Cell Biol* 21:109-25 (2001)). Genes involved in this system include TSC13, ELO2 and ELO3. ELO1 catalyzes the elongation of C12 acyl-CoAs to C16-C18 fatty acids.

Protein	Accession #	GI number	Organism
ELO2	NP_009963.1	6319882	<i>Saccharomyces cerevisiae</i>
ELO3	NP_013476.3	398366027	<i>Saccharomyces cerevisiae</i>
TSC13	NP_010269.1	6320189	<i>Saccharomyces cerevisiae</i>
ELO1	NP_012339.1	6322265	<i>Saccharomyces cerevisiae</i>
ELO1	AAX70671.1	62176566	<i>Trypanosoma brucei</i>
ELO2	AAX70672.1	62176567	<i>Trypanosoma brucei</i>
ELO3	AAX70673.1	62176568	<i>Trypanosoma brucei</i>
ELO4	AAX70768.1	62176665	<i>Trypanosoma brucei</i>
ELO4	AAX69821.1	62175690	<i>Trypanosoma brucei</i>

Those skilled in the art also can obtain nucleic acids encoding any or all of the malonyl-CoA independent FAS pathway or acyl-reduction pathway enzymes by cloning using known sequences from available sources. For example, any or all of the encoding nucleic acids for the malonyl-CoA independent FAS pathway can be readily obtained using methods well known in the art from *E. gracilis* as this pathway has been well characterized in this organism. *E. gracilis* encoding nucleic acids can be isolated from, for example, an *E. gracilis* cDNA library using probes of known sequence. The probes can be designed with whole or partial DNA sequences from the following EST sequences from the publically available sequence database TBestDB (<http://tbestdb.bcm.umontreal.ca>). The nucleic acids generated from this process can be inserted into an appropriate expression vector and transformed into *E. coli* or other microorganisms to generate fatty alcohols, fatty aldehydes or fatty acids production organisms of the invention.

Thiolase (FIG. 2A): ELL00002550, ELL00002493, ELL00000789

3-Hydroxyacyl-CoA dehydrogenase (FIG. 2B): ELL00000206, ELL00002419, ELL00006286, ELL00006656

Enoyl-CoA hydratase (FIG. 2C): ELL00005926, ELL00001952, ELL00002235, ELL00006206

Enoyl-CoA reductase (FIG. 2D): ELL00002199, ELL00002335, ELL00002648

Acyl-CoA reductase (FIG. 2E; 2E/F): ELL00002572, ELL00002581, ELL00000108

Alternatively, the above EST sequences can be used to identify homologue polypeptides in GenBank through BLAST search. The resulting homologue polypeptides and their corresponding gene sequences provide additional encoding nucleic acids for transformation into *E. coli* or other microorganisms to generate the fatty alcohols, fatty aldehydes or fatty acids producing organisms of the invention. Listed below are exemplary homologue polypeptide and their gene accession numbers in GenBank which are applicable for use in the non-naturally occurring organisms of the invention.

Ketoacyl-CoA Acyltransferase (or Ketoacyl-CoA Thiolase)

Protein	GenBank ID	GI number	Organism
Dole_2160	YP_001530041	158522171	<i>Desulfococcus oleovorans</i> Hxd3
DalkDRAFT_1939	ZP_02133627	163726110	<i>Desulfatibacillum alkenivorans</i> AK-01
BSG1_09488	ZP_01860900	149182424	<i>Bacillus</i> sp. SG-1

3-Hydroxyacyl-CoA Dehydrogenase

Protein	GenBank ID	GI number	Organism
AaeL_AAEL002841	XP_001655993	157132312	<i>Aedes aegypti</i>
hadh	NP_001011073	58331907	<i>Xenopus tropicalis</i>
hadh	NP_001003515	51011113	<i>Danio rerio</i>

Enoyl-CoA Hydratase

Protein	GenBank ID	GI number	Organism
Tb927.3.4850	XP_844077	72387305	<i>Trypanosoma brucei</i>
Tc00.1047053509701.10	XP_802711	71399112	<i>Trypanosoma cruzi</i> strain CL Brener
PputGB1_3629	YP_001669856	167034625	<i>Pseudomonas putida</i> GB-1

Enoyl-CoA Reductase

Protein	GenBank ID	GI number	Organism
mecr	XP_642118	66816217	<i>Dictyostelium discoideum</i> AX4
NEMVEDRAFT_v1g228294	XP_001639469	156402181	<i>Nematostella vectensis</i>
AaeL_AAEL003995	XP_001648220	157104018	<i>Aedes aegypti</i>

In addition to the above exemplary encoding nucleic acids, nucleic acids other than those within the MI-FAE cycle, MD-FAE and/or termination pathways of the invention also can be introduced into a host organism for further production of fatty alcohols, fatty aldehydes or fatty acids. For example, the *Ralstonia eutropha* BktB and PhbB genes catalyze the condensation of butyryl-CoA and acetyl-CoA to form β -keto-hexanoyl-CoA and the reduction of β -keto-hexanoyl-CoA to 3-hydroxy-hexanoyl-CoA (Fukui et al., *Biomacromolecules* 3:618-624 (2002)). To improve the production of fatty alcohols, exogenous DNA sequences encoding for these specific enzymes can be expressed in the production host of interest. Furthermore, the above described enzymes can be subjected to directed evolution to generate improved versions of these enzymes with high activity and high substrate specificity. A similar approach also can be utilized with any or all other enzymatic steps in the fatty alcohol, fatty aldehyde or fatty acid producing pathways of the invention to, for example, improve enzymatic activity and/or specificity and/or to generate a fatty alcohol, a fatty aldehyde or a fatty acid of a predetermined chain length or lengths.

Example V

Pathways For Producing Cytosolic Acetyl-CoA from Cytosolic Pyruvate

The following example describes exemplary pathways for the conversion of cytosolic pyruvate and threonine to cytosolic acetyl-CoA, as shown in FIG. 3.

10 Pathways for the conversion of cytosolic pyruvate and threonine to cytosolic acetyl-CoA could enable deployment of a cytosolic fatty alcohol, fatty aldehyde or fatty acid production pathway that originates from acetyl-CoA. Several pathways for converting cytosolic pyruvate to cytosolic acetyl-CoA are shown in FIG. 3. Direct conversion of pyruvate to acetyl-CoA can be catalyzed by pyruvate dehydrogenase, pyruvate formate lyase, pyruvate:NAD(P) oxidoreductase or pyruvate:fenedoxin oxidoreductase. If a

pymvate formate lyase is utilized, the formate byproduct can be further converted to CO₂ by formate dehydrogenase or formate hydrogen lyase.

30 Indirect conversion of pymvate to acetyl-CoA can proceed through several alternate routes. Pyruvate can be converted to acetaldehyde by a pymvate decarboxylase. Acetaldehyde can then be converted to acetyl-CoA by an acylating (CoA-dependent) acetaldehyde dehydrogenase. Alternately, acetaldehyde generated by pymvate decarboxylase can be converted to acetyl-CoA by the "PDH bypass" pathway. In this pathway, acetaldehyde is oxidized by acetaldehyde dehydrogenase to acetate, which is then converted to acetyl-CoA by a CoA ligase, synthetase or transferase. In another embodiment, the acetate intermediate is converted by an acetate kinase to acetyl-phosphate that is then converted to acetyl-CoA by a phosphotransacetylase. In yet another embodiment, pyruvate is directly converted to acetyl-phosphate by a pymvate oxidase (acetyl-phosphate forming). Conversion of pymvate to acetate is also catalyzed by acetate-forming pymvate oxidase.

Cytosolic acetyl-CoA can also be synthesized from threonine by expressing a native or heterologous threonine aldolase (FIG. 6J) (van Mans et al, *AEM* 69:2094-9 (2003)). Threonine aldolase converts threonine into acetaldehyde and glycine. The acetaldehyde product is subsequently converted to acetyl-CoA by various pathways described above.

Gene candidates for the acetyl-CoA forming enzymes shown in FIG. 3 are described below.

Pyruvate oxidase (acetate-forming) (FIG. 3A) or pyruvate:quinone oxidoreductase (PQO) can catalyze the oxidative decarboxylation of pyruvate into acetate, using ubiquinone (EC 1.2.5.1) or quinone (EC 1.2.2.1) as an electron acceptor. The *E. coli* enzyme, PoxB, is localized on the inner membrane (Abdel-Hamid et al., *Microbiol* 147:1483-98 (2001)). The enzyme has thiamin pyrophosphate and flavin adenine dinucleotide (FAD) cofactors (Koland and Gennis, *Biochemistry* 21:4438-4442 (1982)); O'Brien et al., *Biochemistry* 16:3105-3109 (1977); O'Brien and Gennis, *J. Biol. Chem.* 255:3302-3307 (1980)). PoxB has similarity to pyruvate decarboxylase of *S. cerevisiae* and *Zymomonas*

mobilis. The *pqo* transcript of *Corynebacterium glutamicum* encodes a quinone-dependent and acetate-forming pyruvate oxidoreductase (Schreiner et al., *J. Bacteriol.* 188:1341-50 (2006)). Similar enzymes can be inferred by sequence homology.

Protein	GenBank ID	GI Number	Organism
poxB	NP_415392.1	16128839	<i>Escherichia coli</i>
pqo	YP_226851.1	62391449	<i>Corynebacterium glutamicum</i>
poxB	YP_309835.1	74311416	<i>Shigella sonnei</i>
poxB	ZP_03065403.1	194433121	<i>Shigella dysenteriae</i>

The acylation of acetate to acetyl-CoA (FIG. 3B) can be catalyzed by enzymes with acetyl-CoA synthetase, ligase or transferase activity. Two enzymes that can catalyze this reaction are AMP-forming acetyl-CoA synthetase or ligase (EC 6.2.1.1) and ADP-forming acetyl-CoA synthetase (EC 6.2.1.13). AMP-forming acetyl-CoA synthetase (ACS) is the predominant enzyme for activation of acetate to acetyl-CoA. Exemplary ACS enzymes are found in *E. coli* (Brown et al., *J. Gen. Microbiol.* 102:327-336 (1977)), *Ralstonia eutropha* (Priefert and Steinbuchel, *J. Bacteriol.* 174:6590-6599 (1992)), *Methanothermobacter thermautotrophicus* (Ingram-Smith and Smith, *Archaea* 2:95-107 (2007)), *Salmonella enterica* (Gulick et al., *Biochemistry* 42:2866-2873 (2003)) and *Saccharomyces cerevisiae* (Jogl and Tong, *Biochemistry* 43:1425-1431 (2004)). ADP-forming acetyl-CoA synthetases are reversible enzymes with a generally broad substrate range (Musfeldt and Schönheit, *J. Bacteriol.* 184: 636-644 (2002)). Two isozymes of ADP-forming acetyl-CoA synthetases are encoded in the *Archaeoglobus fulgidus* genome by are encoded by AF1211 and AF1983 (Musfeldt and Schönheit, supra (2002)). The enzyme from *Haloarcula marismortui* (annotated as a succinyl-CoA synthetase) also accepts acetate as a substrate and reversibility of the enzyme was demonstrated (Brasen and Schönheit, *Arch. Microbiol.* 182:277-287 (2004)). The ACD encoded by PAE3250 from hyperthermophilic crenarchaeon *Pyrobaculum aerophilum* showed the broadest substrate range of all characterized ACDs, reacting with acetate, isobutyryl-CoA (preferred substrate) and phenylacetyl-CoA (Brasen and Schönheit, supra (2004)). Directed evolution or engineering can be used to modify this enzyme to operate at the physiological temperature of the host organism. The enzymes from *A. fulgidus*, *H. marismortui* and *P. aerophilum* have all been cloned, functionally expressed, and characterized in *E. coli* (Brasen and Schönheit, supra (2004); Musfeldt and Schönheit, supra (2002)). Additional candidates include the succinyl-CoA synthetase encoded by *sucCD* in *E. coli* (Buck et al., *Biochemistry* 24:6245-6252 (1985)) and the acyl-CoA ligase from *Pseudomonas putida* (Fernandez-Valverde et al., *Appl. Environ. Microbiol.* 59:1149-1154 (1993)). The aforementioned proteins are shown below.

Protein	GenBank ID	GI Number	Organism
acs	AAC77039.1	1790505	<i>Escherichia coli</i>
acoE	AAA21945.1	141890	<i>Ralstonia eutropha</i>
acs1	ABC87079.1	86169671	<i>Methanothermobacter thermautotrophicus</i>
acs1	AAL23099.1	16422835	<i>Salmonella enterica</i>
ACS1	Q01574.2	257050994	<i>Saccharomyces cerevisiae</i>
AF1211	NP_070039.1	11498810	<i>Archaeoglobus fulgidus</i>
AF1983	NP_070807.1	11499565	<i>Archaeoglobus fulgidus</i>
scs	YP_135572.1	55377722	<i>Haloarcula marismortui</i>
PAE3250	NP_560604.1	18313937	<i>Pyrobaculum aerophilum</i> str. IM2

-continued

Protein	GenBank ID	GI Number	Organism
sucC	NP_415256.1	16128703	<i>Escherichia coli</i>
sucD	AAC73823.1	1786949	<i>Escherichia coli</i>
paaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>

The acylation of acetate to acetyl-CoA can also be catalyzed by CoA transferase enzymes (FIG. 3B). Numerous enzymes employ acetate as the CoA acceptor, resulting in the formation of acetyl-CoA. An exemplary CoA transferase is acetoacetyl-CoA transferase, encoded by the *E. coli* *atoA* (alpha subunit) and *atoD* (beta subunit) genes (Korolev et al., *Acta Crystallogr. D. Biol. Crystallogr.* 58:2116-2121 (2002); Vanderwinkel et al., 33:902-908 (1968)). This enzyme has a broad substrate range (Sramek et al., *Arch. Biochem. Biophys.* 171:14-26 (1975)) and has been shown to transfer the CoA moiety to acetate from a variety of branched and linear acyl-CoA substrates, including isobutyrate (Matthies et al., *Appl. Environ. Microbiol.* 58:1435-1439 (1992)), valerate (Vanderwinkel et al., *Biochem. Biophys. Res. Commun.* 33:902-908 (1968)) and butanoate (Vanderwinkel et al., *Biochem. Biophys. Res. Commun.* 33:902-908 (1968)). Similar enzymes exist in *Corynebacterium glutamicum* ATCC 13032 (Duncan et al., 68:5186-5190 (2002)), *Clostridium acetobutylicum* (Cary et al., *Appl. Environ. Microbiol.* 56:1576-1583 (1990); Wiesenbom et al., *Appl. Environ. Microbiol.* 55:323-329 (1989)), and *Clostridium saccharoperbutylacetonicum* (Kosaka et al., *Biosci. Biotechnol. Biochem.* 71:58-68 (2007)).

Gene	GI #	Accession No.	Organism
atoA	2492994	P76459.1	<i>Escherichia coli</i>
atoD	2492990	P76458.1	<i>Escherichia coli</i>
actA	62391407	YP_226809.1	<i>Corynebacterium glutamicum</i>
cg0592	62389399	YP_224801.1	<i>Corynebacterium glutamicum</i>
ctfA	15004866	NP_149326.1	<i>Clostridium acetobutylicum</i>
ctfB	15004867	NP_149327.1	<i>Clostridium acetobutylicum</i>
ctfA	31075384	AAP42564.1	<i>Clostridium saccharoperbutylacetonicum</i>
ctfB	31075385	AAP42565.1	<i>Clostridium saccharoperbutylacetonicum</i>

Acetate kinase (EC 2.7.2.1) can catalyze the reversible ATP-dependent phosphorylation of acetate to acetylphosphate (FIG. 3C). Exemplary acetate kinase enzymes have been characterized in many organisms including *E. coli*, *Clostridium acetobutylicum* and *Methanosarcina thermophila* (Ingram-Smith et al., *J. Bacteriol.* 187:2386-2394 (2005); Fox and Roseman, *J. Biol. Chem.* 261:13487-13497 (1986); Winzer et al., *Microbiology* 143 (Pt 10):3279-3286 (1997)). Acetate kinase activity has also been demonstrated in the gene product of *E. coli* *purT* (Marolewski et al., *Biochemistry* 33:2531-2537 (1994)). Some butyrate kinase enzymes (EC 2.7.2.7), for example *buk1* and *buk2* from *Clostridium acetobutylicum*, also accept acetate as a substrate (Hartmanis, M. G., *J. Biol. Chem.* 262:617-621 (1987)). Homologs exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*.

Protein	GenBank ID	GI Number	Organism
ackA	NP_416799.1	16130231	<i>Escherichia coli</i>
Ack	AAB18301.1	1491790	<i>Clostridium acetobutylicum</i>
Ack	AAA72042.1	349834	<i>Methanosarcina thermophila</i>

-continued

Protein	GenBank ID	GI Number	Organism
purT	AAC74919.1	1788155	<i>Escherichia coli</i>
buk1	NP_349675	15896326	<i>Clostridium acetobutylicum</i>
buk2	Q97III	20137415	<i>Clostridium acetobutylicum</i>
ackA	NP_461279.1	16765664	<i>Salmonella typhimurium</i>
ACK1	XP_001694505.1	159472745	<i>Chlamydomonas reinhardtii</i>
ACK2	XP_001691682.1	159466992	<i>Chlamydomonas reinhardtii</i>

The formation of acetyl-CoA from acetyl-phosphate can be catalyzed by phosphotransacetylase (EC 2.3.1.8) (FIG. 3D). The pta gene from *E. coli* encodes an enzyme that reversibly converts acetyl-CoA into acetyl-phosphate (Suzuki, T., *Biochim. Biophys. Acta* 191:559-569 (1969)). Additional acetyltransferase enzymes have been characterized in *Bacillus subtilis* (Rado and Hoch, *Biochim. Biophys. Acta* 321:114-125 (1973), *Clostridium kluyveri* (Stadtman, E., *Methods Enzymol.* 1:5896-599 (1955), and *Thermotoga maritima* (Bock et al., *J. Bacteriol.* 181:1861-1867 (1999)). This reaction can also be catalyzed by some phosphotransbutyrylase enzymes (EC 2.3.1.19), including the ptb gene products from *Clostridium acetobutylicum* (Wiesenbom et al., *App. Environ. Microbiol.* 55:317-322 (1989); Walter et al., *Gene* 134:107-111(1993)). Additional ptb genes are found in butyrate-producing bacterium L2-50 (Louis et al., *J. Bacteriol.* 186:2099-2106 (2004) and *Bacillus megaterium* (Vazquez et al., *Curr. Microbiol.* 42:345-349 (2001). Homologs to the *E. coli* pta gene exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*.

Protein	GenBank ID	GI Number	Organism
Pta	NP_416800.1	71152910	<i>Escherichia coli</i>
Pta	P39646	730415	<i>Bacillus subtilis</i>
Pta	ASN801	146346896	<i>Clostridium kluyveri</i>
Pta	Q9X0L4	6685776	<i>Thermotoga maritima</i>
Ptb	NP_349676	34540484	<i>Clostridium acetobutylicum</i>
Ptb	AAR19757.1	38425288	butyrate-producing bacterium L2-50
Ptb	CAC07932.1	10046659	<i>Bacillus megaterium</i>
Pta	NP_461280.1	16765665	<i>Salmonella enterica</i> subsp. <i>enterica</i> serovar <i>Typhimurium</i> str. LT2
PAT2	XP_001694504.1	159472743	<i>Chlamydomonas reinhardtii</i>
PAT1	XP_001691787.1	159467202	<i>Chlamydomonas reinhardtii</i>

Pyruvate decarboxylase (PDC) is a key enzyme in alcoholic fermentation, catalyzing the decarboxylation of pyruvate to acetaldehyde (FIG. 3E). The PDC1 enzyme from *Saccharomyces cerevisiae* has been extensively studied (Kiltenberg-Jabs et al., *Eur.J.Biochem.* 268:1698-1704 (2001); Li et al., *Biochemistry.* 38:10004-10012 (1999); ter Schure et al., *Appl.Environ.Microbiol.* 64:1303-1307 (1998)). Other well-characterized PDC enzymes are found in *Zymomonas mobilis* (Siegert et al., *Protein Eng Des Sel* 18:345-357 (2005)), *Acetobacter pasteurians* (Chandra et al., 176:443-451(2001)) and *Kluyveromyces lactis* (Krieger et al., 269:3256-3263 (2002)). The PDC1 and PDC5 enzymes of *Saccharomyces cerevisiae* are subject to positive transcriptional regulation by PDC2 (Hohmann et al, *Mol Gen Genet* 241: 657-66 (1993)). Pyruvate decarboxylase activity is also possessed by a protein encoded by CTRL_03826 (GI: 255729208) in *Candida tropicalis*, PDC1 (GI number: 1226007) in *Kluyveromyces lactis*, YALIOD10131g (GI: 50550349) in *Yarrowia lipolytica*, PAS_chr3_0188 (GI: 254570575) in *Pichia pastoris*, pyruvate decarboxylase (GI:

GI:159883897) in *Schizosaccharomyces pombe*, ANI_1_1024084 (GI:145241548), ANI_1_796114 (GI: 317034487), ANI_1_936024 (GI:317026934) and ANI_1_2276014 (GI:317025935) in *Aspergillus niger*.

Protein	GenBank ID	GI Number	Organism
pdc	P06672.1	118391	<i>Zymomonas mobilis</i>
pdc1	P06169	30923172	<i>Saccharomyces cerevisiae</i>
Pdc2	NP_010366.1	6320286	<i>Saccharomyces cerevisiae</i>
Pdc5	NP_013235.1	6323163	<i>Saccharomyces cerevisiae</i>
CTR_03826	XP_002549529	255729208	<i>Candida tropicalis</i> .
CUC29670.1:	CAA90807	159883897	<i>Schizosaccharomyces pombe</i>
585597.587312			
YALIOD10131g	XP_502647	50550349	<i>Yarrowia lipolytica</i>
PAS_chr3_0188	XP_002492397	254570575	<i>Pichia pastoris</i>
15 pdc	Q8L388	20385191	<i>Acetobacter pasteurians</i>
pdc1	Q12629	52788279	<i>Kluyveromyces lactis</i>
ANI_1_1024084	XP_001393420	145241548	<i>Aspergillus niger</i>
ANI_1_796114	XP_001399817	317026934	<i>Aspergillus niger</i>
ANI_1_936024	XP_001396467	317034487	<i>Aspergillus niger</i>
20 ANI_1_2276014	XP_001388598	317025935	<i>Aspergillus niger</i>

Aldehyde dehydrogenase enzymes in EC class 1.2.1 catalyze the oxidation of acetaldehyde to acetate (FIG. 3F). Exemplary genes encoding this activity were described above. The oxidation of acetaldehyde to acetate can also be catalyzed by an aldehyde oxidase with acetaldehyde oxidase activity. Such enzymes can convert acetaldehyde, water and O₂ to acetate and hydrogen peroxide. Exemplary aldehyde oxidase enzymes that have been shown to catalyze this transformation can be found in *Bos taurus* and *Mus musculus* (Garattini et al., *Cell Mol Life Sci* 65:109-48 (2008); Cabre et al., *Biochem Soc Trans* 15:882-3 (1987)). Additional aldehyde oxidase gene candidates include the two flavin- and molybdenum-containing aldehyde oxidases of *Zea mays*, encoded by zmAO-1 and zmAO-2 (Sekimoto et al., *J Biol Chem* 272:15280-85 (1997)).

Gene	GenBank Accession No.	GI No.	Organism
zmAO-1	NP_001105308.1	162458742	<i>Zea mays</i>
zmAO-2	BAA23227.1	2589164	<i>Zea mays</i>
Aox1	O54754.2	20978408	<i>Mus musculus</i>
XDH	DAA24801.1	296482686	<i>Bos taurus</i>

Pyruvate oxidase (acetyl-phosphate forming) can catalyze the conversion of pyruvate, oxygen and phosphate to acetyl-phosphate and hydrogen peroxide (FIG. 3G). This type of pyruvate oxidase is soluble and requires the cofactors thiamin diphosphate and flavin adenine dinucleotide (FAD). Acetyl-phosphate forming pyruvate oxidase enzymes can be found in lactic acid bacteria *Lactobacillus delbrueckii* and *Lactobacillus plantarum* (Lorquet et al., *J Bacteriol* 186: 3749-3759 (2004); Hager et al., *Fed Proc* 13:734-38 (1954)). A crystal structure of the *L. plantarum* enzyme has been solved (Muller et al., (1994)). In *Streptococcus sanguinis* and *Streptococcus pneumoniae*, acetyl-phosphate forming pyruvate oxidase enzymes are encoded by the spxB gene (Spellerberg et al., *Mol Micro* 19:803-13 (1996); Ramos-Montanez et al., *Mol Micro* 67:729-46 (2008)). The SpXR was shown to positively regulate the transcription of spxB in *S. pneumoniae* (Ramos-Montanez et al., supra). A similar regulator in *S. sanguinis* was identified by sequence homology. Introduction or modification of catalase activity can reduce accumulation of the hydrogen peroxide product.

Gene	GenBank Accession No.	GI No.	Organism
poxB	NP_786788.1	28379896	<i>Lactobacillus plantarum</i>
spxB	L39074.1	1161269	<i>Streptococcus pneumoniae</i>
Spd_0969 (spxR)	YP_816445.1	116517139	<i>Streptococcus pneumoniae</i>
spxB	ZP_07887723.1	315612812	<i>Streptococcus sanguinis</i>
spxR	ZP_07887944.1	315613033	<i>Streptococcus sanguinis</i>

The pyruvate dehydrogenase (PDH) complex catalyzes the conversion of pyruvate to acetyl-CoA (FIG. 3H). The *E. coli* PDH complex is encoded by the genes aceEF and lpdA. Enzyme engineering efforts have improved the *E. coli* PDH enzyme activity under anaerobic conditions (Kim et al., *J. Bacteriol.* 190:3851-3858 (2008); Kim et al., *Appl. Environ. Microbiol.* 73:1766-1771 (2007); Zhou et al., *Biotechnol. Lett.* 30:335-342 (2008)). In contrast to the *E. coli* PDH, the *B. subtilis* complex is active and required for growth under anaerobic conditions (Nakano et al., 179:6749-6755 (1997)). The *Klebsiella pneumoniae* PDH, characterized during growth on glycerol, is also active under anaerobic conditions (Menzel et al., 56:135-142 (1997)). Crystal structures of the enzyme complex from bovine kidney (Zhou et al., 98:14802-14807 (2001)) and the E2 catalytic domain from *Azotobacter vinelandii* are available (Mattevi et al., *Science*. 255:1544-1550 (1992)). Some mammalian PDH enzymes complexes can react on alternate substrates such as 2-oxobutanoate. Comparative kinetics of *Rattus norvegicus* PDH and BCKAD indicate that BCKAD has higher activity on 2-oxobutanoate as a substrate (Paxton et al., *Biochem. J.* 234:295-303 (1986)). The *S. cerevisiae* PDH complex can consist of an E2 (LAT1) core that binds E1 (PDA1, PDB1), E3 (LPD1), and Protein X (PDX1) components (Pronk et al., *Yeast* 12:1607-1633 (1996)). The PDH complex of *S. cerevisiae* is regulated by phosphorylation of E1 involving PKP1 (PDH kinase I), PTCS (PDH phosphatase I), PKP2 and PTC6. Modification of these regulators may also enhance PDH activity. Coexpression of lipoyl ligase (Lp/A of *E. coli* and AIM22 in *S. cerevisiae*) with PDH in the cytosol may be necessary for activating the PDH enzyme complex. Increasing the supply of cytosolic lipoate, either by modifying a metabolic pathway or media supplementation with lipoate, may also improve PDH activity.

Gene	Accession No.	GI Number	Organism
aceE	NP_414656.1	16128107	<i>Escherichia coli</i>
aceF	NP_414657.1	16128108	<i>Escherichia coli</i>
lpd	NP_414658.1	16128109	<i>Escherichia coli</i>
lplA	NP_418803.1	16132203	<i>Escherichia coli</i>
pdhA	P21881.1	3123238	<i>Bacillus subtilis</i>
pdhB	P21882.1	129068	<i>Bacillus subtilis</i>
pdhC	P21883.2	129054	<i>Bacillus subtilis</i>
pdhD	P21880.1	118672	<i>Bacillus subtilis</i>
aceE	YP_001333808.1	152968699	<i>Klebsiella pneumoniae</i>
aceF	YP_001333809.1	152968700	<i>Klebsiella pneumoniae</i>
lpdA	YP_001333810.1	152968701	<i>Klebsiella pneumoniae</i>
PdhA1	NP_001004072.2	124430510	<i>Rattus norvegicus</i>
PdhA2	NP_446446.1	16758900	<i>Rattus norvegicus</i>
Dlat	NP_112287.1	78365255	<i>Rattus norvegicus</i>
Dld	NP_955417.1	40786469	<i>Rattus norvegicus</i>
LAT1	NP_014328	6324258	<i>Saccharomyces cerevisiae</i>
PDA1	NP_011105	37362644	<i>Saccharomyces cerevisiae</i>
PDB1	NP_009780	6319698	<i>Saccharomyces cerevisiae</i>
LPD1	NP_116635	14318501	<i>Saccharomyces cerevisiae</i>
PDX1	NP_011709	6321632	<i>Saccharomyces cerevisiae</i>
AIM22	NP_012489.2	83578101	<i>Saccharomyces cerevisiae</i>

As an alternative to the large multienzyme PDH complexes described above, some organisms utilize enzymes in the 2-ketoacid oxidoreductase family (OFOR) to catalyze acylating oxidative decarboxylation of 2-keto-acids. Unlike the PDH complexes, PFOR enzymes contain iron-sulfur clusters, utilize different cofactors and use fenedoxin or flavodoxin as electron acceptors in lieu of NAD(P)H. Pyruvate ferredoxin oxidoreductase (PFOR) can catalyze the oxidation of pyruvate to form acetyl-CoA (FIG. 3H). The PFOR from *Desulfovibrio africanus* has been cloned and expressed in *E. coli* resulting in an active recombinant enzyme that was stable for several days in the presence of oxygen (Pieulle et al., *J. Bacteriol.* 179:5684-5692 (1997)). Oxygen stability is relatively uncommon in PFORs and is believed to be conferred by a 60 residue extension in the polypeptide chain of the *D. africanus* enzyme. The *M. thermoacetica* PFOR is also well characterized (Menon et al., *Biochemistry* 36:8484-8494 (1997)) and was even shown to have high activity in the direction of pyruvate synthesis during autotrophic growth (Furdui et al., *J. Biol. Chem.* 275:28494-28499 (2000)). Further, *E. coli* possesses an uncharacterized open reading frame, ydbK that encodes a protein that is 51% identical to the *M. thermoacetica* PFOR. Evidence for pyruvate oxidoreductase activity in *E. coli* has been described (Blaschkowski et al., *Eur. J. Biochem.* 123:563-569 (1982)). Several additional PFOR enzymes are described in Ragsdale, *Chem. Rev.* 103:2333-2346 (2003). Finally, flavodoxin reductases (e.g., fqrB from *Helicobacter pylori* or *Campylobacter jejuni* (St Maurice et al., *J. Bacteriol.* 189:4764-4773 (2007))) or Rnf-type proteins (Seedorf et al., *Proc. Natl. Acad. Sci. U.S.A.* 105:2128-2133 (2008); Herrmann et al., *J. Bacteriol.* 190:784-791 (2008)) provide a means to generate NADH or NADPH from the reduced ferredoxin generated by PFOR. These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
Por	CAA70873.1	1770208	<i>Desulfovibrio africanus</i>
Por	YP_428946.1	83588937	<i>Moorella thermoacetica</i>
ydbK	NP_415896.1	16129339	<i>Escherichia coli</i>
fqrB	NP_207955.1	15645778	<i>Helicobacter pylori</i>
fqrB	YP_001482096.1	157414840	<i>Campylobacter jejuni</i>
RnfC	EDK33306.1	146346770	<i>Clostridium kluyveri</i>
RnfD	EDK33307.1	146346771	<i>Clostridium kluyveri</i>
RnfE	EDK33308.1	146346772	<i>Clostridium kluyveri</i>
RnfF	EDK33309.1	146346773	<i>Clostridium kluyveri</i>
RnfA	EDK33310.1	146346774	<i>Clostridium kluyveri</i>
RnfB	EDK33311.1	146346775	<i>Clostridium kluyveri</i>

Pyruvate formate-lyase (PFL, EC 2.3.1.54) (FIG. 3H), encoded by pflB in *E. coli*, can convert pyruvate into acetyl-CoA and formate. The activity of PFL can be enhanced by an activating enzyme encoded by pflA (Knappe et al., *Proc. Natl. Acad. Sci. U.S.A.* 81:1332-1335 (1984); Wong et al., *Biochemistry* 32:14102-14110 (1993)). Keto-acid formate-lyase (EC 2.3.1.-), also known as 2-ketobutyrate formate-lyase (KFL) and pyruvate formate-lyase 4, is the gene product of tdcE in *E. coli*. This enzyme catalyzes the conversion of 2-ketobutyrate to propionyl-CoA and formate during anaerobic threonine degradation, and can also substitute for pyruvate formate-lyase in anaerobic catabolism (Simanshu et al., *J. Biosci.* 32:1195-1206 (2007)). The enzyme is oxygen-sensitive and, like NIB, can require post-translational modification by PFL-AE to activate a glycyl radical in the active site (Hesslinger et al., *Mol. Microbiol.* 27:477-492 (1998)). A pyruvate formate-lyase from *Archaeoglobus fulgidus* encoded by pflD has been cloned,

expressed in *E. coli* and characterized (Lehtio et al., *Protein Eng Des Sel* 17:545-552 (2004)). The crystal structures of the *A. fulgidus* and *E. coli* enzymes have been resolved (Lehtio et al., *J Mol. Biol.* 357:221-235 (2006); Leppanen et al., *Structure*. 7:733-744 (1999)). Additional PFL and PFL-AE candidates are found in *Lactococcus lactis* (Melchiorson et al., *Appl Microbiol Biotechnol* 58:338-344 (2002)), and *Streptococcus mutans* (Takahashi-Abbe et al., *Oral. Microbiol Immunol.* 18:293-297 (2003)), *Chlamydomonas reinhardtii* (Hemschemeier et al., *Eukaryot. Cell* 7:518-526

Sfum_2703 through Sfum_2706 in *Syntrophobacter fumaroxidans* (de Bok et al., *Eur J Biochem.* 270:2476-2485 (2003)); Reda et al., *PNAS* 105:10654-10658 (2008)). A similar set of genes presumed to carry out the same function are encoded by CHY_0731, CHY_0732, and CHY_0733 in *C. hydrogenoformans* (Wu et al., *PLoS Genet* 1:e65 (2005)). Formate dehydrogenases are also found many additional organisms including *C. carboxidivorans* P7, *Bacillus methanolicus*, *Burkholderia stabilis*, *Moorella thermoacetica* ATCC 39073, *Candida boidinii*, *Candida methylica*, and *Saccharomyces cerevisiae* S288c.

Protein	GenBank ID	GI Number	Organism
Moth_2312	YP_431142	148283121	<i>Moorella thermoacetica</i>
Moth_2314	YP_431144	83591135	<i>Moorella thermoacetica</i>
Sfum_2703	YP_846816.1	116750129	<i>Syntrophobacter fumaroxidans</i>
Sfum_2704	YP_846817.1	116750130	<i>Syntrophobacter fumaroxidans</i>
Sfum_2705	YP_846818.1	116750131	<i>Syntrophobacter fumaroxidans</i>
Sfum_2706	YP_846819.1	116750132	<i>Syntrophobacter fumaroxidans</i>
CHY_0731	YP_359585.1	78044572	<i>Carboxydotherrmus hydrogenoformans</i>
CHY_0732	YP_359586.1	78044500	<i>Carboxydotherrmus hydrogenoformans</i>
CHY_0733	YP_359587.1	78044647	<i>Carboxydotherrmus hydrogenoformans</i>
CcarbDRAFT_0901	ZP_05390901.1	255523938	<i>Clostridium carboxidivorans</i> P7
CcarbDRAFT_4380	ZP_05394380.1	255527512	<i>Clostridium carboxidivorans</i> P7
fdhA, MGA3_06625	EIJ82879.1	387590560	<i>Bacillus methanolicus</i> MGA3
fdhA, PB1_11719	ZP_10131761.1	387929084	<i>Bacillus methanolicus</i> PB1
fdhD, MGA3_06630	EIJ82880.1	387590561	<i>Bacillus methanolicus</i> MGA3
fdhD, PB1_11724	ZP_10131762.1	387929085	<i>Bacillus methanolicus</i> PB1
fdh	ACF35003.	194220249	<i>Burkholderia stabilis</i>
FDH1	AAC49766.1	2276465	<i>Candida boidinii</i>
fdh	CAA57036.1	1181204	<i>Candida methylica</i>
FDH2	POCF35.1	294956522	<i>Saccharomyces cerevisiae</i> S288c
FDH1	NP_015033.1	6324964	<i>Saccharomyces cerevisiae</i> S288c

(2008b); Atteia et al., *J. Biol. Chem.* 281:9909-9918 (2006)) and *Clostridium pasteurianum* (Weidner et al., *J Bacteriol.* 178:2440-2444 (1996)).

Protein	GenBank ID	GI Number	Organism
pflB	NP_415423	16128870	<i>Escherichia coli</i>
pflA	NP_415422.1	16128869	<i>Escherichia coli</i>
tdcE	AAT48170.1	48994926	<i>Escherichia coli</i>
pflD	NP_070278.1	11499044	<i>Archaeoglobus fulgidus</i>
pfl	CAA03993	2407931	<i>Lactococcus lactis</i>
pfl	BAA09085	1129082	<i>Streptococcus mutans</i>
PFL1	XP_001689719.1	159462978	<i>Chlamydomonas reinhardtii</i>
pflAl	XP_001700657.1	159485246	<i>Chlamydomonas reinhardtii</i>
pfl	Q46266.1	2500058	<i>Clostridium pasteurianum</i>
act	CAA63749.1	1072362	<i>Clostridium pasteurianum</i>

If a pyruvate formate lyase is utilized to convert pyruvate to acetyl-CoA, coexpression of a formate dehydrogenase or formate hydrogen lyase enzyme will convert formate to carbon dioxide. Formate dehydrogenase (FDH) catalyzes the reversible transfer of electrons from formate to an acceptor. Enzymes with FDH activity utilize various electron carriers such as, for example, NADH (EC 1.2.1.2), NADPH (EC 1.2.1.43), quinols (EC 1.1.5.6), cytochromes (EC 1.2.2.3) and hydrogenases (EC 1.1.99.33). FDH enzymes have been characterized from *Moorella thermoacetica* (Andreesen and Ljungdahl, *J Bacteriol* 116:867-873 (1973); Li et al., *J Bacteriol* 92:405-412 (1966); Yamamoto et al., *J Biol Chem.* 258:1826-1832 (1983). The loci, Moth_2312 is responsible for encoding the alpha subunit of formate dehydrogenase while the beta subunit is encoded by Moth_2314 (Pierce et al., *Environ Microbiol* (2008)). Another set of genes encoding formate dehydrogenase activity with a propensity for CO₂ reduction is encoded by

Alternately, a formate hydrogen lyase enzyme can be employed to convert formate to carbon dioxide and hydrogen. An exemplary formate hydrogen lyase enzyme can be found in *Escherichia coli*. The *E. coli* formate hydrogen lyase consists of hydrogenase 3 and formate dehydrogenase-H (Maeda et al., *Appl Microbiol Biotechnol* 77:879-890 (2007)). It is activated by the gene product of fhfA. (Maeda et al., *Appl Microbiol Biotechnol* 77:879-890 (2007)). The addition of the trace elements, selenium, nickel and molybdenum, to a fermentation broth has been shown to enhance formate hydrogen lyase activity (Soini et al., *Microb. Cell Fact.* 7:26 (2008)). Various hydrogenase 3, formate dehydrogenase and transcriptional activator genes are shown below. A formate hydrogen lyase enzyme also exists in the hyperthermophilic archaeon, *Thermococcus litoralis* (Takacs et al., *BMC. Microbiol* 8:88 (2008)). Additional formate hydrogen lyase systems have been found in *Salmonella typhimurium*, *Klebsiella pneumoniae*, *Rhodospirillum rubrum*, *Methanobacterium formicicum* (Vardar-Schara et al., *Microbial Biotechnology* 1:107-125 (2008)).

Protein	GenBank ID	GI number	Organism
hycA	NP_417205	16130632	<i>Escherichia coli</i> K-12MG1655
hycB	NP_417204	16130631	<i>Escherichia coli</i> K-12MG1655
hycC	NP_417203	16130630	<i>Escherichia coli</i> K-12MG1655
hycD	NP_417202	16130629	<i>Escherichia coli</i> K-12MG1655
hycE	NP_417201	16130628	<i>Escherichia coli</i> K-12MG1655
hycF	NP_417200	16130627	<i>Escherichia coli</i> K-12MG1655
hycG	NP_417199	16130626	<i>Escherichia coli</i> K-12MG1655
hycH	NP_417198	16130625	<i>Escherichia coli</i> K-12MG1655
hycI	NP_417197	16130624	<i>Escherichia coli</i> K-12MG1655
fdhF	NP_418503	16131905	<i>Escherichia coli</i> K-12MG1655
fhfA	NP_417211	16130638	<i>Escherichia coli</i> K-12MG1655

-continued

Protein	GenBank ID	GI number	Organism
mhyC	ABW05543	157954626	<i>Thermococcus litoralis</i>
mhyD	ABW05544	157954627	<i>Thermococcus litoralis</i>
mhyE	ABW05545	157954628	<i>Thermococcus litoralis</i>
myhF	ABW05546	157954629	<i>Thermococcus litoralis</i>
myhG	ABW05547	157954630	<i>Thermococcus litoralis</i>
myhH	ABW05548	157954631	<i>Thermococcus litoralis</i>
fdhA	AAB94932	2746736	<i>Thermococcus litoralis</i>
fdhB	AAB94931	157954625	<i>Thermococcus litoralis</i>

Pyruvate:NADP oxidoreductase (PNO) catalyzes the conversion of pyruvate to acetyl-CoA. This enzyme is encoded by a single gene and the active enzyme is a homodimer, in contrast to the multi-subunit PDH enzyme complexes described above. The enzyme from *Euglena gracilis* is stabilized by its cofactor, thiamin pyrophosphate (Nakazawa et al, *Arch Biochem Biophys* 411:183-8 (2003)). The mitochondrial targeting sequence of this enzyme should be removed for expression in the cytosol. The PNO protein of *E. gracilis* protein and other NADP-dependant pyruvate:NADP⁺ oxidoreductase enzymes are listed in the table below.

Protein	GenBank ID	GI Number	Organism
PNO	Q94IN5.1	33112418	<i>Euglena gracilis</i>
cgd_690	XP_625673.1	66356990	<i>Cryptosporidium parvum</i> <i>Iowa II</i>
TPP_PNO	XP_002765111.11	294867463	<i>Perkinsus marinus</i> ATCC 50983

The NAD(P)⁺ dependent oxidation of acetaldehyde to acetyl-CoA (FIG. 31) can be catalyzed by an acylating acetaldehyde dehydrogenase (EC 1.2.1.10). Acylating acetaldehyde dehydrogenase enzymes of *E. coli* are encoded by adhE, eutE, and mhpF (Ferrandez et al, *J Bacteriol* 1179: 2573-81 (1997)). The *Pseudomonas* sp. CF600 enzyme, encoded by dmpF, participates in meta-cleavage pathways and forms a complex with 4-hydroxy-2-oxovalerate aldolase (Shingler et al, *J Bacteriol* 174:711-24 (1992)). Solventogenic organisms such as *Clostridium acetobutylicum* encode bifunctional enzymes with alcohol dehydrogenase and acetaldehyde dehydrogenase activities. The bifunctional *C. acetobutylicum* enzymes are encoded by bdh I and adhE2 (Walter, et al., *J. Bacteriol.* 174:7149-7158 (1992); Fontaine et al., *J. Bacteriol.* 184:821-830 (2002)). Yet another candidate for acylating acetaldehyde dehydrogenase is the ald gene from *Clostridium beijerinckii* (Toth, *Appl. Environ. Microbiol.* 65:4973-4980 (1999)). This gene is very similar to the eutE acetaldehyde dehydrogenase genes of *Salmonella typhimurium* and *E. coli* (Toth, *Appl. Environ. Microbiol.* 65:4973-4980 (1999)).

Protein	Gen Bank ID	GI Number	Organism
adhE	NP_415757.1	16129202	<i>Escherichia coli</i>
mhpF	NP_414885.1	16128336	<i>Escherichia coli</i>
dmpF	CAA43226.1	45683	<i>Pseudomonas</i> sp. CF600
adhE2	AAK09379.1	12958626	<i>Clostridium acetobutylicum</i>
bdh I	NP_349892.1	15896543	<i>Clostridium acetobutylicum</i>
Ald	AAT66436	49473535	<i>Clostridium beijerinckii</i>
eutE	NP_416950	16130380	<i>Escherichia coli</i>
eutE	AAA80209	687645	<i>Salmonella typhimurium</i>

Threonine aldolase (EC 4.1.2.5) catalyzes the cleavage of threonine to glycine and acetaldehyde (FIG. 3J). The *Sac-*

charomyces cerevisiae and *Candida albicans* enzymes are encoded by GLY1 (Liu et al, *Eur J Biochem* 245:289-93 (1997); McNeil et al, *Yeast* 16:167-75 (2000)). The ItaE and glyA gene products of *E. coli* also encode enzymes with this activity (Liu et al, *Eur J Biochem* 255:220-6 (1998)).

Protein	GenBank ID	GI Number	Organism
GLY1	NP_010868.1	6320789	<i>Saccharomyces cerevisiae</i>
GLY1	AAB64198.1	2282060	<i>Candida albicans</i>
ItaE	AAC73957.1	1787095	<i>Escherichia coli</i>
glyA	AAC75604.1	1788902	<i>Escherichia coli</i>

Example VI

Pathways for Producing Acetyl-CoA from PEP and Pyruvate

Pathways for the conversion of cytosolic phosphoenolpyruvate (PEP) and pyruvate to cytosolic acetyl-CoA can also enable deployment of a cytosolic fatty alcohol, fatty aldehyde or fatty acid production pathway from acetyl-CoA. FIG. 4 shows numerous pathways for converting PEP and pyruvate to acetyl-CoA.

The conversion of PEP to oxaloacetate is catalyzed in one, two or three enzymatic steps. Oxaloacetate is further converted to acetyl-CoA via malonate semialdehyde or malonyl-CoA intermediates. In one pathway, PEP carboxylase or PEP carboxykinase converts PEP to oxaloacetate (step A); oxaloacetate decarboxylase converts the oxaloacetate to malonate (step B); and malonate semialdehyde dehydrogenase (acylating) converts the malonate semialdehyde to acetyl-CoA (step C). In another pathway pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); pyruvate carboxylase converts the pyruvate to (step H); oxaloacetate decarboxylase converts the oxaloacetate to malonate (step B); and malonate semialdehyde dehydrogenase (acylating) converts the malonate semialdehyde to acetyl-CoA (step C). In another pathway pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); malic enzyme converts the pyruvate to malate (step L); malate dehydrogenase or oxidoreductase converts the malate to oxaloacetate (step M); oxaloacetate decarboxylase converts the oxaloacetate to malonate (step B); and malonate semialdehyde dehydrogenase (acylating) converts the malonate semialdehyde to acetyl-CoA (step C). In another pathway, PEP carboxylase or PEP carboxykinase converts PEP to oxaloacetate (step A); oxaloacetate decarboxylase converts the oxaloacetate to malonate semialdehyde (step B); malonyl-CoA reductase converts the malonate semialdehyde to malonyl-CoA (step G); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); pyruvate carboxylase converts the pyruvate to oxaloacetate (step H); (oxaloacetate decarboxylase converts the oxaloacetate to malonate semialdehyde (step B); malonyl-CoA reductase converts the malonate semialdehyde to malonyl-CoA (step G); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); malic enzyme converts the pyruvate to malate (step L); malate dehydrogenase or oxidoreductase converts the malate to oxaloacetate (step M); oxaloacetate decarboxylase converts the oxaloacetate to malonate semialdehyde (step B); malonyl-CoA reductase converts the malonate semialdehyde to malonyl-CoA (step G); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D).

nyl-CoA reductase converts the malonate semialdehyde to malonyl-CoA (step G); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, PEP carboxylase or PEP carboxykinase converts PEP to oxaloacetate (step A); oxaloacetate decarboxylase converts the oxaloacetate to malonate semialdehyde (step B); malonate semialdehyde dehydrogenase converts the malonate semialdehyde to malonate (step J); malonyl-CoA synthetase or transferase converts the malonate to malonyl-CoA (step K); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); pyruvate carboxylase converts the pyruvate to oxaloacetate (step H); oxaloacetate decarboxylase converts the oxaloacetate to malonate semialdehyde (step B); malonate semialdehyde dehydrogenase converts the malonate semialdehyde to malonate (step J); malonyl-CoA synthetase or transferase converts the malonate to malonyl-CoA (step K); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); malic enzyme converts the pyruvate to malate (step L); malate dehydrogenase or oxidoreductase converts the malate to oxaloacetate (step M); oxaloacetate decarboxylase converts the oxaloacetate to malonate semialdehyde (step B); malonate semialdehyde dehydrogenase converts the malonate semialdehyde to malonate (step J); malonyl-CoA synthetase or transferase converts the malonate to malonyl-CoA (step K); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, PEP carboxylase or PEP carboxykinase converts PEP to oxaloacetate (step A); oxaloacetate dehydrogenase or oxaloacetate oxidoreductase converts the oxaloacetate to malonyl-CoA (step F); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); pyruvate carboxylase converts the pyruvate to oxaloacetate (step H); oxaloacetate dehydrogenase or oxaloacetate oxidoreductase converts the oxaloacetate to malonyl-CoA (step F); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D). In another pathway, pyruvate kinase or PEP phosphatase converts PEP to pyruvate (step N); malic enzyme converts the pyruvate to malate (step L); malate dehydrogenase or oxidoreductase converts the malate to oxaloacetate (step M); oxaloacetate dehydrogenase or oxaloacetate oxidoreductase converts the oxaloacetate to malonyl-CoA (step F); and malonyl-CoA decarboxylase converts the malonyl-CoA to acetyl-CoA (step D).

Enzymes candidates for the reactions shown in FIG. 4 are described below.

1.1.n.a	Oxidoreductase (alcohol to oxo)	M
1.1.1.d	Malic enzyme	L
1.2.1.a	Oxidoreductase (aldehyde to acid)	J
1.2.1.b	Oxidoreductase (acyl-CoA to aldehyde)	G
1.2.1.f	Oxidoreductase (decarboxylating acyl-CoA to aldehyde)	C
2.7.2.a	Kinase	N
2.8.3.a	CoA transferase	K
3.1.3.a	Phosphatase	N
4.1.1.a	Decarboxylase	A, B, D
6.2.1.a	CoA synthetase	K
6.4.1.a	Carboxylase	D, H

Enzyme candidates for several enzymes in FIG. 4 have been described elsewhere herein. These include acetyl-CoA carboxylase, acetoacetyl-CoA synthase, acetoacetyl-CoA

thiolase, malonyl-CoA reductase (also called malonate semialdehyde dehydrogenase (acylating), malate dehydrogenase. 1.1.n.a Oxidoreductase (Alcohol to Oxo)

Malate dehydrogenase or oxidoreductase catalyzes the oxidation of malate to oxaloacetate. Different carriers can act as electron acceptors for enzymes in this class. Malate dehydrogenase enzymes utilize NADP or NAD as electron acceptors. Malate dehydrogenase (Step M) enzyme candidates are described herein. Malate:quinone oxidoreductase enzymes (EC 1.1.5.4) are membrane-associated and utilize quinones, flavoproteins or vitamin K as electron acceptors. Malate:quinone oxidoreductase enzymes of *E. coli*, *Helicobacter pylori* and *Pseudomonas syringae* are encoded by mqo (Kather et al, J Bacteriol 182:3204-9 (2000); Mellgren et al, J Bacteriol 191:3132-42 (2009)). The Cg12001 gene of *C. glutamicum* also encodes an MQO enzyme (Mitsuhashi et al, Biosci Biotechnol Biochem 70:2803-6 (2006)).

Protein	GenBank ID	GI Number	Organism
mqo	NP_416714.1	16130147	<i>Escherichia coli</i>
mqo	NP_206886.1	15644716	<i>Helicobacter pylori</i>
mqo	NP_790970.1	28868351	<i>Pseudomonas syringae</i>
Cgl2001	NP_601207.1	19553205	<i>Corynebacterium glutamicum</i>

1.1.1.d Malic Enzyme

Malic enzyme (malate dehydrogenase) catalyzes the reversible oxidative carboxylation of pyruvate to malate. *E. coli* encodes two malic enzymes, MaeA and MaeB (Takeo, J. Biochem. 66:379-387 (1969)). Although malic enzyme is typically assumed to operate in the direction of pyruvate formation from malate, the NAD-dependent enzyme, encoded by maeA, has been demonstrated to operate in the carbon-fixing direction (Stols and Donnelly, Appl. Environ. Microbiol. 63(7) 2695-2701(1997)). A similar observation was made upon overexpressing the malic enzyme from *Ascaris suum* in *E. coli* (Stols et al., Appl. Biochem. Biotechnol. 63-65(1), 153-158 (1997)). The second *E. coli* malic enzyme, encoded by maeB, is NADP-dependent and also decarboxylates oxaloacetate and other alpha-keto acids (Iwakura et al., J. Biochem. 85(5):1355-65 (1979)). Another suitable enzyme candidate is me1 from *Zea mays* (Furumoto et al, Plant Cell Physiol 41:1200-1209 (2000)).

Protein	GenBank ID	GI Number	Organism
maeA	NP_415996	90111281	<i>Escherichia coli</i>
maeB	NP_416958	16130388	<i>Escherichia coli</i>
NAD-ME	P27443	126732	<i>Ascaris suum</i>
Me1	P16243.1	126737	<i>Zea mays</i>

1.2.1.a Oxidoreductase (Aldehyde to Acid)

The oxidation of malonate semialdehyde to malonate is catalyzed by malonate semialdehyde dehydrogenase (EC 1.2.1.15). This enzyme was characterized in *Pseudomonas aeruginosa* (Nakamura et al, Biochim Biophys Acta 50:147-52 (1961)). The NADP and NAD-dependent succinate semialdehyde dehydrogenase enzymes of *Euglena gracilis* accept malonate semialdehyde as substrates (Tokunaga et al, Biochem Biophys Acta 429:55-62 (1976)). Genes encoding these enzymes has not been identified to date. Aldehyde dehydrogenase enzymes from eukaryotic organisms such as *S. cerevisiae*, *C. albicans*, *Y. lipolytica* and *A. niger* typically have broad substrate specificity and are suitable candidates. These enzymes and other acid forming aldehyde dehydrogenase and aldehyde oxidase enzymes are described earlier

and listed in Tables 9 and 30. Additional MSA dehydrogenase enzyme candidates include NAD(P)⁺-dependent aldehyde dehydrogenase enzymes (EC 1.2.1.3). Two aldehyde dehydrogenases found in human liver, ALDH-1 and ALDH-2, have broad substrate ranges for a variety of aliphatic, aromatic and polycyclic aldehydes (Klyosov, *Biochemistry* 35:4457-4467 (1996a)). Active ALDH-2 has been efficiently expressed in *E. coli* using the GroEL proteins as chaperonins (Lee et al., *Biochem.Biophys.Res.Comm.* 298:216-224 (2002)). The rat mitochondrial aldehyde dehydrogenase also has a broad substrate range (Siew et al., *Arch.Biochem.Biophys.* 176:638-649 (1976)). The *E. coli* genes astD and aldH encode NAD⁺-dependent aldehyde dehydrogenases. AstD is active on succinic semialdehyde (Kuznetsova et al., *FEMS Microbiol Rev* 29:263-279 (2005)) and aldH is active on a broad range of aromatic and aliphatic substrates (Jo et al, *Appl Microbiol Biotechnol* 81:51-60 (2008)).

Gene	GenBank Accession No.	GI No.	Organism
astD	P76217.1	3913108	<i>Escherichia coli</i>
aldH	AAC74382.1	1787558	<i>Escherichia coli</i>
ALDH-2	P05091.2	118504	<i>Homo sapiens</i>
ALDH-2	NP_115792.1	14192933	<i>Rattus norvegicus</i>

1.2.1.f Oxidoreductase (Decarboxylating acyl-CoA to Aldehyde)

Malonate semialdehyde dehydrogenase (acetylating) (EC 1.2.1.18) catalyzes the oxidative decarboxylation of malonate semialdehyde to acetyl-CoA. Exemplary enzymes are encoded by ddcC of *Halomonas* sp. HINK1 (Todd et al, *Environ Microbiol* 12:237-43 (2010)) and lolA of *Lactobacillus casei* (Yebra et al, *AEM* 73:3850-8 (2007)). The DdcC enzyme has homologs in *A. niger* and *C. albicans*, shown in the table below. The malonate semialdehyde dehydrogenase enzyme in *Rattus norvegicus*, Mmsdh, also converts malonate semialdehyde to acetyl-CoA (U.S. Pat. No. 8,048, 624). A malonate semialdehyde dehydrogenase (acetylating) enzyme has also been characterized in *Pseudomonas fluorescens*, although the gene has not been identified to date (Hayaishi et al, *J Biol Chem* 236:781-90 (1961)). Methylmalonate semialdehyde dehydrogenase (acetylating) enzymes (EC 1.2.1.27) are also suitable candidates, as several enzymes in this class accept malonate semialdehyde as a substrate including Msdh of *Bacillus subtilis* (Stines-Chaumeil et al, *Biochem J* 395:107-15 (2006)) and the methylmalonate semialdehyde dehydrogenase of *R. norvegicus* (Kedishvii et al, *Methods Enzymol* 324:207-18 (2000)).

Protein	GenBank ID	GI Number	Organism
ddcC	ACV84070.1	258618587	<i>Halomonas</i> sp. HTNK1
ANI_1_1120014	XP_001389265.1	145229913	<i>Aspergillus niger</i>
ALD6	XP_710976.1	68490403	<i>Candida albicans</i>
YALI0C01859g	XP_501343.1	50547747	<i>Yarrowia lipolytica</i>
mmsA_1	YP_257876.1	70734236	<i>Pseudomonas fluorescens</i>
mmsA_2	YP_257884.1	70734244	<i>Pseudomonas fluorescens</i>
PA0130	NP_248820.1	15595328	<i>Pseudomonas aeruginosa</i>
Mmsdh	Q02253.1	400269	<i>Rattus norvegicus</i>
msdh	NP_391855.1	16081027	<i>Bacillus subtilis</i>
lolA	ABP57762.1	145309085	<i>Lactobacillus casei</i>

2.72.a Kinase

Pyruvate kinase (Step 10N), also known as phosphoenolpyruvate synthase (EC 2.7.9.2), converts pyruvate and ATP to PEP and AMP. This enzyme is encoded by the PYK1 (Burke et al., *J. Biol. Chem.* 258:2193-2201 (1983)) and PYK2 (Boles et al., *J. Bacteriol* 179:2987-2993 (1997)) genes in *S. cerevisiae*. In *E. coli*, this activity is catalyzed by the gene products of pykF and pykA. Selected homologs of the *S. cerevisiae* enzymes are also shown in the table below.

Protein	GenBank ID	GI Number	Organism
PYK1	NP_009362	6319279	<i>Saccharomyces cerevisiae</i>
PYK2	NP_014992	6324923	<i>Saccharomyces cerevisiae</i>
pykF	NP_416191.1	16129632	<i>Escherichia coli</i>
pykA	NP_416368.1	16129807	<i>Escherichia coli</i>
KLLA0F23397g	XP_456122.1	50312181	<i>Kluyveromyces lactis</i>
CaO19.3575	XP_714934.1	68482353	<i>Candida albicans</i>
CaO19.11059	XP_714997.1	68482226	<i>Candida albicans</i>
YALI0F09185p	XP_505195	210075987	<i>Yarrowia lipolytica</i>
ANI_1_1126064	XP_001391973	145238652	<i>Aspergillus niger</i>

2.83.a CoA Transferase

Activation of malonate to malonyl-CoA is catalyzed by a CoA transferase in EC class 2.8.3.a. Malonyl-CoA:acetate CoA transferase (EC 2.8.3.3) enzymes have been characterized in *Pseudomonas* species including *Pseudomonas fluorescens* and *Pseudomonas putida* (Takamura et al, *Biochem Int* 3:483-91 (1981); Hayaishi et al, *J Biol Chem* 215:125-36 (1955)). Genes associated with these enzymes have not been identified to date. A mitochondrial CoA transferase found in *Rattus norvegicus* liver also catalyzes this reaction and is able to utilize a range of CoA donors and acceptors (Deana et al, *Biochem Int* 26:767-73 (1992)). Several CoA transferase enzymes described herein can also be applied to catalyze step K of FIG. 4. These enzymes include acetyl-CoA transferase, 3-HB CoA transferase, acetoacetyl-CoA transferase, SCOT and other CoA transferases.

3.13.a Phosphatase

Phosphoenolpyruvate phosphatase (EC 3.1.3.60, FIG. 4, Step N) catalyzes the hydrolysis of PEP to pyruvate and phosphate. Numerous phosphatase enzymes catalyze this activity, including alkaline phosphatase (EC 3.1.3.1), acid phosphatase (EC 3.1.3.2), phosphoglycerate phosphatase (EC 3.1.3.20) and PEP phosphatase (EC 3.1.3.60). PEP phosphatase enzymes have been characterized in plants such as *Vigna radiate*, *Bruguiera sexangula* and *Brassica nigra*. The phytase from *Aspergillus fumigatus*, the acid phosphatase from *Homo sapiens* and the alkaline phosphatase of *E. coli* also catalyze the hydrolysis of PEP to pyruvate (Bragger et al, *Appl Microbiol Biotech* 63:383-9 (2004); Hayman et al, *Biochem J* 261:601-9 (1989); et al, *The Enzymes* 3rd Ed. 4:373-415 (1971)). Similar enzymes have been characterized in *Campylobacter jejuni* (van Mourik et al., *Microbiol.* 154:584-92 (2008)), *Saccharomyces cerevisiae* (Oshima et al., *Gene* 179:171-7 (1996)) and *Staphylococcus aureus* (Shah and Blobel, *J. Bacteriol.* 94:780-1 (1967)). Enzyme engineering and/or removal of targeting sequences may be required for alkaline phosphatase enzymes to function in the cytoplasm.

Protein	GenBank ID	GI Number	Organism
phyA	O00092.1	41017447	<i>Aspergillus fumigatus</i>
Acp5	P13686.3	56757583	<i>Homo sapiens</i>
phoA	NP_414917.2	49176017	<i>Escherichia coli</i>
phoX	ZP_01072054.1	86153851	<i>Campylobacter jejuni</i>

-continued

Protein	GenBank ID	GI Number	Organism
PHO8	AAA34871.1	172164	<i>Saccharomyces cerevisiae</i>
SaurJH1_2706	YP_001317815.1	150395140	<i>Staphylococcus aureus</i>

4.1.1.a Decarboxylase

Several reactions in FIG. 4 are catalyzed by decarboxylase enzymes in EC class 4.1.1, including oxaloacetate decarboxylase (Step B), malonyl-CoA decarboxylase (step D) and pyruvate carboxylase or carboxykinase (step A).

Carboxylation of phosphoenolpyruvate to oxaloacetate is catalyzed by phosphoenolpyruvate carboxylase (EC 4.1.1.31). Exemplary PEP carboxylase enzymes are encoded by ppc in *E. coli* (Kai et al., *Arch. Biochem. Biophys.* 414:170-179 (2003)), ppcA in *Methylobacterium extorquens* AM1 (Arps et al., *J. Bacteriol.* 175:3776-3783 (1993)), and ppc in *Corynebacterium glutamicum* (Eikmanns et al., *Mol. Gen. Genet.* 218:330-339 (1989)).

Protein	GenBank ID	GI Number	Organism
Ppc	NP_418391	16131794	<i>Escherichia coli</i>
ppcA	AAB58883	28572162	<i>Methylobacterium extorquens</i>
Ppc	ABB53270	80973080	<i>Corynebacterium glutamicum</i>

An alternative enzyme for carboxylating phosphoenolpyruvate to oxaloacetate is PEP carboxykinase (EC 4.1.1.32, 4.1.1.49), which simultaneously forms an ATP or GTP. In most organisms PEP carboxykinase serves a gluconeogenic function and converts oxaloacetate to PEP at the expense of one ATP. *S. cerevisiae* is one such organism whose native PEP carboxykinase, PCK1, serves a gluconeogenic role (Valdes-Hevia et al., *FEBS Lett.* 258:313-316 (1989)). *E. coli* is another such organism, as the role of PEP carboxykinase in producing oxaloacetate is believed to be minor when compared to PEP carboxylase (Kim et al., *Appl. Environ. Microbiol.* 70:1238-1241 (2004)). Nevertheless, activity of the native *E. coli* PEP carboxykinase from PEP towards oxaloacetate has been recently demonstrated in ppc mutants of *E. coli* K-12 (Kwon et al., *J. Microbiol. Biotechnol.* 16:1448-1452 (2006)). These strains exhibited no growth defects and had increased succinate production at high NaHCO₃ concentrations. Mutant strains of *E. coli* can adopt Pck as the dominant CO₂-fixing enzyme following adaptive evolution (Zhang et al. 2009). In some organisms, particularly rumen bacteria, PEP carboxykinase is quite efficient in producing oxaloacetate from PEP and generating ATP. Examples of PEP carboxykinase genes that have been cloned into *E. coli* include those from *Mannheimia succiniciproducens* (Lee et al., *Biotechnol. Bioprocess Eng* 7:95-99 (2002)), *Anaerobiospirillum succiniciproducens* (Laive-nieks et al., *Appl. Environ. Microbiol.* 63:2273-2280 (1997)), and *Actinobacillus succinogenes* (Kim et al. supra). The PEP carboxykinase enzyme encoded by *Haemophilus influenza* is effective at forming oxaloacetate from PEP. Another suitable candidate is the PEPCK enzyme from *Megathyrus maximus*, which has a low Km for CO₂, a substrate thought to be rate-limiting in the *E. coli* enzyme (Chen et al., *Plant Physiol* 128:160-164 (2002); Cotelesage et al., *Int.J Biochem.Cell Biol.* 39:1204-1210 (2007)). The kinetics of the GTP-dependent pepck gene product from *Cupriavidus necator* favor oxaloacetate formation (U.S. Pat. No. 8,048, 624 and Lea et al., *Amino Acids* 20:225-41(2001)).

Protein	GenBank ID	GI Number	Organism
PCK1	NP_013023	6322950	<i>Saccharomyces cerevisiae</i>
pck	NP_417862.1	16131280	<i>Escherichia coli</i>
pckA	YP_089485.1	52426348	<i>Mannheimia succiniciproducens</i>
pckA	O09460.1	3122621	<i>Anaerobiospirillum succiniciproducens</i>
pckA	Q6W6X5	75440571	<i>Actinobacillus succinogenes</i>
pckA	P43923.1	1172573	<i>Haemophilus influenza</i>
AF532733.1	AAQ10076.1	33329363	<i>Megathyrus maximus</i>
1..1929			
pepck	YP_728135.1	113869646	<i>Cupriavidus necator</i>

Oxaloacetate decarboxylase catalyzes the decarboxylation of oxaloacetate to malonate semialdehyde. Enzymes catalyzing this reaction include kgd of *Mycobacterium tuberculosis* (GenBank ID: 050463.4, GI: 160395583). Enzymes evolved from kgd with improved activity and/or substrate specificity for oxaloacetate have also been described (U.S. Pat. No. 8048624). Additional enzymes useful for catalyzing this reaction include keto-acid decarboxylases shown in the table below.

EC number	Name
4.1.1.1	Pyruvate decarboxylase
4.1.1.7	Benzoylformate decarboxylase
4.1.1.40	Hydroxypyruvate decarboxylase
4.1.1.43	Ketophenylpyruvate decarboxylase
4.1.1.71	Alpha-ketoglutarate decarboxylase
4.1.1.72	Branched chain keto-acid decarboxylase
4.1.1.74	Indolepyruvate decarboxylase
4.1.1.75	2-Ketoarginine decarboxylase
4.1.1.79	Sulfoxyruvate decarboxylase
4.1.1.80	Hydroxyphenylpyruvate decarboxylase
4.1.1.82	Phosphonopyruvate decarboxylase

The decarboxylation of keto-acids is catalyzed by a variety of enzymes with varied substrate specificities, including pyruvate decarboxylase (EC 4.1.1.1), benzoylformate decarboxylase (EC 4.1.1.7), alpha-ketoglutarate decarboxylase and branched-chain alpha-ketoacid decarboxylase. Pyruvate decarboxylase (PDC), also termed keto-acid decarboxylase, is a key enzyme in alcoholic fermentation, catalyzing the decarboxylation of pyruvate to acetaldehyde. The PDC1 enzyme from *Saccharomyces cerevisiae* has a broad substrate range for aliphatic 2-keto acids including 2-ketobutyrate, 2-ketovalemt, 3-hydroxypyruvate and 2-phenylpyruvate (22). This enzyme has been extensively studied, engineered for altered activity, and functionally expressed in *E. coli* (Killenberg-Jabs et al., *Eur.J.Biochem.* 268:1698-1704 (2001); Li et al., *Biochemistry.* 38:10004-10012 (1999); ter Schure et al., *Appl.Environ.Microbiol.* 64:1303-1307 (1998)). The PDC from *Zymomonas mobilis*, encoded by pdc, also has a broad substrate range and has been a subject of directed engineering studies to alter the affinity for different substrates (Siegert et al., *Protein Eng Des Sel* 18:345-357 (2005)). The crystal structure of this enzyme is available (Killenberg-Jabs et al., *Eur.J.Biochem.* 268:1698-1704 (2001)). Other well-characterized PDC candidates include the enzymes from *Acetobacter pasteurians* (Chandra et al., 176:443-451(2001)) and *Kluyveromyces lactis* (Krieger et al., 269:3256-3263 (2002)).

Protein	GenBank ID	GI Number	Organism
pdc	P06672.1	118391	<i>Zymomonas mobilis</i>
pdc1	P06169	30923172	<i>Saccharomyces cerevisiae</i>

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-continued

Protein	GenBank ID	GI Number	Organism
pdc	Q8L388	20385191	<i>Acetobacter pasteurians</i>
pdc1	Q12629	52788279	<i>Kluyveromyces lactis</i>

Like PDC, benzoylformate decarboxylase (EC 4.1.1.7) has a broad substrate range and has been the target of enzyme engineering studies. The enzyme from *Pseudomonas putida* has been extensively studied and crystal structures of this enzyme are available (Polovnikova et al., 42:1820-1830 (2003); Hasson et al., 37:9918-9930 (1998)). Site-directed mutagenesis of two residues in the active site of the *Pseudomonas putida* enzyme altered the affinity (Km) of naturally and non-naturally occurring substrates (Siegert et al., *Protein Eng Des Sel* 18:345-357 (2005)). The properties of this enzyme have been further modified by directed engineering (Lingen et al., *Chembiochem*. 4:721-726 (2003); Lingen et al., *Protein Eng* 15:585-593 (2002)). The enzyme from *Pseudomonas aeruginosa*, encoded by mdIC, has also been characterized experimentally (Barrowman et al., 34:57-60 (1986)). Additional gene candidates from *Pseudomonas stutzeri*, *Pseudomonas fluorescens* and other organisms can be inferred by sequence homology or identified using a growth selection system developed in *Pseudomonas putida* (Henning et al., *Appl. Environ. Microbiol.* 72:7510-7517 (2006)).

Protein	GenBank ID	GI Number	Organism
mdIC	P20906.2	3915757	<i>Pseudomonas putida</i>
mdIC	Q9HUR2.1	81539678	<i>Pseudomonas aeruginosa</i>
dpgB	ABN80423.1	126202187	<i>Pseudomonas stutzeri</i>
ilvB-1	YP_260581.1	70730840	<i>Pseudomonas fluorescens</i>

A third enzyme capable of decarboxylating 2-oxoacids is alpha-ketoglutarate decarboxylase (KGD, EC 4.1.1.71). The substrate range of this class of enzymes has not been studied to date. An exemplarily KDC is encoded by kad in *Mycobacterium tuberculosis* (Tian et al., *PNAS* 102:10670-10675 (2005)). KDC enzyme activity has also been detected in several species of rhizobia including *Bradyrhizobium japonicum* and *Mesorhizobium loti* (Green et al., *J Bacteriol* 182:2838-2844 (2000)). Although the KDC-encoding gene(s) have not been isolated in these organisms, the genome sequences are available and several genes in each genome are annotated as putative KDCs. A KDC from *Euglena gracilis* has also been characterized but the gene associated with this activity has not been identified to date (Shigeoka et al., *Arch. Biochem. Biophys.* 288:22-28 (1991)). The first twenty amino acids starting from the N-terminus were sequenced MTYKAPVKDVKFLDKVFKV (SEQ ID NO: 24) (Shigeoka and Nakano, *Arch. Biochem. Biophys.* 288:22-28 (1991)). The gene could be identified by testing candidate genes containing this N-terminal sequence for KDC activity. A novel class of AKG decarboxylase enzymes has recently been identified in cyanobacteria such as *Synechococcus* sp. PCC 7002 and homologs (Zhang and Bryant, *Science* 334:1551-3 (2011)).

Protein	GenBank ID	GI Number	Organism
kgd	O50463.4	160395583	<i>Mycobacterium tuberculosis</i>
kgd	NP_767092.1	27375563	<i>Bradyrhizobium japonicum</i> USDA110

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Protein	GenBank ID	GI Number	Organism
kgd	NP_105204.1	13473636	<i>Mesorhizobium loti</i>
ilvB	ACB00744.1	169887030	<i>Synechococcus</i> sp. PCC 7002

A fourth candidate enzyme for catalyzing this reaction is branched chain alpha-ketoacid decarboxylase (BCKA). This class of enzyme has been shown to act on a variety of compounds varying in chain length from 3 to 6 carbons (Oku et al., *J Biol Chem.* 263:18386-18396 (1988); Smit et al., *Appl Environ Microbiol* 71:303-311 (2005)). The enzyme in *Lactococcus lactis* has been characterized on a variety of branched and linear substrates including 2-oxobutanoate, 2-oxohexanoate, 2-oxopentanoate, 3-methyl-2-oxobutanoate, 4-methyl-2-oxobutanoate and isocaproate (Smit et al., *Appl Environ Microbiol* 71:303-311 (2005)). The enzyme has been structurally characterized (Berg et al., *Science*. 318:1782-1786 (2007)). Sequence alignments between the *Lactococcus lactis* enzyme and the pyruvate decarboxylase of *Zymomonas mobilis* indicate that the catalytic and substrate recognition residues are nearly identical (Siegert et al., *Protein Eng Des Sel* 18:345-357 (2005)), so this enzyme would be a promising candidate for directed engineering. Several ketoacid decarboxylases of *Saccharomyces cerevisiae* catalyze the decarboxylation of branched substrates, including ARO10, PDC6, PDC5, PDC1 and THI3 (Dickenson et al., *J Biol Chem* 275:10937-42 (2000)). Yet another BCKAD enzyme is encoded by rv0853c of *Mycobacterium tuberculosis* (Werther et al., *J Biol Chem* 283:5344-54 (2008)). This enzyme is subject to allosteric activation by alpha-ketoacid substrates. Decarboxylation of alpha-ketoglutarate by a BCKA was detected in *Bacillus subtilis*; however, this activity was low (5%) relative to activity on other branched-chain substrates (Oku and Kaneda, *J Biol Chem.* 263:18386-18396 (1988)) and the gene encoding this enzyme has not been identified to date. Additional BCKA gene candidates can be identified by homology to the *Lactococcus lactis* protein sequence. Many of the high-scoring BLASTp hits to this enzyme are annotated as indolepyruvate decarboxylases (EC 4.1.1.74). Indolepyruvate decarboxylase (IPDA) is an enzyme that catalyzes the decarboxylation of indolepyruvate to indoleacetaldehyde in plants and plant bacteria. Recombinant branched chain alpha-keto acid decarboxylase enzymes derived from the E1 subunits of the mitochondrial branched-chain keto acid dehydrogenase complex from *Homo sapiens* and *Bos taurus* have been cloned and functionally expressed in *E. coli* (Davie et al., *J. Biol. Chem.* 267:16601-16606 (1992); Wynn et al., *J. Biol. Chem.* 267:12400-12403 (1992); Wynn et al., *J. Biol. Chem.* 267:1881-1887 (1992)). In these studies, the authors found that co-expression of chaperonins GroEL and GroES enhanced the specific activity of the decarboxylase by 500-fold (Wynn et al., *J. Biol. Chem.* 267:12400-12403 (1992)). These enzymes are composed of two alpha and two beta subunits.

Protein	GenBank ID	GI Number	Organism
kdcA	AAS49166.1	44921617	<i>Lactococcus lactis</i>
PDC6	NP_010366.1	6320286	<i>Saccharomyces cerevisiae</i>
PDC5	NP_013235.1	6323163	<i>Saccharomyces cerevisiae</i>
PDC1	P06169	30923172	<i>Saccharomyces cerevisiae</i>
ARO10	NP_010668.1	6320588	<i>Saccharomyces cerevisiae</i>
THI3	NP_010203.1	6320123	<i>Saccharomyces cerevisiae</i>
rv0853c	O53865.1	81343167	<i>Mycobacterium tuberculosis</i>

-continued

Protein	GenBank ID	GI Number	Organism
BCKDHB	NP_898871.1	34101272	<i>Homo sapiens</i>
BCKDHA	NP_000700.1	11386135	<i>Homo sapiens</i>
BCKDHB	P21839	115502434	<i>Bos taurus</i>
BCKDHA	P11178	129030	<i>Bos taurus</i>

3-Phosphonopyruvate decarboxylase (EC 4.1.1.82) catalyzes the decarboxylation of 3-phosphonopyruvate to 2-phosphonoacetaldehyde. Exemplary phosphonopyruvate decarboxylase enzymes are encoded by *dhpF* of *Streptomyces lundus*, *ppd* of *Streptomyces viridochromogenes*, *fom2* of *Streptomyces wedmorensis* and *bcpC* of *Streptomyces hygroscopicus* (Circello et al, *Chem Biol* 17:402-11(2010); Blodgett et al, *FEMS Microbiol Lett* 163:149-57 (2005); Hidaka et al, *Mol Gen Genet* 249:274-80 (1995); Nakashita et al, *Biochim Biophys Acta* 1490:159-62 (2000)). The *Bacteroides fragilis* enzyme, encoded by *aepY*, also decarboxylates pyruvate and sulfofpyruvate (Zhang et al, *J Biol Chem* 278:41302-8 (2003)).

Protein	GenBank ID	GI Number	Organism
<i>dhpF</i>	ACZ13457.1	268628095	<i>Streptomyces lundus</i>
<i>Ppd</i>	CAJ14045.1	68697716	<i>Streptomyces viridochromogenes</i>
<i>Fom2</i>	BAA32496.1	1061008	<i>Streptomyces wedmorensis</i>
<i>aepY</i>	AAG26466.1	11023509	<i>Bacteroides fragilis</i>

Many oxaloacetate decarboxylase enzymes such as the *eda* gene product in *E. coli* (EC 4.1.1.3), act on the terminal acid of oxaloacetate to form pyruvate. Because decarboxylation at the 3-keto acid position competes with the malonate semialdehyde forming decarboxylation at the 2-keto-acid position, this enzyme activity can be knocked out in a host strain with a pathway proceeding through a malonate semialdehyde intermediate.

Malonyl-CoA decarboxylase (EC 4.1.1.9) catalyzes the decarboxylation of malonyl-CoA to acetyl-CoA. Enzymes have been characterized in *Rhizobium leguminosarum* and *Acinetobacter calcoaceticus* (An et al, *Eur J Biochem* 257:395-402 (1998); Koo et al, *Eur J Biochem* 266:683-90 (1999)). Similar enzymes have been characterized in *Streptomyces erythreus* (Hunaiti et al, *Arch Biochem Biophys* 229:426-39 (1984)). A recombinant human malonyl-CoA decarboxylase was overexpressed in *E. coli* (Zhou et al, *Prot Expr Pur* 34:261-9 (2004)). Methylmalonyl-CoA decarboxylase enzymes that decarboxylate malonyl-CoA are also suitable candidates. For example, the *Veillonella parvula* enzyme accepts malonyl-CoA as a substrate (Hilpert et al, *Nature* 296:584-5 (1982)). The *E. coli* enzyme is encoded by *ygfG* (Benning et al., *Biochemistry*. 39:4630-4639 (2000); Haller et al., *Biochemistry*. 39:4622-4629 (2000)). The stereo specificity of the *E. coli* enzyme was not reported, but the enzyme in *Propionigenium modestum* (Bott et al., *Eur.J.Biochem*. 250:590-599 (1997)) and *Veillonella parvula* (Huder et al., *J.Biol.Chem*. 268:24564-24571 (1993)) catalyzes the decarboxylation of the (S)-stereoisomer of methylmalonyl-CoA (Hoffmann et al., *FEBS.Lett*. 220:121-125 (1987)). The enzymes from *P. modestum* and *V. parvula* are comprised of multiple subunits that not only decarboxylate (S)-methylmalonyl-CoA, but also create a pump that transports sodium ions across the cell membrane as a means to generate energy.

Protein	GenBank ID	GI Number	Organism
<i>YgfG</i>	NP_417394	90111512	<i>Escherichia coli</i>
<i>matA</i>	Q9ZIP6	75424899	<i>Rhizobium leguminosarum</i>
<i>mdcD</i>	AAB97628.1	2804622	<i>Acinetobacter calcoaceticus</i>
<i>mdcE</i>	AAF20287.1	6642782	<i>Acinetobacter calcoaceticus</i>
<i>mdcA</i>	AAB97627.1	2804621	<i>Acinetobacter calcoaceticus</i>
<i>mdcC</i>	AAB97630.1	2804624	<i>Acinetobacter calcoaceticus</i>
<i>med</i>	NP_036345.2	110349750	<i>Homo sapiens</i>
<i>mmdA</i>	CAA05137	2706398	<i>Propionigenium modestum</i>
<i>mmdD</i>	CAA05138	2706399	<i>Propionigenium modestum</i>
<i>mmdC</i>	CAA05139	2706400	<i>Propionigenium modestum</i>
<i>mmdB</i>	CAA05140	2706401	<i>Propionigenium modestum</i>
<i>mmdA</i>	CAA80872	415915	<i>Veillonella parvula</i>
<i>mmdC</i>	CAA80873	415916	<i>Veillonella parvula</i>
<i>mmdE</i>	CAA80874	415917	<i>Veillonella parvula</i>
<i>mmdD</i>	CAA80875	415918	<i>Veillonella parvula</i>
<i>mmdB</i>	CAA80876	415919	<i>Veillonella parvula</i>

6.2.1.a CoA Synthetase

Activation of malonate to malonyl-CoA is catalyzed by a CoA synthetase in EC class 6.2.1.a. CoA synthetase enzymes that catalyze this reaction have not been described in the literature to date. Several CoA synthetase enzymes described above can also be applied to catalyze step K of FIG. 4. These enzymes include acetyl-CoA synthetase and ADP forming CoA synthetases.

6.4.1.a Carboxylase

Pyruvate carboxylase (EC 6.4.1.1) converts pyruvate to oxaloacetate at the cost of one ATP (step H). Exemplary pyruvate carboxylase enzymes are encoded by *PYC1* (Walker et al., *Biochem. Biophys. Res. Commun.* 176:1210-1217 (1991) and *PYC2* (Walker et al., supra) in *Saccharomyces cerevisiae*, and *pyc* in *Mycobacterium smegmatis* (Mukhopadhyay and Purwantini, *Biochim. Biophys. Acta* 1475:191-206 (2000)).

Protein	GenBank ID	GI Number	Organism
<i>PYC1</i>	NP_011453	6321376	<i>Saccharomyces cerevisiae</i>
<i>PYC2</i>	NP_009777	6319695	<i>Saccharomyces cerevisiae</i>
<i>Pyc</i>	YP_890857.1	118470447	<i>Mycobacterium smegmatis</i>

Example VII

Pathways for Producing Cytosolic Acetyl-CoA from Mitochondrial Acetyl-CoA

A mechanism for transporting acetyl-CoA from the mitochondrion to the cytosol can facilitate deployment of a cytosolic fatty alcohol, fatty aldehyde or fatty acid production pathway that originates from acetyl-CoA. Exemplary mechanisms for exporting acetyl-CoA include those depicted in FIGS. 5 and 6, which can involve forming citrate from acetyl-CoA and oxaloacetate in the mitochondrion, exporting the citrate from the mitochondrion to the cytosol, and converting the citrate to oxaloacetate and either acetate or acetyl-CoA. In certain embodiments, provided herein are methods for engineering a eukaryotic organism to increase its availability of cytosolic acetyl-CoA by introducing enzymes capable of carrying out the transformations depicted in any one of FIGS. 5 and 6. Exemplary enzymes capable of carrying out the required transformations are also disclosed herein.

The production of cytosolic acetyl-CoA from mitochondrial acetyl-CoA can be accomplished by a number of pathways, for example, in three to five enzymatic steps. In

one exemplary pathway, mitochondrial acetyl-CoA and oxaloacetate are combined into citrate by a citrate synthase and the citrate is exported out of the mitochondrion by a citrate or citrate/oxaloacetate transporter. Enzymatic conversion of the citrate in the cytosol results in cytosolic acetyl-CoA and oxaloacetate. The cytosolic oxaloacetate can then optionally be transported back into the mitochondrion by an oxaloacetate transporter and/or a citrate/oxaloacetate transporter. In another exemplary pathway, the cytosolic oxaloacetate is first enzymatically converted into malate in the cytosol and then optionally transferred into the mitochondrion by a malate transporter and/or a malate/citrate transporter. Mitochondrial malate can then be converted into oxaloacetate with a mitochondrial malate dehydrogenase.

In yet another exemplary pathway, mitochondrial acetyl-CoA can be converted to cytosolic acetyl-CoA via a citramalate intermediate. For example, mitochondrial acetyl-CoA and pyruvate are converted to citramalate by citramalate synthase. Citramalate can then be transported into the cytosol by a citramalate or dicarboxylic acid transporter. Cytosolic acetyl-CoA and pyruvate are then regenerated from citramalate, directly or indirectly, and the pyruvate can re-enter the mitochondria.

Along these lines, several exemplary acetyl-CoA pathways for the production of cytosolic acetyl-CoA from mitochondrial acetyl-CoA are shown in FIGS. 5 and 6. In one embodiment, mitochondrial oxaloacetate is combined with mitochondrial acetyl-CoA to form citrate by a citrate synthase. The citrate is transported outside of the mitochondrion by a citrate transporter, a citrate/oxaloacetate transporter or a citrate/malate transporter. Cytosolic citrate is converted into cytosolic acetyl-CoA and oxaloacetate by an ATP citrate lyase. In another pathway, cytosolic citrate is converted into acetate and oxaloacetate by a citrate lyase. Acetate can then be converted into cytosolic acetyl-CoA by an acetyl-CoA synthetase or transferase. Alternatively, acetate can be converted by an acetate kinase to acetyl phosphate, and the acetyl phosphate can be converted to cytosolic acetyl-CoA by a phosphotransacetylase. Exemplary enzyme candidates for acetyl-CoA pathway enzymes are described below.

The conversion of oxaloacetate and mitochondrial acetyl-CoA is catalyzed by a citrate synthase (FIGS. 5 and 6, step A). In certain embodiments, the citrate synthase is expressed in a mitochondrion of a non-naturally occurring eukaryotic organism provided herein.

Protein	GenBank ID	GI number	Organism
CIT1	NP_014398.1	6324328	<i>Saccharomyces cerevisiae</i> S288c
CIT2	NP_009931.1	6319850	<i>Saccharomyces cerevisiae</i> S288c
CIT3	NP_015325.1	6325257	<i>Saccharomyces cerevisiae</i> S288c
YAL10E02684p	XP_503469.1	50551989	<i>Yarrowia lipolytica</i>
YAL10E00638p	XP_503380.1	50551811	<i>Yarrowia lipolytica</i>
ANI_1_876084	XP_001393983.1	145242820	<i>Aspergillus niger</i> CBS 513.88
ANI_1_1474074	XP_001393195.2	317030721	<i>Aspergillus niger</i> CBS 513.88
ANI_1_2950014	XP_001389414.2	317026339	<i>Aspergillus niger</i> CBS 513.88
ANI_1_1226134	XP_001396731.1	145250435	<i>Aspergillus niger</i> CBS 513.88
gltA	NP_415248.1	16128695	<i>Escherichia coli</i> K-12 MG1655

Transport of citrate from the mitochondrion to the cytosol can be carried out by several transport proteins. Such

proteins either export citrate directly (i.e., citrate transporter, FIGS. 5 and 6, step B) to the cytosol or export citrate to the cytosol while simultaneously transporting a molecule such as malate (i.e., citrate/malate transporter, FIG. 5, step C) or oxaloacetate (i.e., citrate/oxaloacetate transporter FIG. 6, step C) from the cytosol into the mitochondrion as shown in FIGS. 5 and 6. Exemplary transport enzymes that carry out these transformations are provided in the table below.

Protein	GenBank ID	GI number	Organism
CTP1	NP_009850.1	6319768	<i>Saccharomyces cerevisiae</i> S288c
YAL10F26323p	XP_505902.1	50556988	<i>Yarrowia lipolytica</i>
ATEG_09970	EAU29419.1	114187719	<i>Aspergillus terreus</i> NIH2624
KLLA0E18723g	XP_454797.1	50309571	<i>Kluyveromyces lactis</i> NRRL Y-1140
CTRG_02320	XP_002548023.1	255726194	<i>Candida tropicalis</i> MYA-3404
ANI_1_1474094	XP_001395080.1	145245625	<i>Aspergillus niger</i> CBS 513.88
YHM2	NP_013968.1	6323897	<i>Saccharomyces cerevisiae</i> S288c
DTC	CAC84549.1	19913113	<i>Arabidopsis thaliana</i>
DTC1	CAC84545.1	19913105	<i>Nicotiana tabacum</i>
DTC2	CAC84546.1	19913107	<i>Nicotiana tabacum</i>
DTC3	CAC84547.1	19913109	<i>Nicotiana tabacum</i>
DTC4	CAC84548.1	19913111	<i>Nicotiana tabacum</i>
DTC	AAR06239.1	37964368	<i>Citrus junos</i>

ATP citrate lyase (ACL, EC 2.3.3.8, FIGS. 5 and 6, step D), also called ATP citrate synthase, catalyzes the ATP-dependent cleavage of citrate to oxaloacetate and acetyl-CoA. In certain embodiments, ATP citrate lyase is expressed in the cytosol of a eukaryotic organism. ACL is an enzyme of the RTCA cycle that has been studied in green sulfur bacteria *Chlorobium limicola* and *Chlorobium tepidum*. The alpha(4)beta(4) heteromeric enzyme from *Chlorobium limicola* was cloned and characterized in *E. coli* (Kanao et al., *Eur. J. Biochem.* 269:3409-3416 (2002)). The *C. limicola* enzyme, encoded by *aclAB*, is irreversible and activity of the enzyme is regulated by the ratio of ADP/ATP. The *Chlorobium tepidum* a recombinant ACL from *Chlorobium tepidum* was also expressed in *E. coli* and the holoenzyme was reconstituted in vitro, in a study elucidating the role of the alpha and beta subunits in the catalytic mechanism (Kim and Tabita, *J. Bacteriol.* 188:6544-6552 (2006)). ACL enzymes have also been identified in *Balnearium lithotrophicum*, *Sulfurihydrogenibium subterraneum* and other members of the bacterial phylum Aquificae (Hugler et al., *Environ. Microbiol.* 9:81-92 (2007)). This activity has been reported in some fungi as well. Exemplary organisms include *Sordaria macrospora* (Nowrousian et al., *Curr. Genet.* 37:189-93 (2000)), *Aspergillus nidulans* and *Yarrowia lipolytica* (Hynes and Murray, *Eukaryotic Cell*, July: 1039-1048, (2010), and *Aspergillus niger* (Meijer et al. *J. Ind. Microbiol. Biotechnol.* 36:1275-1280 (2009)). Other candidates can be found based on sequence homology. Information related to these enzymes is tabulated below.

Protein	GenBank ID	GI Number	Organism
aclA	BAB21376.1	12407237	<i>Chlorobium limicola</i>
aclB	BAB21375.1	12407235	<i>Chlorobium limicola</i>
aclA	AAM72321.1	21647054	<i>Chlorobium tepidum</i>
aclB	AAM72322.1	21647055	<i>Chlorobium tepidum</i>
aclB	ABI50084.1	114055039	<i>Sulfurihydrogenibium subterraneum</i>

-continued

Protein	GenBank ID	GI Number	Organism
aclA	AAX76834.1	62199504	<i>Sulfurimonas denitrificans</i>
aclB	AAX76835.1	62199506	<i>Sulfurimonas denitrificans</i>
acl1	XP_504787.1	50554757	<i>Yarrowia lipolytica</i>
acl2	XP_503231.1	50551515	<i>Yarrowia lipolytica</i>
SPBC1703.07	NP_596202.1	19112994	<i>Schizosaccharomyces pombe</i>
SPAC22A12.16	NP_593246.1	19114158	<i>Schizosaccharomyces pombe</i>
acl1	CAB76165.1	7160185	<i>Sordaria macrospora</i>
acl2	CAB76164.1	7160184	<i>Sordaria macrospora</i>
aclA	CBF86850.1	259487849	<i>Aspergillus nidulans</i>
aclB	CBF86848	259487848	<i>Aspergillus nidulans</i>

In some organisms the conversion of citrate to oxaloacetate and acetyl-CoA proceeds through a citryl-CoA intermediate and is catalyzed by two separate enzymes, citryl-CoA synthetase (EC 6.2.1.18) and citryl-CoA lyase (EC 4.1.3.34) (Aoshima, M., *Appl. Microbial. Biotechnol.* 75:249-255 (2007)). Citryl-CoA synthetase catalyzes the activation of citrate to citryl-CoA. The *Hydrogenobacter thermophilus* enzyme is composed of large and small subunits encoded by *ccsA* and *ccsB*, respectively (Aoshima et al., *Mol. Microbiol.* 52:751-761(2004)). The citryl-CoA synthetase of *Aquifex aeolicus* is composed of alpha and beta subunits encoded by *sucC1* and *sucD1* (Hugler et al., *Environ. Microbiol.* 9:81-92 (2007)). Citryl-CoA lyase splits citryl-CoA into oxaloacetate and acetyl-CoA. This enzyme is a homotrimer encoded by *ccl* in *Hydrogenobacter thermophilus* (Aoshima et al., *Mol. Microbiol.* 52:763-770 (2004)) and *aq_150* in *Aquifex aeolicus* (Hugler et al., supra (2007)). The genes for this mechanism of converting citrate to oxaloacetate and citryl-CoA have also been reported recently in *Chlorobium tepidum* (Eisen et al., PNAS 99(14): 9509-14 (2002)).

Protein	GenBank ID	GI Number	Organism
<i>ccsA</i>	BAD17844.1	46849514	<i>Hydrogenobacter thermophilus</i>
<i>ccsB</i>	BAD17846.1	46849517	<i>Hydrogenobacter thermophilus</i>
<i>sucC1</i>	AAC07285	2983723	<i>Aquifex aeolicus</i>
<i>sucD1</i>	AAC07686	2984152	<i>Aquifex aeolicus</i>
<i>ccl</i>	BAD17841.1	46849510	<i>Hydrogenobacter thermophilus</i>
<i>aq_150</i>	AAC06486	2982866	<i>Aquifex aeolicus</i>
CT0380	NP_661284	21673219	<i>Chlorobium tepidum</i>
CT0269	NP_661173.1	21673108	<i>Chlorobium tepidum</i>
CT1834	AAM73055.1	21647851	<i>Chlorobium tepidum</i>

Citrate lyase (EC 4.1.3.6, FIGS. 5 and 6, step E) catalyzes a series of reactions resulting in the cleavage of citrate to acetate and oxaloacetate. In certain embodiments, citrate lyase is expressed in the cytosol of a eukaryotic organism. The enzyme is active under anaerobic conditions and is composed of three subunits: an acyl-carrier protein (ACP, gamma), an ACP transferase (alpha), and an acyl lyase (beta). Enzyme activation uses covalent binding and acetylation of an unusual prosthetic group, 2'-(5"-phosphoribosyl)-3'-dephospho-CoA, which is similar in structure to acetyl-CoA. Acylation is catalyzed by CitC, a citrate lyase synthetase. Two additional proteins, CitG and CitX, are used to convert the apo enzyme into the active holo enzyme (Schneider et al., *Biochemistry* 39:9438-9450 (2000)). Wild type *E. coli* does not have citrate lyase activity; however, mutants deficient in molybdenum cofactor synthesis have an active citrate lyase (Clark, *FEMS Microbiol. Lett.* 55:245-249 (1990)). The *E. coli* enzyme is encoded by *citEFD* and the citrate lyase synthetase is encoded by *citC* (Nilekani and

SivaRaman, *Biochemistry* 22:4657-4663 (1983)). The *Leuconostoc mesenteroides* citrate lyase has been cloned, characterized and expressed in *E. coli* (Bekal et al., *J. Bacteriol.* 180:647-654 (1998)). Citrate lyase enzymes have also been identified in enterobacteria that utilize citrate as a carbon and energy source, including *Salmonella typhimurium* and *Klebsiella pneumoniae* (Bott, *Arch. Microbiol.* 167: 78-88 (1997); Bott and Dimroth, *Mol. Microbiol.* 14:347-356 (1994)). The aforementioned proteins are tabulated below.

Protein	GenBank ID	GI Number	Organism
<i>citF</i>	AAC73716.1	1786832	<i>Escherichia coli</i>
<i>citE</i>	AAC73717.2	87081764	<i>Escherichia coli</i>
<i>citD</i>	AAC73718.1	1786834	<i>Escherichia coli</i>
<i>citC</i>	AAC73719.2	87081765	<i>Escherichia coli</i>
<i>citG</i>	AAC73714.1	1786830	<i>Escherichia coli</i>
<i>citX</i>	AAC73715.1	1786831	<i>Escherichia coli</i>
<i>citF</i>	CAA71633.1	2842397	<i>Leuconostoc mesenteroides</i>
<i>citE</i>	CAA71632.1	2842396	<i>Leuconostoc mesenteroides</i>
<i>citD</i>	CAA71635.1	2842395	<i>Leuconostoc mesenteroides</i>
<i>citC</i>	CAA71636.1	3413797	<i>Leuconostoc mesenteroides</i>
<i>citG</i>	CAA71634.1	2842398	<i>Leuconostoc mesenteroides</i>
<i>citX</i>	CAA71634.1	2842398	<i>Leuconostoc mesenteroides</i>
<i>citF</i>	NP_459613.1	16763998	<i>Salmonella typhimurium</i>
<i>citE</i>	AAL19573.1	16419133	<i>Salmonella typhimurium</i>
<i>citD</i>	NP_459064.1	16763449	<i>Salmonella typhimurium</i>
<i>citC</i>	NP_459616.1	16764001	<i>Salmonella typhimurium</i>
<i>citG</i>	NP_459611.1	16763996	<i>Salmonella typhimurium</i>
<i>citX</i>	NP_459612.1	16763997	<i>Salmonella typhimurium</i>
<i>citF</i>	CAA56217.1	565619	<i>Klebsiella pneumoniae</i>
<i>citE</i>	CAA56216.1	565618	<i>Klebsiella pneumoniae</i>
<i>citD</i>	CAA56215.1	565617	<i>Klebsiella pneumoniae</i>
<i>citC</i>	BAH66541.1	238774045	<i>Klebsiella pneumoniae</i>
<i>citG</i>	CAA56218.1	565620	<i>Klebsiella pneumoniae</i>
<i>citX</i>	AAL60463.1	18140907	<i>Klebsiella pneumoniae</i>

The acylation of acetate to acetyl-CoA is catalyzed by enzymes with acetyl-CoA synthetase activity (FIGS. 5 and 6, step F). In certain embodiments, acetyl-CoA synthetase is expressed in the cytosol of a eukaryotic organism. Two enzymes that catalyze this reaction are AMP-forming acetyl-CoA synthetase (EC 6.2.1.1) and ADP-forming acetyl-CoA synthetase (EC 6.2.1.13). AMP-forming acetyl-CoA synthetase (ACS) is the predominant enzyme for activation of acetate to acetyl-CoA. Exemplary ACS enzymes are found in *E. coli* (Brown et *Gen. Microbiol.* 102:327-336 (1977)), *Ralstonia eutropha* (Priefert and Steinbuchel, *J. Bacteriol.* 174:6590-6599 (1992)), *Methanothermobacter thermautotrophicus* (Ingram-Smith and Smith, *Archaea* 2:95-107 (2007)), *Salmonella enterica* (Gulick et al., *Biochemistry* 42:2866-2873 (2003)) and *Saccharomyces cerevisiae* (Jogl and Tong, *Biochemistry* 43:1425-1431(2004)).

Protein	GenBank ID	GI Number	Organism
<i>acs</i>	AAC77039.1	1790505	<i>Escherichia coli</i>
<i>acoE</i>	AAA21945.1	141890	<i>Ralstonia eutropha</i>
<i>acs1</i>	ABC87079.1	86169671	<i>Methanothermobacter thermautotrophicus</i>
<i>acs1</i>	AAL23099.1	16422835	<i>Salmonella enterica</i>
<i>ACS1</i>	Q01574.2	257050994	<i>Saccharomyces cerevisiae</i>

ADP forming acetyl-CoA synthetase (ACD, EC 6.2.1.13) is another candidate enzyme that couples the conversion of acyl-CoA esters to their corresponding acids with the concurrent synthesis of ATP. Several enzymes with broad substrate specificities have been described in the literature. ACD I from *Archaeoglobus fulgidus*, encoded by AF1211, was shown to operate on a variety of linear and branched-chain substrates including acetyl-CoA, propionyl-CoA, butyryl-

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CoA, acetate, propionate, butyrate, isobutyryate, isovakrate, succinate, fumarate, phenylacetate, indoleacetate (Musfeldt et al., *J. Bacteriol.* 184:636-644 (2002)). The enzyme from *Haloarcula marismortui* (annotated as a succinyl-CoA synthetase) accepts propionate, butyrate, and branched-chain acids (isovalerate and isobutyrate) as substrates, and was shown to operate in the forward and reverse directions (Brasen et al., *Arch. Microbiol.* 182:277-287 (2004)). The ACD encoded by PAE3250 from hyperthermophilic crenarchaeon *Pyrobaculum aerophilum* showed the broadest substrate range of all characterized ACDs, reacting with acetyl-CoA, isobutyryl-CoA (preferred substrate) and phenylacetyl-CoA (Brasen et al., supra (2004)). The enzymes from *A. fulgidus*, *H. marismortui* and *P. aerophilum* have all been cloned, functionally expressed, and characterized in *E. coli* (Musfeldt et al., supra; Brasen et al., supra (2004)). Additional candidates include the succinyl-CoA synthetase encoded by sucCD in *E. coli* (Buck et al., *Biochemistry* 24:6245-6252 (1985)) and the acyl-CoA ligase from *Pseudomonas putida* (Fernandez-Valverde et al., *Appl. Environ. Microbiol.* 59:1149-1154 (1993)). Information related to these proteins and genes is shown below.

Protein	GenBank ID	GI number	Organism
AF1211	NP_070039.1	11498810	<i>Archaeoglobus fulgidus</i> DSM 4304
AF1983	NP_070807.1	11499565	<i>Archaeoglobus fulgidus</i> DSM 4304
SCS	YP_135572.1	55377722	<i>Haloarcula marismortui</i> ATCC 43049
PAE3250	NP_560604.1	18313937	<i>Pyrobaculum aerophilum</i> str. IM2
sucC	NP_415256.1	16128703	<i>Escherichia coli</i>
sueD	AAC73823.1	1786949	<i>Escherichia coli</i>
paafF	AAC24333.2	22711873	<i>Pseudomonas putida</i>

An alternative method for adding the CoA moiety to acetate is to apply a pair of enzymes such as a phosphate-transferring acyltransferase and an acetate kinase (FIGS. 5 and 6, Step F). This activity enables the net formation of acetyl-CoA with the simultaneous consumption of ATP. In certain embodiments, phosphotransacetylase is expressed in the cytosol of a eukaryotic organism. An exemplary phosphate-transferring acyltransferase is phosphotransacetylase, encoded by pta. The pta gene from *E. coli* encodes an

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enzyme that can convert acetyl-CoA into acetyl-phosphate, and vice versa (Suzuki, T. *Biochim.Biophys.Acta* 191:559-569 (1969)). This enzyme can also utilize propionyl-CoA instead of acetyl-CoA forming propionate in the process (Hesslinger et al. *Mol.Microbiol.* 27:477-492 (1998)). Homologs exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*.

Protein	GenBank ID	GI number	Organism
Pta	NP_416800.1	16130232	<i>Escherichia coli</i>
Pta	NP_461280.1	16765665	<i>Salmonella enterica</i> subsp. <i>enterica</i> serovar <i>Typhimurium</i> str. LT2
PAT2	XP_001694504.1	159472743	<i>Chlamydomonas reinhardtii</i>
PAT1	XP_001691787.1	159467202	<i>Chlamydomonas reinhardtii</i>

An exemplary acetate kinase is the *E. coli* acetate kinase, encoded by ackA (Skarstedt and Silverstein *J.Biol.Chem.* 251:6775-6783 (1976)). Homologs exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*. Information related to these proteins and genes is shown below:

Protein	GenBankID	GI number	Organism
AckA	NP_416799.1	16130231	<i>Escherichia coli</i>
AckA	NP_461279.1	16765664	<i>Salmonella enterica</i> subsp. <i>enterica</i> serovar <i>Typhimurium</i> str. LT2
ACK1	XP_001694505.1	159472745	<i>Chlamydomonas reinhardtii</i>
ACK2	XP_001691682.1	159466992	<i>Chlamydomonas reinhardtii</i>

In some embodiments, cytosolic oxaloacetate is transported back into a mitochondrion by an oxaloacetate transporter. Oxaloacetate transported back into a mitochondrion can then be used in the acetyl-CoA pathways described herein. Transport of oxaloacetate from the cytosol to the mitochondrion can be carried out by several transport proteins. Such proteins either import oxaloacetate directly (i.e., oxaloacetate transporter) to the mitochondrion or import oxaloacetate to the cytosol while simultaneously transporting a molecule such as citrate (i.e., citrate/oxaloacetate transporter) from the mitochondrion into the cytosol as shown in FIG. 6. Exemplary transport enzymes that carry out these transformations are provided in the table below.

Protein	GenBankID	GI number	Organism
OAC1	NP_012802.1	6322729	<i>Saccharomyces cerevisiae</i> S288c
KLLA0B12826g	XP_452102.1	50304305	<i>Kluyveromyces lactis</i> NRRL Y-1140
YALI0E04048g	XP_503525.1	50552101	<i>Yarrowia lipolytica</i>
CTRG_02239	XP_002547942.1	255726032	<i>Candida tropicalis</i> MYA-3404
DIC1	NP_013452.1	6323381	<i>Saccharomyces cerevisiae</i> S288c
YALI0B03344g	XP_500457.1	50545838	<i>Yarrowia lipolytica</i>
CTRG_02122	XP_002547815.1	255725772	<i>Candida tropicalis</i> MYA-3404
PAS_chr4_0877	XP_002494326.1	254574434	<i>Pichia pastoris</i> GS115
DTC	CAC84549.1	19913113	<i>Arabidopsis thaliana</i>
DTC1	CAC84545.1	19913105	<i>Nicotiana tabacum</i>
DTC2	CAC84546.1	19913107	<i>Nicotiana tabacum</i>
DTC3	CAC84547.1	19913109	<i>Nicotiana tabacum</i>
DTC4	CAC84548.1	19913111	<i>Nicotiana tabacum</i>
DTC	AAR06239.1	37964368	<i>Citrus junos</i>

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In some embodiments, cytosolic oxaloacetate is first converted to malate by a cytosolic malate dehydrogenase (FIGS. 5, step H). Cytosolic malate is transported into a mitochondrion by a malate transporter or a citrate/malate transporter (FIG. 5, step I). Mitochondrial malate is then converted to oxaloacetate by a mitochondrial malate dehydrogenase (FIG. 5, step J). Mitochondrial oxaloacetate can then be used in the acetyl-CoA pathways described herein. Exemplary examples of each of these enzymes are provided below.

Oxaloacetate is converted into malate by malate dehydrogenase (EC 1.1.1.37, FIG. 5, step H). When malate is the dicarboxylate transported from the cytosol to mitochondrion, expression of both a cytosolic and mitochondrial version of malate dehydrogenase, e.g., as shown in FIG. 4, can be used. *S. cerevisiae* possesses three copies of malate dehydrogenase, MDH1 (McAlister-Henn and Thompson, *J.*

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Bacteriol. 169:5157-5166 (1987), MDH2 (Minard and McAlister-Henn, *Mol. Cell. Biol.* 11:370-380 (1991); Gibson and McAlister-Henn, *J. Biol. Chem.* 278:25628-25636 (2003)), and MDH3 (Steffan and McAlister-Henn, *J. Biol. Chem.* 267:24708-24715 (1992)), which localize to the mitochondrion, cytosol, and peroxisome, respectively. Close homologs to the cytosolic malate dehydrogenase, MDH2, from *S. cerevisiae* are found in several organisms including *Kluyveromyces lacti* and *Candida tropicalis*. *E. coli* is also known to have an active malate dehydrogenase encoded by *mdh*. In some embodiments, the exogenous malate dehydrogenase genes are *Rhizopus delemar* malate dehydrogenase genes encoding the amino acid sequence disclosed in WO2013112939 as SEQ ID NO:167 or its variants.

Protein	GenBank ID	GI Number	Organism
MDH1	NP_012838	6322765	<i>Saccharomyces cerevisiae</i>
MDH2	NP_014515	116006499	<i>Saccharomyces cerevisiae</i>
MDH3	NP_010205	6320125	<i>Saccharomyces cerevisiae</i>
Mdh	NP_417703.1	16131126	<i>Escherichia coli</i>
KLLA0E07525p	XP_454288.1	50308571	<i>Kluyveromyces lactis</i> NRRL Y-1140
YALI0D16753g	XP_502909.1	50550873	<i>Yarrowia lipolytica</i>
CTRG_01021	XP_002546239.1	255722609	<i>Candida tropicalis</i> MYA-3404

Transport of malate from the cytosol to the mitochondrion can be carried out by several transport proteins. Such proteins either import malate directly (i.e., malate transporter) to the mitochondrion or import malate to the cytosol while simultaneously transporting a molecule such as citrate (i.e., citrate/malate transporter) from the mitochondrion into the cytosol as shown in FIG. 5. Exemplary transport enzymes that carry out these transformations are provided in the table below.

Protein	GenBank ID	GI number	Organism
OAC1	NP_012802.1	6322729	<i>Saccharomyces cerevisiae</i> S288c
KLLA0B12826g	XP_452102.1	50304305	<i>Kluyveromyces lactis</i> NRRL Y-1140
YALI0E04048g	XP_503525.1	50552101	<i>Yarrowia lipolytica</i>
CTRG_02239	XP_002547942.1	255726032	<i>Candida tropicalis</i> MYA-3404
DIC1	NP_013452.1	6323381	<i>Saccharomyces cerevisiae</i> S288c
YALI0B03344g	XP_500457.1	50545838	<i>Yarrowia lipolytica</i>
CTRG_02122	XP_002547815.1	255725772	<i>Candida tropicalis</i> MYA-3404
PAS_chr4_0877	XP_002494326.1	254574434	<i>Pichia pastoris</i> GS115
DTC	CAC84549.1	19913113	<i>Arabidopsis thaliana</i>
DTC1	CAC84545.1	19913105	<i>Nicotiana tabacum</i>
DTC2	CAC84546.1	19913107	<i>Nicotiana tabacum</i>
DTC3	CAC84547.1	19913109	<i>Nicotiana tabacum</i>
DTC4	CAC84548.1	19913111	<i>Nicotiana tabacum</i>
DTC	AAR06239.1	37964368	<i>Citrus junos</i>

Malate can be converted into oxaloacetate by malate dehydrogenase (EC 1.1.1.37, FIG. 5, step J). When malate is the dicarboxylate transported from the cytosol to mitochondrion, in certain embodiments, both a cytosolic and mitochondrial version of malate dehydrogenase is expressed, as shown in FIGS. 4 and 5. *S. cerevisiae* possesses three copies of malate dehydrogenase, MDH1 (McAlister-Henn and Thompson, *J. Bacteriol.* 169:5157-5166 (1987), MDH2 (Minard and McAlister-Henn, *Mol. Cell. Biol.* 11:370-380 (1991); Gibson and McAlister-Henn, *J. Biol. Chem.* 278: 25628-25636 (2003)), and MDH3 (Steffan and McAlister-Henn, *J. Biol. Chem.* 267:24708-24715 (1992)), which localize to the mitochondrion, cytosol, and peroxisome, respectively. Close homologs to the mitochondrial malate dehydrogenase, MDH1 from *S. cerevisiae* are found in several organisms including *Kluyveromyces lactis*, *Yarrowia lipolytica*, *Candida tropicalis*. *E. coli* is also known to have an active malate dehydrogenase encoded by *mdh*.

Protein	GenBank ID	GI Number	Organism
MDH1	NP_012838	6322765	<i>Saccharomyces cerevisiae</i>
MDH2	NP_014515	116006499	<i>Saccharomyces cerevisiae</i>
MDH3	NP_010205	6320125	<i>Saccharomyces cerevisiae</i>
Mdh	NP_417703.1	16131126	<i>Escherichia coli</i>
KLLA0F25960g	XP_456236.1	50312405	<i>Kluyveromyces lactis</i> NRRL Y-1140
YALJ0D16753g	XP_502909.1	50550873	<i>Yarrowia lipolytica</i>
CTRG_00226	XP_002545445.1	255721021	<i>Candida tropicalis</i> MYA-3404

Example VIII

Utilization of Pathway Enzymes with a Preference for NADH

The production of acetyl-CoA from glucose can generate at most four reducing equivalents in the form of NADH. A straightforward and energy efficient mode of maximizing the yield of reducing equivalents is to employ the Embden-Meyerhof-Parnas glycolysis pathway (EMP pathway). In many carbohydrate utilizing organisms, one NADH molecule is generated per oxidation of each glyceraldehyde-3-phosphate molecule by means of glyceraldehyde-3-phosphate dehydrogenase. Given that two molecules of glyceraldehyde-3-phosphate are generated per molecule of glucose metabolized via the EMP pathway, two NADH molecules can be obtained from the conversion of glucose to pyruvate.

Two additional molecules of NADH can be generated from conversion of pyruvate to acetyl-CoA given that two molecules of pyruvate are generated per molecule of glucose metabolized via the EMP pathway. This could be done by employing any of the following enzymes or enzyme sets to convert pyruvate to acetyl-CoA:

- I. NAD-dependant pyruvate dehydrogenase;
- II. Pyruvate formate lyase and NAD-dependant formate dehydrogenase;
- III. Pyruvate:ferredoxin oxidoreductase and NADH:ferredoxin oxidoreductase;
- IV. Pyruvate decarboxylase and an NAD-dependant acylating acetaldehyde dehydrogenase;
- V. Pyruvate decarboxylase, NAD-dependant acylating acetaldehyde dehydrogenase, acetate kinase, and phosphotransacetylase; and
- VI. Pyruvate decarboxylase, NAD-dependant acylating acetaldehyde dehydrogenase, and acetyl-CoA synthetase.

Overall, four molecules of NADH can be attained per glucose molecule metabolized. In one aspect, the fatty alcohol pathway requires three reduction steps from acetyl-CoA. Therefore, it can be possible that each of these three reduction steps will utilize NADPH or NADH as the reducing agents, in turn converting these molecules to NADP or NAD, respectively. Therefore, in some aspects, it can be desirable that all reduction steps are NADH-dependant in order to maximize the yield of fatty alcohols, fatty aldehydes or fatty acids. High yields of fatty alcohols, fatty aldehydes and fatty acids can thus be accomplished by:

- I. Identifying and implementing endogenous or exogenous fatty alcohol, fatty aldehyde or fatty acid pathway enzymes with a stronger preference for NADH than other reducing equivalents such as NADPH,
- II. Attenuating one or more endogenous fatty alcohol, fatty aldehyde or fatty acid pathway enzymes that contribute NADPH-dependant reduction activity,

- III. Altering the cofactor specificity of endogenous or exogenous fatty alcohol, fatty aldehyde or fatty acid pathway enzymes so that they have a stronger preference for NADH than their natural versions, or
- IV. Altering the cofactor specificity of endogenous or exogenous fatty alcohol, fatty aldehyde or fatty acid pathway enzymes so that they have a weaker preference for NADPH than their natural versions.

Exemplary NADH-dependent enzymes that participate in the elongation cycle are shown in the table below.

Enzyme	Substrate	Gene	Organism
Multifunctional ketoacyl-CoA reductase/epimerase/dehydratase	3-ketoacyl-CoA	fadB Fox2 FOX2	<i>Escherichia coli</i> <i>Candida tropicalis</i> <i>Saccharomyces cerevisiae</i>
3-Ketoacyl-CoA reductase	3-ketoacyl-CoA	paaH1 3HCDH	<i>Ralstonia eutropha</i> <i>Euglena gracilis</i>
Enoyl-CoA reductase	enoyl-CoA	TDE0597 TER ECR1 ECR2 ECR3 acad1 acad acad	<i>Treponema denticola</i> <i>Euglena gracilis</i> <i>Euglena gracilis</i> <i>Euglena gracilis</i> <i>Euglena gracilis</i> <i>Ascaris suum</i> <i>Ascaris suum</i> <i>Mycobacterium smegmatis</i>

The individual enzyme or protein activities from the endogenous or exogenous DNA sequences can be assayed using methods well known in the art. For example, the genes can be expressed in *E. coli* and the activity of their encoded proteins can be measured using cell extracts. Alternatively, the enzymes can be purified using standard procedures well known in the art and assayed for activity. Spectrophotometric based assays are particularly effective.

Several examples and methods of altering the cofactor specificity of enzymes are known in the art. For example,

Khoury et al. (Protein Sci. 2009 October; 18(10): 2125-2138) created several xylose reductase enzymes with an increased affinity for NADH and decreased affinity for NADPH. Ehsani et al (Biotechnology and Bioengineering, Volume 104, Issue 2, pages 381-389, 1 Oct. 2009) drastically decreased activity of 2,3-butanediol dehydrogenase on NADH while increasing activity on NADPH. Machielsen et al (Engineering in Life Sciences, Volume 9, Issue 1, pages 38-44, February 2009) dramatically increased activity of alcohol dehydrogenase on NADH. Khoury et al (Protein Sci. 2009 October; 18(10): 2125-2138) list in Table I several previous examples of successfully changing the cofactor preference of over 25 other enzymes. Additional descriptions can be found in Lutz et al, Protein Engineering Handbook, Volume 1 and Volume 2, 2009, Wiley-VCH Verlag GmbH & Co. KGaA, in particular, Chapter 31: Altering Enzyme Substrate and Cofactor Specificity via Protein Engineering.

Example IX

Determining Cofactor Preference of Pathway Enzymes

This example describes an experimental method for determining the cofactor preference of an enzyme.

Cofactor preference of enzymes for each of the pathway steps can be determined by cloning the individual genes on a plasmid behind a constitutive or inducible promoter and transforming into a host organism such as *Escherichia coli*. For example, genes encoding enzymes that catalyze pathway steps from: 1) acetoacetyl-CoA to 3-hydroxybutyryl-CoA, 2) 3-hydroxybutyryl-CoA to 3-hydroxybutyraldehyde, 3) 3-hydroxybutyraldehyde to 1,3-butanediol (wherein R_1 is C_1 ; R_3 is OH) can be assembled onto the pZ-based expression vectors as described below.

Replacement of the Stiffer Fragment in the pZ-based Expression Vectors. Vector backbones were obtained from Dr. Rolf Lutz of Expressys (<http://www.expressys.de/>). The vectors and strains are based on the pZ Expression System developed by Lutz and Bujard (*Nucleic Acids Res* 25, 1203-1210 (1997)). The pZE131uc, pZA331uc, pZS*131uc and pZE221uc contain the luciferase gene as a stuffer fragment. To replace the luciferase stuffer fragment with a lacZ-alpha fragment flanked by appropriate restriction enzyme sites, the luciferase stuffer fragment is removed from each vector by digestion with EcoRI and XbaI. The lacZ-alpha fragment is PCR amplified from pUC19 with the following primers:

```
lacZalpha-RI
                    (SEQ ID NO: 1)
5' GACGAATTTCGCTAGCAAGAGGAGAAGTCGACATGTCCAATTCCTGG
CGTCGTTTTAC3'

lacZalpha 3'BB
                    (SEQ ID NO: 2)
5' -GACCCTAGGAAGCTTTCTAGAGTCGACCTATGCGGCATCAGAGCAG
A-3'
```

This generates a fragment with a 5' end of EcoRI site, NheI site, a Ribosomal Binding Site, a SalI site and the start codon. On the 3' end of the fragment are the stop codon, XbaI, HindIII, and AvrII sites. The PCR product is digested with EcoRI and AvrII and ligated into the base vectors digested with EcoRI and XbaI (XbaI and AvrII have com-

patible ends and generate a non-site). Because NheI and XbaI restriction enzyme sites generate compatible ends that can be ligated together (but generate a site after ligation that is not digested by either enzyme), the genes cloned into the vectors can be "Biobricked" together (http://openwetware.org/wiki/Synthetic_Biology:BioBricks). Briefly, this method enables joining an unlimited number of genes into the vector using the same 2 restriction sites (as long as the sites do not appear internal to the genes), because the sites between the genes are destroyed after each addition. These vectors can be subsequently modified using the Phusion® Site-Directed Mutagenesis Kit (NEB, IpswichMA, USA) to insert the spacer sequence AATTAA between the EcoRI and NheI sites. This eliminates a putative stem loop structure in the RNA that bound the RBS and start codon.

All vectors have the pZ designation followed by letters and numbers indicating the origin of replication, antibiotic resistance marker and promoter/regulatory unit. The origin of replication is the second letter and is denoted by E for ColE1, A for p15A and S for pSC101 (as well as a lower copy number version of pSC101 designated S*)-based origins. The first number represents the antibiotic resistance marker (1 for Ampicillin, 2 for Kanamycin, 3 for Chloramphenicol). The final number defines the promoter that regulated the gene of interest (1 for PLlacO-1, 2 for PLlacO-1 and 3 for PA1lacO-1). For the work discussed here we employed three base vectors, pZS*13S, pZA33S and pZE13S, modified for the biobricks insertions as discussed above.

Plasmids containing genes encoding pathway enzymes can then be transformed into host strains containing lacI^Q, which allow inducible expression by addition of isopropyl β-D-1-thiogalactopyranoside (IPTG). Activities of the heterologous enzymes are tested in *in vitro* assays, using strain *E. coli* MG1655 lacI^Q as the host for the plasmid constructs containing the pathway genes. Cells can be grown aerobically in LB media (Difco) containing the appropriate antibiotics for each construct, and induced by addition of IPTG at 1 mM when the optical density (OD₆₀₀) reached approximately 0.5. Cells can be harvested after 6 hours, and enzyme assays conducted as discussed below.

In Vitro Enzyme Assays. To obtain crude extracts for activity assays, cells can be harvested by centrifugation at 4,500 rpm (Beckman-Coulter, Allegra X-15R) for 10 min. The pellets are resuspended in 0.3 mL BugBuster (Novagen) reagent with benzonase and lysozyme, and lysis proceeds for about 15 minutes at room temperature with gentle shaking. Cell-free lysate is obtained by centrifugation at 14,000 rpm (Eppendorf centrifuge 5402) for 30 min at 4° C. Cell protein in the sample is determined using the method of Bradford et al., *Anal. Biochem.* 72:248-254 (1976), and specific enzyme assays conducted as described below. Activities are reported in Units/mg protein, where a unit of activity is defined as the amount of enzyme required to convert 1 micromol of substrate in 1 minute at room temperature.

Pathway steps can be assayed in the reductive direction using a procedure adapted from several literature sources (Durre et al., *FEMS Microbiol. Rev.* 17:251-262 (1995); Palosaari and Rogers, *Bacteriol.* 170:2971-2976 (1988) and Welch et al., *Arch. Biochem. Biophys.* 273:309-318 (1989). The oxidation of NADH or NADPH can be followed by reading absorbance at 340 nM every four seconds for a total of 240 seconds at room temperature. The reductive assays can be performed in 100 mM MOPS (adjusted to pH 7.5 with KOH), 0.4 mM NADH or 0.4 mM NADPH, and from 1 to 50 μmol of cell extract. For carboxylic acid reductase-

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like enzymes, ATP can also be added at saturating concentrations. The reaction can be started by adding the following reagents: 100 μ mol of 100 mM acetoacetyl-CoA, 3-hydroxybutyryl-CoA, 3-hydroxybutyrate, or 3-hydroxybutyraldehyde. The spectrophotometer is quickly blanked and then the kinetic read is started. The resulting slope of the reduction in absorbance at 340 nM per minute, along with the molar extinction coefficient of NAD(P)H at 340 nM (6000) and the protein concentration of the extract, can be used to determine the specific activity.

Example X

Methods for Increasing NADPH Availability

In some aspects of the invention, it can be advantageous to employ pathway enzymes that have activity using NADPH as the reducing agent. For example, NADPH-dependant pathway enzymes can be highly specific for MI-FAE cycle, MD-FAE cycle and/or termination pathway intermediates or can possess favorable kinetic properties using NADPH as a substrate. If one or more pathway steps is NADPH dependant, several alternative approaches to increase NADPH availability can be employed. These include:

- 1) Increasing flux relative to wild-type through the oxidative branch of the pentose phosphate pathway comprising glucose-6-phosphate dehydrogenase, 6-phosphogluconolactonase, and 6-phosphogluconate dehydrogenase (decarboxylating). This will generate 2 NADPH molecules per glucose-6-phosphate metabolized. However, the decarboxylation step will reduce the maximum theoretical yield of 1,3-butanediol.
- 2) Increasing flux relative to wild-type through the Entner Doudoroff pathway comprising glucose-6-phosphate dehydrogenase, 6-phosphogluconolactonase, phosphogluconate dehydratase, and 2-keto-3-deoxygluconate 6-phosphate aldolase.
- 3) Introducing a soluble transhydrogenase to convert NADH to NADPH.
- 4) Introducing a membrane-bound transhydrogenase to convert NADH to NADPH.
- 5) Employing an NADP-dependant glyceraldehyde-3-phosphate dehydrogenase.
- 6) Employing any of the following enzymes or enzyme sets to convert pyruvate to acetyl-CoA
 - a) NADP-dependant pyruvate dehydrogenase;
 - b) Pyruvate formate lyase and NADP-dependant formate dehydrogenase;
 - c) Pyruvate:ferredoxin oxidoreductase and NADPH:ferredoxin oxidoreductase;
 - d) Pyruvate decarboxylase and an NADP-dependant acylating acetylaldehyde dehydrogenase;

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- e) Pyruvate decarboxylase, NADP-dependant acetaldehyde dehydrogenase, acetate kinase, and phosphotransacetylase; and
- f) Pyruvate decarboxylase, NADP-dependant acetaldehyde dehydrogenase, and acetyl-CoA synthetase; and optionally attenuating NAD-dependant versions of these enzymes.
- 7) Altering the cofactor specificity of a native glyceraldehyde-3-phosphate dehydrogenase, pyruvate dehydrogenase, formate dehydrogenase, or acylating acetylaldehyde dehydrogenase to have a stronger preference for NADPH than their natural versions.
- 8) Altering the cofactor specificity of a native glyceraldehyde-3-phosphate dehydrogenase, pyruvate dehydrogenase, formate dehydrogenase, or acylating acetylaldehyde dehydrogenase to have a weaker preference for NADH than their natural versions.

The individual enzyme or protein activities from the endogenous or exogenous DNA sequences can be assayed using methods well known in the art. For example, the genes can be expressed in *E. coli* and the activity of their encoded proteins can be measured using cell extracts as described in the previous example. Alternatively, the enzymes can be purified using standard procedures well known in the art and assayed for activity. Spectrophotometric based assays are particularly effective.

Several examples and methods of altering the cofactor specificity of enzymes are known in the art. For example, Khoury et al (Protein Sci. 2009 October; 18(10): 2125-2138) created several xylose reductase enzymes with an increased affinity for NADH and decreased affinity for NADPH. Ehsani et al (Biotechnology and Bioengineering, Volume 104, Issue 2, pages 381-389, 1 Oct. 2009) drastically decreased activity of 2,3-butanediol dehydrogenase on NADH while increasing activity on NADPH. Machielsen et al (Engineering in Life Sciences, Volume 9, Issue 1, pages 38-44, February 2009) dramatically increased activity of alcohol dehydrogenase on NADH. Khoury et al (Protein Sci. 2009 October; 18(10): 2125-2138) list in Table I several previous examples of successfully changing the cofactor preference of over 25 other enzymes. Additional descriptions can be found in Lutz et al, Protein Engineering Handbook, Volume 1 and Volume 2, 2009, Wiley-VCH Verlag GmbH & Co. KGaA, in particular, Chapter 31: Altering Enzyme Substrate and Cofactor Specificity via Protein Engineering.

Enzyme candidates for these steps are provided below. Glucose-6-Phosphate Dehydrogenase

Protein	GenBank ID	GI Number	Organism
ZWF1	NP_014158.1	6324088	<i>Saccharomyces cerevisiae</i> S288c
ZWF1	XP_504275.1	50553728	<i>Yarrowia Uplytica</i>
Zwf	XP_002548953.1	255728055	<i>Candida tropicalis</i> MYA-3404
Zwf	XP_001400342.1	145233939	<i>Aspergillus niger</i> CBS 513.88
KLLA0D19855g	XP_453944.1	50307901	<i>Kluyveromyces lactis</i> NRRL Y-1140

6-Phosphogluconolactonase

Protein	GenBank ID	GI Number	Organism
SOL3	NP_012033.2	82795254	<i>Saccharomyces cerevisiae</i> S288c
SOL4	NP_011764.1	6321687	<i>Saccharomyces cerevisiae</i> S288c
YALI0E11671g	XP_503830.1	50552840	<i>Yarrowia lipolytica</i>
YALI0C19085g	XP_501998.1	50549055	<i>Yarrowia lipolytica</i>
ANI_1_656014	XP_001388941.1	145229265	<i>Aspergillus niger</i> CBS 513.88
CTRG_00665	XP_002545884.1	255721899	<i>Candida tropicalis</i> MYA-3404
CTRG_02095	XP_002547788.1	255725718	<i>Candida tropicalis</i> MYA-3404
KLLA0A05390g	XP_451238.1	50302605	<i>Kluyveromyces lactis</i> NRRL Y-1140
KLLA0C08415g	XP_452574.1	50305231	<i>Kluyveromyces lactis</i> NRRL Y-1140

6-Phosphogluconate Dehydrogenase (Decarboxylating)

Protein	GenBank ID	GI Number	Organism
GNDI	NP_012053.1	6321977	<i>Saccharomyces cerevisiae</i> S288c
GND2	NP_011772.1	6321695	<i>Saccharomyces cerevisiae</i> S288c
ANI_1_282094	XP_001394208.2	317032184	<i>Aspergillus niger</i> CBS 513.88
ANI_1_2126094	XP_001394596.2	317032939	<i>Aspergillus niger</i> CBS 513.88
YALI0B15598g	XP_500938.1	50546937	<i>Yarrowia lipolytica</i>
CTRG_03660	XP_002549363.1	255728875	<i>Candida tropicalis</i> MYA-3404
KLLA0A09339g	XP_451408.1	50302941	<i>Kluyveromyces lactis</i> NRRL Y-1140

Phosphogluconate Sehydratase

Protein	GenBank ID	GI Number	Organism
Edd	AAC74921.1	1788157	<i>Escherichia coli</i> K-12MG1655
Edd	AAG29866.1	11095426	<i>Zymomonas mobilis subsp. mobilis</i> ZM4
Edd	YP_350103.1	77460596	<i>Pseudomonas fluorescens</i> Pf0-1
ANI_1_2126094	XP_001394596.2	317032939	<i>Aspergillus niger</i> CBS 513.88
YALI0B15598g	XP_500938.1	50546937	<i>Yarrowia lipolytica</i>
CTRG_03660	XP_002549363.1	255728875	<i>Candida tropicalis</i> MYA-3404
KLLA0A09339g	XP_451408.1	50302941	<i>Kluyveromyces lactis</i> NRRL Y-1140

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2-Keto-3-deoxygluconate 6-phosphate Aldolase

-

Protein	GenBank ID	GI Number	Organism
Eda	NP_416364.1	16129803	<i>Escherichia coli</i> K-12 MG1655
Eda	Q00384.2	59802878	<i>Zymomonas mobilis subsp. mobilis</i> ZM4
Eda	ABA76098.1	77384585	<i>Pseudomonas fluorescens</i> Pf0-1
SthA	NP_418397.2	90111670	<i>Escherichia coli</i> K-12 MG1655
SthA	YP_002798658.1	226943585	<i>Azotobacter vinelandii</i> DJ
SthA	O05139.3	11135075	<i>Pseudomonas fluorescens</i>

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Soluble Transhydrogenase

Membrane-Bound Transhydrogenase

Protein	GenBank ID	GI Number	Organism
ANI_1_29100	XP_001400109.2	317027842	<i>Aspergillus niger</i> CBS 513.88
Pc21g18800	XP_002568871.1	226943585	<i>Penicillium chrysogenum</i> Wisconsin 54-1255
SthA	O05139.3	11135075	<i>Pseudomonas fluorescens</i>
NCU01140	XP_961047.2	164426165	<i>Neurospora crassa</i> OR74A

NADP-Dependant Glyceraldehyde-3-phosphate Dehydrogenase

Protein	GenBank ID	GI Number	Organism
gapN	AAA91091.1	642667	<i>Streptococcus mutans</i>
NP-GAPDH	AEC07555.1	330252461	<i>Arabidopsis thaliana</i>
GAPN	AAM77679.2	82469904	<i>Triticum aestivum</i>
gapN	CAI56300.1	87298962	<i>Clostridium acetobutylicum</i>
NADP-GAPDH	2D2I A	112490271	<i>Synechococcus elongatus</i> PCC 7942
NADP-GAPDH	CAA62619.1	4741714	<i>Synechococcus elongatus</i> PCC 7942
GDP1	XP_455496.1	50310947	<i>Kluyveromyces lactis</i> NRRL Y-1140
HP1346	NP_208138.1	15645959	<i>Helicobacter pylori</i> 26695

NAD-Dependant Glyceraldehyde-3-phosphate Dehydrogenase

Protein	GenBank ID	GI Number	Organism
TDH1	NP_012483.1	6322409	<i>Saccharomyces cerevisiae</i> s288c
TDH2	NP_012542.1	6322468	<i>Saccharomyces cerevisiae</i> s288c
TDH3	NP_011708.1	632163	<i>Saccharomyces cerevisiae</i> s288c
KLLA0A11858g	XP_451516.1	50303157	<i>Kluyveromyces lactis</i> NRRL Y-1140
KLLA0F20988g	XP_456022.1	50311981	<i>Kluyveromyces lactis</i> NRRL Y-1140
ANI_1_256144	XP_001397496.1	145251966	<i>Aspergillus niger</i> CBS 513.88
YALIO06369g	XP_501515.1	50548091	<i>Yarrowia Upolytica</i>
CTRG_05666	XP_002551368.1	255732890	<i>Candida tropicalis</i> MYA-3404

Mutated LpdA from *E. coli* K-12 MG1655 described in Biochemistry, 1993, 32 (11), pp 2737-2740:

(SEQ ID NO: 3)
MSTEIKTQVVVLGAGPAGYSAAPRCADLGLTIVIVERYNTLGGVCLNVG
CIPSKALLHVAKVIEEAKALAEHGIVFGPKTDIDKIRTWKEKVINQLT
GGLAGMAKGRKVKVNVNGLGKFTGANTLEVEGENGKTVINFDNIIAAGS
RPIQLPFI PHEDPRIWSTDAL ELKEVPERLLVMGGGI IGLEMGTVYHA
LGSQIDVVVRKHQVIRAADKDIVKFTKRISKKNLMLTKVTAVEAKE
DGIYVTMEGKKAPAEQRYDAVLVAIGRVPNGKNLDAGKAGVEVDDRGF
IRVDKQLRTNVPHIFAIGDIVQPMLAHKGVHEGHVAAEVIAGKKHYFD
PKVIPSIAYTEPEVAWVGLTEKEAKEKGISYETATFPWAASGRAIASDC
ADGMTKLIFDKESHVRVIGGAIVGTNGGELLGEIGLAIEMGCD AEDIALT
IHAHPTLHESVGLAAEFVEGSI TDL PNP KAKKK

Mutated LpdA from *E. coli* K-12 MG1655 described in Biochemistry, 1993, 32 (11), pp 2737-2740:

(SEQ ID NO: 4)
MSTEIKTQVVVLGAGPAGYSAAPRCADLGLTIVIVERYNTLGGVCLNVG
CIPSKALLHVAKVIEEAKALAEHGIVFGPKTDIDKIRTWKEKVINQLT
GGLAGMAKGRKVKVNVNGLGKFTGANTLEVEGENGKTVINFDNIIAAGS
RPIQLPFI PHEDPRIWSTDAL ELKEVPERLLVMGGGI IAEEMATVYHA
LGSQIDVVVRKHQVIRAADKDIVKFTKRISKKNLMLTKVTAVEAKE
DGIYVTMEGKKAPAEQRYDAVLVAIGRVPNGKNLDAGKAGVEVDDRGF
IRVDKQLRTNVPHIFAIGDIVQPMLAHKGVHEGHVAAEVIAGKKHYFD
PKVIPSIAYTEPEVAWVGLTEKEAKEKGISYETATFPWAASGRAIASDC

-continued

30 ADGMTKLIFDKESHVRVIGGAIVGTNGGELLGEIGLAIEMGCD AEDIALT
IHAHPTLHESVGLAAEFVEGSI TDL PNP KAKKK

NADP-Dependant Formate Dehydrogenase

Protein	GenBank ID	GI Number	Organism
fdh	ACF35003.	194220249	<i>Bwkholderia stabilis</i>
fdh	ABC20599.2	146386149	<i>Moorella thermoacetica</i> ATCC 39073

40 Mutant *Candida bodinii* enzyme described in Journal of Molecular Catalysis B: Enzymatic, Volume 61, Issues 3-4, December 2009, Pages 157-161:

45 (SEQ ID NO: 5)
MKIVLVLYDAGKHAAD ECKLYGCTENKLGIANWLKDQGH E LITTS DKEG
ETSELDKHIPDADIIITPPHPAYITKERLDKAKNLKLVVAVGVS DHI
DLDYINQTKGKKISVLEVTGSNVVSVAEHVMTMLVLRNFVPAHEQIIN
50 HDWEVAAIAKDAYDIEGKTIATIGAGRIGYRVLERLLFPNPKELLYYQR
QALPKEAEKVGARRVENIEELVAQADIVTVNAPLHAGTKGLINKELLS
KFKKGAWLVNTARGAICVAEDVAAALESGQLRGYGGDVWFPQAPAKDHP
55 WRDMRNKYGAGNAMPHYSGTTLDAQTRYAEGTKNILESFFTGFDFYRP
QDII LLNGEYVTKAYGKHDKK

60 Mutant *Candida bodinii* enzyme described in Journal of Molecular Catalysis B: Enzymatic, Volume 61, Issues 3-4, December 2009, Pages 157-161:

65 (SEQ ID NO: 6)
MKIVLVLYDAGKHAAD ECKLYGCTENKLGIANWLKDQGH E LITTS DKEG
ETSELDKHIPDADIIITPPHPAYITKERLDKAKNLKLVVAVGVS DHI

DLDYINQGTGKKISVLEVTGSNVVSAEHVMTMLVLRNFVPAHEQIIN
 HDWEVAIAKDAYDIEGKTIATIGAGRIGYRVLERLLPFNPKELLYYSP
 QALPKEAEEKVGARRVENIEELVAQADIVTVNAPLHAGTKGLINKELLS
 KFKKGAWLVNTARGAICVAEDVAAALESGQLRGYGGDVWFPQAPKDH
 WRDMRNKYGAGNAMTPHYSGTTLDQTRYAEGTKNILESFFTGFYDYP
 QDIIILLNGEYVTKAYGKHKK

Mutant *Saccharomyces cerevisiae* enzyme described in
 Biochem J. 2002 Nov. 1:367(Pt. 3):841-847:

(SEQ ID NO: 7)
 MSGKGVLLVLYEGGKHAEEQEKLLGCIENELGIRNFIEEQGYELVTTID
 KDPEPTSTVDRELKDAEIVITPPFPAYISRNRIAEAPNLKLCVTAGVG
 SDHVDLEAANERKITVTEVTGSNVVSAEHVMATILVLIRNYNGGHQQA
 INGEWDIAGVAKNEYDLEDKIISTVGAGRIGYRVLERLVAFNPKKLLYY
 ARQELPAEAINRLNEASKLFNGRGDIVQRVEKLEDMVAQSDVVTINCL
 HKDSRGLFNKKLISHMKDGAYLVNTARGAICVAEDVAEAVKSGKLAGYG
 GDVWDKQPAPKDHWPRTMDNKHVGNAMTVHISGTSLDAQRYAGVKN
 ILNSYFSKKFDYRPQDIIVQNGSYATRAYSQKK.

NADPH:Flavin Oxidoreductase

Protein	GenBank ID	GI Number	Organism
petH	YP_171276.1	56750575	<i>Synechococcus elongatus</i> PCC 6301
fpr	NP_457968.1	16762351	<i>Salmonella enterica</i>
fmr1	XP_001697352.1	159478523	<i>Chlamydomonas reinhardtii</i>
rfmr1	NP_567293.1	18412939	<i>Arabidopsis thaliana</i>
aceF	NP_414657.1	6128108	<i>Escherichia coli</i> K-12 MG1655

NADP-Dependant Acylating Acetylaldehyde Dehydroge- nase

Protein	GenBank ID	GI Number	Organism
adhB	AAB06720.1	1513071	<i>Thermoanaerobacter</i> <i>pseudethanolicus</i> ATCC 33223
TheetDRAFT_0840	ZP_08211603.	326390041	<i>Thermoanaerobacter</i> <i>ethanolicus</i> JW 200
Cbei_3832	YP_001310903.1	150018649	<i>Clostridium</i> <i>beijerinckii</i> NCIMB 8052
Cbei_4054	YP_001311120.1	150018866	<i>Clostridium</i> <i>beijerinckii</i> NCIMB 8052
Cbei_4045	YP_001311111.1	150018857	<i>Clostridium</i> <i>beijerinckii</i> NCIMB 8052

Exemplary genes encoding pyruvate dehydrogenase, pyruvate:ferredoxin oxidoreductase, pyruvate formate lyase, pyruvate decarboxylase, acetate kinase, phosphotransacetylase and acetyl-CoA synthetase are described above in Example V.

Engineering *Saccharomyces cerevisiae* for Chemical Production

Eukaryotic hosts have several advantages over prokaryotic systems. They are able to support post-translational modifications and host membrane-anchored and organelle-specific enzymes. Genes in eukaryotes typically have introns, which can impact the timing of gene expression and protein structure.

An exemplary eukaryotic organism well suited for industrial chemical production is *Saccharomyces cerevisiae*. This organism is well characterized, genetically tractable and industrially robust. Genes can be readily inserted, deleted, replaced, overexpressed or underexpressed using methods known in the art. Some methods are plasmid-based whereas others modify the chromosome (Guthrie and Fink. *Guide to Yeast Genetics and Molecular and Cell Biology, Part B, Volume 350*, Academic Press (2002); Guthrie and Fink, *Guide to Yeast Genetics and Molecular and Cell Biology, Part C, Volume 351*, Academic Press (2002)).

Plasmid-mediated gene expression is enabled by yeast episomal plasmids (YEPs). YEPs allow for high levels of expression; however they are not very stable and they require cultivation in selective media. They also have a high maintenance cost to the host metabolism. High copy number plasmids using auxotrophic (e.g., URA3, TRP1, HIS3, LEU2) or antibiotic selectable markers (e.g., Zeo^R or Kan^R) can be used, often with strong, constitutive promoters such as PGK1 or ACT1 and a transcription terminator-polyadenylation region such as those from CYC1 or AOX. Many examples are available for one well-versed in the art. These include pVV214 (a 2 micron plasmid with URA3 selectable marker) and pVV200 (2 micron plasmid with TRP1 selectable marker) (Van et al., *Yeast* 20:739-746 (2003)). Alternatively, low copy plasmids such as centromeric or CEN plasmids can be used. Again, many examples are available for one well-versed in the art. These include pRS313 and pRS315 (Sikorski and Hieter, *Genetics* 122:19-27 (1989)) both of which require that a promoter (e.g., PGK1 or ACT1) and a terminator (e.g., CYC1, AOX) are added.

For industrial applications, chromosomal overexpression of genes is preferable to plasmid-mediated overexpression. Mikkelsen and coworkers have identified 11 integration sites on highly expressed regions of the *S. cerevisiae* genome on chromosomes X, XI and XII (Mikkelsen et al, *Met Eng* 14:104-11(2012)). The sites are separated by essential genes, minimizing the possibility of recombination between sites.

Tools for inserting genes into eukaryotic organisms such as *S. cerevisiae* are known in the art. Particularly useful tools include yeast integrative plasmids (Yips), yeast artificial chromosomes (YACS) and gene targeting/homologous recombination. Note that these tools can also be used to insert, delete, replace, underexpress or otherwise alter the genome of the host.

Yeast integrative plasmids (Yips) utilize the native yeast homologous recombination system to efficiently integrate DNA into the chromosome. These plasmids do not contain an origin of replication and can therefore only be maintained after chromosomal integration. An exemplary construct includes a promoter, the gene of interest, a terminator, and a selectable marker with a promoter, flanked by FRT sites, loxP sites, or direct repeats enabling the removal and recycling of the resistance marker. The method entails the synthesis and amplification of the gene of interest with suitable primers, followed by the digestion of the gene at a

unique restriction site, such as that created by the EcoRI and XhoI enzymes (Vellanki et al., *Biotechnol Lett.* 29:313-318 (2007)). The gene of interest is inserted at the EcoRI and XhoI sites into a suitable expression vector, downstream of the promoter. The gene insertion is verified by PCR and DNA sequence analysis. The recombinant plasmid is then linearized and integrated at a desired site into the chromosomal DNA of *S. cerevisiae* using an appropriate transformation method. The cells are plated on the YPD medium with an appropriate selection marker and incubated for 2-3 days. The transformants are analyzed for the requisite gene insert by colony PCR. To remove the antibiotic marker from a construct flanked by loxP sites, a plasmid containing the Cre recombinase is introduced. Cre recombinase promotes the excision of sequences flanked by loxP sites. (Gueldener et al., *Nucleic Acids Res* 30:e23 (2002)). The resulting strain is cured of the Cre plasmid by successive culturing on media without any antibiotic present. Alternately, the Cre recombinase plasmid has a URA selection marker and the plasmid is efficiently removed by growing cells on 5-FOA which acts as a counter-selection for URA. This method can also be employed for a scarless integration instead of using loxP. One skilled in the art can integrate using URA as a marker, select for integration by growing on URA-minus plates, and then select for URA mutants by growing on 5-FOA plates. 5-FOA is converted to the toxic 5-fluorouracil by the URA gene product. Alternatively, the FLP-FRT system can be used to integrate genes into the chromosome. This system involves the recombination of sequences between short Flipase Recognition Target (FRT) sites by the Flipase recombination enzyme (FLP) derived from the 2 μ plasmid of the yeast *Saccharomyces cerevisiae* (Sadowski, P. D., *Prog.Nucleic.Acid.Res.Mol.Biol.* 51:53-91 (1995); Zhu and Sadowski *J.Biol.Chem.* 270:23044-23054 (1995)). Similarly, gene deletion methodologies will be carried out as described in refs. Baudin et al. *Nucleic.Acids Res.* 21:3329-3330 (1993); Brachmann et al., *Yeast* 14:115-132 (1998); Giaever et al., *Nature* 418:387-391 (2002); Longtine et al., *Yeast* 14:953-961 (1998) Winzeler et al., *Science* 285:901-906 (1999).

Another approach for manipulating the yeast chromosome is gene targeting. This approach takes advantage of the fact that double stranded DNA breaks in yeast are repaired by homologous recombination. Linear DNA fragments flanked by targeting sequences can thus be efficiently integrated into the yeast genome using the native homologous recombination machinery. In addition to the application of inserting genes, gene targeting approaches are useful for genomic DNA manipulations such as deleting genes, introducing mutations in a gene, its promoter or other regulatory elements, or adding a tag to a gene.

Yeast artificial chromosomes (YACs) are artificial chromosomes useful for pathway construction and assembly. YACs enable the expression of large sequences of DNA (100-3000 kB) containing multiple genes. The use of YACs was recently applied to engineer flavenoid biosynthesis in yeast (Naesby et al, *Microb Cell Fact* 8:49-56 (2009)). In this approach, YACs were used to rapidly test randomly assembled pathway genes to find the best combination.

The expression level of a gene can be modulated by altering the sequence of a gene and/or its regulatory regions. Such gene regulatory regions include, for example, promoters, enhancers, introns, and terminators. Functional disruption of negative regulatory elements such as repressors and/or silencers also can be employed to enhance gene expression. RNA based tools can also be employed to

regulate gene expression. Such tools include RNA aptamers, riboswitches, antisense RNA, ribozymes and riboswitches.

For altering a gene's expression by its promoter, libraries of constitutive and inducible promoters of varying strengths are available. Strong constitutive promoters include pTEF1, pADH1 and promoters derived from glycolytic pathway genes. The pGAL promoters are well-studied inducible promoters activated by galactose and repressed by glucose. Another commonly used inducible promoter is the copper inducible promoter pCUP1 (Farhi et al, *Met Eng* 13:474-81 (2011)). Further variation of promoter strengths can be introduced by mutagenesis or shuffling methods. For example, error prone PCR can be applied to generate synthetic promoter libraries as shown by Alper and colleagues (Alper et al, *PNAS* 102:12678-83 (2005)). Promoter strength can be characterized by reporter proteins such as beta-galactosidase, fluorescent proteins and luciferase.

The placement of an inserted gene in the genome can alter its expression level. For example, overexpression of an integrated gene can be achieved by integrating the gene into repeating DNA elements such as ribosomal DNA or long terminal repeats.

For exogenous expression in yeast or other eukaryotic cells, genes can be expressed in the cytosol without the addition of leader sequence, or can be targeted to mitochondrion or other organelles, or targeted for secretion, by the addition of a suitable targeting sequence such as a mitochondrial targeting or secretion signal suitable for the host cells. Thus, it is understood that appropriate modifications to a nucleic acid sequence to remove or include a targeting sequence can be incorporated into an exogenous nucleic acid sequence to impart desirable properties. Genetic modifications can also be made to enhance polypeptide synthesis. For example, translation efficiency is enhanced by substituting ribosome binding sites with an optimal or consensus sequence and/or altering the sequence of a gene to add or remove secondary structures. The rate of translation can also be increased by substituting one coding sequence with another to better match the codon preference of the host.

Example XII

Termination Pathways for Making Fatty Alcohols, Aldehydes and Acids

This example describes enzymes for converting intermediates of the MI-FAE cycle or MD-FAE cycle to products of interest such as fatty alcohols, fatty aldehydes, and fatty acids. Pathways are shown in FIGS. 2 and 8. Enzymes for catalyzing steps A-G are disclosed in Example IV. This example describes enzymes suitable for catalyzing steps H—N.

Enzymes include: A. Thiolase, B. 3-Ketoacyl-CoA reductase, C. β -Hydroxyl-ACP dehydratase, D. Enoyl-CoA reductase, E. Acyl-CoA reductase (aldehyde forming), F. Alcohol dehydrogenase, G. Acyl-CoA reductase (alcohol forming), H. acyl-CoA hydrolase, transferase or synthetase, J. Acyl-ACP reductase, K. Acyl-CoA:ACP acyltransferase, L. Thioesterase, N. Aldehyde dehydrogenase (acid forming) or carboxylic acid reductase.

Pathways for converting an MI-FAE cycle intermediate to a fatty alcohol, fatty aldehyde or fatty acid product are shown in the table below. These pathways are also referred to herein as "termination pathways".

Product	Termination pathway enzymes from FIG. 2
Acid	H K/L E/N K/J/N
Aldehyde	E K/J H/N K/L/N
Alcohol	E/F K/J/F H/N/F K/L/N/F G

Product specificity can be fine-tuned using one or more enzymes shown in FIGS. 2 and 7. Chain length is controlled by one or more enzymes of the elongation pathway in conjunction with one more enzymes of the termination pathway as described above. The structure of the product is controlled by one or more enzymes of the termination pathway. Examples of selected termination pathway enzymes reacting with various pathway intermediates are shown in the table below. Additional examples are described herein.

Enzyme	Substrate	Example
Acyl-CoA reductase	Acyl-CoA 3-Hydroxyacyl-CoA	Acr1 of <i>A. baylii</i> (GenBank AAC45217) PduP of <i>L. reuteri</i> (GenBank CCC03595.1)
Acyl-CoA hydrolase, transferase or synthetase	Acyl-CoA 3-Hydroxyacyl-CoA 3-Oxoacyl-CoA Enoyl-CoA	Mcr of <i>S. tokodaii</i> (GenBank NP_378167) tesB of <i>E. coli</i> (GenBank NP_414986) hibch of <i>R. norvegicus</i> (GenBank Q5XIE6.2) MKS2 of <i>S. lycopersicum</i> (GenBank ACG69783) gctAB of <i>Acidaminococcus fermentans</i> (GenBank CAA57199, CAA57200)
Acyl-ACP acyl-transferase	Acyl-CoA	fabH of <i>E. coli</i> (GenBank AAC74175.1)

Step H. Acyl-CoA Hydrolase, Transferase or Synthase

Acyl-CoA hydrolase, transferase and synthase enzymes convert acyl-CoA moieties to their corresponding acids. Such an enzyme can be utilized to convert, for example, a fatty acyl-CoA to a fatty acid, a 3-hydroxyacyl-CoA to a 3-hydroxyacid, a 3-oxoacyl-CoA to a 3-oxoacid, or an enoyl-CoA to an enoic acid.

CoA hydrolase or thioesterase enzymes in the 3.1.2 family hydrolyze acyl-CoA molecules to their corresponding acids. Several CoA hydrolases with different substrate ranges are suitable for hydrolyzing acyl-CoA, 3-hydroxyacyl-CoA, 3-oxoacyl-CoA and enoyl-CoA substrates to their corresponding acids. For example, the enzyme encoded by acot12 from *Rattus norvegicus* brain (Robinson et al., *Biochem. Biophys. Res. Commun.* 71:959-965 (1976)) can react with butyryl-CoA, hexanoyl-CoA and malonyl-CoA. The human dicarboxylic acid thioesterase, encoded by acot8, exhibits activity on glutaryl-CoA, adipyl-CoA, suberyl-CoA, sebacyl-CoA, and dodecanedioyl-CoA (Westin et al., *J. Bi-*

ol. Chem. 280:38125-38132 (2005)). The closest *E. coli* homolog to this enzyme, tesB, can also hydrolyze a range of CoA thioesters (Naggert et al., *J. Biol. Chem.* 266:11044-11050 (1991)). A similar enzyme has also been characterized in the rat liver (Deana R., *Biochem. Int.* 26:767-773 (1992)). Additional enzymes with hydrolase activity in *E. coli* include ybgC, paaI, and ybdB (Kuznetsova, et al., *FEMS Microbiol. Rev.* 2005, 29(2):263-279; Song et al., *J. Biol. Chem.* 2006, 281(16):11028-38). Though its sequence has not been reported, the enzyme from the mitochondrion of the pea leaf has a broad substrate specificity, with demonstrated activity on acetyl-CoA, propionyl-CoA, butyryl-CoA, palmitoyl-CoA, oleoyl-CoA, succinyl-CoA, and crotonyl-CoA (Zeiger et al., *Plant. Physiol.* 94:20-27 (1990)). The acetyl-CoA hydrolase, ACH1, from *S. cerevisiae* represents another candidate hydrolase (Buu et al., *J. Biol. Chem.* 278:17203-17209 (2003)). Additional enzymes with acyl-CoA hydrolase activity include the palmitoyl-CoA hydrolase of *Mycobacterium tuberculosis* (Wang et al., *Chem. Biol.* 14:543-551 (2007)) and the acyl-CoA hydrolase of *E. coli* encoded by entH (Guo et al., *Biochemistry* 48:1712-1722 (2009)). Additional CoA hydrolase enzymes are described above.

Gene name	GenBank Accession #	GI#	Organism
acot12	NP_570103.1	18543355	<i>Rattus norvegicus</i>
tesB	NP_414986	16128437	<i>Escherichia coli</i>
acot8	CAA15502	3191970	<i>Homo sapiens</i>
acot8	NP_570112	51036669	<i>Rattus norvegicus</i>
tesA	NP_415027	16128478	<i>Escherichia coli</i>
ybgC	NP_415264	16128711	<i>Escherichia coli</i>
paaI	NP_415914	16129357	<i>Escherichia coli</i>
ybdB	NP_415129	16128580	<i>Escherichia coli</i>
ACH1	NP_009538	6319456	<i>Saccharomyces cerevisiae</i>
Rv0098	NP_214612.1	15607240	<i>Mycobacterium tuberculosis</i>
entH	AAC73698.1	1786813	<i>Escherichia coli</i>

CoA hydrolase enzymes active on 3-hydroxyacyl-CoA, 3-oxoacyl-CoA and enoyl-CoA intermediates are also well known in the art. For example, an enzyme for converting enoyl-CoA substrates to their corresponding acids is the glutamate CoA-transferase from *Acidaminococcus fermentans*. This enzyme was transformed by site-directed mutagenesis into an acyl-CoA hydrolase with activity on glutaryl-CoA, acetyl-CoA and 3-butenoyl-CoA (Mack et al., *FEBS. Lett.* 405:209-212 (1997)). Another suitable enzyme is the fdcM thioesterase III of *E. coli*. This enzyme is involved in oleate beta-oxidation and the preferred substrate is 3,5-tetradecadienoyl-CoA (Nie et al., *Biochem.* 47:7744-51(2008)).

Protein	GenBank ID	GI Number	Organism
gctA	CAA57199.1	559392	<i>Acidaminococcus fermentans</i>
gctB	CAA57200.1	559393	<i>Acidaminococcus fermentans</i>
gctA	ACJ24333.1	212292816	<i>Clostridium symbiosum</i>
gctB	ACJ24326.1	212292808	<i>Clostridium symbiosum</i>
gctA	NP_603109.1	19703547	<i>Fusobacterium nucleatum</i>
gctB	NP_603110.1	19703548	<i>Fusobacterium nucleatum</i>
fadM	NP_414977.1	16128428	<i>Escherichia coli</i>

3-Hydroxyisobutyryl-CoA hydrolase is active on 3-hydroxyacyl-CoA substrates (Shimomura et al., *J. Biol. Chem.* 269:14248-14253 (1994)). Genes encoding this enzyme include hibch of *Rattus norvegicus* (Shimomura et al., *Methods Enzymol.* 324:229-240 (2000)) and *Homo sapiens* (Shi-

momura et al., supra). Similar gene candidates can also be identified by sequence homology, including hibch of *Saccharomyces cerevisiae* and BC 2292 of *Bacillus cereus*. An exemplary 3-oxoacyl-CoA hydrolase is MKS2 of *Solanum lycopersicum* (Yu et al., *Plant Physiol* 154:67-77 (2010)). The native substrate of this enzyme is 3-oxo-myristoyl-CoA, which produces a C14 chain length product.

Gene name	GenBank Accession #	GI#	Organism
fadM	NP_414977.1	16128428	<i>Escherichia coli</i>
hibch	Q5XIE6.2	146324906	<i>Rattus norvegicus</i>
hibch	Q6NVY1.2	146324905	<i>Homo sapiens</i>
hibch	P28817.2	2506374	<i>Saccharomyces cerevisiae</i>
BC_2292	AP09256	29895975	<i>Bacillus cereus</i>
MKS2	ACG69783.1	196122243	<i>Solanum lycopersicum</i>

CoA transferases catalyze the reversible transfer of a CoA moiety from one molecule to another. Several transformations require a CoA transferase to activate carboxylic acids to their corresponding acyl-CoA derivatives. CoA transferase enzymes have been described in the open literature and represent suitable candidates for these steps. These are described below.

The gene products of cat1, cat2, and cat3 of *Clostridium kluyveri* have been shown to exhibit succinyl-CoA, 4-hydroxybutyryl-CoA, and butyryl-CoA transferase activity, respectively (Seedorf et al., *Proc.Natl.AcadSci U.S.A.* 105: 2128-2133 (2008); Sohling et al., *J Bacteriol.* 178:871-880 (1996)). Similar CoA transferase activities are also present in *Trichomonas vaginalis*, *Trypanosoma brucei*, *Clostridium aminobutyricum* and *Porphyromonas gingivalis* (Riviere et al., *J.Biol.Chem.* 279:45337-45346 (2004); van Grinsven et al., *J.Biol.Chem.* 283:1411-1418 (2008)).

Protein	GenBank ID	GI Number	Organism
cat1	P38946.1	729048	<i>Clostridium kluyveri</i>
cat2	P38942.2	172046066	<i>Clostridium kluyveri</i>
cat3	EDK35586.1	146349050	<i>Clostridium kluyveri</i>
TVAG_395550	XP_001330176	123975034	<i>Trichomonas vaginalis</i> G3
Tb11.02.0290	XP_828352	71754875	<i>Trypanosoma brucei</i>
cat2	CAB60036.1	6249316	<i>Clostridium aminobutyricum</i>
cat2	NP_906037.1	34541558	<i>Porphyromonas gingivalis</i> W83

A fatty acyl-CoA transferase that utilizes acetyl-CoA as the CoA donor is acetoacetyl-CoA transferase, encoded by the *E. coli* atoA (alpha subunit) and atoD (beta subunit) genes (Korolev et al., *Acta Crystallogr.D.Biol.Crystallogr.* 58:2116-2121(2002); Vanderwinkel et al., 33:902-908 (1968)). This enzyme has a broad substrate range on substrates of chain length C3-C6 (Sramek et al., *Arch Biochem Biophys* 171:14-26 (1975)) and has been shown to transfer the CoA moiety to acetate from a variety of branched and linear 3-oxo and acyl-CoA substrates, including isobutyrate (Matthies et al., *Appl Environ.Microbiol* 58:1435-1439 (1992)), valerate (Vanderwinkel et al., *Biochem.Biophys.Res.Comm.* 33:902-908 (1968)) and butanoate (Vanderwinkel et al., *Biochem.Biophys.Res.Comm.* 33:902-908 (1968)). This enzyme is induced at the transcriptional level by acetoacetate, so modification of regulatory control may be necessary for engineering this enzyme into a pathway (Pauli et al., *Eur.J Biochem.* 29:553-562 (1972)). Similar enzymes exist in *Corynebacterium glutamicum* ATCC 13032 (Duncan et al., 68:5186-5190 (2002)), *Clostridium*

acetobutylicum (Cary et al., *Appl Environ Microbiol* 56:1576-1583 (1990); Wiesenbom et al., *Appl Environ Microbiol* 55:323-329 (1989)), and *Clostridium saccharoperbulaceticum* (Kosaka et al., *Biosci.Biotechnol Biochem.* 71:58-68 (2007)).

Gene	GI #	Accession No.	Organism
atoA	2492994	P76459.1	<i>Escherichia coli</i>
atoD	2492990	P76458.1	<i>Escherichia coli</i>
actA	62391407	YP_226809.1	<i>Corynebacterium glutamicum</i>
cg0592	62389399	YP_224801.1	<i>Corynebacterium glutamicum</i>
ctfA	15004866	NP_149326.1	<i>Clostridium acetobutylicum</i>
ctfB	15004867	NP_149327.1	<i>Clostridium acetobutylicum</i>
ctfA	31075384	AAP42564.1	<i>Clostridium saccharoperbutylacetonicum</i>
ctfB	31075385	AAP42565.1	<i>Clostridium saccharoperbutylacetonicum</i>

Beta-ketoadipyl-CoA transferase, also known as succinyl-CoA:3:oxoacid-CoA transferase, is active on 3-oxoacyl-CoA substrates. This enzyme is encoded by pcal and pcaJ in *Pseudomonas putida* (Kaschabek et al., *J Bacteriol.* 184: 207-215 (2002)). Similar enzymes are found in *Acinetobacter* sp. ADP1 (Kowalchuk et al., *Gene* 146:23-30 (1994)), *Streptomyces coelicolor* and *Pseudomonas knackmussii* (formerly sp. B13) (Gobel et al., *J Bacteriol.* 184: 216-223 (2002); Kaschabek et al., *J Bacteriol.* 184:207-215 (2002)). Additional exemplary succinyl-CoA:3:oxoacid-CoA transferases have been characterized in *Helicobacter pylori* (Corthesy-Theulaz et al., *J Biol.Chem.* 272:25659-25667 (1997)), *Bacillus subtilis* (Stols et al., *Protein Expr.Purif.* 53:396-403 (2007)) and *Homo sapiens* (Fukao, T., et al., *Genomics* 68:144-151(2000); Tanaka, H., et al., *Mol Hum Reprod* 8:16-23 (2002)). Genbank information related to these genes is summarized below.

Gene	GI #	Accession No.	Organism
pcaI	24985644	AAN69545.1	<i>Pseudomonas putida</i>
pcaJ	26990657	NP_746082.1	<i>Pseudomonas putida</i>
pcaI	50084858	YP_046368.1	<i>Acinetobacter</i> sp. ADP1
pcaJ	141776	AAC37147.1	<i>Acinetobacter</i> sp. ADP1
pcaI	21224997	NP_630776.1	<i>Streptomyces coelicolor</i>
pcaJ	21224996	NP_630775.1	<i>Streptomyces coelicolor</i>
catI	75404583	Q8VPE3	<i>Pseudomonas knackmussii</i>
catJ	75404582	Q8VPE2	<i>Pseudomonas knackmussii</i>
HPAG1_0676	108563101	YP_627417	<i>Helicobacter pylori</i>
HPAG1_0677	108563102	YP_627418	<i>Helicobacter pylori</i>
ScoA	16080950	NP_391778	<i>Bacillus subtilis</i>
ScoB	16080949	NP_391777	<i>Bacillus subtilis</i>
OXCT1	NP_000427	4557817	<i>Homo sapiens</i>
OXCT2	NP_071403	11545841	<i>Homo sapiens</i>

The conversion of acyl-CoA substrates to their acid products can be catalyzed by a CoA acid-thiol ligase or CoA synthetase in the 6.2.1 family of enzymes. CoA synthetases that convert ATP to ADP (ADP-forming) are reversible and react in the direction of acid formation, whereas AMP forming enzymes only catalyze the activation of an acid to an acyl-CoA. For fatty acid formation, deletion or attenuation of AMP forming enzymes will reduce backflux. ADP-forming acetyl-CoA synthetase (ACD, EC 6.2.1.13) is an enzyme that couples the conversion of acyl-CoA esters to their corresponding acids with the concomitant synthesis of ATP. ACD I from *Archaeoglobus fulgidus*, encoded by AF1211, was shown to operate on a variety of linear and branched-chain substrates including isobutyrate, isopentanoate, and fumarate (Musfeldt et al., *J Bacteriol.* 184:636-

644 (2002)). A second reversible ACD in *Archaeoglobus fulgidus*, encoded by AF1983, was also shown to have a broad substrate range (Musfeldt and Schonheit, *J Bacteriol.* 184:636-644 (2002)). The enzyme from *Haloarcula marismortui* (annotated as a succinyl-CoA synthetase) accepts propionate, butyrate, and branched-chain acids (isovalerate and isobutyrate) as substrates, and was shown to operate in the forward and reverse directions (Brasen et al., *Arch Microbiol* 182:277-287 (2004)). The ACD encoded by PAE3250 from hyperthermophilic crenarchaeon *Pyrobaculum aerophilum* showed the broadest substrate range of all characterized ACDs, reacting with acetyl-CoA, isobutyryl-CoA (preferred substrate) and phenylacetyl-CoA (Brasen et al, supra). Directed evolution or engineering can be used to modify this enzyme to operate at the physiological temperature of the host organism. The enzymes from *A. fulgidus*, *H. marismortui* and *P. aerophilum* have all been cloned, functionally expressed, and characterized in *E. coli* (Brasen and Schonheit, supra; Musfeldt and Schonheit, *J Bacteriol.* 184:636-644 (2002)). An additional candidate is succinyl-CoA synthetase, encoded by sucCD of *E. coli* and LSCJ and LSC2 genes of *Saccharomyces cerevisiae*. These enzymes catalyze the formation of succinyl-CoA from succinate with the concomitant consumption of one ATP in a reaction which is reversible in vivo (Buck et al., *Biochemistry* 24:6245-6252 (1985)). The acyl CoA ligase from *Pseudomonas putida* has been demonstrated to work on several aliphatic substrates including acetic, propionic, butyric, valeric, hexanoic, heptanoic, and octanoic acids and on aromatic compounds such as phenylacetic and phenoxyacetic acids (Fernandez-Valverde et al., *Appl. Environ. Microbiol.* 59:1149-1154 (1993)). A related enzyme, malonyl CoA synthetase (6.3.4.9) from *Rhizobium leguminosarum* could convert several diacids, namely, ethyl-, propyl-, allyl-, isopropyl-, dimethyl-, cyclopropyl-, cyclopropylmethylene-, cyclobutyl-, and benzyl-malonate into their corresponding monothioesters (Pohl et al., *J. Am. Chem. Soc.* 123:5822-5823 (2001)).

Protein	GenBank ID	GI Number	Organism
AF1211	NP_070039.1	11498810	<i>Archaeoglobus fulgidus</i>
AF1983	NP_070807.1	11499565	<i>Archaeoglobus fulgidus</i>
scs	YP_135572.1	55377722	<i>Haloarcula marismortui</i>
PAE3250	NP_560604.1	18313937	<i>Pyrobaculum aerophilum</i> str. IM2
sucC	NP_415256.1	16128703	<i>Escherichia coli</i>
sucD	AAC73823.1	1786949	<i>Escherichia coli</i>
LSC1	NP_014785	6324716	<i>Saccharomyces cerevisiae</i>
LSC2	NP_011760	6321683	<i>Saccharomyces cerevisiae</i>
paaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>
matB	AAC83455.1	3982573	<i>Rhizobium leguminosarum</i>

Step J. Acyl-ACP Reductase

The reduction of an acyl-ACP to its corresponding aldehyde is catalyzed by an acyl-ACP reductase (AAR). Such a transformation is depicted in step J of FIGS. 2 and 8. Suitable enzyme candidates include the orf1594 gene product of *Synechococcus elongatus* PCC7942 and homologs thereof (Schirmer et al, *Science*, 329: 559-62 (2010)). The *S. elongatus* PCC7942 acyl-ACP reductase is coexpressed with an aldehyde decarbonylase in an operon that appears to be conserved in a majority of cyanobacterial organisms. This enzyme, expressed in *E. coli* together with the aldehyde decarbonylase, conferred the ability to produce alkanes. The

P. marinus AAR was also cloned into *E. coli* and, together with a decarbonylase, demonstrated to produce alkanes (US Application 2011/0207203).

Protein	GenBank ID	GI Number	Organism
orf1594	YP_400611.1	81300403	<i>Synechococcus elongatus</i> PCC7942
PMT9312_0533	YP_397030.1	78778918	<i>Prochlorococcus marinus</i> MIT 9312
10 syc0051_d	YP_170761.1	56750060	<i>Synechococcus elongatus</i> PCC 6301
Ava_2534	YP_323044.1	75908748	<i>Anabaena variabilis</i> ATCC 29413
alr5284	NP_489324.1	17232776	<i>Nostoc</i> sp. PCC 7120
Aazo_3370	YP_003722151.1	298491974	<i>Nostoc azollae</i>
15 Cyan7425_0399	YP_002481152.1	220905841	<i>Cyanothece</i> sp. PCC 7425
N9414_21225	ZP_01628095.1	119508943	<i>Nodularia spumigena</i> CCY9414
L8106_07064	ZP_01619574.1	119485189	<i>Lyngbya</i> sp. PCC 8106

Step K. Acyl-CoA:ACP Acyltransferase

The transfer of an acyl-CoA to an acyl-ACP is catalyzed by acyltransferase enzymes in EC class 2.3.1. Enzymes with this activity are described above.

Step L. Thioesterase

Acyl-ACP thioesterase enzymes convert an acyl-ACP to its corresponding acid. Such a transformation is required in step L of FIG. 2. Exemplary enzymes include the FatA and FatB isoforms of *Arabidopsis thaliana* (Salas et al, *Arch Biochem Biophys* 403:25-34 (2002)). The activities of these two proteins vary with carbon chain length, with FatA preferring oleyl-ACP and FatB preferring palmitoyl-ACP. A number of thioesterases with different chain length specificities are listed in WO 2008/113041 and are included in the table below. For example, it has been shown previously that expression of medium chain plant thioesterases like FatB from *Umbellularia californica* in *E. coli* results in accumulation of high levels of medium chain fatty acids, primarily laurate (C12:0). Similarly, expression of *Cuphea palustris* FatB1 thioesterase in *E. coli* led to accumulation of C8-10:0 products (Dehesh et al, *Plant Physiol* 110:203-10 (1996)). Similarly, *Carthamus tinctorius* thioesterase expressed in *E. coli* leads to >50 fold elevation in C 18:1 chain termination and release as free fatty acid (Knutzon et al, *Plant Physiol* 100:1751-58 (1992)). Methods for altering the substrate specificity of thioesterases are also known in the art (for example, EP1605048).

Protein	GenBank ID	GI Number	Organism
fatA	AEE76980.1	332643459	<i>Arabidopsis thaliana</i>
fatB	AEE28300.1	332190179	<i>Arabidopsis thaliana</i>
55 fatB2	AAC49269.1	1292906	<i>Cuphea hookeriana</i>
fatB3	AAC72881.1	3859828	<i>Cuphea hookeriana</i>
fatB1	AAC49179.1	1215718	<i>Cuphea palustris</i>
M96568.1:94..1251	AAA33019.1	404026	<i>Carthamus tinctorius</i>
fatB1	Q41635.1	8469218	<i>Umbellularia californica</i>
60 tesA	AAC73596.1	1786702	<i>Escherichia coli</i>

Step N. Aldehyde Dehydrogenase (Acid Forming) or Carboxylic Acid Reductase

The conversion of an aldehyde to an acid is catalyzed by an acid-forming aldehyde dehydrogenase. Several *Saccha-*

Saccharomyces cerevisiae enzymes catalyze the oxidation of aldehydes to acids including ALD1 (ALD6), ALD2 and ALD3 (Navarro-Avino et al, Yeast 15:829-42 (1999); Quash et al, *Biochem Pharmacol* 64:1279-92 (2002)). The mitochondrial proteins ALD4 and ALD5 catalyze similar transformations (Wang et al, *J Bacteriol* 180:822-30 (1998); Boubekour et al, *Eur J Biochem* 268:5057-65 (2001)). HFD1 encodes a hexadecanal dehydrogenase. Exemplary acid-forming aldehyde dehydrogenase enzymes are listed in the table below.

Protein	GenBank ID	GI number	Organism
ALD2	NP_013893.1	6323822	<i>Saccharomyces cerevisiae</i> s288c
ALD3	NP_013892.1	6323821	<i>Saccharomyces cerevisiae</i> s288c
ALD4	NP_015019.1	6324950	<i>Saccharomyces cerevisiae</i> s288c
ALD5	NP_010996.2	330443526	<i>Saccharomyces cerevisiae</i> s288c
ALD6	NP_015264.1	6325196	<i>Saccharomyces cerevisiae</i> s288c
HFD1	NP_013828.1	6323757	<i>Saccharomyces cerevisiae</i> s288c
CaO19.8361	XP_710976.1	68490403	<i>Candida albicans</i>
CaO19.742	XP_710989.1	68490378	<i>Candida albicans</i>
YALI0C03025	CAG81682.1	49647250	<i>Yarrowia lipolytica</i>
ANI_1_1334164	XP_001398871.1	145255133	<i>Aspergillus niger</i>
ANI_1_2234074	XP_001392964.2	317031176	<i>Aspergillus niger</i>
ANI_1_226174	XP_001402476.1	145256256	<i>Aspergillus niger</i>
ALDH	P41751.1	1169291	<i>Aspergillus niger</i>
KLLA0D09999	CAH00602.1	49642640	<i>Kluyveromyces lactis</i>

The conversion of an acid to an aldehyde is thermodynamically unfavorable and typically requires energy-rich cofactors and multiple enzymatic steps. For example, in butanol biosynthesis conversion of butyrate to butyraldehyde is catalyzed by activation of butyrate to its corresponding acyl-CoA by a CoA transferase or ligase, followed by reduction to butyraldehyde by a CoA-dependent aldehyde dehydrogenase. Alternately, an acid can be activated to an acyl-phosphate and subsequently reduced by a phosphate reductase. Direct conversion of the acid to aldehyde by a single enzyme is catalyzed by a bifunctional carboxylic acid reductase enzyme in the 1.2.1 family. Exemplary enzymes that catalyze these transformations include carboxylic acid reductase, alpha-aminoacidate reductase and retinoic acid reductase.

Carboxylic acid reductase (CAR), found in *Nocardia iowensis*, catalyzes the magnesium, ATP and NADPH-dependent reduction of carboxylic acids to their corresponding aldehydes (Venkitasubramanian et al., *J Biol.Chem.* 282: 478-485 (2007)). The natural substrate of this enzyme is benzoic acid and the enzyme exhibits broad acceptance of aromatic and aliphatic substrates including fatty acids of length C12-C18 (Venkitasubramanian et al., *Biocatalysis in Pharmaceutical and Biotechnology Industries*. CRC press (2006); WO 2010/135624). CAR requires post-translational activation by a phosphopantetheine transferase (PPTase) that converts the inactive apo-enzyme to the active holo-enzyme (Hansen et al., *Appl.Environ.Microbiol* 75:2765-2774 (2009)). The *Nocardia* CAR enzyme was cloned and functionally expressed in *E. coli* (Venkitasubramanian et al., *J Biol.Chem.* 282:478-485 (2007)). Co-expression of the npt gene, encoding a specific PPTase, improved activity of the enzyme. A related enzyme from *Mycobacterium* sp. strain JLS catalyzes the reduction of fatty acids of length C12-C16. Variants of this enzyme with enhanced activity on fatty acids are described in WO 2010/135624. Alpha-aminoacidate reductase (AAR, EC 1.2.1.31), participates in lysine biosyn-

thesis pathways in some fungal species. This enzyme naturally reduces alpha-aminoacidate to alpha-aminoacidate semialdehyde. The carboxyl group is first activated through the ATP-dependent formation of an adenylate that is then reduced by NAD(P)H to yield the aldehyde and AMP. Like CAR, this enzyme utilizes magnesium and requires activation by a PPTase. Enzyme candidates for AAR and its corresponding PPTase are found in *Saccharomyces cerevisiae* (Morris et al., *Gene* 98:141-145 (1991)), *Candida albicans* (Guo et al., *Mol. Genet. Genomics* 269:271-279 (2003)), and *Schizosaccharomyces pombe* (Ford et al., *Curr. Genet.* 28:131-137 (1995)). The AAR from *S. pombe* exhibited significant activity when expressed in *E. coli* (Guo et al., *Yeast* 21:1279-1288 (2004)). The AAR from *Penicillium chrysogenum* accepts S-carboxymethyl-L-cysteine as an alternate substrate, but did not react with adipate, L-glutamate or diaminopimelate (Hijarrubia et al., *J Biol.Chem.* 278:8250-8256 (2003)). The gene encoding the *P. chrysogenum* PPTase has not been identified to date and no high-confidence hits were identified by sequence comparison homology searching.

Protein	GenBank ID	GI Number	Organism
car	AAR91681.1	40796035	<i>Nocardia iowensis</i>
npt	ABI83656.1	114848891	<i>Nocardia iowensis</i>
car	YP_001070587.1	126434896	<i>Mycobacterium</i> sp. strain JLS
npt	YP_001070355.1	126434664	<i>Mycobacterium</i> sp. strain JLS
LYS2	AAA34747.1	171867	<i>Saccharomyces cerevisiae</i>
LYS5	P50113.1	1708896	<i>Saccharomyces cerevisiae</i>
LYS2	AAC02241.1	2853226	<i>Candida albicans</i>
LYS5	AAO26020.1	28136195	<i>Candida albicans</i>
Lys1p	P40976.3	13124791	<i>Schizosaccharomyces pombe</i>
Lys7p	Q10474.1	1723561	<i>Schizosaccharomyces pombe</i>
Lys2	CAA74300.1	3282044	<i>Penicillium chrysogenum</i>

Additional car and npt genes can be identified based on sequence homology.

Gene name	GI No.	GenBank Accession No.	Organism
fadD9	121638475	YP_978699.1	<i>Mycobacterium bovis</i> BCG
BCG_2812c	121638674	YP_978898.1	<i>Mycobacterium bovis</i> BCG
nfa20150	54023983	YP_118225.1	<i>Nocardia farcinica</i> IFM 10152
nfa40540	54026024	YP_120266.1	<i>Nocardia farcinica</i> IFM 10152
SGR_6790	YP_001828302.1	182440583	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350
SGR_665	YP_001822177.1	182434458	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350
MSMEG_2956	YP_887275.1	118473501	<i>Mycobacterium smegmatis</i> MC2 155
MSMEG_5739	YP_889972.1	118469671	<i>Mycobacterium smegmatis</i> MC2 155
MSMEG_2648	YP_886985.1	118471293	<i>Mycobacterium smegmatis</i> MC2 155
MAP1040c	NP_959974.1	41407138	<i>Mycobacterium avium</i> subsp. <i>paratuberculosis</i> K-10
MAP2899c	NP_961833.1	41408997	<i>Mycobacterium avium</i> subsp. <i>paratuberculosis</i> K-10
MMAR_2117	YP_001850422.1	183982131	<i>Mycobacterium marinum</i> M
MMAR_2936	YP_001851230.1	183982939	<i>Mycobacterium</i>

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-continued

Gene name	GI No.	GenBank Accession No.	Organism
MMAR_1916	YP_001850220.1	183981929	<i>marinum M</i> <i>Mycobacterium</i> <i>marinum M</i>
Tpau_1373	YP_003646340.1	296139097	<i>Tsukamurella</i> <i>paurometabola</i> DSM 20162
Tpau_1726	YP_003646683.1	296139440	<i>Tsukamurella</i> <i>paurometabola</i> DSM 20162
CPCC7001_ 1320	ZP_05045132.1	254431429	<i>Cyanobium</i> PCC7001
DDBDRAFT_ 0187729	XP_636931.1	66806417	<i>Dictyostelium</i> <i>discoideum</i> AX4

An additional enzyme candidate found in *Streptomyces griseus* is encoded by the griC and griD genes. This enzyme is believed to convert 3-amino-4-hydroxybenzoic acid to 3-amino-4-hydroxybenzaldehyde as deletion of either griC or griD led to accumulation of extracellular 3-acetylamin-4-hydroxybenzoic acid, a shunt product of 3-amino-4-hydroxybenzoic acid metabolism (Suzuki, et al., *J. Antibiot.* 60(6):380-387 (2007)). Co-expression of gnC and griD with SGR_665, an enzyme similar in sequence to the *Nocardia iowensis* npt, can be beneficial.

Gene name	GI No.	GenBank Accession No.	Organism
griC	YP_001825755.1	182438036	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350
griD	YP_001825756.1	182438037	<i>Streptomyces griseus</i> subsp. <i>griseus</i> NBRC 13350

Example XIII

Production of 1,3-Butanediol from Glucose in
Saccharomyces cerevisiae

This example illustrates the construction and biosynthetic production of 1,3-BDO from glucose in *Saccharomyces cerevisiae*.

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The pathway for 1,3-BDO production is comprised of two MI-FAE cycle enzymes (thiolase and 3-oxoacyl-CoA reductase), in conjunction with termination pathway enzymes (acyl-CoA reductase (aldehyde forming) and alcohol dehydrogenase). The 1,3-BDO pathway engineered into *S. cerevisiae* is composed of four enzymatic steps which transform acetyl-CoA to 1,3-BDO. The first step entails the condensation of two molecules of acetyl-CoA into acetoacetyl-CoA by an acetoacetyl-CoA thiolase enzyme (THL). In the second step, acetoacetyl-CoA is reduced to 3-hydroxybutyryl-CoA by acetoacetyl-CoA reductase, also called 3-hydroxybutyryl-CoA dehydrogenase (HBD). 3-hydroxybutyryl-CoA reductase (ALD) catalyzes formation of the aldehyde from the acyl-CoA. Further reduction of 3-hydroxybutyraldehyde to 1,3-BDO is catalyzed by 1,3-BDO dehydrogenase (ADH).

To enable 1,3-BDO production in the cytosol, two acetyl-CoA forming pathways were engineered into *S. cerevisiae*. The first pathway entails conversion of pyruvate to acetyl-CoA by pyruvate decarboxylase (FIG. 3E), acetaldehyde dehydrogenase (FIG. 3F) and acetyl-CoA synthetase (FIG.

3B). The second pathway is pyruvate formate lyase (FIG. 3H).

For each enzymatic step of the 1,3-BDO pathway, a list of applicable genes was assembled for corroboration. The genes cloned and assessed in this study are presented below in Table 1, along with the appropriate references and URL citations to the polypeptide sequence.

TABLE 1

Exemplary Step	ID	Gene	NCBI Accession #	GI	Source Organism
Acetoacetyl-CoA thiolase (THL)					
FIG. 2A	1502	thiI	P45359.1	1174677	<i>Clostridium acetobutylicum</i> ATCC 824
FIG. 2A	1491	atoB	NP_416728	16130161	<i>Escherichia coli</i> str. K-12 substr. MG1655
FIG. 2A	560	thiA	NP_349476.1	15896127	<i>Clostridium acetobutylicum</i> ATCC 824
FIG. 2A	1512	phbA	P07097.4	135759	<i>Zoogloea ramigera</i>
FIG. 2A	1501	phbA	P14611.1	135754	<i>Ralstonia eutropha</i> H16
3-Hydroxybutyryl-CoA dehydrogenase (HBD)					
FIG. 2B	1495	hbd	AAM14586.1	20162442	<i>Clostridium beijerinckii</i> NCIMB 8052
3-Hydroxybutyryl-CoA reductase (ALD)					
FIG. 2E	707	Lvis_1603	YP_795711.1	116334184	<i>Lactobacillus brevis</i> ATCC 367
3-Hydroxybutyraldehyde reductase (ADH)					
FIG. 2F	28	bdh	BAF45463.1	124221917	<i>Clostridium saccharoperbutylacetonicum</i>
Pyruvate formate lyase (PflAB)					
FIG. 3H	1799	pflA	NP_415422.1	16128869	<i>Escherichia coli</i> MG1655
FIG. 3H	500	pflB	NP_415423	16128870	<i>Escherichia coli</i> MG1655

TABLE 1-continued

Exemplary Step	ID	Gene	NCBI Accession #	GI	Source Organism
PDH Bypass (aldehyde dehydrogenase, acetyl-CoA synthase)					
FIG. 3F	1849	ALD6	NP_015264.1	6325196	<i>Saccharomyces cerevisiae</i> S288c
FIG. 3B	1845	Acs	AAL23099.1	16422835	<i>Salmonella enterica</i> LT2
FIG. 3B	1845A	Acsm	AAL23099.1	16422835	<i>Salmonella enterica</i> LT2

Genes were cloned via PCR from the genomic DNA of the native or wild-type organism. Primers used to amplify the pathway genes are (from 5' to 3'; underlined sequences are gene specific):

Thl 1502:

FP:

(SEQ ID NO: 8)

TCTAATCTAAGTTTTCTAGAACTAGTAAAGATGAGAGATGTAGTAATAG

TAAGTGCTGTA

RP:

(SEQ ID NO: 9)

GATATCGAATTCCTGCAGCCCGGGGATCCTTAGTCTCTTTCAACTACG

AGAGCTGTT

Thl 1491:

FP:

(SEQ ID NO: 10)

TCTAATCTAAGTTTTCTAGAACTAGTAAAGATGAAAAATTGTGTCATCG

TCAGTG

RP:

(SEQ ID NO: 11)

GATATCGAATTCCTGCAGCCCGGGGATCCTTAATTC AACCGTTCAATC

ACCATCGCAAT

Thl 560:

FP:

(SEQ ID NO: 12)

AATCTAAGTTTTCTAGAACTAGTAAAGATGAAAGAGTTGTAATAGCTA

GTGCAGTAA

RP:

(SEQ ID NO: 13)

TATCGAATTCCTGCAGCCCGGGGATCCTTAATGGTGATGGTGATGATG

GCACTTTTCTA

Thl 1512:

FP:

(SEQ ID NO: 14)

TCTAATCTAAGTTTTCTAGAACTAGTAAAGATGAGCACCCCGTCCATCG

TCA

PR:

(SEQ ID NO: 15)

GATATCGAATTCCTGCAGCCCGGGGATCCTTAAAGGCTCTCGATGCAC

ATCGCC

Thl 1501:

FP:

(SEQ ID NO: 16)

TAAGCTAGCAAGAGGAGAAGTCGACATGACTGACGTTGTCATCGTATCC

GC

10

-continued

RP:

(SEQ ID NO: 17)

GCCTCTAGGAAGCTTCTAGATTATTATTTGCGCTCGACTGCCAGC

15

Hbd 1495:

FP:

(SEQ ID NO: 18)

AAGCATACAATCAACTATCTCATATACAATGAAAAGATTTTGTACTT

GGAGCA

20

RP:

(SEQ ID NO: 19)

AAAAATCATAAATCATAAGAAATTCGCTTATTAGAGTAATCATAGAAT

CCTTTTCCTGA

25

Alc 707:

FP:

(SEQ ID NO: 20)

AATCTAAGTTTTCTAGAACTAGTAAAGATGAACACAGAAACATTGAAC

AAGCCAT

30

RP:

(SEQ ID NO: 21)

TATCGAATTCCTGCAGCCCGGGGATCCCTAAGCCTCCCAAGTCCGTAA

TGAGAACCCTT

35

Adh 28:

FP:

(SEQ ID NO: 22)

CCAAGCATACAATCAACTATCTCATATACAATGAGAAATTTAGATTTA

ATGCATATACA

40

RP:

(SEQ ID NO: 23)

AATAAAATCATAAATCATAAGAAATTCGCTTAAAGGACATTCTAAA

ATTTTATATAC

45

1845A is a sequence variant of the wild type (1845) enzyme. The variation is a point mutation in the residue Leu-641 (L641P), described in Starai and coworkers (Starai et al, *J Biol Chem* 280: 26200-5 (2005)). The function of the mutation, e.g., is to prevent post-translational regulation by acetylation and maintain the Acs enzyme in its active state.

Shuttle plasmids shown in Table 2 were constructed for expression of heterologous genes in *S. cerevisiae*. Plasmids d9, d10, and d11 are empty plasmid controls with the selection marker of Ura, His, and Leu, respectively. Plasmids d12 or d13 contains a single ALD or ADH gene with the URA3 selection marker. Plasmids d14, d16, and d17 contains hbd and thl genes with the HIS3 selection marker.

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TABLE 2

Plasmid	Selection Marker	Gene(s)
pESC-L	URA3	NA
pESC-H	HIS3	NA
pESC-U	LEU2	NA
pY3Hd1	URA3	1799(pflA)-500(pflB)

65

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TABLE 2-continued

Plasmid	Selection Marker	Gene(s)
pY3Hd2	HIS3	1799(pflA)-500(pflB)
pY3Hd3	LEU2	1799(pflA)-500(pflB)
pY3Hd4	URA3	1849(ALD6)-1845(Acs)
pY3Hd5	URA3	1849(ALD6)-1845A(Acsm)
pY3Hd6	URA3	1495(Hbd) - 1491(Thl)
pY3Hd7	URA3	1495(Hbd) - 560(Thl)
pY3Hd8	LEU2	28(ADH)-707(ALD)
pY3Hd9	URA3	NA
pY3Hd10	HIS3	NA
pY3Hd11	LEU2	NA
pY3Hd12	URA3	707(ALD)
pY3Hd13	URA3	28(ADH)
pY3Hd14	HIS3	1495(Hbd) - 1502(Thl)
pY3Hd15	HIS3	1495(Hbd) - 1512(Thl)
pY3Hd16	HIS3	1495(Hbd) - 1491(Thl)
pY3Hd17	HIS3	1495(Hbd) - 560 (Thl)

Yeast host BY4741 [MATa his3Δ0 leu2Δ0 met15Δ0 ura3Δ0] was chosen as the host strain for this work as a wild-type laboratory strain with the appropriate auxotrophic markers to host the pathway plasmids BY4741 was transformed with plasmids containing 1,3-BDO pathway genes alone or along with plasmids that contain PDH bypass genes or pflAB genes. Vector backbones used in this example include p427TEF yeast expression vectors, the pY3H bridging vectors (Sunrise Science) and pESC yeast epitope tagging vectors (Agilent Technologies). The pY3H vector containing a TEF1 promoter, CYC terminator and URA3

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selection marker from *S. cerevisiae* was used to build dual-promoter plasmids with different selection markers. ADH1 promoter and terminator sequences from *S. cerevisiae* were inserted upstream of the TEF1 promoter so the two transcriptional units are in a back-to-back orientation. The SV40 nuclear localization signal sequence was removed during the cloning process. The resulting plasmid was named pY3Hd9. To construct plasmids with a different selection marker, the URA3 gene in pY3Hd9 was replaced with the HIS3 or LEU2 gene from *S. cerevisiae* to produce pY3Hd10 and pY3Hd11, respectively. Two of the four 1,3-BDO pathway genes—Hbd and Thl (see Table 103 for gene numbers)—were cloned into the dual-promoter plasmid with the HIS3 marker such that the expression of the Hbd genes is controlled by the ADH1 promoter while the expression of the Thl gene is controlled by the TEF1 promoter (pY3Hd14~17). Ald and Adh genes were cloned into the dual-promoter plasmid with the LEU2 selection marker such that the ADH1 promoter drives the adh genes and the TEF1 promoter drives the ald genes (pY3Hd8). The PflAB genes or the PDH bypass genes (ALD6 and acs) were cloned into the dual-promoter plasmid with the URA3 marker where pflA or ALD6 is controlled under the ADH1 promoter and pflB or acs is controlled under the TEF1 promoter. Yeast transformation was done using Frozen-EZ Yeast Transformation (Zymo Research).

Tables 3 and 4 show the combinations of plasmids and experimental conditions tested.

TABLE 3

Sample	Plasmid 1	Plasmid 2	plasmid 3	gene 1	gene 2	gene 3	gene 4	gene 5	gene 6	Aeroation	Note
1	pESC-L	pESC-H								Anaerobic	EV2
2	pESC-L	pESC-H								23G	EV2
3	d8	d16		1495	1491	28	707			Anaerobic	BDO
4	d8	d16		1495	1491		707			Anaerobic	BDO
5	d8	d16		1495	1491		707			23G	BDO
6	d8	d16		1495	1491		707			23G	BDO
7	d8	d17		1495	560		707			Anaerobic	BDO
8	d8	d17		1495	560		707			Anaerobic	BDO
9	d8	d17		1495	560		707			23G	BDO
10	d8	d17		1495	560	28	707			23G	BDO
11	pESC-H	pESC-L	pESC-U							Anaerobic	EV3
12	pESC-H	pESC-L	pESC-U							23G	EV3
13	d8	d16	d1	1495	1491	28	707	pflA	pflB	Anaerobic	BDO + pflAB
14	d8	d16	d1	1495	1491	28	707	pflA	pflB	Anaerobic	BDO + pflAB
15	d8	d16	d1	1495	1491	28	707	pflA	pflB	23G	BDO + pflAB
16	d8	d16	d1	1495	1491	28	707	pflA	pflB	23G	BDO + pflAB
17	d8	d17	d1	1495	560	28	707	pflA	pflB	Anaerobic	BDO + pflAB
18	d8	d17	d1	1495	560	28	707	pflA	pflB	Anaerobic	BDO + pflAB
19	d8	d17	d1	1495	560	28	707	pflA	pflB	23G	BDO + pflAB
20	d8	d17	d1	1495	560	28	707	pflA	pflB	23G	BDO + pflAB
21	d8	d16	d5	1495	1491	28	707	ALD6	acsm	Anaerobic	BDO + PDH
22	d8	d16	d5	1495	1491	28	707	ALD6	acsm	Anaerobic	BDO + PDH
23	d8	d16	d5	1495	1491	28	707	ALD6	acsm	23G	BDO + PDH
24	d8	d16	d5	1495	1491	28	707	ALD6	acsm	23G	BDO + PDH
25	d8	d17	d5	1495	560	28	707	ALD6	acsm	Anaerobic	BDO + PDH
26	d8	d17	d5	1495	560	28	707	ALD6	acsm	Anaerobic	BDO + PDH
27	d8	d17	d5	1495	560	28	707	ALD6	acsm	23G	BDO + PDH
28	d8	d17	d5	1495	560	28	707	ALD6	acsm	23G	BDO + PDH

TABLE 4

Plasmid 1	Plasmid 2	plasmid 3	gene 1	gene 2	gene 3	gene 4	gene 5	gene 6	Aeroation	Note
d9	d11								aerobic	EVC
d8	d17		1495	560	28	707			aerobic	BDO
d8	d17	d5	1495	560	28	707	1849	1845A	aerobic	BDO + PDH
d8	d14		1495	1502	28	707			aerobic	BDO
d8	d14	d5	1495	1502	28	707	1849	1845A	aerobic	BDO + PDH

In Table 3, colonies were inoculated in 5 ml of 2% glucose medium with corresponding amino acid dropouts and cultured at 30 degree for approximately 48 hrs. Cells were briefly spun down and re-suspended in 2 ml fresh 2% glucose medium with tween-80 and ergosterol added. Resuspended cultures were added to 10 ml fresh glucose medium in 20 ml bottles to obtain a starting OD of 0.2. For anaerobic cultures, the bottles containing cultures were vacuumed and filled with nitrogen. For micro-aerobic growth, a 23G needle was inserted. All the cultures were incubated at 30 degree with shaking for 24 hours. In Table 4, the experiment was carried out in a 96-well plate and cells grown aerobically in 1.2 ml of medium with varying glucose and acetate concentrations (5% glucose, 10% glucose, 5% glucose+50 mM acetate, and 10% glucose+50 mM acetate).

Concentrations of glucose, 1,3-BDO, alcohols, and other organic acid byproducts in the culture supernatant were determined by HPLC using an HPX-87H column (BioRad).

MI-FAE cycle and termination pathway genes were tested with or without pflAB or PDH bypass. As shown in FIGS. 9-11, these constructs produced 0.3-3.35 mM 1,3-BDO in yeast *S. cerevisiae* BY4741, and ethanol was produced in the tested samples tested. The PDH bypass (here, overexpression of ALD6 and acs or acsm genes) improved production of 1,3-BDO.

Example XIV

Enzymatic Activity of 1,3-Butanediol Pathway Enzymes

This example describes the detection of 1,3-BDO pathway enzyme activity using in vitro assays.

Activity of the heterologous enzymes was tested in in vitro assays, using an internal yeast strain as the host for the plasmid constructs containing the pathway genes. Cells were grown aerobically in yeast media containing the appropriate amino acid for each construct. To obtain crude extracts for activity assays, cells were harvested by centrifugation. The pellets were resuspended in 0.1 mL 100 mM Tris pH 7.0 buffer containing protease inhibitor cocktail. Lysates were prepared using the method of bead beating for 3 min. Following bead beating, the solution was centrifuged at 14,000 rpm (Eppendorf centrifuge 5402) for 15 min at 4° C. Cell protein in the sample was determined using the method of Bradford et al., Anal. Biochem. 72:248-254 (1976), and specific enzyme assays conducted as described below.

Thiolase

Thiolase enzymes catalyze the condensation of two acetyl-CoA to form acetoacetyl-CoA. In the reaction, coenzyme A (CoA) is released and the free CoA can be detected using 5,5'-dithiobis-2-nitrobenzoic acid (DTNB) which absorbs at 410 nm upon reaction with CoA. Five thiolases were tested (see Example XIII, Table 1). Estimated specific activity in *E. coli* crude lysates is shown in FIGS. 13A and 13B.

Among the Thl that showed expressed protein, 1512 and 1502 demonstrated the highest specific activity for acetyl-CoA condensation activity in *E. coli* crude lysates.

Both 1491 and 560 were cloned in dual promoter yeast vectors with 1495, which is the 3-hydroxybutyryl-CoA dehydrogenase (see FIGS. 14A-14D). These thiolases were evaluated for acetyl-CoA condensation activity, and the data is shown in FIGS. 14A-14D. The results indicate that both 560 and 1491 demonstrate an initial burst of activity that is too fast to measure. However, after the initial enzyme rate, the condensation rate of 560 is greater than 1491. Thus, there is protein expression and active enzyme with the yeast dual promoter vectors as indicated by active thiolase activity observed in crude lysates.

3-Hydroxybutyryl-CoA Dehydrogenase (Hbd)

Acetoacetyl-CoA is metabolized to 3-hydroxybutyryl-CoA by 3-hydroxybutyryl-CoA dehydrogenase. The reaction requires oxidation of NADH, which can be monitored by fluorescence at an excitation wavelength at 340 nm and an emission at 460 nm. The oxidized form, NAD⁺, does not fluoresce. This detection strategy was used for all of the dehydrogenase steps. 1495, the Hbd from *Clostridium beijerinckii*, was assayed in the dual promoter yeast vectors that contained either 1491 (vector id=pY3Hd17) or 560 (vector id=pY3Hd16). See Table 1 for GenBank identifiers of each enzyme. The time course data is shown in FIG. 15.

The Hbd rate of 1495 containing 560 was much faster than 1491. The results provided in FIG. 16 show that the Hbd prefers NADH over NADPH. The Hbd enzyme appears to display the fastest catalytic activity among the four pathway enzymes in crude lysates. The Hbd enzyme, i.e. a 3-ketoacyl-CoA reductase, is an example of a MI-FAE cycle or MD-FAE cycle enzyme that preferentially reacts with an NADH cofactor.

Aldehyde Dehydrogenase (Ald)

An aldehyde reductase converts 3-hydroxybutyryl-CoA to 3-hydroxybutyraldehyde. This reaction requires NAD(P)H oxidation, which can be used to monitor enzyme activity. The Ald from *Lactobacillus brevis* (Gene ID 707) was cloned in a dual vector that contained the alcohol dehydrogenase from *Clostridium saccharoperbutylacetonicum* (Gene ID 28). These two enzymes were cloned in another dual promoter yeast vector containing a Leu marker.

The Ald activity data for crude lysates is shown in FIGS. 17A and 17B with a 707 lysate from *E. coli* used as a standard. The results indicate the 707 showed enzyme activity in yeast lysates that is comparable to the lysate from bacteria. In addition, the 707 gene product prefers NADH to NADPH as the cofactor. The 707 gene product, i.e. an acy-CoA reductase (aldehyde forming), is an example of a termination pathway enzyme that preferentially reacts with an NADH cofactor.

Alcohol Dehydrogenase (Adh)

1,3-BDO is formed by an alcohol dehydrogenase (Adh), which reduces 3-hydroxybutyraldehyde in the presence of NAD(P)H. The oxidation of NAD(P)H can be used to monitor the reaction as described above.

The evaluation of ADH (Gene 28) in the dual promoter vector with ALD (Gene 707) is shown in FIG. 18 with butyraldehyde, a surrogate substrate for 3-hydroxybutyraldehyde. The data indicate that Gene 28 have Adh activity similar to the no insert control (EV) with butyraldehyde and NADPH. This is likely caused by endogenous ADH enzymes present in yeast that may function in the same capability as 28.

In summary, candidates for the Thl, Hbd, Ald, and Adh to produce 1,3-BDO showed enzyme activity in yeast crude lysates for the dual promoter vectors constructed.

Example XV

Isopropanol Synthesis Pathway

This example describes enzymes for converting acetyl-CoA to isopropanol. Pathways are shown in FIG. 11. Enzymes for catalyzing steps T-Y are disclosed herein.

Isopropanol production was achieved in recombinant *E. coli* following expression of two heterologous genes from *C. acetobutylicum* (thl and adc encoding acetoacetyl-CoA thiolase and acetoacetate decarboxylase, respectively) and one from *C. beijerinckii* (adh encoding a secondary alcohol dehydrogenase), along with the increased expression of the native atoA and atoD genes which encode acetoacetyl-CoA:

acetate:CoA transferase activity (Hanai et al., *Appl. Environ. Microbiol.* 73:7814-7818 (2007)). The acetoacetyl-CoA thiolase (AtoB) enzymes are described herein.

Acetyl-CoA:acetyl-CoA Acyltransferase (Acetoacetyl-CoA thiolase)—Step V, FIG. 11

Acetoacetyl-CoA thiolase (also known as acetyl-CoA acetyltransferase) converts two molecules of acetyl-CoA into one molecule each of acetoacetyl-CoA and CoA. Exemplary acetoacetyl-CoA thiolase enzymes include the gene products of atoB from *E. coli* (Martin et al., *Nat. Biotechnol.* 21:796-802 (2003)), th1A and th1B from *C. acetobutylicum* (Hanai et al., *Appl. Environ. Microbiol.* 73:7814-7818 (2007); Winzer et al., *J. Mol. Microbiol. Biotechnol.* 2:531-541 (2000),

Acetoacetyl-CoA synthase (EC 2.3.1.194)—Step U, FIG. 11

Acetoacetyl-CoA can also be synthesized from acetyl-CoA and malonyl-CoA by acetoacetyl-CoA synthase (EC 2.3.1.194). This enzyme (FhsA) has been characterized in the soil bacterium *Streptomyces* sp. CL190 where it participates in mevalonate biosynthesis (Okamura et al., *PNAS USA* 107:11265-70 (2010)). As this enzyme catalyzes an essentially irreversible reaction, it is particularly useful for metabolic engineering applications for overproducing metabolites, fuels or chemicals derived from acetoacetyl-CoA such as long chain alcohols. Other acetoacetyl-CoA synthase genes can be identified by sequence homology to fhsA. Acyl-CoA synthase enzymes such as fhsA and homologs can be engineered or evolved to accept longer acyl-CoA substrates by methods known in the art.

Protein	GenBank ID	GI Number	Organism
fhsA	BAJ83474.1	325302227	<i>Streptomyces</i> sp. CL190
AB183750.1:11991 . . . 12971	BAD86806.1	57753876	<i>Streptomyces</i> sp. KO-3988
epzT	ADQ43379.1	312190954	<i>Streptomyces cinnamonensis</i>
ppzT	CAX48662.1	238623523	<i>Streptomyces anulatus</i>
O31_22085	ZP_09840373.1	378817444	<i>Nocardia brasiliensis</i>

and ERG10 from *S. cerevisiae* Hiser et al., *J. Biol. Chem.* 269:31383-31389 (1994)). These genes/proteins are identified in the Table below.

Gene	GenBank ID	GI Number	Organism
AtoB	NP_416728	16130161	<i>Escherichia coli</i>
Th1A	NP_349476.1	15896127	<i>Clostridium acetobutylicum</i>
Th1B	NP_149242.1	15004782	<i>Clostridium acetobutylicum</i>
ERG10	NP_015297	6325229	<i>Saccharomyces cerevisiae</i>

Acetyl-CoA carboxylase (6.4.1.2)—Step T, FIG. 11

The conversion of acetyl-CoA to malonyl-CoA can be carried out by acetyl-CoA carboxylase. The *E. coli* enzyme complex is composed of two catalytic units and one carrier protein, encoded by four different genes. The catalytic units are biotin carboxylase (6.3.4.14), a homodimer encoded by the accC gene, and acetyl-CoA carboxylase (ACCT), an $\alpha_2\beta_2$ tetramer, encoded by the accA and accD genes. The carrier protein is the biotin carboxyl carrier protein, a homodimer encoded by accB. Several such candidates can be found in US20120142979.

Gene	Accession number	GI Number	Organism
accA	AAC73296.1	1786382	<i>Escherichia coli</i> K-12
accB	AAC76287.1	1789653	<i>Escherichia coli</i> K-12
accC	AAC76288.1	1789654	<i>Escherichia coli</i> K-12
accD	AAC75376.1	1788655	<i>Escherichia coli</i> K-12
accA	CAD08690.1	16501513	<i>Salmonella enterica</i>
accB	CAD07894.1	16504441	<i>Salmonella enterica</i>
accC	CAD07895.1	16504442	<i>Salmonella enterica</i>
accD	CAD07598.1	16503590	<i>Salmonella enterica</i>
YMR207C	NP_013934.1	6323863	<i>Saccharomyces cerevisiae</i>
YNR016C	NP_014413.1	6324343	<i>Saccharomyces cerevisiae</i>
YGR037C	NP_011551.1	6321474	<i>Saccharomyces cerevisiae</i>
YKL182W	NP_012739.1	6322666	<i>Saccharomyces cerevisiae</i>
YPL231W	NP_015093.1	6325025	<i>Saccharomyces cerevisiae</i>
accA	ZP_00618306.1	69288468	<i>Kineococcus radiotolerans</i>
accB	ZP_00618387.1	69288621	<i>Kineococcus radiotolerans</i>
accC	ZP_00618040.1/ ZP_00618387.1	69287824/69288621	<i>Kineococcus radiotolerans</i>
accD	ZP_00618306.1	69288468	<i>Kineococcus radiotolerans</i>

Acetoacetyl-CoA Transferase—Step W, FIG. 11

The conversion of acetoacetyl-CoA to acetoacetate can be carried out by an acetoacetyl-CoA transferase. These enzymes conserve the energy stored in the CoA-ester bonds of acetoacetyl-CoA. Many transferases have broad specificity and thus may utilize CoA acceptors as diverse as acetate, succinate, propionate, butyrate, 2-methylacetoacetate, 3-ke-tohexanoate, 3-ketopentanoate, valerate, crotonate, 3-mercaptopropionate, propionate, vinylacetate, butyrate, among others. Acetoacetyl-CoA transferase catalyzes the conversion of acetoacetyl-CoA to acetoacetate while transferring the CoA moiety to a CoA acceptor molecule. Several exemplary transferase enzymes capable of catalyzing this transformation are provided below. These enzymes either naturally exhibit the desired acetoacetyl-CoA transferase activity or they can be engineered via directed evolution to accept acetoacetyl-CoA as a substrate with increased efficiency.

In one embodiment an exemplary acetoacetyl-CoA transferase is acetoacetyl-CoA:acetate-CoA transferase. This enzyme naturally converts acetate to acetyl-CoA while converting acetoacetyl-CoA to acetoacetate. In another embodiment, a succinyl-CoA:3-ketoacid CoA transferase (SCOT)

catalyzes the conversion of the 3-ketoacyl-CoA, acetoacetyl-CoA, to the 3-ketoacid, acetoacetate.

Acetoacetyl-CoA:acetyl-CoA transferase naturally converts acetoacetyl-CoA and acetate to acetoacetate and acetyl-CoA. This enzyme can also accept 3-hydroxybutyryl-CoA as a substrate or could be engineered to do so. Exemplary enzymes include the gene products of atoAD from *E. coli* (Hanai et al., *Appl Environ Microbiol* 73:7814-7818 (2007)), ctfAB from *C. acetobutylicum* (Jojima et al., *Appl Microbiol Biotechnol* 77:1219-1224 (2008)), and ctfAB from *Clostridium saccharoperbutylacetonicum* (Kosaka et al., *Biosci. Biotechnol Biochem.* 71:58-68 (2007)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
AtoA	P76459.1	2492994	<i>Escherichia coli</i>
AtoD	P76458.1	2492990	<i>Escherichia coli</i>
CtfA	NP_149326.1	15004866	<i>Clostridium acetobutylicum</i>
CtfB	NP_149327.1	15004867	<i>Clostridium acetobutylicum</i>
CtfA	AAP42564.1	31075384	<i>Clostridium saccharoperbutylacetonicum</i>
CtfB	AAP42565.1	31075385	<i>Clostridium saccharoperbutylacetonicum</i>

Succinyl-CoA:3-ketoacid-CoA transferase naturally converts succinate to succinyl-CoA while converting a 3-ketoacyl-CoA to a 3-ketoacid. Exemplary succinyl-CoA:3-ketoacid-CoA transferases are present in *Helicobacter pylori* (Corthesy-Theulaz et al., *J. Biol. Chem.* 272:25659-25667 (1997)), *Bacillus subtilis* (Stols et al., *Protein. Expr. Purif.* 53:396-403 (2007)), and *Homo sapiens* (Fukao et al., *Genomics* 68:144-151(2000); Tanaka et al., *Mol. Hum. Reprod* 8:16-23 (2002)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
HPAG1_0676	YP_627417	108563101	<i>Helicobacter pylori</i>
HPAG1_0677	YP_627418	108563102	<i>Helicobacter pylori</i>
ScoA	NP_391778	16080950	<i>Bacillus subtilis</i>
ScoB	NP_391777	16080949	<i>Bacillus subtilis</i>
OXCT1	NP_000427	4557817	<i>Homo sapiens</i>
OXCT2	NP_071403	11545841	<i>Homo sapiens</i>

Additional suitable acetoacetyl-CoA transferases are encoded by the gene products of cat1, cat2, and cat3 of *Clostridium kluyveri*. These enzymes have been shown to exhibit succinyl-CoA, 4-hydroxybutyryl-CoA, and butyryl-CoA transferase activity, respectively (Seedorf et al., *Proc. Natl. Acad. Sci. USA* 105:2128-2133 (2008); Sohling and Gottschalk, *J Bacteriol* 178:871-880 (1996)). Similar CoA

transferase activities are also present in *Trichomonas vaginalis* (van Grinsven et al., *J. Biol. Chem.* 283:1411-1418 (2008)) and *Trypanosoma brucei* (Riviere et al., *J. Biol. Chem.* 279:45337-45346 (2004)). Yet another transferase capable of the desired conversions is butyryl-CoA:acetoacetate CoA-transferase. Exemplary enzymes can be found in *Fusobacterium nucleatum* (Barker et al., *J. Bacteriol.* 152 (1):201-7 (1982)), *Clostridium* SB4 (Barker et al. *J. Biol. Chem.* 253(4):1219-25 (1978)), and *Clostridium acetobutylicum* (Wiesenbom et al., *Appl. Environ. Microbiol.* 55(2):323-9 (1989)). Although specific gene sequences were not provided for butyryl-CoA:acetoacetate CoA-transferase in these references, the genes FN0272 and FN0273 have

been annotated as a butyrate-acetoacetate CoA-transferase (Kapatral et al., *J. Bact.* 184(7) 2005-2018 (2002)). Homologs in *Fusobacterium nucleatum* such as FN1857 and FN1856 also likely have the desired acetoacetyl-CoA transferase activity. FN1857 and FN1856 are located adjacent to many other genes involved in lysine fermentation and are thus very likely to encode an acetoacetate:butyrate CoA transferase (Kreimeyer, et al., *J Biol. Chem.* 282 (10) 7191-7197 (2007)). Additional candidates from *Porphyromonas gingivalis* and *Thermoanaerobacter tengcongensis* can be identified in a similar fashion (Kreimeyer, et al., *J. Biol. Chem.* 282 (10) 7191-7197 (2007)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
Cat1	P38946.1	729048	<i>Clostridium kluyveri</i>
Cat2	P38942.2	1705614	<i>Clostridium kluyveri</i>
Cat3	EDK35586.1	146349050	<i>Clostridium kluyveri</i>
TVAG_395550	XP_001330176	123975034	<i>Trichomonas vaginalis</i> G3
Tb11.02.0290	XP_828352	71754875	<i>Trypanosoma brucei</i>
FN0272	NP_603179.1	19703617	<i>Fusobacterium nucleatum</i>
FN0273	NP_603180.1	19703618	<i>Fusobacterium nucleatum</i>
FN1857	NP_602657.1	19705162	<i>Fusobacterium nucleatum</i>
FN1856	NP_602656.1	19705161	<i>Fusobacterium nucleatum</i>
PG1066	NP_905281.1	34540802	<i>Porphyromonas gingivalis</i> W83
PG1075	NP_905290.1	34540811	<i>Porphyromonas gingivalis</i> W83
TTE0720	NP_622378.1	20807207	<i>Thermoanaerobacter tengcongensis</i> MB4
TTE0721	NP_622379.1	20807208	<i>Thermoanaerobacter tengcongensis</i> MB4

Acetoacetyl-CoA can be hydrolyzed to acetoacetate by acetoacetyl-CoA hydrolase. Many CoA hydrolases (EC 3.1.2.1) have broad substrate specificity and are suitable enzymes for these transformations either naturally or following enzyme engineering. Though the sequences were not reported, several acetoacetyl-CoA hydrolases were identified in the cytosol and mitochondrion of the rat liver (Aragon

and Lowenstein, *J. Biol. Chem.* 258(8):4725-4733 (1983)). Additionally, an enzyme from *Rattus norvegicus* brain (Robinson et al., *Biochem. Biophys. Res. Commun.* 71:959-965 (1976)) can react with butyryl-CoA, hexanoyl-CoA and malonyl-CoA. The acot12 enzyme from the rat liver was shown to hydrolyze C2 to C6 acyl-CoA molecules (Suematsu et al., *Eur. J. Biochem.* 268:2700-2709 (2001)). Though its sequence has not been reported, the enzyme from the mitochondrion of the pea leaf showed activity on acetyl-CoA, propionyl-CoA, butyryl-CoA, palmitoyl-CoA, oleoyl-CoA, succinyl-CoA, and crotonyl-CoA (Zeihner and Randall, *Plant. Physiol.* 94:20-27 (1990)). Additionally, a glutamate CoA-transferase from *Acidaminococcus fermentans* was transformed by site-directed mutagenesis into an acyl-CoA hydrolase with activity on glutaryl-CoA, acetyl-CoA and 3-butenoyl-CoA (Mack and Buckel, *FEBS Lett.* 405:209-212 (1997)). This indicates that the enzymes encoding acetoacetyl-CoA transferases can also be used as hydrolases with certain mutations to change their function. The acetyl-CoA hydrolase, ACH1, from *S. cerevisiae* represents another candidate hydrolase (Buu et al., *J. Biol. Chem.* 278:17203-17209 (2003)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
Acot12	NP_570103.1	18543355	<i>Rattus norvegicus</i>
GctA	CAA57199	559392	<i>Acidaminococcus fermentans</i>
GctB	CAA57200	559393	<i>Acidaminococcus fermentans</i>
ACH1	NP_009538	6319456	<i>Saccharomyces cerevisiae</i>

Another candidate hydrolase is the human dicarboxylic acid thioesterase, acot8, which exhibits activity on glutaryl-CoA, adipyl-CoA, suberyl-CoA, sebacyl-CoA, and dodecanedioyl-CoA (Westin et al., *J. Biol. Chem.* 280:38125-38132 (2005)) and the closest *E. coli* homolog, tesB, which can also hydrolyze a broad range of CoA thioesters (Naggert et al., *J. Biol. Chem.* 266:11044-11050 (1991)) including 3-hydroxybutyryl-CoA (Tseng et al., *Appl. Environ. Microbiol.* 75(10):3137-3145 (2009)). A similar enzyme has also been characterized in the rat liver (Deana, *Biochem. Int.* 26:767-773 (1992)). Other potential *E. coli* thioester hydrolases include the gene products of tesA (Bonner and Bloch, *J. Biol. Chem.* 247:3123-3133 (1972)), ybgC (Kuznetsova et al., *FEMS Microbiol. Rev.* 29:263-279 (2005); Zhuang et al., *FEBS Lett.* 516:161-163 (2002)), pad (Song et al., *J. Biol. Chem.* 281:11028-11038 (2006)), and ybdB (Leduc et al., *J. Bacteriol.* 189:7112-7126 (2007)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
Acot8	CAA15502	3191970	<i>Homo sapiens</i>
TesB	NP_414986	16128437	<i>Escherichia coli</i>
Acot8	NP_570112	51036669	<i>Rattus norvegicus</i>
TesA	NP_415027	16128478	<i>Escherichia coli</i>
YbgC	NP_415264	16128711	<i>Escherichia coli</i>
Paal	NP_415914	16129357	<i>Escherichia coli</i>
YbdB	NP_415129	16128580	<i>Escherichia coli</i>

Additional hydrolase enzymes include 3-hydroxyisobutyryl-CoA hydrolase which has been described to efficiently catalyze the conversion of 3-hydroxyisobutyryl-CoA to 3-hydroxyisobutyrate during valine degradation (Shimomura et al., *J. Biol. Chem.* 269:14248-14253 (1994)). Genes encoding this enzyme include hibch of *Rattus norvegicus* (Shimomura et al., supra (1994); Shimomura et al., *Methods*

Enzymol. 324:229-240 (2000)) and *Homo sapiens* (Shimomura et al., supra (1994)). Candidate genes by sequence homology include hibch of *Saccharomyces cerevisiae* and BC_2292 of *Bacillus cereus*. BC_2292 was shown to demonstrate 3-hydroxybutyryl-CoA hydrolase activity and function as part of a pathway for 3-hydroxybutyrate synthesis when engineered into *Escherichia coli* (Lee et al., *Appl. Microbiol. Biotechnol.* 79:633-641 (2008)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
Hibch	Q5XIE6.2	146324906	<i>Rattus norvegicus</i>
Hibch	Q6NVY1.2	146324905	<i>Homo sapiens</i>
Hibch	P28817.2	2506374	<i>Saccharomyces cerevisiae</i>
BC_2292	AP09256	29895975	<i>Bacillus cereus</i> ATCC 14579

The hydrolysis of acetoacetyl-CoA can alternatively be carried out by a single enzyme or enzyme complex that exhibits acetoacetyl-CoA hydrolase activity. This activity enables the net hydrolysis of the CoA-ester of either molecule, and in some cases, results in the simultaneous generation of ATP. For example, the product of the LSC1 and LSC2 genes of *S. cerevisiae* and the sucC and sucD genes of *E. coli* naturally form a succinyl-CoA synthetase complex that catalyzes the formation of succinyl-CoA from succinate with the concomitant consumption of one ATP, a reaction which is reversible in vivo (Grays et al., U.S. Pat. No. 5,958,745, filed Sep. 28, 1999). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
SucC	NP_415256.1	16128703	<i>Escherichia coli</i>
SucD	AAC73823.1	1786949	<i>Escherichia coli</i>
LSC1	NP_014785	6324716	<i>Saccharomyces cerevisiae</i>
LSC2	NP_011760	6321683	<i>Saccharomyces cerevisiae</i>

Additional exemplary CoA-ligases include the rat dicarboxylate-CoA ligase for which the sequence is yet uncharacterized (Vamecq et al., *Biochemical J.* 230:683-693 (1985)), either of the two characterized phenylacetate-CoA ligases from *P. chrysogenum* (Lamas-Maceiras et al., *Biochem. J.* 395:147-155 (2005); Wang et al., *Biochem. Biophys. Res. Commun.* 360(2):453-458 (2007)), the phenylacetate-CoA ligase from *Pseudomonas putida* (Martinez-Blanco et al., *J. Biol. Chem.* 265:7084-7090 (1990)), and the 6-carboxyhexanoate-CoA ligase from *Bacillus subtilis* (Bower et al., *J. Bacteriol.* 178(14):4122-4130 (1996)). Additional candidate enzymes are acetoacetyl-CoA synthetases from *Mus musculus* (Hasegawa et al., *Biochim. Biophys. Acta* 1779:414-419 (2008)) and *Homo sapiens* (Ohgami et al., *Biochem. Pharmacol.* 65:989-994 (2003)), which naturally catalyze the ATP-dependant conversion of acetoacetate into acetoacetyl-CoA. 4-Hydroxybutyryl-CoA synthetase activity has been demonstrated in *Metallosphaera sedula* (Berg et al., *Science* 318:1782-1786 (2007)). This function has been tentatively assigned to the Msed_1422 gene. Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
Phl	CAJ15517.1	77019264	<i>Penicillium chrysogenum</i>
PhlB	ABS19624.1	152002983	<i>Penicillium chrysogenum</i>
PaaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>
BioW	NP_390902.2	50812281	<i>Bacillus subtilis</i>
AACS	NP_084486.1	21313520	<i>Mus musculus</i>
AACS	NP_076417.2	31982927	<i>Homo sapiens</i>
Msed_1422	YP_001191504	146304188	<i>Metallosphaera sedula</i>

ADP-forming acetyl-CoA synthetase (ACD, EC 6.2.1.13) is another candidate enzyme that couples the conversion of acyl-CoA esters to their corresponding acids with the concurrent synthesis of ATP. Several enzymes with broad substrate specificities have been described in the literature. ACDI from *Archaeoglobus fulgidus*, encoded by AF1211, was shown to operate on a variety of linear and branched-chain substrates including acetyl-CoA, propionyl-CoA, butyryl-CoA, acetate, propionate, butyrate, isobutyrate, isovalerate, succinate, fumarate, phenylacetate, indoleacetate (Musfeldt et al., *J. Bacteriol.* 184:636-644 (2002)). The enzyme from *Haloarcula marismortui* (annotated as a succinyl-CoA synthetase) accepts propionate, butyrate, and branched-chain acids (isovalerate and isobutyrate) as substrates, and was shown to operate in the forward and reverse directions (Brasen et al., *Arch. Microbiol.* 182:277-287 (2004)). The ACD encoded by PAE3250 from hyperthermophilic crenarchaeon *Pyrobaculum aerophilum* showed the broadest substrate range of all characterized ACDs, reacting with acetyl-CoA, isobutyryl-CoA (preferred substrate) and phenylacetyl-CoA (Brasen et al., supra (2004)). The enzymes from *A. fulgidus*, *H. marismortui* and *P. aerophilum* have all been cloned, functionally expressed, and characterized in *E. coli* (Musfeldt et al., supra; Brasen et al., supra (2004)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
AF1211	NP_070039.1	11498810	<i>Archaeoglobus fulgidus</i> DSM 4304
Scs	YP_135572.1	55377722	<i>Haloarcula marismortui</i> ATCC 43049
PAE3250	NP_560604.1	18313937	<i>Pyrobaculum aerophilum</i> str. IM2

An alternative method for removing the CoA moiety from acetoacetyl-CoA is to apply a pair of enzymes such as a phosphate-transferring acyltransferase and a kinase to impart acetoacetyl-CoA synthetase activity. Exemplary names for these enzymes include phosphotransacetoacetylase/acetoacetate kinase which can remove the CoA moiety from acetoacetyl-CoA. This general activity enables the net hydrolysis of the CoA-ester of either molecule with the simultaneous generation of ATP. For example, the butyrate kinase (buk)/phosphotransbutyrylase (ptb) system from *Clostridium acetobutylicum* has been successfully applied to remove the CoA group from 3-hydroxybutyryl-CoA when functioning as part of a pathway for 3-hydroxybutyrate synthesis (Tseng et al., *Appl. Environ. Microbiol.* 75(10): 3137-3145 (2009)). Specifically, the ptb gene from *C. acetobutylicum* encodes an enzyme that can convert an acyl-CoA into an acyl-phosphate (Walter et al. *Gene* 134(1): p. 107-11(1993)); Huang et al. *J Mol Microbiol Biotechnol* 2(1): p. 33-38 (2000). Additional ptb genes can be found in butyrate-producing bacterium L2-50 (Louis et al. *J.Bacteriol.* 186:

2099-2106 (2004)) and *Bacillus megaterium* (Vazquez et al. *Curr.Microbiol* 42:345-349 (2001)). Additional exemplary phosphate-transferring acyltransferases include phosphotransacetylase, encoded by pta. The pta gene from *E. coli* encodes an enzyme that can convert acetyl-CoA into acetyl-phosphate, and vice versa (Suzuki, T. *Biochim.Biophys.Acta* 191:559-569 (1969)). This enzyme can also utilize propionyl-CoA instead of acetyl-CoA forming propionate in the process (Hesslinger et al. *Mol.Microbiol* 27:477-492 (1998)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
Pta	NP_416800.1	16130232	<i>Escherichia coli</i>
Ptb	NP_349676	15896327	<i>Clostridium acetobutylicum</i>
Ptb	AAR19757.1	38425288	butyrate-producing bacterium L2-50
Ptb	CAC07932.1	10046659	<i>Bacillus megaterium</i>

Exemplary kinases include the *E. coli* acetate kinase, encoded by ackA (Skarstedt and Silverstein *J.Bid Chem.* 251:6775-6783 (1976)), the *C. acetobutylicum* butyrate kinases, encoded by buk1 and buk2 ((Walter et al. *Gene* 134(1):107-111(1993); Huang et al. *J Mol Microbiol Biotechnol* 2(1):33-38 (2000)), and the *E. coli* gamma-glutamyl kinase, encoded by proB (Smith et al. *J.Bacteriol.* 157:545-551 (1984)). These enzymes phosphorylate acetate, butyrate, and glutamate, respectively. The ackA gene product from *E. coli* also phosphorylates propionate (Hesslinger et al. *Mol.Microbiol* 27:477-492 (1998)). Information related to these proteins and genes is shown below:

Protein	GENBANK ID	GI NUMBER	ORGANISM
AckA	NP_416799.1	16130231	<i>Escherichia coli</i>
Buk1	NP_349675	15896326	<i>Clostridium acetobutylicum</i>
Buk2	Q97III	20137415	<i>Clostridium acetobutylicum</i>
ProB	NP_414777.1	16128228	<i>Escherichia coli</i>

Acetoacetate Decarboxylase—Step X, FIG. 11

Acetoacetate decarboxylase converts acetoacetate into carbon dioxide and acetone. Exemplary acetoacetate decarboxylase enzymes are encoded by the gene products of adc from *C. acetobutylicum* (Petersen and Bennett, *Appl. Environ. Microbiol.* 56:3491-3498 (1990) and adc from *Clostridium saccharoperbutylacetonicum* (Kosaka et al., *Biosci. Biotechnol. Biochem.* 71:58-68 (2007)). The enzyme from *C. beijerinckii* can be inferred from sequence similarity.

Protein	GenBank ID	GI Number	Organism
Adc	NP_149328.1	15004868	<i>Clostridium acetobutylicum</i>
Adc	AAP42566.1	31075386	<i>Clostridium saccharoperbutylacetonicum</i>
Adc	YP_001310906.1	150018652	<i>Clostridium beijerinckii</i>

Acetone Reductase or Isopropanol Dehydrogenase—Step Y, FIG. 11

The final step in the isopropanol synthesis pathway involves the reduction of acetone to isopropanol. Exemplary alcohol dehydrogenase enzymes capable of this transformation include adh from *C. beijerinckii* (Jojima et al., *Appl. Microbiol. Biotechnol.* 77:1219-1224 (2008); Hanai et al., *Appl. Environ. Microbiol.* 73:7814-7818 (2007) and adh from *Thermoanaerobacter brockii* (Hanai et al., supra; Peretz et al., *Anaerobe* 3:259-270 (1997)). Additional characterized enzymes include alcohol dehydrogenases from *Ralstonia eutropha* (formerly *Alcaligenes eutrophus*) (Steinbuchel and Schlegel, *Eur. J. Biochem.* 141:555-564 (1984) and *Phytomonas* species (Uttaro and Oppendoes, *Mol. Biochem. Parasitol.* 85: 213-219 (1997)).

Protein	GenBank ID	GI Number	Organism
Adh	P14941.1	113443	<i>Thermoanaerobacter brockii</i>
Adh	AAA23199.2	60592974	<i>Clostridium beijerinckii</i>
Adh	YP_299391.1	73539024	<i>Ralstonia eutropha</i>
iPDH	AAP39869.1	31322946	<i>Phytomonas</i> sp.

Example XVI

Production of Fatty Alcohols, Fatty Aldehydes, and Fatty Acids via the Fatty Acyl-ACP Elongation (FAACPE) Cycle and Termination Pathways

This example describes enzymes for converting acetyl-CoA to products of interest such as fatty alcohols, fatty aldehydes, and fatty acids through the FAACPE cycle and termination pathways. Pathways are shown in FIG. 12. Enzymes for catalyzing steps A-O are disclosed herein.

Fatty acid biosynthesis requires several steps. The initiation of fatty acid biosynthesis requires the conversion of acetyl-CoA to malonyl-CoA by an enzyme called acetyl CoA carboxylase (Step A, FIG. 12). Malonyl-CoA is then converted into malonyl-ACP by a CoA-ACP transacylase (Step B, FIG. 12). This is the substrate for the first step of the elongation cycle, namely the condensation step (Steps C and H, FIG. 12). This reaction is called β -keto acyl ACP synthase. The keto group is then reduced to a hydroxyl group by a β -keto acyl ACP reductase (Step E, FIG. 12). The next step

is a dehydration step that involves conversion of the hydroxyl group into an enoyl moiety (Step F, FIG. 12). This is catalyzed by a β -hydroxy acyl ACP dehydratase. The enoyl group is finally reduced to form an acyl-ACP by enoyl-ACP reductase (Step G, FIG. 12). At this point, the acyl-ACP can either be further elongated by the condensation reaction carried out by the β -keto acyl ACP synthase or can be converted into a fatty acid by a thioesterase (Step I, FIG. 12). The acid can be further activated to acyl-CoA by an acyl-CoA synthetase or ligase (Step K, FIG. 12) or can be directly reduced to an acid by a carboxylic acid reductase (CAR) (Step O, FIG. 12). The acyl-CoA can have multiple fates too and can either be converted into an aldehyde by acyl-CoA reductase (Step L, FIG. 12) or can be converted into a fatty alcohol by a fatty alcohol forming acyl-CoA reductase (FAR) (Step N, FIG. 12). The fatty aldehyde can also be converted into a fatty alcohol by a fatty aldehyde reductase (Step M, FIG. 12).

There are two basic types of fatty acid (FAS) biosynthesis mechanisms. The type I system is found in mammals and lower eukaryotes. The mammalian system consists of a single gene product that contains all of the reaction centers required to produce a fatty acid, e.g., the fatty acid synthase from *Homo sapiens*. In lower eukaryotes such as yeast, fatty acid synthase function is catalyzed by two genes (FAS I and FAS II), whose polypeptides form a eukaryotic complex.

Type II systems are found in bacteria and plants (White et al. (2005), *The structural biology of type II fatty acid biosynthesis*, *Annu Rev Biochem.* 74 (791-831)) among other organisms. The reactions in these systems are catalyzed by a series of individual soluble proteins that are each encoded by a discrete gene, and the pathway intermediates are transferred between the enzymes as thioesters of a holo acyl carrier protein (ACP).

Acetyl-CoA Carboxylase (6.4.1.2)—Step A, FIG. 12

The conversion of acetyl-CoA to malonyl-CoA can be carried out by acetyl-CoA carboxylase. The *E. coli* enzyme complex is composed of two catalytic units and one carrier protein, encoded by four different genes. The catalytic units are biotin carboxylase (6.3.4.14), a homodimer encoded by the accC gene, and acetyl-CoA carboxylase (ACCT), an $\alpha_2\beta_2$ tetramer, encoded by the accA and accD genes. The carrier protein is the biotin carboxyl carrier protein, a homodimer encoded by accB. Several such candidates can be found in US20120142979.

Gene	Accession number	GI Number	Organism
accA	AAC73296.1	1786382	<i>Escherichia coli</i> K-12
accB	AAC76287.1	1789653	<i>Escherichia coli</i> K-12
accC	AAC76288.1	1789654	<i>Escherichia coli</i> K-12
accD	AAC75376.1	1788655	<i>Escherichia coli</i> K-12
accA	CAD08690.1	16501513	<i>Salmonella enterica</i>
accB	CAD07894.1	16504441	<i>Salmonella enterica</i>
accC	CAD07895.1	16504442	<i>Salmonella enterica</i>
accD	CAD07598.1	16503590	<i>Salmonella enterica</i>
YMR207C	NP_013934.1	6323863	<i>Saccharomyces cerevisiae</i>
YNR016C	NP_014413.1	6324343	<i>Saccharomyces cerevisiae</i>
YGR037C	NP_011551.1	6321474	<i>Saccharomyces cerevisiae</i>
YKL182W	NP_012739.1	6322666	<i>Saccharomyces cerevisiae</i>

-continued

Gene	Accession number	GI Number	Organism
YPL231W	NP_015093.1	6325025	<i>Saccharomyces cerevisiae</i>
accA	ZP_00618306.1	69288468	<i>Kineococcus radiotolerans</i>
accB	ZP_00618387.1	69288621	<i>Kineococcus radiotolerans</i>
accC	ZP_00618040.1/ ZP_00618387.1	69287824/69288621	<i>Kineococcus radiotolerans</i>
accD	ZP_00618306.1	69288468	<i>Kineococcus radiotolerans</i>

CoA-ACP Acyltransferase (2.3.1.f)—Step B, FIG. 12

The exchange of an ACP moiety for a CoA is catalyzed by enzymes in EC class 2.3.1. Activation of acetyl-CoA to acetyl-ACP and malonyl-CoA to malonyl-ACP are also catalyzed by a CoA:ACP acyltransferase. Enzymes with CoA-ACP acyltransferase activity include acetyl-CoA:ACP transacylase (EC 2.3.1.38) and malonyl-CoA:ACP transacylase (EC 2.3.1.39).

The FabH (KASIII) enzyme of *E. coli* functions as an acyl-CoA:ACP transacylase, in addition to its primary activity of forming acetoacetyl-ACP. Butyryl-ACP is accepted as an alternate substrate of FabH (Prescott et al, Adv. Enzymol. Relat. Areas Mol., 36:269-311(1972)). Acetyl-CoA:ACP transacylase enzymes from *Plasmodium falciparum* and *Streptomyces avermitilis* have been heterologously expressed in *E. coli* (Lobo et al, *Biochem* 40:11955-64 (2001)). A synthetic KASIII (FabH) from *P. falciparum* expressed in a fabH-deficient *Lactococcus lactis* host was able to complement the native fades activity (Du et al, AEM 76:3959-66 (2010)). The acetyl-CoA:ACP transacylase enzyme from *Spinacia oleracea* accepts other acyl-ACP molecules as substrates, including butyryl-ACP (Shimakata et al, Methods Enzym 122:53-9 (1986)). The sequence of this enzyme has not been determined to date. Malonyl-CoA: ACP transacylase enzymes include FabD of *E. coli* and *Brassica napus* (Verwoert et al, J Bacteriol, 174:2851-7 (1992); Simon et al, FEBS Lett 435:204-6 (1998)). FabD of *B. napus* was able to complement fabD-deficient *E. coli*. The multifunctional eukaryotic fatty acid synthase enzyme complexes (described in EC 2.3.1.) also catalyze this activity. More exemplary gene candidates can be found in WO2007136762A2.

Gene	GenBank ID	GI Number	Organism
fabH	AAC74175.1	1787333	<i>Escherichia coli</i>
fadA	NP_824032.1	29829398	<i>Streptomyces avermitilis</i>
fabH	AAC63960.1	3746429	<i>Plasmodium falciparum</i>
Synthetic construct	ACX34097.1	260178848	<i>Plasmodium falciparum</i>
fabH	CAL98359.1	124493385	<i>Lactococcus lactis</i>
fabD	AAC74176.1	1787334	<i>Escherichia coli</i>
fabD	CAB45522.1	5139348	<i>Brassica napus</i>
fabD	ZP_00617602.1	69286751	<i>Kineococcus radiotolerans</i>
fabD	YP_388786.1	78357337	<i>Desulfovibrio alaskensis</i>
fabD	YP_425507	83591755	<i>Rhodospirillum rubrum</i>

Acyl-ACP C-acyltransferase (Decarboxylating) or β -ketoacyl-ACP Synthase (2.3.1.e)—Steps C, D and H, FIG. 12

Acetoacetyl-ACP is formed from malonyl-ACP and either acetyl-CoA or acetyl-ACP. *E. coli* has three ketoacyl-ACP synthases (KAS enzymes), KAS I, KAS II and KAS III, encoded by fabB, fabF and fabH respectively. FabH (KAS III), the key enzyme of initiation of fatty acid biosynthesis in *E. coli*, is selective for the formation of acetoacetyl-ACP from acetyl-CoA and malonyl-ACP. Some gene candidates for this step are shown below.

Gene	GenBank ID	GI Number	Organism
fabH	AAC74175.1	1787333	<i>Escherichia coli</i>
fabH	ZP_00618003.1	69287672	<i>Kineococcus radiotolerans</i>
fabH	YP_388920.1	7835747	<i>Desulfovibrio alaskensis</i>
fabH	YP_425507.1	83591755	<i>Rhodospirillum rubrum</i>

Alternately, acetyl-CoA can first be activated to acetyl-ACP and subsequently condensed to acetoacetyl-ACP by two enzymes, acetyl-CoA:ACP transacylase (EC 2.3.1.38) and acetoacetyl-ACP synthase (EC 2.3.1.41). Acetyl-CoA: ACP transacylase converts acetyl-CoA and an acyl carrier protein to acetyl-ACP, releasing CoA. Enzyme candidates for acetyl-CoA:ACP transacylase are described in section EC 2.3.1.f above. Acetoacetyl-ACP synthase enzymes catalyze the condensation of acetyl-ACP and malonyl-ACP. This activity is catalyzed by FabF and FabB of *E. coli*, as well as the multifunctional eukaryotic fatty acid synthase enzyme complexes described in EC 2.3.1.g. FabB and FabF catalyze the condensation of malonyl-ACP with acyl-ACP substrates (β -ketoacyl-ACP synthase activity) and function primarily in fatty acid elongation. Specifically, a β -ketoacyl-ACP synthase catalyzes the conversion of a saturated fatty acyl ACP and malonyl-ACP into 3-oxoacyl-ACP that is 2 carbons longer than the substrate fatty acyl ACP. When it reacts with acetyl-ACP, it participates in fatty acid initiation. The *Bacillus subtilis* KAS enzymes are similar to FabH but are less selective, accepting branched acyl-CoA substrates (Choi et al, *J Bacteriol* 182:365-70 (2000)).

Gene	GenBank ID	GI Number	Organism
fabB	AAC75383.1	1788663	<i>Escherichia coli</i>
fabF	AAC74179.1	1787337	<i>Escherichia coli</i>
FabHA	NP_389015.1	16078198	<i>Bacillus subtilis</i>
FabHB	NP_388898.1	16078081	<i>Bacillus subtilis</i>

More exemplary gene candidates for acyl-ACP C-acyl transferase can be found in WO2007136762A2 (Production of fatty acids and derivatives thereof). Some of the enzymes

listed below are from US20110250663 (Methods and compositions related to fatty alcohol biosynthetic enzymes). Several more keto-acyl synthases have been identified in these applications. Exemplary acyl-ACP synthases from *E. coli* are described below.

Gene	GenBank ID	GI number	Organism
fabB	ACY27486.1	262176863	<i>Escherichia coli</i> LW1655F+
fabF	ACY27487	262176865	<i>Escherichia coli</i> LW1655F+
fadJ	ACX38989.1	260448567	<i>Escherichia coli</i> DH1
xerC	ACX41768.1	260451346	<i>Escherichia coli</i> DH1
vqeF	ACX38529.1	260448107	<i>Escherichia coli</i> DH1
murQ	ACX38907.1	260448485	<i>Escherichia coli</i> DH1

Oxidoreductase (Oxo to Alcohol) (1.1.1.a)—Step E, FIG. 12

The reduction of 3-oxoacyl-ACP to 3-hydroxyacyl-ACP is catalyzed by 3-oxoacyl-ACP reductase (EC 1.1.1.100). The *E. coli* 3-oxoacyl-ACP reductase is encoded by *fabG*. Key residues responsible for binding the acyl-ACP substrate to the enzyme have been elucidated (Zhang et al, *J Biol Chem* 278:52935-43 (2003)). Additional enzymes with this activity have been characterized in *Bacillus anthracis* (Zaccai et al, *Prot Struct Funct Gen* 70:562-7 (2008)) and *Mycobacterium tuberculosis* (Gurvitz, *Mol Genet Genomics* 282:407-16 (2009)). The beta-ketoacyl reductase (KR) domain of eukaryotic fatty acid synthase also catalyzes this activity (Smith, *FASEB J*, 8:1248-59 (1994)). While many *FabG* enzymes preferentially utilize NADH, NADH-dependent *FabG* enzymes also known in the art and are shown in the table below (Javidpour et al, *AEM* 80: 597-505 (2014)).

Gene	GenBank ID	GI Number	Organism
fabG	P0AEK2.1	84028081	<i>Escherichia coli</i>
fabG	AAP27717.1	30258498	<i>Bacillus anthracis</i>
FabG1	NP_215999.1	15608621	<i>Mycobacterium tuberculosis</i>
FabG4	YP_003030167.1	253797166	<i>Mycobacterium tuberculosis</i>
FabG	EDM75366.1	149815845	<i>Plesiocystis Pacifica</i>
FabG	WP_018008474.1	516633699	<i>Cupriavidus Taiwanensis</i>
FabG	WP_012242413.1	501199395	<i>Acholeplasma Laidlawii</i>
FabG	EDL65432.1	148851283	<i>Bacillus</i> sp SG-1

Hydro-lyase (4.2.1.a)—Step F, FIG. 12

3-Hydroxyacyl-ACP dehydratase enzymes catalyze the conversion of 3-hydroxyacyl-ACP to trans-2-enoyl-ACP. Enzymes with this activity include *FabA* and *FabZ* of *E. coli*, which possess overlapping broad substrate specificities (Heath, *J Biol Chem* 271:1833-6 (1996)). Fatty acid synthase complexes, described above, also catalyze this reaction. The *FabZ* protein from *Plasmodium falciparum* has been crystallized (Kostrew et al, *Protein Sci* 14:1570-80 (2005)). Additional candidates are the mitochondria β -hydroxyacyl-ACP dehydratase encoded by *Htd2p* in yeast and *TbHTD2* in *Homo sapiens* and *Trypanosoma brucei* (Kastanoitis et al, *Mol Micro* 53:1407-21 (2004); Kaija et al, *FEBS Lett* 582:729-33 (2008)).

Gene	GenBank ID	GI Number	Organism
fabA	AAC74040.1	1787187	<i>Escherichia coli</i>
fabZ	AAC73291.1	1786377	<i>Escherichia coli</i>
PfFabZ	AAK83685.1	15080870	<i>Plasmodium falciparum</i>
Htd2p	NP_011934.1	6321858	<i>Saccharomyces cerevisiae</i>
HTD2	P86397.1	281312149	<i>Homo sapiens</i>

Enoyl ACP Reductase (13.1.a)—Step G, FIG. 12

Enoyl-ACP reductase catalyzes the formation of a saturated acyl-ACP by an NAD(P)H-dependent reduction of the enoyl-ACP double bond. The *FabI* protein of *E. coli* is a well-characterized enoyl-ACP reductase that catalyzes the reduction of enoyl substrates of length 4 to 16 carbons (Raft et al, *JBC* 281:39285-93 (2006)). *FabI* utilizes both NADH and NADPH as a cofactor (Bergler et al, *Eur J Biochem* 242:689-94 (1996)) and is inhibited by acyl-ACP via product inhibition (Heath, *J Biol Chem* 271:1833-6 (1996)). *Bacillus subtilis* contains two enoyl-ACP reductase isoforms, *FabI* and *FabL* (Heath et al, *J Biol Chem* 275: 40128-33 (2000)). The *Streptococcus pneumoniae* *FabK* protein is a triclosan-resistant flavoprotein catalyzing the same activity (Heath and Rock, *Nature* 406:145-6 (2000)). An additional candidate is the *Pseudomonas aeruginosa* *FabI* protein, which was recently crystallized (Lee et al, *Acta Cryst Sect F* 67:214-216 (2011)).

Gene	GenBank ID	GI Number	Organism
fabI	P0AEK4.2	84028072	<i>Escherichia coli</i>
fabI	P54616.2	7531269	<i>Bacillus subtilis</i>
fabL	P71079.1	81817482	<i>Bacillus subtilis</i>
fabK	AAF98273.1	9789231	<i>Streptococcus pneumoniae</i>
fabI	Q9ZFE4.1	7531118	<i>Pseudomonas aeruginosa</i>

Fatty Acid Synthase (23.1.2), FIG. 12

Fatty acid synthase or fatty-acyl-CoA synthase are multifunctional enzyme complexes composed of multiple copies of one or more subunits and can together catalyze all the

reactions required for fatty acid synthesis: activation, priming, elongation and termination (Lomakin et al, *Cell* 129: 319-32 (2007)). The fatty acid synthase of *Saccharomyces cerevisiae* is a dodecamer composed of two multifunctional subunits *FAS1* and *FAS2*. This enzyme complex catalyzes the formation of long chain fatty acids from acetyl-CoA and malonyl-CoA. The favored product of eukaryotic FAS systems is palmitic acid (C16). Similar fatty acid synthase complexes are found in *Candida parapsilosis* and *Thermomyces lanuginosus* (Nguyen et al, *PLoS One* 22:e8421 (2009); Jenni et al, *Science* 316:254-61 (2007)). The multifunctional *Fas* enzymes of *Mycobacterium tuberculosis* and mammals such as *Homo sapiens* are also suitable candidates (Fernandes and Kolattukudy, *Gene* 170:95-99 (1996) and Smith et al, *Prog Lipid Res* 42:289-317 (2003)).

Gene	GenBank ID	GI Number	Organism
FAS1	CAA82025.1	486321	<i>Saccharomyces cerevisiae</i>
FAS2	CAA97948.1	1370478	<i>Saccharomyces cerevisiae</i>
Fas1	ABO37973.1	133751597	<i>Thermomyces lanuginosus</i>

Gene	GenBank ID	GI Number	Organism
Fas2	ABO37974.1	133751599	<i>Thermomyces lanuginosus</i>
Fas	AAB03809.1	1036835	<i>Mycobacterium tuberculosis</i>
Fas	NP_004095.4	41872631	<i>Homo sapiens</i>

5

Multiple genes are involved in fatty acid synthesis in bacteria and plants, including: 1. Acetyl-CoA: ACP transacylase (2.3.1.38)—for converting acetyl-CoA to acetyl-ACP, 2. malonyl-CoA:ACP transacylase that converts malonyl-CoA into malonyl-ACP (2.3.1.39), 3. acetyl[acp]:malonyl-[acp] C-acyl transferase (2.3.1.41) and others in fatty acid elongation. Some exemplary gene candidates for the steps are shown below.

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Gene	GenBank ID	GI number	Organism
fabH	AP_001717.1	89107937	<i>Escherichia coli</i> K12
fabB	NP_416826.1	16130258	<i>Escherichia coli</i> K12
fabF	NP_415613.1	16129058	<i>Escherichia coli</i> K12
fabD	NP_415610.1	16129055	<i>Escherichia coli</i> K12
fabI	NP_415804.1	16129249	<i>Escherichia coli</i> K12
fabA	NP_415474.1	16128921	<i>Escherichia coli</i> K12
fabZ	NP_414722.1	16128173	<i>Escherichia coli</i> K12
fabG	NP_415611.1	16129056	<i>Escherichia coli</i> K12
FasII (fatty acid synthase, alpha subunit)	AAA34601.1	171502	<i>Saccharomyces cerevisiae</i>
FasI (fatty acid synthase, beta subunit)	AAA34602.1	171506	<i>Saccharomyces cerevisiae</i>
fas	NP_217040.1	15609661	<i>Mycobacterium tuberculosis</i> H37Rv
fas	AAN25329.1	23326820	<i>Bifidobacterium longum</i> NCC2705
fas	YP_003971698.1	311064972	<i>Bifidobacterium bifidum</i>
fas	AE082252.1	334697455	<i>Corynebacterium ulcerans</i>

Acyl acp Thioesterase (3.1.2.a)—Step F, FIG. 12

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Acyl-ACP thioesterase releases free fatty acids from Acyl-ACPs, thus terminating fatty acid biosynthesis. There are two isoforms of acyl-ACP thioesterase, FatA and FatB. Substrate specificity of these isoforms determines the chain length and level of saturated fatty acids in plants. The highest activity of FatA is with C18:1-ACP. FatA has very low activities towards other acyl-ACPs when compared with C18:1-ACP. FatB has highest activity with C16:0-ACP. It also has significant high activity with C18:1-ACP, followed by C18:0-ACP and C16:1-ACP. Kinetics studies of FatA and FatB indicate that their substrate specificities with different acyl-ACPs came from the Kcat values, rather than from Km. Km values of the two isoforms with different substrates are similar, in the micromolar order.

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Exemplary enzymes include the FatA and FatB isoforms of *Arabidopsis thaliana* (Salas et al, Arch Biochem Biophys 403:25-34 (2002)). A number of thioesterases with different chain length specificities are listed in WO 2008/113041 and are included in the table below [see p 126 Table 2A of patent]. For example, it has been shown previously that expression of medium chain plant thioesterases like FatB from *Umbellularia californica* in *E. coli* results in accumulation of high levels of medium chain fatty acids, primarily laurate (C12:0). Similarly, expression of *Cuphea palustris* FatB1 thioesterase in *E. coli* led to accumulation of C8-10:0 acyl-ACPs (Dehesh et al, Plant Physiol 110:203-10 (1996)). Similarly, *Carthamus tinctorius* thioesterase, when expressed in *E. coli* leads to >50 fold elevation in C 18:1 chain termination and release as free fatty acid (Knutzon et al, Plant Physiol 100:1751-58 (1992)). Methods for altering the substrate specificity of acyl-ACP thioesterases are also known in the art (for example, EP1605048).

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Gene	GenBank ID	GI number	Organism
fatA	AEE76980.1	332643459	<i>Arabidopsis thaliana</i>
fatA	ACC41415	183176305	<i>Mycobacterium marinum</i> M
fatA	AAX54527	61741120	<i>Helianthus annuus</i>
fatA	CAC14164	10944734	<i>Brassica juncea</i>
fatA	ZP_04749108	240170449	<i>Mycobacterium kansasii</i> ATCC 12478
fatA	ZP_04384386.1	229490548	<i>Rhodococcus erythropolis</i> SK121
fatA	YP_885312.1	118472377	<i>Mycobacterium smegmatis</i> str. MC2 155
fatB	AAQ08202.1	33325193	<i>Helianthus annuus</i>
fatB	AEE28300.1	332190179	<i>Arabidopsis thaliana</i>
fatB	ABII8986.1	112455672	<i>Brassica juncea</i>
tesA	NP_415027.1	16128478	<i>Escherichia coli</i> K12
fatB2	AAC49269.1	1292906	<i>Cuphea hookeriana</i>
fatB1	AAC49179.1	1215718	<i>Cuphea palustris</i>
M96568.1:94 . . . 1251	AAA33019.1	404026	<i>Carthamus tinctorius</i>
fatB	Q41635.1	8469218	<i>Umbellularia californica</i>
tesA	AAC73596.1	1786702	<i>Escherichia coli</i>

Several more of these candidates can be found in WO2007136762A2 (Production of fatty acids and derivatives thereof) and are described below.

Gene	GenBank ID	GI number	Source Organism
fatB1	AAA34215.1	170556	<i>Umbellularia California</i>
fatB1	Q39513	8469217	<i>Cuphea hookeriana</i>
fatB	Q39473	8469216	<i>Cinnamomum camphorum</i>
fatB[M141T]	CAA85388	804948	<i>Arabidopsis thaliana</i>
fatA	NP 189147, NP 193041	15230256; 15235555	<i>Arabidopsis thaliana</i>
fatA	CAC39106	14148965	<i>Brassica juncea</i>
fatA	AAC72883	3859832	<i>Cuphea hookeriana</i>

Acyl CoA Synthetase and Acyl CoA Ligase (6.2.13)—Step K, FIG. 12

Fatty acids are often found in the cell in the activated form of an acyl-coA. The activation requires energy in the form of ATP. Acyl-CoAs are used in the biosynthesis of many cellular products and components, including membrane lip-

ids. Acyl CoA cannot move across membranes. Therefore, fatty acids are transported in their free form and converted to acyl-CoAs while crossing the membrane by the enzymes acyl-CoA synthetases (ACS). These enzymes catalyze the esterification of fatty acids into the CoA thioesters concomitant with transport.

Gene	GenBank ID	GI number	Organism
scs	YP_135572.1	55377722	<i>Haloarcula marismortui</i>
sucC	NP_415256.1	16128703	<i>Escherichia coli</i>
sucD	AAC73823.1	1786949	<i>Escherichia coli</i>
LSC1	NP_014785	6324716	<i>Saccharomyces cerevisiae</i>
LSC2	NP_011760	6321683	<i>Saccharomyces cerevisiae</i>
paaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>
matB	AAC83455.1	3982573	<i>Rhizobium leguminosarum</i>
bioW	NP_390902.2	50812281	<i>Bacillus subtilis</i>
bioW	CAA10043.1	3850837	<i>Pseudomonas mendocina</i>
bioW	P22822.1	115012	<i>Bacillus sphaericus</i>
phI	CAJ15517.1	77019264	<i>Penicillium chrysogenum</i>
phIB	ABS19624.1	152002983	<i>Penicillium chrysogenum</i>
paaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>
bioW	NP_390902.2	50812281	<i>Bacillus subtilis</i>
AACS	NP_084486.1	21313520	<i>Mus musculus</i>
AACS	NP_076417.2	31982927	<i>Homo sapiens</i>
acs	AAC77039.1	1790505	<i>Escherichia coli</i>
acoE	AAA21945.1	141890	<i>Ralstonia eutropha</i>
acs1	ABC87079.1	86169671	<i>Methanothermobacter thermotrophicus</i>
acs1	AAL23099.1	16422835	<i>Salmonella enterica</i>
ACS1	Q01574.2	257050994	<i>Saccharomyces cerevisiae</i>
AF1211	NP_070039.1	11498810	<i>Archaeoglobus fulgidus</i>
AF1983	NP_070807.1	11499565	<i>Archaeoglobus fulgidus</i>

Oxidoreductase (acyl-ACP to Aldehyde) 1.2.11—Step J, FIG. 12

The reduction of an acyl-ACP to its corresponding aldehyde is catalyzed by an acyl-ACP reductase (AAR). Such a transformation is depicted in Step J of FIG. 12. Suitable enzyme candidates include the orf1594 gene product of *Synechococcus elongatus* PCC7942 and homologs thereof (Schirmer et al, Science, 329: 559-62 (2010)). The *S. elongatus* PCC7942 acyl-ACP reductase is coexpressed with an aldehyde decarbonylase in an operon that appears to be conserved in a majority of cyanobacterial organisms. This enzyme, expressed in *E. coli* together with the aldehyde decarbonylase, conferred the ability to produce alkanes. The *P. marinus* AAR was also cloned into *E. coli* and, together with a decarbonylase, demonstrated to produce alkanes (US Application 2011/0207203).

Gene	GenBank ID	GI Number	Organism
orf1594	YP_400611.1	81300403	<i>Synechococcus elongatus</i> PCC7942
PMT9312_0533	YP_397030.1	78778918	<i>Prochlorococcus marinus</i> MIT 9312
syc0051_d	YP_170761.1	56750060	<i>Synechococcus elongatus</i> PCC 6301
Ava_2534	YP_323044.1	75908748	<i>Anabaena variabilis</i> ATCC 29413
alr5284	NP_489324.1	17232776	<i>Nostoc</i> sp. PCC 7120
Aazo_3370	YP_003722151.1	298491974	<i>Nostoc azollae</i>
Cyan7425_0399	YP_002481152.1	220905841	<i>Cyanothece</i> sp. PCC 7425
N9414_21225	ZP_01628095.1	119508943	<i>Nodularia spumigena</i> CCY9414
L8106_07064	ZP_01619574.1	119485189	<i>Lyngbya</i> sp. PCC 8106

The gene candidates for Acyl-CoA reductase (Step L), CAR (Step O), FAR (Step N) and fatty aldehyde reductase (Step M) are described elsewhere in this application.

Oxidoreductase (acyl-ACP to Alcohol) Step O, FIG. 2 and Step P, FIG. 12

The reduction of an acyl-ACP to its corresponding alcohol is catalyzed by an acyl-ACP reductase (alcohol forming). Such a transformation is depicted in step P of FIG. 12. Fatty acyl reductase enzymes that use acyl-ACP substrates to produce alcohols are known in the art. Alcohol forming acyl-ACP reductases include Maqu_2220 of *Marinobacter aquaeolei* VT8 and Hch_05075 of *Hahella chejuensis* KCTC2396 (see WO2013/048557). These enzymes convert both acyl-ACP substrates and acyl-CoA substrates to their corresponding alcohols. The *M. aquaeolei* AAR was previously characterized as an aldehyde reductase (Wahlen et al, AEM 75:2758-2764 (2009)) and US 2010/0203614). Alcohol forming acyl-ACP reductase enzymes are shown in the table below.

Protein	GenBank ID	GI Number	Organism
Magu_2220	ABM19299	120324984	<i>Marinobacter aquaeolei</i>
Hch_05075	YP_436183	83647748	<i>Hahella chejuensis</i>
MDG893_11561	ZP_01892457.1	149374683	<i>Marinobacter algicola</i> DG893
HP15_810	ADP96574.1	311693701	<i>Marinobacter adhaerens</i> HP15
RED65_09894	ZP_01305629.1	94499091	<i>Oceanobacter</i> sp. RED65

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Odd Chain Length Fatty acid Biosynthesis

Fatty acids with odd numbers of carbon can be formed by a similar mechanism as shown in FIG. 12. The starting metabolite in this case is propionyl-CoA instead of acetyl-CoA. The product of malonyl-ACP and propionyl-CoA is 3-oxovaleryl-ACP. This reaction is catalyzed by a β -ketoacyl-ACP synthase (EC 2.3.1.180) as shown in FIG. 12. The subsequent steps of fatty acid biosynthesis for an odd-chain fatty acid are the same as shown in FIG. 12. Several exemplary gene candidates for this step have been listed in US20120070868 (Odd chain fatty acid derivatives) and are shown below.

Gene symbol	GenBank ID	GI number	Organism
fabH	AAC74175	1787333	<i>E. coli</i>
fabH1	NP_389015	16078198	<i>B. subtilis</i> 168
fabH2	NP_388898	16078081	<i>B. subtilis</i> 168
fabH	CAB99151	9368919	<i>Streptomyces coelicolor</i>
fabH	AAA99447	870807	<i>Streptomyces glaucescens</i>
fabH3	NP_823466	29828832	<i>Streptomyces avermitilis</i> MA-4680
fabH	YP_002349314	217963636	<i>Listeria monocytogenes</i>
fabH	NP_645682	21282594	<i>Staphylococcus aureus</i> MW2
fabH	AAK74580	14971886	<i>Streptococcus pneumoniae</i>
fabH	NP_722071	24380116	<i>Streptococcus mutans</i> UA159
fabH	NP_266927	15672753	<i>Lactococcus lactis</i> subsp. <i>lactis</i>
fabH	YP_003687907	297626144	<i>Propionibacterium freudenreichii</i> subsp. <i>Shermanii</i>

Example XVII

Production of Propionyl-CoA

This example describes enzymes for production of Propionyl-CoA. Exemplary pathways are described FIG. 22. The pathways for production of propionyl-CoA can proceed via oxaloacetate, which includes conversion of PEP into oxaloacetate either via PEP carboxykinase or PEP carboxylase. Alternatively, PEP is converted first to pyruvate by

15 nase genes that have been cloned into *E. coli* include those from *Mannheimia succiniciproducens* (Lee, *Biotechnol. Bioprocess Eng.* 7:95-99 (2002)), *Anaerobiospirillum succiniciproducens* (Laivenieks, *Appl Environ Microbial* 63:2273-2280 (1997)), and *Actinobacillus succinogenes* (Kim, *Appl Environ Microbial* 70:1238-1241 (2004)). Internal experiments have also found that the PEP carboxykinase enzyme encoded by *Haemophilus influenza* is highly efficient at forming oxaloacetate from PEP. These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
PCK1	NP_013023	6322950	<i>Saccharomyces cerevisiae</i>
pck	NP_417862.1	16131280	<i>Escherichia coli</i>
pckA	YP_089485.1	52426348	<i>Mannheimia succiniciproducens</i>
pckA	O09460.1	3122621	<i>Anaerobiospirillum succiniciproducens</i>
pckA	Q6W6X5	75440571	<i>Actinobacillus succinogenes</i>
pckA	P43923.1	1172573	<i>Haemophilus influenza</i>

pyruvate kinase and then to oxaloacetate by methylmalonyl-CoA carboxytransferase or pyruvate carboxylase. Oxaloacetate is converted to propionyl-CoA by means of the reductive TCA cycle, a methylmutase, a decarboxylase, an epimerase and a decarboxylase.

PEP Carboxykinase

Although the net conversion of phosphoenolpyruvate to oxaloacetate is redox-neutral, the mechanism of this conversion is important to the overall energetics of the co-production pathway. The most desirable enzyme for the conversion of PEP to oxaloacetate is PEP carboxykinase which simultaneously forms an ATP while carboxylating PEP. In most organisms, however, PEP carboxykinase serves a gluconeogenic function and converts oxaloacetate to PEP at the expense of one ATP. *S. cerevisiae* is one such organism whose native PEP carboxykinase, PCK1, serves a gluconeogenic role (Valdes-Hevia, *FEBS. Lett.* 258:313-316 (1989)). *E. coli* is another such organism, as the role of PEP carboxykinase in producing oxaloacetate is believed to be minor when compared to PEP carboxylase, which does not form ATP, possibly due to the higher K_m for bicarbonate of PEP carboxykinase (Kim, *Appl Environ Microbiol* 70:1238-1241 (2004)). Nevertheless, activity of the native *E. coli* PEP carboxykinase from PEP towards oxaloacetate has been recently demonstrated in ppc mutants of *E. coli* K-12 (Kwon, *Journal of Microbiology and Biotechnology* 16:1448-1452 (2006)). These strains exhibited no growth defects and had increased succinate production at high NaHCO_3 concentrations. In some organisms, particularly rumen bacteria, PEP carboxykinase is quite efficient in producing oxaloacetate from PEP and generating ATP. Examples of PEP carboxyki-

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These sequences and sequences for subsequent enzymes listed in this report can be used to identify homologue proteins in GenBank or other databases through sequence similarity searches (e.g. BLASTp). The resulting homologue proteins and their corresponding gene sequences provide additional DNA sequences for transformation into the host organism of choice.

PEP Carboxylase

PEP carboxylase represents an alternative enzyme for the formation of oxaloacetate from PEP. Since the enzyme does not generate ATP upon decarboxylating oxaloacetate, its utilization decreases the maximum ATP yield of the production pathway and represents a less favorable alternative for converting oxaloacetate to PEP. Nevertheless, the maximum theoretical C3 alcohols yield of 1.33 mol/mol will remain unchanged if PEP carboxylase is utilized to convert PEP to oxaloacetate. *S. cerevisiae* does not naturally encode a PEP carboxylase, but exemplary organisms that possess genes that encode PEP carboxylase include *E. coli* (Kai, *Arch. Biochem. Biophys.* 414:170-179 (2003)), *Methylobacterium extorquens* AM1 (Arps, *J. Bacteriol.* 175:3776-3783 (1993)), and *Corynebacterium glutamicum* (Eikmanns, *Mol. Gen. Genet.* 218:330-339 (1989)). These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
ppc	NP_418391	16131794	<i>Escherichia coli</i>
ppcA	AAB58883	28572162	<i>Methylobacterium extorquens</i>
ppc	ABB53270	80973080	<i>Corynebacterium glutamicum</i>

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Pyruvate Kinase and Methylmalonyl-CoA Carboxyltransferase

An additional energetically efficient route to oxaloacetate from PEP requires two enzymatic activities: pyruvate kinase and methylmalonyl-CoA carboxyltransferase. Pyruvate kinase catalyzes the ATP-generating conversion of PEP to pyruvate and is encoded by the PYK1 (Burke, *J. Biol. Chem.* 258:2193-2201 (1983)) and PYK2 (Boles et al., *J. Bacteriol.* 179:2987-2993 (1997)) genes in *S. cerevisiae*. In *E. coli*, this activity is catalyzed by the gene product of pykF and pykA. Methylmalonyl-CoA carboxyltransferase catalyzes the conversion of pyruvate to oxaloacetate. Importantly, this reaction also simultaneously catalyzes the conversion of (S)-methylmalonyl-CoA to propionyl-CoA (see FIG. 22). An exemplary methylmalonyl-CoA carboxyltransferase which is comprised of 1.3S, 5S, and 12S subunits can be found in *Propionibacterium freudenreichii* (Thornton et al., *J. Bacteriol.* 175:5301-5308 (1993)). These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
PYK1	NP_009362	6319279	<i>Saccharomyces cerevisiae</i>
PYK2	NP_014992	6324923	<i>Saccharomyces cerevisiae</i>
pykF	NP_416191.1	16129632	<i>Escherichia coli</i>
pykA	NP_416368.1	16129807	<i>Escherichia coli</i>
1.3S subunit	P02904	114847	<i>Propionibacterium freudenreichii</i>
5S subunit	Q70AC7	62901478	<i>Propionibacterium freudenreichii</i>
12S subunit	Q8GBW6	62901481	<i>Propionibacterium freudenreichii</i>

Pyruvate Kinase and Pyruvate Carboxylase

A combination of enzymes can convert PEP to oxaloacetate with a stoichiometry identical to that of PEP carboxylase. These enzymes are encoded by pyruvate kinase, PYK1 (Burke, *J. Biol. Chem.* 258:2193-2201 (1983)) or PYK2 (Boles et al., *J. Bacteriol.* 179:2987-2993 (1997)) and pyruvate carboxylase, PYC1 (Walker, *Biochem. Biophys. Res. Commun.* 176:1210-1217 (1991)) or PYC2 (Walker, *Biochem. Biophys. Res. Commun.* 176:1210-1217 (1991)). The latter proteins are identified below.

Protein	GenBank ID	GI Number	Organism
PYC1	NP_011453	6321376	<i>Saccharomyces cerevisiae</i>
PYC2	NP_009777	6319695	<i>Saccharomyces cerevisiae</i>
Pyc	YP_890857.1	118470447	<i>Mycobacterium smegmatis</i>

Malate Dehydrogenase, Fumarase, Fumarate Reductase

Oxaloacetate can be converted to succinate by malate dehydrogenase, fumarase and fumarate reductase when the TCA cycle is operating in the reductive cycle. *S. cerevisiae* possesses three copies of malate dehydrogenase, MDH1 (McAlister-Henn, *J. Bacteriol.* 169:5157-5166 (1987)) MDH2 (Minard, *Mol. Cell. Biol.* 11:370-380 (1991); and Gibson, *J. Biol. Chem.* 278:25628-25636 (2003)), and MDH3 (Steffan, *J. Biol. Chem.* 267:24708-24715 (1992)), which localize to the mitochondrion, cytosol, and peroxisome, respectively. *S. cerevisiae* contains one copy of a fumarase-encoding gene, FUM1, whose product localizes to both the cytosol and mitochondrion (Sass, *J. Biol. Chem.* 278:4510945116 (2003)). Fumarate reductase is encoded by two soluble enzymes, FRDS1 (Enomoto, *DNA. Res.* 3:263-267 (1996)) and FRDS2 (Muratsubaki, *Arch. Biochem. Biophys.* 352:175-181 (1998)), which localize to the cytosol and promitochondrion, respectively, and are required for anaerobic growth on glucose (Arikawa, *Microbiol. Lett.*

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165:111-116 (1998)). *E. coli* is known to have an active malate dehydrogenase. It has three fumarases encoded by fumA, B and C, each one of which is active under different conditions of oxygen availability. The fumarate reductase in *E. coli* is composed of four subunits. These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
MDH1	NP_012838	6322765	<i>Saccharomyces cerevisiae</i>
MDH2	NP_014515	116006499	<i>Saccharomyces cerevisiae</i>
MDH3	NP_010205	6320125	<i>Saccharomyces cerevisiae</i>
FUM1	NP_015061	6324993	<i>Saccharomyces cerevisiae</i>
FRDS1	P32614	418423	<i>Saccharomyces cerevisiae</i>
FRDS2	NP_012585	6322511	<i>Saccharomyces cerevisiae</i>
frdA	NP_418578.1	16131979	<i>Escherichia coli</i>
frdB	NP_418577.1	16131978	<i>Escherichia coli</i>
frdC	NP_418576.1	16131977	<i>Escherichia coli</i>
frdD	NP_418475.1	16131877	<i>Escherichia coli</i>

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-continued

Protein	GenBank ID	GI Number	Organism
Mdh	NP_417703.1	16131126	<i>Escherichia coli</i>
FumA	NP_416129.1	16129570	<i>Escherichia coli</i>
FumB	NP_418546.1	16131948	<i>Escherichia coli</i>
FumC	NP_416128.1	16129569	<i>Escherichia coli</i>

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Further exemplary enzymes are found in several organisms including *E. coli*, *Bacillus subtilis*, and *Rhizopus oryzae*.

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Protein	GenBank ID	GI number	Organism
mdh	AAC76268.1	1789632	<i>Escherichia coli</i>
mdh	NP_390790.1	16079964	<i>Bacillus subtilis</i>
MDH	ADG65261.1	296011196	<i>Rhizopus oryzae</i>

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Succinyl-CoA:3-Ketoacid-CoA Transferase

The conversion of succinate to succinyl-CoA is ideally carried by a transferase which does not require the direct consumption of an ATP or GTP. This type of reaction is common in a number of organisms. Perhaps the top candidate enzyme for this reaction step is succinyl-CoA:3-ketoacid-CoA transferase. This enzyme converts succinate to succinyl-CoA while converting a 3-ketoacyl-CoA to a 3-ketoacid. Exemplary succinyl-CoA:3-ketoacid-CoA transferases are present in *Helicobacter pylori* (Corthesy-Theulaz et al., *J. Biol. Chem.* 272:25659-25667 (1997)), *Bacillus subtilis* (Stols et al., *Protein. Expr. Purif.* 53:396-403 (2007)), and *Homo sapiens* (Fukao et al., *Genomics*, 68:144-151 (2000); and Tanaka, *Mol. Hum. Reprod.* 8:16-23 (2002)). These proteins are identified below.

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Protein	GenBank ID	GI Number	Organism
HPAG1__0676	YP_627417	108563101	<i>Helicobacter pylori</i>
HPAG1__0677	YP_627418	108563102	<i>Helicobacter pylori</i>
ScoA	NP_391778	16080950	<i>Bacillus subtilis</i>
ScoB	NP_391777	16080949	<i>Bacillus subtilis</i>
OXCT1	NP_000427	4557817	<i>Homo sapiens</i>
OXCT2	NP_071403	11545841	<i>Homo sapiens</i>

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Succinyl-CoA: Acetyl-CoA Transferase

The conversion of succinate to succinyl-CoA can also be catalyzed by succinyl-CoA: Acetyl-CoA transferase. The gene product of *cat1* of *Clostridium kluyveri* has been shown to exhibit succinyl-CoA: acetyl-CoA transferase activity (Sohling, *J. Bacteriol.* 178:871-880 (1996)). In addition, the activity is present in *Trichomonas vaginalis* (van Grinsven et al., *J. Biol. Chem.* 283:1411-1418 (2008)) and *Trypanosoma brucei* (Riviere et al., *J. Biol. Chem.* 279:45337-45346 (2004)). These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
<i>cat1</i>	P38946.1	729048	<i>Clostridium kluyveri</i>
TVAG__395550	XP_001330176	123975034	<i>Trichomonas vaginalis</i> G3
Tb11.02.0290	XP_828352	71754875	<i>Trypanosoma brucei</i>

Succinyl-CoA Synthetase

The product of the *LSC1* and *LSC2* genes of *S. cerevisiae* and the *sucC* and *sucD* genes of *E. coli* naturally form a succinyl-CoA synthetase complex that catalyzes the formation of succinyl-CoA from succinate with the concomitant consumption of one ATP, a reaction which is reversible in vivo (Przybyla-Zawilask et al., *Eur. J. Biochem.* 258(2):736-743 (1998) and Buck et al., *J. Gen. Microbiol.* 132(6):1753-1762 (1986)). These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
<i>LSC1</i>	NP_014785	6324716	<i>Saccharomyces cerevisiae</i>
<i>LSC2</i>	NP_011760	6321683	<i>Saccharomyces cerevisiae</i>
<i>SucC</i>	NP_415256.1	16128703	<i>Escherichia coli</i>
<i>SucD</i>	AAC73823.1	1786949	<i>Escherichia coli</i>

Methylmalonyl-CoA Mutase

Succinyl-CoA can be converted into (R)-methylmalonyl-CoA by methylmalonyl-CoA mutase (MCM). In *E. coli*, the reversible adenosylcobalamin-dependant mutase participates in a three-step pathway leading to the conversion of succinate to propionate (Haller, *Biochemistry* 39:4622-9 (2000)). MCM is encoded by genes *scpA* in *Escherichia coli* (Haller, *Biochemistry* 39: 4622-4629 (2000); and Bobik, *Anal. Bioanal. Chem.* 375:344-349 (2003)) and *mutA* in *Homo sapiens* (Padovani, *Biochemistry* 45:9300-9306 (2006)). In several other organisms MCM contains alpha and beta subunits and is encoded by two genes. Exemplary gene candidates encoding the two-subunit protein are *Propionibacterium fredenreichii* sp. *shermanii* *mutA* and *mutB* (Korotkova, *J. Biol. Chem.* 279:13652-13658 (2004)) and *Methylobacterium extorquens* *mcmA* and *mcmB* (Korotkova, *J. Biol. Chem.* 279:13652-13658 (2004)). These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
<i>SepA</i>	NP_417392.1	16130818	<i>Escherichia coli</i> K12
<i>MutA</i>	P22033.3	67469281	<i>Homo sapiens</i>
<i>MutA</i>	P11652.3	127549	<i>Propionibacterium fredenreichii</i> sp. <i>shermanii</i>
<i>MutB</i>	P11653.3	127550	<i>Propionibacterium fredenreichii</i> sp. <i>shermanii</i>
<i>mcmA</i>	Q84FZ1	75486201	<i>Methylobacterium extorquens</i>
<i>McmB</i>	Q6TMA2	75493131	<i>Methylobacterium extorquens</i>

Additional enzyme candidates identified based on high homology to the *E. coli* *spcA* gene product are identified below.

Protein	GenBank ID	GI Number	Organism
Sbm	NP_838397.1	30064226	<i>Shigella flexneri</i>
SARI_04585	ABX24358.1	160867735	<i>Salmonella enterica</i>
YfreA_01000861	ZP_00830776.1	77975240	<i>Yersinia frederiksenii</i>

There further exists evidence that genes adjacent to the methylmalonyl-CoA mutase catalytic genes are also required for maximum activity. For example, it has been demonstrated that the *meaB* gene from *M. extorquens* forms a complex with methylmalonyl-CoA mutase, stimulates in vitro mutase activity, and possibly protects it from irreversible inactivation (Korotkova, *J Biol Chem.* 279:13652-13658 (2004)). The *M. extorquens* *meaB* gene product is highly similar to the product of the *E. coli* *argK* gene (BLASTp: 45% identity, e-value: 4e-67) which is adjacent to *scpA* on the chromosome. No sequence for a *meaB* homolog in *P. freudenreichii* is catalogued in GenBank. However, the *Propionibacterium acnes* KPA171202 gene product, YP_055310.1, is 51% identical to the *M. extorquens* *meaB* protein and its gene is also adjacent to the methylmalonyl-CoA mutase gene on the chromosome. These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
ArgK	AAC75955.1	1789285	<i>Escherichia coli</i> K12
KPA171202	YP_055310.1	50842083	<i>Propionibacterium acnes</i>
MeaB	2QM8_B	158430328	<i>Methylobacterium extorquens</i>

Methylmalonyl-CoA Epimerase

Methylmalonyl-CoA epimerase (MMCE) is the enzyme that interconverts (R)-methylmalonyl-CoA and (S)-methylmalonyl-CoA. MMCE is an essential enzyme in the breakdown of odd-numbered fatty acids and of the amino acids valine, isoleucine, and methionine. Methylmalonyl-CoA epimerase is present in organisms such as *Bacillus subtilis* (YqjC) (Haller, *Biochemistry.* 39:4622-4629 (2000)), *Homo sapiens* (YqjC) (Fuller, *Biochem. J* 213:643-650 (1983)), *Rattus norvegicus* (Mcee) (Bobik, *J Biol Chem.* 276:37194-37198 (2001)), *Propionibacterium shermanii* (AF454511) (Haller, *Biochemistry* 39:4622-9 (2000); McCarthy, *Structure* 9:637-46 (2001) and (Fuller, *Biochem. J* 213:643-650 (1983)) and *Caenorhabditis elegans* (mmce) (Kuhnl et al., *FEBS J* 272:1465-1477 (2005)). The additional gene candidate, AE016877 in *Bacillus cereus*, has high sequence homology to the other characterized enzymes. MMCE activity is required if the employed methylmalonyl-CoA decarboxylase or methylmalonyl-CoA carboxytransferase requires the (S) stereoisomer of methylmalonyl-CoA. These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
YqjC	NP_390273	255767522	<i>Bacillus subtilis</i>
MCEE	Q96PE7.1	50401130	<i>Homo sapiens</i>
Mcee_predicted	NP_001099811.1	157821869	<i>Rattus norvegicus</i>
AF454511	AAL57846.1	18042135	<i>Propionibacterium freudenreichii</i> sp. <i>shermanii</i>
Mmce	AAT92095.1	51011368	<i>Caenorhabditis elegans</i>
AE016877	AAP08811.1	29895524	<i>Bacillus cereus</i> ATCC 14579

Methylmalonyl-CoA Decarboxylase

Methylmalonyl-CoA decarboxylase, is a biotin-independent enzyme that catalyzes the conversion of methylmalonyl-CoA to propionyl-CoA in *E. coli* (Benning, *Biochemistry.* 39:4630-4639 (2000); and Haller, *Biochemistry.* 39:4622-4629 (2000)). The stereo specificity of the *E. coli* enzyme was not reported, but the enzyme in *Propionigenium modestum* (Bott et al., *Eur. J. Biochem.* 250:590-599 (1997)) and *Veillonella parvula* (Huder, *J. Biol. Chem.* 268:24564-24571 (1993)) catalyzes the decarboxylation of the (S)-stereoisomer of methylmalonyl-CoA (Hoffmann, *FEBS. Lett.* 220:121-125 (1987)). The enzymes from *P. modestum* and *V. parvula* are comprised of multiple subunits that not only decarboxylate (S)-methylmalonyl-CoA, but also create a pump that transports sodium ions across the cell membrane as a means to generate energy. These proteins are identified below.

nyl-CoA to propionyl-CoA in *E. coli* (Benning, *Biochemistry.* 39:4630-4639 (2000); and Haller, *Biochemistry.* 39:4622-4629 (2000)). The stereo specificity of the *E. coli* enzyme was not reported, but the enzyme in *Propionigenium modestum* (Bott et al., *Eur. J. Biochem.* 250:590-599 (1997)) and *Veillonella parvula* (Huder, *J. Biol. Chem.* 268:24564-24571 (1993)) catalyzes the decarboxylation of the (S)-stereoisomer of methylmalonyl-CoA (Hoffmann, *FEBS. Lett.* 220:121-125 (1987)). The enzymes from *P. modestum* and *V. parvula* are comprised of multiple subunits that not only decarboxylate (S)-methylmalonyl-CoA, but also create a pump that transports sodium ions across the cell membrane as a means to generate energy. These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
YgfG	NP_417394	90111512	<i>Escherichia coli</i>
mmdA	CAA05137	2706398	<i>Propionigenium modestum</i>
mmdD	CAA05138	2706399	<i>Propionigenium modestum</i>
mmdC	CAA05139	2706400	<i>Propionigenium modestum</i>
mmdB	CAA05140	2706401	<i>Propionigenium modestum</i>
mmdA	CAA80872	415915	<i>Veillonella parvula</i>
mmdC	CAA80873	415916	<i>Veillonella parvula</i>
mmdE	CAA80874	415917	<i>Veillonella parvula</i>
mmdD	CAA80875	415918	<i>Veillonella parvula</i>
mmdB	CAA80876	415919	<i>Veillonella parvula</i>

Example XVIII

In Vivo Labeling Assay for Conversion of Methanol to CO₂

This example describes a functional methanol pathway in a microbial organism.

Strains with functional reductive TCA branch and pyruvate formate lyase deletion were grown aerobically in LB medium overnight, followed by inoculation of M9 high-seed

media containing IPTG and aerobic growth for 4 hrs. These strains had methanol dehydrogenase/ACT pairs in the presence and absence of formaldehyde dehydrogenase or formate dehydrogenase. ACT is an activator protein (a Nudix hydrolase). At this time, strains were pelleted, resuspended in fresh M9 medium high-seed media containing 2% $^{13}\text{CH}_3\text{OH}$, and sealed in anaerobic vials. Head space was replaced with nitrogen and strains grown for 40 hours at 37° C. Following growth, headspace was analyzed for $^{13}\text{CO}_2$. Media was examined for residual methanol as well as BDO and byproducts. All constructs expressing methanol dehydrogenase (MeDH) mutants and MeDH/ACT pairs grew to slightly lower ODs than strains containing empty vector controls. This is likely due to the high expression of these constructs (Data not shown). One construct (2315/2317) displayed significant accumulation of labeled CO_2 relative to controls in the presence of FaldH, FDH or no coexpressed protein. This shows a functional MeOH pathway in *E. coli* and that the endogenous glutathione-dependent formaldehyde detoxification genes (frmAB) are sufficient to carry flux generated by the current MeDH/ACT constructs.

2315 is internal laboratory designation for the MeDH from *Bacillus methanolicus* MGA3 (GenBank Accession number: EIJ77596.1; GI number: 387585261), and 2317 is internal laboratory designation for the activator protein from the same organism (locus tag: MGA3_09170; GenBank Accession number: EIJ83380; GI number: 387591061).

Sequence analysis of the NADH-dependent MeDH from *Bacillus methanolicus* places the enzyme in the alcohol dehydrogenase family III. It does not contain any tryptophan residues, resulting in a low extinction coefficient (18,500 M^{-1} , cm^{-1}) and should be detected on SDS gels by Coomassie staining.

The enzyme has been characterized as a multisubunit complex built from 43 kDa subunits containing one Zn and 1-2 Mg atoms per subunit. Electron microscopy and sedimentation studies determined it to be a decamer, in which two rings with five-fold symmetry are stacked on top of each other (Vonck et al., *J. Biol. Chem.* 266:3949-3954, 1991). It is described to contain a tightly but not covalently bound cofactor and requires exogenous NAD^+ as e^- -acceptor to measure activity in vitro. A strong increase (10-40-fold) of in vitro activity was observed in the presence of an activator protein (ACT), which is a homodimer (21 kDa subunits) and contains one Zn and one Mg atom per subunit.

The mechanism of the activation was investigated by Kloosterman et al., *J. Biol. Chem.* 277:34785-34792, 2002, showing that ACT is a Nudix hydrolase and Hektor et al., *J. Biol. Chem.* 277:46966-46973, 2002, demonstrating that mutation of residue S97 to G or Tin MeDH changes activation characteristics along with the affinity for the cofactor. While mutation of residues G15 and D88 had no significant impact, a role of residue G13 for stability as well as of residues G95, D100, and K103 for the activity is suggested. Both papers together propose a hypothesis in which ACT cleaves MeDH-bound NAD^+ . MeDH retains AMP bound and enters an activated cycle with increased turnover.

The stoichiometric ratio between ACT and MeDH is not well defined in the literature. Kloosterman et al., supra determine the ratio of dimeric Act to decameric MeDH for full in vitro activation to be 10:1. In contrast, Arfman et al. *J. Biol. Chem.* 266:3955-3960, 1991 determined a ratio of 3:1 in vitro for maximum and a 1:6 ratio for significant activation, but observe a high sensitivity to dilution. Based on expression of both proteins in *Bacillus*, the authors estimate the ratio in vivo to be around 1:17.5.

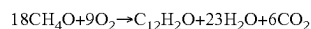
However, our in vitro experiments with purified activator protein (2317A) and methanol dehydrogenase (2315A) showed the ratio of ACT to MeDH to be 10:1. This in vitro test was done with 5 M methanol, 2 mM NAD and 10 μM methanol dehydrogenase 2315A at pH 7.4.

Example XIX

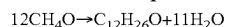
Improving Product Yields on Methanol Using Phosphoketolase-dependent cetyl-CoA Synthesis

Acetyl-CoA is the immediate precursor for the synthesis of isopropanol, fatty acyl-CoA molecules, and fatty acyl-ACP molecules as shown in FIGS. 2, 11, and 12. Phosphoketolase pathways make possible synthesis of acetyl-CoA without requiring decarboxylation of pyruvate (Bogorad et al., *Nature*, 2013, published online 29 Sep. 2013; United States Publication 2006-0040365), which thereby provides higher yields of fatty alcohols, fatty acids, fatty aldehydes, and isopropanol from carbohydrates and methanol than the yields attainable without phosphoketolase enzymes.

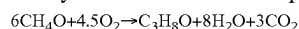
For example, synthesis of an exemplary fatty alcohol, dodecanol, from methanol using methanol dehydrogenase (step A of FIG. 1), a formaldehyde assimilation pathway (steps B, C, D of FIG. 1), the pentose phosphate pathway, and glycolysis can provide a maximum theoretical yield of 0.0556 mole dodecanol/mole methanol.



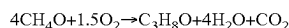
However, if these pathways are combined with a phosphoketolase pathway (steps T, U, V, W, X of FIG. 1), a maximum theoretical yield of 0.0833 mole dodecanol/mole methanol can be obtained if we assume that the pathway is not required to provide net generation of ATP for cell growth and maintenance requirements.



ATP for energetic requirements can be synthesized, at the expense of lowering the maximum theoretical product yield, by oxidizing methanol to CO_2 using several combinations of enzymes depicted in FIG. 10, glycolysis, the TCA cycle, the pentose phosphate pathway, and oxidative phosphorylation. Similarly, synthesis of isopropanol from methanol using methanol dehydrogenase (step A of FIG. 1), a formaldehyde assimilation pathway (steps B, C, D of FIG. 1), the pentose phosphate pathway and glycolysis can provide a maximum theoretical yield of 0.1667 mole isopropanol/mole methanol.



However, if these pathways are applied in combination with a phosphoketolase pathway (steps T, U, V, W, X of FIG. 1), a maximum theoretical yield of 0.250 mole isopropanol/mole methanol can be obtained.



The overall pathway is ATP and redox positive enabling synthesis of both ATP and NAD(P)H from conversion of MeOH to isopropanol. Additional ATP can be synthesized, at the expense of lowering the maximum theoretical product yield, by oxidizing methanol to CO_2 using several combinations of enzymes depicted in FIG. 10, glycolysis, the TCA cycle, the pentose phosphate pathway, and oxidative phosphorylation.

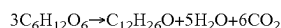
Example XX

Improving Product Yields on Carbohydrates using Phosphoketolase-Dependent Acetyl-CoA Synthesis and Exogenous Reducing Equivalents

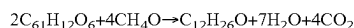
The theoretical yield of fatty acyl-CoA molecules, fatty acyl-ACP molecules, and isopropanol from carbohydrates

including but not limited to glucose, glycerol, sucrose, fructose, xylose, arabinose, and galactose, can also be enhanced by phosphoketolase enzymes, particularly when reducing equivalents are provided by an exogenous source such as hydrogen or methanol. This is because phosphoketolase enzymes provide acetyl-CoA synthesis with 100% carbon conversion efficiency (e.g., 3 acetyl-CoA's per glucose, 2.5 acetyl-CoA's per xylose, 1.5 acetyl-CoA's per glycerol).

For example, synthesis of an exemplary fatty alcohol, dodecanol, from glucose in the absence of phosphoketolase enzymes can reach a maximum theoretical dodecanol yield of 0.3333 mole dodecanol/mole glucose.

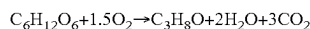


However, if enzyme steps T, U, V, W, X of FIG. 1 are applied in combination with glycolysis, the pentose phosphate pathway, and an external redox source (e.g., methanol, hydrogen) using the pathways shown in FIG. 10, the maximum theoretical yield can be increased to 0.5000 mole dodecanol/mole glucose.

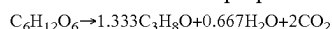


This assumes that the pathway is not required to provide net generation of ATP for cell growth and maintenance requirements. ATP for energetic requirements can be synthesized by oxidizing additional methanol to CO_2 using several combinations of enzymes depicted in FIG. 10.

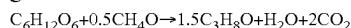
Similarly, synthesis of isopropanol from glucose in the absence of phosphoketolase enzymes can achieve a maximum theoretical isopropanol yield of 1.000 mole isopropanol/mole glucose.



However, if enzyme steps T, U, V, W, X of FIG. 1 are applied in combination with glycolysis and the pentose phosphate pathway, the maximum theoretical yield can be increased to 1.333 mole isopropanol/mole glucose.



If enzyme steps T, U, V, W, X of FIG. 1 are applied in combination with glycolysis, the pentose phosphate pathway, and external redox source (e.g., methanol, hydrogen) using the pathways shown in FIG. 10, the maximum theoretical yield can be increased to 1.500 mole isopropanol/mole glucose.



Example XXI

Phosphoketolase-Dependent Acetyl-CoA Synthesis Enzymes

This Example provides genes that can be used for enhancing carbon flux through acetyl-CoA using phosphoketolase enzymes.

FIG. 1, Step T—Fructose-6-Phosphate Phosphoketolase

Conversion of fructose-6-phosphate and phosphate to acetyl-phosphate and erythrose-5-phosphate can be carried out by fructose-6-phosphate phosphoketolase (EC 4.1.2.22). Conversion of fructose-6-phosphate and phosphate to acetyl-phosphate and erythrose-5-phosphate is one of the key reactions in the *Bifidobacterium* shunt. There is evidence for the existence of two distinct phosphoketolase enzymes in *bifidobacteria* (Sgorbati et al, 1976, Antonie Van Leeuwenhoek, 42(1-2) 49-57; Grill et al, 1995, Curr Microbiol, 31(1); 49-54). The enzyme from *Bifidobacterium dentium* appeared to be specific solely for fructose-6-phosphate (EC: 4.1.2.22) while the enzyme from *Bifidobacterium pseudolongum* subsp. *globosum* is able to utilize both fructose-6-phosphate and D-xylulose 5-phosphate (EC: 4.1.2.9) (Sgorbati et al, 1976, Antonie Van Leeuwenhoek, 42(1-2) 49-57). The enzyme encoded by the xfp gene, originally discovered in *Bifidobacterium animalis lactis*, is the dual-specificity enzyme (Meik et al., 2001, J Bacteriol, 183, 2929-2936; Yin et al, 2005, FEMS Microbiol Lett, 246(2); 251-257). Additional phosphoketolase enzymes can be found in *Leuconostoc mesenteroides* (Lee et al, Biotechnol Lett. 2005 June; 27(12):853-8), *Clostridium acetobutylicum* ATCC 824 (Servinsky et al, Journal of Industrial Microbiology & Biotechnology, 2012, 39, 1859-1867), *Aspergillus nidulans* (Kocharin et al, 2013, Biotechnol Bioeng, 110(8), 2216-2224; Papini, 2012, Appl Microbiol Biotechnol, 95 (4), 1001-1010), *Bifidobacterium breve* (Suziki et al, 2010, Acta Crystallogr Sect F Struct Biol Cryst Commun., 66(Pt 8):941-3), *Lactobacillus paraplantarum* (Jeong et al, 2007, J Microbiol Biotechnol, 17(5), 822-9).

Protein	GENBANK ID	GI NO.	ORGANISM
xfp	YP_006280131.1	386867137	<i>Bifidobacterium animalis lactis</i>
xfp	AAV66077.1	55818565	<i>Leuconostoc mesenteroides</i>
CAC1343	NP_347971.1	15894622	<i>Clostridium acetobutylicum</i> ATCC 824
xpkA	CBF76492.1	259482219	<i>Aspergillus nidulans</i>
xfp	WP_003840380.1	489937073	<i>Bifidobacterium dentium</i> ATCC 27678
xfp	AAR98788.1	41056827	<i>Bifidobacterium pseudolongum</i> subsp. <i>globosum</i>
xfp	WP_022857642.1	551237197	<i>Bifidobacterium pseudolongum</i> subsp. <i>globosum</i>
xfp	ADF97524.1	295314695	<i>Bifidobacterium breve</i>
xfp	AAQ64626.1	34333987	<i>Lactobacillus paraplantarum</i>

FIG. 1, Step U—Xylulose-5-Phosphate Phosphoketolase

Conversion of xylulose-5-phosphate and phosphate to acetyl-phosphate and glyceraldehyde-3-phosphate can be carried out by xylulose-5-phosphate phosphoketolase (EC 4.1.2.9). There is evidence for the existence of two distinct phosphoketolase enzymes in *bifidobacteria* (Sgorbati et al, 1976, Antonie Van Leeuwenhoek, 42(1-2) 49-57; Grill et al, 1995, Curr Microbiol, 31(1); 49-54). The enzyme from *Bifidobacterium dentium* appeared to be specific solely for fructose-6-phosphate (EC: 4.1.2.22) while the enzyme from *Bifidobacterium pseudolongum* subsp. *globosum* is able to utilize both fructose-6-phosphate and D-xylulose 5-phosphate (EC: 4.1.2.9) (Sgorbati et al, 1976, Antonie Van Leeuwenhoek, 42(1-2) 49-57). Many characterized enzymes have dual-specificity for xylulose-5-phosphate and fructose-6-phosphate. The enzyme encoded by the xfp gene, originally discovered in *Bifidobacterium animalis lactis*, is the dual-specificity enzyme (Meik et al., 2001, J Bacteriol, 183, 2929-2936; Yin et al, 2005, FEMS Microbiol Lett, 246(2); 251-257). Additional phosphoketolase enzymes can be found in *Leuconostoc mesenteroides* (Lee et al, Biotechnol

Lett. 2005 June; 27(12):853-8), *Clostridium acetobutylicum* ATCC 824 (Servinsky et al, Journal of Industrial Microbiology & Biotechnology, 2012, 39, 1859-1867), *Aspergillus nidulans* (Kocharin et al, 2013, Biotechnol Bioeng, 110(8), 2216-2224; Papini, 2012, Appl Microbiol Biotechnol, 95 (4), 1001-1010), *Bifidobacterium breve* (Suzuki et al, 2010, Acta Crystallogr Sect F Struct Biol Cryst Commun., 66(Pt 8):941-3), and *Lactobacillus paraplantarum* (Jeong et al, 2007, J Microbiol Biotechnol, 17(5), 822-9).

Protein	GENBANK ID	GI NO.	ORGANISM
xfp	YP_006280131.1	386867137	<i>Bifidobacterium animalis lactis</i>
xfp	AAV66077.1	55818565	<i>Leuconostoc mesenteroides</i>
CAC1343	NP_347971.1	15894622	<i>Clostridium acetobutylicum</i> ATCC 824
xpkA	CBF76492.1	259482219	<i>Aspergillus nidulans</i>
xfp	AAR98788.1	41056827	<i>Bifidobacterium pseudolongum</i> subsp. <i>globosum</i>
xfp	WP_022857642.1	551237197	<i>Bifidobacterium pseudolongum</i> subsp. <i>globosum</i>
xfp	ADF97524.1	295314695	<i>Bifidobacterium breve</i>
xfp	AAQ64626.1	34333987	<i>Lactobacillus paraplantarum</i>

FIG. 1, Step V—Phosphotransacetylase

The formation of acetyl-CoA from acetyl-phosphate can be catalyzed by phosphotransacetylase (EC 2.3.1.8). The pta gene from *E. coli* encodes an enzyme that reversibly converts acetyl-CoA into acetyl-phosphate (Suzuki, T., *Biochim. Biophys. Acta* 191:559-569 (1969)). Additional acetyltransferase enzymes have been characterized in *Bacillus subtilis* (Rado and Hoch, *Biochim. Biophys. Acta* 321:114-125 (1973), *Clostridium kluyveri* (Stadtman, E., *Methods Enzymol.* 1:5896-599 (1955), and *Thermotoga maritima* (Bock et al., *J. Bacteriol.* 181:1861-1867 (1999)). This reaction can also be catalyzed by some phosphotranbutyrylase enzymes (EC 2.3.1.19), including the ptb gene products from *Clostridium acetobutylicum* (Wiesenbom et al., *App. Environ. Microbiol.* 55:317-322 (1989); Walter et al., *Gene* 134:107-111(1993)). Additional ptb genes are found in butyrate-producing bacterium L2-50 (Louis et al., *J. Bacteriol.* 186:2099-2106 (2004) and *Bacillus megaterium* (Vazquez et al., *Curr. Microbiol.* 42:345-349 (2001)). Homologs to the *E. coli* pta gene exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*.

Protein	GenBank ID	GI Number	Organism
Pta	NP_416800.1	71152910	<i>Escherichia coli</i>
Pta	P39646	730415	<i>Bacillus subtilis</i>
Pta	A5N801	146346896	<i>Clostridium kluyveri</i>
Pta	Q9X0L4	6685776	<i>Thermotoga maritima</i>
Ptb	NP_349676	34540484	<i>Clostridium acetobutylicum</i>
Ptb	AAR19757.1	38425288	butyrate-producing bacterium L2-50
Ptb	CAC07932.1	10046659	<i>Bacillus megaterium</i>
Pta	NP_461280.1	16765665	<i>Salmonella enterica</i> subsp. <i>enterica</i> serovar <i>Typhimurium</i> str. LT2
PAT2	XP_001694504.1	159472743	<i>Chlamydomonas reinhardtii</i>
PAT1	XP_001691787.1	159467202	<i>Chlamydomonas reinhardtii</i>

FIG. 1, Step W—Acetate Kinase

Acetate kinase (EC 2.7.2.1) can catalyze the reversible ATP-dependent phosphorylation of acetate to acetylphosphate. Exemplary acetate kinase enzymes have been characterized in many organisms including *E. coli*, *Clostridium acetobutylicum* and *Methanosarcina thermophila* (Ingram-Smith et al., *J. Bacteriol.* 187:2386-2394 (2005); Fox and Roseman, *J. Biol. Chem.* 261:13487-13497 (1986); Winzer et al., *Microbiology* 143 (Pt 10):3279-3286 (1997)). Acetate kinase activity has also been demonstrated in the gene

product of *E. coli* purT (Marolewski et al., *Biochemistry* 33:2531-2537 (1994). Some butyrate kinase enzymes (EC 2.7.2.7), for example buk1 and buk2 from *Clostridium acetobutylicum*, also accept acetate as a substrate (Hartmanis, M. G., *J. Biol. Chem.* 262:617-621 (1987)). Homologs exist in several other organisms including *Salmonella enterica* and *Chlamydomonas reinhardtii*.

Protein	GenBank ID	GI Number	Organism
ackA	NP_416799.1	16130231	<i>Escherichia coli</i>
Ack	AAB18301.1	1491790	<i>Clostridium acetobutylicum</i>
Ack	AAA72042.1	349834	<i>Methanosarcina thermophila</i>
purT	AAC74919.1	1788155	<i>Escherichia coli</i>
buk1	NP_349675	15896326	<i>Clostridium acetobutylicum</i>
buk2	Q97I11	20137415	<i>Clostridium acetobutylicum</i>
ackA	NP_461279.1	16765664	<i>Salmonella typhimurium</i>
ACK1	XP_001694505.1	159472745	<i>Chlamydomonas reinhardtii</i>
ACK2	XP_001691682.1	159466992	<i>Chlamydomonas reinhardtii</i>

FIG. 1, Step X—Acetyl-CoA Transferase, Synthetase, or Ligase

The acylation of acetate to acetyl-CoA can be catalyzed by enzymes with acetyl-CoA synthetase, ligase or transferase activity. Two enzymes that can catalyze this reaction are AMP-forming acetyl-CoA synthetase or ligase (EC 6.2.1.1) and ADP-forming acetyl-CoA synthetase (EC 6.2.1.13). AMP-forming acetyl-CoA synthetase (ACS) is the predominant enzyme for activation of acetate to acetyl-CoA. Exemplary ACS enzymes are found in *E. coli* (Brown et al., *J. Gen. Microbiol.* 102:327-336 (1977)), *Ralstonia eutropha*

(Priefert and Steinbuchel, *J. Bacteriol.* 174:6590-6599 (1992)), *Methanothermobacter thermautotrophicus* (Ingram-Smith and Smith, *Archaea* 2:95-107 (2007)), *Salmonella enterica* (Gulick et al., *Biochemistry* 42:2866-2873 (2003)) and *Saccharomyces cerevisiae* (Jogl and Tong, *Biochemistry* 43:1425-1431(2004)). ADP-forming acetyl-CoA synthetases are reversible enzymes with a generally broad substrate range (Musfeldt and Schonheit, *J. Bacteriol.* 184: 636-644 (2002)). Two isozymes of ADP-forming acetyl-CoA synthetases are encoded in the *Archaeoglobus fulgidus*

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genome by are encoded by AF1211 and AF1983 (Musfeldt and Schonheit, supra (2002)). The enzyme from *Haloarcula marismortui* (annotated as a succinyl-CoA synthetase) also accepts acetate as a substrate and reversibility of the enzyme was demonstrated (Brasen and Schonheit, *Arch. Microbiol.* 182:277-287 (2004)). The ACD encoded by PAE3250 from hyperthermophilic crenarchaeon *Pyrobaculum aerophilum* showed the broadest substrate range of all characterized ACDs, reacting with acetate, isobutyryl-CoA (preferred substrate) and phenylacetyl-CoA (Brasen and Schonheit, supra (2004)). Directed evolution or engineering can be used to

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modify this enzyme to operate at the physiological temperature of the host organism. The enzymes from *A. fulgidus*, *H. marismortui* and *P. aerophilum* have all been cloned, functionally expressed, and characterized in *E. coli* (Brasen and Schonheit, supra (2004); Musfeldt and Schonheit, supra (2002)). Additional candidates include the succinyl-CoA synthetase encoded by sucCD in *E. coli* (Buck et al., *Biochemistry* 24:6245-6252 (1985)) and the acyl-CoA ligase from *Pseudomonas putida* (Fernandez-Valverde et al., *Appl. Environ. Microbiol.* 59:1149-1154 (1993)). The aforementioned proteins are shown below.

Protein	GenBank ID	GI Number	Organism
Acs	AAC77039.1	1790505	<i>Escherichia coli</i>
acoE	AAA21945.1	141890	<i>Ralstonia eutropha</i>
acs1	ABC87079.1	86169671	<i>Methanothermobacter thermautotrophicus</i>
acs1	AAL23099.1	16422835	<i>Salmonella enterica</i>
ACS1	Q01574.2	257050994	<i>Saccharomyces cerevisiae</i>
AF1211	NP_070039.1	11498810	<i>Archaeoglobus fulgidus</i>
AF1983	NP_070807.1	11499565	<i>Archaeoglobus fulgidus</i>
Scs	YP_135572.1	55377722	<i>Haloarcula marismortui</i>
PAE3250	NP_560604.1	18313937	<i>Pyrobaculum aerophilum</i> str. IM2
sucC	NP_415256.1	16128703	<i>Escherichia coli</i>
sucD	AAC73823.1	1786949	<i>Escherichia coli</i>
paaF	AAC24333.2	22711873	<i>Pseudomonas putida</i>

An acetyl-CoA transferase that can utilize acetate as the CoA acceptor is acetoacetyl-CoA transferase, encoded by the *E. coli* atoA (alpha subunit) and atoD (beta subunit) genes (Vanderwinkel et al., *Biochem. Biophys. Res. Commun.* 33:902-908 (1968); Korolev et al., *Acta Crystallogr. D Biol. Crystallogr.* 58:2116-2121 (2002)). This enzyme has also been shown to transfer the CoA moiety to acetate from a variety of branched and linear acyl-CoA substrates, including isobutyrate (Matthies et al., *Appl. Environ. Microbiol.* 58:1435-1439 (1992)), valerate (Vanderwinkel et al., supra) and butanoate (Vanderwinkel et al., supra). Similar enzymes exist in *Corynebacterium glutamicum* ATCC 13032 (Duncan et al., *Appl. Environ. Microbiol.* 68:5186-5190 (2002)), *Clostridium acetobutylicum* (Cary et al., *Appl. Environ. Microbiol.* 56:1576-1583 (1990)), and *Clostridium saccharoperbutylacetonicum* (Kosaka et al., *Biosci. Biotechnol. Biochem.* 71:58-68 (2007)). These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
atoA	P76459.1	2492994	<i>Escherichia coli</i> K12
atoD	P76458.1	2492990	<i>Escherichia coli</i> K12
actA	YP_226809.1	62391407	<i>Corynebacterium glutamicum</i> ATCC 13032
cg0592	YP_224801.1	62389399	<i>Corynebacterium glutamicum</i> ATCC 13032
ctfA	NP_149326.1	15004866	<i>Clostridium acetobutylicum</i>
ctfB	NP_149327.1	15004867	<i>Clostridium acetobutylicum</i>
ctfA	AAP42564.1	31075384	<i>Clostridium saccharoperbutylacetonicum</i>
ctfB	AAP42565.1	31075385	<i>Clostridium saccharoperbutylacetonicum</i>

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Additional exemplary acetyl-CoA transferase candidates are catalyzed by the gene products of cat1, cat2, and cat3 of *Clostridium kluyveri* which have been shown to exhibit succinyl-CoA, 4-hydroxybutyryl-CoA, and butyryl-CoA transferase activity, respectively (Seedorf et al., supra; Sohling et al., *Eur. J. Biochem.* 212:121-127 (1993); Sohling et al., *J. Bacteriol.* 178:871-880 (1996)). Similar CoA transferase activities are also present in *Trichomonas vaginalis* (van Grinsven et al., *J. Biol. Chem.* 283:1411-1418 (2008)) and *Trypanosoma brucei* (Riviere et al., *J. Biol. Chem.* 279:45337-45346 (2004)). These proteins are identified below.

Protein	GenBank ID	GI Number	Organism
cat1	P38946.1	729048	<i>Clostridium kluyveri</i>
cat2	P38942.2	172046066	<i>Clostridium kluyveri</i>
cat3	EDK35586.1	146349050	<i>Clostridium kluyveri</i>
TVAG_395550	XP_001330176	123975034	<i>Trichomonas vaginalis</i> G3
Tb11.02.0290	XP_828352	71754875	<i>Trypanosoma brucei</i>

Example XXII

Attenuation or Disruption of Endogenous Enzymes

This example provides endogenous enzyme targets for attenuation or disruption that can be used for enhancing carbon flux through methanol dehydrogenase and formaldehyde assimilation pathways.

DHA Kinase

Methylotrophic yeasts typically utilize a cytosolic DHA kinase to catalyze the ATP-dependent activation of DHA to DHAP. DHAP together with G3P is combined to form fructose-1,6-bisphosphate (FBP) by FBP aldolase. FBP is then hydrolyzed to F6P by fructose bisphosphatase. The net conversion of DHA and G3P to F6P by this route is energetically costly (1 ATP) in comparison to the F6P aldolase route, described above and shown in FIG. 1. DHA kinase also competes with F6P aldolase for the DHA substrate. Attenuation of endogenous DHA kinase activity will thus improve the energetics of formaldehyde assimilation pathways, and also increase the intracellular availability of DHA for DHA synthase. DHA kinases of *Saccharomyces cerevisiae*, encoded by DAK1 and DAK2, enable the organism to maintain low intracellular levels of DHA (Molin et al, *J Biol Chem* 278:1415-23 (2003)). In methylotrophic yeasts DHA kinase is essential for growth on methanol (Luers et al, *Yeast* 14:759-71(1998)). The DHA kinase enzymes of *Hansenula polymorpha* and *Pichia pastoris* are encoded by DAK (van der Klei et al, *Curr Genet* 34:1-11(1998); Luers et al, supra). DAK enzymes in other organisms can be identified by sequence similarity to known enzymes.

Protein	GenBank ID	GI Number	Organism
DAK1	NP_013641.1	6323570	<i>Saccharomyces cerevisiae</i>
DAK2	NP_116602.1	14318466	<i>Saccharomyces cerevisiae</i>
DAK	AAC27705.1	3171001	<i>Hansenula polymorpha</i>
DAK	AAC39490.1	3287486	<i>Pichia pastoris</i>
DAK2	XP_505199.1	50555582	<i>Yarrowia lipolytica</i>

Methanol Oxidase

Attenuation of redox-inefficient endogenous methanol oxidizing enzymes, combined with increased expression of a cytosolic NADH-dependent methanol dehydrogenase, will enable redox-efficient oxidation of methanol to formaldehyde in the cytosol. Methanol oxidase, also called alcohol oxidase (EC 1.1.3.13), catalyzes the oxygen-dependent oxidation of methanol to formaldehyde and hydrogen peroxide. In eukaryotic organisms, alcohol oxidase is localized in the

10 peroxisome. Exemplary methanol oxidase enzymes are encoded by AOD of *Candida boidinii* (Sakai and Tani, *Gene* 114:67-73 (1992)); and AOX of *H. polymorpha*, *P. methanolica* and *P. pastoris* (Ledeboer et al, *Nucl Ac Res* 13:3063-82 (1985); Koutz et al, *Yeast* 5:167-77 (1989); Nakagawa et al, *Yeast* 15:1223-1230 (1999)).

Protein	GenBank ID	GI Number	Organism
AOX2	AAF02495.1	6049184	<i>Pichia methanolica</i>
AOX1	AAF02494.1	6049182	<i>Pichia methanolica</i>
AOX1	AAB57849.1	2104961	<i>Pichia pastoris</i>
AOX2	AAB57850.1	2104963	<i>Pichia pastoris</i>
AOX	P04841.1	113652	<i>Hansenula polymorpha</i>
AOD1	Q00922.1	231528	<i>Candida boidinii</i>
AOX1	AAQ99151.1	37694459	<i>Ogataea pini</i>

PQQ-Dependent Methanol Dehydrogenase

PQQ-dependent methanol dehydrogenase from *M. extorquens* (mxalF) uses cytochrome as an electron carrier (Nunn et al, *Nucl Acid Res* 16:7722 (1988)). Methanol dehydrogenase enzymes of methanotrophs such as *Methylococcus capsulatus* function in a complex with methane monooxygenase (MMO) (Myronova et al, *Biochem* 45:11905-14 (2006)). Note that of accessory proteins, cytochrome CL and PQQ biosynthesis enzymes are needed for active methanol dehydrogenase. Attenuation of one or more of these required accessory proteins, or retargeting the enzyme to a different cellular compartment, would also have the effect of attenuating PQQ-dependent methanol dehydrogenase activity.

Protein	GenBank ID	GI Number	Organism
MCA0299	YP_112833.1	53802410	<i>Methylococcus capsulatus</i>
MCA0782	YP_113284.1	53804880	<i>Methylococcus capsulatus</i>
mxal	YP_002965443.1	240140963	<i>Methylobacterium extorquens</i>
mxalF	YP_002965446.1	240140966	<i>Methylobacterium extorquens</i>

DHA Synthase and Other Competing Formaldehyde Assimilation and Dissimilation Pathways

Carbon-efficient formaldehyde assimilation can be improved by attenuation of competing formaldehyde assimilation and dissimilation pathways. Exemplary competing assimilation pathways in eukaryotic organisms include the peroxisomal dissimilation of formaldehyde by DHA synthase, and the DHA kinase pathway for converting DHA to

F6P, both described herein. Exemplary competing endogenous dissimilation pathways include one or more of the enzymes shown in FIG. 1.

Methylotrophic yeasts normally target selected methanol assimilation and dissimilation enzymes to peroxisomes during growth on methanol, including methanol oxidase, DHA synthase and S-(hydroxymethyl)-glutathione synthase (see review by Yurimoto et al, supra). The peroxisomal targeting mechanism comprises an interaction between the peroxisomal targeting sequence and its corresponding peroxisomal receptor (Lametschwandtner et al, J Biol Chem 273:33635-43 (1998)). Peroxisomal methanol pathway enzymes in methylotrophic organisms contain a PTS1 targeting sequence which binds to a peroxisomal receptor, such as Pex5p in *Candida boidinii* (Horiguchi et al, J Bacteriol 183:6372-83 (2001)). Disruption of the PTS1 targeting sequence, the Pex5p receptor and/or genes involved in peroxisomal biogenesis would enable cytosolic expression of DHA synthase, S-(hydroxymethyl)-glutathione synthase or other methanol-inducible peroxisomal enzymes. PTS1 targeting sequences of methylotrophic yeast are known in the art (Horiguchi et al, supra). Identification of peroxisomal targeting sequences of unknown enzymes can be predicted using bioinformatic methods (eg. Neuberger et al, J Mol Biol 328:581-92 (2003)).

Example XXIII

Methanol Assimilation via Methanol Dehydrogenase and the Ribulose Monophosphate Pathway

This example shows that co-expression of an active methanol dehydrogenase (MeDH) and the enzymes of the Ribulose Monophosphate (RuMP) pathway can effectively assimilate methanol derived carbon.

An experimental system was designed to test the ability of a MeDH in conjunction with the enzymes H6P synthase (HPS) and 6-phospho-3-hexuloisomerase (PHI) of the RuMP pathway to assimilate methanol carbon into the glycolytic pathway and the TCA cycle. *Escherichia coli* strain ECh-7150 (Δ lacIA, Δ pflB, Δ ptsI, Δ PpckA(pckA), Δ Pglk(glk), glk::glfB, Δ hycE, Δ frmR, Δ frmA, Δ frmB) was constructed to remove the glutathione-dependent formaldehyde detoxification capability encoded by the FrmA and FrmB enzyme. This strain was then transformed with plasmid pZA23S variants that either contained or lacked gene

2616A encoding a fusion of the HPS and PHI enzymes. These two transformed strains were then each transformed with pZS*13S variants that contained gene 2315L (encoding an active MeDH), or gene 2315 RIP2 (encoding a catalytically inactive MeDH), or no gene insertion. Genes 2315 and 2616 are internal nomenclatures for NAD-dependent methanol dehydrogenase from *Bacillus methanolicus* MGA3 and 2616 is a fused phs-lpi constructs as described in Orita et al. (2007) *Appl Microbiol Biotechnol* 76:439-45.

The six resulting strains were aerobically cultured in quadruplicate, in 5 ml minimal medium containing 1% arabinose and 0.6 M ¹³C-methanol as well as 100 ug/ml carbenicillin and 25 μ g/ml kanamycin to maintain selection of the plasmids, and 1 mM IPTG to induce expression of the methanol dehydrogenase and HPS—PHI fusion enzymes. After 18 hours incubation at 37° C., the cell density was measured spectrophotometrically at 600 nM wavelength and a clarified sample of each culture medium was submitted for analysis to detect evidence of incorporation of the labeled methanol carbon into TCA-cycle derived metabolites. The label can be further enriched by deleting the gene araD that competes with ribulose-5-phosphate.

¹³C carbon derived from labeled methanol provided in the experiment was found to be significantly enriched in the metabolites pyruvate, lactate, succinate, fumarate, malate, glutamate and citrate, but only in the strain expressing both catalytically active MeDH 2315L and the HPS—PHI fusion 2616A together (data not shown). Moreover, this strain grew significantly better than the strain expressing catalytically active MeDH but lacking expression of the HPS—PHI fusion (data not shown), suggesting that the HPS—PHI enzyme is capable of reducing growth inhibitory levels of formaldehyde that cannot be detoxified by other means in this strain background. These results show that co-expression of an active MeDH and the enzymes of the RuMP pathway can effectively assimilate methanol derived carbon and channel it into TCA-cycle derived products.

Throughout this application various publications have been referenced. The disclosures of these publications in their entireties, including GenBank and GI number publications, are hereby incorporated by reference in this application in order to more fully describe the state of the art to which this invention pertains. Although the invention has been described with reference to the examples provided above, it should be understood that various modifications can be made without departing from the spirit of the invention.

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 20 25 30

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 35 40 45

Gly Cys Ile Pro Ser Lys Ala Leu Leu His Val Ala Lys Val Ile Glu
 50 55 60

Glu Ala Lys Ala Leu Ala Glu His Gly Ile Val Phe Gly Glu Pro Lys
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Thr Asp Ile Asp Lys Ile Arg Thr Trp Lys Glu Lys Val Ile Asn Gln
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Leu Thr Gly Gly Leu Ala Gly Met Ala Lys Gly Arg Lys Val Lys Val
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Val Asn Gly Leu Gly Lys Phe Thr Gly Ala Asn Thr Leu Glu Val Glu
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Gly Glu Asn Gly Lys Thr Val Ile Asn Phe Asp Asn Ala Ile Ile Ala
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Ala Gly Ser Arg Pro Ile Gln Leu Pro Phe Ile Pro His Glu Asp Pro
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Arg Ile Trp Asp Ser Thr Asp Ala Leu Glu Leu Lys Glu Val Pro Glu
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Val Tyr His Ala Leu Gly Ser Gln Ile Asp Val Val Val Arg Lys His
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Gln Val Ile Arg Ala Ala Asp Lys Asp Ile Val Lys Val Phe Thr Lys
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Arg Ile Ser Lys Lys Phe Asn Leu Met Leu Glu Thr Lys Val Thr Ala
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Val Glu Ala Lys Glu Asp Gly Ile Tyr Val Thr Met Glu Gly Lys Lys
 245 250 255

Ala Pro Ala Glu Pro Gln Arg Tyr Asp Ala Val Leu Val Ala Ile Gly
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355 360 365

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Cys Asp Ala Glu Asp Ile Ala Leu Thr Ile His Ala His Pro Thr Leu
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35 40 45

Gly Cys Ile Pro Ser Lys Ala Leu Leu His Val Ala Lys Val Ile Glu
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Glu Ala Lys Ala Leu Ala Glu His Gly Ile Val Phe Gly Glu Pro Lys
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Val Tyr His Ala Leu Gly Ser Gln Ile Asp Val Val Val Arg Lys His

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	260	265 270
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Gly Lys Lys His Tyr Phe Asp Pro Lys Val Ile Pro Ser Ile Ala Tyr		
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Thr Glu Pro Glu Val Ala Trp Val Gly Leu Thr Glu Lys Glu Ala Lys		
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Phe Asp Lys Glu Ser His Arg Val Ile Gly Gly Ala Ile Val Gly Thr		
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Asn Gly Gly Glu Leu Leu Gly Glu Ile Gly Leu Ala Ile Glu Met Gly		
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Cys Asp Ala Glu Asp Ile Ala Leu Thr Ile His Ala His Pro Thr Leu		
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His	Ile	Asp	Leu	Asp	Tyr	Ile	Asn	Gln	Thr	Gly	Lys	Lys	Ile	Ser	Val	
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Asp	Lys	Asp	Pro	Glu	Pro	Thr	Ser	Thr	Val	Asp	Arg	Glu	Leu	Lys	Asp
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Thr Val Thr Glu Val Thr Gly Ser Asn Val Val Ser Val Ala Glu His	115	120	125
Val Met Ala Thr Ile Leu Val Leu Ile Arg Asn Tyr Asn Gly Gly His	130	135	140
Gln Gln Ala Ile Asn Gly Glu Trp Asp Ile Ala Gly Val Ala Lys Asn	145	150	155
Glu Tyr Asp Leu Glu Asp Lys Ile Ile Ser Thr Val Gly Ala Gly Arg	165	170	175
Ile Gly Tyr Arg Val Leu Glu Arg Leu Val Ala Phe Asn Pro Lys Lys	180	185	190
Leu Leu Tyr Tyr Ala Arg Gln Glu Leu Pro Ala Glu Ala Ile Asn Arg	195	200	205
Leu Asn Glu Ala Ser Lys Leu Phe Asn Gly Arg Gly Asp Ile Val Gln	210	215	220
Arg Val Glu Lys Leu Glu Asp Met Val Ala Gln Ser Asp Val Val Thr	225	230	235
Ile Asn Cys Pro Leu His Lys Asp Ser Arg Gly Leu Phe Asn Lys Lys	245	250	255
Leu Ile Ser His Met Lys Asp Gly Ala Tyr Leu Val Asn Thr Ala Arg	260	265	270
Gly Ala Ile Cys Val Ala Glu Asp Val Ala Glu Ala Val Lys Ser Gly	275	280	285
Lys Leu Ala Gly Tyr Gly Gly Asp Val Trp Asp Lys Gln Pro Ala Pro	290	295	300
Lys Asp His Pro Trp Arg Thr Met Asp Asn Lys Asp His Val Gly Asn	305	310	315
Ala Met Thr Val His Ile Ser Gly Thr Ser Leu Asp Ala Gln Lys Arg	325	330	335
Tyr Ala Gln Gly Val Lys Asn Ile Leu Asn Ser Tyr Phe Ser Lys Lys	340	345	350
Phe Asp Tyr Arg Pro Gln Asp Ile Ile Val Gln Asn Gly Ser Tyr Ala	355	360	365
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Val Phe Lys Val
          20

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What is claimed is:

1. A non-naturally occurring microbial organism having:

- (i) a formaldehyde fixation pathway; and
(ii) a methanol metabolic pathway, and
a fatty acyl-ACP elongation (FAACPE) cycle in combi- 50
nation with a termination pathway,

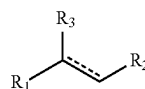
wherein said formaldehyde fixation pathway comprises a
3-hexulose-6-phosphate synthase and a 6-phospho-3-
hexuloisomerase

wherein said methanol metabolic pathway comprises a 55
methanol dehydrogenase,

wherein said FAACPE cycle comprises one or more
 β -ketoacyl-ACP synthase, one or more β -ketoacyl-
ACP reductase, one or more β -hydroxyacyl-ACP
reductase, and one or more enoyl ACP-reductase, 60

wherein said termination pathway comprises a thio-
esterase

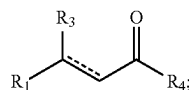
wherein an enzyme of the formaldehyde fixation pathway,
the methanol metabolic pathway, the FAACPE cycle or
the termination pathway is encoded by at least one 65
exogenous nucleic acid and is expressed in a sufficient
amount to produce a compound of Formula (I):



(I)

wherein R_1 is C_{1-24} linear alkyl; R_2 is H; and ----- rep-
resents a single,

wherein the substrate of each of said enzymes of the
FAACPE cycle and the termination pathway are inde-
pendently selected from a compound of Formula (II) or
malonyl-ACP:



(II)

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wherein R_1 is C_{1-24} linear alkyl; R_3 is H, OH, or oxo (=O); R_4 is S-CoA, ACP, OH or H; and ----- represents a single bond with the proviso that the valency of the carbon atom to which R_3 is attached is four, wherein said one or more enzymes of the FAACPE cycle are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no greater than the number of carbon atoms at R_1 of said compound of Formula (I), and wherein said one or more enzymes of the termination pathway are each selective for a compound of Formula (II) having a number of carbon atoms at R_1 that is no less than the number of carbon atoms at R_1 of said compound of Formula (I).

2. The non-naturally occurring microbial organism of claim 1, wherein said microbial organism further comprises a formate assimilation pathway.

3. The non-naturally occurring microbial organism of claim 1, wherein the microbial organism further comprises an acetoacetyl-ACP pathway of:

(1) 12A, 12B, and 12C; or (2) 12A, 12B, and 12D, wherein 12A is an acetyl-CoA carboxylase, wherein 12B is malonyl-CoA ACP transacylase, wherein 12C is an acetoacetyl-ACP synthase, and wherein 12D is a β -ketoacetyl-ACP synthase, or

wherein the microbial organism further comprises a 3-oxovalery-ACP pathway comprising an acetyl-CoA carboxylase, a malonyl-CoA ACP transacylase, and a β -ketoacetyl-ACP synthase.

4. The non-naturally occurring microbial organism of 3, wherein an enzyme of the acetoacetyl-ACP pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce acetoacetyl-ACP, wherein an enzyme of the 3-oxovalery-ACP pathway is encoded by at least one exogenous nucleic acid and is expressed in a sufficient amount to produce 3-oxovalery-ACP, and wherein the acetoacetyl-ACP or the 3-oxovalery-ACP is a β -ketoacetyl-ACP of the FAACPE cycle.

5. The non-naturally occurring microbial organism of claim 1, wherein R_1 is C_9 linear alkyl, C_{10} linear alkyl, C_n linear alkyl, C_{12} linear alkyl or Cis linear alkyl.

6. The non-naturally occurring microbial organism of claim 1, wherein said microbial organism comprises two, three, or four exogenous nucleic acids each encoding an enzyme of said FAACPE cycle or said termination pathway.

7. The non-naturally occurring microbial organism of claim 1, wherein said microbial organism comprises one, two, or three exogenous nucleic acids each encoding a formaldehyde fixation pathway enzyme, or a methanol metabolic pathway enzyme.

8. The non-naturally occurring microbial organism of claim 1, wherein said microbial organism comprises exogenous nucleic acids encoding each of the enzymes 3-hexulose-6-phosphate synthase, 6-phospho-3-hexuloisomerase, and methanol dehydrogenase.

9. The non-naturally occurring microbial organism of claim 1, wherein said organism comprises at least one exogenous nucleic acid encoding the methanol dehydrogenase.

10. The non-naturally occurring microbial organism of claim 1, wherein said microbial organism further comprises 3H or 3P, wherein 3H is a hydrogenase, wherein 3P a carbon monoxide dehydrogenase.

11. The non-naturally occurring microbial organism of claim 1, wherein said at least one exogenous nucleic acid encoding said formaldehyde fixation pathway enzyme, said

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methanol metabolic pathway enzyme, said FAACPE cycle enzyme, or said termination pathway enzyme is a heterologous nucleic acid.

12. The non naturally occurring microbial organism of claim 1, wherein said enzyme of the formaldehyde fixation pathway, methanol metabolic pathway, FAACPE cycle or termination pathway is expressed in a sufficient amount to produce a compound of Formula (XIV):



wherein

R_1 is C_9 linear alkyl, C_{10} linear alkyl, C_{11} linear alkyl, C_{12} linear alkyl or Cis linear alkyl.

13. The non-naturally occurring microbial organism of claim 1, wherein said microbial organism further comprises an acetyl-CoA pathway and at least one exogenous nucleic acid encoding an acetyl-CoA pathway enzyme expressed in a sufficient amount to produce or enhance carbon flux through acetyl-CoA, wherein said acetyl-CoA pathway comprises a pathway selected from:

(1) 3A and 3B; (2) 3A, 3C, and 3D; (3) 3H; (4) 3G and 3D; (5) 3E, 3F and 3B; (6) 3E and 3I; (7) 3J, 3F and 3B; (8) 3J and 3I; (9) 4A, 4B, and 4C; (10) 4A, 4B, 4J, 4K, and 4D; (11) 4A, 4B, 4G, and 4D; (12) 4A, 4F, and 4D; (13) 4N, 4H, 4B and 4C; (14) 4N, 4H, 4B, 4J, 4K, and 4D; (15) 4N, 4H, 4B, 4G, and 4D; (16) 4N, 4H, 4F, and 4D; (17) 4L, 4M, 4B and 4C; (18) 4L, 4M, 4B, 4J, 4K, and 4D; (19) 4L, 4M, 4B, 4G, and 4D; (20) 4L, 4M, 4F, and 4D; (21) 5A, 5B, 5D, 5H, 5I, and 5J; (22) 5A, 5B, 5E, 5F, 5H, 5I, and 5J; (23) 5A, 5B, 5E, 5K, 5L, 5H, 5I, and 5J; (24) 5A, 5C, 5D, 5H, and 5J; (25) 5A, 5C, 5E, 5F, 5H, and 5J; (26) 5A, 5C, 5E, 5K, 5L, 5H, and 5J; (27) 6A, 6B, 6D, and 6G; (28) 6A, 6B, 6E, 6F, and 6G; (29) 6A, 6B, 6E, 6K, 6L, and 6G; (30) 6A, 6C, and 6D; (31) 6A, 6C, 6E, and 6F; (32) 6A, 6C, 6E, 6K, and 6L; (33) 1T and 1V; (34) 1T, 1W, and 1X; (35) 1U and 1V; and (36) 1U, 1W, and 1X,

wherein 3A is a pyruvate oxidase (acetate-forming), wherein 3B is an acetyl-CoA synthetase, an acetyl-CoA ligase or an acetyl-CoA transferase, wherein 3C is an acetate kinase, wherein 3D is a phosphotransacetylase, wherein 3E is a pyruvate decarboxylase, wherein 3F is an acetaldehyde dehydrogenase, wherein 3G is a pyruvate oxidase (acetyl-phosphate forming), wherein 3H is a pyruvate dehydrogenase, a pyruvate: ferredoxin oxidoreductase, a pyruvate: NAD(P)H oxidoreductase or a pyruvate formate lyase, wherein 3I is an acetaldehyde dehydrogenase (acylating), wherein 3J is a threonine aldolase, wherein 4A is a phosphoenolpyruvate (PEP) carboxylase or a PEP carboxykinase, wherein 4B is an oxaloacetate decarboxylase, wherein 4C is a malonate semialdehyde dehydrogenase (acetylating), wherein 4D is an acetyl-CoA carboxylase or a malonyl-CoA decarboxylase, wherein 4F is an oxaloacetate dehydrogenase or an oxaloacetate oxidoreductase, wherein 4G is a malonate semialdehyde dehydrogenase (acylating), wherein 4H is a pyruvate carboxylase, wherein 4J is a malonate semialdehyde dehydrogenase, wherein 4K is a malonyl-CoA synthetase or a malonyl-CoA transferase, wherein 4L is a malic enzyme, wherein 4M is a malate dehydrogenase or a malate oxidoreductase,

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wherein 4N is a pyruvate kinase or a PEP phosphatase, wherein 5A is a citrate synthase, wherein 5B is a citrate transporter, wherein 5C is a citrate/malate transporter, wherein 5D is an ATP citrate lyase, wherein 5E is a citrate lyase, wherein 5F is an acetyl-CoA synthetase or an acetyl-CoA transferase, wherein 5H is a cytosolic malate dehydrogenase, wherein 5I is a malate transporter, wherein 5J is a mitochondrial malate dehydrogenase, wherein 5K is an acetate kinase, wherein 5L is a phosphotransacetylase, wherein 6A is a citrate synthase, wherein 6B is a citrate transporter, wherein 6C is a citrate/oxaloacetate transporter, wherein 6D is an ATP citrate lyase, wherein 6E is a citrate lyase, wherein 6F is an acetyl-CoA synthetase or an acetyl-CoA transferase, wherein 6G is an oxaloacetate transporter, wherein 6K is an acetate kinase, and wherein 6L is a phosphotransacetylase, wherein 1T is a fructose-6-phosphate phosphoketolase, wherein 1U is a xylulose-5-phosphate phosphoketolase, wherein 1V is a phosphotransacetylase, wherein 1W is an acetate kinase, wherein 1X is an acetyl-CoA transferase, an acetyl-CoA synthetase, or an acetyl-CoA ligase.

14. The non-naturally occurring microbial organism of claim 13, wherein said microbial organism comprises two, three, four, five, six, seven or eight exogenous nucleic acids each encoding an acetyl-CoA pathway enzyme.

15. The non-naturally occurring microbial organism of claim 13, wherein said microbial organism comprises exogenous nucleic acids encoding each of the acetyl-CoA pathway enzymes of at least one of the pathways selected from (1)-(36).

16. The non-naturally occurring microbial organism of claim 1, further comprising one or more gene disruptions, said one or more gene disruptions occurring in endogenous genes encoding proteins or enzymes involved in: native production of ethanol, glycerol, acetate, formate, lactate, CO₂, fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native degradation of a FAACPE cycle intermediate or a termination pathway intermediate by said microbial organism, wherein said one or more gene disruptions confer increased production of the compound of Formula (I) in said microbial organism.

17. The non-naturally occurring microbial organism of claim 1, wherein one or more enzymes of the FAACPE cycle or the termination pathway preferentially react with an NADH cofactor or have reduced preference for reacting with an NAD(P)H cofactor, wherein said one or more enzymes of the FAACPE cycle are a 3-ketoacyl-ACP reductase or an enoyl-ACP reductase, and wherein said one or more enzymes of the termination pathway are selected from an acyl-CoA reductase (aldehyde forming), an alcohol dehydrogenase, an acyl-CoA reductase (alcohol forming), a fatty acyl-ACP reductase, and a carboxylic acid reductase.

18. The non-naturally occurring microbial organism of claim 1, further comprising one or more gene disruptions, said one or more gene disruptions occurring in genes encoding proteins or enzymes that result in an increased ratio of NAD(P)H to NAD(P) present in the cytosol of said microbial organism following said disruptions.

19. The non-naturally occurring microbial organism of claim 1, wherein said microbial organism is Crabtree positive and is in culture medium comprising excess glucose, thereby increasing the ratio of NAD(P)H to NAD(P) present in the cytosol of said microbial organism.

20. The non-naturally occurring microbial organism of claim 1, further comprising at least one exogenous nucleic

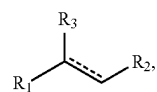
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acid encoding an extracellular transporter or an extracellular transport system for the compound of Formula (I).

21. The non-naturally occurring microbial organism of claim 1, wherein one or more endogenous enzymes involved in: native production of ethanol, glycerol, acetate, formate, lactate, CO₂, fatty acids, or malonyl-CoA by said microbial organism; transfer of pathway intermediates to cellular compartments other than the cytosol; or native degradation of a FAACPE cycle intermediate or a termination pathway intermediate by said microbial organism, has attenuated enzyme activity or expression levels.

22. The non-naturally occurring microbial organism of claim 1, wherein one or more endogenous enzymes involved in the oxidation of NAD(P)H or NADH, has attenuated enzyme activity or expression levels.

23. A method for producing a compound of Formula (I):



(I)

wherein R₁ is C₁₋₂₄ linear alkyl; R₂ is; and ----- represents a single bond, comprising culturing the non-naturally occurring microbial organism of claim 1 under conditions for a sufficient period of time to produce said compound of Formula (I).

24. The method of claim 23, wherein said method further comprises separating the compound of Formula (I) from other components in the culture.

25. The method of claim 24, wherein the separating comprises extraction, continuous liquid-liquid extraction, pervaporation, membrane filtration, membrane separation, reverse osmosis, electrodialysis, distillation, crystallization, centrifugation, extractive filtration, ion exchange chromatography, absorption chromatography, or ultrafiltration.

26. The non-naturally occurring microbial organism of claim 2, wherein said formate assimilation pathway comprises a pathway selected from:

(1) 1E; (2) 1F, and 1G; (3) 1H, 1I, 1J, and 1K; (4) 1H, 1I, 1J, 1L, 1M, and 1N; (5) 1E, 1H, 1I, 1J, 1L, 1M, and 1N; (6) 1F, 1G, 1H, 1I, 1J, 1L, 1M, and 1N; (7) 1K, 1H, 1I, 1J, 1L, 1M, and 1N; and (8) 1H, 1I, 1J, 1O, and 1P,

wherein 1E is a formate reductase, 1F is a formate ligase, a formate transferase, or a formate synthetase, wherein 1G is a formyl-CoA reductase, wherein 1H is a formyltetrahydrofolate synthetase, wherein 1I is a methenyltetrahydrofolate cyclohydrolase, wherein 1J is a methylenetetrahydrofolate dehydrogenase, wherein 1K is a formaldehyde-forming enzyme or spontaneous, wherein 1L is a glycine cleavage system, wherein 1M is a serine hydroxymethyltransferase, wherein 1N is a serine deaminase, wherein 1O is a methylenetetrahydrofolate reductase, and wherein 1P is an acetyl-CoA synthase.

27. The non-naturally occurring microbial organism of claim 1, wherein said termination pathway further comprises an acyl-CoA synthase and an acyl-CoA reductase for production of a fatty aldehyde.

28. The non-naturally occurring microbial organism of claim 27, wherein said termination pathway further comprises a fatty aldehyde reductase for production of a fatty alcohol.

* * * * *